

# Fundamental Theorem of Matrix Product States

## A Formalization Blueprint

Authors

Last update: June 9, 2026

# Contents

|          |                                                                   |           |
|----------|-------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                               | <b>7</b>  |
| <b>2</b> | <b>Matrix Product Vectors</b>                                     | <b>9</b>  |
| 2.1      | Basic definitions                                                 | 9         |
| 2.2      | Site-periodic tensor families                                     | 11        |
| 2.3      | Gauge equivalence and same MPV                                    | 11        |
| 2.4      | Injectivity and normality                                         | 13        |
| 2.5      | Canonical form                                                    | 14        |
| 2.6      | Blocking                                                          | 15        |
| 2.7      | MPV overlap                                                       | 16        |
| <b>3</b> | <b>The Single-Block Fundamental Theorem</b>                       | <b>17</b> |
| 3.1      | The multiplicative linear extension                               | 17        |
| 3.2      | Inner automorphism and the single-block theorem                   | 18        |
| <b>4</b> | <b>Quantum Channels and Positive Maps</b>                         | <b>20</b> |
| 4.1      | Positive maps and channels                                        | 20        |
| 4.2      | Irreducibility                                                    | 21        |
| 4.3      | Cesàro fixed-point theorem                                        | 22        |
| 4.4      | Fixed-point projection                                            | 22        |
| 4.5      | Peripheral spectrum and primitivity                               | 23        |
| 4.5.1    | Periodicity removal by powering                                   | 24        |
| 4.5.2    | The Kadison–Schwarz inequality and the multiplicative domain      | 24        |
| 4.5.3    | Peripheral closure via adjoint fixed point                        | 26        |
| 4.6      | Fixed-point algebra                                               | 27        |
| <b>5</b> | <b>Schwarz Inequalities and Multiplicative Domains</b>            | <b>30</b> |
| 5.1      | Kadison–Schwarz inequality                                        | 30        |
| 5.1.1    | Two-positive maps and the generalized Schwarz inequality          | 31        |
| 5.1.2    | Douglas-type factorization lemmas                                 | 32        |
| 5.2      | Multiplicative domain                                             | 32        |
| 5.2.1    | Full multiplicative-domain structure                              | 33        |
| 5.2.2    | Schwarz inequality for normal operators                           | 34        |
| 5.2.3    | Schwarz inequality for subnormal and commuting-dominant operators | 35        |
| 5.2.4    | Positive maps preserve order and spectral intervals               | 36        |
| 5.2.5    | A positive Schwarz map that is not completely positive            | 36        |

|          |                                                                                   |           |
|----------|-----------------------------------------------------------------------------------|-----------|
| <b>6</b> | <b>Perron–Frobenius Theory for Channels and Transfer Maps</b>                     | <b>38</b> |
| 6.1      | Positive definiteness                                                             | 38        |
| 6.2      | Uniqueness                                                                        | 39        |
| 6.3      | Existence and the Perron–Frobenius theorem                                        | 40        |
| 6.4      | Right- and left-canonical gauges                                                  | 40        |
| 6.5      | Similarity preserves irreducibility                                               | 41        |
| 6.6      | Perron–Frobenius eigenvector existence                                            | 42        |
| 6.7      | Exponential positivity for irreducible CP maps                                    | 43        |
| 6.8      | Ergodicity of irreducible channels                                                | 44        |
| 6.9      | Spectral radius at the Perron eigenvalue                                          | 44        |
| 6.10     | Spectral characterization of irreducibility                                       | 45        |
| <b>7</b> | <b>Transfer-Operator Gaps and Block Separation</b>                                | <b>46</b> |
| 7.1      | Mixed transfer operator                                                           | 46        |
| 7.2      | MPV overlap as transfer trace                                                     | 47        |
| 7.3      | Frobenius norm estimates                                                          | 48        |
| 7.4      | Eigenvalue bound and transfer-operator gap                                        | 48        |
| 7.5      | Transfer-operator gap — rectangular case                                          | 50        |
| 7.6      | MPV overlap decay                                                                 | 51        |
| 7.7      | Block separation                                                                  | 51        |
| 7.8      | Transfer-operator gap under irreducible-TP hypotheses                             | 51        |
| 7.9      | Conditional expectation from a faithful fixed point                               | 53        |
| 7.10     | Stationary support                                                                | 54        |
| 7.10.1   | Faithful compression onto the support sector                                      | 55        |
| 7.11     | Wedderburn decomposition of the fixed-point algebra                               | 55        |
| 7.12     | Further irreducibility and primitivity equivalences                               | 57        |
| 7.13     | Multi-cycle block-permutation structure                                           | 57        |
| 7.14     | Peripheral eigenvalue group structure                                             | 58        |
| 7.14.1   | Cyclic decomposition of irreducible Schwarz maps                                  | 59        |
| 7.14.2   | Group structure of the peripheral spectrum                                        | 60        |
| 7.15     | Primitive overlap convergence (complementary transfer-map gap form)               | 60        |
| <b>8</b> | <b>Wielandt Bound</b>                                                             | <b>62</b> |
| 8.1      | Cumulative span                                                                   | 62        |
| 8.2      | Fitting decomposition                                                             | 63        |
| 8.3      | Nonzero trace product                                                             | 64        |
| 8.4      | Eigenvector extraction                                                            | 65        |
| 8.5      | Eigenvector spreading                                                             | 65        |
| 8.6      | Proof of the cumulative Wielandt bound                                            | 66        |
| 8.7      | Fixed-length matrix spanning                                                      | 69        |
| 8.8      | Blocked tensor and rectangular span                                               | 71        |
| 8.9      | Primitive MPS tensors                                                             | 73        |
| 8.10     | Primitivity and normality                                                         | 73        |
| 8.10.1   | From primitivity to normality                                                     | 74        |
| 8.10.2   | Primitive implies irreducible                                                     | 74        |
| 8.10.3   | From irreducibility to full spanning via Burnside’s theorem                       | 75        |
| 8.10.4   | From primitivity to strong irreducibility and normality                           | 76        |
| 8.11     | Cumulative span to word span                                                      | 76        |
| 8.12     | Block injectivity from TP/primitive/irreducible blocks (relocated from Chapter 9) | 78        |

|           |                                                                            |            |
|-----------|----------------------------------------------------------------------------|------------|
| <b>9</b>  | <b>Canonical Form Reduction</b>                                            | <b>79</b>  |
| 9.1       | Normal tensors and canonical forms                                         | 80         |
| 9.1.1     | Normalization conventions                                                  | 81         |
| 9.2       | Invariant-subspace decomposition and irreducible blocks                    | 81         |
| 9.2.1     | Iterated reduction to irreducible blocks                                   | 83         |
| 9.3       | Normal canonical form                                                      | 84         |
| 9.3.1     | Canonical form from primitivity                                            | 85         |
| 9.4       | Left-canonical and dual diagonalization of irreducible blocks              | 86         |
| 9.5       | Zero-block separation and the trace-preserving gauge                       | 87         |
| 9.5.1     | Trace-preserving gauge for arbitrary tensors                               | 88         |
| 9.6       | Blocking, period removal, and primitive blocks                             | 88         |
| 9.6.1     | Period removal by cyclic sectors                                           | 89         |
| 9.6.2     | Reduction to primitive blocks                                              | 91         |
| 9.7       | Blocking and weighted direct sums                                          | 93         |
| <b>10</b> | <b>Basis of Normal Tensors</b>                                             | <b>95</b>  |
| 10.1      | Bases of normal tensors                                                    | 95         |
| 10.2      | Permutation rigidity for bases of normal tensors                           | 97         |
| 10.3      | Newton–Girard identities and power-sum recovery                            | 99         |
| 10.4      | The BNT canonical form on a sector decomposition                           | 100        |
| 10.4.1    | Sector decompositions and phase matching                                   | 100        |
| 10.4.2    | Prepared-block construction of BNT canonical forms                         | 102        |
| 10.5      | Coefficient comparison and copy-weight recovery                            | 105        |
| 10.5.1    | Matched-sector weight multiset equality                                    | 105        |
| 10.6      | Strong existential matching by exact linear independence                   | 106        |
| 10.6.1    | Bijective matching by symmetry                                             | 107        |
| <b>11</b> | <b>Proof of the Fundamental Theorem</b>                                    | <b>108</b> |
| 11.1      | Coefficient identity, copy weights, and the block-diagonal gauge           | 108        |
| 11.2      | The equal and proportional cases                                           | 113        |
| <b>12</b> | <b>Symmetries and String Order</b>                                         | <b>115</b> |
| 12.1      | Physical Symmetries Give Virtual Gauges                                    | 115        |
| 12.2      | Virtual Representation Theorem                                             | 115        |
| 12.3      | Cohomology Classes of Virtual Symmetry Cocycles                            | 118        |
| 12.3.1    | Non-triviality via the commutator phase                                    | 119        |
| 12.4      | Permutation-Based Physical Symmetry                                        | 119        |
| 12.5      | String Order and Local Symmetry Equivalence                                | 120        |
| 12.6      | SPT Phase Classification                                                   | 125        |
| <b>13</b> | <b>Periodic MPS: irreducible form and the periodic Fundamental Theorem</b> | <b>127</b> |
| 13.1      | Periodic irreducible form and physical-index rotation                      | 127        |
| 13.1.1    | The $Z$ -gauge degree of freedom                                           | 128        |
| 13.2      | Periodic Fundamental Theorem                                               | 129        |
| 13.2.1    | Common-period blocking lemmas                                              | 130        |
| 13.2.2    | Statement of the periodic Fundamental Theorem                              | 130        |
| 13.2.3    | Periodic overlap dichotomy                                                 | 131        |
| 13.3      | Physical symmetries on periodic MPS                                        | 137        |
| 13.4      | Refinement and divisibility                                                | 139        |

|                                                                               |            |
|-------------------------------------------------------------------------------|------------|
| <b>14 The Algebraic Fundamental Theorem</b>                                   | <b>142</b> |
| 14.1 Non-translation-invariant periodic chains                                | 142        |
| 14.2 One-sided inverse and physical realisation of virtual insertions         | 143        |
| 14.2.1 Physical realisation of virtual insertions                             | 144        |
| 14.3 The virtual bond gauge and the proportionality lemma                     | 145        |
| 14.3.1 Two preliminary facts                                                  | 145        |
| 14.3.2 Same state forces a virtual bond gauge                                 | 145        |
| 14.3.3 Aligned gauges force proportionality                                   | 146        |
| 14.4 The Fundamental Theorem for injective chains                             | 147        |
| 14.4.1 From fixed-length chain states to combined-tensor MPV agreement        | 147        |
| 14.4.2 Finite-length MPV agreement                                            | 148        |
| 14.5 Corollaries: translation-invariant, non-uniform, and blocked chains      | 149        |
| 14.5.1 Translation-invariant chains                                           | 149        |
| 14.5.2 Non-translation-invariant descriptions of translation-invariant states | 149        |
| 14.5.3 Blocked chains                                                         | 150        |
| 14.6 Product algebra construction and per-block gauge                         | 150        |
| 14.6.1 Block permutation algebra                                              | 151        |
| 14.6.2 Per-block linear extension and the product automorphism                | 152        |
| 14.6.3 Nondegeneracy of the product trace pairing                             | 153        |
| 14.7 Gauge equivalence for direct sums                                        | 153        |
| 14.7.1 Same-index direct-sum corollaries from the single-block theorem        | 154        |
| 14.7.2 MPV invariance under reindexing                                        | 156        |
| 14.7.3 Converse: gauge equivalence implies same MPV                           | 156        |
| 14.7.4 Canonical-form tensor as a weighted direct sum                         | 156        |
| 14.7.5 Equal-case block matching with differing bond dimensions               | 157        |
| 14.8 Translation-invariant reduction and global symmetry                      | 157        |
| 14.8.1 Global symmetry via gauge composition                                  | 158        |
| <b>15 Fundamental Theorems for Injective and Normal PEPS</b>                  | <b>159</b> |
| 15.1 Normal PEPS                                                              | 187        |
| <b>16 Parent Hamiltonians</b>                                                 | <b>203</b> |
| 16.1 Local ground space $G_L(A)$                                              | 203        |
| 16.2 Parent interaction and chain Hamiltonian                                 | 204        |
| 16.3 Intersection property and unique ground state                            | 206        |
| 16.3.1 Restriction maps                                                       | 206        |
| 16.3.2 Forward direction: ground space restricts                              | 207        |
| 16.3.3 Injectivity of the ground-space map                                    | 207        |
| 16.3.4 The intersection property                                              | 208        |
| 16.3.5 Unique ground state                                                    | 209        |
| 16.3.6 Block-word stripping                                                   | 218        |
| 16.4 Commuting parent Hamiltonians                                            | 231        |
| 16.5 Decorrelation and commuting parent Hamiltonians                          | 232        |
| 16.6 Spectral gap                                                             | 235        |
| 16.7 Ground space of non-injective MPS                                        | 244        |

|                                                                                        |            |
|----------------------------------------------------------------------------------------|------------|
| <b>17 Exponential Decay of Correlations</b>                                            | <b>264</b> |
| 17.1 Connected correlators from the transfer map                                       | 264        |
| 17.2 Spectral expansion and exponential decay                                          | 265        |
| 17.3 Quantitative transfer-map gap bounds                                              | 266        |
| 17.4 Relation to parent Hamiltonians                                                   | 266        |
| 17.5 Renormalization fixed points and zero correlation length                          | 267        |
| <b>18 Concrete Examples</b>                                                            | <b>270</b> |
| 18.1 The GHZ state                                                                     | 270        |
| 18.2 The AKLT state                                                                    | 271        |
| 18.3 The cluster state                                                                 | 273        |
| 18.4 Parent Hamiltonian statements for the examples                                    | 274        |
| <b>19 Channel Representations and Normal Forms</b>                                     | <b>277</b> |
| 19.1 Representations                                                                   | 277        |
| 19.2 Choi–Jamiołkowski isomorphism                                                     | 278        |
| 19.3 Representation corollaries for channel decompositions                             | 278        |
| 19.4 Kraus representation theorem                                                      | 280        |
| 19.5 Stinespring dilation                                                              | 282        |
| 19.6 Ordered CP maps, Radon–Nikodym, and open-system representation                    | 283        |
| 19.7 POVMs and Naimark dilation                                                        | 285        |
| 19.8 Trace-pairing expansion in transfer-matrix form                                   | 287        |
| 19.9 SVD normal form (existence)                                                       | 287        |
| 19.10 Lorentz normal form                                                              | 288        |
| 19.11 Determinant of a quantum channel                                                 | 289        |
| <b>20 Quantum Dynamical Semigroups</b>                                                 | <b>291</b> |
| 20.1 Dynamical semigroups and the exponential form                                     | 291        |
| 20.2 Perturbation theory                                                               | 293        |
| 20.3 GKSL/Lindblad generators                                                          | 295        |
| 20.3.1 Generator decomposition and conditional complete positivity                     | 295        |
| 20.3.2 The Lindblad form                                                               | 295        |
| 20.3.3 Characterization of conditional complete positivity                             | 296        |
| 20.3.4 Completely positive semigroups and conditionally completely positive generators | 296        |
| 20.3.5 Freedom in generator representation                                             | 297        |
| 20.3.6 Uniqueness of the traceless Lindblad form                                       | 297        |
| 20.3.7 Trace-annihilation and trace preservation                                       | 298        |
| 20.3.8 The GKSL/Lindblad theorem                                                       | 298        |
| 20.3.9 Kossakowski matrix form                                                         | 299        |
| 20.4 Primitivity and irreducibility of QDS                                             | 299        |
| 20.4.1 Auxiliary spectral and semigroup lemmas                                         | 299        |
| 20.5 Kernel of the adjoint Liouvillian                                                 | 303        |
| 20.6 Reducibility of quantum dynamical semigroups                                      | 304        |
| 20.6.1 Sufficient conditions for non-reducibility                                      | 306        |

|                                                                        |            |
|------------------------------------------------------------------------|------------|
| <b>21 Operator Convexity and Jensen Inequalities</b>                   | <b>308</b> |
| 21.1 Diagonal Jensen inequality . . . . .                              | 308        |
| 21.2 Trace concavity and convexity of matrix powers . . . . .          | 308        |
| 21.3 Operator Jensen inequality for positive maps . . . . .            | 309        |
| 21.4 Lieb concavity theorem . . . . .                                  | 311        |
| <b>22 Quantum Entropy</b>                                              | <b>312</b> |
| 22.1 Von Neumann entropy . . . . .                                     | 312        |
| 22.2 Tripartite partial traces . . . . .                               | 313        |
| 22.3 Strong subadditivity . . . . .                                    | 314        |
| 22.4 Mutual information . . . . .                                      | 314        |
| 22.5 Entropy formulations . . . . .                                    | 315        |
| 22.6 Mutual information: monotonicity and area-law bound . . . . .     | 315        |
| 22.7 Trivial-factor corollaries . . . . .                              | 316        |
| <b>23 Matrix Product Operators and Density Operators</b>               | <b>318</b> |
| 23.1 MPO tensors . . . . .                                             | 318        |
| 23.2 Transfer maps . . . . .                                           | 320        |
| 23.3 Zero correlation length . . . . .                                 | 320        |
| 23.4 MPDOs and LPDOs . . . . .                                         | 321        |
| 23.5 Pure-state recovery inside the MPO formalism . . . . .            | 322        |
| 23.6 Simple local structure from SAL and ZCL . . . . .                 | 323        |
| 23.7 Horizontal and Vertical Canonical Forms . . . . .                 | 326        |
| 23.8 Fusion isometries . . . . .                                       | 327        |
| 23.9 Algebra structure . . . . .                                       | 328        |
| 23.9.1 Diagonal $\chi$ -matrices and the trace-power formula . . . . . | 331        |
| 23.10 Commuting form and GSNNCH . . . . .                              | 340        |
| 23.11 Blocked-RFP construction for simple MPDOs . . . . .              | 342        |
| 23.12 Saturation of the area law . . . . .                             | 344        |

# Chapter 1

## Introduction

The central result is the Fundamental Theorem for translation-invariant matrix product vectors. The basic object is a family of matrices  $A = \{A^i\}_{i=0}^{d-1}$ , which assigns to each chain length  $N$  the vector  $|V^{(N)}(A)\rangle$  with coefficients indexed by  $\sigma = (i_1, \dots, i_N)$ ,

$$V^{(N)}(A)_\sigma = \text{tr}(A^{i_1} \dots A^{i_N}).$$

The main argument, for translation-invariant tensors, follows the canonical-form and basis-of-normal-tensors argument of [CPGSV16, CPGSV21], with the single-block injective case anchored in the earlier representation theorem of [PGVWC07].

### Notation

|                                             |                                                                                   |
|---------------------------------------------|-----------------------------------------------------------------------------------|
| $\mathbb{C}, \mathbb{N}, \mathbb{Z}$        | complex numbers, natural numbers, integers                                        |
| $M_D(\mathbb{C}), \text{GL}_D(\mathbb{C})$  | complex matrices and invertible complex matrices                                  |
| $\text{tr}(X), \text{span}_{\mathbb{C}}(S)$ | trace and complex linear span                                                     |
| $A = \{A^i\}_{i=0}^{d-1}$                   | a translation-invariant MPS tensor                                                |
| $w = (i_1, \dots, i_L), A^w$                | a word and the product $A^{i_1} \dots A^{i_L}$                                    |
| $ V^{(N)}(A)\rangle$                        | the length- $N$ matrix product vector                                             |
| $V^{(N)}(A)_\sigma$                         | the coefficient of $ V^{(N)}(A)\rangle$ at $\sigma$                               |
| $O_{AB}(N)$                                 | the bilinear overlap $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}$ |
| $\mathcal{E}_A(X)$                          | the transfer map $\sum_i A^i X (A^i)^\dagger$                                     |
| $A^{[L]}$                                   | the length- $L$ blocked tensor                                                    |

In canonical form one writes

$$A^i = \bigoplus_{j=1}^g \mu_j A_j^i, \quad B^i = \bigoplus_{k=1}^h \nu_k B_k^i, \quad (1.1)$$

where the displayed blocks are normal tensors. If  $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$  for every positive chain length  $N$ , then the bases of normal tensors have the same size. After relabelling the blocks of  $B$ , each matched pair satisfies

$$B_j^i = \zeta_j X_j A_j^i X_j^{-1}, \quad (1.2)$$

for some invertible matrix  $X_j$  and nonzero scalar  $\zeta_j$ , and the equal case compares the coefficient power sums attached to the relabelled bases. The block gauges (1.2) and the weight comparison



then combine into a single gauge conjugacy of the two weighted block-diagonal tensors (1.1). For a single injective block this reduces to the familiar conclusion  $B^i = X A^i X^{-1}$  for all physical indices  $i$ , the single-block specialization of (1.2).

Chapters 2–11 carry out this argument. Chapter 2 fixes the matrix product vector notation, word evaluations, transfer maps, gauge equivalence, injectivity, blocking, and site-periodic tensor families. Chapter 3 proves the single-block theorem by trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether. Chapters 4, 5, 6, and 7 develop the positive-map, multiplicative-domain, Perron–Frobenius, and overlap-decay tools used to control normal blocks. Chapters 8 and 9 supply the blocking, primitivity, irreducibility, and canonical-form reductions. Chapter 9 includes the invariant-projection splitting. Chapter 14 records the product-algebra block permutation argument. Chapter 10 supplies bases of normal tensors and the power-sum comparison. The after-blocking reductions in Section 9.6.2 give the common primitive block decompositions used immediately in Chapter 11, which records the conditional theorem statements.

The chapters after Chapter 11 collect applications, alternative formulations, and later tensor-network material. Chapter 12 uses the Fundamental Theorem to obtain virtual symmetry gauges, projective bond representations, and string-order criteria; its Kraus-mixing step appeals only to the later channel-representation chapter, not to any additional input for the Fundamental Theorem. Chapter 13 develops the direct Fundamental Theorem for periodically repeated tensors in irreducible form, together with the compatibility theorems for physical-index rotation (Definition 12.1 in Chapter 12). Chapter 14 gives the fixed-length algebraic Fundamental Theorem of [MGRPG<sup>+</sup>18], and Chapter 15 records the PEPS analogs. Parent Hamiltonians, correlations, and examples are treated after the PEPS material. Channel representations and normal forms are then collected immediately before quantum dynamical semigroups and the operator-convexity material; entropy and matrix product density operators follow. These later topics use the Fundamental Theorem, or develop later finite-dimensional channel theory, rather than enter its proof.

## Chapter 2

# Matrix Product Vectors

Fix a physical dimension  $d$  and a bond dimension  $D$ . This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors here are translation-invariant with periodic boundary conditions (PBC).

### 2.1 Basic definitions

**Definition 2.1** (MPS tensor). [Lean](#) [✓ Stmt](#) A (translation-invariant, PBC) *MPS tensor* with physical dimension  $d$  and bond dimension  $D$  is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \quad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index  $i \in \{0, \dots, d-1\}$ . Such a tensor defines an MPV family. Diagrammatically,



The black node denotes the tensor  $A$ , the horizontal legs are virtual, and the upper leg is the physical index  $i$ .

**Definition 2.2** (Word evaluation). [Lean](#) [✓ Stmt](#) Given a word  $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$ , the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \dots A^{i_L} \in M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity:  $A^\emptyset = \mathbb{1}_D$ . In tensor-network notation,



in which each black node denotes the same local tensor  $A$ , the virtual legs remain open, and the physical legs are labelled by the word  $(i_1, \dots, i_L)$ .

**Lemma 2.3** (Multiplicativity of word evaluation). [Lean](#) [✓ Stmt](#) For any words  $w_1, w_2$ ,  $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$ .

*Proof.* [✓ Proof](#) Immediate from the definition of  $A^w$ . □

**Definition 2.4** (Matrix product vector). [Lean](#) [✓ Stmt](#) The *matrix product vector* (MPV) of a tensor  $A$  at system size  $N$  is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1, \dots, i_N=0}^{d-1} \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

Equivalently, for a configuration  $\sigma = (i_1, \dots, i_N) \in \{0, \dots, d-1\}^N$ , we write

$$V^{(N)}(A)_\sigma := \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}).$$

The coefficient function  $\sigma \mapsto V^{(N)}(A)_\sigma$  gives the components; the displayed ket is the corresponding vector in  $(\mathbb{C}^d)^{\otimes N}$ . For a general word  $w$ , we also write

$$c_w(A) := \text{tr}(A^w).$$

The *MPV family* generated by  $A$  is the collection  $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$ . The coefficient  $V^{(N)}(A)_\sigma$  is the periodic contraction



of  $N$  copies of the local tensor  $A$ , with the outer virtual legs closed by the trace.

**Definition 2.5** (Transfer map). [Lean](#) [✓ Stmt](#) The *transfer map* associated to a tensor  $A$  is the linear map  $\mathcal{E}_A : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger.$$

Diagrammatically, the transfer map is the double-layer contraction



in which the upper node denotes  $A$ , the lower node denotes  $A^\dagger$ , and the physical index is summed over between them.

*Remark 2.6.* [✓ Stmt](#) The transfer map is automatically positive: if  $X \geq 0$  then  $\mathcal{E}_A(X) \geq 0$ , because each summand  $A^i X (A^i)^\dagger$  is positive semidefinite. The abstract positive-map framework appears in Chapter 4.

## 2.2 Site-periodic tensor families

**Definition 2.7** (Site-periodic tensor family). [Lean](#) [✓ Stmt](#) For a fixed period  $m$ , a *site-periodic tensor family* is a family of local tensors indexed by  $\{0, \dots, m-1\}$ , with local physical dimension  $d$  and bond dimension  $D$ .

**Definition 2.8** (Periodic same-state and gauge-equivalence relations). [Lean](#) [Lean](#) [✓ Stmt](#) For site-periodic tensors  $A, B$  at fixed period  $m$ :

- equality of the induced site-periodic states;
- cyclic gauge equivalence.

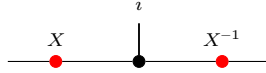
**Lemma 2.9** (Equivalence structures for periodic relations). [Lean](#) [Lean](#) [✓ Stmt](#) Both periodic relations are equivalence relations (reflexive, symmetric, transitive).

## 2.3 Gauge equivalence and same MPV

**Definition 2.10** (Gauge equivalence). [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  of the same bond dimension  $D$  are *gauge equivalent* if there exists an invertible matrix  $X \in \text{GL}_D(\mathbb{C})$  such that, for every  $i \in \{0, \dots, d-1\}$ ,

$$B^i = X A^i X^{-1}.$$

In tensor-network notation,



the black node denotes the tensor named in the surrounding formula, the upper leg is the physical index  $i$ , and the two red side nodes denote the gauge matrices acting on the virtual legs.

**Definition 2.11** (Same MPV). [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  (of the same bond dimension) generate the same MPV family if  $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$  for every system size  $N$ .

**Definition 2.12** (Same MPV - different bond dimensions). [Lean](#) [✓ Stmt](#) For tensors  $A$  and  $B$  of possibly different bond dimensions  $D_1$  and  $D_2$ , we write  $\mathcal{V}(A) = \mathcal{V}(B)$  if  $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$  for every  $N$ .

**Definition 2.13** (Positive-length same MPV - different bond dimensions). [Lean](#) [✓ Stmt](#) For tensors  $A$  and  $B$  of possibly different bond dimensions, we write  $\mathcal{V}^+(A) = \mathcal{V}^+(B)$  if  $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$  for every  $N > 0$ .

**Lemma 2.14** (Length-zero MPV coefficient). [Lean](#) [✓ Stmt](#) For any MPS tensor  $A$  with bond dimension  $D$ , the length-zero MPV coefficient is  $D$ , the trace of the identity on the bond space.

**Lemma 2.15** (Positive-length equality with equal bond dimensions). [Lean](#) [✓ Stmt](#) If two tensors have the same positive-length MPV family and the same bond dimension, then they have the same MPV family at every length.

**Remark 2.16.** [✓ Stmt](#) Definition 2.11 is the same-bond-dimension specialization of Definition 2.12. We keep both because later results use both the same-dimension and different-dimension forms.

**Definition 2.17** (Proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  (of possibly different bond dimensions) generate weakly proportional MPV families if for every  $N$  there exists  $c_N \in \mathbb{C}$  such that  $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$ .

**Definition 2.18** (Nonzero proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  (of possibly different bond dimensions) generate nonzero proportional MPV families if for every  $N$  there is a nonzero scalar  $c_N \in \mathbb{C}$  such that  $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$ . This is the projective reading of the proportionality hypothesis in [CPGSV16, Theorem 2.10].

**Definition 2.19** (Eventually nonzero proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  are eventually nonzero proportional if, for all sufficiently large  $N$ , there is a nonzero scalar  $c_N \in \mathbb{C}$  such that  $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$ . This auxiliary form is used with the eventual linear-independence statement Lemma 10.4.

**Lemma 2.20** (Elementary nonzero proportionality rules). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Nonzero proportionality implies weak proportionality and eventual nonzero proportionality, is symmetric, and equality of MPV families is the special case with scalar 1.*

*Proof.* [✓ Proof](#) Forgetting the nonvanishing condition gives weak proportionality. The eventual form follows by viewing an all-length statement as an eventual one. Symmetry follows by inverting the scalar  $c_N$  at each length, and equality is the case  $c_N = 1$ .  $\square$

**Lemma 2.21** (Scaling of word evaluation). [Lean](#) [✓ Stmt](#) *Let  $\zeta A$  denote the tensor with matrices  $\{\zeta A^i\}_{i=0}^{d-1}$ . Then for any word  $w$ ,*

$$(\zeta A)^w = \zeta^{|w|} A^w,$$

where  $|w|$  is the length of  $w$ .

*Proof.* [✓ Proof](#) Induction on  $w$ : the base case gives  $\zeta^0 = 1$ , and each step picks up one factor of  $\zeta$ .  $\square$

**Lemma 2.22** (Scaling of MPV components). [Lean](#) [✓ Stmt](#) *Let  $\zeta A$  denote the tensor with matrices  $\{\zeta A^i\}_{i=0}^{d-1}$ . Then for every  $N$  and every configuration  $\sigma = (i_1, \dots, i_N)$ ,*

$$V^{(N)}(\zeta A)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

*Proof.* [✓ Proof](#) Apply Lemma 2.21 to the word  $w = (i_1, \dots, i_N)$  of length  $N$ , then take the trace; the scalar  $\zeta^N$  pulls out by linearity.  $\square$

**Definition 2.23** (Gauge-phase equivalence). [Lean](#) [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  are *gauge-phase equivalent* if there exist  $X \in \text{GL}_D(\mathbb{C})$  and  $\zeta \in \mathbb{C} \setminus \{0\}$  such that, for every physical index  $i$ ,

$$B^i = \zeta X A^i X^{-1}.$$

*Proof.* [✓ Proof](#) Gauge equivalence is the special case  $\zeta = 1$ .  $\square$

**Remark 2.24.** [✓ Stmt](#) After spectral-radius normalization one may restrict  $\zeta$  to a unit-modulus phase  $e^{i\phi}$ . We allow an arbitrary nonzero scalar because later canonical-form statements keep the block-scaling factors explicit.

**Lemma 2.25** (Gauge covariance of word evaluation). [Lean](#) [✓ Stmt](#) *If  $B^i = X A^i X^{-1}$  for all  $i$ , then for every word  $w$ ,*

$$B^w = X A^w X^{-1}.$$

*Proof.* ✓ Proof Induction on the word length, using  $XX^{-1} = \mathbb{1}$ . □

**Theorem 2.26** (Gauge invariance of MPVs). Lean ✓ Stmt *If  $A$  and  $B$  are gauge equivalent, then they generate the same MPV family.*

*Proof.* ✓ Proof Let  $B^i = XA^iX^{-1}$ . By Lemma 2.25,  $B^w = XA^wX^{-1}$  for every word  $w$ . Then  $\text{tr}(B^w) = \text{tr}(XA^wX^{-1}) = \text{tr}(A^w)$  by cyclicity of the trace. □

**Theorem 2.27** (Gauge-phase scaling of MPVs). Lean ✓ Stmt *If, for every physical index  $i$ ,*

$$B^i = \zeta XA^iX^{-1},$$

*then for every system size  $N$  and configuration  $\sigma$  one has*

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

*Proof.* ✓ Proof Combine gauge covariance of word evaluation with scalar scaling, then take traces. □

## 2.4 Injectivity and normality

**Definition 2.28** (Injectivity). Lean ✓ Stmt A tensor  $A$  is *injective* if its matrices span the full matrix algebra:  $\text{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$ .

**Definition 2.29** ( $L$ -block injectivity). Lean ✓ Stmt A tensor  $A$  is  *$L$ -block injective* if  $\text{span}_{\mathbb{C}}\{A^{i_1} \dots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$ . An injective tensor is 1-block injective.

**Lemma 2.30** (One-site injectivity gives one-block injectivity). Lean ✓ Stmt *If the one-site matrices  $\{A^i\}_{i=0}^{d-1}$  span  $M_D(\mathbb{C})$ , then the length-one word products span  $M_D(\mathbb{C})$ , so  $A$  is 1-block injective.*

*Proof.* ✓ Proof The length-one words are exactly the letters  $i \in \{0, \dots, d-1\}$ , and evaluating a length-one word gives the matrix  $A^i$ . Thus the length-one word span is the same span as in one-site injectivity. □

**Definition 2.31** (Normal tensor). Lean ✓ Stmt A tensor  $A$  is *normal* [CPGSV21] if there exists a finite  $L_0$  such that  $A$  is  $L_0$ -block injective.

*Remark 2.32.* ✓ Stmt The algebraic characterization of normality in Definition 2.31 — eventual full matrix span, equivalently eventual block injectivity — is equivalent to the spectral characterization used in [CPGSV21, Definition IV.1]: after TP normalization, the transfer map  $\mathcal{E}_A$  is a primitive channel, equivalently irreducible with trivial peripheral spectrum. This equivalence is proved in [SPGWC10, Proposition 3]. In what follows, the directions of the equivalence are supplied by Lemma 8.68, Theorem 8.86, and Theorem 9.30. In particular, every subsequent use of “normal” here refers to this single notion; Definition 9.25 employs the spectral formulation, but it describes the same class of tensors.

## 2.5 Canonical form

**Definition 2.33** (Block-injective canonical form). [Lean](#) [✓ Stmt](#) Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks  $r \geq 1$ ,
2. bond dimensions  $D_1, \dots, D_r$ ,
3. injective tensors  $\{A_k^i\}_{i=0}^{d-1}$  with  $A_k^i \in M_{D_k}(\mathbb{C})$  for each block  $k \in \{1, \dots, r\}$ ,
4. scaling factors  $\mu_1, \dots, \mu_r \in \mathbb{C}$ .

The associated tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

This block-injective formulation of the canonical form is used in [\[PGVWC07\]](#); see also [\[CPGSV21\]](#) for the normal canonical form.

*Remark 2.34.* [✓ Stmt](#) Here we use the block-injective canonical form: each block is injective in the sense of Definition 2.28. This is stronger than the NT-block canonical form of [\[CPGSV21\]](#), where the blocks are only normal in the spectral sense. Chapter 9 treats the normal canonical form needed for the blocking-based multi-block theory, while Chapter 13 develops the separate periodic irreducible-form extension. This section isolates the stronger block-injective form used in the injective and block-injective arguments.

**Definition 2.35** (Block projection). [Lean](#) [✓ Stmt](#) For a finite direct sum  $\bigoplus_k \mathbb{C}^{D_k}$ , the block projection  $P_j$  is the matrix that is the identity on the  $j$ -th summand and zero on all other summands.

**Definition 2.36** (Block-diagonal matrix). [Lean](#) [✓ Stmt](#) A matrix on  $\bigoplus_k \mathbb{C}^{D_k}$  is block diagonal if it is a direct sum of square matrices, one on each summand.

**Lemma 2.37** (Off-block characterization). [Lean](#) [✓ Stmt](#) A matrix on  $\bigoplus_k \mathbb{C}^{D_k}$  is block diagonal if and only if every entry from one summand to a distinct summand is zero.

*Proof.* [✓ Proof](#) The forward direction is immediate from the definition of a direct sum of matrices. Conversely, if all off-block entries vanish, the diagonal blocks reconstruct the original matrix.  $\square$

**Lemma 2.38** (Commutation with block projections). [Lean](#) [✓ Stmt](#) If a matrix on  $\bigoplus_k \mathbb{C}^{D_k}$  commutes with every block projection  $P_j$ , then it is block diagonal.

*Proof.* [✓ Proof](#) Compare the  $(i, j)$  block of the identity  $XP_j = P_jX$  for  $i \neq j$ . The right multiplication by  $P_j$  keeps that block, while the left multiplication by  $P_j$  kills it. Hence every off-block entry is zero, and Lemma 2.37 applies.  $\square$

**Definition 2.39** (Block-diagonal tensor from blocks). [Lean](#) [✓ Stmt](#) Given  $r$  blocks with bond dimensions  $(D_k)_{k=1}^r$ , per-block tensors  $\{A_k^i\}$ , and scaling factors  $(\mu_k)_{k=1}^r$ , the associated *block-diagonal tensor* is the tensor of total bond dimension  $D = \sum_{k=1}^r D_k$  defined by

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

**Lemma 2.40** (Word evaluation of a block-diagonal tensor). [Lean](#) [✓ Stmt](#) For a word  $w$ , the word evaluation of a block-diagonal tensor remains block diagonal:

$$A^w = \bigoplus_k \mu_k^{|w|} A_k^w.$$

*Proof.* [✓ Proof](#) Induct on the word and multiply block-diagonal matrices block by block; every letter contributes one scalar factor  $\mu_k$  on the  $k$ -th block.  $\square$

**Theorem 2.41** (MPV decomposition). [Lean](#) [✓ Stmt](#) The MPV of a block-diagonal tensor decomposes as

$$|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(A_k)\rangle.$$

Equivalently, for each configuration  $\sigma$  one has

$$V^{(N)}(A)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(A_k)_\sigma.$$

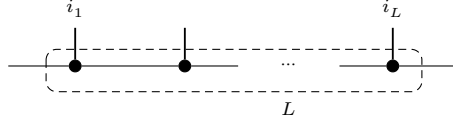
*Proof.* [✓ Proof](#) By Lemma 2.40, the full word evaluation is block diagonal with blocks  $\mu_k^N A_k^w$  for a word  $w$  of length  $N$ . Taking the trace of a block-diagonal matrix gives the sum of the block traces:  $\text{tr}(A^w) = \sum_k \mu_k^N \text{tr}(A_k^w)$ .  $\square$

## 2.6 Blocking

**Definition 2.42** (Blocked tensor). [Lean](#) [✓ Stmt](#) Given a tensor  $A$  and a blocking length  $L$ , the  $L$ -blocked tensor is the tensor with physical index set  $\{0, \dots, d-1\}^L$  (hence physical dimension  $d^L$ ) and the same bond dimension  $D$ , whose matrices are indexed by words  $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$ :

$$(A^{[L]})^{(i_1, \dots, i_L)} = A^{i_1} A^{i_2} \dots A^{i_L}.$$

Blocking coarse-grains  $L$  neighbouring sites into one tensor: the inner virtual bonds are contracted, the  $L$  physical legs merge into a single composite index, and the two outer bonds remain as the bond of  $A^{[L]}$ .



**Lemma 2.43** (Blocked word evaluation). [Lean](#) [✓ Stmt](#) If  $w = (J_1, \dots, J_m)$  is a word in the blocked alphabet, where each  $J_k$  is an  $L$ -tuple in  $\{0, \dots, d-1\}^L$ , then concatenating the block words gives a word  $\tilde{w}$  of length  $mL$  in the original alphabet and one has

$$(A^{[L]})^w = A^{\tilde{w}}.$$

*Proof.* [✓ Proof](#) Induct on the blocked word  $w$  and split off the first block word. Lemma 2.3 turns concatenation of blocks into multiplication of the corresponding evaluations.  $\square$

**Lemma 2.44** (Blocking preserves MPV coefficients). [Lean](#) [✓ Stmt](#) For each basis state  $\sigma = (\sigma_1, \dots, \sigma_N)$  of the blocked system, where each  $\sigma_k$  is an  $L$ -tuple in  $\{0, \dots, d-1\}^L$ , there exists a basis state  $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$  of the original system such that

$$V^{(N)}(A^{[L]})_\sigma = V^{(NL)}(A)_{\tilde{\sigma}}.$$



*Proof.* ✓ Proof Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.43).  $\square$

**Theorem 2.45** (Blocking preserves same MPV). Lean ✓ Stmt *If  $A$  and  $B$  generate the same MPV family, then for any blocking length  $L$ , the blocked tensors  $A^{[L]}$  and  $B^{[L]}$  also generate the same MPV family.*

*Proof.* ✓ Proof By Lemma 2.44, each MPV coefficient of the blocked tensor  $A^{[L]}$  equals a coefficient of  $A$  at the corresponding concatenated configuration. The same holds for  $B^{[L]}$  (with the same concatenated configuration). Since  $A$  and  $B$  agree on all configurations, so do  $A^{[L]}$  and  $B^{[L]}$ .  $\square$

## 2.7 MPV overlap

**Definition 2.46** (MPV as Hilbert-space vector). Lean ✓ Stmt We regard  $|V^{(N)}(A)\rangle$  as an element of  $\mathbb{C}^{d^N}$  indexed by configurations  $\sigma \in \{0, \dots, d-1\}^N$ , with  $\sigma$ -component  $V^{(N)}(A)_\sigma$  as in Definition 2.4. This allows us to use the standard Euclidean inner product on  $\mathbb{C}^{d^N}$ .

**Definition 2.47** (MPV inner product). Lean ✓ Stmt The *MPV inner product* between two tensors  $A$  and  $B$  at system size  $N$  is

$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle := \sum_{\sigma \in \{0, \dots, d-1\}^N} \overline{V^{(N)}(A)_\sigma} V^{(N)}(B)_\sigma,$$

i.e. the Euclidean inner product on  $\mathbb{C}^{d^N}$  (conjugate-linear in the first argument).

**Definition 2.48** (MPV overlap). Lean ✓ Stmt The *MPV overlap* between two tensors  $A$  (of bond dimension  $D_1$ ) and  $B$  (of bond dimension  $D_2$ ) at system size  $N$  is

$$O_{AB}(N) := \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

Diagrammatically, the overlap contracts the MPV ring of  $A$  against the conjugate ring of  $B$  along their shared physical indices, both virtual rings closed by the trace:



This is the bilinear overlap (without complex conjugation) used later. By Lemma 2.49, it is the complex conjugate of the Hilbert-space inner product from Definition 2.47.

**Lemma 2.49** (Overlap equals conjugate inner product). Lean ✓ Stmt  $O_{AB}(N) = \overline{\langle V^{(N)}(A) | V^{(N)}(B) \rangle}$ .

*Proof.* ✓ Proof The overlap sums  $V_\sigma \overline{W_\sigma}$  while the inner product sums  $\overline{V_\sigma} W_\sigma$ . Complex conjugation swaps these.  $\square$

**Lemma 2.50** (Overlap determined by positive-length MPV). Lean ✓ Stmt *If  $V^{(N)}(A)_\sigma = V^{(N)}(A')_\sigma$  and  $V^{(N)}(B)_\sigma = V^{(N)}(B')_\sigma$  for all  $N > 0$  and all  $\sigma$ , then  $O_{AB}(N) = O_{A'B'}(N)$  for all positive  $N$ .*

*Proof.* ✓ Proof Direct pointwise rewriting: if the MPV coefficients agree for all positive chain lengths, the overlap sums are identical.  $\square$

## Chapter 3

# The Single-Block Fundamental Theorem

An injective MPS tensor that generates the same matrix-product vectors as a second tensor of equal bond dimension is conjugate to it [CPGSV21, Corollary IV.5]: there is an invertible  $X$  with  $B^i = XA^iX^{-1}$  for every physical index  $i$ . The argument is purely algebraic. Equality of the matrix-product vectors gives  $\text{tr}(A^w) = \text{tr}(B^w)$  for every word  $w$ ; the length-two and length-three traces produce a linear map  $T$  with  $T(A^i) = B^i$  that is multiplicative and nonzero; a nonzero multiplicative endomorphism of the simple algebra  $M_D(\mathbb{C})$  is a unital automorphism; and by Skolem–Noether every automorphism of  $M_D(\mathbb{C})$  is conjugation by an invertible  $X$ . Evaluating at  $A^i$  closes the proof. The same conjugation, one normal block at a time, is the central step of the multi-block theorem in Chapter 11.

### 3.1 The multiplicative linear extension

**Theorem 3.1** (Nondegeneracy of the trace pairing). Lean ✓ Stmt For  $M \in M_D(\mathbb{C})$ ,  $\text{tr}(MN) = 0$  for all  $N \in M_D(\mathbb{C})$  if and only if  $M = 0$ .

*Proof.* ✓ Proof Testing against the matrix unit  $E_{ji}$  gives  $\text{tr}(E_{ji}M) = M_{ij}$ , so  $\text{tr}(MN) = 0$  for all  $N$  forces every entry of  $M$  to vanish. □

**Lemma 3.2** (Same matrix-product vectors give trace agreement). Lean ✓ Stmt If  $A$  and  $B$  generate the same matrix-product vectors, then  $\text{tr}(A^w) = \text{tr}(B^w)$  for every word  $w$ .

*Proof.* ✓ Proof The configuration  $\sigma = w$  gives  $V^{(|w|)}(A)_w = \text{tr}(A^w)$ . Equality of the matrix-product vectors gives  $V^{(N)}(A) = V^{(N)}(B)$  for all  $N$ , so  $\text{tr}(A^w) = \text{tr}(B^w)$ . □

The trace conditions constrain every product of the matrices  $A^i$ . To bring the algebra structure of  $M_D(\mathbb{C})$  to bear, record how a matrix pairs against the family  $\{A^i\}$  under the trace.

**Definition 3.3** (Trace pairing map). Lean ✓ Stmt For a tensor  $A$ , the trace pairing map  $\Phi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$  is

$$\Phi_A(M)_i = \text{tr}(MA^i).$$

**Theorem 3.4** (The trace pairing map is injective for injective tensors). Lean ✓ Stmt If  $A$  is injective, then  $\ker \Phi_A = \{0\}$ .

*Proof.* ✓ Proof If  $\Phi_A(M) = 0$ , then  $\text{tr}(MA^i) = 0$  for every  $i$ . The  $\{A^i\}$  span  $M_D(\mathbb{C})$ , so  $\text{tr}(MN) = 0$  for all  $N$ , and  $M = 0$  by Theorem 3.1.  $\square$

**Lemma 3.5** (Injectivity transfers across a range inclusion). Lean ✓ Stmt *Let  $\Phi, \Psi : V \rightarrow W$  be  $\mathbb{C}$ -linear maps with  $V$  finite-dimensional. If  $\ker \Phi = \{0\}$  and  $\text{range } \Phi \subseteq \text{range } \Psi$ , then  $\ker \Psi = \{0\}$ .*

*Proof.* ✓ Proof By rank-nullity,  $\ker \Phi = \{0\}$  gives  $\dim \text{range } \Phi = \dim V$ , and the range inclusion gives

$$\dim V = \dim \text{range } \Phi \leq \dim \text{range } \Psi \leq \dim V,$$

so  $\dim \text{range } \Psi = \dim V$  and  $\ker \Psi = \{0\}$ .  $\square$

**Theorem 3.6** (The linear extension exists and is unique). Lean ✓ Stmt *If  $A$  is injective and  $A, B$  generate the same matrix-product vectors, then there is a unique linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  with  $T(A^i) = B^i$  for all  $i$ .*

*Proof.* ✓ Proof Length-two trace agreement gives  $\Phi_A(A^i) = \Phi_B(B^i)$  for every  $i$ . The  $\{A^i\}$  span  $M_D(\mathbb{C})$ , so  $\text{range } \Phi_A \subseteq \text{range } \Phi_B$ . With  $\ker \Phi_A = \{0\}$  (Theorem 3.4), Lemma 3.5 gives  $\ker \Phi_B = \{0\}$ . Let  $g$  be a left inverse of  $\Phi_B$  and set  $T = g \circ \Phi_A$ ; then

$$T(A^i) = g(\Phi_A(A^i)) = g(\Phi_B(B^i)) = B^i.$$

Uniqueness holds because the  $\{A^i\}$  span  $M_D(\mathbb{C})$ .  $\square$

**Theorem 3.7** (The linear extension is multiplicative). Lean ✓ Stmt *Under the hypotheses of Theorem 3.6, the map  $T$  satisfies  $T(MN) = T(M)T(N)$  for all  $M, N \in M_D(\mathbb{C})$ .*

*Proof.* ✓ Proof Length-two trace agreement gives  $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$ ; since the  $\{A^i\}$  span  $M_D(\mathbb{C})$  this extends to  $\Phi_B \circ T = \Phi_A$ , so  $\text{range } \Phi_A = \text{range}(\Phi_B \circ T) \subseteq \text{range } \Phi_B$ . With  $\ker \Phi_A = \{0\}$  (Theorem 3.4), Lemma 3.5 makes  $\Phi_B$  injective. Length-three trace agreement gives, for all  $i, j, k$ ,

$$\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k,$$

so injectivity of  $\Phi_B$  yields  $T(A^i A^j) = B^i B^j$ . Extend in two stages: for fixed  $j$ , the maps  $M \mapsto T(MA^j)$  and  $M \mapsto T(M)B^j$  agree on the spanning  $\{A^i\}$ , hence  $T(MA^j) = T(M)B^j$  for all  $M$ ; then for fixed  $M$ , the maps  $N \mapsto T(MN)$  and  $N \mapsto T(M)T(N)$  agree on the spanning  $\{A^j\}$ , hence  $T(MN) = T(M)T(N)$  for all  $N$ .  $\square$

## 3.2 Inner automorphism and the single-block theorem

The rigidity of the full matrix algebra now finishes the proof. A nonzero multiplicative map on  $M_D(\mathbb{C})$  is forced to be bijective (simplicity), unit-preserving (surjectivity), and conjugation by a single invertible matrix (Skolem–Noether). The next four results establish these steps and the nonvanishing of  $T$ , which the final theorem combines into the gauge  $X$ .

**Theorem 3.8** (A nonzero multiplicative endomorphism of  $M_D(\mathbb{C})$  is bijective). Lean ✓ Stmt *A nonzero multiplicative  $\mathbb{C}$ -linear endomorphism of  $M_D(\mathbb{C})$  is bijective.*

*Proof.* ✓ Proof The kernel of a multiplicative linear map is a two-sided ideal. The algebra  $M_D(\mathbb{C})$  is simple, so the kernel is  $\{0\}$  or all of  $M_D(\mathbb{C})$ ; the latter forces the map to be zero. Hence the map is injective, and surjective by finite-dimensionality.  $\square$

**Lemma 3.9** (A surjective multiplicative map preserves the unit). [Lean] [✓ Stmt] *If  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is a surjective multiplicative  $\mathbb{C}$ -linear map, then  $T(\mathbb{1}) = \mathbb{1}$ , so  $T$  is a  $\mathbb{C}$ -algebra homomorphism.*

*Proof.* [✓ Proof] Choose  $X$  with  $T(X) = \mathbb{1}$ . Then  $T(\mathbb{1}) = T(X)T(\mathbb{1}) = T(X\mathbb{1}) = T(X) = \mathbb{1}$ . □

**Lemma 3.10** (An injective tensor forces a nonzero partner). [Lean] [✓ Stmt] *Let  $D \geq 1$ . If  $A$  is injective and  $A, B$  generate the same matrix-product vectors, then the  $B^i$  are not all zero.*

*Proof.* [✓ Proof] If  $B^i = 0$  for every  $i$ , then Lemma 3.2 on length-one words gives  $\text{tr}(A^i) = 0$  for all  $i$ . The  $\{A^i\}$  span  $M_D(\mathbb{C})$ , so the trace functional vanishes identically, contradicting  $\text{tr}(\mathbb{1}) = D \neq 0$ . □

**Theorem 3.11** (Skolem–Noether). [Lean] [✓ Stmt] *Every  $\mathbb{C}$ -algebra automorphism  $f$  of  $M_D(\mathbb{C})$  is inner: there is  $X \in \text{GL}_D(\mathbb{C})$  with  $f(M) = XMX^{-1}$  for all  $M$ .*

*Proof.* [✓ Proof] Under the identification  $M_D(\mathbb{C}) \cong \text{End}_{\mathbb{C}}(\mathbb{C}^D)$ , every algebra automorphism of  $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$  is conjugation by an invertible linear map. Translating back gives the invertible  $X$ . □

**Theorem 3.12** (Single-block Fundamental Theorem). [Lean] [✓ Stmt] *Let  $A$  be an injective MPS tensor and  $B$  a tensor of the same bond dimension. If  $A$  and  $B$  generate the same matrix-product vectors, then there is  $X \in \text{GL}_D(\mathbb{C})$  with  $B^i = XA^iX^{-1}$  for all  $i$ .*

*Proof.* [✓ Proof] Theorem 3.6 gives the unique linear  $T$  with  $T(A^i) = B^i$ , and Theorem 3.7 makes it multiplicative. It is nonzero: otherwise  $B^i = 0$  for all  $i$ , against Lemma 3.10. By Theorem 3.8,  $T$  is bijective, and by Lemma 3.9 it fixes  $\mathbb{1}$ , so  $T$  is a  $\mathbb{C}$ -algebra automorphism. Theorem 3.11 gives  $X \in \text{GL}_D(\mathbb{C})$  with  $T(M) = XMX^{-1}$ ; evaluating at  $M = A^i$  gives  $B^i = XA^iX^{-1}$ . □

## Chapter 4

# Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

### 4.1 Positive maps and channels

**Definition 4.1** (Positive map). [Lean](#) [✓ Stmt](#) A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is *positive* if  $E(X) \geq 0$  whenever  $X \geq 0$ , where  $X \geq 0$  means that  $X$  is positive semidefinite.

*Remark 4.2.* [✓ Stmt](#) We use the Loewner order on matrices:  $X \leq Y$  means  $Y - X \geq 0$ .

**Definition 4.3** (Trace-preserving map). [Lean](#) [✓ Stmt](#) A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is *trace-preserving* if  $\text{tr}(E(X)) = \text{tr}(X)$  for all  $X$ .

**Definition 4.4** (Completely positive map). [Lean](#) [✓ Stmt](#) A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators  $\{K_i\}_{i=0}^{r-1}$  with  $K_i \in M_D(\mathbb{C})$  such that, for every  $X \in M_D(\mathbb{C})$ ,

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger.$$

By Choi's theorem, this is equivalent to  $E \otimes \text{id}_n$  being positive for all  $n$ .

**Theorem 4.5** (CP maps are positive). [Lean](#) [✓ Stmt](#) *Every completely positive map is positive.*

*Proof.* [✓ Proof](#) If  $X \geq 0$ , then each summand  $K_i X K_i^\dagger \geq 0$  (conjugation preserves positive semidefiniteness), so the sum is PSD.  $\square$

**Theorem 4.6** (Channels are positive). [Lean](#) [✓ Stmt](#) *Every quantum channel is positive.*

*Proof.* [✓ Proof](#) A quantum channel is completely positive by definition, so Theorem 4.5 applies.  $\square$

**Theorem 4.7** (Positive maps preserve Hermiticity). [Lean](#) [✓ Stmt](#) *If  $E$  is positive and  $X = X^\dagger$ , then  $E(X) = E(X)^\dagger$ .*

*Proof.* ✓ Proof Decompose  $X = X_+ - X_-$  into its positive and negative parts. Positivity makes  $E(X_+)$  and  $E(X_-)$  positive semidefinite, hence Hermitian, so their difference  $E(X)$  is Hermitian. □

**Definition 4.8** (Quantum channel). Lean ✓ Stmt A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

**Definition 4.9** (Density matrices). Lean Lean ✓ Stmt The set of *density matrices* in  $M_D(\mathbb{C})$  is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \text{tr}(\rho) = 1\}.$$

**Theorem 4.10** (Density matrices are compact). Lean ✓ Stmt  $\mathcal{D}_D$  is compact (in the entry-wise topology on  $M_D(\mathbb{C})$ ).

*Proof.* ✓ Proof The PSD cone is closed (preimage of the closed non-negative cone under continuous quadratic forms), and if  $\rho \geq 0$  with  $\text{tr}(\rho) = 1$  then each entry satisfies  $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$ : the diagonal entries are non-negative and sum to 1, and the off-diagonal bound is the PSD Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. □

**Theorem 4.11** (Density matrices are convex). Lean ✓ Stmt  $\mathcal{D}_D$  is convex.

*Proof.* ✓ Proof Convex combination of PSD matrices is PSD, and trace is linear. □

**Theorem 4.12** (Density matrices are nonempty). Lean ✓ Stmt For  $D \geq 1$ ,  $\mathcal{D}_D \neq \emptyset$ .

*Proof.* ✓ Proof  $\frac{1}{D}\mathbb{1}_D$  is a density matrix. □

**Theorem 4.13** (Channels preserve density matrices). Lean ✓ Stmt If  $E$  is a quantum channel, then  $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$ .

*Proof.* ✓ Proof Complete positivity implies positivity (Theorem 4.5), so  $E(\rho) \geq 0$ ; trace preservation gives  $\text{tr}(E(\rho)) = \text{tr}(\rho) = 1$ . □

## 4.2 Irreducibility

**Definition 4.14** (Orthogonal projection). Lean ✓ Stmt A matrix  $P \in M_D(\mathbb{C})$  is an *orthogonal projection* if  $P = P^\dagger$  and  $P^2 = P$ .

**Definition 4.15** (Irreducible map). Lean ✓ Stmt A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is *irreducible* if whenever  $P$  is an orthogonal projection satisfying  $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$ , then  $P = 0$  or  $P = \mathbb{1}$ . This is [Wol12, Theorem 6.2(1)]. The definition applies to any linear map; complete positivity is not required.

**Theorem 4.16** (Injectivity implies irreducibility). Lean ✓ Stmt If  $A$  is an injective MPS tensor, then its transfer map  $\mathcal{E}_A$  is irreducible.

*Proof.* ✓ Proof If  $P$  is an invariant projection for  $\mathcal{E}_A$ , then  $(\mathbb{1} - P)A^i P = 0$  for all  $i$ . Since the  $\{A^i\}$  span  $M_D(\mathbb{C})$ ,  $(\mathbb{1} - P)MP = 0$  for all  $M$ , so  $P = 0$  or  $P = \mathbb{1}$ . □

**Theorem 4.17** (Transfer map is CP). Lean ✓ Stmt The transfer map  $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$  is completely positive.

*Proof.* ✓ Proof It is already written in Kraus form, with Kraus operators  $\{A^i\}$ . □

**Theorem 4.18** (Transfer map is a channel). [Lean](#) [✓ Stmt](#) If  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , then the transfer map  $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$  is a quantum channel (CPTP).

*Proof.* [✓ Proof](#) Complete positivity is the observation of Theorem 4.17. Trace preservation:  $\text{tr}(\mathcal{E}_A(X)) = \sum_i \text{tr}(A^i X (A^i)^\dagger) = \text{tr}((\sum_i (A^i)^\dagger A^i)X) = \text{tr}(X)$  by cyclicity and the normalization hypothesis.  $\square$

### 4.3 Cesàro fixed-point theorem

**Definition 4.19** (Cesàro mean). [Lean](#) [✓ Stmt](#) The *Cesàro mean* of a linear map  $E$  at order  $N$ , applied to a starting matrix  $\rho_0$ , is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0).$$

**Theorem 4.20** (Cesàro telescope). [Lean](#) [✓ Stmt](#)  $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$ .

*Proof.* [✓ Proof](#) Telescoping:  $E(\sigma_N) = \frac{1}{N} \sum_{n=1}^N E^n(\rho_0)$  and subtracting  $\sigma_N$  cancels all but the first and last terms.  $\square$

**Theorem 4.21** (Hermitian fixed-point decomposition). [Lean](#) [✓ Stmt](#) Let  $E$  be a quantum channel and  $X = E(X)$  a Hermitian fixed point with decomposition  $X = P_+ - P_-$  into orthogonal positive and negative parts ( $P_\pm \geq 0$ ,  $\text{tr}(P_+ P_-) = 0$ ). Then  $E(P_+) = P_+$  and  $E(P_-) = P_-$ . This is [Wol12, Proposition 6.8].

*Proof.* [✓ Proof](#) Let  $Q$  be the support projection of  $P_+$ . Trace preservation and positivity give  $\text{tr}(QE(P_-)) = 0$  and  $E(P_-) \geq 0$ , so  $QE(P_-) = 0$ . The fixed-point equation  $E(P_+) - E(P_-) = P_+ - P_-$  then identifies  $E(P_\pm) = P_\pm$ .  $\square$

**Theorem 4.22** (Cesàro fixed-point theorem). [Lean](#) [✓ Stmt](#) Every quantum channel  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  (with  $D \geq 1$ ) has a nonzero positive semidefinite fixed point: there exists  $\rho \geq 0$ ,  $\rho \neq 0$ , with  $E(\rho) = \rho$ .

*Proof.* [✓ Proof](#) Start with any  $\rho_0 \in \mathcal{D}_D$  (Theorem 4.12). Each Cesàro mean  $\sigma_N$  lies in  $\mathcal{D}_D$  by applying channel preservation to every iterate, adding positive semidefinite matrices, rescaling by the positive average factor, and using trace linearity. By compactness (Theorem 4.10), extract a convergent subsequence  $\sigma_{N_k} \rightarrow \rho$ . The telescope identity (Theorem 4.20) gives  $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N} \|E^N(\rho_0) - \rho_0\| \rightarrow 0$ , so  $E(\rho) = \rho$  by continuity. In [Wol12, Theorem 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead.  $\square$

### 4.4 Fixed-point projection

**Definition 4.23** (Fixed-point projection). [Lean](#) [✓ Stmt](#) Given a linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  and a fixed point  $\rho$  with  $E(\rho) = \rho$  and  $\text{tr}(\rho) \neq 0$ , the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho.$$

This projects onto the span of  $\rho$  along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

**Theorem 4.24** (Power decomposition). [Lean](#) [✓ Stmt](#) If  $E$  is trace-preserving and  $E(\rho) = \rho$ , then for  $n \geq 1$ ,

$$E^n = P + (E - P)^n,$$

where  $P$  is the fixed-point projection.

*Proof.* [✓ Proof](#)  $P$  is idempotent ( $P^2 = P$ ), the fixed-point relation  $E(\rho) = \rho$  gives  $E \circ P = P$ , and trace preservation gives  $P \circ E = P$ . These imply  $(E - P) \circ P = 0$  and  $P \circ (E - P) = 0$ , so in the binomial expansion of  $(P + (E - P))^n$  all cross terms vanish, leaving  $E^n = P^n + (E - P)^n = P + (E - P)^n$ .  $\square$

## 4.5 Peripheral spectrum and primitivity

**Definition 4.25** (Peripheral spectrum). [Lean](#) [✓ Stmt](#) The *peripheral spectrum* of a bounded linear operator  $T$  is the set of spectral values  $\mu$  with  $|\mu|$  equal to the spectral radius of  $T$ .

**Definition 4.26** (Peripheral eigenvalues). [Lean](#) [✓ Stmt](#) The *peripheral eigenvalues* of a linear map  $E$  on a finite-dimensional space are the eigenvalues  $\lambda$  on the unit circle:  $|\lambda| = 1$ . For a channel, the spectral radius is 1 [Wol12, Proposition 6.1], so these coincide with the eigenvalues of maximal modulus.

The condition  $|\lambda| = 1$  is appropriate for channels, since the spectral radius of a channel is 1; for a general linear map with spectral radius  $r \neq 1$  the condition would generalise to  $|\lambda| = r$ .

**Theorem 4.27** (1 is a peripheral eigenvalue). [Lean](#) [✓ Stmt](#) If  $E$  has a nonzero fixed point  $\rho \neq 0$  with  $E(\rho) = \rho$ , then 1 is a peripheral eigenvalue of  $E$ .

*Proof.* [✓ Proof](#)  $\rho$  is an eigenvector with eigenvalue 1, and  $|1| = 1$ .  $\square$

**Theorem 4.28** (Peripheral eigenvalues lie on the unit circle). [Lean](#) [✓ Stmt](#) Every peripheral eigenvalue  $\lambda$  satisfies  $|\lambda| = 1$ .

*Proof.* [✓ Proof](#) This is the defining condition in Definition 4.26.  $\square$

**Theorem 4.29** (Finiteness of peripheral eigenvalues). [Lean](#) [✓ Stmt](#) On a finite-dimensional space, the set of peripheral eigenvalues is finite.

*Proof.* [✓ Proof](#) Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions.  $\square$

**Theorem 4.30** (Power-stable peripheral eigenvalues are roots of unity). [Lean](#) [✓ Stmt](#) Let  $E$  be a linear endomorphism on a finite-dimensional space. If  $\mu$  is a peripheral eigenvalue of  $E$  and  $\mu^n$  is an eigenvalue of  $E$  for every  $n \geq 1$ , then  $\mu$  is a root of unity.

*Proof.* [✓ Proof](#) Since  $\mu^n$  is an eigenvalue for every  $n$ , the pigeonhole principle on the finite eigenvalue set gives  $\mu^a = \mu^b$  for some  $a \neq b$ , hence  $\mu^{|a-b|} = 1$ .  $\square$

**Theorem 4.31** (Peripheral powers imply root of unity). [Lean](#) [✓ Stmt](#) If the set of peripheral eigenvalues of a linear endomorphism  $E$  on a finite-dimensional space is closed under powers ( $\mu \in \text{peripheral}(E)$  and  $n \geq 1$  imply  $\mu^n \in \text{peripheral}(E)$ ), then every peripheral eigenvalue is a root of unity.

*Proof.* [✓ Proof](#) The closure hypothesis provides  $\mu^n \in \text{peripheral}(E)$  for all  $n$ , which in particular means  $\mu^n$  is an eigenvalue. Theorem 4.30 then gives  $\mu^p = 1$  for some  $p > 0$ .  $\square$



### 4.5.1 Periodicity removal by powering

*Remark 4.32.* ✓ Stmt The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its  $p$ -blocked tensor replaces  $\mathcal{E}_A$  by  $\mathcal{E}_A^p$ , so this powering step is the operator-theoretic form of blocking.

**Lemma 4.33** (Common exponent for a finite set of roots of unity). Lean ✓ Stmt *Let  $s \subseteq \mathbb{C}$  be a finite set. Assume that for every  $\mu \in s$  there exists an exponent  $p_\mu > 0$  with  $\mu^{p_\mu} = 1$ . Then there exists  $p > 0$  such that  $\mu^p = 1$  for all  $\mu \in s$ .*

*Proof.* ✓ Proof Take  $p$  to be the least common multiple of the exponents  $p_\mu$ . □

**Lemma 4.34** (Powering collapses peripheral eigenvalues). Lean ✓ Stmt *Let  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a linear map, and assume that  $E$  has a nonzero fixed point  $\rho \neq 0$ . Let  $p > 0$ . If every peripheral eigenvalue  $\mu$  of  $E$  satisfies  $\mu^p = 1$ , then  $\text{peripheral}(E^p) = \{1\}$ .*

*Proof.* ✓ Proof The fixed point  $\rho$  gives  $1 \in \text{peripheral}(E^p)$ . For the reverse inclusion, let  $\nu \in \text{peripheral}(E^p)$ . Spectral mapping gives  $\nu = \mu^p$  for some  $\mu \in \sigma(E)$ . From  $|\nu| = 1$  and  $p > 0$  we obtain  $|\mu| = 1$ , hence  $\mu \in \text{peripheral}(E)$ . The hypothesis  $\mu^p = 1$  then forces  $\nu = 1$ . □

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

*Remark 4.35* (Bi-canonical special case). ✓ Stmt When a Kraus map happens to be both unital and trace-preserving (the “bi-canonical” case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 4.41) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 4.31. It requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

### 4.5.2 The Kadison–Schwarz inequality and the multiplicative domain

**Definition 4.36** (Kraus maps, adjoints, and normalizations). Lean Lean Lean Lean ✓ Stmt For operators  $\{K_i\}_{i=0}^{d-1} \subseteq M_D(\mathbb{C})$ , the associated *Kraus map* and *adjoint Kraus map* are

$$E(X) = \sum_i K_i X K_i^\dagger, \quad E^*(X) = \sum_i K_i^\dagger X K_i.$$

The Kraus family is *unital* when  $\sum_i K_i K_i^\dagger = \mathbb{1}$  and *trace-preserving* when  $\sum_i K_i^\dagger K_i = \mathbb{1}$ .

**Theorem 4.37** (Kadison–Schwarz inequality). Lean ✓ Stmt *Let  $E(X) = \sum_i K_i X K_i^\dagger$  be a unital Kraus map. Then for every  $X \in M_D(\mathbb{C})$ ,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

*This is [Wol12, Equation (5.2)].*

*Proof.* ✓ Proof Apply the unital completely positive map entrywise to the positive  $2 \times 2$  block matrix  $\begin{pmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{pmatrix}$  and take the Schur complement of the lower-right block. □

**Theorem 4.38** (Hilbert–Schmidt contraction). [Lean](#) [✓ Stmt](#) *If a Kraus map is both unital and trace-preserving, then*

$$\mathrm{tr}(E(X)^\dagger E(X)) \leq \mathrm{tr}(X^\dagger X)$$

for all  $X \in M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) The Kadison–Schwarz gap is positive semidefinite by Theorem 4.37. Trace preservation then identifies the traces of its two terms.  $\square$

**Theorem 4.39** (KS gap decomposition). [Lean](#) [✓ Stmt](#) *For a unital Kraus map,*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_i R_i^\dagger R_i, \quad R_i := XK_i^\dagger - K_i^\dagger E(X).$$

*Proof.* [✓ Proof](#) Expand the right-hand side, distribute the products, and use  $\sum_i K_i K_i^\dagger = \mathbb{1}$  to recover the Kadison–Schwarz gap.  $\square$

**Theorem 4.40** (KS equality implies Kraus commutation). [Lean](#) [✓ Stmt](#) *Let  $E$  be a unital Kraus map. If  $E(X^\dagger X) = E(X)^\dagger E(X)$ , then, for every  $i$ ,*

$$XK_i^\dagger = K_i^\dagger E(X).$$

*Proof.* [✓ Proof](#) The Kadison–Schwarz gap is a sum of positive semidefinite terms  $\sum_i R_i^\dagger R_i$ . If the gap vanishes, each  $R_i$  must vanish.  $\square$

**Theorem 4.41** (KS equality for peripheral eigenvectors). [Lean](#) [✓ Stmt](#) *Let  $E$  be a Kraus map that is both unital and trace-preserving. If  $E(X) = \mu X$  with  $|\mu| = 1$ , then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

*Proof.* [✓ Proof](#) The Kadison–Schwarz gap is positive semidefinite by Theorem 4.37. Trace preservation and the relation  $E(X) = \mu X$  with  $|\mu| = 1$  show that its trace is zero, hence the gap vanishes.  $\square$

**Theorem 4.42** (Left multiplicative identity from KS equality). [Lean](#) [✓ Stmt](#) *Let  $E$  be a unital Kraus map. If  $E(X^\dagger X) = E(X)^\dagger E(X)$ , then for every  $Y \in M_D(\mathbb{C})$ ,*

$$E(X^\dagger Y) = E(X)^\dagger E(Y).$$

*Proof.* [✓ Proof](#) The commutation relation of Theorem 4.40 lets one move  $X^\dagger$  past each Kraus operator inside the sum.  $\square$

**Definition 4.43** (Multiplicative domain). [Lean](#) [✓ Stmt](#) The *multiplicative domain* of a Kraus map  $E$  is the set of matrices that are simultaneously left- and right-multiplicative for  $E$ .

**Theorem 4.44** (Multiplicative domain is a  $*$ -subalgebra). [Lean](#) [✓ Stmt](#) *For a unital Kraus map, the multiplicative domain is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) The multiplicative identities are stable under addition, multiplication, and adjoint, and they contain the identity matrix.  $\square$

### 4.5.3 Peripheral closure via adjoint fixed point

The following results replace the trace-preservation hypothesis by the existence of a positive definite fixed point for the *adjoint* Kraus map, matching the weighted-trace argument in [CPGSV16, Appendix A].

**Lemma 4.45** (Peripheral closure via adjoint fixed point). Lean ✓ Stmt Let  $E(X) = \sum_i K_i X K_i^\dagger$  be a Kraus map that is unital ( $\sum_i K_i K_i^\dagger = \mathbb{1}$ ), and assume:

1. the adjoint Kraus map  $E^*(X) = \sum_i K_i^\dagger X K_i$  has a positive definite fixed point  $\rho > 0$ , and
2.  $E$  is irreducible.

Then  $\text{peripheral}(E)$  is closed under powers: if  $\mu \in \text{peripheral}(E)$  then  $\mu^n \in \text{peripheral}(E)$  for every  $n \in \mathbb{N}$ .

*Proof.* ✓ Proof Let  $E(X) = \mu X$  with  $|\mu| = 1$ , and set  $G := E(X^\dagger X) - E(X)^\dagger E(X)$ . By Kadison–Schwarz,  $G \geq 0$ . Since  $E^*(\rho) = \rho$ ,

$$\text{tr}(\rho G) = \text{tr}(\rho E(X^\dagger X)) - \text{tr}(\rho E(X)^\dagger E(X)) = \text{tr}(E^*(\rho) X^\dagger X) - \text{tr}(\rho E(X)^\dagger E(X)) = \text{tr}(\rho X^\dagger X) - |\mu|^2 \text{tr}(\rho X^\dagger X) = 0.$$

Thus the trace of the Kadison–Schwarz gap against the adjoint fixed point vanishes. Because  $\rho > 0$  and  $G \geq 0$ , this forces  $G = 0$ , so  $E(X^\dagger X) = E(X)^\dagger E(X)$ . Hence  $X^\dagger X$  is a PSD fixed point, which is positive definite by irreducibility (Theorem 6.4). So  $X$  is invertible. By Theorem 4.40 (KS equality implies Kraus commutation),  $X$  commutes with each Kraus operator; iterating via the left multiplicative identity (Theorem 4.42) gives  $E(X^n) = \mu^n X^n$ , so  $\mu^n$  is peripheral.  $\square$

**Theorem 4.46** (Roots of unity via adjoint fixed point). Lean ✓ Stmt Let  $E(X) = \sum_i K_i X K_i^\dagger$  be a Kraus map that is unital, and assume that the adjoint Kraus map  $E^*(X) = \sum_i K_i^\dagger X K_i$  has a positive definite fixed point and that  $E$  is irreducible. Then every peripheral eigenvalue of  $E$  is a root of unity.

*Proof.* ✓ Proof Combine the closure-under-powers result (Lemma 4.45) with Theorem 4.31.  $\square$

**Remark 4.47.** ✓ Stmt Lemma 4.45 and Theorem 4.46 cover the bi-canonical (unital + TP) case as well: trace preservation gives  $E^*(\mathbb{1}) = \mathbb{1}$ , which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV16, Appendix A], which uses a faithful invariant state rather than bi-canonicity.

**Definition 4.48** (Channel period). Lean ✓ Stmt The *period* of a channel  $E$  is the number of peripheral eigenvalues (counted without multiplicity):  $p(E) := |\text{peripheral}(E)|$ .

**Definition 4.49** (Primitive map). Lean ✓ Stmt A linear map is *primitive* if its only peripheral eigenvalue is  $\lambda = 1$ , i.e.  $\text{peripheral}(E) = \{1\}$ . For channels this corresponds to [Wol12, Theorem 6.7]. This definition applies to any linear endomorphism, not only to channels.

**Theorem 4.50** (Primitivity equals period one). Lean ✓ Stmt A linear map with a nonzero fixed point is primitive if and only if its period is 1.

*Proof.* ✓ Proof If  $\text{peripheral}(E) = \{1\}$ , then  $|\text{peripheral}(E)| = 1$ . Conversely, if  $|\text{peripheral}(E)| = 1$ , then since  $1 \in \text{peripheral}(E)$  (Theorem 4.27), the only element is 1.  $\square$

**Theorem 4.51** (Complementary transfer-map gap for primitive maps). [Lean](#) [✓ Stmt](#) Let  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be trace-preserving, and let  $\rho \neq 0$  satisfy  $E(\rho) = \rho$  and  $\text{tr}(\rho) \neq 0$ . Let

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$$

be the fixed-point projection associated to  $\rho$ . Assume that:

1.  $E$  is primitive;
2. every eigenvalue  $\mu$  of  $E$  satisfies  $|\mu| \leq 1$ ;
3. whenever  $E(X) = X$  and  $\text{tr}(X) = 0$ , one has  $X = 0$ .

Then every eigenvalue  $\nu$  of  $E - P$  satisfies  $|\nu| < 1$ .

*Proof.* [✓ Proof](#) Let  $(E - P)(X) = \nu X$  with  $X \neq 0$ . Then

$$E(X) = \nu X + P(X).$$

If  $\text{tr}(X) = 0$ , then  $P(X) = 0$ , so  $E(X) = \nu X$  and hence  $\nu$  is an eigenvalue of  $E$  with  $|\nu| \leq 1$ . If  $|\nu| = 1$ , primitivity forces  $\nu = 1$ , so  $X$  is a trace-zero fixed point of  $E$ ; by assumption this implies  $X = 0$ , a contradiction. Thus  $|\nu| < 1$ .

If instead  $\text{tr}(X) \neq 0$ , trace preservation and  $\text{tr}(P(X)) = \text{tr}(X)$  give

$$\text{tr}(X) = \text{tr}(E(X)) = \nu \text{tr}(X) + \text{tr}(P(X)) = \nu \text{tr}(X) + \text{tr}(X),$$

so  $\nu \text{tr}(X) = 0$  and therefore  $\nu = 0$ . In either case,  $|\nu| < 1$ . □

## 4.6 Fixed-point algebra

**Definition 4.52** (Kraus fixed points). [Lean](#) [✓ Stmt](#) The *fixed-point set* of a Kraus map  $E(X) = \sum_i K_i X K_i^\dagger$  is  $\text{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$ .

**Theorem 4.53** (Fixed points lie in the multiplicative domain). [Lean](#) [✓ Stmt](#) Let  $E$  be a trace-preserving Kraus map whose Schrödinger-picture map has a positive definite fixed point  $\rho > 0$ . Then every adjoint fixed point belongs to the multiplicative domain of the adjoint Kraus family.

*Proof.* [✓ Proof](#) The Kadison–Schwarz equality for fixed points forces both one-sided multiplicative identities, so the fixed point lies in the full multiplicative domain. □

**Theorem 4.54** (Fixed points form a  $*$ -subalgebra). [Lean](#) [✓ Stmt](#) Let  $E$  be a unital Kraus map whose adjoint map  $E^*$  has a positive definite fixed point  $\rho > 0$ . Then  $\text{Fix}(E)$  is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ . This is [Wol12, Theorem 6.12].

*Proof.* [✓ Proof](#) Closure under addition and scalar multiplication is immediate from linearity. Closure under  $\dagger$ : if  $E(X) = X$  then  $E(X^\dagger) = X^\dagger$  because fixed points are stable under adjoint. Closure under multiplication: if  $E(X) = X$  and  $E(Y) = Y$ , the weighted Kadison–Schwarz argument gives  $E(X^\dagger X) = E(X)^\dagger E(X)$  and, applied to  $X^\dagger$ , also  $E(X X^\dagger) = E(X) E(X)^\dagger$ . Thus  $X$  lies in the multiplicative domain, and Theorem 4.42 yields  $E(XY) = E(X)E(Y) = XY$ . □

*Remark 4.55.* [✓ Stmt](#) Theorem 4.53 is the trace-preserving / Heisenberg-picture counterpart of the same multiplicative-domain argument.

**Theorem 4.56** (Projected elements lie in the multiplicative domain). [Lean](#) [✓ Stmt](#) *Let  $E$  be a unital Kraus map whose adjoint map has a positive definite fixed point, and suppose  $E^2 = E$ . Then  $E(X)$  lies in the multiplicative domain of  $E$  for every  $X \in M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) Idempotence gives  $E(E(X)) = E(X)$ , so  $E(X)$  is a fixed point. By the argument in Theorem 4.54, the Kadison–Schwarz gap for the fixed point  $E(X)$  vanishes, hence

$$E(E(X)^\dagger E(X)) = E(X)^\dagger E(X).$$

Fixed points are closed under  $\dagger$ , so the same argument applied to  $E(X)^\dagger$  gives

$$E(E(X)E(X)^\dagger) = E(X)E(X)^\dagger.$$

The vanishing of both Kadison–Schwarz gaps, again by the argument of Theorem 4.54, places  $E(X)$  in the multiplicative domain.  $\square$

**Theorem 4.57** (Choi–Effros left absorption). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, for all  $X, Y \in M_D(\mathbb{C})$ ,*

$$E(E(X)Y) = E(E(X)E(Y)).$$

*Proof.* [✓ Proof](#) The previous theorem places  $E(X)$  in the multiplicative domain, so  $E(E(X)Y) = E(E(X))E(Y) = E(X)E(Y)$ . Applying the same multiplicativity to  $E(X)$  and  $E(Y)$  gives  $E(E(X)E(Y)) = E(E(X))E(E(Y))$ , and idempotence rewrites this as  $E(X)E(Y)$ .  $\square$

**Theorem 4.58** (Choi–Effros right absorption). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, for all  $X, Y \in M_D(\mathbb{C})$ ,*

$$E(XE(Y)) = E(E(X)E(Y)).$$

*Proof.* [✓ Proof](#) The previous theorem places  $E(Y)$  in the multiplicative domain, so  $E(XE(Y)) = E(X)E(E(Y)) = E(X)E(Y)$ . Applying the same multiplicativity to  $E(X)$  and  $E(Y)$  gives  $E(E(X)E(Y)) = E(E(X))E(E(Y))$ , and idempotence rewrites this as  $E(X)E(Y)$ .  $\square$

**Theorem 4.59** (Choi–Effros symmetric absorption). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, for all  $X, Y \in M_D(\mathbb{C})$ ,*

$$E(E(X)Y) = E(XE(Y)).$$

*Proof.* [✓ Proof](#) Both sides equal  $E(E(X)E(Y))$  by the two preceding theorems.  $\square$

**Theorem 4.60** (Choi–Effros projected product). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, for all  $X, Y \in M_D(\mathbb{C})$ ,*

$$E(E(X)E(Y)) = E(X)E(Y).$$

*Proof.* [✓ Proof](#) The previous theorem places  $E(X)$  and  $E(Y)$  in the multiplicative domain. Therefore multiplicativity gives

$$E(E(X)E(Y)) = E(E(X))E(E(Y)).$$

Since  $E$  is idempotent under the standing hypotheses,  $E(E(X)) = E(X)$  and  $E(E(Y)) = E(Y)$ , so

$$E(E(X)E(Y)) = E(X)E(Y).$$

$\square$

**Definition 4.61** (Adjoint fixed points). [Lean](#) [✓ Stmt](#) *The adjoint fixed-point set  $\text{Fix}(E^*) = \{X \in M_D(\mathbb{C}) : E^*(X) = X\}$ , where  $E^*(X) = \sum_i K_i^\dagger X K_i$ .*

**Theorem 4.62** (Fixed points form a  $*$ -subalgebra in the Heisenberg picture). [Lean](#) [Lean](#) [✓ Stmt](#) *If the Kraus map is trace-preserving and has a positive definite fixed point  $\rho > 0$  with  $E(\rho) = \rho$ , then  $\text{Fix}(E^*)$  is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) Apply Theorem 4.54 to the adjoint family  $\{K_i^\dagger\}$ . □

**Definition 4.63** (Kraus commutant). [Lean](#) [Lean](#) [✓ Stmt](#) The *Kraus commutant* is  $\{K_i, K_i^\dagger\}' = \{X \in M_D(\mathbb{C}) : [X, K_i] = [X, K_i^\dagger] = 0 \forall i\}$ . It is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ .

**Theorem 4.64** (Adjoint fixed points equal the Kraus commutant). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 4.62, the adjoint fixed-point  $*$ -subalgebra equals the Kraus commutant:*

$$\text{Fix}(E^*) = \{K_i, K_i^\dagger\}'.$$

*This is [Wol12, Theorem 6.13].*

*Proof.* [✓ Proof](#) ( $\supseteq$ ): if  $X$  commutes with all  $K_i$  and  $K_i^\dagger$ , then  $E^*(X) = \sum_i K_i^\dagger X K_i = X \sum_i K_i^\dagger K_i = X$  by trace preservation. ( $\subseteq$ ): any  $*$ -subalgebra inside  $\text{Fix}(E^*)$  lies in the Kraus commutant, because for  $X \in \text{Fix}(E^*)$  with  $X^\dagger X \in \text{Fix}(E^*)$  the Kadison–Schwarz gap vanishes, which forces  $XK_i = K_iX$  and  $XK_i^\dagger = K_i^\dagger X$ . □

**Theorem 4.65** (Kraus commutant inside the adjoint fixed points). [Lean](#) [✓ Stmt](#) *Under the hypotheses above, the Kraus commutant is the largest  $*$ -subalgebra contained in the adjoint fixed-point set.*

*Proof.* [✓ Proof](#) By Theorem 4.64, every  $*$ -subalgebra inside the adjoint fixed points is automatically contained in the Kraus commutant, while the Kraus commutant itself is such a  $*$ -subalgebra. □

## Chapter 5

# Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory of [Wol12, Chapter 5]: the Kadison–Schwarz inequality for Kraus maps, the multiplicative domain as a  $*$ -subalgebra on which the map restricts to a  $*$ -homomorphism, the extension to normal, subnormal, and commuting-dominant inputs for positive subunital maps, order and spectral-interval contractivity, and the transpose-trace map as a Schwarz map that is not completely positive. The peripheral-eigenvector and normal-block consequences feed the Fundamental Theorem of Chapter 11. The operator-convexity and Jensen material from [Wol12, Chapter 5] is treated separately in Chapter 21.

### 5.1 Kadison–Schwarz inequality

A linear map is completely positive (Definition 4.4) if and only if it has a Kraus form. The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map  $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$  is a Kraus map (Theorem 4.17), so these results apply directly to transfer maps.

**Definition 5.1** (Kraus map). Lean ✓ Stmt Given operators  $\{K_i\}_{i=0}^{d-1}$  with  $K_i \in M_D(\mathbb{C})$ , the *Kraus map* is  $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$ . A linear map is completely positive (Definition 4.4) if and only if it can be written in this form.

**Definition 5.2** (Adjoint Kraus map). Lean ✓ Stmt The *adjoint Kraus map* is  $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$ . When the  $\{K_i\}$  are the matrices of an MPS tensor  $A$ , this is the transfer map of the conjugate-transposed family  $i \mapsto (A^i)^\dagger$ .

**Definition 5.3** (Unital Kraus map). Lean ✓ Stmt A Kraus map is *unital* if  $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$ . Equivalently,  $E(\mathbb{1}) = \mathbb{1}$ .

**Definition 5.4** (Trace-preserving Kraus map). Lean ✓ Stmt A Kraus map is *trace-preserving* if  $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$ . Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

**Theorem 5.5** (Kadison–Schwarz inequality). [Lean](#) [✓ Stmt](#) Let  $E(X) = \sum_i K_i X K_i^\dagger$  be a unital Kraus map (i.e.  $\sum_i K_i K_i^\dagger = \mathbb{1}$ ). Then for all  $X \in M_D(\mathbb{C})$ ,

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Equation (5.2)].

*Proof.* [✓ Proof](#) The  $2 \times 2$  block matrix  $\begin{bmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{bmatrix}$  is PSD ( $= vv^\dagger$  with  $v = [X^\dagger, \mathbb{1}]^T$ ). Applying the unital CP map blockwise preserves positive semidefiniteness (each block  $K_i(\cdot)K_i^\dagger$  preserves PSD, and unitality ensures the  $(2,2)$ -block maps to  $\mathbb{1}$ ). The Schur complement of the  $(2,2)$ -block  $\mathbb{1}$  gives the result.  $\square$

**Theorem 5.6** (Kadison–Schwarz for the adjoint channel). [Lean](#) [✓ Stmt](#) If  $\sum_i K_i^\dagger K_i = \mathbb{1}$  (trace-preserving), then the adjoint Kraus map satisfies  $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$ .

*Proof.* [✓ Proof](#) The adjoint operators  $\{K_i^\dagger\}$  satisfy  $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$ , so they form a unital family. Apply Theorem 5.5 to  $\{K_i^\dagger\}$ .  $\square$

### 5.1.1 Two-positive maps and the generalized Schwarz inequality

**Definition 5.7** ( $k$ -positive map). [Lean](#) [✓ Stmt](#) A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is  $k$ -positive if  $E \otimes \text{id}_k$  is positive on  $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$ .

**Definition 5.8** (Two-positive map). [Lean](#) [✓ Stmt](#) A linear map is *two-positive* if it is 2-positive.

**Theorem 5.9** (Completely positive maps are  $k$ -positive). [Lean](#) [✓ Stmt](#) Every completely positive map is  $k$ -positive for all  $k \geq 1$ .

**Theorem 5.10** ( $(k+1)$ -positive implies  $k$ -positive). [Lean](#) [✓ Stmt](#) Positivity under amplification is monotone in  $k$ .

**Theorem 5.11** (Completely positive implies two-positive). [Lean](#) [✓ Stmt](#) Every completely positive map is two-positive.

**Theorem 5.12** (Two-positive implies positive). [Lean](#) [✓ Stmt](#) Every two-positive map is positive.

**Definition 5.13** (Unital linear map). [Lean](#) [✓ Stmt](#) A linear map  $E$  is unital if  $E(\mathbb{1}) = \mathbb{1}$ .

**Theorem 5.14** (Kadison–Schwarz for unital two-positive maps). [Lean](#) [✓ Stmt](#) If  $E$  is unital and two-positive, then for all  $X \in M_D(\mathbb{C})$ ,

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

*Proof.* Form the  $2 \times 2$  block matrix  $\begin{pmatrix} \mathbb{1} & X \\ X^\dagger & X^\dagger X \end{pmatrix} \geq 0$ . Apply  $E \otimes \text{id}_2$  (positive by 2-positivity) and read off the  $(2,2)$ -block Schur complement:  $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$ .  $\square$

**Theorem 5.15** (Kraus-form Kadison–Schwarz as a special case). [Lean](#) [✓ Stmt](#) The Kraus-map Kadison–Schwarz inequality is recovered from the two-positive formulation.



### 5.1.2 Douglas-type factorization lemmas

**Theorem 5.16** (Range inclusion implies right factorization). [Lean](#) [✓ Stmt](#) If  $\text{ran}(A) \subseteq \text{ran}(B)$ , then there exists  $C$  such that  $A = BC$ .

*Proof.* In finite dimensions,  $\text{ran}(A) \subseteq \text{ran}(B)$  implies  $A = B(B^\dagger B)^\dagger B^\dagger A$ , where  $(B^\dagger B)^\dagger$  denotes the Moore–Penrose pseudoinverse. Set  $C = (B^\dagger B)^\dagger B^\dagger A$ .  $\square$

**Theorem 5.17** (Vectorwise range criterion implies factorization). [Lean](#) [✓ Stmt](#) If  $Av \in \text{ran}(B)$  for every vector  $v$ , then there exists  $C$  with  $A = BC$ .

**Theorem 5.18** (Hilbert–Schmidt contraction). [Lean](#) [✓ Stmt](#) If a Kraus map is both unital and trace-preserving, then  $\text{tr}(E(X)^\dagger E(X)) \leq \text{tr}(X^\dagger X)$  for all  $X \in M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) By Theorem 5.5,  $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$ , so  $\text{tr}(E(X^\dagger X)) \geq \text{tr}(E(X)^\dagger E(X))$ . Trace preservation gives  $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$ .  $\square$

**Lemma 5.19** (Trace pairing for Kraus map and adjoint map). [Lean](#) [✓ Stmt](#) For all  $X, Y \in M_D(\mathbb{C})$ ,

$$\text{tr}(X E(Y)) = \text{tr}(E^*(X) Y).$$

**Lemma 5.20** (Positive-definite trace test). [Lean](#) [✓ Stmt](#) If  $A > 0$ ,  $X \geq 0$ , and  $\text{tr}(AX) = 0$ , then  $X = 0$ .

**Theorem 5.21** (Fixed-point peripheral eigenvectors satisfy KS equality). [Lean](#) [✓ Stmt](#) Let  $E$  be unital and let  $E^*$  be trace-preserving with a positive definite fixed point  $\rho > 0$ . If  $E(X) = \mu X$  with  $|\mu| = 1$ , then

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

## 5.2 Multiplicative domain

**Theorem 5.22** (KS gap decomposition). [Lean](#) [✓ Stmt](#) For a unital Kraus map  $E$ , the Kadison–Schwarz gap decomposes as

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where  $R_i := XK_i^\dagger - K_i^\dagger E(X)$ .

*Proof.* [✓ Proof](#) Expand the right-hand side  $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$ , distribute, and use unitality  $\sum_i K_i K_i^\dagger = \mathbb{1}$  to identify the result with the KS gap.  $\square$

**Theorem 5.23** (KS equality implies Kraus commutation). [Lean](#) [✓ Stmt](#) Let  $E$  be a unital Kraus map. If  $E(X^\dagger X) = E(X)^\dagger E(X)$  for some  $X$ , then  $XK_i^\dagger = K_i^\dagger E(X)$  for all  $i$ .

*Proof.* [✓ Proof](#) By Theorem 5.22, the KS gap is  $\sum_i R_i^\dagger R_i$  with  $R_i = XK_i^\dagger - K_i^\dagger E(X)$ . If the gap vanishes,  $\sum_i R_i^\dagger R_i = 0$ . Each summand  $R_i^\dagger R_i$  is PSD, so each must be zero, giving  $R_i = 0$ .  $\square$

**Theorem 5.24** (KS equality for peripheral eigenvectors). [Lean](#) [✓ Stmt](#) Let  $E$  be a Kraus map that is both unital and trace-preserving. If  $E(X) = \mu X$  with  $|\mu| = 1$ , then the KS gap vanishes:  $E(X^\dagger X) = E(X)^\dagger E(X)$ .

*Proof.* ✓ Proof The gap  $G := E(X^\dagger X) - E(X)^\dagger E(X)$  is PSD by Theorem 5.5. Its trace satisfies  $\text{tr}(G) = \text{tr}(E(X^\dagger X)) - \text{tr}(E(X)^\dagger E(X))$ . Trace preservation gives  $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$ . Using  $E(X) = \mu X$  and  $|\mu| = 1$ ,  $\text{tr}(E(X)^\dagger E(X)) = |\mu|^2 \text{tr}(X^\dagger X) = \text{tr}(X^\dagger X)$ . So  $\text{tr}(G) = 0$ , and a PSD matrix with zero trace must be zero.  $\square$

**Theorem 5.25** (Left multiplicative identity from KS equality). Lean ✓ Stmt Let  $E$  be a unital Kraus map. If  $E(X^\dagger X) = E(X)^\dagger E(X)$ , then  $E(X^\dagger Y) = E(X)^\dagger E(Y)$  for all  $Y \in M_D(\mathbb{C})$ .

*Proof.* ✓ Proof From Theorem 5.23,  $XK_i^\dagger = K_i^\dagger E(X)$  for all  $i$ . Taking conjugate transposes:  $K_i X^\dagger = E(X)^\dagger K_i$ . Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \end{aligned}$$

$\square$

**Theorem 5.26** (Right multiplicative identity from KS equality). Lean ✓ Stmt Let  $E$  be a unital Kraus map. If  $E(X^\dagger X) = E(X)^\dagger E(X)$ , then  $E(YX) = E(Y)E(X)$  for all  $Y \in M_D(\mathbb{C})$ .

*Proof.* ✓ Proof From Theorem 5.23,  $XK_i^\dagger = K_i^\dagger E(X)$  for all  $i$ . Then

$$\begin{aligned} E(YX) &= \sum_i K_i Y X K_i^\dagger \\ &= \sum_i K_i Y K_i^\dagger E(X) \\ &= E(Y)E(X). \end{aligned}$$

$\square$

### 5.2.1 Full multiplicative-domain structure

**Definition 5.27** (Right and left multiplicative domains). Lean Lean ✓ Stmt For a Kraus map  $E$ , define

$$\mathcal{A}_R(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}), E(XY) = E(X)E(Y)\},$$

and

$$\mathcal{A}_L(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}), E(YX) = E(Y)E(X)\}.$$

We follow the convention that  $\mathcal{A}_R(E)$  controls right multiplication and  $\mathcal{A}_L(E)$  controls left multiplication. These are the two one-sided multiplicative domains from [Wol12, Equations (5.11)–(5.12)].

**Definition 5.28** (Full multiplicative domain). Lean ✓ Stmt The *multiplicative domain* of  $E$  is

$$\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E).$$

**Theorem 5.29** (Multiplicative-domain characterization). Lean Lean ✓ Stmt Let  $E$  be a unital Kraus map. Then

$$X \in \mathcal{A}_R(E) \iff E(XX^\dagger) = E(X)E(X)^\dagger,$$

and

$$X \in \mathcal{A}_L(E) \iff E(X^\dagger X) = E(X)^\dagger E(X).$$

Together these are the two one-sided characterizations from [Wol12, Theorem 5.7].

*Proof.* ✓ Proof If  $X$  lies in a one-sided multiplicative domain, evaluate the defining identity at  $Y = X^\dagger$ . Conversely, the left-domain implication is exactly Theorem 5.25, and the right-domain implication follows by applying Theorem 5.25 to  $X^\dagger$ .  $\square$

**Theorem 5.30** (One-sided multiplicative domains are subalgebras). Lean Lean ✓ Stmt For a unital Kraus map  $E$ , both  $\mathcal{A}_R(E)$  and  $\mathcal{A}_L(E)$  are unital subalgebras of  $M_D(\mathbb{C})$ .

*Proof.* ✓ Proof Closure under addition follows from linearity of  $E$ . Closure under multiplication: if  $X, Y \in \mathcal{A}_R(E)$ , then  $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$  for all  $Z$ , using Theorem 5.29 iteratively, and the  $\mathcal{A}_L(E)$  case is analogous. Containment of  $\mathbb{1}$  follows from unitality of  $E$ .  $\square$

**Theorem 5.31** (Multiplicative domain is a  $*$ -subalgebra). Lean ✓ Stmt For a unital Kraus map  $E$ , the full multiplicative domain  $\mathcal{A}(E)$  is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ . This is the algebraic content of [Wol12, Theorem 5.7].

*Proof.* ✓ Proof If  $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$ , then  $E(X^\dagger X) = E(X)^\dagger E(X)$  and  $E(XX^\dagger) = E(X)E(X)^\dagger$ . The characterization of Theorem 5.29 shows  $X^\dagger \in \mathcal{A}(E)$ . Combined with Theorem 5.30, this gives a  $*$ -subalgebra.  $\square$

**Theorem 5.32** (Restriction to multiplicative domain is a  $*$ -homomorphism). Lean ✓ Stmt If  $E$  is unital, then the restricted map  $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \rightarrow M_D(\mathbb{C})$  is a  $*$ -algebra homomorphism. This is the homomorphism conclusion of [Wol12, Theorem 5.7].

*Proof.* ✓ Proof Multiplicativity on  $\mathcal{A}(E)$  is immediate from the definition. For  $X \in \mathcal{A}(E)$ , the Kraus commutation relation from Theorem 5.23 gives  $E(X^\dagger) = E(X)^\dagger$ , so the restriction preserves adjoints.  $\square$

## 5.2.2 Schwarz inequality for normal operators

**Theorem 5.33** (CP Schwarz inequality for normal operators). Lean Lean ✓ Stmt Let  $E^*(X) = \sum_i K_i^\dagger X K_i$  be the adjoint Kraus map of a trace-preserving Kraus family, and let  $A \in M_D(\mathbb{C})$  be normal. Then the Schwarz gap

$$E^*(A^\dagger A) - E^*(A^\dagger)E^*(A) \quad (5.1)$$

is positive semidefinite. Equivalently,

$$E^*(A^\dagger)E^*(A) \leq E^*(A^\dagger A). \quad (5.2)$$

This is the completely positive case of [Wol12, Proposition 5.1].

*Proof.* ✓ Proof The positivity statement is exactly Theorem 5.6. The second formulation (5.2) is the Loewner-order reformulation of the first (5.1).  $\square$

**Theorem 5.34** (Schwarz inequality for normal operators). Lean ✓ Stmt Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a positive linear map with  $T(\mathbb{1}) \leq \mathbb{1}$ . If  $A \in M_D(\mathbb{C})$  is normal, then

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Proposition 5.1].

*Proof.* ✓ Proof Since  $A$  is normal, diagonalize  $A = UDU^\dagger$  with  $D$  diagonal and  $U$  unitary. Express  $A^\dagger A = U|D|^2U^\dagger$  as a sum  $\sum_{i,j} \tilde{d}_i d_j P_i P_j$  where  $P_i = U e_i e_i^\dagger U^\dagger$ . Each  $P_i$  is rank-one PSD, and the images  $T(P_i)$  pairwise commute because they are images of mutually orthogonal projectors under a positive map. The Schwarz gap then reduces to a block quadratic form in the images, which is shown non-negative by a simultaneous-diagonalization argument: positivity of  $T$  ensures the scalar PSD matrix  $(\langle v_k, T(P_i P_j) v_k \rangle)_{i,j}$  dominates the product  $(\langle v_k, T(P_i) v_k \rangle \langle v_k, T(P_j) v_k \rangle)_{i,j}$ , and combining over a common eigenbasis of the  $T(P_i)$  yields the result.  $\square$

*Remark 5.35.* The proof follows the positivity-on-abelian argument of [Wol12, Proposition 1.6]: a positive map restricted to the commutative  $*$ -subalgebra generated by the spectral projectors of  $A$  satisfies the Schwarz inequality directly, without invoking the Arveson extension theorem.

### 5.2.3 Schwarz inequality for subnormal and commuting-dominant operators

**Theorem 5.36** (Schwarz inequality for subnormal operators). Lean ✓ Stmt Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a positive linear map with  $T(1) \leq 1$ . If  $A \in M_D(\mathbb{C})$  is subnormal (i.e. there exists a normal operator on a larger space whose compression to  $\mathbb{C}^D$  is  $A$ ), then

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Theorem 5.5].

*Proof.* ✓ Proof Embed  $A$  as the northwest corner of a normal operator  $N$  on  $\mathbb{C}^D \oplus \mathbb{C}^E$ . Extend  $T$  to a positive subunitary map  $\tilde{T}$  on the larger space (acting as  $T$  on the northwest block, zero elsewhere). Apply Theorem 5.34 to  $\tilde{T}$  and  $N$ , then compress the resulting inequality back to  $\mathbb{C}^D$ .  $\square$

**Theorem 5.37** (Schwarz inequality for commuting-dominant operators). Lean ✓ Stmt Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a positive linear map with  $T(1) \leq 1$ . If  $A \in M_D(\mathbb{C})$  admits a positive semidefinite dominant  $B \geq 0$  with  $A^\dagger A \leq B$  and  $B$  commuting with  $A$  ( $BA = AB$ ), then

$$T(A^\dagger)T(A) \leq T(B) \quad \text{and} \quad T(A)T(A^\dagger) \leq T(B).$$

This is [Wol12, Theorem 5.6].

*Proof.* ✓ Proof First take  $B > 0$ . Since  $B$  commutes with  $A$  and  $A^\dagger A \leq B$ , also  $AA^\dagger \leq B$ , so  $D_R = (B - A^\dagger A)^{1/2}$  and  $D_L = (B - AA^\dagger)^{1/2}$  are positive semidefinite. The block matrix  $N = \begin{pmatrix} A & D_L \\ D_R & -A^\dagger \end{pmatrix}$  is normal, with  $N^\dagger N = NN^\dagger = B \oplus B$ . As in the subnormal case, extend  $T$  to a positive subunitary map on  $M_{2D}(\mathbb{C})$ , apply the normal-operator Schwarz inequality (Theorem 5.34) to that extension and  $N$ , and compress back to  $\mathbb{C}^D$ . Since  $N$  is normal, both  $T(A^\dagger)T(A) \leq T(B)$  and  $T(A)T(A^\dagger) \leq T(B)$  follow. For general  $B \geq 0$ , apply this to  $B_\varepsilon = B + \varepsilon 1 > 0$  and let  $\varepsilon \rightarrow 0$ .  $\square$

**Theorem 5.38** (CP variant: Kadison–Schwarz for commuting-dominant inputs). Lean ✓ Stmt Let  $E^*$  be the adjoint of a trace-preserving Kraus map. If  $A \in M_D(\mathbb{C})$  admits a positive semidefinite dominant  $B$  commuting with  $A$  and satisfying  $A^\dagger A \leq B$ , then  $E^*(A^\dagger)E^*(A) \leq E^*(B)$  and  $E^*(A)E^*(A^\dagger) \leq E^*(B)$ .

*Proof.* ✓ Proof Apply Theorem 5.37 to the adjoint Kraus map (which is unital under the trace-preservation hypothesis).  $\square$

### 5.2.4 Positive maps preserve order and spectral intervals

**Theorem 5.39** (Positive maps are monotone). [Lean](#) [✓ Stmt](#) *If  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is positive and  $A \leq B$ , then  $T(A) \leq T(B)$ .*

*Proof.* [✓ Proof](#) Since  $B - A \geq 0$ , positivity gives  $T(B - A) \geq 0$ . By linearity, this is  $T(B) - T(A) \geq 0$ .  $\square$

**Theorem 5.40** (Positive maps preserve adjoints). [Lean](#) [✓ Stmt](#) *If  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is positive, then  $T(A^\dagger) = T(A)^\dagger$  for all  $A \in M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) Write  $A = B + iC$  with  $B$  and  $C$  Hermitian. Positive maps send Hermitian matrices to Hermitian matrices, so they preserve this decomposition and commute with taking adjoints.  $\square$

**Theorem 5.41** (Order intervals are preserved). [Lean](#) [✓ Stmt](#) *Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be positive with  $T(\mathbb{1}) \leq \mathbb{1}$ . If  $a \leq 0 \leq b$  and  $a\mathbb{1} \leq A \leq b\mathbb{1}$ , then*

$$a\mathbb{1} \leq T(A) \leq b\mathbb{1}.$$

*This is the matrix form of [Wol12, Equation (5.21)].*

*Proof.* [✓ Proof](#) By monotonicity,  $T(a\mathbb{1}) \leq T(A) \leq T(b\mathbb{1})$ . For the lower bound,  $T(a\mathbb{1}) = aT(\mathbb{1})$ . Since  $a \leq 0$  and  $T(\mathbb{1}) \leq \mathbb{1}$ , we have  $aT(\mathbb{1}) \geq a\mathbb{1}$ . For the upper bound,  $T(b\mathbb{1}) = bT(\mathbb{1}) \leq b\mathbb{1}$  since  $b \geq 0$ . Therefore  $a\mathbb{1} \leq T(A) \leq b\mathbb{1}$ .  $\square$

**Theorem 5.42** (Spectral interval contractivity). [Lean](#) [✓ Stmt](#) *Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be positive with  $T(\mathbb{1}) \leq \mathbb{1}$ , and let  $A \in M_D(\mathbb{C})$  be Hermitian. If  $\text{spec}(A) \subseteq [a, b]$  with  $a \leq 0 \leq b$ , then*

$$\text{spec}(T(A)) \subseteq [a, b].$$

*This is [Wol12, Equation (5.21)].*

*Proof.* [✓ Proof](#) For Hermitian matrices, the inclusion  $\text{spec}(A) \subseteq [a, b]$  is equivalent to the order bounds  $a\mathbb{1} \leq A \leq b\mathbb{1}$ . Apply Theorem 5.41 and translate the resulting inequalities back into spectral bounds.  $\square$

### 5.2.5 A positive Schwarz map that is not completely positive

*Example 5.43* (Transpose-trace map on  $M_2(\mathbb{C})$ ). [Lean](#) [✓ Stmt](#) Consider the linear map  $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$  defined by

$$T(A) = \frac{1}{2}A^T + \frac{1}{4}\text{tr}(A)\mathbb{1}.$$

This is the map from [Wol12, Example 5.3].

**Theorem 5.44** (The transpose-trace map is positive). [Lean](#) [✓ Stmt](#) *The map from Example 5.43 is positive.*

*Proof.* [✓ Proof](#) It is a non-negative linear combination of the transpose map and the map  $A \mapsto \text{tr}(A)\mathbb{1}$ , and both of these maps send positive semidefinite matrices to positive semidefinite matrices.  $\square$

**Theorem 5.45** (The transpose-trace map satisfies the Schwarz inequality). [Lean](#) [✓ Stmt](#) For every  $A \in M_2(\mathbb{C})$ ,

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0,$$

where  $T$  is the map from Example 5.43.

*Proof.* [✓ Proof](#) Write the Schwarz gap as  $F(A)$ . One checks that  $F(A + \lambda \mathbb{1}) = F(A)$  for every scalar  $\lambda$ , so it is enough to treat the case  $\text{tr}(A) = 0$ . In that case a direct  $2 \times 2$  computation shows that  $F(A)$  dominates  $\frac{1}{2}(A^\dagger A)^T$ , hence is positive semidefinite.  $\square$

**Theorem 5.46** (The transpose-trace map is not completely positive). [Lean](#) [✓ Stmt](#) The map from Example 5.43 is not completely positive.

*Proof.* [✓ Proof](#) Compute the Choi matrix of  $T$  and test it on the antisymmetric vector  $|01\rangle - |10\rangle$ . The resulting expectation value is negative, so the Choi matrix is not positive semidefinite.  $\square$

## Chapter 6

# Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes the Perron–Frobenius theory: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, ergodicity of irreducible channels, the spectral-radius identification of Perron eigenvalues, and Wolf’s spectral characterization of irreducibility. The conceptual source is Wolf’s treatment of irreducible positive maps [Wol12, Chapter 6, especially Theorems 6.2–6.5 and Corollary 6.3] and [EHK78].

**Theorem 6.1** (Brouwer fixed point on density matrices). Lean Lean Lean Lean Lean ✓ Stmt  
*Let  $D > 0$ . Every continuous map from the compact convex set of density matrices in  $M_D(\mathbb{C})$  to itself has a fixed point.*

*Proof.* ✓ Proof The proof starts from Brouwer’s theorem for products of simplices, transfers it to closed cubes, and then to compact retracts of finite-dimensional real normed spaces. The density-matrix set is such a compact retract, using the explicit density retraction.  $\square$

### 6.1 Positive definiteness

**Remark 6.2.** ✓ Stmt Wolf’s general theory treats irreducible positive maps first ([Wol12, Theorem 6.2]), then derives non-degeneracy and positive definiteness ([Wol12, Theorem 6.3]). The injective-tensor version (Theorem 6.3) uses a shorter direct kernel argument; the irreducible CP-map version (Theorem 6.4) follows via the support-projection argument of [Wol12, Theorem 6.2].

**Theorem 6.3** (PSD fixed point is positive definite). Lean ✓ Stmt *Let  $A$  be an injective MPS tensor and let  $\rho \geq 0$ ,  $\rho \neq 0$ , be a fixed point of the transfer map  $\mathcal{E}_A(\rho) = \rho$ . Then  $\rho$  is positive definite.*

*Proof.* ✓ Proof Suppose  $v^\dagger \rho v = 0$  for some  $v \neq 0$ . Since  $\rho \geq 0$ , this implies  $\rho v = 0$ . The fixed-point equation gives  $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$ . Each term is non-negative; if the sum is zero, then  $\rho (A^i)^\dagger v = 0$  for all  $i$ : the kernel of  $\rho$  is invariant under all  $(A^i)^\dagger$ . Since the  $\{A^i\}$  span  $M_D(\mathbb{C})$ , we have  $\rho M v = 0$  for every matrix  $M$ . Given any vector  $w$ , choose  $M$  with  $M v = w$ . Then  $\rho w = 0$  for every  $w$ , so  $\rho = 0$ , contradicting  $\rho \neq 0$ . In [Wol12, Theorem 6.3(2)], the same conclusion is reached via the growth condition  $(\mathbb{1} + \mathcal{E}_A)^{D-1}(\rho) > 0$  of [Wol12, Theorem 6.2, condition (2)]; here the direct kernel argument is shorter under injectivity.  $\square$

**Theorem 6.4** (PSD fixed point is positive definite — irreducible version). Lean ✓ Stmt *If  $\mathcal{E}_A$  is irreducible and  $\rho \geq 0$ ,  $\rho \neq 0$  is a fixed point, then  $\rho$  is positive definite.*

*Proof.* ✓ Proof Suppose  $\rho$  is not positive definite. Then  $\rho$  has a nontrivial kernel; let  $Q$  be the orthogonal projection onto the support of  $\rho$  (the complement of the kernel). A direct computation using the fixed-point equation  $\mathcal{E}_A(\rho) = \rho$  shows that  $Q$  is an invariant projection for  $\mathcal{E}_A$ : it satisfies  $Q\mathcal{E}_A(QXQ)Q = \mathcal{E}_A(QXQ)$  for all  $X$ . By irreducibility of  $\mathcal{E}_A$  (Definition 4.15), every invariant projection is 0 or  $\mathbb{1}$ . If  $Q = 0$  then  $\rho = 0$ , contradicting  $\rho \neq 0$ . If  $Q = \mathbb{1}$  then  $\rho$  is positive definite, contradicting our assumption. Hence  $\rho$  must be positive definite. This is the support-projection direction of [Wol12, Theorem 6.2(1)].  $\square$

**Theorem 6.5** (Orthogonal trace condition). Lean ✓ Stmt *Let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ , and let  $A, B \geq 0$  be nonzero positive semidefinite matrices with  $\text{tr}(BA) = 0$ . Then there exists  $t$  with  $1 \leq t \leq D - 1$  such that*

$$\text{tr}(B \cdot E^t(A)) > 0.$$

*Proof.* ✓ Proof The growth theorem for irreducible CP maps ([Wol12, Theorem 6.2(2)]) gives  $(\mathbb{1} + E)^{D-1}(A) > 0$  (positive definite). Since  $B \geq 0$  is nonzero,  $\text{tr}(B \cdot (\mathbb{1} + E)^{D-1}(A)) > 0$ . Expanding by the binomial theorem:

$$\text{tr} \left( B \cdot \sum_{k=0}^{D-1} \binom{D-1}{k} E^k(A) \right) > 0.$$

Each term  $\text{tr}(B \cdot E^k(A)) \geq 0$  since both factors are PSD. The  $k = 0$  term vanishes by hypothesis ( $\text{tr}(BA) = 0$ ). Hence at least one  $k \in \{1, \dots, D - 1\}$  contributes a strictly positive term. This is [Wol12, Theorem 6.2(4)]: item (2) (growth condition) implies item (4) by a binomial expansion and PSD positivity argument.  $\square$

## 6.2 Uniqueness

**Theorem 6.6** (Uniqueness of PSD fixed point). Lean ✓ Stmt *Let  $A$  be an injective MPS tensor. If  $\rho, \sigma \geq 0$  are both nonzero fixed points of  $\mathcal{E}_A$ , then  $\sigma = c\rho$  for some  $c \in \mathbb{C}$ .*

*Proof.* ✓ Proof Both  $\rho$  and  $\sigma$  are positive definite by Theorem 6.3. Write  $\rho = SS^\dagger$  via a spectral square root and set  $H = (S^\dagger)^{-1}\sigma S^{-1}$ , which is Hermitian and positive definite. Let  $c_0$  be the minimal eigenvalue of  $H$ . Then  $\tau := \sigma - c_0\rho$  is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of  $c_0$ ). By Theorem 6.3,  $\tau$  must be zero, giving  $\sigma = c_0\rho$ . In [Wol12, Theorem 6.3, item 2] uniqueness is proved via the growth condition of [Wol12, Theorem 6.2]; the same minimal-eigenvalue argument applies here, using the injective-tensor positive-definiteness (Theorem 6.3) in place of the growth condition.  $\square$

*Remark 6.7.* ✓ Stmt The proof produces a positive real scalar  $c_0 > 0$ : it is the minimal eigenvalue of the positive definite Hermitian matrix  $H = (S^\dagger)^{-1}\sigma S^{-1}$ .

**Theorem 6.8** (Irreducible fixed-point uniqueness). Lean ✓ Stmt *Let  $A$  be an MPS tensor whose transfer map is irreducible. If  $\rho, \sigma \geq 0$  are fixed points of  $\mathcal{E}_A$ , and if  $\rho$  is nonzero, then  $\sigma = c\rho$  for some scalar  $c$ .*

*Proof.* ✓ Proof The irreducible positive-definiteness theorem (Theorem 6.4) upgrades every nonzero positive semidefinite fixed point to a positive definite one. The same minimal eigenvalue argument used in Theorem 6.6 then applies.  $\square$



**Theorem 6.9** (Uniqueness of positive eigenvalue for irreducible CP maps). [Lean](#) [✓ Stmt](#) Let  $D \geq 1$  and let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ . Suppose  $\rho, \sigma \geq 0$  are nonzero positive semidefinite matrices satisfying  $E(\rho) = r_1 \rho$  and  $E(\sigma) = r_2 \sigma$  for real scalars  $r_1, r_2 > 0$ . Then  $r_1 = r_2$ .

*Proof.* [✓ Proof](#) Extract Kraus operators  $K$  for  $E$ . Irreducibility of  $E$  passes to the adjoint transfer map  $E^\dagger(\cdot) = \sum_i K_i^\dagger(\cdot) K_i$ , which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 6.22) together with the irreducible positive-definiteness upgrade (Theorem 6.4) gives a positive definite eigenvector  $\tau > 0$  with eigenvalue  $t > 0$ :  $E^\dagger(\tau) = t \tau$ . For any nonzero PSD  $X$  with  $E(X) = sX$ , the trace pairing identity yields

$$s \cdot \text{tr}(\tau X) = \text{tr}(\tau \cdot E(X)) = \text{tr}(E^\dagger(\tau) \cdot X) = t \cdot \text{tr}(\tau X).$$

Since  $\tau > 0$  and  $X \geq 0$ ,  $X \neq 0$ , the scalar  $\text{tr}(\tau X) > 0$ , so  $s = t$ . Applying this to both  $(\rho, r_1)$  and  $(\sigma, r_2)$  gives  $r_1 = t = r_2$ . This is [Wol12, Theorem 6.3, item 3], the non-degeneracy of the spectral-radius eigenvalue for irreducible CP maps; the proof follows Wolf’s dual-map trace argument.  $\square$

### 6.3 Existence and the Perron–Frobenius theorem

**Theorem 6.10** (Existence of PSD fixed point). [Lean](#) [✓ Stmt](#) Assume  $D \geq 1$ . Let  $A$  be an MPS tensor with  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  (so that  $\mathcal{E}_A$  is trace-preserving). Then there exists  $\rho \geq 0$ ,  $\rho \neq 0$ , with  $\mathcal{E}_A(\rho) = \rho$ .

*Proof.* [✓ Proof](#) Under the normalization  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , the transfer map is a quantum channel (Theorem 4.18). The Cesàro fixed-point theorem (Theorem 4.22) then provides the fixed point; injectivity is not needed for this step.  $\square$

**Theorem 6.11** (Quantum Perron–Frobenius). [Lean](#) [✓ Stmt](#) Assume  $D \geq 1$ . Let  $A$  be an injective MPS tensor with  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ . Then the transfer map  $\mathcal{E}_A$  has a unique PSD fixed point  $\rho$  (up to scaling), and  $\rho$  is positive definite.

*Proof.* [✓ Proof](#) Combine Theorems 6.10, 6.3, and 6.6.  $\square$

**Theorem 6.12** (Quantum Perron–Frobenius for irreducible transfer maps). [Lean](#) [✓ Stmt](#) Assume  $D \geq 1$ . Let  $A$  be an MPS tensor such that its transfer map is irreducible and  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  (so that  $\mathcal{E}_A$  is trace-preserving). Then  $\mathcal{E}_A$  has a unique positive definite fixed point, up to scalar multiple.

*Proof.* [✓ Proof](#) Existence is the channel fixed-point theorem (Theorem 6.10). Irreducibility upgrades the fixed point to positive definite, and Theorem 6.8 gives uniqueness up to scalar multiple.  $\square$

### 6.4 Right- and left-canonical gauges

**Theorem 6.13** (Right-canonical (unital) gauge). [Lean](#) [✓ Stmt](#) Let  $\mathcal{E}_A(\rho) = \rho$  and let  $S$  be an invertible matrix with  $SS^\dagger = \rho$ . Define the gauged operators  $A'^i = S^{-1}A^iS$ . Then  $\sum_i A'^i(A'^i)^\dagger = \mathbb{1}$ , i.e. the gauged Kraus map is unital. This is the right-canonical normalization.

*Proof.* [✓ Proof](#) Expand each summand:  $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i \underbrace{SS^\dagger}_\rho (A^i)^\dagger (S^\dagger)^{-1}$ . Summing over  $i$  and using  $\sum_i A^i \rho (A^i)^\dagger = \rho$  gives  $S^{-1} \rho (S^\dagger)^{-1} = S^{-1} SS^\dagger (S^\dagger)^{-1} = \mathbb{1}$ .  $\square$

**Theorem 6.14** (Left-canonical (trace-preserving) gauge). [Lean](#) [✓ Stmt](#) Let  $\sigma$  be a fixed point of the adjoint transfer map  $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ , and let  $S$  be invertible with  $S^\dagger S = \sigma$ . Define  $A^i = S A^i S^{-1}$ . Then  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization.

*Proof.* [✓ Proof](#) Expand each summand:  $(S A^i S^{-1})^\dagger (S A^i S^{-1}) = (S^\dagger)^{-1} (A^i)^\dagger \underbrace{S^\dagger S}_\sigma A^i S^{-1}$ . Summing over  $i$  and using  $\sum_i (A^i)^\dagger \sigma A^i = \sigma$  gives  $(S^\dagger)^{-1} \sigma S^{-1} = (S^\dagger)^{-1} S^\dagger S S^{-1} = \mathbb{1}$ .  $\square$

**Theorem 6.15** (Square-root trace-preserving gauge from an adjoint fixed point). [Lean](#) [✓ Stmt](#) Let  $\sigma$  be positive definite and satisfy  $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ . Define

$$B^i = \sigma^{1/2} A^i \sigma^{-1/2}.$$

Then  $B$  is trace-preserving:  $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ .

*Proof.* [✓ Proof](#) Since  $\sigma^{1/2}$  is self-adjoint and  $\sigma^{1/2} \sigma^{1/2} = \sigma$ , each summand becomes  $(\sigma^{1/2})^{-1} (A^i)^\dagger \sigma A^i \sigma^{-1/2}$ . Summing over  $i$  and using the fixed-point equation leaves  $(\sigma^{1/2})^{-1} \sigma \sigma^{-1/2} = \mathbb{1}$ .  $\square$

**Lemma 6.16** (Square-root trace-preserving gauge is a gauge equivalence). [Lean](#) [✓ Stmt](#) If  $\sigma$  is positive definite and  $B^i = \sigma^{1/2} A^i \sigma^{-1/2}$ , then  $A$  and  $B$  are gauge-equivalent.

*Proof.* [✓ Proof](#) The gauge matrix is  $\sigma^{1/2}$ , which is invertible because  $\sigma$  is positive definite.  $\square$

*Remark 6.17.* [✓ Stmt](#) The right-canonical gauge of Theorem 6.13 and the left-canonical gauge of Theorem 6.14 are generally different similarity transforms: one is built from a fixed point of  $\mathcal{E}_A$ , the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

## 6.5 Similarity preserves irreducibility

**Lemma 6.18** (Similarity transform preserves irreducibility). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ , let  $C \in M_D(\mathbb{C})$  be invertible ( $\det C \neq 0$ ), and let  $c > 0$ . Define the similarity-transformed map

$$E'(X) = c \cdot C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}.$$

Then  $E'$  is also irreducible.

*Proof.* [✓ Proof](#) Suppose  $Q$  is an invariant projection for  $E'$ :  $Q \neq 0$ ,  $Q \neq \mathbb{1}$ , and  $(\mathbb{1}-Q)E'(QXQ)Q = 0$  for all  $X$ . Setting  $R = CQC^\dagger$ , the support projection  $P$  of  $R$  is an invariant projection for  $E$ . Irreducibility of  $E$  forces  $P = 0$  or  $P = \mathbb{1}$ . The invertibility of  $C$  then forces  $Q = 0$  or  $Q = \mathbb{1}$ , a contradiction. This is the full similarity version of [Wol12, Proposition 6.6]; the scalar case ( $C = \mathbb{1}$ ) gives the scaling result stated next.  $\square$

**Lemma 6.19** (Similarity is an involution). [Lean](#) [✓ Stmt](#) For any invertible  $C \in M_D(\mathbb{C})$  and any linear map  $E$  on  $M_D(\mathbb{C})$ , the similarity transforms by  $C$  and  $C^{-1}$  compose to the identity:

$$X \mapsto C(C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}) C^\dagger = E(X).$$

*Proof.* [✓ Proof](#) Direct computation: the inner conjugation  $C \cdot (C^{-1} X (C^\dagger)^{-1}) \cdot C^\dagger$  collapses to  $X$  using  $CC^{-1} = \mathbb{1}$  and  $(C^\dagger)^{-1} C^\dagger = \mathbb{1}$ , and the outer conjugation by  $C(\cdot)C^\dagger$  then cancels the inner  $C^{-1}(\cdot)(C^\dagger)^{-1}$  on  $E(X)$  for the same reason.  $\square$

**Lemma 6.20** (Irreducibility is invariant under similarity). Lean ✓ Stmt For any invertible  $C \in M_D(\mathbb{C})$  and any linear map  $E$  on  $M_D(\mathbb{C})$ , the similarity transform  $X \mapsto C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}$  is irreducible if and only if  $E$  is.

*Proof.* ✓ Proof The forward direction follows from Lemma 6.18 applied with the inverse  $C^{-1}$  (whose determinant is also nonzero): if the similarity transform of  $E$  by  $C$  is irreducible, then so is its similarity transform by  $C^{-1}$ , which equals  $E$  by Lemma 6.19. The reverse direction is Lemma 6.18 itself (with  $c = 1$ ).  $\square$

**Theorem 6.21** (Scaling preserves irreducibility). Lean ✓ Stmt If  $E$  is an irreducible map and  $c \neq 0$ , then  $cE$  is also irreducible. This is the scalar special case ( $C = \mathbb{1}$ ) of [Wol12, Proposition 6.6].

*Proof.* ✓ Proof An invariant projection for  $cE$  is an invariant projection for  $E$  (scale out the nonzero constant from the commutation relation).  $\square$

## 6.6 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Theorem 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Theorem 6.3]. The TP-gauge application follows [CPGSV16, Appendix A].

**Theorem 6.22** (PSD eigenvector for positive maps). Lean ✓ Stmt Let  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a positive linear map with  $D > 0$ , and assume that  $E(\sigma) \neq 0$  for every nonzero PSD matrix  $\sigma$ . Then there exist a nonzero PSD matrix  $\rho$  and a positive real  $r > 0$  such that  $E(\rho) = r\rho$ .

This is the finite-dimensional Perron–Frobenius theorem for positive maps on matrix algebras ([Wol12, Theorem 6.5]; see also [EHK78]). The proof applies Brouwer’s fixed-point theorem to the normalization map  $\rho \mapsto E(\rho)/\text{tr}(E(\rho))$  on the compact convex set of density matrices.

**Theorem 6.23** (Adjoint transfer map is nonzero). Lean ✓ Stmt If some Kraus operator  $A^i \neq 0$ , then the adjoint transfer map  $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$  is nonzero.

*Proof.* ✓ Proof If  $\mathcal{E}_A^\dagger = 0$  then  $\mathcal{E}_A^\dagger(\mathbb{1}) = 0$ , so  $\sum_i (A^i)^\dagger A^i = 0$ . Since each summand  $(A^i)^\dagger A^i$  is PSD, each  $(A^i)^\dagger = 0$  and hence each  $A^i = 0$ .  $\square$

**Theorem 6.24** (Positive-definite adjoint eigenvector for irreducible tensors). Lean ✓ Stmt Let  $A$  be an irreducible MPS tensor with  $D > 0$  and some  $A^i \neq 0$ . Then there exist a positive definite matrix  $\sigma$  and a positive real  $r > 0$  such that  $\mathcal{E}_A^\dagger(\sigma) = r\sigma$ .

This combines the general PF existence ([Wol12, Theorem 6.5]) with the irreducible upgrade to positive definiteness ([Wol12, Theorem 6.3(2)]); the application to the adjoint transfer map follows [CPGSV16, Appendix A].

*Proof.* ✓ Proof By Theorem 6.23,  $\mathcal{E}_A^\dagger \neq 0$ . Since  $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$  is a Kraus map, it is positive. Irreducibility of  $A$  transfers to the adjoint transfer map. More precisely: if  $P$  is invariant for  $\mathcal{E}_A$  (i.e.  $(\mathbb{1} - P)K_i P = 0$  for all Kraus operators  $K_i$ ), then taking adjoints gives  $P K_i^\dagger (\mathbb{1} - P) = 0$ , which means  $\mathbb{1} - P$  is invariant for  $\mathcal{E}_A^\dagger$ . Hence  $\mathcal{E}_A$  is irreducible if and only if  $\mathcal{E}_A^\dagger$  is irreducible, and together with  $\mathcal{E}_A^\dagger \neq 0$  this implies  $\mathcal{E}_A^\dagger(\tau) \neq 0$  for every nonzero PSD matrix  $\tau$ . Hence Theorem 6.22 applies to  $\mathcal{E}_A^\dagger$ , giving  $\sigma_0 \geq 0$  (PSD, nonzero) and  $r > 0$  with  $\mathcal{E}_A^\dagger(\sigma_0) = r\sigma_0$ . Rescale: set  $T^i = r^{-1/2}(A^i)^\dagger$ . Then  $\mathcal{E}_T(\sigma_0) = \sigma_0$ , and  $\mathcal{E}_T$  is irreducible (by

Theorem 6.21, since irreducibility of  $\mathcal{E}_A$  transfers to the adjoint and is preserved under scaling). Since  $\mathcal{E}_T$  is irreducible with a PSD fixed point, the positive-definiteness upgrade (Theorem 6.4) gives  $\sigma_0$  positive definite.  $\square$

**Theorem 6.25** (TP-gauge existence for irreducible tensors). [Lean] [✓ Stmt] *Let  $A$  be an irreducible MPS tensor with  $D > 0$  and some  $A^i \neq 0$ . Then there exist a positive real  $r$ , a positive definite matrix  $\sigma$ , and a gauge-equivalent tensor  $B$  such that*

$$B^i = \sigma^{1/2} \cdot (r^{-1/2} A^i) \cdot \sigma^{-1/2}$$

*is trace-preserving:  $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ .*

*This is the “spectral rescaling + TP gauge” step of [CPGSV16, Appendix A]. The similarity is generally non-unitary.*

*Proof.* [✓ Proof] By Theorem 6.24, obtain  $\sigma$  positive definite and  $r > 0$  with  $\mathcal{E}_A^\dagger(\sigma) = r\sigma$ . Set  $c = r^{-1/2}$  and  $A'^i = cA^i$ . Then  $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$ . By the square-root trace-preserving gauge construction (Theorem 6.15),  $B^i = \sigma^{1/2} A'^i \sigma^{-1/2}$  satisfies  $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ . Lemma 6.16 gives the required gauge equivalence.  $\square$

**Theorem 6.26** (Unital-gauge existence for irreducible tensors). [Lean] [✓ Stmt] *Let  $A$  be an irreducible MPS tensor with  $D > 0$  and some  $A^i \neq 0$ . Then there exist a positive real  $r$ , a positive definite matrix  $\rho$ , and a gauge-equivalent tensor  $B$  such that*

$$B^i = r^{-1/2} \rho^{-1/2} A^i \rho^{1/2}$$

*is unital:  $\sum_i B^i (B^i)^\dagger = \mathbb{1}$ .*

*This is the Perron–Frobenius unital-gauge orientation used in [PGVWC07, Theorem 4, lines 765–770], stated for one irreducible nonzero block.*

*Proof.* [✓ Proof] Apply Theorem 6.24 to the conjugate-transposed Kraus family. Irreducibility transfers to that family, and the nonzero Kraus hypothesis is preserved by taking adjoints. Translating the resulting adjoint eigenvector equation back gives  $\mathcal{E}_A(\rho) = r\rho$  with  $\rho$  positive definite and  $r > 0$ . The unital gauge theorem then gives the displayed unital representative, and the same square-root similarity gives gauge equivalence to  $r^{-1/2} A$ .  $\square$

## 6.7 Exponential positivity for irreducible CP maps

**Theorem 6.27** (Exponential truncation is positive definite). [Lean] [✓ Stmt] *Let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ , let  $A \geq 0$  be nonzero, and let  $t > 0$ . Then the finite exponential truncation*

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A)$$

*is positive definite. This is the finite-sum core of [Wol12, Theorem 6.2(3)].*

*Proof.* [✓ Proof] If a nonzero vector  $v$  annihilated this quadratic form, then every term  $E^k(A)$  with  $0 \leq k \leq D-1$  would annihilate  $v$ . The growth theorem for irreducible CP maps (that is,  $(\mathbb{1} + E)^{D-1}(A) > 0$  for irreducible  $E$  and nonzero PSD  $A$ , as established in the proof of Theorem 6.5) then forces  $(\mathbb{1} + E)^{D-1}(A)$  to annihilate  $v$ , contradicting its positive definiteness.  $\square$

**Theorem 6.28** (Exponential positivity for irreducible CP maps). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ , let  $A \geq 0$  be nonzero, and let  $t > 0$ . Then

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is [Wol12, Theorem 6.2(3)].

*Proof.* [✓ Proof](#) By Theorem 6.27, the first  $D$  terms already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail.  $\square$

## 6.8 Ergodicity of irreducible channels

**Theorem 6.29** (Unique density-matrix fixed point for an irreducible channel). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible quantum channel on  $M_D(\mathbb{C})$  with  $D > 0$ . Then there exists a unique density matrix  $\sigma$  such that  $E(\sigma) = \sigma$ . Moreover  $\sigma$  is positive definite. This is the qualitative form of [Wol12, Corollary 6.3].

*Proof.* [✓ Proof](#) Every Cesàro mean of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges. Its limit is a fixed point by the telescoping argument, positive definiteness follows from irreducibility, and uniqueness comes from the uniqueness of positive semidefinite Perron eigenvectors at eigenvalue 1.  $\square$

**Theorem 6.30** (Cesàro convergence for irreducible channels). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible quantum channel on  $M_D(\mathbb{C})$  with  $D > 0$ , and let  $\rho$  be any density matrix. Then the Cesàro means

$$\frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho)$$

converge to the unique positive definite density-matrix fixed point of  $E$ . This is [Wol12, Corollary 6.3].

*Proof.* [✓ Proof](#) Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 6.29, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit.  $\square$

## 6.9 Spectral radius at the Perron eigenvalue

**Theorem 6.31** (Spectral radius is the Perron eigenvalue). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ . Suppose  $\rho > 0$  and  $E(\rho) = r\rho$  with  $r > 0$ . Then the spectral radius of  $E$  is  $r$ . This is [Wol12, Theorem 6.3(4)].

*Proof.* [✓ Proof](#) Choose a positive definite eigenvector for the adjoint map with the same eigenvalue, gauge by its square root, and rescale by  $r^{-1}$ . The resulting map is trace-preserving and has a positive definite fixed point, so 1 is an eigenvalue. The spectral radius of a trace-preserving map  $F$  satisfies  $\rho(F) \leq 1$ , since  $\text{tr}(F^n(X)) = \text{tr}(X)$  for all  $n$  bounds the growth of iterates. Hence its spectral radius is 1. Undoing the similarity and scalar rescaling recovers the spectral radius of  $E$ .  $\square$

**Theorem 6.32** (Real spectral-radius identity). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 6.31, the real-valued spectral radius of  $E$  is also equal to  $r$ .*

*Proof.* [✓ Proof](#) This is the real-valued reformulation of Theorem 6.31. □

## 6.10 Spectral characterization of irreducibility

**Definition 6.33** (Spectral properties). [Lean](#) [✓ Stmt](#) A completely positive map  $E$  on  $M_D(\mathbb{C})$  has the *spectral properties* of [Wol12, Theorem 6.4] if there exist a positive real number  $r$ , a positive definite right eigenvector  $\rho$ , and a positive definite left eigenvector  $\sigma$  for the adjoint map such that  $E(\rho) = r\rho$ ,  $E^\dagger(\sigma) = r\sigma$ , every positive semidefinite right eigenvector for eigenvalue  $r$  is a scalar multiple of  $\rho$ , and the spectral radius of  $E$  is equal to  $r$ .

**Theorem 6.34** (Irreducibility gives the spectral properties). [Lean](#) [✓ Stmt](#) *Every nonzero irreducible completely positive map on  $M_D(\mathbb{C})$  has the spectral properties of Definition 6.33.*

*Proof.* [✓ Proof](#) Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 6.31. □

**Theorem 6.35** (Spectral properties imply irreducibility). [Lean](#) [✓ Stmt](#) *If a completely positive map on  $M_D(\mathbb{C})$  has the spectral properties of Definition 6.33, then it is irreducible.*

*Proof.* [✓ Proof](#) Gauge by the positive definite left eigenvector and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 6.33. □

**Theorem 6.36** (Irreducibility iff the spectral properties hold). [Lean](#) [✓ Stmt](#) *Let  $E$  be a nonzero completely positive map on  $M_D(\mathbb{C})$ . Then  $E$  is irreducible if and only if it has the spectral properties of Definition 6.33. This is [Wol12, Theorem 6.4].*

*Proof.* [✓ Proof](#) Combine Theorems 6.34 and 6.35. □

## Chapter 7

# Transfer-Operator Gaps and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows. It also proves the corresponding self-overlap convergence to 1 under the complementary transfer-map gap hypothesis that encodes primitivity in the cases of interest. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

This chapter concerns gaps in the spectrum of transfer operators, writing  $\rho_{\text{spec}}(\cdot)$  for the spectral radius:  $\rho_{\text{spec}}(F_{AB}) < 1$  for mixed transfer operators, and  $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$  after removing the fixed-point projection. The unqualified phrase *spectral gap* is reserved for the energy gap of a many-body Hamiltonian in Chapter 16.

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}. \quad (7.1)$$

### 7.1 Mixed transfer operator

**Definition 7.1** (Mixed transfer operator). [Lean](#) [✓ Stmt](#) The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors  $A$  and  $B$  of the same bond dimension  $D$  is the linear map  $F_{AB} : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

**Definition 7.2** (Rectangular mixed transfer operator). [Lean](#) [✓ Stmt](#) For MPS tensors  $A$  of bond dimension  $D_1$  and  $B$  of bond dimension  $D_2$  (possibly  $D_1 \neq D_2$ ), the *rectangular mixed transfer operator* is  $F_{AB}^{\text{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \rightarrow M_{D_1 \times D_2}(\mathbb{C})$  defined by

$$F_{AB}^{\text{rect}}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

**Theorem 7.3** (Self-transfer). [Lean](#) [✓ Stmt](#) Setting  $B = A$  recovers the ordinary transfer map:  $F_{AA} = \mathcal{E}_A$ .

*Proof.* ✓ Proof Substituting  $B = A$  into the defining sum  $F_{AB}(X) = \sum_i A^i X(B^i)^\dagger$  gives  $\sum_i A^i X(A^i)^\dagger = \mathcal{E}_A(X)$ .  $\square$

**Lemma 7.4** (Iterated mixed transfer). Lean ✓ Stmt For all  $X \in M_D(\mathbb{C})$  and  $N \geq 0$ ,

$$F_{AB}^N(X) = \sum_{\sigma \in \{0, \dots, d-1\}^N} A^\sigma X(B^\sigma)^\dagger,$$

where  $A^\sigma = A^{\sigma_1} \dots A^{\sigma_N}$  denotes the word evaluation (Definition 2.2).

*Proof.* ✓ Proof Induction on  $N$ : the base case is  $F^0(X) = X$ ; the inductive step applies  $F_{AB}$  to each summand and re-indexes via  $\{0, \dots, d-1\}^{N+1} \cong \{0, \dots, d-1\} \times \{0, \dots, d-1\}^N$ .  $\square$

**Lemma 7.5** (Matrix trace of iterated mixed transfer at the identity). Lean ✓ Stmt For  $N \geq 0$ ,

$$\text{tr}(F_{AB}^N(\mathbb{1})) = \sum_{\sigma} \text{tr}(A^\sigma (B^\sigma)^\dagger).$$

This is the matrix trace of  $F_{AB}^N$  evaluated at the identity matrix. It should not be confused with the operator trace  $\text{Tr}(F_{AB}^N)$ ; its relation to the MPV overlap is proved in the next section.

*Proof.* ✓ Proof Apply the matrix trace to each summand in Lemma 7.4 with  $X = \mathbb{1}$ :  $\text{tr}(F_{AB}^N(\mathbb{1})) = \sum_{\sigma} \text{tr}(A^\sigma \cdot \mathbb{1} \cdot (B^\sigma)^\dagger) = \sum_{\sigma} \text{tr}(A^\sigma (B^\sigma)^\dagger)$ .  $\square$

## 7.2 MPV overlap as transfer trace

Recall from Definition 2.48 that

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This section rewrites that scalar as the operator trace of the mixed transfer power.

**Lemma 7.6** (Trace expansion over matrix units). Lean ✓ Stmt For any linear endomorphism  $T$  on  $M_{D_1 \times D_2}(\mathbb{C})$ , the operator trace can be expanded as

$$\text{Tr}(T) = \sum_{p=0}^{D_1-1} \sum_{q=0}^{D_2-1} (T(E_{pq}))_{pq},$$

where  $E_{pq} \in M_{D_1 \times D_2}(\mathbb{C})$  is the matrix with 1 in position  $(p, q)$  and 0 elsewhere.

*Proof.* ✓ Proof The operator trace  $\text{Tr}(T)$  equals the sum of  $\text{tr}(T(E_{pq}) \cdot E_{qp})$  over all  $(p, q)$ , and  $\text{tr}(M \cdot E_{qp}) = M_{pq}$ .  $\square$

**Lemma 7.7** (Overlap equals transfer trace). Lean ✓ Stmt For tensors  $A, B$  of bond dimension  $D \geq 1$ ,

$$O_{AB}(N) = \text{Tr}(F_{AB}^N).$$

*Proof.* ✓ Proof Expand the operator trace using Lemma 7.6 as a sum over matrix units  $E_{pq}$ . Apply Lemma 7.4 to evaluate  $F_{AB}^N(E_{pq})$ , extract the  $(p, q)$ -entry, and reassemble. The resulting double sum over  $(p, q)$  and  $\sigma$  equals  $\sum_{\sigma} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N)$ .  $\square$



**Lemma 7.8** (Rectangular overlap as transfer trace). [Lean](#) [✓ Stmt](#) For tensors  $A$  of bond dimension  $D_1 \geq 1$  and  $B$  of bond dimension  $D_2 \geq 1$ ,

$$O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N).$$

*Proof.* [✓ Proof](#) Expand the operator trace over the rectangular matrix-unit basis. Then apply the rectangular iterated-transfer formula and sum over the matrix-unit indices exactly as in Lemma 7.7.  $\square$

**Lemma 7.9** (Word trace as entrywise inner product). [Lean](#) [✓ Stmt](#) For a word  $\sigma \in \{0, \dots, d-1\}^N$ ,

$$\text{tr}(A^\sigma (B^\sigma)^\dagger) = \sum_{j,k=0}^{D-1} (A^\sigma)_{jk} \overline{(B^\sigma)_{jk}}.$$

*Proof.* [✓ Proof](#) Expand the matrix trace and the  $(j, j)$ -entry of  $A^\sigma (B^\sigma)^\dagger$ .  $\square$

### 7.3 Frobenius norm estimates

The transfer-operator gap proofs use the Frobenius (Hilbert–Schmidt) norm as the natural norm for the finite-dimensional matrix algebra. We write  $\|\cdot\|_F$  for this norm; it coincides with, and we use it interchangeably with, the Hilbert–Schmidt norm  $\|\cdot\|_{\text{HS}}$  on finite-dimensional spaces. We use the squared version and an isometric embedding into Euclidean space.

**Definition 7.10** (Frobenius norm squared). [Lean](#) [✓ Stmt](#) For a (possibly rectangular) matrix  $X \in M_{m \times n}(\mathbb{C})$ , the *Frobenius norm squared* is

$$\|X\|_F^2 := \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |X_{ij}|^2.$$

**Lemma 7.11** (Trace formula for Frobenius norm). [Lean](#) [✓ Stmt](#)  $\|X\|_F^2 = \text{Re tr}(X^\dagger X)$ .

*Proof.* [✓ Proof](#) Expand the matrix product and trace entry by entry.  $\square$

**Definition 7.12** (Euclidean-space embedding). [Lean](#) [✓ Stmt](#) The map  $\text{vec} : M_{m \times n}(\mathbb{C}) \rightarrow \mathbb{C}^{mn}$  that flattens matrix entries is an isometric embedding with respect to the Frobenius norm:  $\|\text{vec}(X)\|^2 = \|X\|_F^2$ .

**Lemma 7.13** (Norm of embedded matrix). [Lean](#) [✓ Stmt](#)  $\|\text{vec}(X)\|^2 = \|X\|_F^2$ .

*Proof.* [✓ Proof](#) Expand both sides as sums over entries and use  $|z|^2 = \bar{z}z$ .  $\square$

### 7.4 Eigenvalue bound and transfer-operator gap

The eigenvalue bound comes from a uniform Frobenius-norm estimate obtained directly from the trace-preserving normalization (cf. [Wol12, Proposition 6.1] for the general operator-norm version via Russo–Dye). For the rigidity theorem, one gauges  $A$  and  $B$  separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument ([Wol12, Theorem 5.3]) to a block embedding of a mixed-transfer eigenvector.

**Theorem 7.14** (Eigenvalue bound). Lean ✓ Stmt *Let  $A, B$  be normalized MPS tensors. Every eigenvalue  $\mu$  of  $F_{AB}$  satisfies  $|\mu| \leq 1$ .*

*Proof.* ✓ Proof Expand  $F_{AB}^n(X)$  as a sum over words and use Cauchy–Schwarz, together with  $\sum_i (A^i)^\dagger A^i = \sum_i (B^i)^\dagger B^i = \mathbb{1}$  (Equation (7.1), applied to both tensors), to obtain a uniform Frobenius-norm bound  $\|F_{AB}^n(X)\|_{\text{HS}} \leq C_D \|X\|_{\text{HS}}$  for all  $n$ , where  $C_D$  depends only on  $D$ . If  $F_{AB}(X) = \mu X$  and  $X \neq 0$ , then  $|\mu|^n \|X\|_{\text{HS}} = \|F_{AB}^n(X)\|_{\text{HS}} \leq C_D \|X\|_{\text{HS}}$  for every  $n$ . Hence  $|\mu| \leq 1$ .

The argument uses only the trace-preserving normalization (7.1) and does not appeal to Perron–Frobenius theory.  $\square$

**Theorem 7.15** (Rectangular eigenvalue bound). Lean ✓ Stmt *Let  $A$  and  $B$  be normalized MPS tensors of bond dimensions  $D_1 \geq 1$  and  $D_2 \geq 1$ . Every eigenvalue  $\mu$  of  $F_{AB}^{\text{rect}}$  satisfies  $|\mu| \leq 1$ .*

*Proof.* ✓ Proof The word-expansion and Frobenius-norm estimate used in Theorem 7.14 apply equally to rectangular matrices: if  $F_{AB}^{\text{rect}}(X) = \mu X$  and  $X \neq 0$ , then  $|\mu|^n \|X\|_{\text{HS}} \leq D_1 \|X\|_{\text{HS}}$  for every  $n$ . Hence  $|\mu| \leq 1$ .  $\square$

**Lemma 7.16** (Spectral radius bound). Lean ✓ Stmt  $\rho_{\text{spec}}(F_{AB}) \leq 1$  for normalized tensors.

*Proof.* ✓ Proof The spectral radius is the supremum of  $|\mu|$  over all eigenvalues, and each satisfies  $|\mu| \leq 1$  by Theorem 7.14.  $\square$

**Theorem 7.17** (Spectral radius  $\geq 1$  implies gauge-phase equivalence). Lean ✓ Stmt *Let  $A, B$  be injective normalized MPS tensors. If  $\rho_{\text{spec}}(F_{AB}) \geq 1$ , then  $A$  and  $B$  are gauge-phase equivalent.*

*Proof.* ✓ Proof Since  $\rho_{\text{spec}}(F_{AB}) \leq 1$  (Lemma 7.16) and  $\rho_{\text{spec}}(F_{AB}) \geq 1$  by hypothesis, there exists an eigenvalue  $\mu$  with  $|\mu| = 1$  and eigenvector  $X \neq 0$ :  $F_{AB}(X) = \mu X$ . The proof proceeds in five steps.

*Step 1 (Separate gauging).* Choose positive definite fixed points  $\rho_A, \rho_B$  of the transfer maps  $\mathcal{E}_A, \mathcal{E}_B$  (Theorem 6.11) and gauge each tensor separately to a unital Kraus family (Theorem 6.13): set  $A'^i = \rho_A^{-1/2} A^i \rho_A^{1/2}$  and similarly for  $B$ . Let  $X' = \rho_A^{-1/2} X \rho_B^{-1/2}$  denote the correspondingly gauged eigenvector.

*Step 2 (Block Kraus family and off-diagonal embedding).* Form the block Kraus operators  $C^i = \begin{pmatrix} A'^i & 0 \\ 0 & B'^i \end{pmatrix}$ . These constitute a unital Kraus family on  $M_{2D}(\mathbb{C})$ . The off-diagonal matrix  $Y = \begin{pmatrix} 0 & X' \\ 0 & 0 \end{pmatrix}$  satisfies  $E_C(Y) = \mu Y$ .

*Step 3 (KS equality and Kraus commutation).* Because  $|\mu| = 1$  and  $E_C$  is unital and trace-preserving, the Kadison–Schwarz equality for peripheral eigenvectors (Theorem 5.24) gives  $E_C(Y^\dagger Y) = E_C(Y)^\dagger E_C(Y)$ . By the Kraus commutation relation (Theorem 5.23),  $Y$  commutes with every block Kraus operator:  $C^i Y = Y C^i$  for all  $i$ .

*Step 4 (Intertwining identity and invertibility).* Expanding the commutation relation block by block gives  $A'^i X' = X' B'^i$  for every  $i$ . Since  $\{A'^i\}$  span  $M_D(\mathbb{C})$  (injectivity of  $A$ ),  $X'$  is a nonzero intertwiner from  $M_D(\mathbb{C})$  to  $M_D(\mathbb{C})$ . The left multiplicative identity (Theorem 5.25) gives  $E_C(Y^\dagger Y) = E_C(Y^\dagger) E_C(Y)$ , so  $X'^\dagger X'$  is a positive semi-definite fixed point of  $\mathcal{E}_{B'}$ , positive definite by injectivity. Hence  $X'$  is invertible.

*Step 5 (Skolem–Noether conclusion).* The map  $M \mapsto X' M (X')^{-1}$  is a  $\mathbb{C}$ -algebra automorphism of  $M_D(\mathbb{C})$  sending  $B'^i$  to  $A'^i$ . By Skolem–Noether (Theorem 3.11) this is an inner automorphism. Undoing the separate gauges gives  $B^i = \bar{\mu} X A^i X^{-1}$  for all  $i$ , establishing gauge-phase equivalence (Definition 2.23).

The gauging is applied to the individual transfer maps  $\mathcal{E}_A, \mathcal{E}_B$  separately; the mixed transfer map  $F_{AB}$  need not be simultaneously unital and trace-preserving.  $\square$

**Lemma 7.18** (Gauged peripheral eigenvector yields Kraus intertwining). [Lean](#) [✓ Stmt](#) Let  $A$  and  $B$  be normalized tensors, and let  $X \neq 0$  satisfy  $F_{AB}(X) = \mu X$  with  $|\mu| = 1$ . Choose positive-definite fixed points  $\rho_A, \rho_B$  of the transfer maps and gauge to unital families  $A'^i = S_A^{-1} A^i S_A$  and  $B'^i = S_B^{-1} B^i S_B$  with  $S_A S_A^\dagger = \rho_A$  and  $S_B S_B^\dagger = \rho_B$ . Then the transported matrix  $X' = S_A^{-1} X (S_B^\dagger)^{-1}$  is nonzero, the families  $A'$  and  $B'$  are unital, and for every  $i$  one has

$$X'(B'^i)^\dagger = \mu(A'^i)^\dagger X', \quad A'^i X' = \mu X' B'^i.$$

*Proof.* [✓ Proof](#) Block-embed the gauged families and transported eigenvector into a single unital Kraus map, apply Kadison–Schwarz equality for the modulus-one eigenvalue, then extract the intertwining identities from the block structure.  $\square$

**Lemma 7.19** (Intertwining yields gauge-phase equivalence). [Lean](#) [✓ Stmt](#) Let  $A$  and  $B$  be tensors of the same bond dimension. Suppose there are invertible gauges taking them to families  $A'$  and  $B'$ , an invertible matrix  $X'$ , and a scalar  $\mu$  with  $|\mu| = 1$  such that, for every physical index  $i$ ,

$$A'^i X' = \mu X' B'^i.$$

Then  $A$  and  $B$  are gauge-phase equivalent.

*Proof.* [✓ Proof](#) Compose the two gauge matrices with the intertwiner to obtain a single invertible matrix conjugating  $A$  to a scalar multiple of  $B$ .  $\square$

**Remark 7.20** (Two gauge conventions). [✓ Stmt](#) The right-canonical gauge used here,  $A'^i = \rho^{-1/2} A^i \rho^{1/2}$ , makes  $\{A'^i\}$  a unital Kraus family ( $\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$ ). A different convention,  $\tilde{A}^i = \rho^{1/2} A^i \rho^{-1/2}$ , makes the family trace-preserving ( $\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$ ). Both conventions appear in practice: the unital version via direct matrix square-root inversion (used in the transfer-operator gap proof above), and the trace-preserving version (used in the canonical-form reduction of Chapter 9).

**Theorem 7.21** (Strict transfer-operator gap). [Lean](#) [✓ Stmt](#) Let  $A, B$  be injective normalized MPS tensors that are not gauge-phase equivalent. Then  $\rho_{\text{spec}}(F_{AB}) < 1$ .

*Proof.* [✓ Proof](#)  $\rho_{\text{spec}}(F_{AB}) \leq 1$  by Lemma 7.16. If  $\rho_{\text{spec}}(F_{AB}) = 1$ , Theorem 7.17 gives gauge-phase equivalence, contradicting the hypothesis.  $\square$

## 7.5 Transfer-operator gap — rectangular case

**Lemma 7.22** (Rectangular intertwining forces equal bond dimension). [Lean](#) [✓ Stmt](#) Let  $A$  and  $B$  be injective tensors of bond dimensions  $D_1$  and  $D_2$ . If there exist  $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$  and  $\mu \in \mathbb{C}$  such that, for every physical index  $i$ ,

$$X(B^i)^\dagger = \mu(A^i)^\dagger X, \quad A^i X = \mu X B^i,$$

then  $D_1 = D_2$ .

*Proof.* [✓ Proof](#) The intertwining relation shows that  $\ker(X)$  is invariant under all generators of  $B$ . Since  $B$  is injective, its generators span the full matrix algebra, so  $\ker(X)$  is trivial and  $D_2 \leq D_1$ . Applying the same argument to  $X^\dagger$  gives  $D_1 \leq D_2$ .  $\square$

**Theorem 7.23** (Rectangular transfer-operator gap). [Lean](#) [✓ Stmt](#) Let  $A$  be an injective normalized tensor of bond dimension  $D_1$  and  $B$  an injective normalized tensor of bond dimension  $D_2$ , with  $D_1 \neq D_2$ . Then  $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$ .

*Proof.* ✓ Proof The same word-expansion argument as in Theorem 7.15 gives  $|\mu| \leq 1$  for every eigenvalue of  $F_{AB}^{\text{rect}}$ . If  $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) = 1$ , choose a modulus-one eigenvector  $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ .

Gauge  $A, B$  to unital families  $A', B'$  and transport  $X$  to a nonzero rectangular intertwiner  $X'$  with  $A'^i X' = \mu X' B'^i$  for every  $i$ . This is the hypothesis of Lemma 7.22, so  $D_1 = D_2$ , contradicting the hypothesis.  $\square$

**Theorem 7.24** (Rectangular overlap decay). Lean ✓ Stmt *If  $D_1 \neq D_2$  and both tensors are injective and normalized, then  $O_{AB}(N) \rightarrow 0$  as  $N \rightarrow \infty$ .*

*Proof.* ✓ Proof The transfer-operator gap  $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$  (Theorem 7.23) implies  $(F_{AB}^{\text{rect}})^N \rightarrow 0$ , so the operator trace converges to 0. By Lemma 7.8,  $O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N) \rightarrow 0$ .  $\square$

## 7.6 MPV overlap decay

**Theorem 7.25** (Transfer powers converge to zero). Lean ✓ Stmt *Let  $A, B$  be injective normalized tensors that are not gauge-phase equivalent. Then  $F_{AB}^n(X) \rightarrow 0$  for every  $X \in M_D(\mathbb{C})$ .*

*Proof.* ✓ Proof The transfer-operator gap  $\rho_{\text{spec}}(F_{AB}) < 1$  (Theorem 7.21) implies  $\|F_{AB}^n\|_{\text{op}} \rightarrow 0$  by the Gelfand formula, so  $F_{AB}^n(X) \rightarrow 0$  for every  $X$ .  $\square$

**Theorem 7.26** (Overlap decay). Lean ✓ Stmt *Let  $A, B$  be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then  $O_{AB}(N) \rightarrow 0$  as  $N \rightarrow \infty$ .*

*Proof.* ✓ Proof By Lemma 7.7,  $O_{AB}(N) = \text{Tr}(F_{AB}^N)$ . Expand the operator trace over matrix units (Lemma 7.6):  $\text{Tr}(F_{AB}^N) = \sum_{p,q} (F_{AB}^N(E_{pq}))_{pq}$ . By Theorem 7.25, each  $F_{AB}^N(E_{pq}) \rightarrow 0$  entry-wise. Since the sum is finite,  $O_{AB}(N) \rightarrow 0$ .  $\square$

## 7.7 Block separation

The block-separation results below record the overlap asymptotics used in the canonical-form separation of Chapter 9.

**Theorem 7.27** (Cross-correlation decay). Lean ✓ Stmt *If  $A$  and  $B$  are injective normalized tensors that are not gauge-phase equivalent, then for every  $X \in M_D(\mathbb{C})$  one has  $\text{tr}(F_{AB}^N(X)) \rightarrow 0$  as  $N \rightarrow \infty$ .*

*Proof.* ✓ Proof By Theorem 7.25,  $F_{AB}^N(X) \rightarrow 0$  in operator norm, so  $\text{tr}(F_{AB}^N(X)) \rightarrow 0$ .  $\square$

**Theorem 7.28** (Self-correlation persists). Lean ✓ Stmt *If  $\rho$  is a fixed point of  $\mathcal{E}_A$ , then  $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$  for all  $N \geq 0$ . In particular this applies to the distinguished QPF fixed point from Theorem 6.11.*

*Proof.* ✓ Proof Since  $\rho$  is a fixed point,  $\mathcal{E}_A^N(\rho) = \rho$  for every  $N$ , so  $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$ .  $\square$

## 7.8 Transfer-operator gap under irreducible-TP hypotheses

The preceding transfer-operator gap results require *injectivity* of both tensors. The following variants weaken this to *irreducibility* combined with the trace-preserving normalization (7.1), following Cirac et al. [CPGSV17] and the irreducibility theory of [Wol12, Section 6.2]. The

argument replaces the Kraus-commutation step with a Cauchy–Schwarz rigidity argument for the Perron–Frobenius fixed point.

**Theorem 7.29** (Gauge-equivalence from irreducible-TP spectral radius). Lean ✓ Stmt *Let  $A, B$  be irreducible trace-preserving normalized MPS tensors of bond dimension  $D$ . If  $\rho_{\text{spec}}(F_{AB}) \geq 1$ , then  $A$  and  $B$  are gauge-phase equivalent.*

*Proof.* ✓ Proof Choose a modulus-one eigenvector as in Theorem 7.17. The proof follows the five-step structure of Theorem 7.17, with one key adaptation. Steps 1–3 (separate gauging, block Kraus family, KS equality and Kraus commutation) proceed identically: the gauging uses the unique positive definite fixed points guaranteed by irreducibility (Theorem 6.11) and the right-canonical gauge (Theorem 6.13). The change is in Step 4: the Kadison–Schwarz equality gives  $X'^\dagger X'$  as a nonzero positive semidefinite fixed point of  $\mathcal{E}_{B'}$ . Under injectivity this is immediately positive definite (the fixed point is unique up to scaling). Under the weaker hypothesis of irreducibility, Theorem 6.4 upgrades it to positive definite, giving invertibility of  $X'$ . Formally, Lemma 7.18 combines the separately gauged eigenvector into the required intertwining relations. Once  $X'$  is invertible, Lemma 7.19 upgrades the intertwining to gauge-phase equivalence.  $\square$

**Theorem 7.30** (Strict transfer-operator gap (irreducible-TP)). Lean ✓ Stmt *Let  $A, B$  be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then  $\rho_{\text{spec}}(F_{AB}) < 1$ .*

*Proof.* ✓ Proof Contrapositive of Theorem 7.29, combined with  $\rho_{\text{spec}}(F_{AB}) \leq 1$  (Lemma 7.16).  $\square$

**Theorem 7.31** (Transfer power decay (irreducible-TP)). Lean ✓ Stmt *Under the hypotheses of Theorem 7.30,  $F_{AB}^n(X) \rightarrow 0$  for every  $X \in M_D(\mathbb{C})$ .*

*Proof.* ✓ Proof By the Gelfand formula,  $\rho_{\text{spec}}(F_{AB}) < 1$  implies  $\|F_{AB}^n\|_{\text{op}} \rightarrow 0$ .  $\square$

**Theorem 7.32** (Overlap decay (irreducible-TP)). Lean ✓ Stmt *Let  $A, B$  be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then  $O_{AB}(N) \rightarrow 0$  as  $N \rightarrow \infty$ .*

*Proof.* ✓ Proof Combine the transfer-operator gap (Theorem 7.30) with the overlap–trace identity (Lemma 7.7).  $\square$

**Theorem 7.33** (Rectangular transfer-operator gap (irreducible-TP)). Lean ✓ Stmt *Let  $A$  (bond dimension  $D_1$ ) and  $B$  (bond dimension  $D_2$ ) be irreducible trace-preserving normalized tensors with  $D_1 \neq D_2$ . Then  $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$ .*

*Proof.* ✓ Proof Assume for contradiction that  $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) = 1$ . Choose a modulus-one eigenvector  $X \in M_{D_1 \times D_2}(\mathbb{C})$ . Gauge  $A, B$  separately to unital families  $A', B'$  and transport  $X$  to  $X' \neq 0$ . The matrix  $X'X'^\dagger$  ( $D_1 \times D_1$ ) is a nonzero positive-semidefinite fixed point of  $\mathcal{E}_{A'}$ , so by Theorem 6.4 it is positive definite, hence invertible; thus  $X'$  has full row rank and  $D_1 \leq D_2$ . Symmetrically  $X'^\dagger X'$  ( $D_2 \times D_2$ ) is positive definite, giving full column rank and  $D_2 \leq D_1$ . Hence  $D_1 = D_2$ , contradicting the hypothesis.  $\square$

**Theorem 7.34** (Rectangular overlap decay (irreducible-TP)). Lean ✓ Stmt *If  $D_1 \neq D_2$  and both tensors are irreducible and trace-preserving normalized, then  $O_{AB}(N) \rightarrow 0$  as  $N \rightarrow \infty$ .*

*Proof.* ✓ Proof Combine Theorem 7.33 with Lemma 7.8.  $\square$

## 7.9 Conditional expectation from a faithful fixed point

This section follows [Wol12, Section 6.4]; the conditional expectation is built from the fixed-point algebra of [Wol12, Theorem 6.14] via [Wol12, (1.40)]. The scalar conditional expectation  $E_\sigma$  is the special case where the fixed-point  $*$ -subalgebra is the scalar algebra  $\mathbb{C} \cdot \mathbb{1}$  (the primitive-channel case). The general irreducible case with nontrivial period requires the Wedderburn block decomposition of [Wol12, Theorem 6.14].

**Definition 7.35** (Conditional expectation). Lean ✓ Stmt Given a  $*$ -subalgebra  $S \subseteq M_D(\mathbb{C})$ , a linear map  $P : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is a conditional expectation onto  $S$  if it is idempotent, unital, has range contained in  $S$ , and satisfies  $P(X) = X$  for all  $X \in S$ .

**Definition 7.36** (Scalar conditional expectation). Lean ✓ Stmt For  $\sigma \geq 0$  with  $\text{tr}(\sigma) \neq 0$ , define

$$E_\sigma(X) := \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}.$$

**Lemma 7.37** (Evaluation formula). Lean ✓ Stmt  $E_\sigma(X) = \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}$ .

**Lemma 7.38** (Unitality). Lean ✓ Stmt  $E_\sigma(\mathbb{1}) = \mathbb{1}$ .

**Lemma 7.39** (Idempotence). Lean ✓ Stmt  $E_\sigma(E_\sigma(X)) = E_\sigma(X)$  for all  $X$ .

**Lemma 7.40** (Range is scalar). Lean ✓ Stmt For every  $X$ , there exists  $c \in \mathbb{C}$  with  $E_\sigma(X) = c \mathbb{1}$ .

**Lemma 7.41** (Fixes scalar operators). Lean ✓ Stmt For  $c \in \mathbb{C}$ ,  $E_\sigma(c \mathbb{1}) = c \mathbb{1}$ .

**Lemma 7.42** (Absorbs the adjoint map). Lean ✓ Stmt If  $T(\sigma) = \sigma$ , then  $E_\sigma \circ T^* = E_\sigma$ .

**Lemma 7.43** (Adjoint map absorbs scalar projection). Lean ✓ Stmt If  $T$  is trace-preserving, then  $T^* \circ E_\sigma = E_\sigma$ .

**Lemma 7.44** (Image lies in adjoint fixed points). Lean ✓ Stmt For all  $X$ ,  $E_\sigma(X)$  is fixed by  $T^*$ .

**Theorem 7.45** (Scalar map is a conditional expectation). Lean ✓ Stmt Let  $K$  be trace-preserving, let  $\rho > 0$  be a positive-definite fixed point of the associated channel  $T(\rho) = \sum_i K_i \rho K_i^\dagger$  (so  $T(\rho) = \rho$ ), and assume every adjoint fixed point of  $T^*$  is a scalar multiple of  $\mathbb{1}$ . Then  $E_\rho$  is a conditional expectation onto the adjoint fixed-point  $*$ -subalgebra.

**Lemma 7.46** (Complete positivity of the scalar conditional expectation). Lean ✓ Stmt For any positive semidefinite  $\sigma$ , the scalar conditional expectation  $E_\sigma$  is a completely positive map.

*Proof.* ✓ Proof Write  $S = \sqrt{\sigma}$  and define the Kraus family  $K_{j,i} = |j\rangle\langle i| S$  indexed by  $(j, i) \in \{1, \dots, D\}^2$ . Using  $|j\rangle\langle i| M |i\rangle\langle j| = M_{ii} |j\rangle\langle j|$  and summing first over  $i$  and then over  $j$ ,

$$\sum_{j,i} K_{j,i} X K_{j,i}^\dagger = \sum_j |j\rangle\langle j| \text{tr}(S X S) = \text{tr}(\sigma X) \mathbb{1},$$

where we used cyclicity of the trace and  $S^2 = \sigma$ . Hence  $E_\sigma = (\text{tr}(\sigma))^{-1} \cdot (X \mapsto \sum_{j,i} K_{j,i} X K_{j,i}^\dagger)$  is the Kraus map scaled by the non-negative real number  $(\text{tr}(\sigma))^{-1}$ , and therefore completely positive.  $\square$

**Lemma 7.47** ( $\sigma$ -trace preservation of the scalar conditional expectation). Lean ✓ Stmt If  $\text{tr}(\sigma) \neq 0$ , then for every  $X$

$$\text{tr}(\sigma E_\sigma(X)) = \text{tr}(\sigma X).$$

*Proof.* ✓ Proof Direct computation:  $\text{tr}(\sigma \cdot \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}) = \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \text{tr}(\sigma) = \text{tr}(\sigma X)$ .  $\square$

## 7.10 Stationary support

This section develops the support-projection theory behind stationary states, following [Wol12, Section 6.4, Propositions 6.10–6.11].

**Lemma 7.48** (Lower-triangular Kraus condition implies corner invariance). [Lean](#) [✓ Stmt](#) *Let  $A$  be an MPS tensor and let  $P$  be an orthogonal projection. If, for every physical index  $i$ ,*

$$(\mathbb{1} - P)A^iP = 0$$

*then*

$$P\mathcal{E}_A(PXP)P = \mathcal{E}_A(PXP)$$

*for all  $X \in M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) Expand the transfer map and use  $(\mathbb{1} - P)A^iP = 0$  to reduce each Kraus term to the  $P$ -corner.  $\square$

**Lemma 7.49** (Support projection invariance). [Lean](#) [✓ Stmt](#) *Let  $E$  be a quantum channel and  $\rho \geq 0$  a positive semidefinite fixed point ( $E(\rho) = \rho$ ). Let  $P$  denote the support projection of  $\rho$ . Then*

$$PE(PXP)P = E(PXP)$$

*for all  $X \in M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) Write  $E$  in Kraus form  $E(X) = \sum_i K_i X K_i^\dagger$ . From  $E(\rho) = \rho$  and  $\rho \geq 0$  one deduces  $(\mathbb{1} - P)K_iP = 0$  for all  $i$ . Apply Lemma 7.48.  $\square$

**Definition 7.50** (Stationary state). [Lean](#) [✓ Stmt](#) For an irreducible channel  $E$  on  $M_D(\mathbb{C})$  (with  $D \geq 1$ ), the *stationary state* is the unique density-matrix fixed point  $\rho_\infty$  satisfying  $E(\rho_\infty) = \rho_\infty$ ,  $\rho_\infty > 0$ , and  $\text{tr}(\rho_\infty) = 1$ . Uniqueness follows from the quantum Perron–Frobenius theorem (Theorem 6.11).

**Definition 7.51** (Stationary support). [Lean](#) [✓ Stmt](#) The *stationary support* of an irreducible channel  $E$  is the support projection of the stationary state (Definition 7.50).

**Theorem 7.52** (Stationary support is full). [Lean](#) [✓ Stmt](#) *For an irreducible channel  $E$ , the stationary support equals the identity:  $\text{supp}(\rho_\infty) = \mathbb{1}$ .*

*Proof.* [✓ Proof](#) The support projection  $P$  of  $\rho_\infty$  is an orthogonal projection satisfying  $PE(PXP)P = E(PXP)$  for all  $X$  (Lemma 7.49). By irreducibility,  $P = 0$  or  $P = \mathbb{1}$ . Since  $\rho_\infty \neq 0$ , we have  $P \neq 0$ , so  $P = \mathbb{1}$ .  $\square$

**Remark 7.53** (Stationary-support formulations). The non-trivial formulations needed here are the following results from Wolf:

- the converse direction of [Wol12, Proposition 6.10]: if the support projection of every positive semidefinite fixed point is full, then the channel is irreducible;
- [Wol12, Proposition 6.11]: minimality of the stationary support among invariant projections, without assuming irreducibility.

These statements are not used elsewhere in this chapter.



### 7.10.1 Faithful compression onto the support sector

This subsection establishes the results on the corner algebra,  $*$ -structure, and compression of a PSD fixed point onto its support sector needed for [Wol12, Corollary 6.6]: the corner carries a  $\mathbb{C}$ -algebra and  $*$ -structure, the two standard presentations of the corner are identified, and the compression of a PSD fixed point onto its support sector is positive definite.

**Lemma 7.54** (Support projection annihilates the kernel). [Lean](#) [✓ Stmt](#) *Let  $\rho \succeq 0$  on  $\mathbb{C}^D$  with support projection  $P = \text{supp}(\rho)$ . For every  $v \in \mathbb{C}^D$ , if  $\rho v = 0$ , then  $Pv = 0$ .*

*Proof.* [✓ Proof](#) Diagonalize  $\rho = U \text{diag}(\lambda) U^\dagger$  with  $\lambda_j \geq 0$ . Writing  $w := U^\dagger v$ , the hypothesis  $\rho v = 0$  gives  $\lambda_j w_j = 0$  for every  $j$ ; hence  $w_j = 0$  whenever  $\lambda_j > 0$ . The support projection acts as  $\text{diag}(\chi_{\lambda_j > 0})$  in the eigenbasis, so  $Pv = U \text{diag}(\chi_{\lambda_j > 0}) w = 0$ .  $\square$

**Definition 7.55** (Corner submodule / idempotent-corner identification). [Lean](#) [✓ Stmt](#) For an idempotent  $P \in M_D(\mathbb{C})$ , the subspace  $\{X \mid PXP = X\}$  and the corner  $\{PXP \mid X \in M_D(\mathbb{C})\}$  are identified as  $\mathbb{C}$ -linear spaces by the identity on underlying matrices.

*Remark 7.56* ( $\mathbb{C}$ -algebra and  $*$ -structure on the corner). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The corner algebra associated to an idempotent already carries a ring structure with unit  $P$ . In the matrix case it also has  $\mathbb{C}$ -module and  $\mathbb{C}$ -algebra structures and, under the additional assumption  $P^\dagger = P$ , star, star ring, and star module structures over  $\mathbb{C}$ .

**Theorem 7.57** (Faithful compression onto the support). [Lean](#) [✓ Stmt](#) *Let  $\rho \succeq 0$  on  $\mathbb{C}^D$  with support projection  $P = \text{supp}(\rho)$ , and let  $V : \mathbb{C}^D \rightarrow \mathbb{C}^k$  be a compression isometry satisfying  $VV^\dagger = \mathbb{1}_k$  and  $V^\dagger V = P$ . Then the compression*

$$V \rho V^\dagger \in M_k(\mathbb{C})$$

*is positive definite.*

*Proof.* [✓ Proof](#) Hermiticity follows from Hermiticity of  $\rho$  and the product-star identity. For strict positivity, let  $w \neq 0$ , set  $u := V^\dagger w$ , and note that  $VV^\dagger = \mathbb{1}_k$  forces  $u \neq 0$ , while  $V^\dagger V = P$  forces  $Pu = u$ . The quadratic form computes as  $w^\dagger (V \rho V^\dagger) w = u^\dagger \rho u$ , which is non-negative by positive semidefiniteness. If it vanishes, then  $\rho u = 0$ , and by Lemma 7.54 also  $Pu = 0$ , contradicting  $Pu = u \neq 0$ .  $\square$

## 7.11 Wedderburn decomposition of the fixed-point algebra

This section proves the structure theorem for the fixed-point algebra of a trace-preserving Kraus map, following [Wol12, Theorems 6.12–6.14]. The proof uses the Jacobson radical characterization of semisimplicity together with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field.

**Theorem 7.58** (Fixed-point algebra is semisimple). [Lean](#) [✓ Stmt](#) *The adjoint-fixed-point  $*$ -subalgebra of a trace-preserving Kraus map on  $M_D(\mathbb{C})$  is a semisimple ring.*

*Proof.* [✓ Proof](#) The subalgebra is finite-dimensional (as a subalgebra of  $M_D(\mathbb{C})$ ), hence Artinian. Following [Wol12, Theorem 6.14], it suffices to show the Jacobson radical  $J$  vanishes. If  $x \in J$ , then  $x^*x \in J$  (left-ideal closure). The radical of an Artinian ring is nilpotent, so  $x^*x$  is nilpotent. A self-adjoint nilpotent matrix satisfies  $\|x^*x\|^{2^k} = \|(x^*x)^{2^k}\| = 0$ , hence  $x^*x = 0$ , and the  $\mathbb{C}^*$ -identity gives  $x = 0$ .  $\square$



**Theorem 7.59** (Abstract Wedderburn–Artin decomposition). [Lean](#) [✓ Stmt](#) *There exist  $n \in \mathbb{N}$  and dimensions  $d_1, \dots, d_n \geq 1$  such that the adjoint-fixed-point  $*$ -subalgebra is  $\mathbb{C}$ -algebra isomorphic to*

$$\prod_{k=1}^n M_{d_k}(\mathbb{C}).$$

*Proof.* [✓ Proof](#) Combine semisimplicity (Theorem 7.58) with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field.  $\square$

**Theorem 7.60** (Fixed-point algebra decomposition with multiplicities). [Lean](#) [✓ Stmt](#) *There exist integers  $n$ , block sizes  $d_1, \dots, d_n \geq 1$ , multiplicities  $m_1, \dots, m_n \geq 1$ , and a  $\mathbb{C}$ -algebra isomorphism from the adjoint-fixed-point  $*$ -subalgebra to*

$$\prod_{k=1}^n M_{d_k}(\mathbb{C})$$

*such that*

$$\sum_{k=1}^n d_k m_k \leq D.$$

*Proof.* [✓ Proof](#) Apply the abstract Wedderburn decomposition (Theorem 7.59). Since the adjoint-fixed-point algebra is realized as a subalgebra of the ambient matrix algebra  $M_D(\mathbb{C})$ , this realization also yields multiplicities  $m_k$  and the bound  $\sum_k d_k m_k \leq D$ .  $\square$

**Remark 7.61** (Block form of the fixed-point algebra). Wolf’s block form identifies the adjoint-fixed-point algebra, up to a unitary change of basis, with

$$0 \oplus \bigoplus_k M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$$

together with the density-block form  $M_{d_k}(\mathbb{C}) \otimes \rho_k$ . The theorem above gives the product decomposition, multiplicities, and dimension bound needed from this block form.

**Theorem 7.62** (Weighted fixed points form a  $*$ -subalgebra). [Lean](#) [✓ Stmt](#) *Let  $T(X) = \sum_i K_i X K_i^\dagger$  be trace-preserving, and let  $\rho > 0$  satisfy  $T(\rho) = \rho$ . If*

$$F_T := \{X \in M_D(\mathbb{C}) \mid T(X) = X\},$$

*then  $\rho^{-1/2} F_T \rho^{-1/2}$  is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) Set  $B_i = \rho^{-1/2} K_i \rho^{1/2}$ . The equation  $T(\rho) = \rho$  implies that the family  $\{B_i\}$  is unital, and trace preservation of  $T$  implies that  $\rho$  is a positive-definite fixed point of the adjoint map of the gauged family. Apply Theorem 4.54 to  $B$ . The identity

$$\sum_i B_i X B_i^\dagger = X \iff T(\rho^{1/2} X \rho^{1/2}) = \rho^{1/2} X \rho^{1/2}$$

identifies the fixed points of the gauged map with  $\rho^{-1/2} F_T \rho^{-1/2}$ .  $\square$

## 7.12 Further irreducibility and primitivity equivalences

This section collects several criteria and equivalences from [Wol12, Theorems 6.2, 6.7, 6.8] that are used later.

**Theorem 7.63** (Exponential positivity condition). Lean Lean ✓ Stmt *Let  $E$  be an irreducible completely positive map,  $A \geq 0$  with  $A \neq 0$ , and  $t > 0$ . Then  $\exp(tE)(A)$  is positive definite. Conversely, if  $\exp(tE)(A)$  is positive definite for every such  $t, A$ , then  $E$  is irreducible (full equivalence Lean).*

**Theorem 7.64** (Kraus-span equivalence). Lean ✓ Stmt *Under normalization, primitivity in Wolf's sense is equivalent to eventual full Kraus rank.*

**Theorem 7.65** (Combined primitivity equivalence). Lean ✓ Stmt *Under normalization, primitivity in Wolf's sense is equivalent to the conjunction of eventual full Kraus rank, normality, and strong irreducibility.*

*Remark 7.66* (Pairwise primitivity equivalences). Lean Lean Lean Lean ✓ Stmt These are the pairwise equivalences in [Wol12, Theorem 6.8].

**Theorem 7.67** (Scalar fixed-point algebra case). Lean ✓ Stmt *In the scalar fixed-point algebra setting, the weighted map  $X \mapsto \frac{\text{tr}(\rho X)}{\text{tr}(\rho)} \mathbb{1}$  is a conditional expectation onto the adjoint fixed-point  $*$ -subalgebra.*

## 7.13 Multi-cycle block-permutation structure

This section develops the block-permutation structure underlying the cycle decomposition of [Wol12, Theorem 6.16]. The multi-cycle decomposition refines the single-cycle case to handle arbitrary peripheral periods.

**Definition 7.68** (Block-permutation structure). Lean ✓ Stmt Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ . A *block-permutation structure* for  $T$  consists of a finite index set  $\iota$ , a family of orthogonal projections  $P_k \in M_D(\mathbb{C})$  for  $k \in \iota$ , and a permutation  $\sigma \in \text{Sym}(\iota)$  (not necessarily a single cycle—in general  $\sigma$  may have multiple disjoint cycles), such that

$$T(P_{\sigma(k)}) = P_k, \quad T(P_k X) = T(P_k) T(X), \quad T(X P_k) = T(X) T(P_k)$$

for every  $k \in \iota$  and  $X \in M_D(\mathbb{C})$ .

**Definition 7.69** (Multi-cycle block-permutation structure). Lean ✓ Stmt A *multi-cycle decomposition* of  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  consists of a finite cycle index  $\iota$ , a per-cycle period  $m : \iota \rightarrow \mathbb{N}_{>0}$ , a projection family  $P_{c,k} \in M_D(\mathbb{C})$  for  $c \in \iota$  and  $k \in \{0, \dots, m(c)-1\}$  consisting of orthogonal projections. For every  $c \in \iota$  and  $k \in \{0, \dots, m(c)-1\}$ , the per-cycle cyclic action is

$$T(P_{c,k+1}) = P_{c,k}.$$

The multiplicative-domain factorisations are  $T(P_{c,k} X) = T(P_{c,k}) T(X)$  and  $T(X P_{c,k}) = T(X) T(P_{c,k})$ .

**Lemma 7.70** (Per-cycle corner preservation). Lean ✓ Stmt *For every cycle  $c \in \iota$  and index  $k \in \{0, \dots, m(c)-1\}$ , the iterate  $T^{m(c)}$  preserves the corner  $P_{c,k} M_D(\mathbb{C}) P_{c,k}$ .*

*Proof.* ✓ Proof By induction on  $n$ , the  $n$ -th iterate  $T^n$  sends  $P_{c,k+n} X P_{c,k+n}$  to  $P_{c,k} T^n(X) P_{c,k}$ , using the multiplicative-domain factorisations and the cyclic action  $T(P_{c,j+1}) = P_{c,j}$  at each induction step. Specialising to  $n = m(c)$  and using that the indices  $k+m(c) = k$  in  $\{0, \dots, m(c)-1\}$  yields corner preservation of  $T^{m(c)}$ .  $\square$

**Lemma 7.71** (Common-period corner preservation). [Lean](#) [✓ Stmt](#) If  $N \in \mathbb{N}$  is divisible by every per-cycle period  $m(c)$ , then  $T^N$  preserves every corner  $P_{c,k} M_D(\mathbb{C}) P_{c,k}$  of the decomposition.

*Proof.* [✓ Proof](#) Write  $N = m(c)q$ . Corner preservation is stable under taking powers, so  $T^N = (T^{m(c)})^q$  preserves each corner by Lemma 7.70.  $\square$

**Definition 7.72** (Flattening to a single-index block-permutation structure). [Lean](#) [✓ Stmt](#) A multi-cycle decomposition gives a single-index block-permutation structure on the sigma index  $\Sigma_{c \in \iota} \{0, \dots, m(c)-1\}$ : the permutation is the sigma-product of per-cycle cyclic shifts  $k \mapsto k+1$  (whose cycle decomposition has one cycle per  $c \in \iota$ ), the projections are  $(c, k) \mapsto P_{c,k}$ , and the multiplicative-domain factorisations descend componentwise. The resulting map forgets the explicit cycle-indexing.

## 7.14 Peripheral eigenvalue group structure

This section proves the peripheral spectrum structure of [Wol12, Theorem 6.6] for irreducible Schwarz maps: the peripheral eigenvalues form a finite cyclic group. The trace-preserving refinement that the order divides the bond dimension is proved in Theorem 7.76.

**Lemma 7.73** (Peripheral eigenvectors are invertible). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible unital Kraus map on  $M_D(\mathbb{C})$  with a positive-definite adjoint fixed point. If  $X \neq 0$  satisfies  $E(X) = \mu X$  with  $|\mu| = 1$ , then  $X$  is invertible.

*Proof.* [✓ Proof](#) The KS equality gives  $E(X^\dagger X) = X^\dagger X$ , so  $X^\dagger X$  is a nonzero PSD fixed point. Irreducibility upgrades it to positive definite, hence invertible, so  $\det(X) \neq 0$ .  $\square$

**Lemma 7.74** (Peripheral eigenvalues: product closure). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible unital Kraus map on  $M_D(\mathbb{C})$  with a positive-definite adjoint fixed point. If  $\mu, \nu$  are peripheral eigenvalues of  $E$ , then so is  $\mu\nu$ .

*Proof.* [✓ Proof](#) Take eigenvectors  $X, Y$  for  $\mu, \nu$ . The KS equality at  $X$  puts  $X$  in the multiplicative domain, so  $E(YX) = E(Y)E(X) = (\nu\mu)(YX)$ . Since  $X, Y$  are units (Lemma 7.73),  $YX \neq 0$  and  $|\mu\nu| = 1$ .  $\square$

**Theorem 7.75** (Peripheral eigenvalues: cyclic structure). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible unital Schwarz map on  $M_D(\mathbb{C})$  with a positive-definite adjoint fixed point (trace-preservation is not required). Then there exist  $m \geq 1$  and a primitive  $m$ -th root of unity  $\gamma$  such that the peripheral eigenvalues of  $E$  are exactly  $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$ .

*Proof.* [✓ Proof](#) The peripheral eigenvalues form a finite subset of  $\mathbb{C}$ , contain 1, and are closed under multiplication and inversion, so they form a finite subgroup of  $\mathbb{C}^\times$ . Any finite subgroup of  $\mathbb{C}^\times$  is cyclic. The generator  $\gamma$  has order  $m = |\text{peripheral spectrum}|$ .  $\square$

**Theorem 7.76** (Peripheral eigenvalues form a cyclic group). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible unital trace-preserving Schwarz map on  $M_D(\mathbb{C})$  with a positive-definite adjoint fixed point. Then there exist  $m \geq 1$  and a primitive  $m$ -th root of unity  $\gamma$  such that  $m \mid D$  and the peripheral eigenvalues of  $E$  are exactly  $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$ .

*Proof.* [✓ Proof](#) Apply Theorem 7.75 for the cyclic structure. Divisibility  $m \mid D$  follows from the cyclic decomposition:  $m$  mutually orthogonal projections summing to  $\mathbb{1}$  each have equal trace (by trace preservation), so  $m \cdot \text{tr}(P_0) = D$  with  $\text{tr}(P_0) \in \mathbb{N}$ .  $\square$

**Theorem 7.77** (Period divides bond dimension). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible unital trace-preserving Schwarz map on  $M_D(\mathbb{C})$  with a positive-definite adjoint fixed point. Let  $\gamma$  be a primitive  $m$ -th root of unity such that the peripheral eigenvalues are exactly  $\{\gamma^k : k \in \{0, \dots, m-1\}\}$ . Then  $m \mid D$ .

*Proof.* [✓ Proof](#) The cyclic decomposition gives orthogonal projections  $P_0, \dots, P_{m-1}$  with  $\sum_k P_k = \mathbb{1}$  and  $E(P_{k+1}) = P_k$ . Trace preservation gives  $\text{tr}(P_{k+1}) = \text{tr}(P_k)$  for every  $k$ , hence

$$D = \text{tr}(\mathbb{1}) = \sum_{k=0}^{m-1} \text{tr}(P_k) = m \text{tr}(P_0).$$

Since the trace of an orthogonal projection is an integer,  $m \mid D$ . □

### 7.14.1 Cyclic decomposition of irreducible Schwarz maps

**Definition 7.78** (Corner preservation). [Lean](#) [✓ Stmt](#) For an orthogonal projection  $P$  and linear map  $E$ , the corner  $PM_D(\mathbb{C})P$  is preserved if  $E(PXP) = PE(PXP)P$  for all  $X$ .

**Definition 7.79** (Corner subspace). [Lean](#) [✓ Stmt](#) The corner subspace associated with  $P$  is  $\{PXP : X \in M_D(\mathbb{C})\}$ .

**Definition 7.80** (Restricted corner map). [Lean](#) [✓ Stmt](#) The restriction of  $E$  to an invariant corner  $PM_D(\mathbb{C})P$ .

**Definition 7.81** (Irreducible restriction on a corner). [Lean](#) [✓ Stmt](#) Irreducibility of the corner-restricted map.

**Definition 7.82** (Rank of an orthogonal projection). [Lean](#) [✓ Stmt](#) The rank of an orthogonal projection  $P \in M_D(\mathbb{C})$ , i.e. the dimension of the corner algebra  $PM_D(\mathbb{C})P$ .

**Lemma 7.83** (Compression isometry for an orthogonal projection). [Lean](#) [✓ Stmt](#) Let  $P \in M_D(\mathbb{C})$  be an orthogonal projection of rank  $n$ . The corner algebra  $PM_D(\mathbb{C})P$  is linearly isomorphic to the full matrix algebra  $M_n(\mathbb{C})$ . The isomorphism is built from the spectral diagonalisation of  $P$  by conjugating the top-left  $n \times n$  block by the unitary diagonalising  $P$ .

**Lemma 7.84** (Rank equals trace for an orthogonal projection). [Lean](#) [✓ Stmt](#) The rank of an orthogonal projection  $P$  equals its trace:  $n = \text{tr } P$ .

**Lemma 7.85** (Scalar fixed points for irreducible unital maps). [Lean](#) [✓ Stmt](#) If  $E$  is irreducible and unital, then every fixed point of  $E$  is a scalar multiple of  $\mathbb{1}$ .

*Proof.* [✓ Proof](#) If  $E(X) = X$ , then the Hermitian matrices

$$X + X^\dagger, \quad i(X - X^\dagger)$$

are also fixed by  $E$ . The positive-semidefinite fixed-point uniqueness theorem gives  $X + X^\dagger = a\mathbb{1}$  and  $i(X - X^\dagger) = b\mathbb{1}$ , hence

$$X = \frac{1}{2}(a - ib)\mathbb{1}.$$

□

**Lemma 7.86** (Peripheral unitary eigenvector). [Lean](#) [✓ Stmt](#) For an irreducible unital Schwarz map with faithful adjoint fixed point, each peripheral eigenvalue admits a unitary eigenvector.

**Lemma 7.87** (Powers on a peripheral-unitary orbit). [Lean](#) [✓ Stmt](#) If  $E(U) = \mu U$  with  $|\mu| = 1$  and  $U$  unitary, then  $E(U^n) = \mu^n U^n$  for all  $n$ .

**Lemma 7.88** (Normalized peripheral unitary). [Lean](#) [✓ Stmt](#) Peripheral unitary eigenvectors can be normalized compatibly with the faithful fixed-point state.

**Lemma 7.89** (Cyclic projections from a peripheral unitary). [Lean](#) [✓ Stmt](#) A primitive  $m$ -th peripheral root yields  $m$  orthogonal projections summing to  $\mathbb{1}$  and cyclically permuted by the channel.

**Theorem 7.90** (Cyclic decomposition for irreducible Schwarz maps). [Lean](#) [✓ Stmt](#) Under irreducibility and faithful fixed point hypotheses, there is a full cyclic projection decomposition realizing the peripheral period.

**Lemma 7.91** (Power maps preserve each cyclic sector). [Lean](#) [✓ Stmt](#) Appropriate powers of the channel preserve each sector corner.

**Lemma 7.92** (Irreducibility of restricted sector dynamics). [Lean](#) [✓ Stmt](#) Each sector restriction is irreducible.

**Lemma 7.93** (Primitivity of restricted sector dynamics). [Lean](#) [✓ Stmt](#) Each sector restriction is primitive.

**Lemma 7.94** (Corner invariance at the period). [Lean](#) [✓ Stmt](#) The channel raised to the period preserves every cyclic corner.

## 7.14.2 Group structure of the peripheral spectrum

**Lemma 7.95** (Peripheral product closure). [Lean](#) [✓ Stmt](#) The product of two peripheral eigenvalues is peripheral.

**Lemma 7.96** (Peripheral inverse closure). [Lean](#) [✓ Stmt](#) The inverse of a peripheral eigenvalue is peripheral.

**Theorem 7.97** (Peripheral eigenvalue multiplicity one). [Lean](#) [✓ Stmt](#) Each peripheral eigenvalue has algebraic multiplicity one.

*Proof.* [✓ Proof](#) Given a peripheral unitary eigenvector  $U$  for  $\gamma$ , the multiplicative-domain identity gives  $E(XU^\dagger) = XU^\dagger$  for any  $\gamma$ -eigenvector  $X$ . By the scalar fixed-point lemma for irreducible unital maps,  $XU^\dagger = c \cdot \mathbb{1}$ , hence  $X = c \cdot U$  and the eigenspace is one-dimensional.  $\square$

## 7.15 Primitive overlap convergence (complementary transfer-map gap form)

The theorems in this section are stated directly in terms of the complementary map  $E - P$ . This is the analytic form needed for the later overlap argument, corresponding to the primitive-case specialization of [Wol12, Theorem 6.7] where  $T_\phi$  is a rank-one projection. For primitive normalized transfer maps, Chapter 4 supplies this complementary transfer-map input under explicit hypotheses.

**Theorem 7.98** (Complementary transfer-map gap implies trace convergence to 1). [Lean](#) [✓ Stmt](#) Let  $E$  be trace-preserving with fixed point  $\rho$  ( $\text{tr}(\rho) \neq 0$ ) and let  $P$  be the fixed-point projection (Definition 4.23). If  $\rho_{\text{spec}}(E - P) < 1$ , then  $\text{Tr}(E^n) \rightarrow 1$  as  $n \rightarrow \infty$ .

*Proof.* ✓ Proof By the power decomposition (Theorem 4.24),  $E^n = P + (E - P)^n$  for  $n \geq 1$ . Since  $\rho_{\text{spec}}(E - P) < 1$ , the Gelfand formula gives  $(E - P)^n \rightarrow 0$ , so  $\text{Tr}(E^n) \rightarrow \text{Tr}(P)$ . The operator trace of  $P$  equals 1: since  $P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)}\rho$ , we have  $\text{Tr}(P) = \frac{\text{tr}(\rho)}{\text{tr}(\rho)} = 1$ .  $\square$

**Theorem 7.99** (Self-overlap convergence from a complementary transfer-map gap). Lean ✓ Stmt *Let  $A$  be a normalized MPS tensor with a nonzero PSD fixed point  $\rho$  of the transfer map. If  $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$  (where  $P$  is the fixed-point projection), then  $O_{AA}(N) \rightarrow 1$  as  $N \rightarrow \infty$ .*

*Proof.* ✓ Proof By Lemma 7.7 (with  $A = B$ ),  $O_{AA}(N) = \text{Tr}(\mathcal{E}_A^N)$ . By Theorem 7.98,  $\text{Tr}(\mathcal{E}_A^N) \rightarrow 1$ .  $\square$

**Lemma 7.100** (Norm of self-overlap convergence). Lean ✓ Stmt *If  $A$  is an MPS tensor and  $O_{AA}(N) \rightarrow 1$  in  $\mathbb{C}$  as  $N \rightarrow \infty$ , then  $\|O_{AA}(N)\| \rightarrow 1$  as  $N \rightarrow \infty$ .*

*Proof.* ✓ Proof Follows from the continuity of the norm:  $O_{AA}(N) \rightarrow 1$  implies  $\|O_{AA}(N)\| \rightarrow \|1\| = 1$ .  $\square$

## Chapter 8

# Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension  $D$ , the cumulative span of matrix products stabilises after at most  $D^2$  steps. The results follow [SPGWC10].

### 8.1 Cumulative span

**Definition 8.1** (Word span). [Lean](#) [✓ Stmt](#) The *word span* at length  $n$  is the subspace

$$S_n(A) = \text{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, Section II].

**Definition 8.2** (Cumulative span). [Lean](#) [✓ Stmt](#) The *cumulative span* at level  $n$  is

$$T_n(A) = S_0(A) + S_1(A) + \dots + S_n(A).$$

**Lemma 8.3** (Monotonicity of cumulative span). [Lean](#) [✓ Stmt](#)  $T_n(A) \leq T_{n+1}(A)$  for all  $n$ .

*Proof.* [✓ Proof](#)  $T_{n+1}$  includes all words of length  $\leq n+1$ , which contains all words of length  $\leq n$ . □

**Lemma 8.4** (Dimension bound). [Lean](#) [✓ Stmt](#)  $\dim T_n(A) \leq D^2$  for all  $n$ .

*Proof.* [✓ Proof](#)  $T_n(A) \subseteq M_D(\mathbb{C})$  and  $\dim M_D(\mathbb{C}) = D^2$ . □

**Theorem 8.5** (Cumulative span stabilisation). [Lean](#) [✓ Stmt](#) If  $T_n(A) = T_{n+1}(A)$ , then  $T_m(A) = T_n(A)$  for all  $m \geq n$ .

*Proof.* [✓ Proof](#) If  $T_n = T_{n+1}$ , then  $S_{n+1} \subseteq T_n$ . Left-multiplying any element of  $S_{n+1}$  by  $A^i$  gives an element of  $S_{n+2}$ , so  $S_{n+2} \subseteq T_{n+1} = T_n$ . By induction,  $S_m \subseteq T_n$  for all  $m > n$ , hence  $T_m = T_n$ . □

**Theorem 8.6** (Dimension growth). [Lean](#) [✓ Stmt](#) If  $T_n(A) \neq T_{n+1}(A)$ , then  $\dim T_{n+1}(A) > \dim T_n(A)$ .

*Proof.* [✓ Proof](#)  $T_n \leq T_{n+1}$  (Lemma 8.3) and  $T_n \neq T_{n+1}$  imply strict submodule inclusion, hence strict rank increase. □

**Lemma 8.7** (Word span characterises block injectivity). [Lean](#) [✓ Stmt](#)  $S_L(A) = M_D(\mathbb{C})$  if and only if  $A$  is  $L$ -block injective.

*Proof.* [✓ Proof](#) Both conditions assert that the span of all products of length  $L$  equals  $M_D(\mathbb{C})$ .  $\square$

**Lemma 8.8** (Zero-length word span). [Lean](#) [✓ Stmt](#) If  $D \geq 2$ , then  $S_0(A) \neq M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) The zero-length products consist only of the identity, so  $S_0(A) = \text{span}_{\mathbb{C}}\{\mathbb{1}\}$  has dimension 1. Since  $\dim M_D(\mathbb{C}) = D^2 \geq 4$ , this span cannot be all of  $M_D(\mathbb{C})$ .  $\square$

**Corollary 8.9** (0-block injectivity). [Lean](#) [✓ Stmt](#) If  $D \geq 1$  and  $A$  is 0-block injective, then  $D = 1$ .

*Proof.* [✓ Proof](#) By Lemma 8.7, 0-block injectivity gives  $S_0(A) = M_D(\mathbb{C})$ . Lemma 8.8 excludes  $D \geq 2$ , hence the nonzero bond dimension must be 1.  $\square$

**Lemma 8.10** (Two-sided span from one nonzero matrix). [Lean](#) [✓ Stmt](#) If  $C \in M_D(\mathbb{C})$  is nonzero, then

$$\text{span}_{\mathbb{C}}\{RCS : R, S \in M_D(\mathbb{C})\} = M_D(\mathbb{C}). \quad (8.1)$$

This is the matrix-factorization input used in [PGVWC07, Lemma 3].

*Proof.* [✓ Proof](#) Choose a nonzero entry  $C_{pq}$ . Then every matrix unit  $E_{ij}$  is a scalar multiple of  $E_{ip}CE_{qj}$ , so the span in (8.1) contains the standard matrix basis.  $\square$

**Lemma 8.11** (Exact words around a nonzero middle matrix). [Lean](#) [✓ Stmt](#) If  $A$  is  $L$ -block injective and  $X \neq 0$ , then

$$\text{span}_{\mathbb{C}}\{A^u X A^v : |u| = |v| = L\} = M_D(\mathbb{C}).$$

*Proof.* [✓ Proof](#) By  $L$ -block injectivity, the length- $L$  word products span  $M_D(\mathbb{C})$ . Bilinearity of  $(R, S) \mapsto RXS$  reduces the preceding lemma to products with  $R$  and  $S$  chosen among those word products.  $\square$

## 8.2 Fitting decomposition

**Definition 8.12** (Fitting decomposition). [Lean](#) [✓ Stmt](#) A *Fitting decomposition* of a linear endomorphism  $f : V \rightarrow V$  on a finite-dimensional vector space over an algebraically closed field consists of:

1.  $f$  is nilpotent on the generalized 0-eigenspace,
2.  $f$  is invertible on each generalized  $\mu$ -eigenspace for  $\mu \neq 0$ ,
3. the generalized eigenspaces span  $V$ ,
4. the generalized eigenspaces are linearly independent.

**Theorem 8.13** (Fitting decomposition exists). [Lean](#) [✓ Stmt](#) Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.

*Proof.* [✓ Proof](#) Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because  $f - \mu$  is nilpotent there, so  $f = \mu(1 - (1 - f/\mu))$  is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead.  $\square$



**Theorem 8.14** (Nilpotent power bound). Lean ✓ Stmt *If  $f$  is nilpotent on a space of dimension  $n$ , then  $f^n = 0$ .*

*Proof.* ✓ Proof A nilpotent endomorphism on an  $n$ -dimensional space has nilpotency index  $\leq n$ . □

### 8.3 Nonzero trace product

**Theorem 8.15** (Cumulative span reaches  $M_D(\mathbb{C})$  — explicit bound). Lean ✓ Stmt *If  $A$  is normal, then there exists  $n \leq D^2$  with  $T_n(A) = M_D(\mathbb{C})$ .*

*Proof.* ✓ Proof By Theorem 8.6, if  $T_n \neq T_{n+1}$  then  $\dim T_{n+1} > \dim T_n$ . Since  $\dim T_n \leq D^2$  (Lemma 8.4), after at most  $D^2$  steps either  $T_n = M_D(\mathbb{C})$  or  $T_n$  has stabilised. Stabilisation before reaching  $M_D(\mathbb{C})$  contradicts normality (which requires  $S_{L_0} = M_D(\mathbb{C})$  for some  $L_0$ , hence  $T_{L_0} = M_D(\mathbb{C})$ ). □

**Theorem 8.16** (Cumulative span reaches  $M_D(\mathbb{C})$ ). Lean ✓ Stmt *If  $A$  is normal, then  $T_{D^2}(A) = M_D(\mathbb{C})$ .*

*Proof.* ✓ Proof Take  $n$  from Theorem 8.15; since  $n \leq D^2$  and  $T$  is monotone,  $T_{D^2} = M_D(\mathbb{C})$ . □

**Theorem 8.17** (Nonzero trace product). Lean ✓ Stmt *If  $A$  is normal, then there exists a word  $w$  of length  $\leq D^2$  such that  $\text{tr}(A^w) \neq 0$ . This is [SPGWC10, Lemma 1]. For the trace argument below, the nontrivial case is  $D \geq 1$ ; when  $D = 0$  the normality hypothesis is vacuous.*

*Proof.* ✓ Proof By Theorem 8.16,  $\mathbb{1} \in T_{D^2}(A)$ , so  $\mathbb{1}$  is a linear combination of word products of length  $\leq D^2$ . In the nontrivial case  $D \geq 1$ , at least one has nonzero trace (since  $\text{tr}(\mathbb{1}) = D \neq 0$ ). □

**Theorem 8.18** (Cumulative span reaches  $M_D(\mathbb{C})$  — sharp bound). Lean ✓ Stmt *If  $A$  is normal, then  $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$ , where  $d' = \dim S_1(A) = \text{kr}(A)$ . This is the sharp form of [SPGWC10, Lemma 1]: the cumulative span already reaches  $M_D(\mathbb{C})$  by index  $D^2 - d' + 1 \leq D^2$ .*

*Proof.* ✓ Proof The key observation is that if  $T_n(A) \neq T_{n+1}(A)$ , the dimension strictly increases. Since  $S_1(A) \subseteq T_1(A)$  and  $\dim S_1(A) = d'$ , we have  $\dim T_1(A) \geq d'$ . After at most  $D^2 - d'$  additional strict-growth steps the dimension reaches  $D^2 = \dim M_D(\mathbb{C})$ . Stabilisation before reaching  $M_D(\mathbb{C})$  contradicts normality. Hence  $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$ . □

**Theorem 8.19** (Sharp nonzero trace product). Lean ✓ Stmt *If  $A$  is normal, then there exists a word  $w$  of length  $\leq D^2 - \text{kr}(A) + 1$  such that  $\text{tr}(A^w) \neq 0$ . This is [SPGWC10, Lemma 1] with the sharp length bound.*

*Proof.* ✓ Proof By Theorem 8.18, the identity  $\mathbb{1}$  lies in  $T_{D^2-d'+1}(A)$ , hence is a linear combination of word products of length  $\leq D^2 - d' + 1$ . At least one such product has nonzero trace. □

**Theorem 8.20** (Sharp nonzero trace product — positive length). Lean ✓ Stmt *If  $A$  is normal and  $D \geq 2$ , then there exists a word  $w$  of positive length  $1 \leq |w| \leq D^2 - \text{kr}(A) + 1$  such that  $\text{tr}(A^w) \neq 0$ . This strengthens Theorem 8.19 by requiring  $|w| \geq 1$ , a feature needed for the blocking argument in the general Wielandt bound below.*

*Proof.* ✓ Proof Define the *positive-level* cumulative span  $V_n = \text{span}\{A^w : 1 \leq |w| \leq n\}$ . For  $D \geq 2$ , the span  $S_0(A) = \text{span}\{\mathbb{1}\}$  has dimension 1, so  $V_1 \supseteq S_1(A)$  has dimension  $\geq d' \geq 1$ . The same stabilization-or-strict-growth dichotomy shows  $V_{D^2-d'+1} = M_D(\mathbb{C})$ : if  $V_n$  stabilises and is not  $M_D(\mathbb{C})$ , then  $V_n$  is closed under left-multiplication by each  $A^i$ , so  $S_N(A) \subseteq V_n \neq M_D(\mathbb{C})$  for all  $N \geq 1$ , contradicting normality. Since  $V_{D^2-d'+1} = M_D(\mathbb{C})$ , the identity  $\mathbb{1}$  lies in  $V_{D^2-d'+1}$  and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace (since  $\text{tr}(\mathbb{1}) = D \geq 2 \neq 0$ ).  $\square$

## 8.4 Eigenvector extraction

**Theorem 8.21** (Nonzero trace implies eigenvector). Lean ✓ Stmt If  $M \in M_D(\mathbb{C})$  has  $\text{tr}(M) \neq 0$ , then  $M$  has a nonzero eigenvalue  $\mu \neq 0$  and a corresponding eigenvector  $\varphi \neq 0$  with  $M\varphi = \mu\varphi$ .

*Proof.* ✓ Proof The trace is the sum of the eigenvalues of  $M$ , counted with multiplicity, so a nonzero trace gives a nonzero eigenvalue  $\mu$ . Choosing a nonzero vector  $\varphi \in \ker(M - \mu\mathbb{1})$  gives  $M\varphi = \mu\varphi$ .  $\square$

**Theorem 8.22** (Eigenvalue extraction). Lean ✓ Stmt If  $A$  is normal, then there exist a word  $w_0$  with  $|w_0| \leq D^2$ , a nonzero scalar  $\mu \neq 0$ , and a nonzero vector  $\varphi \neq 0$  such that  $A^{w_0}\varphi = \mu\varphi$ .

*Proof.* ✓ Proof By Theorem 8.17, there is a word  $w_0$  with  $\text{tr}(A^{w_0}) \neq 0$  and  $|w_0| \leq D^2$ . By Theorem 8.21,  $A^{w_0}$  has a nonzero eigenvector.  $\square$

## 8.5 Eigenvector spreading

**Definition 8.23** (Cumulative vector span). Lean ✓ Stmt Given an MPS tensor  $A$  and a vector  $\varphi \in \mathbb{C}^D$ , the *cumulative vector span* is

$$K_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w\varphi : |w| \leq n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

**Theorem 8.24** (Vector span stabilisation). Lean ✓ Stmt If  $K_n(A, \varphi) = K_{n+1}(A, \varphi)$ , then  $K_m(A, \varphi) = K_n(A, \varphi)$  for all  $m \geq n$ .

*Proof.* ✓ Proof Same argument as Theorem 8.5: left-multiplication by  $A^i$  cannot escape the stabilised span.  $\square$

**Lemma 8.25** (Vector span from matrix span). Lean ✓ Stmt If  $T_N(A) = M_D(\mathbb{C})$  and  $\varphi \neq 0$ , then  $K_N(A, \varphi) = \mathbb{C}^D$ .

*Proof.* ✓ Proof Every matrix  $M \in M_D(\mathbb{C})$  is in  $T_N(A)$ , so in particular every rank-one matrix sending  $\varphi$  to any target vector  $v$  is in  $T_N$ . Applying such matrices to  $\varphi$  recovers all of  $\mathbb{C}^D$ .  $\square$

**Theorem 8.26** (Eigenvector spreading). Lean ✓ Stmt Let  $A$  be a normal MPS tensor and let  $A^{i_0}\varphi = \mu\varphi$  with  $\mu \neq 0$  and  $\varphi \neq 0$ . Then  $K_{D-1}(A, \varphi) = \mathbb{C}^D$ . This is [SPGWC10, Lemma 2(a)].

*Proof.* ✓ Proof The vector span  $K_n$  is monotone and  $\dim K_n \leq D$ . If  $K_n = K_{n+1}$  for some  $n < D - 1$ , the span stabilises (Theorem 8.24). But  $T_{D^2}(A) = M_D(\mathbb{C})$  by Theorem 8.16, so Lemma 8.25 gives  $K_{D^2}(A, \varphi) = \mathbb{C}^D$ , contradicting early stabilisation. Hence  $\dim K_n$  grows by  $\geq 1$  at each step, reaching  $D$  by step  $D - 1$ .  $\square$

## 8.6 Proof of the cumulative Wielandt bound

**Theorem 8.27** (Wielandt chain). [Lean](#) [✓ Stmt](#) Let  $A$  be a normal MPS tensor with bond dimension  $D \geq 1$ . Then the following hold simultaneously:

1.  $T_{D^2}(A) = M_D(\mathbb{C})$  (cumulative span stabilises);
2. there exists a word  $w_0$  with  $|w_0| \leq D^2$  and  $\text{tr}(A^{w_0}) \neq 0$ ;
3. there exist  $w_0, \mu \neq 0, \varphi \neq 0$  with  $|w_0| \leq D^2$  and  $A^{w_0}\varphi = \mu\varphi$ ;
4. for every  $\varphi \neq 0$ ,  $K_{D^2}(A, \varphi) = \mathbb{C}^D$ .

*Proof.* [✓ Proof](#) Items (1)–(3) follow from Theorems 8.16 and 8.22. Item (4) combines these with the eigenvector spreading bound (Theorem 8.26).  $\square$

**Theorem 8.28** (Wielandt bound). [Lean](#) [✓ Stmt](#) If  $A$  is a normal MPS tensor with bond dimension  $D$ , then  $T_{D^2}(A) = M_D(\mathbb{C})$ . In particular, the cumulative span stabilises at index at most  $D^2$ . This is the cumulative-span part of [SPGWC10, Theorem 1]. The cited theorem is stronger: via Lemma 2(b) it upgrades this to a fixed-length conclusion  $S_N(A) = M_D(\mathbb{C})$ , stated below as a separate statement.

*Proof.* [✓ Proof](#) Immediate from Theorem 8.16.  $\square$

**Remark 8.29** (Explicit blocking-length estimates). [✓ Stmt](#) Theorem 8.28 gives the  $D^2$  cumulative-span bound. The full-rank index bound  $\iota(A) \leq (D^2 - d + 1) \cdot D^2$  ([SPGWC10, Theorem 1, case (1)]) is proved in Theorem 8.42 below. The left-canonical normal blocked-injectivity estimate  $D^4$  is proved below in Theorem 8.49. The sharper bi-canonical blocking estimate  $3D^5$  from [SPGWC10] is not proved here.

**Definition 8.30** (One-step augmentation). [Lean](#) [✓ Stmt](#) Let  $A$  be an MPS tensor with Kraus family  $\{A^i\}_{i=0}^{d-1}$ , and let  $X \in M_D(\mathbb{C})$ . The *one-step augmentation* of  $A$  by  $X$  is the tensor  $\tilde{A}$  with physical dimension  $d + 1$  whose first Kraus operator is  $X$  and whose remaining Kraus operators are those of  $A$ :

$$\tilde{A}^0 = X, \quad \tilde{A}^{i+1} = A^i.$$

**Lemma 8.31** (Kraus operators lie in the one-step span). [Lean](#) [✓ Stmt](#) Every Kraus operator  $A^i$  belongs to the one-step span  $S_1(A)$ .

*Proof.* [✓ Proof](#) This is the length-one word with letter  $i$ .  $\square$

**Lemma 8.32** (Redundant one-step augmentation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let  $X \in S_1(A)$ , and let  $\tilde{A}$  be the one-step augmentation of  $A$  by  $X$ . Then

$$S_n(\tilde{A}) = S_n(A)$$

for every  $n \geq 0$ .

*Proof.* [✓ Proof](#) The new generator  $X$  is already in  $S_1(A)$ , and each original generator remains in the augmented family. Hence the one-step spans agree. Multiplying word spans and arguing by induction on the word length gives equality at every length.  $\square$

**Lemma 8.33** (Normality and Kraus rank under one-step augmentation). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $X \in S_1(A)$ , and let  $\tilde{A}$  be the one-step augmentation of  $A$  by  $X$ . If  $A$  is normal, then  $\tilde{A}$  is normal, and

$$\text{kr}(\tilde{A}) = \text{kr}(A).$$

*Proof.* ✓ Proof Since all exact word spans are unchanged by Lemma 8.32, eventual full word span for  $A$  gives eventual full word span for  $\tilde{A}$ . The equality of Kraus ranks is the equality of the dimensions of the common one-step span.  $\square$

**Theorem 8.34** (Wielandt case (2), invertible generator). Lean Lean ✓ Stmt *Let  $A$  be a normal MPS tensor with bond dimension  $D$ , and suppose that some Kraus operator  $A^{i_0}$  is invertible. Then*

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2 - \text{kr}(A) + 1.$$

*Proof.* ✓ Proof The dimensions of the exact word spans cannot decrease, because left multiplication by the invertible Kraus operator is injective. If this dimension growth stopped before the full matrix algebra was reached, the same argument applied at all later lengths would contradict normality. Since  $\dim S_1(A) = \text{kr}(A)$  and  $\dim M_D(\mathbb{C}) = D^2$ , the full algebra is reached by length  $D^2 - \text{kr}(A) + 1$ . The index bound follows from the definition of  $\iota(A)$ .  $\square$

**Theorem 8.35** (Wielandt case (2), one-step invertible element). Lean ✓ Stmt *Let  $A$  be an MPS tensor with bond dimension  $D \geq 1$ , satisfying  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , and suppose that  $A$  is primitive with a positive-definite fixed point. If  $S_1(A)$  contains an invertible matrix  $X$ , then*

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}). \tag{8.2}$$

*In the terminology of [SPGWC10], this is the case where the first application space  $S_1(A)$  contains an invertible operator. This is the exact word-span form of [SPGWC10, Theorem 1, case (2)].*

*Proof.* ✓ Proof Adjoin  $X$  as an extra first Kraus operator, obtaining  $\tilde{A}$ . Primitivity with a positive-definite fixed point and normalization give normality of  $A$ , hence normality of  $\tilde{A}$ ; the Kraus rank and all exact word spans are unchanged. The first Kraus operator of  $\tilde{A}$  is invertible, so Theorem 8.34 gives  $S_{D^2 - \text{kr}(\tilde{A}) + 1}(\tilde{A}) = M_D(\mathbb{C})$ . Transfer this equality back to  $A$ .  $\square$

**Corollary 8.36** (Wielandt case (2), index bound). Lean ✓ Stmt *Under the hypotheses of Theorem 8.35,*

$$\iota(A) \leq D^2 - \text{kr}(A) + 1,$$

*where  $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$ .*

*Proof.* ✓ Proof The word-span equality (8.2) gives an admissible value in the defining minimum for  $\iota(A)$ .  $\square$

**Corollary 8.37** (Wielandt case (2), invertible Kraus operator). Lean Lean ✓ Stmt *Let  $A$  be an MPS tensor with bond dimension  $D \geq 1$ , satisfying  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , and suppose that  $A$  is primitive with a positive-definite fixed point. If some Kraus operator  $A^{i_0}$  is invertible, then*

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2 - \text{kr}(A) + 1.$$

*Proof.* ✓ Proof Apply the one-step case to  $X = A^{i_0}$ , which belongs to  $S_1(A)$  by Lemma 8.31.  $\square$

**Lemma 8.38** (Rank-one extraction from a non-invertible eigenvector). Lean ✓ Stmt *Let  $A$  be a normal MPS tensor with bond dimension  $D \geq 1$ . Suppose that a Kraus operator  $A^{i_0}$  is non-invertible and that there exists a vector  $\varphi \in \mathbb{C}^D$  and a nonzero scalar  $\mu$  such that  $A^{i_0}\varphi = \mu\varphi$ . Then, for every  $\psi \in \mathbb{C}^D$ ,*

$$|\varphi\rangle\langle\psi| \in S_{D^2 - D + 1}(A).$$

*Proof.* ✓ Proof Let  $r$  be the nilpotent index of left multiplication by  $A^{i_0}$ . The dimension-growth argument gives a level  $n_0$  for which

$$(A^{i_0})^r S_{n_0}(A) = \{(A^{i_0})^r M : M \in M_D(\mathbb{C})\}.$$

Since  $A^{i_0}\varphi = \mu\varphi$  and  $\mu \neq 0$ , iterating gives  $(A^{i_0})^r \varphi = \mu^r \varphi$ , hence

$$\varphi = \mu^{-r} (A^{i_0})^r \varphi \in \text{range}((A^{i_0})^r).$$

Hence, for every  $\psi \in \mathbb{C}^D$ ,

$$|\varphi\rangle\langle\psi| \in S_{r+n_0}(A).$$

The nilpotent-index estimate gives

$$r + n_0 \leq D^2 - D + 1.$$

Since

$$A^{i_0} |\varphi\rangle\langle\psi| = |A^{i_0}\varphi\rangle\langle\psi| = \mu |\varphi\rangle\langle\psi|$$

and  $\mu \neq 0$ , the equality

$$|\varphi\rangle\langle\psi| = \mu^{-1} A^{i_0} |\varphi\rangle\langle\psi|$$

implies

$$|\varphi\rangle\langle\psi| \in S_k(A) \implies |\varphi\rangle\langle\psi| \in S_{k+1}(A)$$

because left multiplication by  $A^{i_0}$  maps  $S_k(A)$  into  $S_{k+1}(A)$ . Iterating from  $k = r + n_0$  and using  $r + n_0 \leq D^2 - D + 1$  gives

$$|\varphi\rangle\langle\psi| \in S_{D^2-D+1}(A).$$

□

**Theorem 8.39** (Wielandt case (3), one-step non-invertible element). Lean ✓ Stmt *Let  $A$  be an MPS tensor with bond dimension  $D \geq 1$ , satisfying  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , and suppose that  $A$  is primitive with a positive-definite fixed point. If  $S_1(A)$  contains a non-invertible matrix  $X$  with a nonzero eigenvalue, then*

$$S_{D^2}(A) = M_D(\mathbb{C}).$$

*More explicitly, it is enough to have  $X\varphi = \mu\varphi$  with  $\varphi \neq 0$  and  $\mu \neq 0$ . In the terminology of [SPGWC10], this is the case where the first application space  $S_1(A)$  contains a non-invertible operator with a nonzero eigenvalue. This is the exact word-span form of [SPGWC10, Theorem 1, case (3)].*

*Proof.* ✓ Proof Adjoin  $X$  as an extra first Kraus operator. The augmented tensor is normal and has the same exact word spans as  $A$ . The eigenvector relation for  $X$  is now a one-generator eigenvector relation. The eigenvector-spreading argument gives full fixed-length vector span by length  $D - 1$ , and Lemma 8.38 gives the rank-one operators  $|\varphi\rangle\langle\psi|$  in length  $D^2 - D + 1$ . Multiplying these two fixed-length spans gives  $S_{D^2}$  for the augmented tensor, hence for  $A$ . □

**Corollary 8.40** (Wielandt case (3), index bound). Lean ✓ Stmt *Under the hypotheses of Theorem 8.39,*

$$\iota(A) \leq D^2.$$

*Proof.* ✓ Proof The equality  $S_{D^2}(A) = M_D(\mathbb{C})$  gives an admissible value in the defining minimum for  $\iota(A)$ . □

**Corollary 8.41** (Wielandt case (3), non-invertible Kraus operator). [Lean](#) [Lean](#) [Stmt](#) Let  $A$  be an MPS tensor with bond dimension  $D \geq 1$ , satisfying  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , and suppose that  $A$  is primitive with a positive-definite fixed point. If some Kraus operator  $A^{i_0}$  is non-invertible and has an eigenvector  $\varphi \neq 0$  with nonzero eigenvalue  $\mu$ , then

$$S_{D^2}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2.$$

*Proof.* [Proof](#) Apply the one-step case to  $X = A^{i_0}$ , which belongs to  $S_1(A)$  by Lemma 8.31.  $\square$

**Theorem 8.42** (Wielandt's inequality — general bound). [Lean](#) [Stmt](#) Let  $A$  be a normalized MPS tensor ( $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ ) that is primitive with a positive-definite fixed point. Then

$$\iota(A) \leq (D^2 - \text{kr}(A) + 1) \cdot D^2,$$

where  $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$  is the full-Kraus-rank index. This is [SPGWC10, Theorem 1, case (1)].

*Proof.* [Proof](#)

1.  $D = 1$ : In this case  $\iota(A) = 0$ , so the bound holds trivially.
2.  $D \geq 2$ : By Theorem 8.20, there exists a *positive-length* word  $w$  with  $|w| \leq D^2 - d' + 1$  and  $\text{tr}(A^w) \neq 0$ , where  $d' = \text{kr}(A)$ . The matrix  $A^w$  has nonzero trace, hence a nonzero eigenvalue  $\mu$  and eigenvector  $\varphi$  (Theorem 8.21). Set  $n = |w|$  and let  $B = A^{[n]}$  be the blocked tensor. Then  $B$  is normal (Theorem 8.53), and the encoding of  $w$  as a single block index gives  $B^{i_0} = A^w$ .

- (a) If  $A^w$  is invertible (invertible case),  $S_{D^2}(B) = M_D(\mathbb{C})$ , whence  $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$ .
- (b) If  $A^w$  is non-invertible (non-invertible case), the eigenvector  $\varphi$  spans a proper subspace, and the rank-one extraction argument gives  $S_{D^2}(B) = M_D(\mathbb{C})$ , whence  $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$ .

In the invertible case, the invertible Kraus operator  $B^{i_0}$  allows direct dimension growth, giving  $S_{D^2 - k_B + 1}(B) = M_D(\mathbb{C})$  where  $k_B = \text{kr}(B)$ . In the non-invertible case, the bound  $S_{D^2}(B) = M_D(\mathbb{C})$  follows from tracking indices through the rank-one extraction argument of [SPGWC10, Lemma 2(b)]; the separate qualitative fixed-length conclusion stated below does not supply this explicit bound. In both cases  $\iota(A) \leq D^2 \cdot n \leq D^2 \cdot (D^2 - d' + 1)$ .  $\square$

**Remark 8.43** (Quantitative and qualitative Wielandt bounds). [Stmt](#) The quantitative bound  $\iota(A) \leq (D^2 - d' + 1) \cdot D^2$  is cited from [SPGWC10, Theorem 1, case (1)]. The blueprint's own lemmas prove the qualitative conclusion used on the fundamental-theorem route, namely Theorem 8.59: there exists  $N$  with  $S_N(A) = M_D(\mathbb{C})$ . The quantitative bound is not on the fundamental-theorem critical path.

## 8.7 Fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span  $K_{D-1}(A, \varphi) = \mathbb{C}^D$ . The next stage is to pass to a span at one fixed length, namely to find  $n_0$  such that  $S_{n_0}(A) = M_D(\mathbb{C})$  (i.e. word products of *exactly* one length span all matrices).

**Definition 8.44** (Fixed-length vector span). [Lean](#) [✓ Stmt](#) The *fixed-length vector span* at length  $n$  is

$$H_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is  $S_n(A)|\varphi\rangle$  in the notation of [\[SPGWC10\]](#).

**Lemma 8.45** (Word span products). [Lean](#) [✓ Stmt](#)  $S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$ : the product of word spans at lengths  $m$  and  $n$  is contained in the word span at length  $m + n$ .

*Proof.* [✓ Proof](#) Every product  $A^{w_1}A^{w_2}$  with  $|w_1| = m$  and  $|w_2| = n$  equals  $A^{w_1w_2}$  with  $|w_1w_2| = m + n$ .  $\square$

**Lemma 8.46** (Eigenvector padding). [Lean](#) [✓ Stmt](#) Suppose  $A^{i_0} \varphi = \mu \varphi$  with  $\mu \neq 0$ . Then for all  $n$ ,

$$K_n(A, \varphi) = H_n(A, \varphi).$$

In particular, if  $K_n(A, \varphi) = \mathbb{C}^D$  then already  $H_n(A, \varphi) = \mathbb{C}^D$ .

*Proof.* [✓ Proof](#) Any word  $w$  with  $|w| \leq n$  can be padded to exact length  $n$  by appending  $n - |w|$  copies of the index  $i_0$ . The eigenvector relation gives  $A^{w \cdot i_0^k} \varphi = \mu^k \cdot A^w \varphi$ , so the padded vector is a nonzero scalar multiple of the original. Hence every generator of  $K_n$  lies in  $H_n$ . The reverse inclusion  $H_n \leq K_n$  is immediate.  $\square$

**Theorem 8.47** (Fixed-length vector spanning implies matrix spanning). [Lean](#) [✓ Stmt](#) Assume:

1.  $H_n(A, \varphi) = \mathbb{C}^D$  (length- $n$  word products applied to  $\varphi$  span all of  $\mathbb{C}^D$ ), and
2. for each standard basis vector  $e_j$ , the rank-one operator  $|\varphi\rangle\langle e_j|$  lies in  $S_m(A)$ .

Then  $S_{n+m}(A) = M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) Fix a matrix unit  $E_{ij}$ . By hypothesis (1), there exists  $M_i \in S_n(A)$  with  $M_i \varphi = e_i$ . By hypothesis (2),  $|\varphi\rangle\langle e_j| \in S_m(A)$ . Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i \varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and  $E_{ij} \in S_{n+m}(A)$  by Lemma 8.45. Since the matrix units span  $M_D(\mathbb{C})$ ,  $S_{n+m}(A) = M_D(\mathbb{C})$ .  $\square$

**Lemma 8.48** (Blocking transfer). [Lean](#) [✓ Stmt](#) If the blocked tensor  $A^{[L]}$  satisfies  $S_n(A^{[L]}) = M_D(\mathbb{C})$ , then  $S_{nL}(A) = M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) Each length- $n$  word in the blocked alphabet  $\{0, \dots, d^L - 1\}$  decodes to a length- $nL$  word in the original alphabet  $\{0, \dots, d - 1\}$ , so  $S_n(A^{[L]}) \subseteq S_{nL}(A)$ .  $\square$

**Theorem 8.49** (Bounded injective blocking for left-canonical normal tensors). [Lean](#) [✓ Stmt](#) Let  $A$  be a left-canonical normal MPS tensor with bond dimension  $D > 0$ . Then there is a positive blocking length  $L \leq D^4$  such that the blocked tensor  $A^{[L]}$  is one-site injective.

This is the blocked-injectivity form of the quantum Wielandt input used in the Fundamental Theorem for translation-invariant tensors. It corresponds to the statement in [\[CPGSV16, Section II, line 332\]](#) that every normal tensor becomes injective after at most  $D^4$  blockings, using the index estimate of [\[SPGWC10, Theorem 1\]](#). The Lean statement includes the left-canonical/trace-preserving hypothesis because that is the canonical-form context in which the blocked-injectivity input is consumed.



*Proof.* ✓ Proof The proof uses the full-Kraus-rank index bound from the quantum Wielandt inequality. Since  $A$  is normal, the index is attained; the one-step Kraus rank is nonzero in the left-canonical case, so the bound  $(D^2 - \text{rank } S_1(A) + 1)D^2$  is at most  $D^4$ . The passage from exact word-span fullness to injectivity of the blocked tensor is Lemma 8.48.  $\square$

**Corollary 8.50** (Uniform injective blocking for separated normal-canonical families). Lean  
Lean ✓ Stmt Let  $(A_x)_{x \in X}$  be a finite separated normal-canonical family whose blocks have positive virtual bond dimension. Then there is a positive integer  $L$  such that every blocked tensor  $A_x^{[L]}$  is one-site injective. The same conclusion holds simultaneously for two finite separated normal-canonical families whose blocks have positive virtual bond dimension.

Here the family is a basis of normal-canonical representatives in the sense of Definition 10.9; the blocks are distinct and the weight moduli are non-increasing but need not be strictly ordered. It assumes that the representative family has already been selected; it does not reconstruct the full CPSV sector decomposition with repeated copies inside one sector. The restriction is recorded in `docs/paper-gaps/ft_one_copy_scope_restriction.tex`.

*Proof.* ✓ Proof Apply Theorem 8.49 to each block. Taking the product of the finitely many resulting positive blocking lengths gives a common positive multiple, and injectivity persists under positive further blocking.  $\square$

*Remark 8.51* (Relation with the rank-one extraction lemma). ✓ Stmt In [SPGWC10], Lemma 2(b) combines a rank-one extraction for fixed-length spans with a product argument turning fixed-length vector spanning into fixed-length matrix spanning. Theorem 8.47 is the second step. Theorem 8.58 supplies the rank-one hypothesis in blocked form, and Theorem 8.59 then gives a blocked, fixed-length word-span saturation result.

*Remark 8.52* (Use of the quantum Wielandt theorem in the MPS route). The source papers use the quantum Wielandt theorem as an input converting normality into injectivity after blocking. In the notation of this chapter, the input has two distinct conclusions. First, Theorem 8.28 gives cumulative saturation  $T_{D^2}(A) = M_D(\mathbb{C})$ . This alone is not the injectivity statement used in canonical form, because injectivity requires products of one fixed length. Second, Lemma 2(b) of [SPGWC10], formalized here through Theorem 8.59, supplies a length  $N$  for which  $S_N(A) = M_D(\mathbb{C})$ . This is the statement that matches the injectivity of  $\Gamma_N$  and can be transported through blocking by Lemma 8.48. Thus any later use of Wielandt in the Fundamental Theorem must specify whether it is using cumulative span, exact-length span, or the blocked version of the exact-length conclusion.

## 8.8 Blocked tensor and rectangular span

This section extends the Lemma 2(b) argument to the *blocked tensor*  $A^{[L]}$  and develops the “rectangular span” used to produce rank-one elements.

**Theorem 8.53** (Blocking preserves normality). Lean ✓ Stmt If  $A$  is normal and  $L > 0$ , then the blocked tensor  $A^{[L]}$  is also normal.

*Proof.* ✓ Proof Choose  $N$  with  $S_N(A) = M_D(\mathbb{C})$ . Then  $\mathbb{1} \in S_N(A)$ . For any  $M \in S_N(A)$ , write  $\mathbb{1} = \sum_r c_r A^{w_r}$  with each  $|w_r| = N$ . Then

$$M = M \cdot \mathbb{1} = \sum_r c_r M A^{w_r} \in S_N(A) \cdot S_N(A) \subseteq S_{2N}(A) \quad (8.3)$$



by Lemma 8.45. Iterating the inclusion (8.3) gives  $S_N(A) \subseteq S_{kN}(A)$  for every  $k \geq 1$ ; taking  $k = L$  yields  $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$ . By Lemma 8.54,  $S_{NL}(A) \subseteq S_N(A^{[L]})$ , so  $S_N(A^{[L]}) = M_D(\mathbb{C})$ .  $\square$

**Lemma 8.54** (Chunking: word span embeds into blocked word span). Lean ✓ Stmt  $S_{nL}(A) \subseteq S_n(A^{[L]})$  for all  $n, L$ .

*Proof.* ✓ Proof Every length- $nL$  word in the original alphabet can be split into  $n$  consecutive chunks of length  $L$ , each of which is a single blocked Kraus operator. By induction on  $n$ , using the word span product lemma (Lemma 8.45).  $\square$

**Theorem 8.55** (Word eigenvector gives blocked eigenvector). Lean ✓ Stmt If  $A^{w_0}\varphi = \mu\varphi$  for a word  $w_0$  of length  $L$ , then  $(A^{[L]})^{j_0}\varphi = \mu\varphi$ , where  $j_0$  is the encoded blocked index corresponding to  $w_0$ .

*Proof.* ✓ Proof By definition,  $(A^{[L]})^{j_0} = A^{w_0}$ .  $\square$

**Theorem 8.56** (Blocked fixed-length matrix spanning). Lean ✓ Stmt Suppose  $A$  is normal and  $L > 0$ , and we have:

1. a word  $w_0$  of length  $L$  with eigenvector  $A^{w_0}\varphi = \mu\varphi$  ( $\mu \neq 0, \varphi \neq 0$ ),
2. a word  $\tau_0$  of length  $L$  with transpose eigenvector  $(A^{\tau_0})^T\psi = \nu\psi$  ( $\nu \neq 0, \psi \neq 0$ ),
3. a rank-one element  $\varphi\psi^T \in S_m(A^{[L]})$  for some  $m$ .

Then there exists  $N$  such that  $S_N(A) = M_D(\mathbb{C})$ .

*Remark.* The proof gives the explicit bound  $N = ((D-1) + m + (D-1)) \cdot L$ ; the statement above gives only the existential conclusion.

*Proof.* ✓ Proof Apply Theorem 8.47 to the blocked tensor  $A^{[L]}$ , which is normal by Theorem 8.53. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of  $A^{[L]}$  by Theorem 8.55. Transfer back to the original tensor via Lemma 8.48.  $\square$

**Theorem 8.57** (Blocked fixed-length matrix spanning from word eigenvectors). Lean ✓ Stmt Suppose  $A$  is normal and  $L > 0$ . If a length- $L$  word has a nonzero eigenvector and a length- $L$  word has a nonzero transpose eigenvector, then there exists  $N$  such that  $S_N(A) = M_D(\mathbb{C})$ .

*Proof.* ✓ Proof The rank-one extraction theorem supplies the blocked rank-one element required by Theorem 8.56; apply the blocked assembly theorem.  $\square$

**Theorem 8.58** (Rank-one extraction for blocked tensor). Lean ✓ Stmt Suppose  $A$  is normal and  $N_0 \geq 1$  with  $S_{N_0}(A) = M_D(\mathbb{C})$ . Then there exist words  $\sigma_0, \tau_0$  of length  $N_0$ , nonzero vectors  $\varphi, \psi$ , nonzero scalars  $\mu, \nu$ , and  $m \in \mathbb{N}$  such that:

1.  $A^{\sigma_0}\varphi = \mu\varphi$ ,
2.  $(A^{\tau_0})^T\psi = \nu\psi$ ,
3.  $\varphi\psi^T \in S_m(A^{[N_0]})$ .

*Proof.* ✓ Proof Set  $B = A^{[N_0]}$ . Since  $S_{N_0}(A) = M_D(\mathbb{C})$ , we have  $S_1(B) = M_D(\mathbb{C})$  by definition of the blocked tensor. Hence  $\mathbb{1} \in S_1(B)$ , so in the nontrivial case  $D \geq 1$  not all blocked generators can have trace zero. Choose  $j_0$  with  $\text{tr}(B^{j_0}) \neq 0$ . By Theorem 8.21,  $B^{j_0}$  has a nonzero eigenvector  $\varphi$  with eigenvalue  $\mu \neq 0$ . Writing  $j_0$  as the blocked index corresponding to a word  $\sigma_0$  of length  $N_0$  gives  $A^{\sigma_0}\varphi = \mu\varphi$ . Applying the same argument to the transpose blocked tensor gives a word  $\tau_0$  of length  $N_0$ , a nonzero vector  $\psi$ , and  $\nu \neq 0$  with  $(A^{\tau_0})^T\psi = \nu\psi$ . Since  $A$  is normal and  $N_0 \geq 1$ , the blocked tensor  $B$  is normal by Theorem 8.53. Therefore there exists  $M$  with  $S_M(B) = M_D(\mathbb{C})$ . In particular  $\varphi\psi^T \in S_M(B)$ . Taking  $m = M$  proves (3).  $\square$

**Theorem 8.59** (Fixed-length word-span saturation via blocking). Lean ✓ Stmt *If  $A$  is a normal MPS tensor with bond dimension  $D$ , then there exists  $N$  such that  $S_N(A) = M_D(\mathbb{C})$ . This gives the fixed-length spanning conclusion corresponding to [SPGWC10, Lemma 2(b)].*

*Proof.* ✓ Proof By normality, some  $N_0$  has  $S_{N_0}(A) = M_D(\mathbb{C})$ . If  $N_0 = 0$  we are done. Otherwise, Theorem 8.58 produces eigenvectors and a rank-one element  $\varphi\psi^T \in S_m(A^{[N_0]})$ . Theorem 8.56 then gives  $S_{((D-1)+m+(D-1)) \cdot N_0}(A) = M_D(\mathbb{C})$ .  $\square$

*Remark 8.60* (Blocking argument for fixed-length spanning). ✓ Stmt Theorem 8.59 is obtained by first extracting a rank-one operator in a blocked tensor (Theorem 8.58) and then applying the fixed-length matrix-spanning theorem (Theorem 8.56). Thus the fixed-length span of  $A$  is reached through a blocked auxiliary tensor rather than by a direct unblocked argument.

## 8.9 Primitive MPS tensors

**Definition 8.61** (Primitive MPS tensor). Lean ✓ Stmt Assume  $D > 0$ . A tensor  $A$  together with a matrix  $\rho \in M_D(\mathbb{C})$  is *primitive* if

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \quad \rho \geq 0, \quad \rho \neq 0, \quad \mathcal{E}_A(\rho) = \rho,$$

and, if  $P$  is the fixed-point projection associated to  $\rho$ , then  $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ . When the choice of  $\rho$  is irrelevant, we simply say that  $A$  is a primitive MPS tensor. This is weaker than the primitive condition of [SPGWC10], which requires  $\rho > 0$ ; see Remark 8.67. The stronger positive-definite hypothesis is imposed separately in Lemma 8.68 and Theorem 8.78.

**Theorem 8.62** (Primitive overlap convergence). Lean ✓ Stmt *If  $A$  is a primitive MPS tensor, then  $O_{AA}(N) \rightarrow 1$ .*

*Proof.* ✓ Proof Immediate from the self-overlap convergence under a complementary transfer-map gap (Theorem 7.99).  $\square$

## 8.10 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 8.61: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius  $< 1$ .

**Lemma 8.63** (Primitive MPS tensors have nonzero word products). Lean ✓ Stmt *If  $A$  is a primitive MPS tensor, then for every  $n \geq 0$  there exists a word  $w$  of length exactly  $n$  with  $A^w \neq 0$ .*

*Proof.* ✓ Proof The PSD fixed point  $\rho$  of  $\mathcal{E}_A$  satisfies  $\mathcal{E}_A^n(\rho) = \rho \neq 0$ . Since  $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$ , at least one summand is nonzero, hence some  $A^w \neq 0$ .  $\square$

**Lemma 8.64** (Iterated transfer does not vanish). [Lean](#) [✓ Stmt](#) *If  $A$  is a primitive MPS tensor, then  $\mathcal{E}_A^n \neq 0$  for all  $n \geq 0$ .*

*Proof.* [✓ Proof](#) By Lemma 8.63, there exists a word  $w$  of length  $n$  with  $A^w \neq 0$ , so  $\mathcal{E}_A^n(\rho)$  has a nonzero summand.  $\square$

### 8.10.1 From primitivity to normality

The primitivity condition of Definition 8.61 combines TP normalization, a nonzero PSD fixed point, and a complementary transfer-map gap. It is connected to normality through the following results.

**Lemma 8.65** (Fixed-point uniqueness from a complementary transfer-map gap). [Lean](#) [✓ Stmt](#) *If  $A$  is a primitive MPS tensor with PSD fixed point  $\rho$ , then every fixed point  $\sigma$  of  $\mathcal{E}_A$  is proportional to  $\rho$ :  $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$ .*

*Proof.* [✓ Proof](#) Set  $\sigma' = \sigma - \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$ . Then  $\text{tr}(\sigma') = 0$  and  $(\mathcal{E}_A - P_\rho)(\sigma') = \sigma'$ . If  $\sigma' \neq 0$ , then 1 is an eigenvalue of  $\mathcal{E}_A - P_\rho$ , contradicting the complementary transfer-map gap  $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$ .  $\square$

**Lemma 8.66** (Complement powers tend to zero). [Lean](#) [✓ Stmt](#) *If  $A$  is a primitive MPS tensor with PSD fixed point  $\rho$ , then  $(\mathcal{E}_A - P_\rho)^n \rightarrow 0$  in operator norm.*

*Proof.* [✓ Proof](#) The spectral radius of  $\mathcal{E}_A - P_\rho$  is strictly less than 1 by the complementary transfer-map gap hypothesis. Apply the general result that powers of an operator with spectral radius  $< 1$  tend to zero.  $\square$

*Remark 8.67* (Primitivity: transfer-map gap vs. positive definiteness). [✓ Stmt](#) The primitivity condition used in Section 8.10.1 has three parts: TP normalization  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , a nonzero PSD fixed point  $\rho$ , and the complementary transfer-map gap  $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$ . The notion of “primitive” in [SPGWC10, Proposition 3] additionally requires  $\rho$  to be positive definite (full rank). Without positive definiteness, the transfer-map gap condition alone does *not* imply irreducibility or normality. A counterexample is  $D = 2$ ,  $\rho = \text{diag}(1, 0)$ .

With the additional hypothesis  $\rho > 0$ , Lemma 8.68 gives irreducibility. Together with the Burnside argument below (Section 8.10.3) and Theorem 8.85, one obtains

$$\text{primitive} + \rho \text{ positive definite} \Rightarrow \text{irreducible} \Rightarrow \text{alg}(A) = M_D(\mathbb{C}) \Rightarrow \text{normal}.$$

The last step uses the aperiodicity condition  $\mathbb{1} \in S_1(A)$  from Remark 8.79. After restoring the positive-definite hypothesis, [SPGWC10, Proposition 3] identifies that primitive/strongly irreducible condition with eventual full Kraus rank, i.e. with our notion of normality.

### 8.10.2 Primitive implies irreducible

**Lemma 8.68** (Primitive tensors with positive-definite fixed point are irreducible). [Lean](#) [✓ Stmt](#) *If  $A$  is a primitive MPS tensor with PSD fixed point  $\rho$  and  $\rho$  is positive definite, then  $A$  is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV16, Section 2.3]).*

*Proof.* [✓ Proof](#) By contrapositive. Assume  $A$  has an invariant projection  $P$  ( $P \neq 0$ ,  $P \neq \mathbb{1}$ ,  $(\mathbb{1} - P)A^iP = 0$  for all  $i$ ). Set  $\sigma = P\rho P$ .

1. *Invariance of  $\sigma$  under  $\mathcal{E}_A^n$ :* the projection relation  $(\mathbb{1} - P)A^iP = 0$  implies  $(\mathbb{1} - P) \cdot \mathcal{E}_A^n(\sigma) = 0$  for all  $n$  (by induction on  $n$ ).

2. *Convergence*: By Lemma 8.66,  $\mathcal{E}_A^n(\sigma) \rightarrow \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$ . Hence  $(\mathbb{1} - P) \cdot \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho = 0$ .
3. *Nonzero scalar*: Since  $P \neq 0$  and  $\rho$  is positive definite,  $\text{tr}(P\rho P) > 0$ , so  $\frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \neq 0$ . Therefore  $(\mathbb{1} - P)\rho = 0$ .
4. *Contradiction*: Since  $\rho$  is positive definite, it is a unit in  $M_D(\mathbb{C})$ . From  $(\mathbb{1} - P)\rho = 0$  we get  $\mathbb{1} - P = 0$ , i.e.  $P = \mathbb{1}$ , contradicting  $P \neq \mathbb{1}$ .

□

### 8.10.3 From irreducibility to full spanning via Burnside's theorem

The irreducibility condition (Definition 9.16) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of  $\mathbb{C}^D$  under the Kraus operators.

**Definition 8.69** (Algebra span). Lean ✓ Stmt For an MPS tensor  $A$ , let  $\text{alg}(A)$  be the unital  $\mathbb{C}$ -subalgebra of  $M_D(\mathbb{C})$  generated by the matrices  $\{A^i\}_{i=0}^{d-1}$ :

$$\text{alg}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d-1 \rangle \subseteq M_D(\mathbb{C}).$$

**Definition 8.70** (Invariant submodule). Lean ✓ Stmt A subspace  $W \leq \mathbb{C}^D$  is *A-invariant* if  $A^i W \subseteq W$  for every  $i$ .

**Definition 8.71** (Irreducible action). Lean ✓ Stmt The Kraus operators of  $A$  act *irreducibly* on  $\mathbb{C}^D$  if every  $A$ -invariant subspace is trivial, i.e. equal to 0 or to  $\mathbb{C}^D$ .

**Theorem 8.72** (Irreducible tensor implies irreducible action). Lean ✓ Stmt If  $A$  is irreducible as a tensor (Definition 9.16), then the Kraus operators act irreducibly on  $\mathbb{C}^D$  (Definition 8.71).

*Proof.* ✓ Proof Let  $W \leq \mathbb{C}^D$  be an  $A$ -invariant subspace. In finite dimension the orthogonal projection  $P$  onto  $W$  exists. Invariance of  $W$  implies  $(\mathbb{1} - P)A^i P = 0$  for all  $i$ . Thus  $P$  is a nontrivial invariant projection unless  $W = 0$  or  $W = \mathbb{C}^D$ , contradicting irreducibility of the tensor. □

**Theorem 8.73** (Burnside's theorem for Kraus algebras). Lean ✓ Stmt Assume  $D > 0$ . If  $A$  acts irreducibly on  $\mathbb{C}^D$ , then the generated unital subalgebra is the full matrix algebra:

$$\text{alg}(A) = M_D(\mathbb{C}).$$

This is the complex finite-dimensional case of Burnside's theorem (equivalently, Jacobson's density theorem); see [Jac09].

*Proof.* ✓ Proof Transport  $\text{alg}(A)$  along the algebra equivalence  $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \text{End}_{\mathbb{C}}(\mathbb{C}^D)$ . The irreducible-action hypothesis makes  $\mathbb{C}^D$  a simple module for this algebra. Over the algebraically closed field  $\mathbb{C}$ , Schur's lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of  $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$ . □

**Remark 8.74.** ✓ Stmt This Burnside/Jacobson-density step is a substantial external algebraic ingredient in the chapter. We use it here as the finite-dimensional complex Burnside theorem, namely the statement that an irreducible matrix algebra action already generates the full endomorphism algebra.

**Theorem 8.75** (Irreducible action implies eventual cumulative spanning). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . If  $A$  acts irreducibly on  $\mathbb{C}^D$ , then there exists  $N$  such that  $T_N(A) = M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) By Theorem 8.73,  $\text{alg}(A) = M_D(\mathbb{C})$ . Every element of the algebra span is a linear combination of word products, hence belongs to  $T_n(A)$  for some  $n$ . Since  $M_D(\mathbb{C})$  is finite-dimensional, the ascending chain  $T_0(A) \leq T_1(A) \leq \dots$  stabilizes, so one can choose a uniform  $N$  with  $T_N(A) = M_D(\mathbb{C})$ .  $\square$

#### 8.10.4 From primitivity to strong irreducibility and normality

The results in this subsection remove the aperiodicity assumption from the normality conclusions. They connect primitivity in Definition 8.61, formulated via a complementary transfer-map gap, directly to normality (eventually full Kraus rank), using the convergence of the transfer-map iterates to the projection onto the fixed-point subspace.

**Lemma 8.76** (Primitive tensors with positive-definite fixed point have irreducible transfer map). [Lean](#) [✓ Stmt](#) If  $A$  is a primitive MPS tensor with positive-definite fixed point  $\rho$ , then the transfer map  $\mathcal{E}_A$  is irreducible.

The proof uses fixed-point uniqueness (Lemma 8.65): every PSD fixed point is proportional to  $\rho$ , so Wolf's criterion for irreducibility (positive-definite fixed point implies irreducibility) applies directly.

*Proof.* [✓ Proof](#) By Lemma 8.65, if  $\sigma \geq 0$  and  $\mathcal{E}_A(\sigma) = \sigma$ , then  $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$ . Since  $\rho > 0$ , Wolf's uniqueness criterion for irreducibility applies.  $\square$

**Lemma 8.77** (Primitive tensors with positive-definite fixed point are strongly irreducible). [Lean](#) [✓ Stmt](#) If  $A$  is a primitive MPS tensor with positive-definite fixed point  $\rho$ , then  $A$  is strongly irreducible in the sense of [SPGWC10, Proposition 3(c)]:  $\mathcal{E}_A$  is irreducible, the fixed point  $\rho$  is positive definite, and the peripheral spectrum of  $\mathcal{E}_A$  is  $\{1\}$ .

*Proof.* [✓ Proof](#) Combine Lemma 8.76 (irreducibility) with the peripheral-spectrum primitivity from the transfer-map gap hypothesis (Theorem 8.62).  $\square$

**Theorem 8.78** (Primitive MPS with positive-definite fixed point is normal). [Lean](#) [✓ Stmt](#) If  $A$  is a primitive MPS tensor with positive-definite fixed point  $\rho$ , then  $A$  is normal (has eventually full Kraus rank). No aperiodicity assumption is needed.

This is the key connection between primitivity in Definition 8.61, formulated via a complementary transfer-map gap, and normality; it removes the need for the aperiodicity-based argument.

*Proof.* [✓ Proof](#) By Lemma 8.77,  $A$  is strongly irreducible, i.e. primitive in Wolf's sense. Under normalization, Theorem 7.65 identifies this primitivity condition directly with normality (equivalently, eventual full Kraus rank).  $\square$

### 8.11 Cumulative span to word span

Cumulative spanning ( $T_N(A) = M_D(\mathbb{C})$ , the union of word spans of all lengths  $\leq N$ ) is weaker than normality ( $S_L(A) = M_D(\mathbb{C})$  for a single length  $L$ ). An aperiodicity hypothesis upgrades cumulative spanning to a single-length word span.

**Remark 8.79** (Aperiodicity condition). [✓ Stmt](#) We say  $A$  satisfies the *aperiodicity condition* if  $\mathbb{1} \in S_1(A)$ , i.e. the identity matrix lies in the span of the Kraus operators  $\{A^i\}_{i=0}^{d-1}$ . This is an additional hypothesis. The one-sided TP normalization  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  does not imply

$\mathbb{1} \in S_1(A)$ ; it only gives a quadratic identity among the Kraus operators. The aperiodicity condition is kept explicit in this section for generality. For the primitive condition with positive-definite fixed point (peripheral-spectrum primitivity together with a positive-definite fixed point), normality follows directly from Wolf's primitivity equivalence (Theorem 7.65) without passing through aperiodicity.

**Remark 8.80** (Counterexample: cumulative spanning does not imply normality). ✓ Stmt In  $M_2(\mathbb{C})$ , let  $A^0 = E_{12}$  and  $A^1 = E_{21}$  (the off-diagonal matrix units). Then  $\text{alg}(A) = M_2(\mathbb{C})$  and  $T_2(A) = M_2(\mathbb{C})$ , but the word spans alternate:  $S_n(A)$  contains only off-diagonal matrices for odd  $n$  and only diagonal matrices for even  $n$ . Hence  $S_n(A) \neq M_2(\mathbb{C})$  for every  $n$ , and  $A$  is not normal. The aperiodicity condition  $\mathbb{1} \in S_1(A)$  fails:  $S_1(A) = \text{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$ .

**Theorem 8.81** (Word span monotonicity from aperiodicity). Lean ✓ Stmt If  $\mathbb{1} \in S_1(A)$ , then  $S_n(A) \leq S_{n+1}(A)$  for all  $n$ .

*Proof.* ✓ Proof For any  $M \in S_n(A)$ ,  $M = M \cdot \mathbb{1} \in S_n(A) \cdot S_1(A) \subseteq S_{n+1}(A)$  by Lemma 8.45. □

**Theorem 8.82** (Homogeneous word-span propagation under normalization). Lean Lean Lean ✓ Stmt Suppose

$$\sum_a A_a A_a^\dagger = \mathbb{1} \quad \text{and} \quad S_L(A) = M_D(\mathbb{C}).$$

Then, for every  $m \geq L$ ,

$$S_m(A) = M_D(\mathbb{C}).$$

Equivalently,  $L$ -block injectivity propagates to every larger homogeneous length.

*Proof.* ✓ Proof If  $S_n(A) = M_D(\mathbb{C})$ , then for any  $X \in M_D(\mathbb{C})$ ,

$$X = \left( \sum_a A_a A_a^\dagger \right) X = \sum_a A_a (A_a^\dagger X).$$

Since  $A_a^\dagger X \in S_n(A)$ , each summand lies in  $S_1(A)S_n(A) \subseteq S_{n+1}(A)$ . Thus  $S_{n+1}(A) = M_D(\mathbb{C})$ . Iterating this one-step implication gives the result for all  $m \geq L$ . □

**Theorem 8.83** (Cumulative span equals word span under aperiodicity). Lean ✓ Stmt If  $\mathbb{1} \in S_1(A)$ , then  $T_n(A) = S_n(A)$  for all  $n$ .

*Proof.* ✓ Proof By Theorem 8.81, the word spans are monotone:  $S_0 \leq S_1 \leq \dots$ . The supremum  $T_n = S_0 + \dots + S_n$  therefore equals the largest term  $S_n$ . □

**Theorem 8.84** (Cumulative spanning plus aperiodicity implies normality). Lean ✓ Stmt If  $T_N(A) = M_D(\mathbb{C})$  and  $\mathbb{1} \in S_1(A)$ , then  $A$  is normal.

*Proof.* ✓ Proof By Theorem 8.83,  $S_N(A) = T_N(A) = M_D(\mathbb{C})$ . □

**Theorem 8.85** (Algebra span plus aperiodicity implies normality). Lean ✓ Stmt If  $\text{alg}(A) = M_D(\mathbb{C})$  and  $\mathbb{1} \in S_1(A)$ , then  $A$  is normal.

*Proof.* ✓ Proof Since  $\text{alg}(A) = M_D(\mathbb{C})$ , there exists  $N$  with  $T_N(A) = M_D(\mathbb{C})$  (by the ascending chain condition on finite-dimensional subspaces). Apply Theorem 8.84. □

**Theorem 8.86** (Irreducible tensor plus aperiodicity implies normality). [Lean](#) [✓ Stmt](#) *If  $A$  is an irreducible tensor,  $D > 0$ , and  $\mathbb{1} \in S_1(A)$ , then  $A$  is normal. This assembles the full chain:*

$$\text{irreducible tensor} \xrightarrow{8.72} \text{irreducible action} \xrightarrow{\text{Burnside}} \text{alg}(A) = M_D(\mathbb{C}) \xrightarrow{8.85} \text{normal}.$$

*This corresponds to the irreducibility-to-normality direction of [SPGWC10, Proposition 3]. The Burnside step is supplied by Theorems 8.72 and 8.73.*

*Proof.* [✓ Proof](#) By Theorem 8.72,  $A$  acts irreducibly on  $\mathbb{C}^D$ . By Theorem 8.73,  $\text{alg}(A) = M_D(\mathbb{C})$ . By Theorem 8.85 with the aperiodicity hypothesis,  $A$  is normal.  $\square$

*Remark 8.87* (Normal canonical form avoids the aperiodicity step). The aperiodicity results in Section 8.11 are needed only for the block-injective argument through Definition 2.31, where one upgrades cumulative spanning to eventual block injectivity. The normal-canonical theory used later in Chapters 9–10 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, places them in normal canonical form in the sense of [CPGSV16]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are auxiliary here and we do not list them separately.

## 8.12 Block injectivity from TP/primitive/irreducible blocks (relocated from Chapter 9)

**Theorem 8.88** (TP-primitive irreducible tensor is injective after blocking). [Lean](#) [✓ Stmt](#) *Let  $A$  be a left-canonical MPS tensor with  $D > 0$ , irreducible tensor, and primitive transfer map. Then some positive blocking  $A^{[L]}$  is one-site injective.*

*Proof.* [✓ Proof](#) If  $D = 1$ , trace preservation gives a nonzero one-site Kraus matrix, and every nonzero scalar matrix spans  $M_1(\mathbb{C})$ ; hence  $A$  is one-site injective, so Lemma 2.30 gives the positive witness  $L = 1$ . Suppose now that  $D \geq 2$ . The normality theorem (Theorem 9.46) gives a block-injectivity witness  $L$ . If  $L = 0$ , Corollary 8.9 would force  $D = 1$ , a contradiction. Thus  $L > 0$ , and the blocked-chain equivalence identifies this  $L$ -block injectivity with one-site injectivity of the blocked tensor  $A^{[L]}$ .  $\square$

*Remark 8.89.* See Theorem 8.53 for the formalized statement that blocking preserves normality.

## Chapter 9

# Canonical Form Reduction

This is the first of three chapters that together prove the Fundamental Theorem of matrix product states. Here an arbitrary translation-invariant MPS tensor is reduced to a weighted family of primitive blocks; Chapter 10 constructs the basis of normal tensors carried by those blocks and compares their coefficients, and Chapter 11 proves the theorem itself. The reduction follows [PGVWC07], [CPGSV16], [CPGSV21], and [Wol12, Chapter 6]. Its goal is to expose the primitive blocks inside the tensor. The transfer operator of an arbitrary tensor can be reducible and can have a period greater than one; the reduction splits it along its invariant subspaces and brings the period of each block down to one, so that each block is primitive, with peripheral spectrum  $\{1\}$ . The blocks are returned in a fixed gauge, and the sections below carry out the reduction. The periodic decomposition theorem of [PGVWC07, Theorem 5] also gives an explicit decomposition of the state vector into  $p$ -periodic summands; here we use only its blocked cyclic-sector consequence (Theorem 9.43), and that explicit state-vector decomposition lies outside this chapter.

Two normalizations appear. The unital orientation  $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$  is used in [PGVWC07, Theorem 4]; the trace-preserving (left-canonical) orientation  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$  is used in the TP-gauge and after-blocking reductions below. They exchange under the conjugate-transposed Kraus family  $K_i := (A^i)^\dagger$ , whose transfer map is the Frobenius adjoint of  $\mathcal{E}_A$ .

**Theorem 9.1** (Translation-invariant canonical form). Lean Stmt *Let  $A$  be a translation-invariant MPS representation of bond dimension  $D$ . There is a positive scalar  $s$  and a finite family of blocks  $A_k$ , possibly empty, such that the globally rescaled tensor  $s^{-1}A$  has the same MPV coefficients on every positive-length ring as*

$$\bigoplus_k \mu_k A_k^i.$$

*The weights are positive and satisfy  $1 \geq |\mu_k|$ ; if the family is nonempty, then some weight has norm 1. For each block,*

$$\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}, \quad \sum_i (A_k^i)^\dagger \Lambda_k A_k^i = \Lambda_k,$$

*where  $\Lambda_k$  is diagonal, positive, and full-rank. Each block has positive bond dimension. Moreover the only fixed points of  $\mathcal{E}_{A_k}(X) = \sum_i A_k^i X (A_k^i)^\dagger$  are the scalar multiples of  $\mathbb{1}$ : for every  $X$ ,*

$$\mathcal{E}_{A_k}(X) = X \implies \exists c \in \mathbb{C}, X = c \mathbb{1}.$$



The total bond dimension of the decomposition is at most  $D$ .

This is the positive-length form of [PGVWC07, Theorem 4], with the global spectral-radius normalization step made explicit.

*Remark 9.2* (Source proof order). The proof follows [PGVWC07, Theorem 4] in order: normalize the spectral radius and use a full-rank positive fixed point to obtain the unital gauge; derive the invariant support projection from a singular positive fixed point by the positivity contradiction; split the finite-ring trace into the supported and orthogonal summands; iterate, using a non-scalar fixed point to split further until each block has only scalar fixed points; and finally repeat the fixed-point argument for the dual map and diagonalize the resulting full-rank positive fixed point by a unitary gauge. The primitive-block results below are used only after their hypotheses have been obtained from this argument.

*Remark 9.3* (CPSV blocked canonical form (source statement)). The blocked canonical-form reduction of [CPGSV16, Section 2.3] and [CPGSV21, Section IV.A] states that an arbitrary translation-invariant MPS tensor becomes, after blocking by some period  $p \geq 1$ , the direct sum of an all-zero summand and a weighted family of normal tensors  $\bigoplus_k \mu_k B_k$  with  $|\mu_k| \leq 1$  and at least one  $|\mu_k| = 1$ . The closest proved results here are Theorem 9.47 and Theorem 10.40; the source upper normalization  $|\mu_k| \leq 1$  with a unit-modulus witness is the one remaining input, recorded in Remark 10.41.

## 9.1 Normal tensors and canonical forms

This section records the CPSV definitions of *normal tensor* and *canonical form* from [CPGSV16]. The basis of normal tensors is stated in Chapter 10. Write

$$\sigma_{\partial}(\mathcal{E}) = \{\lambda \in \sigma(\mathcal{E}) : |\lambda| = r(\mathcal{E})\}$$

for the peripheral spectrum of a transfer map  $\mathcal{E}$ , where  $r(\mathcal{E})$  is its spectral radius. The stronger predicates used later (Definition 9.25 and the canonical-form predicate of Chapter 10) add left-canonical normalization and spectral primitivity. The weight moduli are taken non-increasing, not strictly decreasing; equal-modulus and repeated-copy sectors are retained and compared by the coefficient argument of Chapter 10.

**Definition 9.4** (CPSV normal tensor (NT)). Lean ✓ Stmt A tensor  $A$  of physical dimension  $d$  and bond dimension  $D$  is a *normal tensor* if, after the spectral-radius normalization of [CPGSV16], (i)  $A$  admits no nontrivial invariant orthogonal projection and (ii) the associated CPM  $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$  has peripheral spectrum  $\sigma_{\partial}(\mathcal{E}_A) = \{1\}$ .

*Remark 9.5.* Condition (i) cannot be dropped. A reducible tensor may have peripheral spectrum  $\{1\}$  while still admitting a nontrivial invariant projection, so condition (ii) alone does not characterize normality.

The CPSV *canonical form* (CF) of a tensor  $A$  is the normal-block decomposition

$$A^i \cong \bigoplus_{k=1}^r \mu_k A_k^i$$

with weights normalized by  $|\mu_k| \leq 1$  and with at least one  $|\mu_k| = 1$ ; see [CPGSV16, Section 2.3]. In the statements below the structural part is supplied by the stronger normal canonical form of Definition 9.25, and the global weight normalization is invoked as a separate source hypothesis where needed.

*Remark 9.6* (Relation to the strong predicates). A tensor  $A$  which is both irreducible (Definition 9.16) and has primitive transfer map is normal in the sense of Definition 9.4. The reverse implications from the stronger predicates of Definition 9.25 and of Chapter 10 require the fundamental-theorem step or the algebraic-to-spectral normality equivalence, established later.

### 9.1.1 Normalization conventions

*Remark 9.7* (Scalar normalization conventions). The reductions use four scalar facts: scaling preserves injectivity; the transfer map under a scalar  $\zeta$  rescales by  $|\zeta|^2$ ; unit-modulus phase rescaling preserves  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ ; and the MPV of a weighted block-diagonal tensor factorizes as  $\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma$  with  $\eta_k := \mu_k / |\mu_k|$ .

Left-canonical normalization is required before the blocks can be compared. Without it the implication  $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k) \Rightarrow \forall k, \mathcal{V}(A_k) = \mathcal{V}(B_k)$  already fails for two scalar blocks: for  $\mu = (1, 2)$ ,  $A = (1, 3/2)$ ,  $B = (3, 1/2)$  the two direct sums generate identical MPV families but the first blocks differ at  $N = 1$ . Left-canonical normalization removes this particular ambiguity; the sectors are then matched, up to a permutation, by the exact fixed-length linear independence of Chapter 10, which requires no ordering of the weight moduli.

## 9.2 Invariant-subspace decomposition and irreducible blocks

The block decomposition rests on one principle: a positive semidefinite fixed point of  $\mathcal{E}_A$  that is not full rank has a proper invariant support, which block-diagonalizes the Kraus operators without changing the MPV family. Iterating the resulting splitting decomposes any tensor into irreducible blocks.

**Theorem 9.8** (Unitary conjugation preserves the MPV family). [Lean](#) [✓ Stmt](#) *If  $U$  is a unitary matrix, then  $A$  and  $U^\dagger A U$  generate the same MPV family.*

*Proof.* [✓ Proof](#) By cyclicity of the trace,  $\text{tr}((U^\dagger A^{i_1} U) \cdots (U^\dagger A^{i_N} U)) = \text{tr}(A^{i_1} \cdots A^{i_N})$ .  $\square$

**Theorem 9.9** (Invariant projection gives a two-block decomposition). [Lean](#) [✓ Stmt](#) *Suppose  $P$  is an orthogonal projection satisfying, for every physical index  $i$ ,*

$$(\mathbb{1} - P)A^i P = 0.$$

*Then there exist  $n, m$  with  $n + m = D$  and MPS tensors  $A_1, A_2$  of bond dimensions  $n$  and  $m$  such that  $A$  and the two-block tensor  $A_1 \oplus A_2$  generate the same MPV family in the sense of Definition 2.12.*

*Proof.* [✓ Proof](#) Diagonalize  $P$  by a unitary  $U$ . Since  $P^2 = P$ , the diagonal form of  $P$  has only 0- and 1-eigenvalues, so the basis splits as  $n + m = D$ . Conjugating  $A$  by  $U$  preserves the MPV family by Theorem 9.8. In this basis the relation  $(\mathbb{1} - P)A^i P = 0$  makes each letter upper block triangular. Write

$$B^i = \begin{pmatrix} A_1^i & * \\ 0 & A_2^i \end{pmatrix}$$

for the conjugated letters in this split basis. Since  $B^{i_1} \cdots B^{i_N}$  is again upper block triangular, its trace satisfies

$$\text{tr}(B^{i_1} \cdots B^{i_N}) = \text{tr}(A_1^{i_1} \cdots A_1^{i_N}) + \text{tr}(A_2^{i_1} \cdots A_2^{i_N}).$$

Hence the block-diagonal tensor  $A_1 \oplus A_2$  has the same MPV coefficients as the conjugated tensor.  $\square$

**Theorem 9.10** (Nontrivial invariant projection gives strict two-block reduction). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 9.9, assume in addition that  $P \neq 0$  and  $P \neq \mathbb{1}$ . Then there exist  $n, m$  with  $n + m = D$ ,  $n < D$ , and  $m < D$ , together with block tensors  $A_1, A_2$ , such that  $A$  and  $A_1 \oplus A_2$  generate the same MPV family in the sense of Definition 2.12.

*Proof.* [✓ Proof](#) Apply Theorem 9.9. In the diagonal form of  $P$ , the 1-eigenspace is nonempty because  $P \neq 0$ , and the 0-eigenspace is nonempty because  $P \neq \mathbb{1}$ . Hence both block sizes are positive. Since  $n + m = D$ , this implies  $n < D$  and  $m < D$ .  $\square$

**Definition 9.11** (Support projection). [Lean](#) [✓ Stmt](#) For a positive semidefinite matrix  $\rho \geq 0$ , the *support projection*  $P$  is the orthogonal projection onto the range of  $\rho$ . Via the spectral decomposition  $\rho = U \text{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$ ,

$$P = U \text{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger,$$

where  $\mathbf{1}_{\lambda_j > 0}$  is 1 if  $\lambda_j > 0$  and 0 otherwise.

**Definition 9.12** (Invariant projection predicate). [Lean](#) [✓ Stmt](#) An MPS tensor  $A$  has an *invariant projection* if there exists an orthogonal projection  $P$  with  $P \neq 0$ ,  $P \neq \mathbb{1}$ , and  $(\mathbb{1} - P)A^i P = 0$  for all  $i$ . A tensor is *irreducible* if it has no nontrivial invariant projection (Definition 9.16).

**Theorem 9.13** (PSD fixed point gives invariant projection). [Lean](#) [✓ Stmt](#) Let  $\rho \geq 0$  be a fixed point of  $\mathcal{E}_A$ , and let  $P$  be its support projection. Then  $(\mathbb{1} - P)A^i P = 0$  for all  $i$ : the range of  $P$  is invariant under every  $A^i$ .

*Proof.* [✓ Proof](#) From  $\mathcal{E}_A(\rho) = \rho$  and positivity of each summand  $A^i \rho (A^i)^\dagger$ , one gets  $(\mathbb{1} - P)A^i \rho (A^i)^\dagger (\mathbb{1} - P) = 0$  for each  $i$ . Since  $\rho$  is supported on the range of  $P$ , this forces  $(\mathbb{1} - P)A^i P = 0$ .

This is the finite-dimensional case of [Wol12, Proposition 6.10]: the support projection of any stationary state is sub-harmonic. The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Theorem 6.2]; see also [CPGSV21, Section IV.A.1].  $\square$

**Theorem 9.14** (Two-block decomposition). [Lean](#) [✓ Stmt](#) If  $\mathcal{E}_A$  has a nonzero PSD fixed point  $\rho$  that is not positive definite, then there exist MPS tensors  $A_1, A_2$  of smaller bond dimensions with  $\mathcal{V}(A) = \mathcal{V}(A_1 \oplus A_2)$ .

*Proof.* [✓ Proof](#) The support projection  $P$  of  $\rho$  is neither 0 nor  $\mathbb{1}$  (since  $\rho \neq 0$  and  $\rho$  is not positive definite). Theorem 9.13 gives  $(\mathbb{1} - P)A^i P = 0$ , so a unitary change of basis block-diagonalizing  $P$  puts  $A^i$  in upper-triangular block form with diagonal blocks  $B_{11}^i, B_{22}^i$ . Theorem 9.8 preserves the MPV under the conjugation, and the trace of an upper-triangular product equals the sum of the diagonal-block traces, so deleting the off-diagonal blocks preserves all MPV coefficients.  $\square$

**Theorem 9.15** (Non-scalar fixed point gives a further split). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose  $A$  is unital,  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ . If  $X$  is a Hermitian fixed point of  $\mathcal{E}_A$  which is not a scalar multiple of  $\mathbb{1}$ , then there is a nonzero PSD fixed point  $\rho$  which is not positive definite, and consequently  $A$  admits a two-block MPV-equivalent decomposition with both block dimensions strictly smaller than  $D$ . This is the fixed-point shift and further decomposition step in [PGVWC07, Theorem 4].

*Proof.* [✓ Proof](#) Let  $\lambda_{\max}$  be the largest eigenvalue of  $X$ . Then  $\rho := \lambda_{\max} \mathbb{1} - X$  is positive semidefinite, singular, and nonzero (since  $X$  is not scalar), and unitality with the fixed-point equation for  $X$  gives  $\mathcal{E}_A(\rho) = \rho$ . Apply Theorem 9.14. The non-scalar hypothesis is needed because scalar multiples of  $\mathbb{1}$  are fixed by every unital map and yield no nontrivial support projection.  $\square$

### 9.2.1 Iterated reduction to irreducible blocks

Iterating the two-block splitting on bond dimension decomposes any tensor into irreducible blocks; the remaining lemmas record the spectral facts about irreducible unital blocks and the algebra-spanning input used later.

**Definition 9.16** (Irreducible tensor). [Lean](#) [✓ Stmt](#) An MPS tensor  $A$  is *irreducible* if it has no nontrivial invariant projection.

**Theorem 9.17** (Irreducible block decomposition). [Lean](#) [✓ Stmt](#) Every MPS tensor  $A$  admits a block-diagonal tensor  $\bigoplus_{k=1}^r A_k$  with each  $A_k$  irreducible and  $\mathcal{V}(A) = \mathcal{V}(\bigoplus_k A_k)$ .

*Proof.* [✓ Proof](#) By strong induction on  $D$ . If  $A$  is irreducible, take  $r = 1$ . Otherwise a nontrivial invariant projection  $P$  gives the upper-triangular splitting of Theorem 9.14; deleting the off-diagonal part preserves the MPV family and yields two blocks of strictly smaller bond dimension. Apply the induction hypothesis to each block.

Both [PGVWC07, Theorem 4] and [CPGSV21, Section IV.A.1] obtain the block decomposition by the same invariant-projection splitting.  $\square$

**Theorem 9.18** (Scalar fixed points in an irreducible unital block). [Lean](#) [✓ Stmt](#) If  $A$  is irreducible and unital,  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ , and  $\mathcal{E}_A(X) = X$ , then  $X$  is a scalar multiple of  $\mathbb{1}$ . This is the final fixed-point assertion in the proof of [PGVWC07, Theorem 4].

*Proof.* [✓ Proof](#) Tensor irreducibility gives irreducibility of the completely positive transfer map. By Wolf’s Theorem 6.6 for irreducible unital Kraus maps, every fixed point is scalar.  $\square$

**Theorem 9.19** (Unital gauge from a full-rank Perron–Frobenius eigenvector). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let  $\rho$  be positive definite with  $\mathcal{E}_A(\rho) = r\rho$  for some  $r > 0$ . For  $B^i := r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$ ,

$$\sum_i B^i (B^i)^\dagger = \mathbb{1}. \quad (9.1)$$

If  $r = 1$ , then the unscaled gauge  $B^i := \rho^{-1/2}A^i\rho^{1/2}$  is unital and generates the same MPV family as  $A$ . This is the spectral-radius normalization and full-rank fixed-point gauge of [PGVWC07, Theorem 4].

*Proof.* [✓ Proof](#) The positive square root of  $\rho$  is invertible. Substituting  $B^i = r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$  into  $\sum_i B^i (B^i)^\dagger$  and using  $\mathcal{E}_A(\rho) = r\rho$  gives (9.1). For  $r = 1$  the change from  $A$  to  $B$  is a similarity, so cyclicity of the trace gives equality of all MPV coefficients.  $\square$

**Theorem 9.20** (Unitary diagonal positive fixed point in TP gauge). [Lean](#) [✓ Stmt](#) Let  $A$  be an irreducible MPS tensor with  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  and  $D > 0$ . Then there exist a unitary  $U$  and a diagonal positive definite  $\Lambda$  such that, for  $B^i := U^\dagger A^i U$ ,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1}, \quad \mathcal{E}_B(\Lambda) = \Lambda.$$

*Proof.* [✓ Proof](#) The TP hypothesis makes  $\mathcal{E}_A$  a channel with a nonzero PSD fixed point. Irreducibility makes  $\mathcal{E}_A$  an irreducible CP map, and Theorem 6.4 upgrades the fixed point to positive definite. The spectral theorem diagonalizes it by a unitary conjugation, which preserves the TP normalization.  $\square$

The passage from irreducible tensors to full algebra spanning uses Burnside’s theorem: an irreducible subalgebra of  $M_D(\mathbb{C})$  is all of  $M_D(\mathbb{C})$ . It is developed alongside the Wielandt material in Chapter 8; see Theorem 8.75.

**Lemma 9.21** (The algebra span reaches a finite cumulative span). [Lean](#) [✓ Stmt](#) *If the generated unital algebra satisfies  $\text{alg}(A) = M_D(\mathbb{C})$ , then there exists  $N$  such that  $T_N(A) = M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) Every element of  $\text{alg}(A)$  lies in some cumulative span  $T_n(A)$ : the generators lie in  $T_1(A)$ , scalar multiples of the identity lie in  $T_0(A)$ , and the family  $\{T_n(A)\}_{n \geq 0}$  is closed under addition and under multiplication with lengths adding. Since  $M_D(\mathbb{C})$  is finite-dimensional, the ascending chain  $T_0(A) \subseteq T_1(A) \subseteq \dots$  stabilizes, so one can choose a single  $N$  with  $T_N(A) = M_D(\mathbb{C})$ .  $\square$

**Lemma 9.22** (Irreducible action gives an irreducible tensor). [Lean](#) [✓ Stmt](#) *If the letters of  $A$  act irreducibly on  $\mathbb{C}^D$ , then  $A$  has no nontrivial invariant orthogonal projection.*

*Proof.* [✓ Proof](#) If  $P$  were a nontrivial invariant orthogonal projection, then its range would be invariant under every letter of  $A$ . Irreducibility of the action forces this range to be either 0 or all of  $\mathbb{C}^D$ , so  $P$  must be 0 or  $\mathbb{1}$ , a contradiction.  $\square$

### 9.3 Normal canonical form

This section fixes the two canonical-form predicates used downstream: the block-injective form of Definition 9.23 and the spectral *normal* canonical form of Definition 9.25. It then records how the self-overlap clause and gauge-phase equivalence follow from primitivity, and closes with the periodicity-removal step that produces primitive transfer maps.

Recall from Definition 2.48 and Lemma 7.7 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \text{Tr}(F_{AB}^N).$$

This notation enters the self-overlap clause of the canonical-form predicate below.

**Definition 9.23** (Canonical form conditions). [Lean](#) [✓ Stmt](#) A collection of scaling factors  $(\mu_k)_{k=1}^r$  and block tensors  $(A_k)_{k=1}^r$  satisfies the *canonical form conditions* if:

1. each  $A_k$  is injective,
2. each  $A_k$  satisfies the TP normalization  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ ,
3. the moduli are non-increasing:  $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$ ,
4. all  $\mu_k \neq 0$ ,
5. the self-overlap converges:  $O_{A_k A_k}(N) \rightarrow 1$  for each  $k$ .

Only this one-sided normalization is assumed; no unital condition is assumed here. Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Theorem 6.7(3)]),  $T^n(\rho) \rightarrow \rho_\infty$  for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Theorem 6.8] for the completely-positive characterisation and Chapter 7 for the transfer-operator gap proof.

*Remark 9.24* (Normalization convention). [PGVWC07, Theorem 4] uses the unital gauge  $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$ ; here the blocks are in the dual TP gauge  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ . A positive-definite fixed point of the adjoint transfer map provides the change of gauge between the two (Theorem 6.14).

**Definition 9.25** (Normal canonical form conditions). [Lean](#) [✓ Stmt](#) Scaling factors  $(\mu_k)_{k=1}^r$  and block tensors  $(A_k)_{k=1}^r$  satisfy the *normal canonical form conditions* if:

1. each  $A_k$  is irreducible,
2. each  $A_k$  satisfies  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ ,
3. each  $\mathcal{E}_{A_k}$  is primitive (equivalently,  $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$ ),
4.  $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$ ,
5. all  $\mu_k \neq 0$ ,
6. all bond dimensions are positive.

Here “normal” is the spectral sense of [CPGSV16], which by [SPGWC10, Proposition 3] is equivalent to the eventual block-injectivity formulation of Definition 2.31.

**Theorem 9.26** (Self-overlap is derived in normal canonical form). Lean ✓ Stmt *If  $(\mu_k, A_k)$  satisfies the normal canonical form conditions, then  $O_{A_k A_k}(N) \rightarrow 1$  for every block  $k$ .*

*Proof.* ✓ Proof Fix a block  $k$ . Irreducibility and the TP normalization give a nonzero positive semidefinite fixed point  $\rho$  of  $\mathcal{E}_{A_k}$  with fixed-point projector  $P$ ; primitivity gives the complementary gap  $\rho_{\text{spec}}(\mathcal{E}_{A_k} - P) < 1$  (Theorem 4.51). Theorem 7.99 then yields  $O_{A_k A_k}(N) \rightarrow 1$ .  $\square$

*Remark 9.27.* See Theorem 7.29 for the modulus-one eigenvalue rigidity statement for irreducible trace-preserving blocks.

**Lemma 9.28** (Mixed-transfer spectral radius from unit-modulus overlap). Lean ✓ Stmt *Let  $A$  and  $B$  have the same bond dimension. If  $\|\langle V^{(N)}(A) | V^{(N)}(B) \rangle\| \rightarrow 1$ , then the mixed transfer map has spectral radius at least 1.*

*Proof.* ✓ Proof If the mixed transfer spectral radius were  $< 1$ , the iterated mixed transfer map would tend to zero in operator norm, hence so would its trace; via  $\text{Tr}(F_{AB}^N) = \langle V^{(N)}(A) | V^{(N)}(B) \rangle$  the overlap would tend to 0, contradicting the hypothesis.  $\square$

**Theorem 9.29** (Gauge-phase equivalence from unit-modulus overlap). Lean ✓ Stmt *Let  $A$  and  $B$  be irreducible MPS tensors of the same bond dimension with  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  and  $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ . If  $\|\langle V^{(N)}(A) | V^{(N)}(B) \rangle\| \rightarrow 1$ , then  $A$  and  $B$  are gauge-phase equivalent.*

*Proof.* ✓ Proof By Lemma 9.28 the mixed transfer spectral radius is at least 1, and Theorem 7.29 then gives gauge-phase equivalence.  $\square$

### 9.3.1 Canonical form from primitivity

A normal block needs a primitive transfer map; an irreducible TP block may still have peripheral spectrum larger than  $\{1\}$ , namely the roots of unity coming from its period. Blocking by the least common multiple of those periods raises the transfer map to a power whose peripheral spectrum is  $\{1\}$ , removing the periodicity. In [CPGSV21] this is the passage from irreducibility to primitivity by blocking to period one.

**Theorem 9.30** (Blocking yields a peripheral-spectrum primitive transfer map). Lean ✓ Stmt *Let  $A$  be an irreducible MPS tensor with  $D > 0$  and  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ . Then there exists a blocking length  $p > 0$  such that  $\sigma_\partial(\mathcal{E}_{A^{[p]}}) = \{1\}$ .*

*Proof.* ✓ Proof Pass to the conjugate-transposed Kraus family  $K_i := (A^i)^\dagger$ , which is unital with irreducible Kraus map. The TP fixed-point theorem for  $\mathcal{E}_A$  provides a positive definite fixed point for the adjoint of  $K$ , so Theorem 4.46 shows every peripheral eigenvalue is a root of unity. Since  $\sigma_\partial(\mathcal{E}_A)$  is finite (Theorem 4.29), Lemma 4.33 gives a common exponent  $p$  with  $\mu^p = 1$  for all  $\mu \in \sigma_\partial(\mathcal{E}_A)$ , and Lemma 4.34 gives  $\sigma_\partial(\mathcal{E}_A^p) = \{1\}$ . Blocking by  $p$  takes the  $p$ th power of the transfer map, so  $\mathcal{E}_{A[p]}$  is primitive.  $\square$

*Remark 9.31.* Theorem 9.30 is the periodicity-removal-by-blocking step of [CPGSV16, Appendix A] for an irreducible block already in TP gauge.

## 9.4 Left-canonical and dual diagonalization of irreducible blocks

Each irreducible block in trace-preserving gauge admits the PGVWC07 unital orientation together with a diagonal positive-definite dual fixed point (Theorem 9.32, the Perron–Frobenius gauge of Chapter 6, and Theorem 9.18); the full PGVWC07 canonical form is Theorem 9.1.

**Theorem 9.32** (Left-canonical normalization for irreducible TP blocks). Lean ✓ Stmt *Let  $A$  be an irreducible MPS tensor in TP gauge ( $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ ) with  $D > 0$ . Then there exist a unitary  $U$  and a diagonal positive definite  $\Lambda$  such that:*

1.  $B^i := U^\dagger A^i U$  is TP,
2.  $\mathcal{E}_B(\Lambda) = \Lambda$ ,
3.  $\Lambda$  is diagonal and positive definite,
4.  $A$  and  $B$  generate the same MPV family.

*This is the left-canonical normalization step (Canonical Form II) of [CPGSV16, Appendix A]. The unitary conjugation is a second step inside TP gauge, performed only after TP normalization.*

*Proof.* ✓ Proof Theorem 9.20 provides the unitary and the diagonal PD fixed point, and unitary conjugation preserves trace-preservation and the MPV family (Theorem 9.8).  $\square$

**Theorem 9.33** (Dual diagonalization for irreducible unital blocks). Lean ✓ Stmt *Let  $A$  be an irreducible MPS tensor in unital gauge  $\sum_i A^i (A^i)^\dagger = \mathbb{1}$  with  $D > 0$ . Then there exist a unitary  $U$  and a diagonal positive definite  $\Lambda$  such that, for  $B^i := U^\dagger A^i U$ ,*

$$\sum_i B^i (B^i)^\dagger = \mathbb{1}, \quad \sum_i (B^i)^\dagger \Lambda B^i = \Lambda, \quad (9.2)$$

*and  $A$ ,  $B$  generate the same MPV family. This is the dual-map fixed-point and unitary-diagonalization step of [PGVWC07, Theorem 4].*

*Proof.* ✓ Proof Apply Theorem 9.32 to the conjugate-transposed family  $i \mapsto (A^i)^\dagger$ : unitality of  $A$  is trace preservation for this family, whose transfer map is the dual of  $\mathcal{E}_A$ . Translating back gives the equations (9.2).  $\square$

**Lemma 9.34** (Common scalar rescaling of block weights). Lean ✓ Stmt *Multiplying every block weight in a block-diagonal MPS tensor by a scalar  $c$  multiplies every length- $N$  MPV coefficient by  $c^N$ .*



*Proof.* ✓ Proof Expand the block-diagonal MPV coefficient as a finite sum over blocks and factor  $c^N$  out of the sum.  $\square$

**Theorem 9.35** (Normal canonical form from a primitive block decomposition). Lean ✓ Stmt Suppose  $A$  generates the same MPV family as  $\bigoplus_k \mu_k A_k$ , and that each  $A_k$  is irreducible, satisfies  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ , and has primitive transfer map, all  $\mu_k \neq 0$ , and all bond dimensions are positive. Then there exist a blocking length  $p > 0$ , weights  $(\nu_k)$ , and blocked tensors  $(B_k)$  such that  $A^{[p]}$  generates the same MPV family as  $\bigoplus_k \nu_k B_k$ , and  $(\nu_k, B_k)$  satisfies the normal canonical form conditions. The weight moduli need not be distinct; equal-modulus blocks are allowed.

*Proof.* ✓ Proof The given blocks are already primitive, so take  $p = 1$ . Reorder the blocks by non-increasing weight modulus, with ties allowed, and set  $\nu_k := \mu_k$  and  $B_k := A_k$  in that order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are preserved, so the reordered family satisfies Definition 9.25 and generates the same MPV family as  $A$ .  $\square$

*Remark 9.36* (Scope of Theorem 9.35). Theorem 9.35 assumes all six block hypotheses in advance and produces the normal canonical form by reordering. It is not the arbitrary-input existence theorem of [CPGSV16, Section 2.3]; that primitive-block reduction, including the zero-tail count, is Theorem 9.47 in Section 9.6.2.

## 9.5 Zero-block separation and the trace-preserving gauge

In an irreducible block decomposition some blocks may have all-zero Kraus operators. An all-zero block contributes  $V^{(N)} = 0$  for every  $N \geq 1$ , so it adds nothing to the positive-length MPV family; at  $N = 0$  the MPV coefficient is the bond dimension. The all-zero summand therefore enters the output not as a tensor but as a single count  $D_0$  in the length-zero identity  $D = D_0 + \sum_k D_k$ , and all positive-length content is an equality of MPV families. This is the positive-length convention used throughout the reduction: comparisons of weighted block sums are stated for  $N \geq 1$ , with the zero block carried only by  $D_0$ . After separating the zero summand, each nonzero block is placed in the trace-preserving gauge, the furthest unconditional reduction available before periodicity removal.

**Theorem 9.37** (Zero-block separation). Lean ✓ Stmt Every MPS tensor  $A$  admits a block decomposition into a zero-block bond dimension  $D_0$  and a finite family of nonzero blocks  $(B_k)_{k=1}^r$ , each irreducible, with at least one nonzero Kraus operator, and of positive bond dimension, recorded by two facts. The positive-length equality of MPV families: for every  $N \geq 1$  and every  $\sigma$ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r B_k)_\sigma.$$

The length-zero dimension identity

$$D = D_0 + \sum_{k=1}^r D_k,$$

where  $D_0$  is the total bond dimension of the all-zero blocks.

*Proof.* ✓ Proof Apply the irreducible block decomposition (Theorem 9.17) and partition the blocks by whether each has a nonzero Kraus operator;  $D_0$  accumulates the bond dimensions of the all-zero blocks. For every all-zero irreducible block  $C_j$  and every  $N \geq 1$ ,  $V^{(N)}(C_j)_\sigma = 0$ , so

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r B_k)_\sigma.$$



At  $N = 0$  the MPV coefficient is the bond dimension, giving  $D = D_0 + \sum_{k=1}^r D_k$ .  $\square$

**Theorem 9.38** (Blockwise TP-gauge normalization of irreducible blocks). [Lean](#) [✓ Stmt](#) Suppose  $A$  generates the same MPV family as  $\bigoplus_{k=1}^r B_k$ , where each  $B_k$  is irreducible with at least one nonzero Kraus operator. Then there are nonzero weights  $(\mu_k)_{k=1}^r$  and blocks  $(C_k)_{k=1}^r$  such that, for every  $N$  and every  $\sigma$ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k C_k)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(C_k)_\sigma,$$

each  $C_k$  irreducible and left-canonical,  $\sum_i (C_k^i)^\dagger C_k^i = \mathbb{1}$ , with positive bond dimension. This is the blockwise form used in the arbitrary-tensor trace-preserving gauge below.

*Proof.* [✓ Proof](#) The Perron–Frobenius TP-gauge construction is applied to each irreducible block. Gauge equivalence preserves irreducibility and the MPV family, and the positive scaling factors are absorbed into the weights  $\mu_k$ .  $\square$

### 9.5.1 Trace-preserving gauge for arbitrary tensors

**Theorem 9.39** (TP-gauge reduction for arbitrary tensors with zero-block separation). [Lean](#) [✓ Stmt](#) For any MPS tensor  $A$ , there exist a zero-block bond dimension  $D_0$  and nontrivial blocks  $(B_k)_{k=1}^r$  with nonzero weights  $(\mu_k)_{k=1}^r$ , each block irreducible, left-canonical ( $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$ ), and of positive bond dimension, such that for every  $N \geq 1$  and every  $\sigma$ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k B_k)_\sigma,$$

and the zero-block contribution is recorded by the length-zero identity

$$D = D_0 + \sum_{k=1}^r D_k.$$

*Proof.* [✓ Proof](#) Theorem 9.37 supplies  $D_0$  and the nonzero irreducible blocks  $(C_k)_{k=1}^r$ . For every  $N \geq 1$  and every  $\sigma$ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r C_k)_\sigma.$$

The blockwise TP-gauge theorem then replaces each  $C_k$  by a left-canonical block  $B_k$  with weight  $\mu_k$ . Again for every  $N \geq 1$  and every  $\sigma$ ,

$$V^{(N)}(\bigoplus_{k=1}^r C_k)_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

The length-zero identity  $D = D_0 + \sum_{k=1}^r D_k$  is preserved.  $\square$

## 9.6 Blocking, period removal, and primitive blocks

Blocking by an integer  $p$  carries MPV equalities, weights, transfer maps, and identifications of the blocked physical alphabet. The first consequence is the existence of a common blocking period under which every transfer map in a finite family becomes primitive; the period-removal and primitive-block reductions then proceed from that common period.

**Theorem 9.40** (Common blocking period for a block family). [Lean](#) [✓ Stmt](#) Let  $(B_k)_{k=1}^r$  be a finite family of positive-bond-dimension MPS tensors, each admitting a blocking period  $p_k$  making its transfer map primitive. Then  $p = \text{lcm}(p_1, \dots, p_r)$  is a common period:  $\mathcal{E}_{B_k^{[p]}}$  is primitive for every  $k$ .

*Proof.* ✓ Proof Each  $p_k$  divides  $p$ , so primitivity of  $\mathcal{E}_{B_k^{[p_k]}}$  carries over to  $\mathcal{E}_{B_k^{[p]}}$  via the divisibility identity for transfer-map powers.  $\square$

The following stability statements are used when a primitive irreducible block is blocked again for a common-refinement or injectivity length. This later blocking is distinct from the period-removal length that produced the sector block.

**Theorem 9.41** (Blocking preserves irreducibility for primitive irreducible blocks). Lean ✓ Stmt *Let  $A$  be left-canonical, irreducible, primitive, of positive bond dimension. For every  $P > 0$ , the blocked tensor  $A^{[P]}$  is irreducible: there is no orthogonal projection  $Q$  with  $Q \neq 0$ ,  $Q \neq \mathbb{1}$  such that, for every  $P$ -blocked index  $\alpha$ ,*

$$(\mathbb{1} - Q)(A^{[P]})^\alpha Q = 0.$$

*Proof.* ✓ Proof The blocked transfer map is the corresponding positive power of the original one:

$$\mathcal{E}_{A^{[P]}}(X) = \mathcal{E}_A^P(X).$$

Hence a positive-definite fixed point of  $\mathcal{E}_A$  remains fixed by  $\mathcal{E}_{A^{[P]}}$ . Primitivity gives uniqueness of positive semidefinite fixed points for the blocked map, hence its transfer map is irreducible, which is the irreducibility criterion for  $A^{[P]}$ .  $\square$

**Theorem 9.42** (Extra blocking of primitive irreducible sector blocks). Lean ✓ Stmt *Let  $A$  be trace-preserving, primitive, and tensor-irreducible. For every  $k > 0$ ,*

$$\sum_{\alpha} ((A^{[k]})^\alpha)^\dagger (A^{[k]})^\alpha = \mathbb{1}, \quad \sigma_{\partial}(\mathcal{E}_{A^{[k]}}) = \{1\},$$

*and  $A^{[k]}$  is tensor-irreducible. This  $k$  is a later common-refinement or injectivity length, distinct from the period-removal length that produced the sector block.*

*Proof.* ✓ Proof Trace preservation follows from blocking a trace-preserving tensor, primitivity from a positive power of a primitive transfer map, and irreducibility from Theorem 9.41.  $\square$

### 9.6.1 Period removal by cyclic sectors

The step needed below is the period-removing blocking described in [CPGSV16, Section 2.3]: after blocking each irreducible TP block by the least common multiple of its peripheral periods, the result has no nontrivial  $p$ -periodic vectors. The cyclic-sector decomposition comes from the peripheral spectrum of the adjoint transfer map [Wol12, Theorem 6.6].

**Theorem 9.43** (Period removal by blocking). Lean ✓ Stmt *Let  $A$  be a left-canonical irreducible tensor. Then there is a period-removing length  $m \geq 1$  such that  $A^{[m]}$  admits:*

1. left-canonical sector tensors  $(C_k)_{k=1}^m$  of nonzero bond dimension,
2. a unit-weight direct-sum MPV decomposition  $A^{[m]} \sim \bigoplus_{k=1}^m C_k$ ,
3. orthogonal projections  $(P_k)_{k=1}^m$  summing to  $\mathbb{1}$  with  $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$ ,
4. commutation relations  $P_k A_i^{[m]} = A_i^{[m]} P_k$  for every blocked letter  $i$ ,
5. the sector trace formula  $f_{C_k}(\sigma) = \text{tr}(P_k A_{\sigma_1}^{[m]} \cdots A_{\sigma_N}^{[m]})$ ,

6. linear isomorphisms onto the compression corners  $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$ ,
7. transfer-intertwining identities  $\varphi_k(\mathcal{E}_{C_k^\dagger}(X)) = \mathcal{E}_{(P_k A^{[m]})^\dagger}(\varphi_k(X))$ ,
8. multiplicativity and adjoint compatibility  $\varphi_k(XY) = \varphi_k(X)\varphi_k(Y)$  and  $\varphi_k(X^\dagger) = \varphi_k(X)^\dagger$ .

This is the cyclic-sector part of the period-removal step of [CPGSV16, Section 2.3]; the projections  $P_k$  come from the peripheral spectrum of the adjoint transfer map ([Wol12, Theorem 6.6]).

*Proof.* ✓ Proof The cyclic decomposition of the adjoint transfer map gives projections  $(P_k)$  with  $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$ . Iterating the cyclic relation  $m$  times gives  $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$ , and the blocked-transfer identity  $\mathcal{E}_{(A^{[m]})^\dagger} = (\mathcal{E}_{A^\dagger})^m$  identifies these with the adjoint transfer map of  $A^{[m]}$ . Apply the block decomposition from adjoint-fixed projections.  $\square$

**Theorem 9.44** (Unconditional primitive blocked sectors). Lean ✓ Stmt Let  $A$  be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum  $\{\gamma^j : 0 \leq j < m\}$  for a primitive  $m$ th root of unity  $\gamma$ , with blocked cyclic-sector decomposition  $(C_u)_{u=1}^m$ ,  $(P_u)_{u=1}^m$ , and linear isomorphisms  $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$ . Assume explicitly that the  $P_u$  are orthogonal projections summing to  $\mathbb{1}$  and satisfying  $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$ , that each sector has nonzero bond dimension, and that the compression maps satisfy

$$\varphi_u(\mathcal{E}_{C_u^\dagger}(X)) = \mathcal{E}_{(P_u A^{[m]})^\dagger}(\varphi_u(X)), \quad \varphi_u(XY) = \varphi_u(X)\varphi_u(Y), \quad \varphi_u(X^\dagger) = \varphi_u(X)^\dagger.$$

Here  $P_u A^{[m]}$  denotes the blocked tensor with letters  $P_u(A^{[m]})^\alpha$ . Then every sector tensor satisfies  $\sigma_\partial(\mathcal{E}_{C_u}) = \{1\}$  and is tensor-irreducible.

*Proof.* ✓ Proof Theorem 9.62 gives irreducibility of  $(\mathcal{E}_A^\dagger)^m$  on each corner  $P_u$ . The compression equivalences  $\varphi_u$  intertwine the adjoint transfer map of  $C_u$  with the corresponding corner restriction of  $(\mathcal{E}_A^\dagger)^m$ . The peripheral spectrum of the blocked adjoint map is

$$\sigma_\partial((\mathcal{E}_A^\dagger)^m) = \{(\gamma^j)^m : 0 \leq j < m\} = \{1\},$$

because  $\gamma$  is a primitive  $m$ th root of unity. Hence each nonzero irreducible corner restriction is primitive. Primitivity and irreducibility are preserved across  $\varphi_u$ ; then use Theorem 9.54 to pass from the adjoint blocked map to the primal transfer map of  $C_u$ .  $\square$

**Theorem 9.45** (Period removal with primitive irreducible sectors). Lean ✓ Stmt Let  $A$  be a left-canonical irreducible tensor. Then there is a period  $m \geq 1$  and a unit-weight decomposition of  $A^{[m]}$  into sector blocks  $(C_k)_{k=1}^m$ , each of which is left-canonical, has primitive transfer map, is tensor-irreducible, and has nonzero bond dimension. Equivalently, the cyclic-sector decomposition of  $A^{[m]}$  already satisfies the trace-preservation, primitivity, and irreducibility hypotheses needed for the common-blocking step (Lemma 9.61).

*Proof.* ✓ Proof Left-canonicity makes  $K_i := (A^i)^\dagger$  unital, irreducibility passes to the adjoint map, and the quantum Perron–Frobenius theorem provides a positive-definite fixed point; the cyclic peripheral-spectrum theorem and Theorem 9.43 then produce the cyclic-sector decomposition. Keep the projections and compression equivalences, and apply Theorem 9.44 to the sector blocks.  $\square$

### 9.6.2 Reduction to primitive blocks

This subsection completes the reduction after period removal. A TP-primitive-irreducible block is normal, and an arbitrary tensor becomes, after a positive blocking length, a weighted family of left-canonical primitive blocks with the zero-block dimension recorded separately. For two tensors with the same MPV family, the cyclic sectors can be expressed over one common blocked alphabet, which is the form used by the Fundamental Theorem.

**Theorem 9.46** (TP-primitive irreducible tensor is normal). Lean ✓ Stmt *Let  $A$  be a left-canonical MPS tensor with  $D > 0$ , irreducible transfer map, and peripheral spectrum  $\{1\}$ . Then  $A$  is normal.*

*Proof.* ✓ Proof Irreducibility upgrades the PSD fixed point to positive definite (Theorem 6.4), and Theorem 8.78 then gives normality.  $\square$

An arbitrary translation-invariant tensor reduces, after blocking, to a weighted direct sum of primitive irreducible blocks.

**Theorem 9.47** (Arbitrary tensor to primitive block decomposition). Lean ✓ Stmt *For any MPS tensor  $A$ , there exist  $p \geq 1$ , a zero-tail bond dimension  $D_0$ , nonzero weights  $(\mu_k)_{k=1}^r$ , and blocks  $(B_k)_{k=1}^r$  such that, for every blocked  $\sigma$  of positive length  $N$ ,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

*The zero-block contribution is recorded separately by the length-zero identity  $D_0 + \sum_{k=1}^r D_k = D$ . Each block is left-canonical and primitive,*

$$\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}, \quad \sigma_\partial(\mathcal{E}_{B_k}) = \{1\},$$

*and each bond dimension is positive.*

*Proof.* ✓ Proof Theorem 9.39 supplies the zero-block dimension, the left-canonical nontrivial blocks, the positive-length equality, and the identity  $D = D_0 + \sum_k D_k$ . Theorem 9.40 provides a common blocking period  $p$  making all transfer maps primitive, Lemma 9.60 keeps the blocked weights nonzero, and Lemma 9.58 carries the positive-length equality through blocking.  $\square$

**Theorem 9.48** (Cyclic sectors after the zero-block split). Lean ✓ Stmt *If  $A$  and  $B$  generate the same MPV family, then after the zero-block split and the trace-preserving gauge there are weighted nonzero-block tensors  $\bigoplus_j \mu_j^A A_j$  and  $\bigoplus_k \mu_k^B B_k$  with, for every  $N \geq 1$  and  $\sigma$ ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_j \mu_j^A A_j)_\sigma, \quad V^{(N)}(B)_\sigma = V^{(N)}(\bigoplus_k \mu_k^B B_k)_\sigma,$$

*and the two nonzero parts agree:*

$$V^{(N)}(\bigoplus_j \mu_j^A A_j)_\sigma = V^{(N)}(\bigoplus_k \mu_k^B B_k)_\sigma.$$

*Each nonzero block has primitive irreducible cyclic sectors: for each  $j$ , there is  $m_j \geq 1$  with a unit-weight sector decomposition  $A_j^{[m_j]} \sim \bigoplus_{u=1}^{m_j} C_{j,u}^A$ , and similarly for  $B_k$ .*

*Proof.* ✓ Proof Apply Theorem 9.39 separately to  $A$  and  $B$ . Its positive-length conclusion gives, for  $N \geq 1$ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_j \mu_j^A A_j)_\sigma, \quad V^{(N)}(B)_\sigma = V^{(N)}(\bigoplus_k \mu_k^B B_k)_\sigma.$$

Chaining these through  $V(A) = V(B)$  identifies the two nonzero parts. Finally apply Theorem 9.45 to every trace-preserving irreducible nonzero-weight block.  $\square$

**Theorem 9.49** (Common blocking length for cyclic sectors). [Lean](#) [✓ Stmt](#) *If two tensors generate the same MPV family, then the cyclic-sector decompositions on the two sides can be expressed at one common positive blocking length  $p$ . For every  $N \geq 1$  and every blocked  $\sigma$ , each blocked tensor equals its powered nonzero part,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\bigoplus_j (\mu_j^A)^p (A_j)^{[p]})_\sigma, \quad V^{(N)}(B^{[p]})_\sigma = V^{(N)}(\bigoplus_k (\mu_k^B)^p (B_k)^{[p]})_\sigma,$$

and the two powered nonzero parts agree,

$$V^{(N)}(\bigoplus_j (\mu_j^A)^p (A_j)^{[p]})_\sigma = V^{(N)}(\bigoplus_k (\mu_k^B)^p (B_k)^{[p]})_\sigma.$$

*Proof.* [✓ Proof](#) Start from Theorem 9.48. With  $p_A, p_B$  the least common multiples of the period-removal lengths on the two sides, use  $p = \text{lcm}(p_A, p_B)$ . The prescribed-common-multiple lemma constructs both one-sided sector families at this length, and the positive-length blocking lemmas carry the one-sided tensor/nonzero-part equalities over to the common alphabet; for every  $N \geq 1$ ,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\bigoplus_j (\mu_j^A)^p (A_j)^{[p]})_\sigma.$$

The same applies on the  $B$  side, and the two-sided blocking lemma gives equality of the two powered nonzero parts.  $\square$

**Theorem 9.50** (Unconditional common primitive block decompositions). [Lean](#) [✓ Stmt](#) *Let  $A$  and  $B$  have the same MPV family. Then there is a positive blocking length at which both blocked tensors have the same positive-length MPV family as weighted direct sums of trace-preserving, primitive, tensor-irreducible blocks with positive bond dimensions and nonzero weights. The two weighted nonzero-sector families have the same positive-length MPV family.*

*Proof.* [✓ Proof](#) Apply Theorem 9.49 to choose the common blocking length and the two common cyclic-sector families. For either one of these families, write  $G_k$  for its cyclic-sector blocks, and write  $(\nu_x, C_x)$  for the weights and blocks of the flattened common-sector family from Lemma 9.68. The common-alphabet nonzero-part identity gives, for every length  $N$  and blocked word  $\sigma$ ,

$$V^{(N)}(\bigoplus_k \mu_k^p G_k^{[p]})_\sigma = V^{(N)}(\bigoplus_x \nu_x C_x)_\sigma,$$

Applying this on the two sides of the comparison yields a positive length  $p$  and weighted primitive block families such that, for every  $N \geq 1$  and blocked word  $\sigma$ ,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\bigoplus_x \mu_x^A A_x)_\sigma, \quad V^{(N)}(B^{[p]})_\sigma = V^{(N)}(\bigoplus_y \mu_y^B B_y)_\sigma,$$

with the two weighted nonzero parts equal at every positive length. The structural properties and the nonvanishing of the weights come from the common-sector families.  $\square$

A weighted block sum with distinct sectors is the simplest sector decomposition: one block per sector, each of multiplicity one. Equal-modulus and repeated-copy sectors are not single scalar powers, already for  $A = C \oplus (-C)$  with  $V^{(N)}(A)_\sigma = (1 + (-1)^N) V^{(N)}(C)_\sigma$ ; they are compared by the basis-of-normal-tensors construction of Chapter 10.

**Definition 9.51** (Single-copy sector decomposition). [Lean](#) [✓ Stmt](#) Given nonzero weights  $(\mu_k)_{k=1}^r$  and blocks  $(B_k)_{k=1}^r$ , the *single-copy sector decomposition* has  $r$  sectors, one basis block  $B_k$  per sector, multiplicity 1, and sector weight  $\mu_k$ . As a finite sum,

$$\sum_{k=1}^r \mu_k^N V^{(N)}(B_k)_\sigma = V^{(N)}(\bigoplus_k \mu_k B_k)_\sigma.$$

**Lemma 9.52** (Single-copy sector decomposition preserves the represented family). Lean ✓ Stmt The single-copy sector decomposition of  $(\mu_k, B_k)_k$  represents the same MPS family as  $\bigoplus_k \mu_k B_k$ .

*Proof.* ✓ Proof Expanding the sector decomposition gives  $\sum_k \mu_k^N V^{(N)}(B_k)_\sigma$ , the finite block-sum expression for  $\bigoplus_k \mu_k B_k$ .  $\square$

## 9.7 Blocking and weighted direct sums

Blocking by a length  $p$  commutes with weighted direct sums up to the  $p$ th power of each weight: for nonzero weights  $(\mu_k)$  and blocks  $(B_k)$ ,

$$(\bigoplus_k \mu_k B_k)^{[p]} \sim \bigoplus_k \mu_k^p B_k^{[p]},$$

where  $\sim$  is equality of MPV families. When two block families are reblocked to a common length  $p = m_k e_k$ , the  $p$ -blocked alphabet is identified with the  $e_k$ -fold tensor power of the  $m_k$ -blocked alphabet, so each sector reblocked through its own period agrees with the directly  $p$ -blocked block.

**Lemma 9.53** (Canonical form from peripheral primitivity). Not ready Let  $(\mu_k, A_k)_{k=1}^r$  be nonzero weights and injective left-canonical blocks with positive bond dimensions, non-increasing moduli  $|\mu_1| \geq \dots \geq |\mu_r| > 0$ , and  $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$ . Then  $(\mu_k, A_k)_{k=1}^r$  satisfies the canonical-form conditions of Definition 9.23, and  $O_{A_k A_k}(N) \rightarrow 1$  for every  $k$ .

**Lemma 9.54** (Adjoint primitivity equivalence). Lean ✓ Stmt Let  $\mathcal{E}$  be a finite-dimensional completely positive map. Then  $\mathcal{E}$  is primitive if and only if its Frobenius adjoint  $\mathcal{E}^\dagger$  is primitive; equivalently,  $\sigma_\partial(\mathcal{E}) = \{1\}$  iff  $\sigma_\partial(\mathcal{E}^\dagger) = \{1\}$ , since the peripheral spectrum of  $\mathcal{E}^\dagger$  is the complex conjugate of that of  $\mathcal{E}$ .

**Lemma 9.55** (Primitivity under blocking by a multiple). Lean ✓ Stmt Let  $A$  have positive bond dimension and let  $p_0, p \geq 1$  with  $p_0 \mid p$ . If  $\mathcal{E}_{A^{[p_0]}}$  is primitive, then  $\mathcal{E}_{A^{[p]}}$  is primitive.

**Lemma 9.56** (Blocking the weighted block sum). Lean ✓ Stmt For nonzero weights  $(\mu_k)_{k=1}^r$ , blocks  $(B_k)_{k=1}^r$ , and any  $p \geq 1$ ,

$$(\bigoplus_k \mu_k B_k)^{[p]} \sim \bigoplus_k \mu_k^p B_k^{[p]},$$

where  $\sim$  denotes generation of the same MPV family at every length.

**Lemma 9.57** (Positive-length equality survives blocking). Lean ✓ Stmt If two tensors have the same MPV coefficients at every positive length, then so do their blocked tensors after any positive blocking length.

**Lemma 9.58** (Positive-length blocking of a weighted block sum). Lean ✓ Stmt If  $A$  and  $\bigoplus_k \mu_k B_k$  have the same MPV coefficients at every positive length, then after any positive blocking length  $p$ ,  $A^{[p]}$  and  $\bigoplus_k \mu_k^p B_k^{[p]}$  have the same MPV coefficients at every positive length.

**Lemma 9.59** (Positive-length equality of powered block sums). Lean ✓ Stmt If  $\bigoplus_j \mu_j^A A_j$  and  $\bigoplus_k \mu_k^B B_k$  have the same MPV coefficients at every positive length, then after any positive common blocking length  $p$ ,  $\bigoplus_j (\mu_j^A)^p A_j^{[p]}$  and  $\bigoplus_k (\mu_k^B)^p B_k^{[p]}$  have the same MPV coefficients at every positive length.

**Lemma 9.60** (Nonzero weights survive blocking). [Lean](#) [✓ Stmt](#) If  $\mu \in \mathbb{C}$  is nonzero and  $p \geq 1$ , then  $\mu^p \neq 0$ ; blocking preserves the nonzero-weight hypothesis of a weighted block decomposition.

**Lemma 9.61** (Common reblocking of cyclic-sector families). [Lean](#) [✓ Stmt](#) Given a finite family  $(B_k)_{k=1}^r$ , each with primitive irreducible cyclic sectors, there is a common blocking length  $p$  such that each  $B_k^{[p]}$  admits a unit-weight direct-sum decomposition over a single common blocked alphabet  $d^{[p]}$  into left-canonical blocks with primitive transfer maps, tensor irreducibility, and positive bond dimensions.

**Lemma 9.62** (Sector irreducibility for the adjoint blocked map). [Lean](#) [✓ Stmt](#) Let  $A$  be a left-canonical irreducible MPS tensor with  $D > 0$ , let  $m$  be the period of its cyclic-sector decomposition, and let  $(P_u)_{u=1}^m$  be the cyclic projections of the adjoint transfer map with  $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$  and  $\sum_u P_u = \mathbb{1}$ . Then the restriction of  $(\mathcal{E}_{A^\dagger})^m$  to each corner  $P_u M_D(\mathbb{C}) P_u$  is irreducible.

**Lemma 9.63** (Iterated blocking at the transfer-map level). [Lean](#) [✓ Stmt](#) For any MPS tensor  $A$  and  $m, n \geq 0$ ,

$$\mathcal{E}_{(A^{[m]})^{[n]}} = \mathcal{E}_{A^{[mn]}}.$$

**Definition 9.64** (Primitive irreducible cyclic-sector decomposition). [Lean](#) [✓ Stmt](#)  $A$  has primitive irreducible cyclic sectors of period  $m \geq 1$  if there is a family  $(C_k)_{k=1}^m$  with  $A^{[m]} \sim \bigoplus_{k=1}^m C_k$  (unit weights), each  $C_k$  left-canonical, with primitive transfer map, tensor-irreducible, and of positive bond dimension.

**Lemma 9.65** (Common reblocking at a prescribed multiple). [Lean](#) [✓ Stmt](#) With hypotheses as in Lemma 9.61, and given any common multiple  $p$  of the per-block period-removal lengths  $(m_k)$ , the common reblocking can be performed at the prescribed length  $p$ .

**Lemma 9.66** (Structural properties of the common reblocked sectors). [Lean](#) [✓ Stmt](#) Each sector tensor in the common reblocked cyclic-sector family of Lemma 9.61 is trace-preserving ( $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ ), has primitive transfer map ( $\sigma_\partial(\mathcal{E}_B) = \{1\}$ ), is tensor-irreducible, and has positive bond dimension.

**Lemma 9.67** (Common-alphabet flattening). [Lean](#) [✓ Stmt](#) With  $p = m_k e_k$  for each  $k$ , the iterated blocked tensor  $(B_k^{[m_k]})^{[e_k]}$  agrees with  $B_k^{[p]}$  in MPV family, after the canonical identification of the  $p$ -blocked alphabet  $d^{[p]}$  with the  $e_k$ -fold tensor power of the  $m_k$ -blocked alphabet  $d^{[m_k]}$ .

**Lemma 9.68** (Reindexed nonzero-part identity). [Lean](#) [✓ Stmt](#) The powered nonzero-part decomposition  $\bigoplus_k \mu_k^p B_k^{[p]}$  carries over under the common-alphabet identification of Lemma 9.67: the common-alphabet sectors have the same nonzero-part decomposition as the direct  $p$ -blocked tensors.

**Lemma 9.69** (Direct blocking in the common alphabet). [Lean](#) [✓ Stmt](#) For every original block  $B_k$ , direct blocking at the common length  $p = m_k e_k$  agrees, as an MPV family, with the same block transported to the common blocked alphabet. Write  $\tilde{B}_k$  for this common-alphabet image. Then

$$V^{(N)}(B_k^{[p]})_\sigma = V^{(N)}(\tilde{B}_k)_\sigma.$$



# Chapter 10

## Basis of Normal Tensors

A *basis of normal tensors* (BNT) for an MPS tensor  $A$  is a finite family of normal blocks  $\{A_j\}$  whose MPV states  $\{|V^{(N)}(A_j)\rangle\}$  span  $|V^{(N)}(A)\rangle$  at every length and are eventually linearly independent. Two payoffs follow. First, equal MPV families force a bijective gauge-phase matching of the basis sectors  $B_k = e^{i\phi_k} X_k A_j X_k^{-1}$ . Second, the sector coefficients  $c_N^{(j)} = \sum_q \mu_{j,q}^N$  recover the per-copy weight multisets by Newton–Girard. The references are [PGVWC07], [CPGSV16], and [CPGSV21].

### 10.1 Bases of normal tensors

**Definition 10.1** (Basis of Normal Tensors (BNT)). Lean ✓ Stmt A *basis of normal tensors* (BNT) for a total tensor  $A_{\text{tot}}$  consists of  $g$  block tensors  $(A_1, \dots, A_g)$  with bond dimensions  $(D_1, \dots, D_g)$  such that each  $A_j$  is normal (Definition 2.31); at each system size  $N$  the MPV of  $A_{\text{tot}}$  lies in the span of the block MPVs  $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$ ; and for sufficiently large  $N$  those block MPVs are linearly independent. This is [CPGSV21, Definition IV.2]. In the canonical-form characterization of [CPGSV16, Section 2.3], a BNT is a minimal family of representatives for  $\tilde{A}_k^i = e^{i\phi_k} X_k A_j^i X_k^{-1}$  among the normal blocks.

**Definition 10.2** (Basis of normal tensors, coefficient form). Lean ✓ Stmt The coefficient form of a basis of normal tensors: each block  $A_j$  is a CPSV normal tensor (Definition 9.4), the MPV of  $A$  is given at every length by explicit coefficients  $c_j^{(N)} \in \mathbb{C}$  with  $|V^{(N)}(A)\rangle = \sum_{j=1}^g c_j^{(N)} |V^{(N)}(A_j)\rangle$ , and the block MPVs  $\{|V^{(N)}(A_j)\rangle\}$  are eventually linearly independent.

The eventual linear-independence clause is produced from the asymptotic overlaps: diagonal overlaps tending to one and off-diagonal overlaps tending to zero make the Gram matrix converge to the identity.

**Lemma 10.3** (Eventual linear independence from finite-index overlap orthonormality). Lean ✓ Stmt Let  $I$  be a finite index set and let  $(A_i)_{i \in I}$  be a finite family of tensors such that  $O_{A_i A_i}(N) \rightarrow 1$  for each  $i$ , and  $O_{A_i A_j}(N) \rightarrow 0$  whenever  $i \neq j$ . Then the MPV states  $\{|V^{(N)}(A_i)\rangle\}_{i \in I}$  are linearly independent for all sufficiently large  $N$ .

*Proof.* ✓ Proof The Gram matrix of the family  $\{|V^{(N)}(A_i)\rangle\}_{i \in I}$  converges entrywise to the identity matrix. Hence for all sufficiently large  $N$  it is invertible, so the MPV states are linearly independent. □



**Lemma 10.4** (Eventual linear independence from overlap orthonormality). [Lean](#) [✓ Stmt](#) Let  $(A_j)_{j=1}^g$  be a finite family of tensors such that  $O_{A_j A_j}(N) \rightarrow 1$  for each  $j$  and  $O_{A_i A_j}(N) \rightarrow 0$  for  $i \neq j$ . Then the MPV states  $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$  are linearly independent for all sufficiently large  $N$ .

*Proof.* [✓ Proof](#) Apply Lemma 10.3 with  $I = \{1, \dots, g\}$ .  $\square$

The same conclusion is often more convenient with an explicit threshold rather than the eventual quantifier.

**Theorem 10.5** (Existential BNT independence from overlap limits). [Lean](#) [✓ Stmt](#) If the self-overlaps of a finite block family tend to 1 and the cross-overlaps of distinct blocks tend to 0, then the MPV states are linearly independent for all sufficiently large system sizes.

*Proof.* [✓ Proof](#) This is Lemma 10.4, restated with an explicit threshold  $N_0$ .  $\square$

Eventual linear independence is also what converts an equality of two finite linear combinations into equality of their coefficients; this is the mechanism behind the coefficient identities of Chapter 11.

**Lemma 10.6** (Eventual coefficient extraction from linear independence). [Lean](#) [✓ Stmt](#) If a finite family of vectors is linearly independent for all sufficiently large  $N$ , and two finite linear combinations of that family are equal for all sufficiently large  $N$ , then the corresponding coefficients are equal for all sufficiently large  $N$ .

*Proof.* [✓ Proof](#) Move the two linear combinations to one side. Linear independence forces each coefficient difference to vanish at every sufficiently large length.  $\square$

When two BNT families must be compared against each other, the same Gram-matrix criterion applies to their disjoint union, provided the mixed overlaps also decay. This is the form used in the exact sector matching of Section 10.6.

**Lemma 10.7** (Eventual linear independence for two families). [Lean](#) [✓ Stmt](#) Let  $(A_j)_{j=1}^{g_A}$  and  $(B_k)_{k=1}^{g_B}$  be finite families of tensors. Suppose the overlaps inside each family are asymptotically orthonormal, and suppose every mixed overlap  $O_{A_j B_k}(N)$  tends to 0. Then the combined MPV family

$$\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g_A} \cup \{|V^{(N)}(B_k)\rangle\}_{k=1}^{g_B}$$

is linearly independent for all sufficiently large  $N$ .

*Proof.* [✓ Proof](#) Apply the finite-index Gram-matrix criterion to the disjoint union of the two index sets. The within-family hypotheses give the diagonal and off-diagonal limits inside each block, and the mixed overlap hypothesis gives the off-diagonal limits between the two blocks.  $\square$

We now isolate the separation condition that supplies the off-diagonal decay and combine it with the canonical-form hypotheses.

**Definition 10.8** (No gauge-phase-equivalent distinct blocks). [Lean](#) [✓ Stmt](#) A finite block family has no gauge-phase-equivalent distinct blocks when, for any two distinct indices with the same bond dimension, the corresponding blocks are not gauge-phase equivalent.

**Definition 10.9** (Normal canonical form with BNT separation). [Lean](#) [✓ Stmt](#) A family is in normal canonical form with BNT separation if it satisfies Definition 9.25 and distinct blocks of the same bond dimension are not gauge-phase equivalent. The weight moduli are non-increasing but need not be strictly decreasing, so equal-modulus blocks are allowed; the grouping into a

BNT is governed by gauge-phase equivalence, not by distinctness of moduli. Repeated equal-modulus copies of one basis tensor are not separate blocks here; they appear as the weights  $\mu_{j,q}$  in  $c_N^{(j)} = \sum_q \mu_{j,q}^N$  in the sector decomposition below.

The off-diagonal decay needed for the BNT criterion follows from the separation condition together with the overlap-decay theorems of the preceding chapter.

**Theorem 10.10** (Cross-overlap decay from explicit BNT hypotheses). Lean ✓ Stmt *Let  $(A_j)_{j=1}^r$  be a family in which each  $A_j$  is injective and left-canonical,  $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ , and distinct equal-dimension blocks are not gauge-phase equivalent. Then  $O_{A_j A_k}(N) \rightarrow 0$  for every pair  $j \neq k$ .*

*Proof.* ✓ Proof If the bond dimensions differ, Theorem 7.24 gives the decay. If the bond dimensions agree, the non-equivalence hypothesis excludes gauge-phase equivalence, so Theorem 7.26 gives the decay. □

**Lemma 10.11** (BNT from explicit blockwise hypotheses). Lean ✓ Stmt *Let  $(\mu_k, A_k)_{k=1}^r$  be a weighted block family. Assume each  $A_k$  is injective and left-canonical,  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ , that  $O_{A_k A_k}(N) \rightarrow 1$  for each  $k$ , and that distinct equal-dimension blocks are not gauge-phase equivalent. Then the blocks form a basis of normal tensors for the block-diagonal tensor  $\bigoplus_k \mu_k A_k$ .*

*Proof.* ✓ Proof Injective blocks are normal. The block-diagonal decomposition theorem gives the MPV expansion of  $\bigoplus_k \mu_k A_k$  in terms of the block MPVs. The self-overlap limits are part of the hypotheses, and Theorem 10.10 supplies the off-diagonal decay. Therefore Lemma 10.4 gives the eventual linear independence required in Definition 10.1. □

The normal-canonical formulation encodes normality spectrally (primitive transfer map), the BNT definition algebraically (eventual block injectivity).

*Remark 10.12* (Comparing the two normality conditions). The spectral normality of Definition 9.25 and the algebraic normality of Definition 2.31 agree: the Wielandt theory of Chapter 8 supplies the implication from primitivity to eventual block injectivity used here.

**Lemma 10.13** (Restricted normal-CF-BNT hypotheses give a BNT). Lean ✓ Stmt *If a family is in normal canonical form with BNT separation and each block is also normal in the algebraic sense, then the blocks form a basis of normal tensors for the associated block-diagonal tensor. The blocks here are distinct representatives.*

*Proof.* ✓ Proof The normal-canonical hypotheses already give the block-diagonal MPV expansion, the self-overlap normalization, and the eventual linear independence coming from irreducible trace-preserving overlap decay. Supplying algebraic normality for each block adds the remaining clause in Definition 10.1. □

## 10.2 Permutation rigidity for bases of normal tensors

When two bases of normal tensors generate the same MPV subspace at every length, the blocks must agree up to a permutation and per-block gauge phases. The argument turns the equality of spans into an invertible change-of-basis matrix, then analyses its asymptotics through the overlaps.

**Lemma 10.14** (Invertible change of basis). [Lean](#) [✓ Stmt](#) Let  $v_1, \dots, v_g$  and  $w_1, \dots, w_g$  be two linearly independent families in a complex vector space with the same span. Then there exists an invertible matrix  $U \in \text{GL}_g(\mathbb{C})$  such that

$$w_j = \sum_{i=1}^g U_{ij} v_i$$

for every  $j$ .

*Proof.* [✓ Proof](#) Because the two families span the same subspace, each  $w_j$  has a unique expansion in the basis  $\{v_i\}_{i=1}^g$ ; these coefficients form the columns of  $U$ . If  $Ua = 0$ , then the corresponding linear relation among the  $w_j$  is trivial because the  $w_j$  are linearly independent. Thus  $U$  is injective, hence invertible.  $\square$

**Lemma 10.15** (Eventual change of basis from equal spans). [Lean](#) [✓ Stmt](#) If two finite families of blocks have asymptotically orthonormal overlaps within each family and their MPV spans agree for every system size  $N$ , then for all sufficiently large  $N$  there exists an invertible matrix  $U_N$  expressing the MPV states of one family as linear combinations of the MPV states of the other.

*Proof.* [✓ Proof](#) By Lemma 10.4, both families are linearly independent for all sufficiently large  $N$ . At such an  $N$ , the span equality lets us apply Lemma 10.14.  $\square$

The change-of-basis matrix is now used to match the blocks. We first treat equal block counts, then remove that restriction.

**Theorem 10.16** (Permutation rigidity for equal block counts). [Lean](#) [✓ Stmt](#) Let  $(A_j)_{j=1}^g$  and  $(B_j)_{j=1}^g$  be injective families of positive bond dimension satisfying trace-preserving normalization, with self-overlaps tending to 1 and off-diagonal overlaps tending to 0 within each family. If the spans of the MPV states agree for every system size  $N$ , then there is a permutation  $\pi$  of  $\{1, \dots, g\}$  such that  $A_j$  and  $B_{\pi(j)}$  have the same bond dimension and are gauge-phase equivalent.

*Proof.* [✓ Proof](#) For large  $N$ , Lemma 10.15 gives an invertible matrix  $U_N$  relating the two MPV families. Fix  $j$  and write  $|V^{(N)}(B_j)\rangle = \sum_i u_{ij}^{(N)} |V^{(N)}(A_i)\rangle$ . With  $G_A(N) = (O_{A_i A_\ell}(N))_{i,\ell}$  and  $b_j(N) = (O_{A_i B_j}(N))_i$ , this gives

$$G_A(N) \overline{u_j(N)} = b_j(N), \quad O_{B_j B_j}(N) = u_j(N)^\top G_A(N) \overline{u_j(N)}.$$

If all mixed overlaps  $O_{A_i B_j}(N)$  tended to 0, then  $b_j(N) \rightarrow 0$ ; since  $G_A(N) \rightarrow I$ , one obtains  $\overline{u_j(N)} \rightarrow 0$  and hence  $u_j(N) \rightarrow 0$ . Thus  $O_{B_j B_j}(N) \rightarrow 0$ , contradicting  $O_{B_j B_j}(N) \rightarrow 1$ . Thus each  $B_j$  has a matched block  $A_{f(j)}$  with non-decaying mixed overlap. Theorems 7.24 and 7.26 then force equal bond dimension and gauge-phase equivalence for each match. Off-diagonal decay within the  $B$ -family makes  $f$  injective, hence bijective, and its inverse is the required permutation.  $\square$

**Lemma 10.17** (Permutation rigidity with asymptotically orthonormal overlaps). [Lean](#) [✓ Stmt](#) Let  $(A_j)_{j=1}^{g_A}$  and  $(B_k)_{k=1}^{g_B}$  be two families of injective tensors of positive bond dimension satisfying the TP normalization  $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$  and  $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$ , whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If for every system size  $N$ ,

$$\text{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g_A\} = \text{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \leq k \leq g_B\},$$

then  $g_A = g_B$ , and there is a permutation  $\pi$  of the common block index set such that  $A_j$  and  $B_{\pi(j)}$  have the same bond dimension and are gauge-phase equivalent for each  $j$ .

*Proof.* ✓ Proof By Lemma 10.4, both families are linearly independent for all sufficiently large  $N$ . At such an  $N$ , the common span has the same dimension when computed from the  $A$ -family or the  $B$ -family, so  $g_A = g_B$ . After identifying the two index sets, Theorem 10.16 yields the permutation conclusion.  $\square$

The two preceding results obtain the block matching from the asymptotic orthonormality of the overlaps. The same matching is obtained by the exact sector comparison of Section 10.6, which pairs the sectors of two canonical forms through a bijection fixed by their bond dimensions and gauge-phase equivalences. The proof of the Fundamental Theorem in Chapter 11 proceeds through the sector comparison.

The same overlap dichotomy also shows that the MPV states of distinct normal blocks are non-proportional at arbitrarily large lengths; this is an input to the exact sector matching later in the chapter.

**Theorem 10.18** (Non-equivalent irreducible TP blocks are eventually non-proportional). Lean

Lean ✓ Stmt *Let  $A$  and  $B$  be irreducible trace-preserving blocks whose self-overlaps converge to 1. If they are not gauge-phase equivalent, then for every lower bound  $N_0$  there is  $N \geq N_0$  such that  $|V^{(N)}(A)\rangle$  is not proportional to  $|V^{(N)}(B)\rangle$ . Consequently, in an already separated same-dimensional BNT block family, any two distinct blocks have such non-proportional MPV states at arbitrarily large lengths.*

*Proof.* ✓ Proof Suppose that for all sufficiently large  $N$  one had  $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$ . Then

$$O_{AA}(N) = |c_N|^2 O_{BB}(N), \quad O_{AB}(N) = c_N O_{BB}(N). \quad (10.1)$$

Since  $O_{AA}(N) \rightarrow 1$  and  $O_{BB}(N) \rightarrow 1$ , the first identity in (10.1) gives  $|c_N| \rightarrow 1$ , and hence  $|O_{AB}(N)| \rightarrow 1$ . The irreducible trace-preserving overlap decay theorem gives  $O_{AB}(N) \rightarrow 0$  when  $A \not\sim_{\text{gp}} B$ , a contradiction.  $\square$

### 10.3 Newton–Girard identities and power-sum recovery

The coefficients  $c_N^{(j)} = \sum_q \mu_{j,q}^N$  that appear in the two-layer BNT expansion are power sums of the copy weights. Recovering the individual weights from these sums is a Newton–Girard problem: equal power sums determine equal characteristic polynomials, hence equal multisets of roots.

**Theorem 10.19** (Newton–Girard trace recursion). Lean Lean ✓ Stmt *If  $A, B \in M_n(\mathbb{C})$  satisfy  $\text{tr}(A^k) = \text{tr}(B^k)$  for all  $k \geq 1$ , then  $A$  and  $B$  have the same characteristic polynomial. It is enough to assume this equality for  $1 \leq k \leq n$ .*

*Proof.* ✓ Proof Let  $e_m$  denote the  $m$ th elementary symmetric polynomial in the eigenvalues, with  $e_0 = 1$ . The Newton–Girard recursion

$$m e_m = \sum_{i=1}^m (-1)^{i-1} e_{m-i} \text{tr}(A^i)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces  $\text{tr}(A^i)$ . Equal power sums for  $A$  and  $B$  therefore give equal coefficients, hence equal characteristic polynomials.  $\square$

The matched-sector comparison below applies the power-sum identities to a single matched pair, after the BNT permutation and phase gauges have identified which basis tensors are compared: the coefficient  $\sum_q \mu_{j,q}^N$  then equals  $\sum_q (\nu_{j,q} e^{i\phi_j})^N$ , and Newton–Girard recovers the weight multiset with multiplicity.

**Theorem 10.20** (Finite-range equal power sums imply equal multisets). [Lean](#) [Lean](#) [Lean](#) [Lean](#)  
 $\checkmark$  *Stmt* Let  $\alpha, \beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$ . If  $\sum_i \alpha_i^k = \sum_i \beta_i^k$  for  $1 \leq k \leq n$ , then the multisets  $\{\alpha_i\}$  and  $\{\beta_i\}$  are equal.

Two variants relax the equal-cardinality assumption. If  $\alpha : \{0, \dots, m-1\} \rightarrow \mathbb{C}$  and  $\beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$  have no zero entries and their power sums agree for  $1 \leq k \leq \max(m, n)$ , then  $m = n$  and the multisets are equal. Without a nonzero hypothesis, the list form  $\sum_{k=1}^{x_a} \lambda_{a,k}^N = \sum_{k=1}^{x_b} \lambda_{b,k}^N$  for  $1 \leq N \leq \max(x_a, x_b)$  gives equality of the nonzero multisets; if it also holds at  $N = 0$  then  $x_a = x_b$  and the full lists agree.

*Proof.*  $\checkmark$  *Proof* Apply the finite-range Newton–Girard theorem to  $\text{diag}(\alpha)$  and  $\text{diag}(\beta)$  and extract the roots of  $\prod_i (X - \alpha_i)$  and  $\prod_i (X - \beta_i)$ . For unequal cardinalities, pad both lists with zeros to length  $\max(m, n)$ ; the nonzero hypothesis makes the padded-zero counts  $\max(m, n) - m$  and  $\max(m, n) - n$ , forcing  $m = n$ . Without that hypothesis, delete the zero entries first (unchanged for positive  $N$ ); the equality at  $N = 0$  gives  $x_a = x_b$ .  $\square$

**Theorem 10.21** (Equal power sums imply equal multisets). [Lean](#)  $\checkmark$  *Stmt* Let  $\alpha, \beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$ . If  $\sum_i \alpha_i^k = \sum_i \beta_i^k$  for all  $k \geq 1$ , then the multisets  $\{\alpha_i\}$  and  $\{\beta_i\}$  are equal.

*Proof.*  $\checkmark$  *Proof* Construct diagonal matrices  $\text{diag}(\alpha)$  and  $\text{diag}(\beta)$ . The power-sum hypothesis gives  $\text{tr}(\text{diag}(\alpha)^k) = \text{tr}(\text{diag}(\beta)^k)$ . By Theorem 10.19, the characteristic polynomials agree:  $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$ . Extracting roots gives multiset equality.  $\square$

## 10.4 The BNT canonical form on a sector decomposition

This section introduces the BNT canonical-form predicate of [CPGSV16, Section 2], defined on a sector decomposition with raw sector weights  $\mu_{j,q}$  and coefficients  $c_N^{(j)} = \sum_q \mu_{j,q}^N$ ; no equal-modulus factorization is assumed.

### 10.4.1 Sector decompositions and phase matching

The sector decomposition and the MPV phase-equivalence predicates introduced here feed the BNT canonical-form predicate below.

**Definition 10.22** (Sector weights and sector decomposition). [Lean](#) [Lean](#)  $\checkmark$  *Stmt* A sector-weight tuple over  $g$  basis blocks consists of multiplicities  $r_j > 0$ , nonzero weights  $\mu_{j,q}$  for  $q \in \{1, \dots, r_j\}$ , and the sector coefficients

$$c_{S,j}^{(N)} := \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

A sector decomposition consists of the basis blocks  $(A_j)_{j=1}^g$  together with these multiplicities and weights.

**Lemma 10.23** (Sector decomposition formula). [Lean](#)  $\checkmark$  *Stmt* For a sector decomposition  $S$  with associated tensor  $A_{\text{tot}}$ , the MPV satisfies

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^g c_{S,j}^{(N)} V^{(N)}(A_j)_\sigma.$$

Two blocks are identified for canonical-form purposes when their MPV families differ by a scalar power. The next three definitions formalize this relation, the common cover it generates between two families, and the phase classes of a single family.

**Definition 10.24** (MPV phase equivalence for nonzero-weight blocks). [Lean](#) [✓ Stmt](#) Two blocks  $A$  and  $B$  are MPV-phase equivalent if there is a nonzero scalar  $\zeta$  such that, for every system size  $N$  and every physical word,

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

The two bond dimensions are allowed to differ.

**Definition 10.25** (Common MPV phase cover). [Lean](#) [✓ Stmt](#) For two finite nonzero-weight block families, a common MPV phase cover consists of a third finite family of blocks, a map from each nonzero-weight block family onto that common family, and MPV-phase equivalences between each block and its common representative. Both maps are required to be surjective, so the common family contains no unused representative.

**Definition 10.26** (MPV phase classes). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a finite family of blocks, put  $j \sim k$  when there is a nonzero scalar factor  $\zeta$  such that

$$|V^{(N)}(B_k)\rangle = \zeta^N |V^{(N)}(B_j)\rangle$$

for every length  $N$ . The associated tuple enumerates the finite quotient, chooses one representative per class, records the scalar relating the representative to each member, and proves that distinct representatives are not MPV-phase equivalent.

**Definition 10.27** (BNT canonical form). [Lean](#) [✓ Stmt](#) Given a sector decomposition  $P$ , the BNT canonical-form predicate records the minimal BNT hypotheses of [CPGSV16, CPGSV21] without any equal-modulus or strict-ordering condition on the raw sector weights. It asserts that each basis block has positive bond dimension, is irreducible, and left-canonical; the normalized self-overlap of every basis block converges to 1; the basis MPV family is eventually linearly independent; distinct same-dimension basis blocks are not gauge-phase equivalent after identifying equal bond dimensions; and the [CPGSV16, Section 2.3] canonical-form normalization holds,

$$\forall(j, q), |\mu_{j,q}| \leq 1, \quad \exists(j_*, q_*) : |\mu_{j_*, q_*}| = 1.$$

The MPV expansion takes the raw two-layer form

$$V^{(N)}(P)_\sigma = \sum_{j=1}^g \left( \sum_{q=1}^{r_j} \mu_{j,q}^N \right) V^{(N)}(P_j)_\sigma,$$

matching [CPGSV16, Section 2.3] and [CPGSV21, Definition IV.2]. The canonical-form normalization admits every example of the source paper (in particular  $C \oplus D$ ,  $C \oplus (-C)$ ,  $C \oplus (1/2)C$ , and  $C \oplus (-C) \oplus (1/2)C$ ) and is not derivable from the per-block normality hypotheses alone.

The matching arguments below read off three facts from a BNT canonical form: the raw coefficient identity, the cross-overlap decay between distinct basis blocks, and the combined-family linear independence.

**Lemma 10.28** (Raw two-layer sector coefficient identity). [Lean](#) [✓ Stmt](#) For every sector decomposition  $P$ , length  $N$ , and basis index  $j$ ,

$$c_N^{(j)} = \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

This is [CPGSV16, Section 2.3].

*Proof.* ✓ **Proof** Immediate from the definition of the sector coefficient. □

**Lemma 10.29** (Cross-overlap decay between distinct basis blocks). Lean ✓ **Stmt** *If  $P$  is a BNT canonical form and  $j \neq k$ , then  $O_{P_j P_k}(N) \rightarrow 0$ .*

*Proof.* ✓ **Proof** Case on the bond dimensions. If they differ, the dimension-mismatch overlap-decay theorem applies. If they agree, the basis-distinctness clause rules out gauge-phase equivalence, so the equal-dimension overlap-decay theorem applies. □

**Lemma 10.30** (Combined-family eventual linear independence). Lean ✓ **Stmt** *Let  $P$  and  $Q$  each be a BNT canonical form, and suppose every mixed overlap  $O_{P_j Q_k}(N)$  tends to 0. Then the union family*

$$\{|V^{(N)}(P_j)\rangle\}_{j=1}^{g_P} \cup \{|V^{(N)}(Q_k)\rangle\}_{k=1}^{g_Q}$$

*is linearly independent for all sufficiently large  $N$ . This is [CPGSV16, Appendix A].*

*Proof.* ✓ **Proof** Feed the per-family normalized self-overlaps, the within-family cross-overlap decay (Lemma 10.29), and the inter-family decay hypothesis into the two-family overlap-orthonormality criterion (Lemma 10.7). □

**Lemma 10.31** (Unit-modulus weight witness from the canonical-form normalization). Lean ✓ **Stmt** *Under the BNT canonical form, the [CPGSV16, Section 2.3] unit-modulus existential provides indices  $(j_*, q_*)$  with  $|\mu_{j_*, q_*}| = 1$ .*

*Proof.* ✓ **Proof** The canonical-form normalization includes a unit-modulus weight  $|\mu_{j_*, q_*}| = 1$ . □

**Example 10.32** ( $C \oplus (1/2)C$ , unequal-modulus admissibility). Lean ✓ **Stmt** The single-sector two-copy decomposition with raw weights  $(1, 1/2)$  is a BNT canonical form, so unequal-modulus copies are admissible:  $|1/2| \leq 1$  and the unit witness is the first copy.

## 10.4.2 Prepared-block construction of BNT canonical forms

The construction below separates the after-blocking preparation of suitable blocks from the normalized-weight BNT construction: the prepared blocks are trace-preserving, primitive, irreducible, and injective; normalized weights then quotient gauge-phase-equivalent blocks into sectors to produce the canonical form of Definition 10.27.

**Lemma 10.33** (Common injective blocking for primitive irreducible families). Lean ✓ **Stmt** *Let  $(B_k)_{k=1}^r$  be a finite family of positive-bond-dimension MPS tensors, each trace-preserving, primitive, and tensor-irreducible. Then there is a single  $L \geq 1$  such that each  $B_k^{[L]}$  is one-site injective, and this blocking preserves trace preservation, primitivity, and irreducibility for every block.*

*Proof.* ✓ **Proof** For each block, Theorem 8.88 gives a positive length at which it becomes one-site injective; the product of these lengths is a common length. Injectivity persists under positive further blocking, and Theorem 9.42 preserves the other hypotheses. □

**Theorem 10.34** (Prepared BNT blocks after blocking). Lean ✓ **Stmt** *Let  $A$  be an MPS tensor. Then there is a blocking length  $p \geq 1$ , nonzero weights  $\mu_k$ , and blocks  $B_k$  over the  $p$ -blocked alphabet such that:*

1. *each block has positive bond dimension;*
2. *each block is trace-preserving, primitive, tensor-irreducible, and one-site injective;*



3.  $A^{[p]}$  and  $\bigoplus_k \mu_k B_k$  generate the same MPV family at every positive length.

No weight-modulus normalization is asserted at this stage.

*Proof.* ✓ Proof Apply the common primitive irreducible block decomposition to  $(A, A)$ , giving a positive blocking length and a nonzero weighted family of trace-preserving primitive irreducible blocks with the positive-length MPV identity. Lemma 10.33 supplies one common further blocking making every block one-site injective while preserving the other hypotheses; finally identify the iterated blocked alphabet with the direct one.  $\square$

With the prepared blocks in hand, normalized weights produce a BNT canonical form by quotienting gauge-phase-equivalent blocks into sectors.

**Theorem 10.35** (BNT from normal blocks with normalized weights). Lean ✓ Stmt Let  $r \in \mathbb{N}$ , and for each  $k = 1, \dots, r$  let  $D_k \in \mathbb{N}$ ,  $\mu_k \in \mathbb{C}$ , and let  $B_k$  be an MPS tensor of physical dimension  $d$  and bond dimension  $D_k$ . Assume:

1. **positive bond dimensions:**  $D_k > 0$  for all  $k$ ,
2. **TP normalization:**  $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$ ,
3. **primitivity:**  $\mathcal{E}_{B_k}$  is primitive for each  $k$ ,
4. **irreducibility:** each  $B_k$  is irreducible,
5. **nonzero weights:**  $\mu_k \neq 0$  for all  $k$ ,
6. **bounded weights:**  $|\mu_k| \leq 1$  for all  $k$ ,
7. **global unit witness:**  $|\mu_k| = 1$  for some  $k$ .

Then there exists a sector decomposition  $P$  such that  $P$  and  $\bigoplus_{k=1}^r \mu_k B_k$  generate the same MPV family at every length, and  $P$  satisfies the basis-of-normal-tensors conditions (Definition 10.2) together with the multiplicity, span, and weight-bound conditions of Definition 10.27.

*Proof.* ✓ Proof The phase-class quotient groups gauge-phase-equivalent blocks into sectors with unit-modulus per-copy scalars. Under the TP, primitive, and irreducible hypotheses the phase-class construction applies, and for distinct representatives  $B_j, B_{j'}$  ( $j \neq j'$ ) the cross-overlap decays,

$$\lim_{N \rightarrow \infty} O_{B_j B_{j'}}(N) = 0, \quad \lim_{N \rightarrow \infty} O_{B_j B_j}(N) = 1,$$

which gives the eventual linear-independence condition with no injectivity required. The resulting sector decomposition inherits the span and multiplicity conditions, and the bounded-weight and global-unit hypotheses give the weight-norm conditions.  $\square$

The BNT construction preserves total bond dimension because CF summands assigned to the same BNT representative have equal bond dimension. Indeed, suppose trace-preserving primitive irreducible NTs  $X$  and  $Y$  of positive bond dimension have MPV families related by  $|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$  with  $\zeta \neq 0$ . The normalized self-overlaps give  $|O_{XX}(N)| \rightarrow 1$ . The phase relation gives the cross-overlap identity

$$O_{YX}(N) = \zeta^N O_{XX}(N), \quad |O_{YX}(N)| = |\zeta|^N |O_{XX}(N)|.$$

Since the same normalization forces  $|\zeta| = 1$ , this gives  $|O_{YX}(N)| \rightarrow 1$ . If their bond dimensions differed, the rectangular overlap-decay theorem would give

$$O_{YX}(N) \rightarrow 0,$$

a contradiction.



**Lemma 10.36** (Total dimension after grouping by BNT representatives). [Lean](#) [✓ Stmt](#) *Suppose CF summands are grouped by BNT representative, every original copy weight is nonzero, and every summand assigned to a representative has that representative's bond dimension. Then the total bond dimension of the resulting BNT sector decomposition is the sum of the original block dimensions.*

*Proof.* [✓ Proof](#) The collapsed decomposition counts one representative dimension for each assigned copy. Let  $e(j, q)$  denote the original CF summand assigned to the  $q$ th copy of representative  $P_j$ . Since  $(j, q) \mapsto e(j, q)$  reindexes the original summands and each assigned summand has the representative's bond dimension,

$$\sum_{j=1}^g \sum_{q=1}^{r_j} \dim(P_j) = \sum_{j=1}^g \sum_{q=1}^{r_j} D_{e(j,q)} = \sum_{k=1}^r D_k.$$

This is the equality between the collapsed total dimension and the original direct-sum dimension.  $\square$

**Lemma 10.37** (Prepared phase-equivalent NTs have the same bond dimension). [Lean](#) [✓ Stmt](#) *Let  $X$  and  $Y$  be trace-preserving, primitive, irreducible NTs of positive bond dimension. If their MPV families differ by a nonzero scalar power, then  $X$  and  $Y$  have the same bond dimension.*

*Proof.* [✓ Proof](#) Choose  $\zeta \neq 0$  with  $|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$  for every length  $N$ . The normalized self-overlaps give  $|O_{XX}(N)| \rightarrow 1$ , and the phase relation gives

$$O_{YX}(N) = \zeta^N O_{XX}(N), \quad |O_{YX}(N)| = |\zeta|^N |O_{XX}(N)|.$$

Since the same normalization forces  $|\zeta| = 1$ , this gives  $|O_{YX}(N)| \rightarrow 1$ . If the two bond dimensions differed, the rectangular overlap-decay theorem would force

$$O_{YX}(N) \rightarrow 0,$$

contradicting the preceding limit.  $\square$

**Lemma 10.38** (Prepared BNT representative classes preserve bond dimension). [Lean](#) [✓ Stmt](#) *In a prepared family of positive-bond-dimension trace-preserving primitive irreducible NTs, every CF summand assigned to a BNT representative has that representative's bond dimension.*

*Proof.* [✓ Proof](#) Apply Lemma 10.37 to each representative and each assigned summand.  $\square$

**Lemma 10.39** (Prepared collapsed BNT total dimension). [Lean](#) [✓ Stmt](#) *For prepared trace-preserving primitive irreducible NTs of positive bond dimension with all copy weights nonzero, grouping CF summands by BNT representative preserves the total bond dimension: the collapsed sector decomposition has total bond dimension equal to the sum of the original block dimensions.*

*Proof.* [✓ Proof](#) Every assigned summand shares the bond dimension of its representative, by the equal-dimension fact above. The displayed regrouping identity in Lemma 10.36 then identifies the collapsed sum over pairs  $(j, q)$  with  $\sum_{k=1}^r D_k$ , the original direct-sum dimension.  $\square$

Combining the after-blocking preparation with the normalized-weight construction gives the arbitrary-input statement, conditional on the weight normalization.

**Theorem 10.40** (Conditional BNT form after blocking for an arbitrary tensor). [Lean](#) [Lean](#) [✓ Stmt](#) *Let  $A$  be an MPS tensor. Then there exists a blocking length  $p \geq 1$ , nonzero weights  $\mu_k$ , and blocks  $B_k$  over the blocked alphabet, such that:*

1. every block has positive bond dimension;
2. every block is TP-normalized, primitive, irreducible, and injective;
3.  $A^{[p]}$  and  $\bigoplus_k \mu_k B_k$  generate the same MPV family at every positive length.

Moreover, under  $|\mu_k| \leq 1$  for every  $k$  and  $|\mu_k| = 1$  for at least one  $k$ , there exists a sector decomposition  $P$  over the blocked alphabet in the BNT canonical form of Definition 10.27 such that  $A^{[p]}$  and  $P$  generate the same MPV family at every positive length; the identity upgrades to all lengths once the total bond dimension of  $P$  equals that of  $A^{[p]}$ .

*Proof.* ✓ Proof The after-blocking reduction supplies the TP-normalized primitive irreducible injective blocks and the positive-length MPV identity. The normalized-weight hypotheses are the choice that [CPGSV16, Section 2.3] makes; Theorem 10.35 then produces the sector decomposition, following the BNT selection of [CPGSV16, Appendix A]. At length zero the MPV coefficient is the bond dimension, so equal total bond dimensions supply the missing length-zero coefficient. The remaining rescaling step is recorded in Remark 10.41.  $\square$

*Remark 10.41* (Arbitrary-input reduction). The reduction is conditional on the two weight-normalization hypotheses of Theorem 10.40.

## 10.5 Coefficient comparison and copy-weight recovery

The exact matching of Section 10.6 produces, on each matched pair, a coefficient identity. This section turns that identity into equality of the per-copy weight multisets [CPGSV16, Appendix A]. No analytic non-decay estimate enters: the fixed-length linear independence isolates each sector coefficient exactly.

### 10.5.1 Matched-sector weight multiset equality

Given a matched pair  $(j_0, \tilde{k}_0)$  and a nonzero scalar  $\zeta$  with  $c_N^{(j_0)}(P) = \zeta^N c_N^{(\tilde{k}_0)}(Q)$  past a threshold  $N_0$ , this section records two conclusions: the multiset equality of the per-copy weights from power-sum recovery (Theorem 10.21), and its upgrade to an explicit per-copy permutation. Together they give the  $\mu_{j,q} = \nu_{j,q} e^{i\varphi_j}$  recovery of [CPGSV16, Appendix A]. The coefficient identity is supplied, in the equal-MPV case, by Lemma 11.3 of Chapter 11; in the proportional case it is the remaining hypothesis of the conditional proportional global gauge (Theorem 11.10).

**Lemma 10.42** (Matched-sector weight multiset equality). Lean ✓ Stmt *Let  $P, Q$  be sector decompositions with weights  $\mu_{j,q}$  and  $\nu_{k,q'}$ , and fix indices  $j_0, \tilde{k}_0$ , and a nonzero scalar  $\zeta \in \mathbb{C} \setminus \{0\}$ . Suppose that for every  $N > N_0$ ,*

$$c_N^{(j_0)}(P) = \zeta^N \cdot c_N^{(\tilde{k}_0)}(Q). \quad (10.2)$$

*Then the per-sector multiplicities agree,  $r_{j_0}(P) = r_{\tilde{k}_0}(Q)$ , and the rescaled per-copy weight multisets coincide. This is [CPGSV16, Appendix A] read on a single matched sector.*

*Proof.* ✓ Proof Expand both sides of the coefficient identity (10.2) into per-copy power sums  $\sum_q \mu_{j_0,q}^N$  and  $\sum_{q'} \nu_{\tilde{k}_0,q'}^N$ , and distribute  $\zeta^N$  through the right-hand sum to obtain, for every  $N > N_0$ ,

$$\sum_q \mu_{j_0,q}^N = \sum_{q'} (\zeta \cdot \nu_{\tilde{k}_0,q'})^N.$$

Extend to all positive exponents and apply the unequal-cardinality multiset-recovery lemma (Theorem 10.20), using  $\zeta \neq 0$  together with the weight non-vanishing assumption to ensure the input weights are nonzero. This yields the multiplicity match and the multiset equality.  $\square$

**Lemma 10.43** (Matched-sector weight-permutation extractor). [Lean] [✓ Stmt] *Under the hypotheses of Lemma 10.42, the per-sector multiplicities agree and there is a permutation  $\tau$  of the per-copy index set identifying the individual weights up to the gauge phase: for every  $q$ ,*

$$\nu_{k_0, \tau(q)}^{-1} = \zeta^{-1} \cdot \mu_{j_0, q}.$$

*This is the per-copy realisation of [CPGSV16, Appendix A]: the multiset equality from Lemma 10.42 is upgraded to a concrete indexing between the matched  $P$ - and  $Q$ -copies.*

*Proof.* [✓ Proof] Invoke Lemma 10.42 to obtain the cardinality match and the multiset equality. From this multiset-map equality extract a permutation matching the elements pointwise (proved by induction on the cardinality). Compose with the cardinality identification and cancel  $\zeta$  via  $\zeta \neq 0$  to convert the gauge factor on the right of the pointwise identity to  $\zeta^{-1}$  on the left.  $\square$

## 10.6 Strong existential matching by exact linear independence

The exact matching reads [CPGSV16, Appendix A] as a statement at fixed length: equal MPV families force each basis sector of one form to pair with a basis sector of the other. The non-vanishing of the sector coefficients isolates the matched sector exactly, with no limit of a single-sector projection.

**Lemma 10.44** (Sector coefficient is not eventually zero). [Lean] [✓ Stmt] *For a BNT canonical form  $P$  and any sector  $j$ , the sector coefficient  $c_{P,j}^{(N)} = \sum_q \mu_{j,q}^N$  is not eventually zero in  $N$ . Only nonvanishing of the copy weights is used, not any modulus condition.*

*Proof.* [✓ Proof] If  $c_{P,j}^{(N)} = 0$  for all  $N \geq M$ , the signed geometric-sequence vanishing lemma forces  $\sum_q \mu_{j,q}^k = 0$  at every exponent  $k$ , in particular  $\sum_q 1 = r_j = 0$  at  $k = 0$ , contradicting  $r_j > 0$ . Repeated ratios are handled by the same extrapolation, so distinct weights are not needed.  $\square$

**Theorem 10.45** (Exact single-sector matching). [Lean] [✓ Stmt] *Let  $P, Q$  be two BNT canonical forms with the same MPV family (at all positive lengths). For every sector  $j$  of  $P$  there exist a sector  $k$  of  $Q$  with matched bond dimension  $\dim_P(j) = \dim_Q(k)$ , a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap. No per-sector unit-modulus hypothesis is required.*

*Proof.* [✓ Proof] Suppose  $j$  decays against every sector of  $Q$ , and let  $T$  be the set of  $P$ -sectors with this property. For  $j' \notin T$  the overlap dichotomy turns a non-decaying overlap with some  $Q_{k(j')}$  into a gauge-phase equivalence, hence a scalar  $\omega_{j'} \in \mathbb{C}^\times$  with  $V^{(N)}(P_{j'}) = \omega_{j'}^N V^{(N)}(Q_{k(j')})$ , so  $V^{(N)}(P_{j'})$  lies in the span of  $\{V^{(N)}(Q_k)\}_k$ . The family  $\{P_{j'} : j' \in T\} \cup \{Q_k\}_k$  then has all pairwise cross-overlaps decaying, hence is eventually linearly independent (Lemma 10.30). Rewriting the common-MPV identity over this family,

$$\sum_{j' \in T} c_N^{(j')} V^{(N)}(P_{j'}) = \sum_k \left( c_N^{(k)} - \sum_{j' \notin T, k(j')=k} c_N^{(j')} \omega_{j'}^N \right) V^{(N)}(Q_k),$$

exact coefficient extraction forces  $c_N^{(j)} = 0$  at each fixed large  $N$ , contradicting Lemma 10.44. Hence a matched sector  $k$  exists; the gauge-phase equivalence is recovered from the non-decaying overlap by the irreducible trace-preserving overlap dichotomy.  $\square$

**Theorem 10.46** (Strong existential matching, full-basis form). Lean ✓ Stmt *Let  $P, Q$  be two BNT canonical forms with the same MPV family. Then for every sector  $k$  of  $Q$  there exist a sector  $j$  of  $P$  with matched bond dimension  $\dim_P(j) = \dim_Q(k)$ , a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap.*

*Proof.* ✓ Proof Apply Theorem 10.45 with the roles of  $P$  and  $Q$  swapped and the same-MPV hypothesis flipped by symmetry of gauge-phase equivalence and overlap convergence.  $\square$

### 10.6.1 Bijective matching by symmetry

Applying the exact matching once more with  $P$  and  $Q$  swapped gives the reverse direction; the basis-distinctness clause forces injectivity both ways, so  $g_a = g_b$  and the source relabelling  $B_k = e^{i\phi_k} X_k A_{j_k} X_k^{-1}$  holds on the full basis [CPGSV16, Appendix A].

**Theorem 10.47** (Bijective matching by symmetry). Lean ✓ Stmt *Let  $P, Q$  be two BNT canonical forms with the same MPV family. Then there is a bijection*

$$\beta : J(Q) \simeq J(P)$$

*such that for every  $k \in J(Q)$  the bond dimensions match, the basis tensors  $P_{\beta(k)}$  and  $Q_k$  are gauge-phase equivalent after identifying the matched bond dimensions, and the basis cross-overlap does not tend to zero.*

*Proof.* ✓ Proof Theorem 10.45 applied to  $(Q, P)$  gives a matching map  $m : J(Q) \rightarrow J(P)$ , and applied to  $(P, Q)$  a matching map  $J(P) \rightarrow J(Q)$ .

*Injectivity of  $m$ .* If  $m(k_1) = m(k_2) = j$ , the two witnesses give a common centre  $P_j$  gauge-phase equivalent, after identifying bond dimensions, to both  $Q_{k_1}$  and  $Q_{k_2}$ ; transitivity yields a gauge-phase equivalence  $Q_{k_1} \sim Q_{k_2}$  under  $\dim_Q(k_1) = \dim_Q(k_2)$ . The basis-distinctness clause on  $Q$  then forces  $k_1 = k_2$ ; the reverse map is injective by symmetry.

*Bijection.* The two injections give equal finite sector counts, so  $m$  is a bijection  $\beta : J(Q) \simeq J(P)$ .  $\square$

## Chapter 11

# Proof of the Fundamental Theorem

For BNT canonical forms  $P$  and  $Q$ , write the weighted direct sums over normal-tensor sectors  $P^i = \bigoplus_j \mu_j P_j^i$  and  $Q^i = \bigoplus_k \nu_k Q_k^i$ . The proportional theorem gives a bijection  $\beta$  of the normal-tensor sectors with unit-modulus phases  $\zeta_k$ , invertible matrices  $X_k$ , and  $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ ; the equal-MPV corollary collapses the per-block phases into a single  $X$  with  $Q_{\text{tot}}^i = X P_{\text{tot}}^i X^{-1}$ . Given the sector matching of Chapter 10, extract the per-sector coefficient identities  $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$  from eventual linear independence, recover the copy weights by finite power-sum comparison, and combine the block gauges into  $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$ .

Throughout, “eventually” abbreviates “for all sufficiently large  $N$ ”: there is  $N_0$  such that the stated relation holds for every  $N > N_0$ . The comparison is made at a fixed such  $N$ , where BNT linear independence detects a vanishing coefficient even when  $|\mu| < 1$  lets it decay.

### 11.1 Coefficient identity, copy weights, and the block-diagonal gauge

The bijective matching theorem of Chapter 10 (Theorem 10.47) gives the sector phases  $\zeta_k$  and block gauges  $X_k$ . The coefficient comparison of [CPGSV16, Appendix MPV proof, lines 1187–1188] then turns the equality of total MPV families into the per-sector coefficient identity  $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ . Neither step uses a per-sector unit-modulus hypothesis.

**Lemma 11.1** (Unit phase from matched normalized BNT blocks). [Lean] [✓ Stmt] *Let  $P$  and  $Q$  be BNT canonical forms. If, for every length  $N$  and word  $\sigma$ , a sector  $P_j$  and a sector  $Q_k$  have MPV families related by*

$$V^{(N)}(Q_k)_\sigma = \zeta^N V^{(N)}(P_j)_\sigma \quad (11.1)$$

*then  $|\zeta| = 1$ .*

*Proof.* [✓ Proof] The BNT normalization gives self-overlap limits of modulus 1 for both sectors. The MPV identity (11.1) gives, for every  $N$ ,

$$O_{Q_k Q_k}(N) = |\zeta|^{2N} O_{P_j P_j}(N).$$

Since  $O_{Q_k Q_k}(N) \rightarrow 1$  and  $O_{P_j P_j}(N) \rightarrow 1$ , taking limits forces  $|\zeta|^{2N} \rightarrow 1$  and hence  $|\zeta| = 1$ .  $\square$

**Theorem 11.2** (Proportional sector matching with unit phases). [Lean] [✓ Stmt] *Let  $P$  and  $Q$  be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection  $\beta : J(Q) \simeq J(P)$ , bond-dimension equalities  $\dim P_{\beta(k)} = \dim Q_k$ , scalars  $\zeta_k \in \mathbb{C}$ , and block gauges  $X_k$  such that  $|\zeta_k| = 1$  and, for every sector  $k$  and physical index  $i$ ,*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1} \quad (11.2)$$

*Equivalently, for every length  $N$  and word  $\sigma$ ,*

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma.$$

*This is [CPGSV16, Appendix MPV theorem statement, lines 1167–1170, and proof line 1182].*

*Proof.* [✓ Proof] Apply the proportional sector-matching theorem to obtain the bijection and dimension-compatible gauge-phase equivalences. Choosing the gauges and scalars gives  $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ , and therefore

$$O_{Q_k Q_k}(N) = |\zeta_k|^{2N} O_{P_{\beta(k)} P_{\beta(k)}}(N).$$

The normalized self-overlap limits give  $O_{Q_k Q_k}(N) \rightarrow 1$  and  $O_{P_{\beta(k)} P_{\beta(k)}}(N) \rightarrow 1$ , so  $|\zeta_k|^{2N} \rightarrow 1$  and hence  $|\zeta_k| = 1$ . The MPV scalar-power identity follows from the same gauge equation (11.2):  $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$ . □

**Lemma 11.3** (Full-basis coefficient identity from matched sectors). [Lean] [✓ Stmt] *Let  $P$  and  $Q$  be BNT canonical forms with the same MPV family, matched by a bijection  $\beta : J(Q) \simeq J(P)$  with bond-dimension equalities and a gauge-phase equivalence between  $Q_k$  and  $P_{\beta(k)}$  for every  $k$ . Then each sector carries a unit-modulus phase  $\zeta_k$ , and eventually*

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q). \quad (11.3)$$

*This is the equal-MPV coefficient comparison of [CPGSV16, Appendix MPV proof, lines 1187–1188].*

*Proof.* [✓ Proof] For each matched sector, the gauge-phase equivalence gives  $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$ . Lemma 11.1 gives  $|\zeta_k| = 1$ . Applying Lemma 11.4 to the same phases gives  $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$  for all sufficiently large  $N$ . □

**Lemma 11.4** (Full-basis coefficient identity from matched phases). [Lean] [✓ Stmt] *Let  $P$  be a BNT canonical form, let  $Q$  be a sector decomposition, and assume  $P_{\text{tot}}$  and  $Q_{\text{tot}}$  have the same MPV family. Suppose that there is a bijection*

$$\beta : J(Q) \simeq J(P)$$

*and scalars  $\zeta_k \in \mathbb{C}$  such that, for every  $k \in J(Q)$ , every length  $N$ , and every spin configuration  $\sigma$  of length  $N$ ,*

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma. \quad (11.4)$$

*Then for every  $k \in J(Q)$  there is  $N_0$  such that, for all  $N > N_0$ ,*

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q).$$

*This is the coefficient-comparison step of [CPGSV16, Appendix MPV proof, lines 1187–1188] in full-basis form.*

*Proof.* ✓ Proof For every  $N$ , expand the equality of total MPVs as

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = \sum_{k \in J(Q)} c_N^{(k)}(Q) |V^{(N)}(Q_k)\rangle.$$

The phase relation (11.4) gives  $|V^{(N)}(Q_k)\rangle = \zeta_k^N |V^{(N)}(P_{\beta(k)})\rangle$ , so reindexing the  $Q$ -sum by  $\beta$  onto the  $P$ -basis yields

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = \sum_{j \in J(P)} \zeta_{\beta^{-1}(j)}^N c_N^{(\beta^{-1}(j))}(Q) |V^{(N)}(P_j)\rangle.$$

For all sufficiently large  $N$  the  $P$ -basis MPV family  $\{|V^{(N)}(P_j)\rangle\}_{j \in J(P)}$  is linearly independent, by the BNT eventual linear independence of  $P$ . Lemma 10.6 then extracts the coefficient equalities  $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$  from this eventual linear independence, for all sufficiently large  $N$ .  $\square$

The coefficient identity (11.3) is now converted, sector by sector, into a matching of the individual copy weights.

**Definition 11.5** (Copy-weight matching for matched BNT sectors). Lean ✓ Stmt Let the sectors of two decompositions  $P$  and  $Q$  be matched by  $\beta : J(Q) \simeq J(P)$ , with sector phases  $\zeta_k$ . A copy-weight matching consists of copy permutations  $\tau_k$  such that, for all sectors  $k$  and copies  $q$ ,

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}.$$

This is the copy-level conclusion of [CPGSV16, Appendix MPV proof, line 1188].

**Lemma 11.6** (Copy-weight matching from matched-sector coefficient identities). Lean ✓ Stmt Let the sectors of  $P$  and  $Q$  be matched by  $\beta : J(Q) \simeq J(P)$ , with nonzero sector phases  $\zeta_k$ . If the matched-sector coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q) \tag{11.5}$$

hold eventually, then the phases  $\zeta_k$  determine a copy-weight matching  $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$ . This is, sector by sector, [CPGSV16, Appendix MPV proof, line 1188].

*Proof.* ✓ Proof Eventually the coefficient identity (11.5) in sector  $k$  reads

$$\sum_r \mu_{\beta(k), r}^N = \zeta_k^N \sum_q \nu_{k,q}^N = \sum_q (\zeta_k \nu_{k,q})^N.$$

Newton–Girard finite power-sum comparison (Lemma 10.43) applies sector by sector, producing a copy permutation  $\tau_k$  with  $\mu_{\beta(k), \tau_k(q)} = \zeta_k \nu_{k,q}$ ; since  $\zeta_k \neq 0$  this is  $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$ .  $\square$

The matched block gauges and copy weights now combine into a single block-diagonal gauge.

**Theorem 11.7** (Global gauge from matched sectors and copy weights). Lean ✓ Stmt Suppose that the sectors of two sector decompositions  $P$  and  $Q$  are matched by a bijection  $\beta : J(Q) \simeq J(P)$ , with bond-dimension equalities, copy permutations  $\tau_k$ , nonzero phases  $\zeta_k$ , and block gauges  $X_k$ . If, for every sector  $k$ , copy  $q$ , and physical index  $i$ ,

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}, \quad \nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}, \tag{11.6}$$

then, in the flattened  $Q$ -coordinates,

$$Q_{\text{flat}}^i = X P_{\text{matched}}^i X^{-1}, \quad X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

This is the direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192], separated from the coefficient-comparison step that supplies the weight identities.

*Proof.* ✓ Proof For each matched copy,

$$\nu_{k,q} Q_k^i = (\zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}) (\zeta_k X_k P_{\beta(k)}^i X_k^{-1}) = \mu_{\beta(k), \tau_k(q)} X_k P_{\beta(k)}^i X_k^{-1}.$$

Thus every flattened weighted block is conjugated by its sector gauge  $X_k$ . Lemma 14.48 combines these block conjugacies into the direct-sum gauge  $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$ .  $\square$

**Theorem 11.8** (Global gauge from a copy-weight matching). Lean ✓ Stmt *Suppose that the sectors of  $P$  and  $Q$  are matched, with bond-dimension equalities, nonzero phases, and block gauges satisfying*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}.$$

*If the same phases also give a copy-weight matching, then the flattened  $Q$ -tensor is obtained from the matched-coordinate  $P$ -tensor by the direct-sum gauge*

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

*This is only a reformulation of Theorem 11.7.*

*Proof.* ✓ Proof The copy-weight matching supplies  $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$ . Together with  $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ , these are exactly the two displayed hypotheses (11.6) of Theorem 11.7.  $\square$

We can now state the proportional global gauge, first from a copy-weight matching and then from the coefficient identities that produce one.

**Theorem 11.9** (Conditional proportional global gauge from a copy-weight matching). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then the proportional sector-matching theorem gives  $\beta$ , the bond-dimension equalities, the unit phases  $\zeta_k$ , and the block gauges  $X_k$ . If the remaining proportional coefficient comparison supplies a copy-weight matching for these same phases, then the flattened  $Q$ -tensor is obtained from the matched-coordinate  $P$ -tensor by the direct-sum gauge*

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

*Proof.* ✓ Proof The proportional sector-matching theorem supplies  $\beta$ ,  $X_k$ , and unit phases  $\zeta_k$  satisfying  $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ . Since  $|\zeta_k| = 1$ , each  $\zeta_k$  is nonzero. A copy-weight matching gives  $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$ , so Theorem 11.8 gives the direct-sum gauge.  $\square$

**Theorem 11.10** (Conditional proportional global gauge from coefficient identities). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection  $\beta : J(Q) \simeq J(P)$ , bond-dimension equalities, unit-modulus phases  $\zeta_k$ , and block gauges  $X_k$  such that*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$$

*for every sector  $k$  and physical index  $i$ . If these same phases satisfy the matched-sector coefficient identities*

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$$

*for all sufficiently large  $N$ , then they determine a copy-weight matching and hence the flattened  $Q$ -tensor is obtained from the matched-coordinate  $P$ -tensor by the direct-sum gauge*

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$



*Proof.* ✓ Proof The sector-matching theorem gives  $\beta$ ,  $X_k$ , and  $|\zeta_k| = 1$  with  $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ . Since  $\zeta_k \neq 0$ , the coefficient identities  $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$  produce  $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$  by Lemma 11.6. Theorem 11.9 then applies to this copy-weight matching.  $\square$

In the equal-MPV case all ingredients are unconditional.

**Lemma 11.11** (Equal-MPV matched copy-weight witnesses). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms whose total tensors have the same MPV family. Then there are a bijection  $\beta : J(Q) \simeq J(P)$ , bond-dimension equalities, unit-modulus phases  $\zeta_k$ , and block gauges  $X_k$  such that*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}.$$

*For the same phases there is a copy-weight matching: after a permutation of the copies inside each matched sector,*

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}.$$

*Proof.* ✓ Proof The full-basis matching theorem gives the bijection and block gauges. The self-overlap comparison

$$\langle V_N(Q_k), V_N(Q_k) \rangle = |\zeta_k|^{2N} \langle V_N(P_{\beta(k)}), V_N(P_{\beta(k)}) \rangle$$

together with the BNT normalization limits on both sides forces  $|\zeta_k| = 1$ . The matched MPV phase identities also give the eventual coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q).$$

Lemma 11.6 applies the finite power-sum comparison in each sector and gives the copy-weight identities.  $\square$

**Theorem 11.12** (Full-basis global gauge in matched coordinates). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms whose total tensors have the same MPV family. Then there exist:*

- a bijection  $\beta : J(Q) \simeq J(P)$ ,
- bond-dimension equalities  $\dim P_{\beta(k)} = \dim Q_k$ ,
- copy-number equalities  $r_{\beta(k)}(P) = r_k(Q)$  and copy permutations  $\tau_k$ ,
- unit-modulus phases  $\zeta_k$  with  $|\zeta_k| = 1$  and block gauges  $X_k$ ,

*such that, for every sector  $k$ , copy  $q$ , and physical index  $i$ ,*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}, \quad \nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}.$$

*Moreover, if*

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$$

*is the flattened block-diagonal gauge, then, for every physical index  $i$ , the flattened  $Q$ -coordinates satisfy*

$$Q_{\text{flat}}^i = X P_{\text{matched}}^i X^{-1}.$$

*This is the explicit direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192].*

*Proof.* ✓ Proof Lemma 11.11 gives  $\beta$ , the bond-dimension identifications, the phases  $\zeta_k$ , the block gauges  $X_k$ , and copy permutations  $\tau_k$  satisfying

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}.$$

Theorem 11.8 combines the block conjugacies into the flattened direct-sum conjugation by  $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$ . □

**Theorem 11.13** (Literal equal-MPV gauge after sector-coordinate permutation). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms whose total tensors have the same MPV family. Then the total bond dimensions agree; write their common value as  $D$ . There is*

$$Y \in \text{GL}_D(\mathbb{C}) \tag{11.7}$$

such that, for every physical index  $i$ ,

$$Q_{\text{tot}}^i = Y P_{\text{tot}}^i Y^{-1},$$

where  $P_{\text{tot}}^i$  and  $Q_{\text{tot}}^i$  are both regarded as matrices in  $M_D(\mathbb{C})$  using the total bond-dimension equality. This is the BNT form of [CPGSV16, Corollary II.2 and Appendix MPV proof, lines 1189–1192].

*Proof.* ✓ Proof Apply Theorem 11.12 to obtain the flattened gauge after the sectors of  $P$  have been reindexed by the bijection between basis blocks of  $P$  and  $Q$ . The sector-coordinate permutation  $\rho$ , induced by this bijection, the copy permutations, and the bond-dimension equalities, identifies the reindexed  $P$ -side direct sum with the total tensor  $P_{\text{tot}}$ , transported along the total bond-dimension equality. Composing the permutation gauge for  $\rho$  with the sector-permuted gauge gives the single matrix  $Y$  of (11.7). □

## 11.2 The equal and proportional cases

**Theorem 11.14** (Multi-block fundamental theorem for normal tensors). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms (Definition 10.27), with bases of normal tensors  $(P_j)_{j=1}^{g_a}$  and  $(Q_k)_{k=1}^{g_b}$ . If the total tensors  $P_{\text{tot}}, Q_{\text{tot}}$  generate eventually nonzero proportional MPV families — for all sufficiently large  $N$  there is a nonzero scalar  $c_N$  with  $V^{(N)}(P_{\text{tot}}) = c_N V^{(N)}(Q_{\text{tot}})$  — then  $g_a = g_b$ . After relabelling the basis of normal tensors of  $P$  by a bijection  $\beta : J(Q) \simeq J(P)$ , there are scalars  $\zeta_k \in \mathbb{C}$  with  $|\zeta_k| = 1$  and invertible matrices  $Y_k$  such that, for every  $k$  and every physical index  $i$ ,*

$$Q_k^i = \zeta_k Y_k P_{\beta(k)}^i Y_k^{-1}.$$

Thus the paired normal tensors have the same bond dimension ( $\dim P_{\beta(k)} = \dim Q_k$ ) and the same gauge-phase class.

*Proof.* ✓ Proof The proportional sector-matching Theorem 11.2, applied to the BNT canonical forms  $P$  and  $Q$ , supplies the bijection  $\beta$ , the bond-dimension equalities, the unit phases  $\zeta_k$ , and the block gauges  $Y_k$  with  $Q_k^i = \zeta_k Y_k P_{\beta(k)}^i Y_k^{-1}$ ; the matching needs no per-sector unit-modulus hypothesis. □

This is [CPGSV16, Theorem 2.10]; see also [CPGSV21, Theorem IV.4 and Corollary IV.5]. The statement is given on the BNT canonical-form surface (Definition 10.27), which is the paper’s “ $A$  and  $B$  in canonical form” with chosen bases of normal tensors; the reduction of an arbitrary proportional-MPV pair to a common primitive block decomposition is Theorem 9.49 of Chapter 9.

**Theorem 11.15** (Equal BNT canonical forms are globally conjugate). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms (Definition 10.27) whose total tensors  $P_{\text{tot}}, Q_{\text{tot}}$  generate the same MPV family for every system size  $N$ . Then their total bond dimensions coincide; writing the common value as  $D$ , there is one invertible matrix  $X \in \text{GL}_D(\mathbb{C})$  such that, for every physical index  $i$ ,*

$$Q_{\text{tot}}^i = X P_{\text{tot}}^i X^{-1}.$$

*This is [CPGSV16, Corollary 2.11] on the BNT canonical-form surface; the canonical form of [CPGSV16, Section 2.3], a direct sum of normal tensors with scalar weights, is a BNT canonical form. The reduction of an arbitrary pair of same-MPV tensors to a common BNT canonical form is Chapter 9.*

*Proof.* ✓ Proof This is Theorem 11.13. The BNT comparison of Chapter 10 provides the basis relabelling, the per-block gauge-phase equivalences, and the copy-weight multiset matching, which combine into the block-diagonal global gauge. The exact linear-independence matching needs no per-sector unit-modulus hypothesis, and the BNT canonical form needs no one-site injectivity. □

This is also [CPGSV21, Corollary IV.5].

## Chapter 12

# Symmetries and String Order

This chapter collects the symmetry-theoretic consequences of the MPS Fundamental Theorem [PGVWC07], following the symmetry analysis of [PGWS<sup>+</sup>08]. For injective tensors, physical on-site symmetries determine virtual gauges, those gauges define projective representations on the bond space, and virtual intertwiners enter the string-order criteria.

### 12.1 Physical Symmetries Give Virtual Gauges

**Definition 12.1** (Physical-index rotation). [Lean](#) [✓ Stmt](#) Given a matrix  $M \in M_d(\mathbb{C})$  and an MPS tensor  $A$ , the *physical-index rotation* is the tensor  $(A_M)^i := \sum_j M_{ij} A^j$ . When the matrix is a unitary  $u$ , the rotation is written  $A_u$ .

**Corollary 12.2** (Periodic symmetry yields gauge equivalence up to a root of unity). [Lean](#) [✓ Stmt](#) Suppose that for any two tensors in irreducible form II with the same MPV family there is  $m > 0$  making them gauge equivalent up to an  $m$ -th root of unity (Theorem 13.24). Let  $A$  be in irreducible form II and let  $M$  be a matrix such that  $B := A_M$  is also in irreducible form II and  $\mathcal{V}(A) = \mathcal{V}(B)$ . Then there is  $m > 0$  such that  $A$  and  $B$  are gauge equivalent up to an  $m$ -th root of unity.

*Proof.* [✓ Proof](#) Apply the assumed equivalence (Theorem 13.24) to  $A$  and  $B$ , which are in irreducible form II with  $\mathcal{V}(A) = \mathcal{V}(B)$ . □

### 12.2 Virtual Representation Theorem

With a full group of on-site symmetries (rather than just a single unitary), the resulting gauges determine a projective representation on the bond space.

**Definition 12.3** (Twisted tensor). [Lean](#) [✓ Stmt](#) Given a group representation  $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$  on the physical index, the  *$g$ -twisted tensor* is defined, for each  $i \in \{0, \dots, d-1\}$ , by

$$\tilde{A}_g^i := \sum_{j=0}^{d-1} U(g)_{ij} A^j.$$

**Definition 12.4** (On-site symmetry). [Lean](#) [✓ Stmt](#) A tensor  $A$  is *on-site symmetric* under a group representation  $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$  if  $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$  for every  $g \in G$ .

**Lemma 12.5** (Functoriality of twisting). [Lean](#) [✓ Stmt](#) *The twisted tensor satisfies the composition law*

$$\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)}_g,$$

i.e. twisting by  $gh$  is the same as first twisting by  $h$  and then by  $g$ .

*Proof.* [✓ Proof](#) Expanding both sides using  $U(gh) = U(g)U(h)$ , then swapping the order of summation.  $\square$

**Lemma 12.6** (Identity twist). [Lean](#) [✓ Stmt](#) *Twisting by the identity group element is trivial:  $\widetilde{A}_1 = A$ .*

*Proof.* [✓ Proof](#) Since  $U(1) = \mathbb{1}_d$ , each component reduces to  $\sum_j \delta_{ij} A^j = A^i$ .  $\square$

**Lemma 12.7** (Symmetric twists are gauge equivalent). [Lean](#) [✓ Stmt](#) *If  $A$  is injective and on-site symmetric under  $U$ , then for every  $g \in G$  the twisted tensor  $\widetilde{A}_g$  is gauge equivalent to  $A$ .*

*Proof.* [✓ Proof](#) On-site symmetry gives  $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$ , and the single-block Fundamental Theorem turns this equality of MPV families into gauge equivalence.  $\square$

*Remark 12.8* (Fundamental-Theorem route for injective symmetry). The injective virtual-representation theorem is not proved from string-order rigidity. The on-site symmetry condition first identifies  $A$  and  $\widetilde{A}_g$  as the same MPV family; the single-block Fundamental Theorem gives the gauge in Lemma 12.7. Gauge uniqueness and the twist composition law then give Theorem 12.15. The string-order results later in this chapter use this virtual representation as an input.

**Theorem 12.9** (Virtual symmetry equation). [Lean](#) [✓ Stmt](#) *If  $A$  is injective and on-site symmetric under  $U$ , then for every  $g \in G$  there exist an invertible matrix  $X(g) \in \text{GL}_D(\mathbb{C})$  and a nonzero scalar  $\varphi(g) \in \mathbb{C}^\times$  (in fact  $\varphi = 1$  in the single-block case) such that  $\widetilde{A}_g^i = \varphi(g) X(g) A^i X(g)^{-1}$  for all  $i$ .*

*Proof.* [✓ Proof](#) The physical action is transferred to the virtual level by the same local move that replaces the twisted tensor by a gauge-conjugated one:



Apply Lemma 12.7 to obtain  $X(g)$ , then set  $\varphi(g) = 1$ .  $\square$

**Lemma 12.10** (Gauge uniqueness up to scalar). [Lean](#) [✓ Stmt](#) *Let  $A$  be an injective tensor. If  $X, Y \in \text{GL}_D(\mathbb{C})$  both satisfy  $B^i = X A^i X^{-1} = Y A^i Y^{-1}$  for all  $i$ , then there exists a nonzero scalar  $u \in \mathbb{C}^\times$  such that  $Y = u \cdot X$ .*

*Proof.* [✓ Proof](#) The matrix  $Z = Y^{-1}X$  commutes with every  $A^i$ . Since  $A$  is injective,  $\{A^i\}$  spans  $M_D(\mathbb{C})$ , so  $Z$  lies in the centre of  $M_D(\mathbb{C})$ . By Lemma 14.12,  $Z$  is a scalar matrix. Since  $Z$  is invertible, that scalar is nonzero, and  $Y = u \cdot X$ .  $\square$

**Theorem 12.11** (Gauge uniqueness from equality of MPV families). [Lean](#) [✓ Stmt](#) *Let  $A$  be an injective tensor and assume that  $A$  and  $B$  generate the same MPV family. If  $X, Y \in \text{GL}_D(\mathbb{C})$  both satisfy  $B^i = X A^i X^{-1} = Y A^i Y^{-1}$  for all  $i$ , then there is a nonzero scalar  $u \in \mathbb{C}^\times$  such that  $Y = u \cdot X$ .*

*Proof.* ✓ Proof This is exactly Lemma 12.10; the extra MPV hypothesis specifies the setting in which the two gauges are usually produced.  $\square$

**Definition 12.12** (Scalar 2-cochain). Lean ✓ Stmt A *scalar 2-cochain* on a group  $G$  is a function  $\omega : G \times G \rightarrow \mathbb{C}^\times$ .

**Definition 12.13** (Projective representation). Lean ✓ Stmt Given a scalar 2-cochain  $\omega$ , a *projective representation* of  $G$  on  $\mathbb{C}^D$  is a map  $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$  satisfying, for all  $g, h \in G$ ,

$$\rho(g) \rho(h) = \omega(g, h) \rho(gh).$$

**Lemma 12.14** (Associativity forces the cocycle condition). Lean ✓ Stmt If  $\rho$  is a projective representation with factor system  $\omega$  and  $D > 0$ , then  $\omega$  satisfies the 2-cocycle identity:

$$\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k).$$

*Proof.* ✓ Proof Expanding  $\rho(g) (\rho(h) \rho(k))$  and  $(\rho(g) \rho(h)) \rho(k)$  using the projective multiplication law and equating via matrix associativity. The hypothesis  $D > 0$  allows cancellation of the invertible matrix factor.  $\square$

**Theorem 12.15** (Virtual representation theorem for injective MPS). Lean ✓ Stmt Let  $A$  be an injective MPS tensor that is on-site symmetric under a group representation  $U : G \rightarrow \text{GL}_d(\mathbb{C})$ . Then there exist:

1. a scalar 2-cochain  $\omega : G \times G \rightarrow \mathbb{C}^\times$ , and
2. a map  $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$  satisfying, for all  $g, h \in G$ ,

$$\rho(g) \rho(h) = \omega(g, h) \rho(gh).$$

such that, defining the gauge matrices  $X(g) := \rho(g^{-1})$ , one has, for every  $g \in G$  and  $i \in \{0, \dots, d-1\}$ ,

$$\tilde{A}_g^i = X(g) A^i X(g)^{-1}.$$

In tensor-network notation, for each fixed  $g$  this is the local gauge relation



with the black node denoting the tensor named in the surrounding formula and the two red side nodes acting on the virtual legs. In particular,  $\rho$  is a projective representation of  $G$  on the bond space  $\mathbb{C}^D$ .

*Proof.* ✓ Proof Since  $A$  is injective and on-site symmetric, each twisted tensor  $\tilde{A}_g$  is gauge equivalent to  $A$  (Lemma 12.7). Choose  $X(g) \in \text{GL}_D(\mathbb{C})$  with  $\tilde{A}_g^i = X(g) A^i X(g)^{-1}$  for all  $i$ .

By functoriality (Lemma 12.5), the product  $X(h) X(g)$  also intertwines  $A$  with  $\tilde{A}_{gh}$ . Since  $X(gh)$  does the same, gauge uniqueness (Lemma 12.10) gives a nonzero scalar  $\omega'(h, g)$  with  $X(h) X(g) = \omega'(h, g) X(gh)$ .

Defining  $\rho(g) := X(g^{-1})$  converts this to the standard law:  $\rho(g) \rho(h) = \omega(g, h) \rho(gh)$  where  $\omega(g, h) := \omega'(h^{-1}, g^{-1})$ .  $\square$

**Corollary 12.16** (2-cocycle identity for the factor system). Lean ✓ Stmt If  $D > 0$ , the factor system  $\omega$  from Theorem 12.15 satisfies the 2-cocycle identity for all  $g, h, k \in G$ :

$$\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k).$$

*Proof.* ✓ Proof Theorem 12.15 gives  $\rho$  and  $\omega$ . By Lemma 12.14, the cocycle identity follows from associativity of matrix multiplication and invertibility of the representation matrices.  $\square$

## 12.3 Cohomology Classes of Virtual Symmetry Cocycles

Two cocycles may differ by a coboundary yet represent the same projective-representation class. The following definitions and results make this precise and establish the quotient  $H^2(G, \mathbb{C}^\times)$ .

**Definition 12.17** (Coboundary). [Lean](#) [✓ Stmt](#) A scalar 2-cocycle  $\omega$  is a *coboundary* if there exists  $\varphi: G \rightarrow \mathbb{C}^\times$  such that, for all  $g, h \in G$ ,

$$\omega(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}.$$

**Definition 12.18** (Cohomologous cocycles). [Lean](#) [✓ Stmt](#) Two cocycles  $\omega_1, \omega_2$  are *cohomologous* (written  $\omega_1 \sim \omega_2$ ) if there exists  $\varphi: G \rightarrow \mathbb{C}^\times$  such that, for all  $g, h \in G$ ,

$$\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h).$$

**Theorem 12.19** (Cohomology relation is an equivalence relation). [Lean](#) [✓ Stmt](#) The relation “cohomologous to” is reflexive, symmetric, and transitive on the set of scalar 2-cocycles.

*Proof.* [✓ Proof](#) Reflexivity: take  $\varphi = 1$ . Symmetry: replace  $\varphi$  by  $\varphi^{-1}$ . Transitivity: compose the witnesses  $\varphi$  and  $\psi$  pointwise.  $\square$

**Lemma 12.20** (Coboundary iff cohomologous to the trivial cocycle). [Lean](#) [✓ Stmt](#) A cocycle  $\omega$  is a coboundary if and only if  $\omega \sim 1$ , where  $1(g, h) = 1$  for all  $g, h$ .

*Proof.* [✓ Proof](#) Both directions follow by absorbing or introducing the factor  $1(g, h) = 1$  via  $x \cdot 1 = x$ .  $\square$

**Theorem 12.21** (Gauge independence of the cocycle class). [Lean](#) [✓ Stmt](#) If the bond dimension satisfies  $D \geq 1$  and two projective representations  $\rho_1, \rho_2$  (with cocycles  $\omega_1, \omega_2$ ) both arise as virtual representations of the same injective MPS tensor  $A$  under an on-site symmetry  $U$ , then  $\omega_1 \sim \omega_2$ .

*Proof.* [✓ Proof](#) For each  $g$ , gauge uniqueness gives a nonzero scalar  $f(g)$  with  $X_2(g^{-1}) = f(g) X_1(g^{-1})$ . Equivalently, for each fixed  $g$  the two local gauges

$$\begin{array}{c} i \\ | \\ \text{---} \bullet \text{---} \end{array} \begin{array}{c} X_1(g^{-1}) \\ \bullet \\ X_1(g^{-1})^{-1} \end{array} \quad \text{and} \quad \begin{array}{c} i \\ | \\ \text{---} \bullet \text{---} \end{array} \begin{array}{c} X_2(g^{-1}) \\ \bullet \\ X_2(g^{-1})^{-1} \end{array}$$

describe the same twisted tensor, so they differ by a scalar. Substituting into the projective law and cancelling  $X_1(g \cdot h)$  yields  $\omega_2(g, h) = f(g)f(h)f(gh)^{-1}\omega_1(g, h)$ .  $\square$

**Definition 12.22** (Multiplicative 2-cocycle condition). [Lean](#) [✓ Stmt](#) A scalar 2-cochain  $\omega: G \times G \rightarrow \mathbb{C}^\times$  is a *2-cocycle* if it satisfies, for all  $g, h, k \in G$ ,

$$\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k).$$

**Definition 12.23** (Second cohomology quotient). [Lean](#) [✓ Stmt](#) The second cohomology set is the quotient

$$H^2(G, \mathbb{C}^\times) := Z^2(G, \mathbb{C}^\times)/\sim,$$

where  $\sim$  is the coboundary-equivalence relation.

**Definition 12.24** (Projective equivalence). [Lean](#) [✓ Stmt](#) Two projective representations are *projectively equivalent* when their factor systems are cohomologous.

**Theorem 12.25** (Projective equivalence and cohomologous cocycles). [Lean](#) [✓ Stmt](#) Let  $\rho_1, \rho_2$  be projective representations with factor systems  $\omega_1, \omega_2$ . Then

$$\rho_1 \sim_{\text{proj}} \rho_2 \iff \omega_1 \sim \omega_2.$$

*Proof.* [✓ Proof](#) This is immediate from the definition of projective equivalence as cohomology-class equality for factor systems.  $\square$

### 12.3.1 Non-triviality via the commutator phase

For an abelian symmetry group the cohomology class of a factor system is detected by an anti-symmetric pairing, which gives a computable test for non-triviality.

**Definition 12.26** (Commutator phase). [Lean](#) [✓ Stmt](#) The *commutator phase* of a 2-cochain  $\omega$  at a pair  $g, h$  is  $\beta_\omega(g, h) := \omega(g, h) \omega(h, g)^{-1}$ . It is defined for any  $\mathbb{C}^\times$ -valued 2-cochain, the cocycle condition is not needed.

**Theorem 12.27** (The commutator phase is a class invariant). [Lean](#) [✓ Stmt](#) If  $\omega_1 \sim \omega_2$  and  $gh = hg$ , then  $\beta_{\omega_1}(g, h) = \beta_{\omega_2}(g, h)$ .

*Proof.* [✓ Proof](#) Write  $\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h)$  for a coboundary witness  $\varphi$ . Substituting into the ratio and using  $gh = hg$  and the commutativity of  $\mathbb{C}^\times$ ,

$$\beta_{\omega_1}(g, h) = \frac{\varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h)}{\varphi(h)\varphi(g)\varphi(hg)^{-1}\omega_2(h, g)} = \frac{\omega_2(g, h)}{\omega_2(h, g)} = \beta_{\omega_2}(g, h),$$

the  $\varphi$ -factors cancelling since  $\varphi(gh) = \varphi(hg)$ .  $\square$

**Definition 12.28** (Non-trivial cohomology class). [Lean](#) [✓ Stmt](#) A factor system  $\omega$  has a *non-trivial class* when it is not cohomologous to the constant cocycle 1.

**Theorem 12.29** (Commutator-phase non-triviality test). [Lean](#) [✓ Stmt](#) If a commuting pair  $g, h$  has  $\beta_\omega(g, h) \neq 1$ , then  $\omega$  has a non-trivial class.

*Proof.* [✓ Proof](#) The trivial cocycle has commutator phase 1 at every pair, so by Theorem 12.27 any cocycle cohomologous to it has  $\beta = 1$  at every commuting pair. Contrapositively,  $\beta_\omega(g, h) \neq 1$  for one commuting pair forces  $\omega$  to be non-cohomologous to the trivial cocycle.  $\square$

## 12.4 Permutation-Based Physical Symmetry

The linear-combination twist  $\tilde{A}_g^i = \sum_j U(g)_{ij} A^j$  is the standard form used in the virtual representation theorem. The following alternative formulation replaces the matrix action by a permutation of the physical index; it is useful for symmetries that act by reordering basis states rather than by a general linear map.

**Definition 12.30** (Permutation symmetry datum). [Lean](#) [✓ Stmt](#) An *on-site symmetry datum* for a group  $G$  on a physical space of dimension  $d$  consists of a matrix-valued map  $u : G \rightarrow M_d(\mathbb{C})$  and a physical-index action  $\sigma : G \rightarrow \text{End}(\{0, \dots, d-1\})$ .

**Definition 12.31** (Permutation-twisted tensor). [Lean](#) [✓ Stmt](#) Given an on-site symmetry datum  $(u, \sigma)$ , the *permutation-twisted tensor* at group element  $g$  is

$$(A^{\sigma_g})^i := A^{\sigma(g)(i)}.$$

In tensor-network notation the local tensor is unchanged and only the physical leg is relabelled:





**Lemma 12.32** (Identity permutation twist). [Lean](#) [✓ Stmt](#) If  $\sigma(1) = \text{id}$ , then  $(A^{\sigma_1})^i = A^i$  for all  $i$ .

*Proof.* [✓ Proof](#) Immediate from  $\sigma(1)(i) = i$  for all  $i$ . □

*Composition order.* The linear-combination twist (Lemma 12.5) composes as  $\widetilde{(\widetilde{A}_h)_g} = \widetilde{A}_{gh}$ , applying  $h$  first then  $g$ . The permutation twist below uses the same left-action convention  $\sigma(gh) = \sigma(g) \circ \sigma(h)$ , so  $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$ ; the inner permutation  $\sigma(h)$  acts first on the index.

**Lemma 12.33** (Composition of permutation twists). [Lean](#) [✓ Stmt](#) If  $\sigma(gh) = \sigma(g) \circ \sigma(h)$  for all  $g, h \in G$ , then

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}.$$

*Proof.* [✓ Proof](#) Expanding both sides:  $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$ . Diagrammatically this is just two consecutive relabellings of the same local tensor:



□

## 12.5 String Order and Local Symmetry Equivalence

The string order parameter [PGWS<sup>+</sup>08] detects whether an on-site unitary  $u$  acts as a local symmetry on the state generated by an injective MPS tensor.

**Definition 12.34** (Twisted transfer map). [Lean](#) [✓ Stmt](#) Given an MPS tensor  $A$  and a physical-index matrix  $u$ , the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n,n'} \langle n' | u | n \rangle A_n X A_{n'}^\dagger.$$

Diagrammatically one inserts the physical operator  $u$  on the doubled local transfer cell:



The upper and lower black nodes denote  $A_n$  and  $A_{n'}^\dagger$ , and the red node labelled  $u$  couples the physical legs.

**Definition 12.35** (String order parameter). [Lean](#) [✓ Stmt](#) The *string order parameter* for twist  $u$  and stationary state  $\Lambda$  at length  $L$  is

$$R_L(u) := \text{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})).$$

In tensor-network notation this is the length- $L$  doubled chain with a copy of  $u$  inserted on each physical rung:



The boundary matrix  $\Lambda$  is contracted into the virtual boundary, the red nodes labelled  $u$  sit on the physical rungs, and the right boundary is traced.

**Definition 12.36** (Boundary string order parameter). [Lean](#) [✓ Stmt](#) For a boundary state  $\Lambda$  and boundary matrices  $X, Y \in M_D(\mathbb{C})$ , the *boundary string order parameter* is

$$R_L(u; X, Y) := \text{tr}(\Lambda X \mathcal{E}_u^L(Y)).$$

The ordinary string order parameter is the special case  $X = Y = \mathbb{1}$ .

**Definition 12.37** (Local symmetry). [Lean](#) [✓ Stmt](#) An MPS tensor  $A$  has *local symmetry* under a unitary  $u$  with respect to a boundary state  $\Lambda$  if there exist a unitary matrix  $V$  and a phase  $\mu$  with  $|\mu| = 1$  such that  $V^\dagger \Lambda V = \Lambda$  and, for every physical index  $i$ ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger.$$

The unitary  $V$  intertwines the physical action of  $u$  with a virtual conjugation, and the boundary state  $\Lambda$  is invariant under  $V$ .

**Definition 12.38** (String order). [Lean](#) [✓ Stmt](#) String order *exists* for  $u$  with respect to a boundary state  $\Lambda$  if there exist boundary matrices  $X, Y$  and a positive constant  $c > 0$  such that  $c \leq \|R_L(u; X, Y)\|$  for all  $L$ , where  $R_L(u; X, Y)$  is the boundary-refined string-order parameter. Thus some boundary-refined string-order sequence does not decay to zero.

**Lemma 12.39** (String order prevents decay to zero). [Lean](#) [✓ Stmt](#) If  $A$  has string order for  $u$  with respect to a boundary state  $\Lambda$ , then for some boundary matrices  $X, Y$  the sequence  $R_L(u; X, Y)$  does not tend to 0 as  $L \rightarrow \infty$ .

*Proof.* [✓ Proof](#) A uniform lower bound  $c \leq \|R_L(u; X, Y)\|$  is incompatible with convergence to 0.  $\square$

**Definition 12.40** (Local physical intertwining). [Lean](#) [✓ Stmt](#) The local physical intertwining relation is  $\sum_j u_{ij} A^j = V A^i V^\dagger$ . Locally, the physical action of  $u$  can therefore be pushed to the virtual legs:



**Definition 12.41** (Transfer-map covariance). [Lean](#) [✓ Stmt](#) Transfer-map covariance is the identity that the transfer map commutes with virtual conjugation:  $\mathcal{E}(V X V^\dagger) = V \mathcal{E}(X) V^\dagger$ . In doubled-layer notation, the virtual operators slide through the local transfer cell:



**Definition 12.42** (Commutation in the doubled transfer picture). [Lean](#) [✓ Stmt](#) In the transfer-map language, the commutation of the doubled transfer matrix  $E = \sum_i A_i \otimes \bar{A}_i$  with  $V \otimes \bar{V}$  is

$$V \mathcal{E}_1(X) V^\dagger = \mathcal{E}_1(VXV^\dagger).$$

**Theorem 12.43** (Transfer-map covariance and doubled commutation are equivalent). [Lean](#) [✓ Stmt](#) *Transfer-map covariance and commutation in the doubled transfer picture are equivalent for any matrix  $V$ .*

*Proof.* [✓ Proof](#) Both sides express the same local picture with opposite orientation:



The two formulas are the same identity read from opposite sides.  $\square$

**Lemma 12.44** (Unitary Kraus mixing). [Lean](#) [✓ Stmt](#) *If  $u$  is unitary, then replacing Kraus operators  $\{A_i\}$  by their unitary mixtures  $\{\sum_j u_{ij} A_j\}$  preserves the channel:*

$$\sum_i \left( \sum_j u_{ij} A_j \right) Y \left( \sum_j u_{ij} A_j \right)^\dagger = \sum_i A_i Y A_i^\dagger.$$

*Proof.* [✓ Proof](#) This is exactly the usual invariance of a Kraus decomposition under a unitary change of Kraus operators; see Theorem 19.20.  $\square$

**Theorem 12.45** (Local physical intertwining implies transfer-map covariance). [Lean](#) [✓ Stmt](#) *If  $V$  and  $u$  are unitary and the local physical intertwining relation holds, then the transfer map is covariant.*

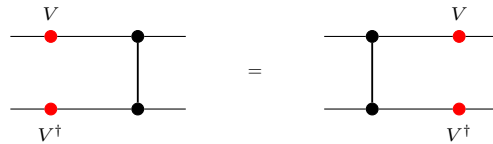
*Proof.* [✓ Proof](#) Starting from



one inserts  $V^\dagger V = \mathbb{1}$  so that each local transfer cell is written in terms of conjugated Kraus operators, then replaces those operators using the intertwining relation. After that, the physical  $u$ -boxes disappear by the unitary Kraus-mixing identity.  $\square$

**Theorem 12.46** (Transfer-map covariance implies local physical intertwining). [Lean](#) [✓ Stmt](#) *For an injective MPS, if  $V$  is unitary and the transfer map is covariant, then there exists a unitary  $u$  satisfying the local physical intertwining relation.*

*Proof.* [✓ Proof](#) Set  $B^i := V A^i V^\dagger$ . The transfer-map covariance relation



implies that the two Kraus families  $B$  and  $A$  define the same transfer map. Kraus freedom for equal channels converts  $\mathcal{E}_B = \mathcal{E}_A$  into a unitary Kraus mixing matrix  $u$  with  $B^i = \sum_j u_{ij} A^j$ .  $\square$

**Definition 12.47** (Twisted companion tensor). [Lean](#) [✓ Stmt](#) For an MPS tensor  $A$  and a physical-index matrix  $u$ , define the companion tensor  $B_u$  by

$$B_u^n := \sum_{n'=0}^{d-1} \overline{u_{n'n}} A^{n'}.$$

**Lemma 12.48** (Twisted transfer as a mixed transfer operator). [Lean](#) [✓ Stmt](#) The twisted transfer map of  $A$  is the mixed transfer operator of the pair  $(A, B_u)$ :

$$\mathcal{E}_u = F_{A, B_u}.$$

*Proof.* [✓ Proof](#) Expand both definitions and rearrange the double sum.  $\square$

**Theorem 12.49** (Twisted-transfer eigenvalues have modulus at most one). [Lean](#) [✓ Stmt](#) Let  $A$  be an injective MPS tensor with  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ , and let  $u$  satisfy  $uu^\dagger = \mathbb{1}$ . If

$$\mathcal{E}_u(X) = \lambda X$$

for some nonzero  $X \in M_D(\mathbb{C})$ , then  $|\lambda| \leq 1$ .

*Proof.* [✓ Proof](#) By Lemma 12.48, the twisted transfer map is the mixed transfer operator  $F_{A, B_u}$ . Because  $B_u$  is obtained from  $A$  by unitary mixing of the Kraus operators, the transfer maps of  $A$  and  $B_u$  coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. Theorem 7.14 then applies to the mixed transfer operator in that gauge, and similarity preserves the eigenvalue  $\lambda$ .  $\square$

**Theorem 12.50** (Modulus-one twisted eigenvalue implies gauge-phase equivalence). [Lean](#) [✓ Stmt](#) Let  $A$  be an injective MPS tensor with  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ , and let  $u$  satisfy  $uu^\dagger = \mathbb{1}$ . If

$$\mathcal{E}_u(X) = \lambda X \tag{12.1}$$

for some nonzero  $X \in M_D(\mathbb{C})$  with  $|\lambda| = 1$ , then  $A$  is gauge-phase equivalent to the companion tensor  $B_u$ .

*Proof.* [✓ Proof](#) By Lemma 12.48, the twisted transfer map is the mixed transfer operator  $F_{A, B_u}$ . Because  $B_u$  is obtained from  $A$  by unitary mixing of the Kraus operators, the transfer maps of  $A$  and  $B_u$  coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. In that gauge the eigenvalue equation (12.1) becomes a modulus-one peripheral eigenvalue for an irreducible mixed transfer operator, so Theorem 7.29 yields gauge-phase equivalence. Undoing the common trace-preserving gauge gives the claim for  $A$  and  $B_u$ .  $\square$

**Lemma 12.51** (Spectral radius  $< 1$  forces boundary-string decay). [Lean](#) [✓ Stmt](#) If  $\rho(\mathcal{E}_u) < 1$ , then, as  $L \rightarrow \infty$ , for all boundary matrices  $X, Y, \Lambda \in M_D(\mathbb{C})$  one has

$$R_L(u; X, Y) \rightarrow 0.$$

*Proof.* [✓ Proof](#) In finite dimension,  $\rho(\mathcal{E}_u) < 1$  implies  $\mathcal{E}_u^L \rightarrow 0$  in operator norm. Composing with the fixed trace pairing  $Z \mapsto \text{tr}(\Lambda X Z)$  gives the stated scalar convergence.  $\square$

**Theorem 12.52** (String order gives gauge-phase equivalence with the companion tensor). [Lean](#) [✓ Stmt](#) *Let  $A$  be injective, let  $\Lambda \in M_D(\mathbb{C})$ , and assume  $uu^\dagger = \mathbb{1}$  and  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ . If string order exists for  $u$  with boundary parameter  $\Lambda$ , then  $A$  is gauge-phase equivalent to the companion tensor  $B_u$ .*

*Proof.* [✓ Proof](#) If  $\rho(\mathcal{E}_u) < 1$ , Lemma 12.51 forces every boundary string-order sequence to decay to zero, contradicting Definition 12.38. Hence  $\rho(\mathcal{E}_u) \geq 1$ . Theorem 12.49 shows that every eigenvalue has modulus at most 1, so some peripheral eigenvalue has modulus exactly 1. Applying Theorem 12.50 then gives the gauge-phase equivalence.  $\square$

**Theorem 12.53** (Gauge-phase equivalence with the companion tensor yields a virtual unitary). [Lean](#) [✓ Stmt](#) *Let  $A$  be injective, assume  $uu^\dagger = \mathbb{1}$  and  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ . If  $A$  is gauge-phase equivalent to the companion tensor  $B_u$ , then there exist a unitary  $V$  and a phase  $\mu$  with  $|\mu| = 1$  such that, for every physical index  $i$ ,*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger. \quad (12.2)$$

*Proof.* [✓ Proof](#) Start from an invertible intertwiner  $X$  between  $A$  and  $B_u$ . The positive matrix  $Q := XX^\dagger$  is a positive fixed point of  $\mathcal{E}_A$ . By uniqueness of positive fixed points for an injective normalized tensor,  $Q$  is a positive scalar multiple of the identity. Rescaling  $X$  by the square root of that scalar produces a unitary  $V$ , and the original intertwining relation becomes the phased covariance equation (12.2), with  $|\mu| = 1$ .  $\square$

**Theorem 12.54** (Virtual unitary gives a twisted-transfer eigenmatrix). [Lean](#) [✓ Stmt](#) *Assume  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$  and that, for every physical index  $i$ ,*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger$$

*with  $V$  unitary. Then  $V$  is an eigenmatrix of the twisted transfer map:*

$$\mathcal{E}_u(V) = \mu V.$$

*Proof.* [✓ Proof](#) Expand  $\mathcal{E}_u(V)$ , insert the intertwining relation into each local term, and use  $V^\dagger V = \mathbb{1}$  together with  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ .  $\square$

**Theorem 12.55** (Boundary-state invariance of the virtual unitary). [Lean](#) [✓ Stmt](#) *Let  $A$  be injective, assume  $uu^\dagger = \mathbb{1}$ , and let  $\Lambda$  be a positive definite boundary state with  $\text{tr}(\Lambda) = 1$  and  $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$ . If  $V$  is a unitary and  $\mu$  a phase satisfying, for every physical index  $i$ ,*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger, \quad (12.3)$$

*then*

$$V^\dagger \Lambda V = \Lambda.$$

*Proof.* [✓ Proof](#) Relation (12.3) shows that  $V^\dagger \Lambda V$  is another positive fixed point of the adjoint transfer map. Since  $\Lambda$  is the normalized positive fixed point, uniqueness forces  $V^\dagger \Lambda V = \Lambda$ .  $\square$

**Lemma 12.56** (Local symmetry implies string order). [Lean](#) [✓ Stmt](#) *Assume  $\text{tr}(\Lambda) = 1$  and  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ . If  $u$  is a local symmetry of  $A$  with respect to  $\Lambda$ , then string order exists for  $u$ .*

*Proof.* ✓ Proof Choose the boundary matrices  $X = V^\dagger$  and  $Y = V$ , where  $V$  is the unitary from the local-symmetry datum. By Theorem 12.54, one has  $\mathcal{E}_u^L(V) = \mu^L V$ . Therefore

$$R_L(u; V^\dagger, V) = \mu^L \text{tr}(\Lambda) = \mu^L,$$

whose norm is constantly 1. □

**Theorem 12.57** (Local symmetry iff a unitary peripheral eigenmatrix exists). Lean ✓ Stmt *Let  $A$  be injective, and assume  $uu^\dagger = \mathbb{1}$ ,  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ ,  $\Lambda$  is positive definite,  $\text{tr}(\Lambda) = 1$ , and  $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$ . Then  $u$  is a local symmetry if and only if there exist a unitary  $V \in M_D(\mathbb{C})$  and a phase  $\mu$  with  $|\mu| = 1$  such that*

$$\mathcal{E}_u(V) = \mu V.$$

*By Theorem 12.49, this witness form is equivalent to the peripheral-spectrum condition  $\rho(\mathcal{E}_u) = 1$ .*

*Proof.* ✓ Proof The forward direction is Theorem 12.54. For the reverse direction, start from a unitary peripheral eigenmatrix. Theorem 12.50 gives gauge-phase equivalence with the companion tensor; then Theorem 12.53 yields a virtual unitary and phase satisfying the local covariance relation, and Theorem 12.55 gives the boundary-state invariance needed for local symmetry. □

**Theorem 12.58** (String order  $\Leftrightarrow$  local symmetry). Lean ✓ Stmt *For an injective MPS, assume  $uu^\dagger = \mathbb{1}$ ,  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ ,  $\Lambda$  is positive definite,  $\text{tr}(\Lambda) = 1$ , and  $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$ . Then string order for  $u$  exists iff  $u$  is a local symmetry.*

*Proof.* ✓ Proof If string order exists, Theorem 12.52 gives a gauge-phase equivalence with the companion tensor. Theorem 12.53 extracts a virtual unitary and phase from that equivalence, and Theorem 12.55 shows that the same unitary preserves the boundary state, so one obtains local symmetry. Conversely, Lemma 12.56 turns local symmetry into a non-decaying boundary string-order sequence. □

**Theorem 12.59** (Virtual unitary from string order). Lean ✓ Stmt *For an injective MPS, let  $\Lambda \in M_D(\mathbb{C})$ , and assume  $uu^\dagger = \mathbb{1}$  and  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ . If string order exists for  $u$  with boundary parameter  $\Lambda$ , then there exists a virtual unitary  $V$  and a unit-modulus scalar  $\mu$  such that  $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$  for all  $i$ . Diagrammatically this is the same local intertwining relation, but only up to an overall unit scalar on the virtual side:*

*Proof.* ✓ Proof String order first gives gauge-phase equivalence with the companion tensor by Theorem 12.52. Theorem 12.53 then normalizes the resulting intertwiner to a unitary and produces the phase  $\mu$ . □

## 12.6 SPT Phase Classification

Two injective symmetric tensors belong to the same symmetry-protected topological phase when their virtual representation cocycles are cohomologous [CGW11].

**Definition 12.60** (Same SPT phase). Lean ✓ Stmt *Injective tensors  $A$  and  $B$ , both symmetric under an on-site representation  $U: G \rightarrow \text{GL}_d(\mathbb{C})$ , are in the same SPT phase if there exist virtual cocycles  $\omega_A, \omega_B$  with corresponding projective representations that intertwine the respective twisted tensors and satisfy  $\omega_A \sim \omega_B$  (cohomologous).*

**Theorem 12.61** (String order holds for injective symmetric MPS). Lean ✓ Stmt *Let  $A$  be an injective tensor, symmetric under a unitary representation  $U$ . Assume  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ ,  $\Lambda$  is positive definite,  $\text{tr}(\Lambda) = 1$ , and  $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$ . Then string order exists for every group element  $g$ .*

*Proof.* ✓ Proof The virtual representation theorem produces  $\rho(g^{-1}) \in \text{GL}_D(\mathbb{C})$  satisfying  $\sum_j U(g)_{ij} A^j = \rho(g^{-1}) A^i \rho(g^{-1})^{-1}$ . A direct computation using  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$  shows that  $\mathcal{E}_{U(g)}(\rho(g^{-1})) = \rho(g^{-1})$ , so  $\rho(g^{-1})$  is a nonzero eigenvector with eigenvalue 1. Theorem 12.50 gives gauge-phase equivalence with the companion tensor, and Theorem 12.53 extracts a virtual unitary and phase. By Theorem 12.55, this unitary preserves the boundary state, so Lemma 12.56 gives string order.  $\square$

**Theorem 12.62** (String order is an SPT invariant). Lean ✓ Stmt *If  $A$  and  $B$  are injective, both on-site symmetric under a unitary representation  $U$ , and if  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ ,  $\mathcal{E}_B(\mathbb{1}) = \mathbb{1}$ ,  $\Lambda_A$  and  $\Lambda_B$  are positive definite,  $\text{tr}(\Lambda_A) = 1$ ,  $\text{tr}(\Lambda_B) = 1$ ,  $\mathcal{E}_A^\dagger(\Lambda_A) = \Lambda_A$ , and  $\mathcal{E}_B^\dagger(\Lambda_B) = \Lambda_B$ , then for every  $g \in G$ , string order exists for  $A$  with twist  $U(g)$  if and only if it exists for  $B$ . In particular, since Definition 12.60 implies on-site symmetry for both tensors, this theorem applies to tensors in the same SPT phase.*

*Proof.* ✓ Proof Both directions follow from Theorem 12.61, which shows that string order holds universally for injective symmetric tensors with canonical normalisations.  $\square$

## Chapter 13

# Periodic MPS: irreducible form and the periodic Fundamental Theorem

This chapter develops the periodic equal-case Fundamental Theorem of MPS (Theorem 13.24) following [DICCSPG17], and gives its two applications: the symmetry corollary lifting a physical on-site symmetry of a tensor in irreducible form  $\Pi$  to a virtual  $Z$ -gauge, and the equivalence between  $p$ -refinability of the matrix-product-vector family and  $p$ -divisibility of the transfer map.

### 13.1 Periodic irreducible form and physical-index rotation

**Definition 13.1** (Periodic tensor). [Lean](#) [✓ Stmt](#) An irreducible MPS tensor  $A$  of bond dimension  $D$  is *m-periodic* (for some  $m \geq 1$ ) if the peripheral spectrum of its transfer map  $\mathcal{E}_A$  is exactly the  $m$ -th roots of unity  $\{e^{2\pi i k/m} : 0 \leq k < m\}$ . A 1-periodic tensor is exactly an irreducible tensor with primitive transfer map (Definition 4.49).

**Definition 13.2** (Irreducible form). [Lean](#) [✓ Stmt](#) A tensor  $A$  is in *irreducible form* if it is block-diagonal

$$A^i = \bigoplus_{j \in J} \mu_j A_j^i,$$

where each block  $A_j$  is an  $m_j$ -periodic tensor, each weight  $\mu_j \neq 0$ , and the blocks are pairwise non-repeated (no two distinct  $j, j'$  are related by a gauge transformation and a unit-modulus scalar). This is the standard form introduced in [DICCSPG17, Section II]. It extends the normal canonical form (Definition 9.25), which requires the stronger condition that every block is 1-periodic.

**Remark 13.3** (Relation to normal canonical form). [Not ready](#) Every tensor in normal canonical form (Definition 9.25) is in irreducible form with all periods equal to 1. The irreducible form is strictly more general: it accommodates blocks such as the antiferromagnet, which are 2-periodic. Physical-index rotation was introduced in Definition 12.1 (Chapter 12). The following compatibility theorems are proved in the present chapter.

**Theorem 13.4** (Transfer map preserved by physical rotation). [Lean](#) [✓ Stmt](#) If  $uu^\dagger = I$ , then the transfer map is invariant under physical-index rotation:  $\mathcal{E}_{A_u} = \mathcal{E}_A$ .



*Proof.* [✓ Proof](#) This is the unitary freedom of Kraus representations from [Wol12, Theorem 2.1(4)].  $\square$

**Theorem 13.5** (Left-canonical preserved by physical rotation). [Lean](#) [✓ Stmt](#) If  $uu^\dagger = I$  and  $A$  is left-canonical, then  $A_u$  is left-canonical.

*Proof.* [✓ Proof](#) Expand  $\sum_i B_i^\dagger B_i$  where  $B_i = \sum_j u_{ij} A_j$ , swap sums, apply orthogonality  $u^\dagger u = I$ , and collapse to  $\sum_j A_j^\dagger A_j = I$ .  $\square$

**Theorem 13.6** (Irreducibility preserved by physical rotation). [Lean](#) [✓ Stmt](#) If  $uu^\dagger = I$  and  $A$  is irreducible, then  $A_u$  is irreducible.

*Proof.* [✓ Proof](#) Contrapositive: any nontrivial invariant projection  $P$  for the rotated tensor yields  $(1 - P) A_k P = 0$  for all  $k$  via the orthogonality of  $u$ , contradicting irreducibility of  $A$ .  $\square$

**Theorem 13.7** (Periodicity preserved by physical rotation). [Lean](#) [✓ Stmt](#) If  $uu^\dagger = I$  and  $A$  is periodic with period  $m$ , then  $A_u$  is periodic with the same period.

*Proof.* [✓ Proof](#) Assembles transfer-map invariance, left-canonical preservation, and irreducibility preservation.  $\square$

**Theorem 13.8** (Equality of MPV families preserved by physical rotation). [Lean](#) [✓ Stmt](#) If  $\mathcal{V}(A) = \mathcal{V}(B)$ , then  $\mathcal{V}(A_u) = \mathcal{V}(B_u)$ .

*Proof.* [✓ Proof](#) Induction on the word length with a strengthened hypothesis carrying an arbitrary prefix matrix, reducing each step to the hypothesis via word concatenation.  $\square$

**Theorem 13.9** (Physical rotation distributes over block-diagonal composition). [Lean](#) [✓ Stmt](#) The rotated block-diagonal tensor satisfies  $(\bigoplus_k \mu_k A_k)_u = \bigoplus_k \mu_k (A_k)_u$ .

*Proof.* [✓ Proof](#) Pointwise matrix-entry computation using linearity of the block-diagonal embedding and reindexing.  $\square$

**Theorem 13.10** (Irreducible form preserved by physical rotation). [Lean](#) [✓ Stmt](#) If  $uu^\dagger = I$  and  $A$  is in irreducible form, then  $A_u$  is in irreducible form with the same block structure and rotated blocks.

*Proof.* [✓ Proof](#) Each block remains periodic by Theorem 13.7, equality of MPV families transfers via Theorem 13.8, and block-diagonal compatibility follows from Theorem 13.9.  $\square$

### 13.1.1 The Z-gauge degree of freedom

**Definition 13.11** (Entrywise periodic Z-gauge ratio). [Lean](#) [✓ Stmt](#) Given complex weights  $\mu, \nu$ , define the entrywise Z-gauge ratio by

$$z(\mu, \nu) := \mu / \nu.$$

**Lemma 13.12** (Matched powers imply root-of-unity ratio). [Lean](#) [✓ Stmt](#) If  $\mu^m = \nu^m$  and  $\nu \neq 0$ , then

$$z(\mu, \nu)^m = 1.$$

**Lemma 13.13** (Ratio rescales denominator back to numerator). [Lean](#) [✓ Stmt](#) If  $\nu \neq 0$ , then

$$z(\mu, \nu)\nu = \mu.$$

**Definition 13.14** (Diagonal periodic  $Z$ -gauge matrix). [Lean](#) [✓ Stmt](#) For matched weight families  $(\mu_i)_i$  and  $(\nu_i)_i$  on a finite index type, define the diagonal matrix

$$Z := \text{diag}(z(\mu_i, \nu_i))_i.$$

**Lemma 13.15** (Diagonal  $Z$ -gauge has finite order under matched powers). [Lean](#) [✓ Stmt](#) If  $\mu_i^m = \nu_i^m$  and  $\nu_i \neq 0$  for all  $i$ , then

$$Z^m = \mathbb{1}.$$

**Lemma 13.16** (Diagonal  $Z$ -gauge transports diagonal weights). [Lean](#) [✓ Stmt](#) Under  $\nu_i \neq 0$  for all  $i$ ,

$$Z \text{ diag}(\nu) = \text{diag}(\mu).$$

**Definition 13.17** ( $Z$ -gauge equivalence). [Lean](#) [✓ Stmt](#) Let  $A$  be an  $m$ -periodic irreducible tensor of bond dimension  $D$ . A  $Z$ -gauge transformation of  $A$  is a diagonal unitary matrix  $Z \in M_D(\mathbb{C})$  satisfying:

1.  $Z$  commutes with every Kraus operator:  $[A^i, Z] = 0$  for all physical indices  $i$ , and
2.  $Z^m = \mathbb{1}_D$  (all diagonal entries are  $m$ -th roots of unity).

Replacing  $A$  by  $ZA$  (i.e.  $A^i \mapsto ZA^i$ ) leaves the MPV family invariant for lengths  $N$  that are multiples of  $m$ , because the factor  $\text{tr}(Z^N) = \text{tr}(\mathbb{1}) = D$  for  $N \equiv 0 \pmod{m}$ . Two tensors  $A, B$  are  $Z$ -gauge equivalent if there exists an invertible  $Y$  and a  $Z$ -gauge transformation  $Z$  for  $A$  such that  $ZA^i = YB^iY^{-1}$  for all  $i$ .

*Remark 13.18* ( $Z$ -gauge is trivial for period-1 blocks). [Lean](#) [✓ Stmt](#) When  $m = 1$  the condition  $Z^1 = \mathbb{1}$  forces  $Z = \mathbb{1}$ , so  $Z$ -gauge equivalence reduces to ordinary gauge equivalence (Definition 2.10). The  $Z$ -gauge is therefore a genuine new degree of freedom that appears only for  $m > 1$ .

**Theorem 13.19** (Blocking a  $\mathbb{Z}_m$ -gauge gives an ordinary gauge). [Lean](#) [✓ Stmt](#) If  $A$  and  $B$  are  $\mathbb{Z}_m$ -gauge equivalent, then their  $m$ -blocked tensors  $A^{[m]}$  and  $B^{[m]}$  are gauge equivalent in the ordinary sense.

*Proof.* [✓ Proof](#) Write the  $\mathbb{Z}_m$ -gauge relation as  $ZA^i = YB^iY^{-1}$  with  $Z^m = \mathbb{1}$  and  $[A^i, Z] = 0$ . For a blocked letter, commute one factor of  $Z$  past each physical letter in the associated length- $m$  word. The total prefactor becomes  $Z^m = \mathbb{1}$ , so the entire blocked word is related by the same ordinary gauge  $Y$ .  $\square$

**Corollary 13.20** (Blocking a  $\mathbb{Z}_m$ -gauge preserves the MPV family). [Lean](#) [✓ Stmt](#) If  $A$  and  $B$  are  $\mathbb{Z}_m$ -gauge equivalent, then the blocked tensors  $A^{[m]}$  and  $B^{[m]}$  have the same MPV family.

*Proof.* [✓ Proof](#) Apply Theorem 13.19 and then the gauge invariance of MPV families.  $\square$

## 13.2 Periodic Fundamental Theorem

The theorems in Chapter 11 all require that each block has a *primitive* transfer map, i.e. peripheral spectrum exactly  $\{1\}$ . This is achieved by blocking the original tensor by a common period. [DCCSPG17] develops a framework that avoids this blocking step, giving a fundamental theorem that applies directly to periodic tensors.

### 13.2.1 Common-period blocking lemmas

**Definition 13.21** (LCM period for a finite block family). [Lean](#) [✓ Stmt](#) Given a finite family of blocking periods  $(p_i)$  indexed by a finite set, their least common multiple serves as the shared blocking period.

**Definition 13.22** (Common-period blocked family). [Lean](#) [✓ Stmt](#) Given a finite family of tensors  $(B_i)$  with designated periods  $(p_i)$ , block each  $B_i$  to the shared period  $\text{lcm}(p_i)$ , so every block lives in the same physical space.

**Theorem 13.23** (Common-period blocking preserves transfer-map primitivity). [Lean](#) [✓ Stmt](#) *If each tensor  $B_i$  has a primitive transfer map after blocking by its own positive period  $p_i$ , then blocking the entire family to the common LCM period preserves primitivity of each member's transfer map.*

*Proof.* [✓ Proof](#) Each  $p_i$  divides  $\text{lcm}(p_i)$ . Apply the divisibility monotonicity theorem for primitive blocked transfer maps to each family member separately.  $\square$

### 13.2.2 Statement of the periodic Fundamental Theorem

The following is Theorem 3.8 (“equal case”) of [\[DICCSPG17\]](#). The statements recorded here are conditional components of that theorem: the final comparison still assumes periodic-overlap and power-equality hypotheses, so the literature theorem has not yet been obtained here as a single unconditional result.

**Theorem 13.24** (Periodic Fundamental Theorem). [Not ready](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) *Let  $A$  and  $B$  be MPS tensors in irreducible form, with bases of periodic blocks  $\{A_j\}_{j \in J}$  and  $\{B_k\}_{k \in K}$  respectively. Suppose that  $\mathcal{V}(A) = \mathcal{V}(B)$  (equal MPV families for all lengths  $N$ ).*

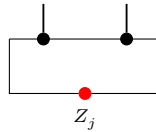
*Then  $|J| = |K|$  and there is a bijection  $\pi : J \rightarrow K$  such that  $A_j$  and  $B_{\pi(j)}$  are  $m_j$ -periodic for the same period  $m_j$ . Moreover, for each  $j \in J$  there exists:*

1. *an invertible matrix  $Y_j$  of size  $D_j \times D_{\pi(j)}$ , and*
2. *a diagonal unitary matrix  $Z_j$  of size  $D_j \times D_j$  satisfying  $Z_j^{m_j} = \mathbb{1}$  and  $[A_j^i, Z_j] = 0$  for all  $i$ ,*

*such that, for every physical index  $i$ ,*

$$Z_j A_j^i = Y_j B_{\pi(j)}^i Y_j^{-1}$$

*For each periodic block, the extra freedom is encoded by*



*rather than by an ordinary gauge alone: the operator  $Z_j$  sits on the closing virtual bond and satisfies  $Z_j^{m_j} = \mathbb{1}$ . Equivalently, setting  $Y = \bigoplus_j Y_j \otimes \mathbb{1}_{r_j}$  and  $Z = \bigoplus_j Z_j \otimes \mathbb{1}_{r_j}$ , one has  $ZA^i = YB^iY^{-1}$  for all  $i$ , with  $[A^i, Z] = 0$  and  $\mathcal{V}(A) = \mathcal{V}(ZA)$ .*

*Remark. The proof is conditional on the periodic-overlap dichotomy [\[DICCSPG17, Proposition 3.3\]](#), which is left here as a separate hypothesis.*

### 13.2.3 Periodic overlap dichotomy

**Definition 13.25** (Cyclic sector decomposition). [Lean] [Stmt] A family of compressed sector tensors  $(C_k)_{k=0}^{m-1}$  on corner bond spaces forms a *cyclic sector decomposition* of the blocked tensor  $A^{(m)}$  if there exist orthogonal projections  $P_0, \dots, P_{m-1}$  on the ambient bond space such that  $\sum_k P_k = \mathbb{1}$ , the projections satisfy the cyclic-shift relation  $\mathcal{E}_A^\dagger(P_{k+1}) = P_k$  with the index  $k+1$  taken cyclically modulo  $m$ , each  $P_k$  commutes with every blocked letter  $P_k A_i^{(m)} = A_i^{(m)} P_k$ , and  $V^{(N)}(C_k)_\sigma = \text{tr}(P_k(A^{(m)})^\sigma)$  for every word  $\sigma$ , and each block carries a  $*$ -algebra isomorphism  $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k \cdot M_D(\mathbb{C}) \cdot P_k$  (multiplicative and adjoint-preserving) intertwining the compressed adjoint transfer map with the sector adjoint transfer map on the corner of  $P_k$ . This is the inverse cyclic indexing of the off-diagonal convention in [DICCSPG17, Appendix A]. There the same adjoint transfer map is written  $E^*$ , and the source projectors satisfy  $E^*(P_u) = P_{u+1}$  together with  $A^i = \sum_u P_u A^i P_{u+1}$ . Here  $P_k$  corresponds to the source sector  $u = -k \pmod{m}$ , which turns the source shift into the displayed adjoint-transfer relation.

**Theorem 13.26** (Single-site cyclic grading). [Lean] [Stmt] Let  $A$  be left-canonical, and let  $P_0, \dots, P_{m-1}$  be orthogonal projections on the bond space. If the adjoint transfer map satisfies

$$\mathcal{E}_A^\dagger(P_{k+1}) = P_k$$

for all  $k$ , with indices taken modulo  $m$ , then for every physical letter  $i$ ,

$$P_{k+1} A^i = A^i P_k.$$

This is the single-site grading behind [DICCSPG17, eq. Aoffdiag]; the compressed blocks of [DICCSPG17, eq. Auprop] use the inverse cyclic indexing convention of Definition 13.25.

*Proof.* [Proof] Put  $K_i = (A^i)^\dagger$  and let  $\Phi_K$  be the Kraus map of the family  $K$ . Since  $A$  is left-canonical,  $\Phi_K$  is unital. The projection identities and the shift relation give

$$\Phi_K(P_{k+1}^\dagger P_{k+1}) = \Phi_K(P_{k+1})^\dagger \Phi_K(P_{k+1}),$$

because both sides are equal to  $P_k$ . The equality case in the Kadison–Schwarz inequality therefore gives

$$P_{k+1} K_i^\dagger = K_i^\dagger \Phi_K(P_{k+1}).$$

Substituting  $K_i^\dagger = A^i$  and  $\Phi_K(P_{k+1}) = P_k$  gives  $P_{k+1} A^i = A^i P_k$ . □

**Theorem 13.27** (Off-diagonal reconstruction). [Lean] [Stmt] Under the hypotheses of Theorem 13.26, assume in addition that

$$\sum_{u=0}^{m-1} P_u = \mathbb{1}.$$

Then every physical letter decomposes as

$$A^i = \sum_{u=0}^{m-1} P_{u+1} A^i P_u.$$

This is [DICCSPG17, eq. Aoffdiag], written in the same inverse cyclic indexing convention.

*Proof.* [Proof] Insert  $\sum_u P_u = \mathbb{1}$  on the right of  $A^i$ , then use Theorem 13.26 and  $P_u^2 = P_u$  term by term. □

**Theorem 13.28** (Word cyclic grading). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 13.26, a word  $w = i_1 \cdots i_\ell$  satisfies

$$P_{k+\ell} A^w = A^w P_k.$$

Thus the single-site grading accumulates one cyclic step for each letter of the word.

*Proof.* [✓ Proof](#) Induct on the length of  $w$ , using Theorem 13.26 for the first letter and the induction hypothesis for the remaining word.  $\square$

**Theorem 13.29** (Cyclic-sector single-site grading). [Lean](#) [✓ Stmt](#) If a periodic tensor admits a cyclic sector decomposition, then for every  $k$  and every physical letter  $i$  there exists a family  $P_0, \dots, P_{m-1}$  such that

$$P_u^\dagger = P_u, \quad P_u^2 = P_u \quad (0 \leq u < m), \quad \sum_{u=0}^{m-1} P_u = \mathbb{1}$$

and

$$P_{k+1} A^i = A^i P_k.$$

*Proof.* [✓ Proof](#) Choose the projection family contained in the cyclic sector decomposition. It satisfies

$$P_u^\dagger = P_u, \quad P_u^2 = P_u, \quad \sum_u P_u = \mathbb{1}, \quad \mathcal{E}_A^\dagger(P_{u+1}) = P_u.$$

Apply Theorem 13.26.  $\square$

**Theorem 13.30** (Cyclic-sector off-diagonal reconstruction). [Lean](#) [✓ Stmt](#) If a periodic tensor admits a cyclic sector decomposition, then for every physical letter  $i$  there exists a family  $P_0, \dots, P_{m-1}$  such that

$$P_u^\dagger = P_u, \quad P_u^2 = P_u \quad (0 \leq u < m), \quad \sum_{u=0}^{m-1} P_u = \mathbb{1}$$

and

$$A^i = \sum_{u=0}^{m-1} P_{u+1} A^i P_u.$$

*Proof.* [✓ Proof](#) Choose the projection family contained in the cyclic sector decomposition. It satisfies

$$P_u^\dagger = P_u, \quad P_u^2 = P_u, \quad \sum_u P_u = \mathbb{1}, \quad \mathcal{E}_A^\dagger(P_{u+1}) = P_u.$$

Apply Theorem 13.27.  $\square$

**Theorem 13.31** (Cyclic sector decomposition of a periodic tensor). [Lean](#) [✓ Stmt](#) If  $A$  is periodic with period  $m$ , then the blocked tensor  $A^{(m)}$  admits a cyclic sector decomposition. Thus there are projectors  $P_u$  and corner tensors  $C_u = P_u A^{(m)}$  which reproduce the blocked matrix-product vectors, are left-canonical, have nonzero bond dimensions, and realize the cyclic adjoint-transfer relation of Lemma bdcf in [DICCSPG17, lines 404–423]. The normality and non-repetition conclusions of Lemma bdcf are recorded separately in the component lemmas below.

*Proof.* [✓ Proof](#) This is the cyclic-sector existence part of Lemma bdcf, translated into the inverse indexing convention of Definition 13.25.  $\square$

**Lemma 13.32** (Off-diagonal eigenvectors of the period transfer vanish). [Lean](#) [✓ Stmt](#) Let  $A$  be periodic of period  $m$ . Let  $P_0, \dots, P_{m-1}$  be orthogonal projections with

$$\sum_{k=0}^{m-1} P_k = 1, \quad \mathcal{E}_A^*(P_{k+1}) = P_k.$$

If

$$U = P_u U P_v, \quad P_u P_v = 0, \quad \mathcal{E}_A^m(U) = \zeta U, \quad |\zeta| = 1,$$

then  $U = 0$ .

*Proof.* [✓ Proof](#) Since the peripheral spectrum of  $\mathcal{E}_A$  consists of  $m$ -th roots of unity, Lemma 4.34 gives

$$\text{Spec}_{\text{per}}(\mathcal{E}_A^m) = \{1\},$$

hence  $\zeta = 1$ . The cyclic trace gives

$$\text{tr } U = \text{tr}(P_u U P_v) = \text{tr}(P_v P_u U) = 0.$$

Put

$$W = \sum_{t=0}^{m-1} \mathcal{E}_A^t(U).$$

The equation  $\mathcal{E}_A^m(U) = U$  gives  $\mathcal{E}_A(W) = W$ , and trace preservation gives  $\text{tr } W = 0$ . Irreducibility of  $A$  therefore gives  $W = 0$ . The shift relation

$$P_{k+1} A^i = A^i P_k$$

moves  $\mathcal{E}_A^t(U)$  into the  $(u+t, v+t)$  corner, so only the  $t = 0$  term contributes after multiplying by  $P_u$  and  $P_v$ . Thus

$$U = P_u W P_v = 0.$$

□

**Theorem 13.33** (Self-overlap along period multiples). [Lean](#) [✓ Stmt](#) If  $A$  is  $m$ -periodic, then  $\lim_{N \rightarrow \infty} O_{AA}(mN) = m$ .

**Theorem 13.34** (Different periods imply asymptotic orthogonality). [Lean](#) [✓ Stmt](#) If  $A$  and  $B$  have periods  $m_A \neq m_B$ , then  $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$ .

*Proof.* [✓ Proof](#) If  $D_A \neq D_B$ , use Theorem 13.40. Assume  $D_A = D_B$  and  $A \sim_{\text{gp}} B$ , say  $B^i = \zeta X A^i X^{-1}$  with  $X \in \text{GL}_D(\mathbb{C})$  and  $\zeta \neq 0$ . The transfer maps satisfy

$$\mathcal{E}_B(Y) = \zeta \bar{\zeta} X \mathcal{E}_A(X^{-1} Y X^{-*}) X^*.$$

With  $\mathcal{E}_A(\rho_A) = \rho_A$ , put  $\sigma = X \rho_A X^*$ . Then  $\mathcal{E}_B(\sigma) = |\zeta|^2 \sigma$ , while  $\mathcal{E}_B(\rho_B) = \rho_B$  for a nonzero positive fixed point  $\rho_B$ . Perron–Frobenius eigenvalue uniqueness [Wol12, Theorem 6.3] gives  $|\zeta|^2 = 1$ . Hence  $\mathcal{E}_B$  is conjugate to  $\mathcal{E}_A$ , so  $\sigma_{\text{per}}(\mathcal{E}_A) = \sigma_{\text{per}}(\mathcal{E}_B)$ . Since

$$\sigma_{\text{per}}(\mathcal{E}_A) = \{e^{2\pi i r/m_A} : 0 \leq r < m_A\}, \quad \sigma_{\text{per}}(\mathcal{E}_B) = \{e^{2\pi i r/m_B} : 0 \leq r < m_B\}, \quad (13.1)$$

equality of the two finite sets in (13.1) gives  $m_A = m_B$ , a contradiction. Thus  $A \not\sim_{\text{gp}} B$ , and the irreducible trace-preserving overlap dichotomy gives  $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$ . □

**Lemma 13.35** (Sector-block normality for periodic tensors). [Lean](#) [✓ Stmt](#) For a periodic tensor equipped with a cyclic sector decomposition, each nonzero blocked sector tensor is normal.

**Theorem 13.36** (No sector match implies asymptotic orthogonality). [Lean](#) [✓ Stmt](#) Suppose  $A$  and  $B$  have the same total bond dimension and the same period  $m$ . Let  $A_u$  and  $B_v$  be period-blocked sector tensors of virtual dimensions  $d_u$  and  $e_v$ , indexed by  $u, v \in \mathbb{Z}_m$ , such that

$$\bar{A} \sim \bigoplus_{u \in \mathbb{Z}_m} A_u, \quad \bar{B} \sim \bigoplus_{v \in \mathbb{Z}_m} B_v$$

with unit weights, and the two decompositions are cyclic sector decompositions. Assume each sector is left-canonical,

$$\sum_{\alpha} (A_u^{\alpha})^{\dagger} A_u^{\alpha} = I_{d_u}, \quad \sum_{\alpha} (B_v^{\alpha})^{\dagger} B_v^{\alpha} = I_{e_v},$$

Assume  $\forall u, d_u \neq 0$  and  $\forall v, e_v \neq 0$ , and whenever  $d_u = e_v$ , the corresponding sector tensors are not gauge-phase equivalent after identifying their virtual spaces:

$$d_u = e_v \implies A_u \not\sim_{\text{gp}} B_v.$$

Then  $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$ .

*Proof.* [✓ Proof](#) Write  $\bar{A} \sim \bigoplus_{u \in \mathbb{Z}_m} A_u$  and  $\bar{B} \sim \bigoplus_{v \in \mathbb{Z}_m} B_v$ . For every  $N$ ,

$$O_{\bar{A}\bar{B}}(N) = \sum_{u \in \mathbb{Z}_m} \sum_{v \in \mathbb{Z}_m} O_{A_u B_v}(N). \quad (13.2)$$

For each sector pair  $(u, v)$ ,

$$d_u = e_v, \quad A_u \sim_{\text{gp}} B_v \implies \lim_{N \rightarrow \infty} O_{A_u B_v}(N) = 0,$$

and

$$d_u \neq e_v \implies \lim_{N \rightarrow \infty} O_{A_u B_v}(N) = 0.$$

The finite double sum (13.2) therefore gives

$$\lim_{N \rightarrow \infty} O_{\bar{A}\bar{B}}(N) = \sum_{u \in \mathbb{Z}_m} \sum_{v \in \mathbb{Z}_m} \lim_{N \rightarrow \infty} O_{A_u B_v}(N) = 0.$$

If  $A \sim_{\text{gp}} B$ , then  $\bar{A} \sim_{\text{gp}} \bar{B}$ , so for some  $X \in \text{GL}_D(\mathbb{C})$  and  $\zeta \neq 0$ ,  $(\bar{B})^{\alpha} = \zeta X (\bar{A})^{\alpha} X^{-1}$ . Thus  $V^{(N)}(\bar{B})_{\sigma} = \zeta^N V^{(N)}(\bar{A})_{\sigma}$ , and

$$O_{\bar{A}\bar{B}}(N) = \bar{\zeta}^N O_{\bar{A}\bar{A}}(N), \quad O_{\bar{B}\bar{B}}(N) = |\zeta|^{2N} O_{\bar{A}\bar{A}}(N).$$

Since  $\lim_{N \rightarrow \infty} O_{\bar{A}\bar{A}}(N) = \lim_{N \rightarrow \infty} O_{\bar{B}\bar{B}}(N) = m > 0$ ,

$$1 = \lim_{N \rightarrow \infty} \frac{O_{\bar{B}\bar{B}}(N)}{O_{\bar{A}\bar{A}}(N)} = \lim_{N \rightarrow \infty} |\zeta|^{2N},$$

hence  $|\zeta| = 1$ . Therefore

$$\lim_{N \rightarrow \infty} |O_{\bar{A}\bar{B}}(N)| = \lim_{N \rightarrow \infty} O_{\bar{A}\bar{A}}(N) = m \neq 0,$$

contradicting  $\lim_{N \rightarrow \infty} O_{\bar{A}\bar{B}}(N) = 0$ . Thus  $A \not\sim_{\text{gp}} B$ , and the irreducible trace-preserving overlap dichotomy gives  $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$ .  $\square$

**Lemma 13.37** (Sector-match propagation under translation). Lean ✓ Stmt Let  $A$  and  $B$  be left-canonical tensors with period  $m$ , and let  $\bar{A} \sim \bigoplus_u A_u$  and  $\bar{B} \sim \bigoplus_v B_v$  be unit-weight cyclic sector decompositions whose sectors are left-canonical. Suppose  $d_{u_0} = e_{v_0}$ ,  $d_{u_0} \neq 0$ , and, after identifying the virtual spaces,  $A_{u_0} \sim_{\text{gp}} B_{v_0}$ . Then for every  $l \in \mathbb{Z}_m$  there is an equality  $d_{u_0+l} = e_{v_0+l}$  and, after this identification,

$$A_{u_0+l} \sim_{\text{gp}} B_{v_0+l}.$$

**Lemma 13.38** (Per-site proportionality from blocked sector match). Lean Not ready Let  $A$  and  $B$  be left-canonical tensors with period  $m$ , and let  $\bar{A} \sim \bigoplus_u A_u$  and  $\bar{B} \sim \bigoplus_v B_v$  be unit-weight cyclic sector decompositions whose sectors are left-canonical. Assume  $d_u \neq 0$  and  $A_u$  is normal for every  $u$ . Fix  $q \in \mathbb{Z}_m$  and assume that for every  $u$  there are  $d_u = e_{u+q}$ ,  $\zeta_u \neq 0$ , and  $X_u \in \text{GL}_{d_u}(\mathbb{C})$  such that  $B_{u+q}^\alpha = \zeta_u X_u A_u^\alpha X_u^{-1}$  for every blocked physical index  $\alpha$ . The source obtains a phase-only product identity after absorbing these sector gauges into the  $B$ -sectors: with  $Q_v$  the cyclic sector projectors of  $B$ , it writes

$$\tilde{B}_v^i = U_v Q_v B^i Q_{v+1} U_{v+1}^\dagger.$$

The contraction input is then, for every physical block  $(i_1, \dots, i_m)$ ,

$$A_u^{i_1} A_{u+1}^{i_2} \cdots A_{u+m-1}^{i_m} = e^{i\lambda_{u+q}} \tilde{B}_{u+q}^{i_1} \tilde{B}_{u+q+1}^{i_2} \cdots \tilde{B}_{u+q+m-1}^{i_m}.$$

Then there are a phase  $\xi$  and an invertible matrix  $U$  such that  $A^i = e^{i\xi} U B^i U^{-1}$  for every physical index  $i$ .

**Theorem 13.39** (Sector match yields repeated-block equivalence). Lean ✓ Stmt Let  $A$  and  $B$  be periodic tensors with the same period  $m$ . Let  $\bar{A} \sim \bigoplus_u A_u$  and  $\bar{B} \sim \bigoplus_v B_v$  be unit-weight cyclic sector decompositions whose sectors are left-canonical, and assume every sector  $A_u$  has nonzero virtual dimension. If there are sectors  $u_0, v_0$  with  $d_{u_0} = e_{v_0}$  and  $A_{u_0} \sim_{\text{gp}} B_{v_0}$  after identifying their virtual spaces, then  $A$  and  $B$  are equivalent up to repeated blocks.

**Theorem 13.40** (Different bond dimensions imply asymptotic orthogonality). Lean ✓ Stmt If two periodic tensors have different bond dimensions, then  $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$  as  $N \rightarrow \infty$ .

**Theorem 13.41** (Periodic overlap dichotomy). Lean ✓ Stmt Let  $A$  and  $B$  be periodic tensors. Then at least one of the following alternatives holds:

1.  $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$  as  $N \rightarrow \infty$ ;
2.  $A$  and  $B$  are equivalent up to repeated blocks.

*Proof.* Not ready If  $m_A \neq m_B$ , then  $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$  by Theorem 13.34. Assume  $m_A = m_B = m$ . If  $D_A \neq D_B$ , then  $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$  by Theorem 13.40. Assume  $D_A = D_B$ .

Choose period-blocked sector decompositions

$$\bar{A} \sim \bigoplus_{u \in \mathbb{Z}_m} A_u, \quad \bar{B} \sim \bigoplus_{v \in \mathbb{Z}_m} B_v.$$

If no pair  $(u, v)$  satisfies

$$d_u = e_v, \quad A_u \sim_{\text{gp}} B_v,$$

then Theorem 13.36 gives  $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$ .

Otherwise there are  $u_0, v_0 \in \mathbb{Z}_m$  such that

$$d_{u_0} = e_{v_0}, \quad A_{u_0} \sim_{\text{gp}} B_{v_0}.$$

Theorem 13.39 then gives repeated-block equivalence between  $A$  and  $B$ . □



**Theorem 13.42** (Eventual linear independence of periodic vectors). [Lean](#) [Stmt](#) For a finite family  $\{A_j\}_{j \in J}$  of pairwise non-repeated periodic tensors, in the sense that no two are related by a gauge transformation and a unit-modulus scalar, whose periods divide a common positive integer  $p$ , there is  $N_0$  such that for every  $N \geq N_0$ , the map

$$\Phi_N : \mathbb{C}^J \rightarrow (\mathbb{C}^d)^{\otimes pN}, \quad (c_j)_{j \in J} \mapsto \sum_{j \in J} c_j |V^{(pN)}(A_j)\rangle$$

satisfies

$$\ker \Phi_N = \{0\}.$$

*Proof.* [Not ready](#) Let

$$G_N(j, k) = \langle V^{(pN)}(A_j), V^{(pN)}(A_k) \rangle$$

be the Gram matrix of the periodic vectors. The diagonal entries have positive limits,

$$G_N(j, j) \rightarrow m_j > 0,$$

by Theorem 13.33. If  $j \neq k$ , the second alternative in Theorem 13.41 would make  $A_j$  and  $A_k$  repeated blocks, that is, gauge equivalent up to a unit-modulus scalar. This contradicts the non-repetition hypothesis, and hence

$$G_N(j, k) \rightarrow 0.$$

Hence

$$G_N \rightarrow \text{diag}(m_j)_{j \in J}.$$

Since  $m_j > 0$  for every  $j$ , the limit matrix  $\text{diag}(m_j)_{j \in J}$  is positive definite. Hence  $G_N$  is positive definite for all sufficiently large  $N$ . Thus

$$\Phi_N(c) = 0 \implies 0 = \langle \Phi_N(c), \Phi_N(c) \rangle = c^* G_N c \implies c = 0,$$

and  $\ker \Phi_N = \{0\}$ . □

*Remark 13.43* (Comparison with the blocking approach). [Not ready](#) Theorem 13.24 and the equal-MPV corollary of Section 11.2 both characterize the gauge freedom between two tensors with equal MPV families, but they do so at different levels:

1. *Blocking approach* (Section 11.2, following [CPGSV16]): for a common multiple  $p$  of the periods,

$$A^{[p]} \sim \bigoplus_a C_a, \quad B^{[p]} \sim \bigoplus_b D_b,$$

with all  $C_a, D_b$  normal. The Fundamental Theorem gives, after a permutation of the normal summands,

$$D_{\pi(a)}^\alpha = X_a C_a^\alpha X_a^{-1}.$$

This is a gauge statement for  $A^{[p]}$  and  $B^{[p]}$ ; the cyclic phases have been absorbed into the period- $p$  words.

2. *Periodic approach* (Theorem 13.24, following [DICCSPG17]): keep the cyclic sectors

$$A^i = \sum_{u \in \mathbb{Z}_m} P_u A^i P_{u+1}, \quad B^i = \sum_{v \in \mathbb{Z}_m} Q_v B^i Q_{v+1}.$$

Sector matching gives phases  $\phi_v$ , a shift  $q$ , and unitaries  $U_v$  such that

$$P_u A^i P_{u+1} = e^{i\xi} e^{i\phi_{u+q}} U_{u+q} Q_{u+q} B^i Q_{u+q+1} U_{u+q+1}^{-1} e^{-i\phi_{u+q+1}}.$$

In the equal-MPV theorem the proportional phase is absorbed, so one obtains  $ZA^i = UB^iU^{-1}$  with  $Z^m = \mathbb{1}$ ; the trace only sees powers of the corresponding  $m$ -th roots of unity.

Chapters 9 and 11 follow only the blocking argument: remove periodicity by blocking, then apply the Fundamental Theorem. The periodic extension treats periodic blocks directly.

**Theorem 13.44** (Peripheral proportional case from exact MPV equality). Lean Not ready *Let  $A$  and  $B$  be periodic tensors with the same matrix product vectors. Then their bond dimensions agree, and after identifying the bond spaces,  $A$  and  $B$  are equivalent up to repeated blocks.*

*Proof.* Not ready By Theorem 13.41, either the overlap  $\langle V^{(N)}(A) | V^{(N)}(B) \rangle$  tends to 0 or the bond dimensions agree and  $A$  and  $B$  are equivalent up to repeated blocks. In the decay case, equality of the MPV families gives, for every  $N$ ,

$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle = \langle V^{(N)}(A) | V^{(N)}(A) \rangle. \quad (13.3)$$

Along multiples of the period  $m_a$ , (13.3) would imply

$$0 = \lim_{N \rightarrow \infty} \langle V^{(m_a N)}(A) | V^{(m_a N)}(B) \rangle = \lim_{N \rightarrow \infty} \langle V^{(m_a N)}(A) | V^{(m_a N)}(A) \rangle = m_a,$$

contradicting  $m_a > 0$ . □

**Theorem 13.45** (Peripheral proportional case from phase rescaling). Lean Not ready *Assume that whenever two periodic tensors have proportional MPV families, one can rescale one tensor by a unit-modulus phase so that the MPV families agree exactly. Then proportional periodic MPV families already force the tensors to agree up to repeated blocks.*

*Proof.* Not ready Apply the phase-rescaling hypothesis to replace  $A$  by a periodic tensor  $A'$  with the same MPV family as  $B$ . Theorem 13.44 then yields repeated-block equivalence between  $A'$  and  $B$ . Since  $A'$  differs from  $A$  only by a unit-modulus phase,  $A$  and  $A'$  are themselves equivalent up to repeated blocks, and transitivity gives the claim. □

*Remark 13.46* (Proportional periodic FT). Not ready There is also a proportional periodic Fundamental Theorem (Theorem 3.4, “proportional case”, in [DICCSPG17]). The former formulation of this theorem assumed externally supplied coefficient arrays with nonzero limits as hypotheses. That is not the hypothesis set of the source theorem. A faithful statement must instead derive the nondecaying cross-overlap witnesses from the proportional MPV hypothesis and the periodic BNT structure.

### 13.3 Physical symmetries on periodic MPS

**Theorem 13.47** (Physical symmetry yields a virtual  $Z$ -gauge). Lean ✓ Stmt *Let  $A$  be an MPS tensor in irreducible form  $\Pi$  and let  $U : G \rightarrow \text{GL}_d(\mathbb{C})$  be a representation of a group  $G$  on the physical leg, acting unitarily ( $U(g)U(g)^\dagger = \mathbb{1}$ ). Assume moreover the periodic equal-case Fundamental Theorem of MPS (Theorem 13.24) as an explicit hypothesis on the pair  $(d, D)$ , expressed by the hypothesis Lean  $dD$ . If  $A$  is on-site symmetric under  $U$  — that is, the twisted tensor  $\tilde{A}_g$  has the same MPV family as  $A$  for every  $g \in G$  — then for each  $g \in G$  there exists a positive period  $m_g$  and a  $\mathbb{Z}_{m_g}$ -gauge equivalence between  $A$  and  $\tilde{A}_g$ .*

Concretely there are matrices  $Z_g$  and  $Y_g$ , with  $Z_g^{m_g} = \mathbb{1}$  and  $[A^i, Z_g] = 0$ , satisfying, for every physical index  $i$ ,

$$Z_g A^i = Y_g \tilde{A}_g^i Y_g^{-1}$$

*Proof.* ✓ Proof The twisted tensor  $\tilde{A}_g$  coincides with the physical-index rotation  $A_{U(g)}$  of Definition 12.1. By Theorem 13.10 the rotated tensor remains in irreducible form II, and the on-site symmetry hypothesis supplies the equality  $\mathcal{V}(A) = \mathcal{V}(A_{U(g)})$ . Apply the periodic equal-case Fundamental Theorem (Theorem 13.24) to obtain the asserted  $\mathbb{Z}_{m_g}$ -gauge equivalence.  $\square$

**Definition 13.48** (Periodic projective-rigidity hypothesis). Lean ✓ Stmt Let  $A$  be an MPS tensor with physical dimension  $d$  and bond dimension  $D$ , and assume it is on-site symmetric under a group action  $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$ . The tensor  $A$  satisfies the *periodic projective-rigidity hypothesis* if one may choose a family of virtual gauges  $Y : G \rightarrow \mathrm{GL}_D(\mathbb{C})$  and a scalar 2-cochain  $u : G \times G \rightarrow \mathbb{C}^\times$  such that

1. every  $Y_g$  realises the intertwining from [DICCSPG17, Corollary 4.1] with the twisted tensor  $\tilde{A}_g$  for some accompanying  $Z_g$  (with  $Z_g^{m_g} = \mathbb{1}$  and  $[A^i, Z_g] = 0$ ); and
2. the family obeys the projective multiplication law on the bond space:  $Y_h Y_g = u(g, h) Y_{gh}$ .

The injective case reduces the rigidity to scalar uniqueness of gauges (cf. §12.2); in the genuinely periodic setting the bond-space commutant of  $A$  is generated by the  $Z_g$  matrices, so coherence of the factor system  $u$  is an independent analytic assumption.

**Theorem 13.49** (Projective-representation upgrade). Lean ✓ Stmt Under the hypotheses of Corollary 13.47 together with the periodic projective-rigidity hypothesis (Definition 13.48), the  $Y_g$  family defines a projective representation of  $G$  on the bond space: there exist a scalar 2-cochain  $\omega : G \times G \rightarrow \mathbb{C}^\times$  and virtual gauges  $X : G \rightarrow \mathrm{GL}_D(\mathbb{C})$  satisfying

$$X(g) X(h) = \omega(g, h) X(gh),$$

while for each  $g$  the matrix  $X(g^{-1})$  still realises the  $\mathbb{Z}_{m_g}$ -intertwining from [DICCSPG17, Corollary 4.1] between  $A$  and the twisted tensor  $\tilde{A}_g$ .

*Proof.* ✓ Proof Unpack the rigidity hypothesis to extract the family  $(Y, u)$  together with the intertwiners from [DICCSPG17, Corollary 4.1]. Set  $X(g) := Y(g^{-1})$  and  $\omega(g, h) := u(h^{-1}, g^{-1})$ . The projective multiplication law  $X(g) X(h) = \omega(g, h) X(gh)$  is then exactly the  $(h^{-1}, g^{-1})$ -instance of the factor-system identity, using  $(gh)^{-1} = h^{-1}g^{-1}$ . The intertwining relation between  $A$  and  $\tilde{A}_g$  transports unchanged.  $\square$

**Theorem 13.50** (Cocycle identity for the upgraded factor system). Lean ✓ Stmt If the bond dimension  $D$  is positive, the factor system  $\omega$  produced by Theorem 13.49 satisfies the multiplicative 2-cocycle identity  $\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k)$ .

*Proof.* ✓ Proof Associativity of matrix multiplication applied to  $X(g) X(h) X(k)$ , combined with the projective multiplication law, gives

$$\omega(g, h) \omega(gh, k) X((gh)k) = \omega(g, hk) \omega(h, k) X(g(hk)). \quad (13.4)$$

Since  $(gh)k = g(hk)$ , both sides of (13.4) are scalar multiples of the same matrix  $X((gh)k)$ . The matrix lies in  $\mathrm{GL}_D(\mathbb{C})$  and is therefore invertible, but the scalar cancellation leading from an equality of scalar multiples of a matrix to equality of the scalar coefficients additionally requires  $D > 0$  so that the underlying index set is inhabited. Cancelling then yields  $\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k)$  in  $\mathbb{C}^\times$ , hence in  $\mathrm{Units}(\mathbb{C})$ .  $\square$

*Remark 13.51* (SPT classification). The cohomology class of  $\omega$  in  $H^2(G, U(1))$  classifies the SPT phase of the symmetric MPS (cf. Chen–Gu–Wen and Pérez-García et al.). This classification statement is not established here; only the underlying 2-cocycle identity of Theorem 13.50 is available.

**Theorem 13.52** (Projective representation and cocycle from symmetry). [Lean](#) [✓ Stmt](#) Theorems 13.49 and 13.50 combine into a single statement: from the periodic projective-rigidity hypothesis (Definition 13.48), one obtains a projective representation of  $G$  on the bond space whose factor system is a genuine 2-cocycle whenever  $D > 0$ .

## 13.4 Refinement and divisibility

The next theorem characterises when the matrix-product-vector family of a periodic tensor admits a finer description in terms of a smaller-period tensor whose blocks have been merged. This was the original motivation for the irreducible-form framework in [DICCSPG17].

**Definition 13.53** ( $p$ -divisible CP map). [Lean](#) [✓ Stmt](#) A linear endomorphism  $\mathcal{E}$  of  $M_D(\mathbb{C})$  is  $p$ -divisible if there exists a CPTP map  $\mathcal{E}'$  of  $M_D(\mathbb{C})$  such that  $\mathcal{E} = (\mathcal{E}')^p$ .

**Definition 13.54** ( $p$ -refinable MPS tensor). [Lean](#) [✓ Stmt](#) An MPS tensor  $B$  with physical dimension  $d$  and bond dimension  $D$  is  $p$ -refinable if there exist another such tensor  $A$  and an isometry  $W : \mathbb{C}^d \rightarrow (\mathbb{C}^d)^{\otimes p}$  (encoded as a matrix  $W \in M_{d^p \times d}(\mathbb{C})$  with  $W^\dagger W = \mathbb{1}$ ) such that the  $p$ -blocked tensor  $A^{[p]}$  matches the  $W$ -image of  $B$  at the level of MPV coefficients:

$$V^{(N)}(A^{[p]})_\tau = \sum_{\sigma \in \{0, \dots, d-1\}^N} \left( \prod_k W_{\tau(k), \sigma(k)} \right) V^{(N)}(B)_\sigma$$

for every  $N \in \mathbb{N}$  and every  $\tau \in \{0, \dots, d^p - 1\}^N$ .

In Dirac notation, this encodes  $|V_{pN}(A)\rangle = W^{\otimes N} |V_N(B)\rangle$  for all  $N$ .

**Theorem 13.55** (Pullback of a refinement witness). [Lean](#) [✓ Stmt](#) Let  $B$  be an MPS tensor in irreducible form II. If  $B$  is  $p$ -refinable, then there exist a tensor  $A$ , an isometry  $W : \mathbb{C}^d \rightarrow (\mathbb{C}^d)^{\otimes p}$ , and the pullback tensor

$$C^\alpha := \sum_{i=0}^{d-1} W_{\alpha, i} B^i \quad (\alpha \in \{0, \dots, d^p - 1\})$$

such that  $C$  is again in irreducible form II,  $\mathcal{E}_C = \mathcal{E}_B$ , and  $C$  has the same MPV family as  $A^{[p]}$ .

*Proof.* [✓ Proof](#) Choose  $(A, W)$  from the refinement data. The pullback construction gives  $\mathcal{E}_C = \mathcal{E}_B$  and equality of MPV families between  $C$  and  $A^{[p]}$ . To see that  $C$  is still in irreducible form II, transport each periodic block  $B_j$  of  $B$  through the same physical isometry  $W$ . The left-canonical normalization is preserved because  $W^\dagger W = \mathbb{1}$ , while irreducibility and the peripheral spectrum are preserved because the transfer map of each transported block agrees with that of  $B_j$ . Reassembling these transported blocks with the same weights gives an irreducible-form decomposition of  $C$ .  $\square$

**Definition 13.56** (Blocked equal-case Z-gauge extraction hypothesis). [Lean](#) [✓ Stmt](#) Fix  $(d, D, p)$ . This hypothesis says that whenever a blocked tensor  $C$  in irreducible form II has the same MPV family as a blocked root  $A^{[p]}$ , there exists an integer  $m \geq 1$  and a  $\mathbb{Z}_m$ -gauge equivalence between  $C$  and  $A^{[p]}$ .

**Definition 13.57** (Blocked-to-root reconstruction hypothesis from Z-gauge). [Lean](#) [✓ Stmt](#) Fix  $(d, D, p)$ . This hypothesis says that if  $m \geq 1$  and a blocked tensor  $C$  in irreducible form II is  $\mathbb{Z}_m$ -gauge equivalent to  $A^{[p]}$ , then there exists a tensor  $A'$  with  $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$  and  $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$ .

**Definition 13.58** (Blocked equal-case root hypothesis). [Lean](#) [✓ Stmt](#) Fix  $(d, D, p)$ . This states the combined root-existence statement: whenever a blocked tensor  $C$  in irreducible form II has the same MPV family as a blocked root  $A^{[p]}$ , one can replace  $A$  by a left-canonical tensor  $A'$  with  $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$ .

**Theorem 13.59** (Blocked equal-case root from Z-gauge extraction). [Lean](#) [✓ Stmt](#) *If the blocked equal-case Z-gauge extraction hypothesis and the blocked-to-root reconstruction hypothesis both hold, then the combined blocked equal-case root hypothesis holds.*

*Proof.* [✓ Proof](#) Apply the Z-gauge extraction hypothesis to obtain a blocked  $\mathbb{Z}_m$ -gauge witness between  $C$  and  $A^{[p]}$ , and then apply the reconstruction hypothesis to that witness.  $\square$

**Theorem 13.60** (Reduction to the blocked equal-case root hypothesis). [Lean](#) [✓ Stmt](#) *Assume the blocked equal-case root hypothesis. Then every  $p$ -refinable tensor  $B$  in irreducible form II admits a left-canonical root  $A'$  with  $\mathcal{E}_B = \mathcal{E}_{A'^{[p]}}$ .*

*Proof.* [✓ Proof](#) Apply Theorem 13.55 to the refinement witness of  $B$ . This produces a blocked tensor  $C$  in irreducible form II with  $\mathcal{E}_C = \mathcal{E}_B$  and the same MPV family as  $A^{[p]}$ . The blocked equal-case root hypothesis then gives a left-canonical tensor  $A'$  with  $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$ . Combining the two transfer-map identities yields  $\mathcal{E}_B = \mathcal{E}_{A'^{[p]}}$ .  $\square$

**Theorem 13.61** (Forward direction from the blocked equal-case root hypothesis). [Lean](#) [✓ Stmt](#) *Assume the blocked equal-case root hypothesis. Then for every tensor  $B$  in irreducible form II,  $p$ -refinability of  $B$  implies  $p$ -divisibility of  $\mathcal{E}_B$ .*

*Proof.* [✓ Proof](#) Combine Theorem 13.60 with the witness-level forward implication: once  $\mathcal{E}_B$  is identified with  $\mathcal{E}_{A'^{[p]}}$  for a left-canonical  $A'$ , the blocking identity gives  $\mathcal{E}_B = (\mathcal{E}_{A'})^p$ .  $\square$

The following conditional statements isolate the refinability–divisibility equivalence from [DICCSPG17, Theorem 4.1].

**Theorem 13.62** ( $p$ -refinability iff transfer-map  $p$ -divisibility). [Lean](#) [✓ Stmt](#) *Let  $B$  be an MPS tensor in irreducible form II and let  $p \geq 1$ . Assume the two root hypotheses used in the one-way statements: the forward left-canonical root hypothesis for refinability  $\Rightarrow$  divisibility and the reverse transfer-map root hypothesis for divisibility  $\Rightarrow$  refinability. Then  $B$  is  $p$ -refinable iff its transfer map  $\mathcal{E}_B$  is  $p$ -divisible.*

*Proof.* [✓ Proof](#) Bundle the forward implication (Theorem 13.63) with the reverse implication (Theorem 13.66) into a single biconditional. The forward direction assumes the left-canonical root hypothesis; the reverse direction assumes only the transfer-map root hypothesis. Together they give the full equivalence under those hypotheses.  $\square$

**Theorem 13.63** ( $p$ -refinability implies transfer-map  $p$ -divisibility). [Lean](#) [✓ Stmt](#) *Let  $B$  be an MPS tensor in irreducible form II. Assume a left-canonical root hypothesis: every refinement witness gives a tensor  $A$  with  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  and  $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$ . Then  $p$ -refinability of  $B$  implies  $p$ -divisibility of  $\mathcal{E}_B$ .*

*Proof.* [✓ Proof](#) The left-canonical root hypothesis supplies a witness  $A$  with  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  (left-canonical) and  $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$ . The left-canonical property makes  $\mathcal{E}_A$  a CPTP channel, and the blocking identity  $\mathcal{E}_{A^{[p]}} = (\mathcal{E}_A)^p$  then identifies  $\mathcal{E}_B$  with the  $p$ -th power of the channel  $\mathcal{E}_A$ .  $\square$

**Definition 13.64** (Blocked-to-root channel-root hypothesis from Z-gauge). Lean ✓ Stmt Fix  $(d, D, p)$ . This sharpened reconstruction hypothesis asks for a CPTP map  $\mathcal{E}'$  such that  $\mathcal{E}_C = (\mathcal{E}')^p$  and the Choi/Kraus rank of  $\mathcal{E}'$  is at most  $d$ , whenever  $C$  is related to  $A^{[p]}$  by a blocked  $\mathbb{Z}_m$ -gauge witness.

**Theorem 13.65** (Channel-root criterion for blocked-to-root reconstruction). Lean ✓ Stmt *If the blocked-to-root channel-root hypothesis holds, then the tensor-level blocked-to-root reconstruction hypothesis holds as well.*

*Proof.* ✓ Proof Choose a tensor  $A'$  realizing the bounded Kraus family of the root channel. Because the channel is trace-preserving, the Kraus operators of  $A'$  satisfy  $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$ . The identity  $\mathcal{E}_C = (\mathcal{E}_{A'})^p$  is then equivalent to  $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$  by the blocking formula.  $\square$

**Theorem 13.66** (Transfer-map  $p$ -divisibility implies  $p$ -refinability). Lean ✓ Stmt *Let  $B$  be an MPS tensor in irreducible form  $II$  and let  $p \geq 1$ . Assume the reverse transfer-map root hypothesis: there is a tensor  $A$  with  $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$  whenever  $\mathcal{E}_B$  is  $p$ -divisible. Then  $p$ -divisibility of  $\mathcal{E}_B$  implies  $p$ -refinability of  $B$ .*

*Proof.* ✓ Proof From the reverse transfer-map root hypothesis, extract  $A$  with  $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$ . This matches two finite Kraus families of the same completely positive map: the blocked family  $A^{[p]}$  with  $d^p$  operators and the original family  $B$  with  $d$  operators. The cardinality comparison  $d \leq d^p$  holds whenever  $p \geq 1$ , so [Wol12, Theorem 2.1(4)] supplies an isometry  $V \in M_{d^p \times d}(\mathbb{C})$  with  $V^\dagger V = \mathbb{1}$  and  $(A^{[p]})^\alpha = \sum_j V_{\alpha,j} B^j$ . Expanding the coefficient of  $A^{[p]}$  at the word  $\tau$  by linearity of matrix multiplication and of the trace, the contributions reorganise into the  $V$ -weighted sum over length- $N$  words of  $B$  required by Definition 13.54, exhibiting  $V$  as the isometry  $W$  in the statement of  $p$ -refinability.  $\square$

*Remark 13.67* (Continuum-limit application). Not ready The original motivation for Theorem 13.62 is a continuum-limit construction in the spirit of [DICCSPG17]: refinability along arbitrarily large  $p$  is equivalent to infinite divisibility of the transfer map, which selects out continuum limits of translationally invariant MPS. The further consequences are explored elsewhere.

## Chapter 14

# The Algebraic Fundamental Theorem

This chapter develops the one-dimensional MPS results of [MGRPG<sup>+</sup>18]: the Fundamental Theorem for injective and normal MPS on a finite periodic chain of fixed length. The argument is purely algebraic, working directly inside  $M_D(\mathbb{C})$ . Two lemmas carry the whole fixed-length argument: the virtual bond gauge (Lemma 14.13) and the proportionality lemma (Lemma 14.14). From them follow the main chain theorem and its corollaries for translation-invariant chains, non-translation-invariant descriptions of translation-invariant states, and blocked chains. A second, parallel development treats block-diagonal tensors through a product-algebra automorphism and its block-permutation decomposition, leading to the direct-sum gauge-equivalence statements and the equal-case block matching with differing bond dimensions. Appendix A of [MGRPG<sup>+</sup>18] sharpens the normal-MPS bound from  $n \geq 3L$  to  $n \geq 2L + 1$  by a more refined blocking argument. The same two lemmas (Lemma 14.13 and Lemma 14.14) underlie the injective PEPS theorem of Chapter 15 and the normal PEPS theorem on arbitrary graphs, where the graph geometry enters only through the blocking step that reduces to a three-site chain.

Throughout,  $D$  is the bond dimension and  $d$  is the physical dimension with  $d \geq D^2$ . The extensions to injective and normal PEPS on arbitrary graphs and the improved system-size bound ( $2L + 1$  in place of  $3L$ ) are not pursued here.

### 14.1 Non-translation-invariant periodic chains

The algebraic proof of [MGRPG<sup>+</sup>18] keeps a fixed periodic chain length and lets the local tensors vary from site to site. The necessary objects—the coefficient of a configuration, the same-state relation, sitewise injectivity, and cyclic gauge equivalence—are recorded first.

**Definition 14.1** (Periodic non-TI chain data). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix a chain length  $n$ . A *non-translation-invariant periodic MPS chain* consists of sitewise tensors

$$A_k = \{A_k^i\}_{i=0}^{d-1}, \quad k = 0, \dots, n-1,$$

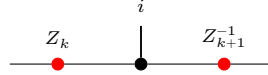
where each  $A_k^i \in M_D(\mathbb{C})$ . For a basis configuration  $\sigma = (\sigma_0, \dots, \sigma_{n-1})$ , its coefficient is

$$c_{A,\sigma} := \text{tr}(A_0^{\sigma_0} A_1^{\sigma_1} \dots A_{n-1}^{\sigma_{n-1}}).$$

Two such chains generate the *same chain state* if these coefficients agree for every  $\sigma$ . The chain is *injective* if each local tensor  $A_k$  is injective in the usual single-site sense. Finally, two chains are *cyclically gauge equivalent* if there are invertible bond matrices  $Z_0, \dots, Z_{n-1}$  such that

$$B_k^i = Z_k A_k^i Z_{k+1}^{-1}$$

for every site  $k$  and physical index  $i$ , with indices understood modulo  $n$  so that  $Z_n = Z_0$ . Locally, this gauge relation is represented by



where the black node denotes the site tensor named in the surrounding formula and the two red side nodes denote the bond gauges on the neighboring virtual legs.

**Lemma 14.2** (Cyclic gauge equivalence is an equivalence relation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *On fixed-length periodic chains, cyclic gauge equivalence is reflexive, symmetric, and transitive.*

*Proof.* [✓ Proof](#) Reflexivity uses the identity gauge on every bond. Symmetry inverts each bond gauge, and transitivity multiplies the gauges pointwise along the chain.  $\square$

## 14.2 One-sided inverse and physical realisation of virtual insertions

If  $A$  is injective, the linear combination map  $\Phi_A(c) := \sum_i c_i A^i : \mathbb{C}^d \rightarrow M_D(\mathbb{C})$  is surjective and admits a right inverse. The resulting decomposition map expresses any matrix as a linear combination of the spanning family  $\{A^i\}$ , and its multiplicative refinement re-expresses a virtual insertion on the bond as a physical operation on the local index.

**Lemma 14.3** (Existence of a right inverse). [Lean](#) [✓ Stmt](#) *If  $A$  is injective, then there exists a  $\mathbb{C}$ -linear map  $\Psi : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$  satisfying  $\Phi_A \circ \Psi = \text{id}_{M_D(\mathbb{C})}$ .*

*Proof.* [✓ Proof](#) Surjectivity of  $\Phi_A$  yields a linear section by choice.  $\square$

**Definition 14.4** (Decomposition map). [Lean](#) [✓ Stmt](#) Let  $A$  be an injective MPS tensor. The *decomposition map*  $\Psi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$  is a fixed choice of right inverse of  $\Phi_A$ .

**Lemma 14.5** (Decomposition property). [Lean](#) [✓ Stmt](#) *If  $A$  is injective, then for every  $X \in M_D(\mathbb{C})$  there exist coefficients  $c \in \mathbb{C}^d$  such that*

$$X = \sum_{i=0}^{d-1} c_i A^i.$$

*Proof.* [✓ Proof](#) Take  $c = \Psi_A(X)$ . Then  $\Phi_A(c) = \Phi_A(\Psi_A(X)) = X$ .  $\square$

**Remark 14.6** (Non-uniqueness of the decomposition). When  $d > D^2$ , the map  $\Phi_A$  has a nontrivial kernel, so the section  $\Psi_A$  is not unique. The statements that follow depend only on  $\Phi_A \circ \Psi_A = \text{id}$ , so the choice does not affect the conclusions.



### 14.2.1 Physical realisation of virtual insertions

For an injective tensor  $A$ , the physical realisation map encodes the effect of inserting a virtual matrix  $X$  on the bond as a  $d \times d$  operation on the physical indices. Its multiplicativity (Theorem 14.9) turns the bond algebra into an algebra acting on the physical leg, and is the source of the algebra homomorphism used in the virtual bond gauge.

**Definition 14.7** (Physical realisation map). [Lean](#) [Stmt](#) Let  $A$  be an injective tensor. For  $X \in M_D(\mathbb{C})$ , define the  $d \times d$  matrix  $O_A(X) \in M_d(\mathbb{C})$  by

$$O_A(X)_{ij} := (\Psi_A(A^i X))_j,$$

so that row  $i$  of  $O_A(X)$  contains the decomposition coefficients of  $A^i X$  in the spanning set  $\{A^j\}$ . Diagrammatically, the inserted virtual matrix is re-expressed as a physical operation on the same local tensor:

The map  $X \mapsto O_A(X)$  is  $\mathbb{C}$ -linear.

**Lemma 14.8** (Defining property of the physical realisation map). [Lean](#) [Stmt](#) For every  $X \in M_D(\mathbb{C})$  and every physical index  $i$ ,

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

*Proof.* [Proof](#) By definition,  $O_A(X)_{ij} = (\Psi_A(A^i X))_j$ . Applying  $\Phi_A$  gives  $\sum_j O_A(X)_{ij} A^j = \Phi_A(\Psi_A(A^i X)) = A^i X$ , where the last step uses  $\Phi_A \circ \Psi_A = \text{id}$ .  $\square$

**Theorem 14.9** (Physical realisation is multiplicative). [Lean](#) [Stmt](#) Let  $A$  be an injective MPS tensor. For all  $X, Y \in M_D(\mathbb{C})$ ,

$$O_A(XY) = O_A(X) O_A(Y).$$

*Proof.* [Proof](#) Applying Lemma 14.8 to  $X$  yields

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

Multiply by  $Y$  on the right and apply the linear map  $\Psi_A$ :

$$\Psi_A(A^i XY) = \sum_{j=0}^{d-1} O_A(X)_{ij} \Psi_A(A^j Y).$$

Taking the  $k$ th component of both sides gives

$$O_A(XY)_{ik} = \sum_{j=0}^{d-1} O_A(X)_{ij} O_A(Y)_{jk} = (O_A(X) O_A(Y))_{ik}.$$

Since this holds for all  $i$  and  $k$ , the matrices are equal.  $\square$

**Remark 14.10** (General injective case). When  $d > D^2$ , the decomposition map  $\Psi_A$  need not be unique. Nevertheless, Theorem 14.9 is already the general injective statement: for the fixed choice of  $\Psi_A$  in Definition 14.4, the associated physical realisation map  $O_A$  is multiplicative. This supersedes the earlier basis-only formulation.

## 14.3 The virtual bond gauge and the proportionality lemma

The two structural lemmas of [MGRPG<sup>+</sup>18] carry the entire fixed-length argument. The virtual bond gauge (Lemma 14.13) turns equality of the combined MPV families into the virtual-bond conjugacy  $X \mapsto Z^{-1}XZ$  by a fixed invertible matrix  $Z$ ; the proportionality lemma (Lemma 14.14) shows that two tensor pairs agreeing under all virtual insertions are proportional. Both rest on two elementary facts: injectivity is preserved by blocking, and the centre of  $M_D(\mathbb{C})$  consists of scalars.

### 14.3.1 Two preliminary facts

**Lemma 14.11** (Blocking preserves injectivity). [Lean] ✓ Stmt *Let  $A_0, \dots, A_{m-1}$  be injective MPS tensors with compatible bond dimensions. Then the blocked tensor  $(A_0 \otimes \dots \otimes A_{m-1})^{(i_0, \dots, i_{m-1})} := A_0^{i_0} A_1^{i_1} \dots A_{m-1}^{i_{m-1}}$  is again injective.*

*Proof.* ✓ Proof The span of the blocked matrices contains all products  $A_0^{i_0} \dots A_{m-1}^{i_{m-1}}$ . Since each  $\{A_k^{i_k}\}_i$  spans  $M_D(\mathbb{C})$ , these products span all of  $M_D(\mathbb{C})$ .  $\square$

**Lemma 14.12** (Centre of  $M_D(\mathbb{C})$ ). [Lean] ✓ Stmt *If  $Z \in M_D(\mathbb{C})$  commutes with every element of  $M_D(\mathbb{C})$ , then  $Z = \lambda \mathbb{1}_D$  for some  $\lambda \in \mathbb{C}$ .*

*Proof.* ✓ Proof Standard:  $Z$  commutes with every matrix unit  $E_{ij}$ , which forces all off-diagonal entries to vanish and all diagonal entries to agree.  $\square$

### 14.3.2 Same state forces a virtual bond gauge

The 3-site virtual-bond statement is used later in the proof of the PEPS fundamental theorem. Its hypothesis is the all-length MPV equality of the combined tensors, and its conclusion is the conjugacy of the middle-bond insertion data.

**Lemma 14.13** (Virtual bond gauge – 3-site case). [Lean] ✓ Stmt *Let  $A = (A_0, A_1, A_2)$  and  $B = (B_0, B_1, B_2)$  be two 3-site injective periodic chains with common bond dimension  $D \geq 1$  and  $\mathcal{V}(A) = \mathcal{V}(B)$  (same MPV family). Then at the middle bond there exists an invertible matrix  $Z \in \text{GL}_D(\mathbb{C})$  such that the virtual-insertion trace coefficients agree after conjugation:*

$$\text{tr}(A_0^{i_0} X A_1^{i_1} A_2^{i_2}) = \text{tr}(B_0^{i_0} Z^{-1} X Z B_1^{i_1} B_2^{i_2})$$

for all  $X \in M_D(\mathbb{C})$  and all physical indices  $i_0, i_1, i_2$ .

In [MGRPG<sup>+</sup>18, Lemma 1], the statement is given for arbitrary chain length  $n \geq 3$  and any bond position, with uniqueness of  $Z$  up to scalar. The chain-level generalization to arbitrary  $n \geq 3$  and any bond position is not pursued here.

*Proof.* ✓ Proof The  $\mathcal{V}(A) = \mathcal{V}(B)$  hypothesis gives, for every  $X \in M_D(\mathbb{C})$ , equality of the virtual-insertion trace coefficients:  $\text{tr}(A_0^{i_0} X A_1^{i_1} A_2^{i_2}) = \text{tr}(B_0^{i_0} Y B_1^{i_1} B_2^{i_2})$  for some  $Y$  depending on  $X$ . Since all three site tensors on each side are injective, the physical realisation maps  $O_A, O_B$  are algebra homomorphisms by Theorem 14.9. Setting  $T(X) = Y$  defines a  $\mathbb{C}$ -linear multiplicative map, hence a nonzero  $\mathbb{C}$ -algebra endomorphism of  $M_D(\mathbb{C})$ , which is bijective by Theorem 3.8. Skolem–Noether (Theorem 3.11) gives  $Z \in \text{GL}_D(\mathbb{C})$  with  $T(X) = Z^{-1}XZ$ .  $\square$

### 14.3.3 Aligned gauges force proportionality

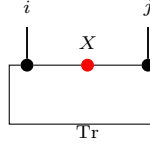
After applying Lemma 14.13 to each bond and absorbing the gauge matrices, one obtains two chains whose virtual-insertion operations agree on every bond. Lemma 2 of [MGRPG<sup>+</sup>18] shows that this agreement forces the local tensors to be proportional.

**Lemma 14.14** (Aligned gauges force proportionality – Lemma 2 of [MGRPG<sup>+</sup>18]). Lean ✓ Stmt  
 Let  $A_1, A_2$  and  $B_1, B_2$  be injective MPS tensors with the same bond dimension  $D$ . Suppose that for all  $X \in M_D(\mathbb{C})$  (on the internal bond), all  $Y \in M_D(\mathbb{C})$  (on the external bond), and all physical indices  $i, j$ ,

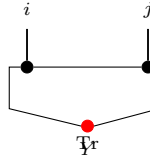
$$\text{tr}(A_1^i X A_2^j) = \text{tr}(B_1^i X B_2^j) \quad (14.1)$$

$$\text{tr}(A_2^j Y A_1^i) = \text{tr}(B_2^j Y B_1^i) \quad (14.2)$$

Diagrammatically, inserting  $X$  on the bond between the two sites (Eq. 14.1):



while the insertion  $Y$  on the outer trace bond (Eq. 14.2) is:



so the two hypotheses distinguish the internal and external virtual loops of the two-site periodic chain.

Then there exists a nonzero scalar  $\lambda \in \mathbb{C}^\times$  such that

$$A_1^i = \lambda B_1^i \quad \text{and} \quad A_2^j = \lambda^{-1} B_2^j$$

for all physical indices  $i, j$ .

*Proof.* ✓ Proof By cyclicity and non-degeneracy of the trace pairing, the conditions in Equations (14.1) and (14.2) give, for all physical indices  $i, j$ ,

$$A_2^j A_1^i = B_2^j B_1^i, \quad (14.3)$$

$$A_1^i A_2^j = B_1^i B_2^j. \quad (14.4)$$

Write  $\mathbb{1}_D = \sum_j c_j A_2^j$  using the decomposition map  $\Psi_{A_2}$ , and define  $Z := \sum_j c_j B_2^j \in M_D(\mathbb{C})$ . This is the point where the two-site comparison collapses to a single boundary matrix:



The first product identity (14.3) gives  $A_1^i = Z B_1^i$ , and the second (14.4) gives  $A_1^i = B_1^i Z$ . Hence  $Z$  commutes with every  $B_1^i$ ; since  $\{B_1^i\}$  spans  $M_D(\mathbb{C})$ ,  $Z$  is central. By Lemma 14.12,  $Z = \lambda \mathbb{1}_D$  for some  $\lambda \in \mathbb{C}^\times$  (nonzero since the tensors are nonzero). Thus  $A_1^i = \lambda B_1^i$ , and substituting into the first product identity (14.3) forces  $A_2^j = \lambda^{-1} B_2^j$ . □

## 14.4 The Fundamental Theorem for injective chains

The chain theorem combines the site and physical indices into a single physical index and applies the single-block Fundamental Theorem to the resulting tensor. Its hypothesis is all-length MPV equality of the combined tensors; the fixed-length same-state form is obtained from a separate same-state reduction hypothesis, treated below.

**Theorem 14.15** (Fundamental Theorem for injective chains). [Lean](#) [✓ Stmt](#) Let  $\mathbf{A} = (A_0, \dots, A_{n-1})$  and  $\mathbf{B} = (B_0, \dots, B_{n-1})$  be two  $n$ -site chains with common bond dimension  $D$ , and assume that  $\mathbf{A}$  is injective. Suppose that, after combining the site index and physical index into a single MPS tensor, the resulting two tensors generate the same MPV family. Then  $\mathbf{A}$  and  $\mathbf{B}$  are cyclically gauge equivalent.

*Proof.* [✓ Proof](#) Choose one site of  $\mathbf{A}$ ; its injectivity implies injectivity of the combined tensor by Lemma 14.11. Apply the single-block Fundamental Theorem (Theorem 3.12) to the two combined tensors. This yields one invertible matrix  $X$  such that every combined matrix of  $\mathbf{B}$  equals the corresponding combined matrix of  $\mathbf{A}$  conjugated by  $X$ . Reading this identity back at each site gives  $B_k^i = X A_k^i X^{-1}$  for all  $k$  and  $i$ , so the constant choice  $Z_k = X$  is a cyclic gauge.  $\square$

**Lemma 14.16** (Uniform site rescaling rescales the combined tensor). [Lean](#) [✓ Stmt](#) Let  $\mathbf{A} = (A_0, \dots, A_{n-1})$  be an  $n$ -site chain and let  $\zeta \in \mathbb{C}$ . If  $\zeta \mathbf{A}$  denotes the chain obtained by replacing each site tensor  $A_k$  with  $\zeta A_k$ , and if  $\widetilde{\mathbf{A}}$  denotes the combined tensor of  $\mathbf{A}$ , then

$$\widetilde{\zeta \mathbf{A}} = \zeta \widetilde{\mathbf{A}}.$$

*Proof.* [✓ Proof](#) Combining the site and physical indices does not change which local matrix is read at the component  $(k, i)$ , so both sides evaluate to  $\zeta A_k^i$  there.  $\square$

**Theorem 14.17** (Injective-chain Fundamental Theorem up to a global phase). [Lean](#) [✓ Stmt](#) Let  $\mathbf{A} = (A_0, \dots, A_{n-1})$  and  $\mathbf{B} = (B_0, \dots, B_{n-1})$  be two  $n$ -site chains with common bond dimension  $D$ , and assume that  $\mathbf{A}$  is injective. Suppose that the combined tensors  $\widetilde{\mathbf{A}}$  and  $\widetilde{\mathbf{B}}$  are gauge-phase equivalent. Then there exist invertible matrices  $Z_0, \dots, Z_{n-1} \in \text{GL}_D(\mathbb{C})$  and a nonzero scalar  $\zeta \in \mathbb{C}^\times$  such that

$$B_k^i = \zeta Z_k A_k^i Z_{k+1}^{-1} \tag{14.5}$$

for all sites  $k$  and physical indices  $i$ , where  $k+1$  is interpreted cyclically modulo  $n$ .

*Proof.* [✓ Proof](#) If the combined tensor of  $\mathbf{B}$  differs from that of  $\mathbf{A}$  by a gauge and a nonzero scalar  $\zeta$ , rescale each site of  $\mathbf{B}$  by  $\zeta^{-1}$ . Lemma 14.16 removes this common scalar from the combined tensor, so Theorem 14.15 applies to the rescaled chain. Multiplying the resulting sitewise relations by  $\zeta$  gives the conclusion (14.5).  $\square$

### 14.4.1 From fixed-length chain states to combined-tensor MPV agreement

The chain theorem assumes equality of the *combined-tensor* MPV families. Converting the fixed-length same-state hypothesis into this form is isolated as a separate hypothesis, which the source proof establishes by the virtual bond gauge and proportionality lemmas above.

**Definition 14.18** (Same-state reduction hypothesis). [Lean](#) [✓ Stmt](#) A *same-state reduction hypothesis* for dimensions  $(d, D)$  is the assertion that whenever  $\mathbf{A}$  is an injective  $n$ -site chain with  $n \geq 3$  and  $\mathbf{A}, \mathbf{B}$  generate the same chain state, then the combined tensors of  $\mathbf{A}$  and  $\mathbf{B}$  generate the same MPV family.

**Lemma 14.19** (Same state gives combined-tensor MPV agreement). [Lean](#) [✓ Stmt](#) Assume a same-state reduction hypothesis for  $(d, D)$ . Let  $\mathbf{A}$  and  $\mathbf{B}$  be  $n$ -site chains with common bond dimension  $D$ , with  $n \geq 3$ , and assume that  $\mathbf{A}$  is injective. If  $\mathbf{A}$  and  $\mathbf{B}$  generate the same chain state, then the combined tensors of  $\mathbf{A}$  and  $\mathbf{B}$  generate the same MPV family.

*Proof.* [✓ Proof](#) This is exactly the defining property of the same-state reduction hypothesis.  $\square$

**Theorem 14.20** (Fundamental Theorem for injective chains from same state). [Lean](#) [✓ Stmt](#) Assume a same-state reduction hypothesis for  $(d, D)$ . Let  $\mathbf{A}$  and  $\mathbf{B}$  be  $n$ -site chains with common bond dimension  $D$ , with  $n \geq 3$ , and assume that  $\mathbf{A}$  is injective. If  $\mathbf{A}$  and  $\mathbf{B}$  generate the same chain state, then  $\mathbf{A}$  and  $\mathbf{B}$  are cyclically gauge equivalent.

*Proof.* [✓ Proof](#) Apply Lemma 14.19 to convert the fixed-length chain-state hypothesis into equality of the combined-tensor MPV families, then invoke Theorem 14.15.  $\square$

*Remark 14.21* (Relationship to the translation-invariant all-sizes theorem). Theorem 14.15 treats a fixed chain length and assumes equality of the combined-tensor MPV families. The single-block Fundamental Theorem (Theorem 3.12) instead studies one tensor at all system sizes and concludes exact gauge equivalence  $B^i = XA^iX^{-1}$ . The chain theorem is recovered by converting the chain into one combined tensor; the price is that the hypothesis is all-length MPV equality for that combined tensor rather than equality of the chain state at one fixed length.

## 14.4.2 Finite-length MPV agreement

The single-block theorem can be strengthened by weakening its hypothesis from all-length to finite-length MPV agreement: for an injective tensor, agreement from any threshold  $N_0$  onward already implies full agreement.

**Definition 14.22** (Finite-length MPV agreement). [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  of the same bond dimension satisfy *finite-length MPV agreement from  $N_0$  onward* if they generate the same MPV family for all system sizes  $N \geq N_0$ , meaning that for every  $N \geq N_0$  and  $\sigma \in [d]^N$ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(B)_\sigma$$

**Theorem 14.23** (Finite-length agreement implies full agreement). [Lean](#) [✓ Stmt](#) Let  $A$  be an injective tensor of bond dimension  $D \geq 1$ , and let  $B$  be a tensor of the same bond dimension. If  $A$  and  $B$  generate the same MPV family for every system size  $N \geq N_0$ , then they generate the same MPV family for every system size.

*Proof.* [✓ Proof](#) Write  $1 = \sum_i c_i A^i$  (by injectivity) and set  $S = \sum_i c_i B^i$ .

1. Show that the length- $N_0$  word span of  $B$  is all of  $M_D(\mathbb{C})$  via the linear relation between the length- $N_0$  word-evaluation maps of  $A$  and  $B$  induced by trace agreement at length  $N_0 + 1$ , together with a finrank transfer argument.
2. Derive  $S = 1$  by trace nondegeneracy: for every  $M \in M_D(\mathbb{C})$ ,  $\text{tr}(M(S - 1)) = 0$ .
3. Downward induction on  $k = N_0 - |w|$ : decompose  $\text{tr}(w_A)$  and  $\text{tr}(w_B)$  using  $1 = \sum c_i A^i$  and  $S = 1$  respectively, then apply the inductive hypothesis at length  $|w| + 1$ .

$\square$

**Theorem 14.24** (Strengthened single-block FT (finite-length)). Lean ✓ Stmt *Let  $A$  be an injective tensor of bond dimension  $D \geq 1$ , and let  $B$  be a tensor of the same bond dimension. If  $A$  and  $B$  agree on all MPV coefficients for system sizes  $N \geq N_0$  (for any threshold  $N_0$ ), then  $A$  and  $B$  are gauge equivalent.*

*Proof.* ✓ Proof Apply Theorem 14.23 to obtain equality of the MPV families for all system sizes, then invoke the single-block Fundamental Theorem (Theorem 3.12).  $\square$

## 14.5 Corollaries: translation-invariant, non-uniform, and blocked chains

The chain theorem specializes to three settings: constant translation-invariant chains, non-uniform chains whose state happens to be translation-invariant, and constant chains of blocked tensors.

### 14.5.1 Translation-invariant chains

**Theorem 14.25** (Single gauge for translation-invariant chains). Lean ✓ Stmt *Let  $A$  and  $B$  be MPS tensors of bond dimension  $D$ , and fix a chain length  $n \geq 1$ . Assume that  $A$  is injective and that the combined tensors of the two constant chains  $(A, \dots, A)$  and  $(B, \dots, B)$  generate the same MPV family. Then there exists an invertible matrix  $X \in \text{GL}_D(\mathbb{C})$  such that*

$$B^i = X A^i X^{-1}$$

for every physical index  $i$ .

*Proof.* ✓ Proof Bundle the constant chains into one tensor. Injectivity of  $A$  implies injectivity of the combined tensor by Lemma 14.11, and the combined-tensor MPV equality is exactly the hypothesis needed for the single-block Fundamental Theorem. The resulting gauge matrix is independent of the site because every site carries the same local tensor.  $\square$

**Corollary 14.26** (Translation-invariant collapse to one gauge). Lean ✓ Stmt *Let  $A$  and  $B$  be MPS tensors of bond dimension  $D$ , and fix a chain length  $n \geq 1$ . Assume that  $A$  is injective and that the combined tensors of the two constant chains  $(A, \dots, A)$  and  $(B, \dots, B)$  generate the same MPV family. Then there exist an invertible matrix  $Z \in \text{GL}_D(\mathbb{C})$  and a scalar  $\lambda \in \mathbb{C}^\times$  with  $\lambda^n = 1$  such that*

$$B^i = \lambda Z^{-1} A^i Z \tag{14.6}$$

for every physical index  $i$ .

*Proof.* ✓ Proof Apply Theorem 14.25 to obtain  $B^i = X A^i X^{-1}$ . Set  $Z = X^{-1}$  and  $\lambda = 1$ . Then  $\lambda^n = 1$  and the identity (14.6) is exactly the same gauge relation rewritten in the form used by the corollary.  $\square$

### 14.5.2 Non-translation-invariant descriptions of translation-invariant states

**Corollary 14.27** (Translation-invariant states from nonuniform chains). Not ready *Assume a same-state reduction hypothesis for  $(d, D)$ . Let  $\mathbf{A} = (A_0, \dots, A_{n-1})$  be an injective non-translation-invariant chain of length  $n \geq 3$ . Suppose the state generated by  $\mathbf{A}$  is translation-invariant, i.e. it equals the state generated by the cyclically shifted chain  $(A_1, \dots, A_{n-1}, A_0)$ . Then  $\mathbf{A}$  admits a*

translation-invariant description with the same bond dimension: there exists an injective tensor  $B$  such that the constant chain  $(B, \dots, B)$  generates the same state as  $\mathbf{A}$ . Explicitly,  $B^i = A_0^i Z_0^{-1}$  (right multiplication, not conjugation), where  $Z_0$  is extracted from the cyclic shift.

*Proof.* Apply Theorem 14.20 to  $\mathbf{A}$  and its cyclic shift. This gives invertible gauges  $Z_0, \dots, Z_{n-1}$  with  $A_{k+1}^i = Z_k^{-1} A_k^i Z_{k+1}$ . Iterating these identities yields  $A_k^i = L_k^{-1} A_0^i R_k$ , where the products  $L_k, R_k$  satisfy  $R_k L_{k+1}^{-1} = Z_0^{-1}$ . Substituting into the chain trace shows that the same state is generated by the constant tensor  $B^i := A_0^i Z_0^{-1}$ . Since  $A_0$  is injective and  $Z_0$  is invertible,  $B$  is injective.  $\square$

Symmetry-theoretic consequences of the Fundamental Theorem, including virtual representations and string order, are developed in Chapter 12.

### 14.5.3 Blocked chains

**Lemma 14.28** (Block injectivity and blocked-tensor injectivity). [Lean](#) [✓ Stmt](#) A tensor  $A$  is  $L$ -block injective if and only if its  $L$ -blocked tensor  $A^{[L]}$  is injective.

*Proof.* [✓ Proof](#) The blocked physical alphabet is just a reindexing of words of length  $L$ . Decoding blocked indices therefore identifies the two spanning families, so the two injectivity conditions are equivalent.  $\square$

**Definition 14.29** (Constant blocked chain). [Lean](#) [✓ Stmt](#) For an MPS tensor  $A$ , a blocking length  $L$ , and a chain length  $n$ , the *blocked chain* is the constant  $n$ -site chain whose local tensor is the blocked tensor  $A^{[L]}$ .

**Lemma 14.30** (Blocked chains are injective). [Lean](#) [✓ Stmt](#) If  $A$  is  $L$ -block injective, then the constant blocked chain built from  $A^{[L]}$  is injective at every site.

*Proof.* [✓ Proof](#) By Lemma 14.28, the local blocked tensor  $A^{[L]}$  is injective, and therefore every site of the constant blocked chain is injective.  $\square$

**Theorem 14.31** (Fundamental Theorem for blocked chains). [Lean](#) [✓ Stmt](#) Let  $A$  and  $B$  be MPS tensors of bond dimension  $D$ , let  $L$  be a blocking length, and let  $n$  be a chain length. Assume that  $A$  is  $L$ -block injective and that the combined tensors of the two constant blocked chains built from  $A^{[L]}$  and  $B^{[L]}$  generate the same MPV family. Then the two blocked chains are gauge equivalent.

*Proof.* [✓ Proof](#) By Lemma 14.30, the blocked chain built from  $A^{[L]}$  is injective. Apply Theorem 14.15 to the two constant blocked chains.  $\square$

*Remark 14.32.* The theorem above is the blocked-chain result established here. Recovering site-wise gauges for the original unblocked tensors gives the normal-MPS corollary of [MGRPG<sup>+</sup>18]; that step is outside this blocked-chain statement.

## 14.6 Product algebra construction and per-block gauge

The chain Fundamental Theorem (Theorem 14.15) handles non-translation-invariant periodic chains. A parallel development addresses block-diagonal tensors: given a family of injective block tensors  $\{A_k^i\}_{k=0}^{r-1}$  with block dimensions  $D_k$  and a second family  $\{B_k^i\}$  with per-block  $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ , one seeks per-block gauge equivalence  $B_k^i = X_k A_k^i X_k^{-1}$  for each  $k$  independently, and a global block-permutation plus inner-automorphism decomposition. Throughout this section, assume  $D_k \geq 1$  for every block  $k$ .

### 14.6.1 Block permutation algebra

**Definition 14.33** (Block ideal). [Lean](#) [✓ Stmt](#) Let  $\{R_k\}_{k=0}^{r-1}$  be simple rings. The  $k$ -th block ideal in  $\prod_{j=0}^{r-1} R_j$  is

$$I_k := \{x \in \prod_{j=0}^{r-1} R_j : x_j = 0 \text{ for } j \neq k\}.$$

**Theorem 14.34** (Each block ideal is an atom). [Lean](#) [✓ Stmt](#) Each block ideal  $I_k$  is an atom in the lattice of two-sided ideals of  $\prod_{j=0}^{r-1} R_j$ .

*Proof.* [✓ Proof](#) Under the order isomorphism between two-sided ideals of  $\prod_{j=0}^{r-1} R_j$  and products of two-sided ideal lattices,  $I_k$  corresponds to the tuple with  $\top$  in the  $k$ -th coordinate and  $\perp$  elsewhere. This tuple is an atom, hence so is  $I_k$ .  $\square$

**Theorem 14.35** (Ring isomorphisms permute block ideals). [Lean](#) [✓ Stmt](#) Let  $\{R_k\}_{k=0}^{r-1}$  be simple rings. Every ring isomorphism  $T : \prod_k R_k \rightarrow \prod_k R_k$  permutes the block ideals: there exists a permutation  $\pi$  such that  $T$  maps the  $k$ -th block ideal to the  $\pi(k)$ -th block ideal.

*Proof.* [✓ Proof](#) By Theorem 14.34, each block ideal is an atom in the lattice of two-sided ideals of  $\prod_k R_k$ . A ring isomorphism induces an order automorphism of this lattice, so it permutes the atoms. Since the atoms are exactly the block ideals, this gives the required permutation  $\pi$ .  $\square$

**Theorem 14.36** (Dimension preservation). [Lean](#) [✓ Stmt](#) Assume  $D_k \geq 1$  for all  $k$ . If  $T$  is an algebra automorphism of  $\prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$ , and  $\pi$  is the induced permutation, then  $D_{\pi(k)} = D_k$  for all  $k$ .

*Proof.* [✓ Proof](#) The restriction of  $T$  to the  $k$ -th block ideal induces a  $\mathbb{C}$ -algebra isomorphism  $M_{D_k}(\mathbb{C}) \cong M_{D_{\pi(k)}}(\mathbb{C})$ , which forces  $D_k = D_{\pi(k)}$  by comparing dimensions as  $\mathbb{C}$ -vector spaces.  $\square$

**Theorem 14.37** (Per-block component maps are bijections). [Lean](#) [✓ Stmt](#) Assume  $D_k \geq 1$  for all  $k$ . Let  $T$  be a ring automorphism of  $\prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$ , and let  $\pi$  be the induced permutation from Theorem 14.35. For each block index  $k$ , the map

$$M \mapsto (T(0, \dots, 0, M, 0, \dots, 0))_{\pi(k)}$$

from  $M_{D_k}(\mathbb{C})$  to  $M_{D_{\pi(k)}}(\mathbb{C})$  is bijective.

*Proof.* [✓ Proof](#) Theorem 14.35 forces a single-support element in the  $k$ -th block ideal to remain supported on the  $\pi(k)$ -th block after applying  $T$ . Injectivity follows from injectivity of  $T$ . For surjectivity, apply the same argument to  $T^{-1}$  and the inverse permutation.  $\square$

**Theorem 14.38** (Decomposition of automorphisms of  $\prod M_{D_k}(\mathbb{C})$ ). [Lean](#) [✓ Stmt](#) Assume  $D_k \geq 1$  for all  $k$ . Every  $\mathbb{C}$ -algebra automorphism  $T$  of  $\prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$  decomposes as per-block inner automorphisms followed by a block permutation  $\pi$ : there exist  $\pi \in S_r$  with  $D_{\pi(k)} = D_k$  and invertible matrices  $X_k \in \text{GL}_{D_k}(\mathbb{C})$  such that, for every tuple  $M = (M_0, \dots, M_{r-1})$  and every block index  $k$ ,  $T(M)_{\pi(k)} = X_k M_k X_k^{-1}$ , where  $T(M)_j$  denotes the  $j$ -th block component of  $T(M)$ .

*Proof.* [✓ Proof](#) Theorem 14.35 gives the permutation  $\pi$ . Theorem 14.36 identifies the matrix sizes  $D_{\pi(k)} = D_k$ . For each  $k$ , Theorem 14.37 gives a bijective multiplicative map from  $M_{D_k}(\mathbb{C})$  to



$M_{D_{\pi(k)}}(\mathbb{C})$  extracted from the elements supported on one factor of  $T$ ; after reindexing by the dimension equality, this is a  $\mathbb{C}$ -algebra automorphism of  $M_{D_k}(\mathbb{C})$ . Skolem–Noether (Theorem 3.11) makes each such automorphism inner.

This is a standard result in the theory of semisimple algebras; see [CPGSV21, Section IV.A] for the MPS context.  $\square$

*Remark 14.39* (Ring level versus algebra level). Theorem 14.35 is a ring-level statement: it only uses the ideal structure of  $\prod_k R_k$ . Theorem 14.37 stays at this ring level: once the block ideals are permuted, one can extract the corresponding single-block maps without using scalar multiplication. Theorems 14.36 and 14.38 require an upgrade to  $\mathbb{C}$ -algebra automorphisms. The reason is threefold: dimension preservation compares the blocks as  $\mathbb{C}$ -vector spaces; Skolem–Noether applies to the resulting central simple  $\mathbb{C}$ -algebras; and the final per-block maps must commute with scalar multiplication. Thus the argument separates the ring-level permutation step from the later algebra-level decomposition.

## 14.6.2 Per-block linear extension and the product automorphism

Per-block equality of MPV families produces a unique linear map on each matrix block; assembling these maps componentwise yields an algebra automorphism of the product, to which the block-permutation decomposition applies.

**Definition 14.40** (Per-block linear extension). Lean ✓ Stmt Let  $A_k, B_k$  be MPS tensors of bond dimension  $D_k$  with  $A_k$  injective and  $\mathcal{V}(A_k) = \mathcal{V}(B_k)$  for each block  $k \in \{0, \dots, r-1\}$ . The *per-block linear extension*  $T_k : M_{D_k}(\mathbb{C}) \rightarrow M_{D_k}(\mathbb{C})$  is the unique  $\mathbb{C}$ -linear map satisfying  $T_k(A_k^i) = B_k^i$  for every physical index  $i$ .

**Lemma 14.41** (Per-block multiplicativity). Lean ✓ Stmt For each block  $k$ , the per-block linear extension  $T_k$  is multiplicative:  $T_k(MN) = T_k(M)T_k(N)$  for all  $M, N \in M_{D_k}(\mathbb{C})$ .

*Proof.* ✓ Proof Since  $\{A_k^i\}$  spans  $M_{D_k}(\mathbb{C})$ , it suffices to check on spanning elements, where the result follows from the MPV identity.  $\square$

**Lemma 14.42** (Per-block bijectivity). Lean ✓ Stmt Each  $T_k$  is bijective.

*Proof.* ✓ Proof A nonzero multiplicative linear endomorphism of  $M_{D_k}(\mathbb{C})$  is automatically bijective: injectivity follows from the simplicity of  $M_{D_k}(\mathbb{C})$  (the kernel is a proper ideal, hence zero), and surjectivity from finite dimension. Non-vanishing of  $T_k$  is established via the trace pairing: if  $T_k = 0$ , then all  $B_k^i = 0$ , contradicting the non-vanishing of the trace pairing for injective tensors.  $\square$

**Definition 14.43** (Product algebra automorphism). Lean ✓ Stmt The *product algebra automorphism* is the  $\mathbb{C}$ -algebra equivalence

$$T : \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C}) \xrightarrow{\sim} \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$$

defined by  $T(M)_k = T_k(M_k)$ , where each  $T_k$  is the per-block linear extension. It is an algebra automorphism since each  $T_k$  is a multiplicative, unital, bijective linear map. Diagrammatically, this is the componentwise map on the product algebra obtained by applying the corresponding extension to each summand.

**Theorem 14.44** (Block-permutation decomposition). Lean ✓ Stmt *The product algebra automorphism  $T$  decomposes as a block permutation composed with per-block inner automorphisms: there exist a permutation  $\sigma \in S_r$  (with  $D_{\sigma(k)} = D_k$  for all  $k$ ) and invertible matrices  $X_k \in \text{GL}_{D_k}(\mathbb{C})$  such that*

$$(T(M))_{\sigma(k)} = X_k M X_k^{-1}$$

*for all  $M \in M_{D_k}(\mathbb{C})$  (viewed as supported in block  $k$ ) and all  $k$ . Equivalently, the product-algebra automorphism first permutes the simple matrix blocks and then acts inside each block by an inner conjugation.*

*Proof.* ✓ Proof Apply Theorem 14.38 to the product-algebra automorphism of Definition 14.43. The theorem writes the nonzero input block as  $\pi(k)$  in output component  $k$ ; for the statement above we set  $\sigma := \pi^{-1}$ , so an element supported in block  $k$  lands in block  $\sigma(k)$ .  $\square$

### 14.6.3 Nondegeneracy of the product trace pairing

**Lemma 14.45** (Nondegeneracy of the product trace pairing). Lean ✓ Stmt *Let  $M = (M_0, \dots, M_{r-1}) \in \prod_k M_{D_k}(\mathbb{C})$ . If  $\sum_k \text{tr}(M_k N_k) = 0$  for all  $N = (N_0, \dots, N_{r-1})$ , then  $M = 0$ .*

*Proof.* ✓ Proof Choose  $N$  supported on a single block  $k$  (all other components zero). The hypothesis reduces to  $\text{tr}(M_k N_k) = 0$  for all  $N_k$ , so  $M_k = 0$  by nondegeneracy of the matrix trace pairing on  $M_{D_k}(\mathbb{C})$ . Since  $k$  was arbitrary,  $M = 0$ .  $\square$

**Lemma 14.46** (Injectivity of the product Gram map). Lean ✓ Stmt *Let  $\{A_k\}$  be a family of injective block tensors. The product Gram map  $M \mapsto (k, i \mapsto \text{tr}(M_k \cdot A_k^i))$  is injective.*

*Proof.* ✓ Proof Reduces to per-block injectivity of the Gram map  $M_k \mapsto (i \mapsto \text{tr}(M_k \cdot A_k^i))$ , which holds by injectivity of  $A_k$  (the matrices  $\{A_k^i\}$  span  $M_{D_k}(\mathbb{C})$ , so the trace pairing against them separates points).  $\square$

## 14.7 Gauge equivalence for direct sums

The first lemmas in this section are elementary facts about block-diagonal tensors: they assemble already-known block gauges, and they record the easy consequences of applying the injective single-block theorem to each block separately. They do not derive the block matching appearing in [CPGSV16, Theorem 2.10]. In the source proof, that matching is obtained from BNT overlap, independence, and coefficient comparison; the corresponding global gauge assembly is Theorem 11.12.

**Definition 14.47** (Flattened block-diagonal gauge). Lean Lean ✓ Stmt Given per-block gauges  $X_k \in \text{GL}_{D_k}(\mathbb{C})$ , first form the dependent block-diagonal element  $\bigoplus_k X_k$  and then reindex the dependent block-coordinate index to the standard index  $\{0, \dots, \sum_k D_k - 1\}$  of the assembled tensor. The resulting element of  $\text{GL}_{\sum_k D_k}(\mathbb{C})$  is the global gauge used in the conjugation formula for  $\bigoplus_k \mu_k A_k$ .

**Lemma 14.48** (Explicit direct-sum conjugation by the flattened gauge). Lean Lean ✓ Stmt *If each block satisfies  $B_k^i = X_k A_k^i X_k^{-1}$ , then the weighted direct sums satisfy, for every physical index  $i$ ,*

$$\bigoplus_k \mu_k B_k^i = X \left( \bigoplus_k \mu_k A_k^i \right) X^{-1}, \quad (14.7)$$

*where  $X$  is the flattened block-diagonal gauge of Definition 14.47. Consequently the two assembled tensors are gauge equivalent with this explicit witness.*

*Proof.* ✓ Proof The block-diagonal matrix product is computed blockwise. The scalar weight  $\mu_k$  commutes with the per-block conjugation, so each diagonal block of the right-hand side is  $\mu_k B_k^i$ . Reindexing from the dependent block coordinates to the standard bond index preserves multiplication and inverses. The gauge equivalence follows directly from the conjugation identity (14.7).  $\square$

**Lemma 14.49** (Gauge equivalence of direct sums from per-block gauge equivalence). Lean  
✓ Stmt *If each pair  $(A_k, B_k)$  is gauge equivalent, then the weighted block-diagonal tensors  $\bigoplus_k \mu_k A_k$  and  $\bigoplus_k \mu_k B_k$  are gauge equivalent.*

*Proof.* ✓ Proof Choose per-block gauge matrices via the hypothesis; then apply Lemma 14.48.  $\square$

**Lemma 14.50** (Gauge equivalence of direct sums with phase-weight absorption). Lean ✓ Stmt  
*Let  $A_k, B_k$  be block families, and let  $X_k \in \text{GL}_{D_k}(\mathbb{C})$  and scalars  $\zeta_k \in \mathbb{C}$  satisfy  $B_k^i = \zeta_k \cdot X_k A_k^i X_k^{-1}$  for every block  $k$  and physical index  $i$ . If the weights satisfy  $\mu_A(k) = \mu_B(k) \zeta_k$ , then the weighted block-diagonal tensors  $\bigoplus_k \mu_A(k) A_k$  and  $\bigoplus_k \mu_B(k) B_k$  are gauge equivalent.*

*Proof.* ✓ Proof Per-block, the weight identity  $\mu_A(k) = \mu_B(k) \zeta_k$  rewrites the scaled relation as  $\mu_B(k) B_k^i = X_k (\mu_A(k) A_k^i) X_k^{-1}$ . Each weighted block then satisfies an ordinary conjugation, and the block-diagonal conjugation formula assembles the per-block  $X_k$  into the flattened global gauge  $\bigoplus_k X_k$  relating  $\bigoplus_k \mu_A(k) A_k$  to  $\bigoplus_k \mu_B(k) B_k$ .  $\square$

### 14.7.1 Same-index direct-sum corollaries from the single-block theorem

*Remark 14.51.* See Definition 2.39 for the formalized block-diagonal tensor construction. The results in this subsection assume that the block correspondence has already been fixed; they do not prove [CPGSV16, Theorem 2.10]. They are therefore retained here as same-index assembly corollaries, not among the BNT preparation steps used in the source proof or in the after-blocking canonical-form reduction.

**Lemma 14.52** (Per-block application of the single-block theorem). Lean ✓ Stmt *If all block tensors  $A_k$  are injective and the per-block same-MPV condition holds ( $A_k$  and  $B_k$  generate the same MPV family for each  $k$ ), then each pair  $(A_k, B_k)$  is gauge equivalent.*

*Proof.* ✓ Proof Apply the single-block Fundamental Theorem (Theorem 3.12) to each block independently.  $\square$

*Remark 14.53* (Per-block restatement, not the multi-block paper theorem). This is a block-sum reformulation of the per-block injective single-block FT: the block pairing  $(A_k, B_k)$  is assumed already given and the blocks already matched. It is not the multi-block Fundamental Theorem of [CPGSV16, Theorem 2.10, lines 349–352], which derives the block matching and permutation from only the global same-MPV hypothesis.

**Lemma 14.54** (Per-block same MPV implies global same MPV). Lean ✓ Stmt *If  $A_k$  and  $B_k$  generate the same MPV family for each block  $k$ , then the block-diagonal tensors  $\bigoplus_k \mu_k A_k$  and  $\bigoplus_k \mu_k B_k$  also generate the same MPV family.*

*Proof.* ✓ Proof By the MPV decomposition (Theorem 2.41), each MPV is  $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$ . Per-block equality of MPV coefficients gives global equality.  $\square$

*Remark 14.55* (Direct-sum congruence). This is a direct-sum congruence statement: per-block MPV equality lifts to global MPV equality via the block-diagonal decomposition formula.

**Lemma 14.56** (Gauge equivalence for same-index direct sums). [Lean](#) [✓ Stmt](#) *If each block  $A_k$  is injective and each pair  $(A_k, B_k)$  generates the same MPV family, then the block-diagonal tensors  $\bigoplus_k \mu_k A_k$  and  $\bigoplus_k \mu_k B_k$  are gauge equivalent.*

*Proof.* [✓ Proof](#) Apply the per-block single-block Fundamental Theorem (Lemma 14.52) to obtain blockwise gauge equivalences from injectivity and equality of MPV families, then combine them into a global gauge for the block-diagonal tensors.  $\square$

*Remark 14.57* (Block-diagonal gauge assembly, not the FT block-matching theorem). This combines per-block gauges into a block-diagonal gauge for the global assembled tensor, given pre-matched same-index blocks. It is not the FT block-matching theorem: it does not derive the block pairing or permutation from a global same-MPV hypothesis; the block correspondence is assumed in advance.

**Lemma 14.58** (Gauge equivalence of direct sums from per-block MPV equality). [Lean](#) [✓ Stmt](#) *Let  $\{A_k\}_{k=0}^{r-1}$  and  $\{B_k\}_{k=0}^{r-1}$  be two families of MPS tensors with block dimensions  $D_k$ , each  $A_k$  injective, and  $\mathcal{V}(A_k) = \mathcal{V}(B_k)$  for all  $k$ . Then:*

1. *Per-block gauge equivalence:  $B_k^i = X_k A_k^i X_k^{-1}$  for some  $X_k \in \text{GL}_{D_k}(\mathbb{C})$ , for every  $k$  and  $i$ .*
2. *Global gauge equivalence: the block-diagonal tensors  $\bigoplus_k \mu_k A_k$  and  $\bigoplus_k \mu_k B_k$  are gauge equivalent.*

Explicitly, for  $X = \bigoplus_k X_k$ ,

$$\bigoplus_k \mu_k B_k^i = X \left( \bigoplus_k \mu_k A_k^i \right) X^{-1}. \quad (14.8)$$

*Proof.* [✓ Proof](#) Per-block gauge equivalence follows from applying the single-block Fundamental Theorem to each block, using injectivity of  $A_k$  and  $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ . With  $X = \bigoplus_k X_k$ , the identity (14.8) follows by multiplying block-diagonal matrices block by block.  $\square$

**Lemma 14.59** (Explicit per-block gauges). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Lemma 14.58, there exist invertible matrices  $X_k \in \text{GL}_{D_k}(\mathbb{C})$  such that*

$$B_k^i = X_k A_k^i X_k^{-1}$$

*for every block  $k$  and every physical index  $i$ .*

*Proof.* [✓ Proof](#) Extract the gauge matrix from the per-block part of Lemma 14.58 for each block.  $\square$

**Lemma 14.60** (Product-algebra decomposition of per-block MPV equality). [Lean](#) [✓ Stmt](#) *Assume that every block dimension is nonzero. Under the hypotheses of Lemma 14.58, the product-algebra automorphism determined by the per-block linear extensions decomposes into a block permutation together with inner conjugation on each block: there exist a permutation  $\sigma$  of the blocks, compatible block-dimension equalities, and invertible matrices  $X_k$  such that, after reindexing the target block  $\sigma(k)$  back to block  $k$ , the action on the  $k$ th matrix block is  $M \mapsto X_k M X_k^{-1}$ .*

*Proof.* [✓ Proof](#) Apply Theorem 14.44 to the product-algebra automorphism built from the per-block linear extensions.  $\square$

### 14.7.2 MPV invariance under reindexing

**Lemma 14.61** (MPV invariance under block permutation). [Lean] [✓ Stmt] *Permuting the block index set and transporting the weights correspondingly does not change the  $\text{SameMPV}_2$  family of the weighted block-diagonal tensor.*

*Proof.* [✓ Proof] The sum of per-block MPV contributions is invariant under rearrangement of the summands. □

**Lemma 14.62** (MPV invariance under dimension cast). [Lean] [✓ Stmt] *If two block-dimension families agree pointwise, the corresponding weighted block-diagonal tensors generate the same  $\text{SameMPV}_2$  family after transporting the blocks across the dimension equalities.*

*Proof.* [✓ Proof] Substituting the equality of dimensions makes the two tensor families identical. □

### 14.7.3 Converse: gauge equivalence implies same MPV

**Lemma 14.63** (Gauge equivalence of direct sums implies same MPV). [Lean] [✓ Stmt] *If the weighted block-diagonal tensors  $\bigoplus_k \mu_k A_k$  and  $\bigoplus_k \mu_k B_k$  are gauge equivalent, then they generate the same MPV family.*

*Proof.* [✓ Proof] Gauge equivalence preserves all traces of word evaluations, hence all MPV coefficients. □

### 14.7.4 Canonical-form tensor as a weighted direct sum

**Lemma 14.64** (Canonical-form tensor agrees with block sum). [Lean] [✓ Stmt] *For a canonical-form record  $C$ , the assembled tensor  $T_C$  is definitionally equal to the weighted block-diagonal tensor  $\bigoplus_k \mu_k^C A_k^C$ , where  $\mu_k^C$  are the block weights and  $A_k^C$  are the block tensors stored in  $C$ .*

*Proof.* [✓ Proof] The equality holds by definition;  $T_C$  is defined as the weighted block-diagonal tensor formed from the weights and block tensors of  $C$ . □

**Theorem 14.65** (Fundamental theorem for canonical forms with the same block structure). [Lean] [✓ Stmt] *Let  $C$  be a canonical-form record with block weights  $\mu_k^C$  and block tensors  $A_k^C$ . Let  $\{B_k\}$  be a family of MPS tensors on the same block dimensions, and assume that each  $A_k^C$  is injective and generates the same MPV family as  $B_k$ . Then  $T_C$  and  $\bigoplus_k \mu_k^C B_k$  are gauge equivalent.*

*Proof.* [✓ Proof] Rewrite  $T_C$  as a weighted block-diagonal tensor via Lemma 14.64, then apply the multi-block global gauge-equivalence lemma (Lemma 14.56). □

**Lemma 14.66** (Per-block equality of MPV families and per-block gauge equivalence). [Lean] [✓ Stmt] *Let  $\{A_k\}$  be a family of injective block tensors. Then  $\mathcal{V}(A_k) = \mathcal{V}(B_k)$  for all  $k$  if and only if  $A_k$  and  $B_k$  are gauge equivalent for all  $k$ .*

*Proof.* [✓ Proof] The forward direction applies the single-block Fundamental Theorem to each block. The reverse direction holds because gauge equivalence preserves all traces of word evaluations. □

### 14.7.5 Equal-case block matching with differing bond dimensions

The same-index lemmas above assume a fixed block correspondence. The equal-case theorem on the BNT canonical form instead derives the sector matching from the same-MPV hypothesis alone, allowing different sector counts and bond dimensions before matching.

**Lemma 14.67** (Equal-case block matching with differing bond dimensions). [Lean](#) [✓ Stmt](#) *Let  $P$  and  $Q$  be two BNT canonical forms (Definition 10.27), each admitting a unit-modulus copy weight in every basis sector, possibly with different sector counts and different bond dimensions before matching. If their block-diagonal tensors generate the same MPV family (Definition 2.12), then the basis sectors are bijectively matched by a permutation  $\beta$ , each matched basis pair is gauge-phase equivalent (Definition 2.23), the matched copy multiplicities agree, and the copy weights agree up to a unit-modulus phase  $\zeta$  and a copy-index reindexing. This is the sector-data form of the equal-case theorem [CPGSV16, Corollary 2.11, lines 354–361 and appendix 1172–1192]; the corresponding global gauge corollary is recorded in Chapter 11.*

*Proof.* [✓ Proof](#) On the BNT canonical form, the proof proceeds in three steps: bijective block matching from non-decaying cross-overlaps, coefficient identity via the global-gauge phase  $\zeta$ , and multiset comparison of the scalar power sums recovering matched copy multiplicities and weights up to  $\zeta^{-1}$ .  $\square$

## 14.8 Translation-invariant reduction and global symmetry

The single-gauge corollary for translation-invariant chains specializes to a reduction statement: equal states force one tensor to be a gauge of the other and inherit injectivity. Composing two gauges of the same tensor then yields the scalar law underlying the global-symmetry consequences.

**Theorem 14.68** (Translation-invariant reduction corollary). [Lean](#) [✓ Stmt](#) *Let  $A$  and  $B$  be MPS tensors of bond dimension  $D$  with  $A$  injective. Consider the two constant translation-invariant chains  $(A, \dots, A)$  and  $(B, \dots, B)$  of length  $n \geq 1$ . If the combined tensors of the two chains generate the same MPV family, then:*

1.  $B$  is gauge equivalent to  $A$ : there exists  $X \in \text{GL}_D(\mathbb{C})$  with  $B^i = X A^i X^{-1}$  for all  $i$ .
2.  $B$  is injective.

*Proof.* [✓ Proof](#) Theorem 14.25 gives a single gauge matrix  $X$  with  $B^i = X A^i X^{-1}$  for all  $i$ . Since gauge equivalence preserves injectivity,  $B$  is injective as well.  $\square$

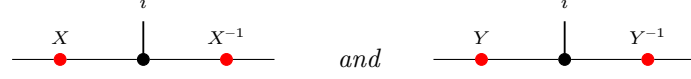
**Theorem 14.69** (Translation-invariant reduction from same state). [Lean](#) [✓ Stmt](#) *Assume a same-state reduction hypothesis for  $(d, D)$ . Let  $A$  and  $B$  be MPS tensors of bond dimension  $D$ , and let  $n \geq 3$ . If the two constant chains  $(A, \dots, A)$  and  $(B, \dots, B)$  generate the same chain state, with  $A$  injective, then the same conclusion as in Theorem 14.68 holds:  $B$  is gauge equivalent to  $A$ , and  $B$  is injective.*

*Proof.* [✓ Proof](#) The same-state reduction hypothesis upgrades equality of the two constant chain states to equality of their combined-tensor MPV families. Then Theorem 14.68 applies.  $\square$

### 14.8.1 Global symmetry via gauge composition

The virtual representation theorem (Theorem 12.15) relies on the fact that composing two gauge transformations yields another gauge up to a scalar factor.

**Lemma 14.70** (Gauge ratio commutes with tensor matrices). [Lean](#) [✓ Stmt](#) *Diagrammatically, the hypothesis is that the two local gauges*



produce the same target tensor  $B^i$  from the same source tensor  $A^i$ . If  $X, Y \in \text{GL}_D(\mathbb{C})$  both conjugate an MPS tensor  $A$  to the same tensor  $B$  (i.e.  $B^i = X A^i X^{-1} = Y A^i Y^{-1}$  for all  $i$ ), then  $Y^{-1}X$  commutes with every  $A^i$ .

*Proof.* [✓ Proof](#) Direct matrix calculation: conjugating both gauge relations and composing gives  $(Y^{-1}X)A^i = A^i(Y^{-1}X)$ .  $\square$

**Lemma 14.71** (Gauge ratio is scalar for injective tensors). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, if  $A$  is additionally injective, then  $Y^{-1}X = \lambda \mathbb{1}_D$  for some scalar  $\lambda \in \mathbb{C}$ .*

*Proof.* [✓ Proof](#) The same two-gauge picture shows that the ratio  $Y^{-1}X$  acts invisibly on the local tensor. By Lemma 14.70,  $Y^{-1}X$  commutes with all  $A^i$ . Since  $\{A^i\}$  spans  $M_D(\mathbb{C})$ , the matrix  $Y^{-1}X$  commutes with all of  $M_D(\mathbb{C})$ . By Lemma 14.12, it is a scalar matrix.  $\square$

**Theorem 14.72** (Projective multiplication law for gauge matrices). [Lean](#) [✓ Stmt](#) *Let  $A$  be an injective MPS tensor with on-site symmetry under a group representation  $U : G \rightarrow \text{GL}_d(\mathbb{C})$ . For each  $g \in G$ , let  $X(g) \in \text{GL}_D(\mathbb{C})$  satisfy  $\tilde{A}_g^i = X(g) A^i X(g)^{-1}$  for all  $i$ . Then for all  $g, h \in G$ , there exists  $c(g, h) \in \mathbb{C}$  such that*

$$X(h) X(g) = c(g, h) X(g \cdot h).$$

In fact  $c(g, h) \neq 0$  since both sides are invertible, so  $c(g, h) \in \mathbb{C}^\times$ . Diagrammatically, the two candidate gauges are competing local replacements for the same twisted tensor:



and both sides conjugate  $A^i$  to the same twisted tensor  $\tilde{A}_{gh}^i$ .

*Proof.* [✓ Proof](#) By the composition law  $\tilde{A}_{gh} = \widetilde{(\tilde{A}_h)_g}$  (Lemma 12.5), both  $X(h) X(g)$  and  $X(gh)$  conjugate  $A$  to  $\tilde{A}_{gh}$ . Lemma 14.71 then gives the scalar factor.  $\square$



## Chapter 15

# Fundamental Theorems for Injective and Normal PEPS

The following definitions and results extend the Fundamental Theorem to injective and normal PEPS, following Theorems 2 and 3 of [MGRPG<sup>+</sup>18].

**Definition 15.1** (Graph edge for PEPS). [Lean](#) [✓ Stmt](#) For a finite simple graph  $G$ , an *edge* is an ordered pair  $(u, v)$  with  $u < v$  and  $u$  adjacent to  $v$ . The ordering is only used to avoid double counting.

**Definition 15.2** (Incident edges at a vertex). [Lean](#) [✓ Stmt](#) For a vertex  $v$ , an *incident edge* is an edge of  $G$  having  $v$  as one of its endpoints.

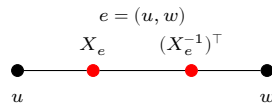
**Definition 15.3** (PEPS tensor on a graph). [Lean](#) [✓ Stmt](#) A PEPS tensor on  $G$  with physical dimension  $d$  consists of a bond dimension  $D_e$  for every edge  $e$ , together with a local tensor at each vertex  $v$  whose virtual indices are indexed by the edges incident to  $v$  and whose physical index lies in  $\{0, \dots, d - 1\}$ .

**Definition 15.4** (Virtual configuration). [Lean](#) [✓ Stmt](#) A *virtual configuration* assigns to every edge  $e$  one basis index in the corresponding bond space of dimension  $D_e$ .

**Definition 15.5** (PEPS state coefficient). [Lean](#) [✓ Stmt](#) For a physical configuration  $\sigma$  on the vertices, the PEPS state coefficient is obtained by summing over all virtual configurations and, for each such configuration, multiplying the local tensor entries over all vertices.

**Definition 15.6** (Equality of PEPS states). [Lean](#) [✓ Stmt](#) Two PEPS tensors represent the same state if all their state coefficients agree for every physical configuration.

**Definition 15.7** (Edge gauge at a vertex). [Lean](#) [✓ Stmt](#) For an edge  $e = (u, w)$  with  $u < w$  and gauge  $X_e \in \text{GL}(D_e)$ , the gauge matrix applied at vertex  $v$  is  $X_e$  if  $v = u$  and  $(X_e^{-1})^\top$  if  $v = w$ . Diagrammatically, the ordered edge carries opposite endpoint actions:



**Definition 15.8** (Gauge vertex action). [Lean](#) [✓ Stmt](#) The gauged tensor at vertex  $v$  is  $A_v^X$ . For each incident edge  $ie$ , write  $M_{ie}$  for the matrix specified by Definition 15.7, applied at the vertex  $v$



to the gauge on the underlying edge of  $ie$ . Then  $A_v^X(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) A_v(\eta', \sigma)$ . In tensor-network notation, one inserts the appropriate gauge matrix on each incident virtual leg of the local tensor:



**Definition 15.9** (Apply gauge). [Lean](#) [✓ Stmt](#) The PEPS tensor obtained by applying edge-gauge matrices to  $A$ , with vertex tensors  $A_v^X$ .

**Definition 15.10** (Absorb edge gauges). [Lean](#) [✓ Stmt](#) Given a tensor family  $B$  and an oriented gauge matrix  $X_e$  on every edge, the absorbed tensor family is  $\tilde{B} = B^X$ ; at each vertex, its local tensor is obtained by inserting the oriented endpoint matrices on the incident virtual legs.

**Theorem 15.11** (Absorbed gauges keep the bond spaces). [Lean](#) [✓ Stmt](#) The absorbed tensor family has  $D_{\tilde{B}}(e) = D_B(e)$  for every edge  $e$ .

*Proof.* [✓ Proof](#) Since  $\tilde{B} = B^X$  is defined with the same bond-dimension function as  $B$ , one has  $D_{\tilde{B}}(e) = D_{B^X}(e) = D_B(e)$  for every edge  $e$ .  $\square$

**Theorem 15.12** (Local formula for absorbed gauges). [Lean](#) [✓ Stmt](#) If  $M_{ie}$  is the oriented endpoint matrix attached to an incident edge  $ie$  at  $v$ , then  $\tilde{B}_v(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) B_v(\eta', \sigma)$ .

*Proof.* [✓ Proof](#) Expanding  $\tilde{B} = B^X$  at a vertex gives  $\tilde{B}_v(\eta, \sigma) = (B^X)_v(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) B_v(\eta', \sigma)$ .  $\square$

**Definition 15.13** (Gauge equivalence for PEPS). [Lean](#) [✓ Stmt](#) Two PEPS tensors  $A, B$  are gauge-equivalent if they have the same bond dimensions and  $B$  is obtained from  $A$  by applying invertible gauge matrices on each edge.

**Theorem 15.14** (Gauge invariance of PEPS state). [Lean](#) [✓ Stmt](#) Applying a gauge to a PEPS tensor preserves the state coefficients. Along each internal edge, the two endpoint insertions cancel:

$$\begin{array}{ccc}
 X_e(j, i) ((X_e^{-1})^\top)(j, k) & \sum_j X_e(j, i) ((X_e^{-1})^\top)(j, k) = \delta(i, k) & \\
 \bullet \text{---} \bullet & = & \bullet \text{---} \bullet
 \end{array}$$

**Theorem 15.15** (Gauge equivalence implies same state). [Lean](#) [✓ Stmt](#) Gauge equivalence implies that the two PEPS tensors represent the same state.

**Definition 15.16** (Local tensor evaluation). [Lean](#) [✓ Stmt](#) The local tensor evaluated at vertex  $v$  with virtual-index weighting  $f$ , computing  $\sum_{\eta} (\prod_{ie} f(ie)(\eta(ie))) \cdot A_v(\eta, \sigma)$ . This is multilinear in the components of  $f$ , not linear in the full tuple.

**Definition 15.17** (Vertex injectivity). [Lean](#) [✓ Stmt](#) A PEPS tensor  $A$  is *vertex-injective* if, at every vertex  $v$ , the family of physical vectors

$$(A_v(\eta, \cdot) \in \mathbb{C}^d)_\eta$$

indexed by virtual configurations  $\eta$  on the edges incident to  $v$  is linearly independent in  $\mathbb{C}^d$ . Equivalently, extending  $A_v$  by linearity to a map from the free vector space on virtual configurations into the physical space  $\mathbb{C}^d$  gives a linear map with trivial kernel. This matches the injectivity formulation of [MGRPG<sup>+</sup>18, Section 3], where the local tensor is interpreted as a map from virtual configurations to the physical space.

**Definition 15.18** (Local tensor map). [Lean](#) [✓ Stmt](#) For a fixed vertex  $v$ , the family of local physical vectors  $\eta \mapsto A_v(\eta, \cdot)$  extends by linearity to a map from the coefficient space on local virtual configurations at  $v$  into  $\mathbb{C}^d$ .

**Lemma 15.19** (Trivial kernel in explicit coordinates). [Lean](#) [✓ Stmt](#) If  $A$  is vertex-injective and  $R$  is a coefficient function on the local virtual configurations at  $v$  satisfying, for every physical index  $\tau$ ,

$$\sum_{\eta} R(\eta) A_v(\eta, \tau) = 0,$$

then  $R = 0$ .

*Proof.* [✓ Proof](#) The displayed hypothesis says that the local tensor map sends  $R$  to the zero vector. Vertex injectivity makes this map injective, hence  $R = 0$ .  $\square$

**Definition 15.20** (Local left inverse). [Lean](#) [✓ Stmt](#) If  $A$  is vertex-injective, one may choose a linear map  $\Psi_v$  from  $\mathbb{C}^d$  to the coefficient space on local virtual configurations at  $v$  such that  $\Psi_v \circ \Phi_v = \mathbb{1}$ .

**Definition 15.21** (Physical realization of a local virtual operator). [Lean](#) [✓ Stmt](#) For an endomorphism  $T$  of the coefficient space on local virtual configurations at  $v$ , define the local physical operator

$$O_T = \Phi_v \circ T \circ \Psi_v.$$

**Theorem 15.22** (Local virtual operations on the image). [Lean](#) [✓ Stmt](#) For every coefficient function  $c$  on the local virtual configurations at  $v$ ,

$$O_T(\Phi_v(c)) = \Phi_v(Tc).$$

*Proof.* [✓ Proof](#) This is immediate from  $\Psi_v \circ \Phi_v = \mathbb{1}$ .  $\square$

**Theorem 15.23** (Injectivity of local physical realization). [Lean](#) [✓ Stmt](#) If two virtual operations have the same canonical physical realization, then they agree.

*Proof.* [✓ Proof](#) Evaluate both physical realizations on vectors in the image of the local tensor map and use vertex injectivity.  $\square$

**Definition 15.24** (Local projector). [Lean](#) [✓ Stmt](#) The local projector is the physical realization of the identity virtual operator:

$$P_v = \Phi_v \circ \Psi_v.$$

**Definition 15.25** (Virtual pullback of a local physical operator). [Lean](#) [✓ Stmt](#) For a physical operator  $O$  on the local physical space at  $v$ , define its virtual pullback by

$$\Psi_v \circ O \circ \Phi_v.$$

This is the local injectivity step used in the physical-to-virtual direction of the insertion-algebra argument in [MGRPG<sup>+</sup>18, Section 3].

**Theorem 15.26** (Virtual pullback realizes the projected physical action). [Lean](#) [✓ Stmt](#) For every coefficient function  $c$ , applying the local tensor map after the virtual pullback gives the projection of  $O(\Phi_v(c))$  onto the image of  $\Phi_v$ .

*Proof.* [✓ Proof](#) Expand the definitions. The chosen left inverse gives  $\Psi_v \Phi_v = \mathbb{1}$ , and the remaining map is the local projector.  $\square$

**Theorem 15.27** (Virtual pullback for image-preserving physical actions). [Lean](#) [✓ Stmt](#) If  $O$  sends every vector in the image of  $\Phi_v$  back into that image, then its virtual pullback realizes exactly the action of  $O$  on  $\Phi_v(c)$ .

*Proof.* [✓ Proof](#) Use the preceding theorem and the image-preservation hypothesis.  $\square$

**Theorem 15.28** (Recovery of a virtual operation from a physical realization). [Lean](#) [✓ Stmt](#) If a physical operator  $O$  realizes a virtual operation  $T$  on every vector in the image of  $\Phi_v$ , then pulling back  $O$  recovers  $T$ .

*Proof.* [✓ Proof](#) After applying the local tensor map, both virtual operations have the same image. Vertex injectivity makes the local tensor map injective, so the virtual operations agree.  $\square$

**Theorem 15.29** (Extensionality of physical-to-virtual pullback). [Lean](#) [✓ Stmt](#) Two physical endpoint operations have the same virtual pullback exactly when their projected actions agree on every vector in the image of the local tensor map:

$$T_O = T_{O'} \iff P_v O \Phi_v(c) = P_v O' \Phi_v(c) \text{ for all } c.$$

*Proof.* [✓ Proof](#) Apply the local tensor map to the two virtual pullbacks and use the pullback specification. The converse follows from vertex injectivity; the forward direction is obtained by rewriting the two virtual pullbacks.  $\square$

**Theorem 15.30** (Pullback of the canonical physical realization). [Lean](#) [✓ Stmt](#) Pulling back the canonical physical realization of a virtual operation recovers the original virtual operation.

*Proof.* [✓ Proof](#) The canonical physical realization agrees with the virtual operation on the image of the local tensor map. Apply the preceding recovery theorem.  $\square$

**Theorem 15.31** (Physical realization of a pulled-back physical operator). [Lean](#) [✓ Stmt](#) Let  $P_v = \Phi_v \circ \Psi_v$  be the local projector, and let  $T_O = \Psi_v \circ O \circ \Phi_v$  be the virtual pullback of a physical operator  $O$ . Then

$$\Phi_v \circ T_O \circ \Psi_v = P_v \circ O \circ P_v.$$

*Proof.* [✓ Proof](#) Expand the definitions of the physical realization, the virtual pullback, and the local projector.  $\square$

**Theorem 15.32** (Projected local recovery). [Lean](#) [✓ Stmt](#) Suppose that the compressed physical action  $P_v O P_v$  agrees with the canonical physical realization of a virtual operation  $T$ . Then the virtual pullback of  $O$  is  $T$ .

*Proof.* [✓ Proof](#) The physical realization of the virtual pullback of  $O$  is  $P_v O P_v$ . By hypothesis this is the physical realization of  $T$ . Injectivity of the canonical physical realization gives equality of the virtual operations.  $\square$

**Theorem 15.33** (Projected local recovery equivalence). [Lean](#) [✓ Stmt](#) The compressed physical action  $P_v O P_v$  is the canonical physical realization of a virtual operation  $T$  if and only if the virtual pullback of  $O$  is  $T$ .

*Proof.* ✓ Proof One implication follows by realizing both virtual operations physically and using the formula for the physical realization of a pulled-back operator. The converse is the preceding recovery theorem.  $\square$

**Definition 15.34** (Matrix action on one incident virtual edge). Lean ✓ Stmt A matrix on one edge incident to  $v$  acts on the coefficient space of local virtual configurations by applying the matrix to that edge coordinate and summing over the input edge index while keeping the residual virtual configuration fixed.

**Theorem 15.35** (One-edge matrix action on a basis configuration). Lean ✓ Stmt *If the coefficient function is the unit basis function at one local virtual configuration, and the matrix acts on the distinguished incident edge, the result is the sum over the new distinguished edge index with the residual local boundary held fixed.*

*Proof.* ✓ Proof Evaluate both sides on a local virtual configuration and split into the two cases according to whether the residual boundary agrees with the fixed residual boundary.  $\square$

**Theorem 15.36** (Local tensor form of a one-edge basis action). Lean ✓ Stmt *Applying the local tensor map after a one-edge matrix action on a basis virtual configuration gives the corresponding linear combination of local tensor components.*

*Proof.* ✓ Proof Use the preceding coefficient identity and linearity of the local tensor map.  $\square$

**Theorem 15.37** (Physical realization of one-edge virtual matrix action). Lean ✓ Stmt *If  $A$  is vertex-injective, then a matrix acting on one edge incident to  $v$  is realized by a physical operator on the physical space at  $v$ .*

*Proof.* ✓ Proof Apply the physical realization of local virtual operations to the one-edge matrix action.  $\square$

**Definition 15.38** (Split at one incident edge). Lean ✓ Stmt Choosing one incident edge at  $v$  identifies the set of local virtual configurations with the product of the bond index on that edge and the residual indices on the remaining incident edges.

**Definition 15.39** (Middle region of an edge). Lean ✓ Stmt For an edge  $e = (u, v)$ , the middle region is the complement  $V \setminus \{u, v\}$ .

**Theorem 15.40** (Edge-centered vertex partition). Lean ✓ Stmt *For an edge  $e = (u, v)$ , the three regions  $\{u\}$ ,  $V \setminus \{u, v\}$ , and  $\{v\}$  cover the whole vertex set.*

*Proof.* ✓ Proof Every vertex is either  $u$ ,  $v$ , or different from both.  $\square$

**Theorem 15.41** (Nonempty middle region from a cardinal bound). Lean ✓ Stmt *If  $2 < |V|$ , then for every edge  $e = (u, v)$  the middle region  $V \setminus \{u, v\}$  is nonempty.*

*Proof.* ✓ Proof The middle region has cardinality  $|V| - 2$ . Hence  $2 < |V|$  gives  $|V| - 2 > 0$ .  $\square$

**Theorem 15.42** (Product splitting at an edge). Lean ✓ Stmt *For any edge  $e = (u, v)$  and any function  $f$  on the vertex set with values in a commutative monoid  $M$ ,*

$$\prod_{x \in V} f(x) = f(u) \left( \prod_{x \in V \setminus \{u, v\}} f(x) \right) f(v).$$

*Proof.* ✓ Proof Isolate the two distinguished endpoints in the product over  $V$  and group the remaining factors into the middle region.  $\square$

**Theorem 15.43** (PEPS coefficient splitting at an edge). [Lean](#) [✓ Stmt](#) For any edge  $e = (u, v)$ , each summand in the PEPS coefficient factors into the tensor contribution at  $u$ , the product over the middle region  $V \setminus \{u, v\}$ , and the tensor contribution at  $v$ .

*Proof.* [✓ Proof](#) Expand the state coefficient and apply the product splitting at the chosen edge to the per-vertex product inside each summand.  $\square$

**Definition 15.44** (Edge-centered boundary data). [Lean](#) [✓ Stmt](#) For an edge  $e = (u, v)$ , the boundary data consist of the virtual index on  $e$ , the remaining virtual indices incident to  $u$ , and the remaining virtual indices incident to  $v$ . These are the indices over which the endpoint tensors couple to the blocked middle region.

**Definition 15.45** (Boundary data read from a virtual configuration). [Lean](#) [✓ Stmt](#) A global virtual configuration determines edge-centered boundary data by reading the index on the distinguished edge and the residual endpoint indices.

**Definition 15.46** (Boundary compatibility at an edge). [Lean](#) [✓ Stmt](#) A global virtual configuration is compatible with fixed edge-centered boundary data if it has the prescribed index on the distinguished edge and the prescribed residual indices at both endpoint stars.

**Definition 15.47** (Middle configurations over fixed boundary data). [Lean](#) [✓ Stmt](#) For fixed boundary data  $\beta$ , the middle configurations are the subtype of global virtual configurations satisfying the compatibility condition with  $\beta$ .

**Definition 15.48** (Virtual configurations fibred by edge boundary data). [Lean](#) [✓ Stmt](#) A global virtual configuration is equivalently a choice of edge-centered boundary data together with a middle configuration in the corresponding fibre.

**Definition 15.49** (Left endpoint local configuration from boundary data). [Lean](#) [✓ Stmt](#) The left endpoint local virtual configuration is reconstructed from the edge index and the residual data on the left endpoint star.

**Definition 15.50** (Right endpoint local configuration from boundary data). [Lean](#) [✓ Stmt](#) The right endpoint local virtual configuration is reconstructed from the edge index and the residual data on the right endpoint star.

**Theorem 15.51** (Left endpoint reconstruction for compatible boundary data). [Lean](#) [✓ Stmt](#) If a global virtual configuration is compatible with boundary data, then its restriction to the left endpoint star is the local configuration reconstructed from those data.

**Theorem 15.52** (Right endpoint reconstruction for compatible boundary data). [Lean](#) [✓ Stmt](#) If a global virtual configuration is compatible with boundary data, then its restriction to the right endpoint star is the local configuration reconstructed from those data.

**Definition 15.53** (Boundary data with an inserted edge matrix). [Lean](#) [✓ Stmt](#) For an edge  $e = (u, v)$ , inserted-edge boundary data consist of two independent indices on the bond  $e$ , together with the residual virtual indices incident to  $u$  and to  $v$ . These are the boundary indices used when a matrix is inserted on the distinguished bond.

**Definition 15.54** (Diagonal inserted-edge boundary data). [Lean](#) [✓ Stmt](#) Ordinary edge-centered boundary data determine inserted-edge boundary data by using the same bond index at both endpoints. This is the diagonal boundary datum selected by the identity matrix in the edge-insertion coefficient.

**Definition 15.55** (Right-endpoint inserted-boundary reindexing). [Lean](#) [✓ Stmt](#) Ordinary edge-boundary data together with an independent right endpoint bond index are equivalent to inserted-edge boundary data. The ordinary boundary supplies the left bond index and the two residual endpoint boundaries.

**Definition 15.56** (Left-endpoint inserted-boundary reindexing). [Lean](#) [✓ Stmt](#) Ordinary edge-boundary data together with an independent left endpoint bond index are equivalent to inserted-edge boundary data. The ordinary boundary supplies the right bond index and the two residual endpoint boundaries.

**Definition 15.57** (Diagonal inserted-boundary configurations). [Lean](#) [✓ Stmt](#) The diagonal inserted-boundary configurations are those inserted-edge boundary data for which the two distinguished bond indices agree. This is the support of the identity matrix inside the inserted-boundary sum.

**Definition 15.58** (Diagonal boundary reindexing). [Lean](#) [✓ Stmt](#) Ordinary edge-boundary data are equivalent to diagonal inserted-boundary data. This is the finite reindexing map used after the identity matrix has removed all off-diagonal inserted-boundary summands.

**Definition 15.59** (Open middle tensor at an edge). [Lean](#) [✓ Stmt](#) Fixing the residual endpoint data but leaving the bond  $e$  out of the middle region, the open middle tensor is the contraction over all virtual indices away from  $e$  that are compatible with those endpoint residual data.

**Definition 15.60** (Middle physical configurations at an edge). [Lean](#) [Lean](#) [✓ Stmt](#) For an edge  $e = (u, v)$ , the middle physical configurations are the physical indices on the vertices in  $V \setminus \{u, v\}$ . A global physical configuration restricts to such a middle configuration.

**Definition 15.61** (Middle tensor with middle physical index). [Lean](#) [Lean](#) [✓ Stmt](#) The ordinary and open middle contractions may be written using only the middle physical configuration, rather than a physical configuration on all vertices. This is the middle tensor in the three-site chain obtained after the edge blocking in arXiv:1804.04964, Section 3.

**Theorem 15.62** (Middle restriction of the blocked tensor). [Lean](#) [Lean](#) [✓ Stmt](#) *The middle contractions written with a global physical configuration are the corresponding middle-indexed contractions evaluated on the restricted middle physical configuration.*

*Proof.* [✓ Proof](#) Rewrite the product over  $V \setminus \{u, v\}$  as the product over the corresponding finite set of middle vertices. □

**Definition 15.63** (Restricting middle configurations away from the edge). [Lean](#) [✓ Stmt](#) An ordinary middle configuration for the edge-blocked contraction restricts to a virtual configuration on all edges except the distinguished edge. This is the first reindexing map needed to compare the identity insertion with the ordinary edge-blocked middle tensor.

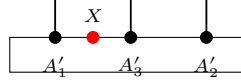
**Definition 15.64** (Restoring the distinguished edge index). [Lean](#) [✓ Stmt](#) Conversely, a compatible open middle configuration becomes an ordinary middle configuration after the fixed edge-boundary index is restored on the distinguished edge. Together with the restriction map, this gives the finite reindexing underlying the identity-insertion specialization.

**Definition 15.65** (Open-middle reindexing equivalence). [Lean](#) [✓ Stmt](#) For fixed edge-boundary data, ordinary middle configurations are equivalent to compatible open-middle configurations. This is the finite reindexing used when the inserted matrix is the identity.

**Theorem 15.66** (Middle-indexed open tensor reindexing). [Lean](#) [✓ Stmt](#) *The ordinary middle tensor and the open middle tensor agree after restoring the fixed distinguished-edge index and reindexing the finite sum, with the physical index already restricted to the middle block.*

*Proof.* [✓ Proof](#) Reindex the ordinary middle configurations by the equivalence with compatible open-middle configurations, then compare each local tensor entry.  $\square$

**Definition 15.67** (Edge insertion coefficient). [Lean](#) [✓ Stmt](#) For an edge  $e = (u, v)$  and a matrix  $X$  acting on the bond space of  $e$ , the edge insertion coefficient is the contraction obtained by inserting  $X$  between the endpoint tensors and the open middle tensor:



**Theorem 15.68** (Middle tensor reindexing for an opened edge). [Lean](#) [✓ Stmt](#) *Restoring the fixed distinguished-edge index identifies the ordinary blocked middle tensor with the open middle tensor. This is the finite reindexing used before the identity matrix collapses the two inserted bond indices to the diagonal.*

**Theorem 15.69** (Diagonal summand of the identity insertion). [Lean](#) [✓ Stmt](#) *On the diagonal boundary datum selected by the identity matrix, the inserted-edge summand is the ordinary edge-blocked summand. Thus the remaining identity-insertion step is purely a finite summation statement: the off-diagonal inserted boundary data vanish and the diagonal data reindex the ordinary edge-boundary sum.*

**Theorem 15.70** (Off-diagonal summands of the identity insertion). [Lean](#) [✓ Stmt](#) *If the two inserted bond indices are distinct, then the corresponding summand in the identity insertion is zero. This is the pointwise off-diagonal vanishing used before the inserted-boundary sum is restricted to its diagonal part.*

**Theorem 15.71** (Identity insertion recovers the edge-blocked coefficient). [Lean](#) [✓ Stmt](#) *If the inserted matrix on the distinguished edge is the identity, then the edge insertion coefficient is the ordinary edge-blocked coefficient.*

*Proof.* [✓ Proof](#) Split the inserted-boundary sum into the diagonal part, where the two inserted bond indices agree, and the off-diagonal part. The off-diagonal summands vanish for the identity matrix, while the diagonal summands reindex through the equivalence with ordinary edge-boundary data.  $\square$

**Definition 15.72** (Blocked middle tensor at an edge). [Lean](#) [✓ Stmt](#) Fixing the edge-centered boundary data, the blocked middle tensor is the sum over global virtual configurations compatible with those data of the product of local tensor entries over the middle region  $V \setminus \{u, v\}$ . The graph-local contraction is read as a three-site chain:





**Definition 15.73** (Edge-blocked PEPS coefficient). [Lean](#) [✓ Stmt](#) The edge-blocked coefficient is the three-block contraction obtained from the left endpoint tensor, the blocked middle tensor, and the right endpoint tensor at the chosen edge.

**Theorem 15.74** (Edge-blocked coefficient equals the PEPS coefficient). [Lean](#) [✓ Stmt](#) For every edge  $e$ , the edge-blocked three-block coefficient is exactly the original PEPS state coefficient.

*Proof.* [✓ Proof](#) Write  $\text{Coeff}_A(\sigma)$  for the PEPS state coefficient and  $C_A(e; \sigma)$  for the edge-blocked coefficient, and put  $e = (u, v)$  and  $M_e = V \setminus \{u, v\}$ . The equivalence  $\omega \leftrightarrow (\beta, \omega_{M_e})$  between global virtual configurations and an edge boundary configuration with a compatible middle configuration rewrites the PEPS coefficient as

$$\sum_{\omega} A_u(\omega|_u, \sigma_u) \left( \prod_{x \in M_e} A_x(\omega|_x, \sigma_x) \right) A_v(\omega|_v, \sigma_v) = \sum_{\beta} A_u(\beta_u, \sigma_u) C_{A, M_e}(\beta, \sigma|_{M_e}) A_v(\beta_v, \sigma_v).$$

Under the equivalence,  $\omega|_u = \beta_u$  and  $\omega|_v = \beta_v$ ; the inner sum over  $\omega_{M_e}$  is exactly  $C_{A, M_e}(\beta, \sigma|_{M_e})$ . Thus  $C_A(e; \sigma) = \text{Coeff}_A(\sigma)$ .  $\square$

**Theorem 15.75** (Identity insertion equals the PEPS coefficient). [Lean](#) [✓ Stmt](#) If the inserted matrix on the distinguished edge is the identity, then the edge insertion coefficient is the original PEPS state coefficient.

*Proof.* [✓ Proof](#) Let  $e = (u, v)$  and put  $M_e = V \setminus \{u, v\}$ . With  $\mathbb{1}$  inserted on  $e$ , the inserted-boundary sum is

$$C_A(e, \mathbb{1}; \sigma) = \sum_{a, b, \lambda, \rho} \delta_{a, b} A_u(a, \lambda; \sigma_u) C_{A, M_e}^{\text{open}}(\lambda, \rho; \sigma|_{M_e}) A_v(b, \rho; \sigma_v).$$

Here  $a, b$  are the two copies of the distinguished bond index, while  $\lambda, \rho$  are the remaining virtual configurations at  $u$  and  $v$ . The factor  $\delta_{a, b}$  leaves only the diagonal  $a = b$ , and the open-middle reindexing gives  $C_{A, M_e}^{\text{open}}(\lambda, \rho; \sigma|_{M_e}) = C_{A, M_e}(a, \lambda, \rho; \sigma|_{M_e})$ . Hence  $C_A(e, \mathbb{1}; \sigma) = C_A(e; \sigma) = \text{Coeff}_A(\sigma)$ .  $\square$

**Theorem 15.76** (Same state gives the same edge-blocked coefficients). [Lean](#) [✓ Stmt](#) If two PEPS tensors have the same state, then their edge-blocked coefficients agree at every edge and every physical configuration.

*Proof.* [✓ Proof](#) For every physical configuration  $\sigma$ ,

$$C_A(e; \sigma) = \text{Coeff}_A(\sigma) = \text{Coeff}_B(\sigma) = C_B(e; \sigma).$$

$\square$

**Theorem 15.77** (Same state gives the same identity insertions). [Lean](#) [✓ Stmt](#) If two PEPS tensors have the same state, then their edge-centered identity insertions agree at every edge and every physical configuration.

*Proof.* [✓ Proof](#) For every physical configuration  $\sigma$ ,

$$C_A(e, \mathbb{1}; \sigma) = \text{Coeff}_A(\sigma) = \text{Coeff}_B(\sigma) = C_B(e, \mathbb{1}; \sigma).$$

$\square$



**Definition 15.78** (Middle tensor family at an edge). [Lean](#) [Lean](#) [✓ Stmt](#) For the edge-blocked three-site chain at  $e = (u, v)$ , the middle tensor is the family obtained by fixing the two residual endpoint boundaries and evaluating the open middle contraction on the middle physical index. Its injectivity is the middle-block part of the assertion following the edge-blocking diagram.

**Lemma 15.79** (Middle tensor vanishes for an empty complement configuration). [Lean](#) [✓ Stmt](#) *If the virtual configuration space on the complement of the distinguished edge is empty, then for every residual boundary label  $\rho$  the vector  $T_{A, M_e}^\rho$  in the edge-middle tensor family is zero.*

*Proof.* [✓ Proof](#) The finite sum defining the open middle tensor is taken over the empty complement configuration space, and is therefore zero.  $\square$

**Lemma 15.80** (Failure of middle injectivity for an empty complement configuration). [Lean](#) [✓ Stmt](#) *If the complement configuration space is empty and the residual boundary-label set is nonempty, then the edge-middle tensor family is not injective.*

*Proof.* [✓ Proof](#) Choose a residual boundary label. By the preceding lemma its middle-tensor vector is the zero vector. A linearly independent family cannot contain a zero vector.  $\square$

**Definition 15.81** (Edge-blocked three-site injectivity). [Lean](#) [✓ Stmt](#) For a chosen edge  $e = (u, v)$ , the edge-blocked three-site chain is injective when the left endpoint tensor map, the middle tensor family, and the right endpoint tensor map are all injective.

**Definition 15.82** (Singleton blocked region). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a singleton block  $\{v\}$ , the blocked physical index is the original physical index and the blocked tensor is

$$T_{A, \{v\}}(\eta, \sigma) = A_v(\eta, \sigma).$$

Its injectivity is  $\text{inj}(T_{A, \{v\}})$ . The comparison with a region-injectivity predicate  $\kappa$  records the implication

$$\text{inj}(T_{A, \{v\}}) \implies \kappa(\{v\}).$$

**Theorem 15.83** (Singleton comparison for a vertex-injective tensor). [Lean](#) [✓ Stmt](#) *Let  $A$  be vertex-injective with every bond dimension positive. Then the concrete region-injectivity predicate of  $A$  satisfies the singleton comparison: injectivity of a singleton blocked tensor implies injectivity of the corresponding singleton region.*

*Proof.* [✓ Proof](#) Under vertex injectivity and positive bond dimensions every finite region is injective. In particular the singleton region  $\{v\}$  is injective, so the comparison holds independently of the singleton-tensor hypothesis.  $\square$

**Theorem 15.84** (Vertex injectivity gives singleton-block injectivity). [Lean](#) [✓ Stmt](#) *If  $A$  is vertex-injective, then every singleton blocked tensor is injective. For every vertex  $v$ ,*

$$\text{inj}(T_{A, \{v\}}).$$

*Proof.* [✓ Proof](#) For  $R = \{v\}$ , the contraction defining  $T_{A, R}$  has one factor. Hence

$$T_{A, \{v\}}(\eta, \sigma) = A_v(\eta, \sigma).$$

Thus  $\text{inj}(A_v)$  is exactly  $\text{inj}(T_{A, \{v\}})$ . Vertex injectivity gives  $\forall v, \text{inj}(A_v)$ , and therefore  $\forall v, \text{inj}(T_{A, \{v\}})$ . For a fixed  $v$ , this is the displayed assertion.  $\square$

**Theorem 15.85** (Singleton comparison gives the region base case). [Lean](#) [Lean](#) [✓ Stmt](#) Assume that a region-injectivity predicate  $\kappa$  satisfies, for every vertex  $v$ ,

$$\text{inj}(T_{A,\{v\}}) \implies \kappa(\{v\}).$$

Then, for every vertex  $v$ ,

$$A \text{ vertex-injective} \implies \kappa(\{v\}).$$

*Proof.* [✓ Proof](#) For each  $v$ , the preceding theorem gives  $A$  vertex-injective  $\implies \text{inj}(T_{A,\{v\}})$ . The comparison hypothesis gives  $\text{inj}(T_{A,\{v\}}) \implies \kappa(\{v\})$ . Their composition is the claimed implication.  $\square$

**Theorem 15.86** (Finite regions from singleton regions and union closure). [Lean](#) [✓ Stmt](#) Assume that  $\kappa$  satisfies singleton comparison and union closure:  $\text{inj}(T_{A,\{v\}}) \implies \kappa(\{v\})$  for every vertex  $v$ , and  $\kappa(R) \wedge \kappa(S) \implies \kappa(R \cup S)$ . If  $A$  is vertex-injective, then every nonempty finite region  $R$  satisfies  $\kappa(R)$ .

*Proof.* [✓ Proof](#) For  $R \neq \emptyset$ , write  $R = \bigcup_{v \in R} \{v\}$ . The preceding theorem turns singleton comparison and vertex injectivity into  $\kappa(\{v\})$  for every  $v \in R$ . Applying the finite-union consequence of union closure to these singleton regions gives  $\kappa(R)$ .  $\square$

**Definition 15.87** (Region injectivity supplies the edge-middle tensor). [Lean](#) [✓ Stmt](#) A comparison from finite-region injectivity to the edge-middle tensor records the following assertion: if the middle vertex region  $V \setminus \{u, v\}$  is injective after blocking, then the middle tensor in the edge-blocked three-site chain is injective. This isolates the finite-region contraction claim in [MGRPG<sup>+</sup>18, Section 3].

**Theorem 15.88** (Endpoint injectivity in the edge-blocked chain). [Lean](#) [✓ Stmt](#) If the original PEPS tensor is vertex-injective, then the two endpoint tensor maps in the edge-blocked three-site chain are injective.

*Proof.* [✓ Proof](#) If  $e = (u, v)$ , vertex injectivity gives  $\text{inj}(T_{A,u})$  and  $\text{inj}(T_{A,v})$ , which are the left and right endpoint tensor maps of the edge-blocked chain.  $\square$

**Theorem 15.89** (Endpoint injectivity at all edges). [Lean](#) [✓ Stmt](#) If the original PEPS tensor is vertex-injective, then the two endpoint tensor maps in the edge-blocked three-site chain are injective at every edge.

*Proof.* [✓ Proof](#) For each  $e = (u, v)$ , Theorem 15.88 gives  $\text{inj}(T_{A,u}) \wedge \text{inj}(T_{A,v})$ .  $\square$

**Theorem 15.90** (Reducing edge-blocked injectivity to the middle block). [Lean](#) [✓ Stmt](#) For a vertex-injective PEPS tensor, injectivity of the edge-blocked middle tensor gives the full injectivity statement for the edge-blocked three-site chain.

*Proof.* [✓ Proof](#) Let  $e = (u, v)$  and put  $M_e = V \setminus \{u, v\}$ . The endpoint theorem supplies  $\text{inj}(T_{A,u})$  and  $\text{inj}(T_{A,v})$ , while the hypothesis is  $\text{inj}(T_{A,M_e})$ . Hence

$$\text{inj}(T_{A,u}) \wedge \text{inj}(T_{A,M_e}) \wedge \text{inj}(T_{A,v}).$$

$\square$

**Theorem 15.91** (All edge-blocked chains from middle-tensor injectivity). [Lean](#) [✓ Stmt](#) If every edge-middle tensor is injective, then a vertex-injective PEPS tensor gives an injective edge-blocked three-site chain at every edge.

*Proof.* ✓ Proof For each  $e = (u, v)$ , put  $M_e = V \setminus \{u, v\}$ . Vertex injectivity gives  $\text{inj}(T_{A,u})$  and  $\text{inj}(T_{A,v})$ , while the hypothesis gives  $\text{inj}(T_{A,M_e})$ . Hence

$$\text{inj}(T_{A,u}) \wedge \text{inj}(T_{A,M_e}) \wedge \text{inj}(T_{A,v})$$

is the injectivity assertion for the edge-blocked three-site chain at  $e$ . □

**Theorem 15.92** (Region injectivity gives edge-blocked injectivity). Lean ✓ Stmt *Suppose the middle region  $V \setminus \{u, v\}$  is injective after blocking, and suppose this region-injectivity assertion has been identified with injectivity of the corresponding edge-middle tensor family. Then a vertex-injective PEPS tensor gives an injective edge-blocked three-site chain at the edge  $e = (u, v)$ .*

*Proof.* ✓ Proof Let  $e = (u, v)$  and put  $M_e = V \setminus \{u, v\}$ . The comparison gives  $\kappa(M_e) \Rightarrow \text{inj}(T_{A,M_e})$ . Vertex injectivity supplies  $\text{inj}(T_{A,u})$  and  $\text{inj}(T_{A,v})$ . The preceding reduction then gives

$$\text{inj}(T_{A,u}) \wedge \text{inj}(T_{A,M_e}) \wedge \text{inj}(T_{A,v}).$$

□

**Theorem 15.93** (Middle tensors from all middle regions). Lean ✓ Stmt *Suppose every edge-middle region  $V \setminus \{u, v\}$  is injective after blocking, and suppose finite-region injectivity has been identified with injectivity of the corresponding edge-middle tensor family. Then every edge-middle tensor family is injective.*

*Proof.* ✓ Proof For each edge  $e = (u, v)$ , put  $M_e = V \setminus \{u, v\}$ . The hypotheses give  $\kappa(M_e)$ . The comparison implication  $\kappa(M_e) \Rightarrow \text{inj}(T_{A,M_e})$  yields  $\text{inj}(T_{A,M_e})$  for every edge. □

**Theorem 15.94** (All edge-blocked chains from middle-region injectivity). Lean ✓ Stmt *Suppose every edge-middle region  $V \setminus \{u, v\}$  is injective after blocking, and suppose finite-region injectivity has been identified with injectivity of the corresponding edge-middle tensor family. Then a vertex-injective PEPS tensor gives an injective edge-blocked three-site chain at every edge.*

*Proof.* ✓ Proof For every edge  $e = (u, v)$ , put  $M_e = V \setminus \{u, v\}$ . The region hypothesis gives  $\kappa(M_e)$ , and the comparison gives  $\kappa(M_e) \Rightarrow \text{inj}(T_{A,M_e})$ . Vertex injectivity gives  $\text{inj}(T_{A,u})$  and  $\text{inj}(T_{A,v})$ . Thus

$$\text{inj}(T_{A,u}) \wedge \text{inj}(T_{A,M_e}) \wedge \text{inj}(T_{A,v}).$$

□

**Theorem 15.95** (Edge-middle tensor from singleton regions, nonempty middle case). Lean

✓ Stmt *Let  $e = (u, v)$  and put  $M_e = V \setminus \{u, v\}$ . Assume  $M_e \neq \emptyset$ . Assume also singleton comparison, union closure for  $\kappa$ , and the comparison  $\kappa(M_e) \Rightarrow \text{inj}(T_{A,M_e})$ . If  $A$  is vertex-injective, then the edge-middle tensor family  $T_{A,M_e}$  is injective.*

*Proof.* ✓ Proof The finite-region theorem applied to  $M_e = \bigcup_{w \in M_e} \{w\}$  gives  $\kappa(M_e)$ . The edge-middle comparison sends this to  $\text{inj}(T_{A,M_e})$ . □

**Theorem 15.96** (One edge-blocked chain from singleton regions, nonempty middle case). Lean

✓ Stmt *Let  $e = (u, v)$  and put  $M_e = V \setminus \{u, v\}$ . Under the hypotheses of the preceding theorem, vertex injectivity of  $A$  gives injectivity of the edge-blocked three-site chain at  $e$ .*

*Proof.* ✓ Proof The preceding theorem gives  $\text{inj}(T_{A,M_e})$ . The endpoint reduction combines this middle-tensor injectivity with vertex injectivity at  $u$  and  $v$ . □

**Theorem 15.97** (All edge-blocked chains from singleton regions, nonempty middle case). [Lean](#)

[✓ Stmt](#) If  $M_e = V \setminus \{u, v\}$  is nonempty for every edge  $e = (u, v)$ , then the preceding one-edge statement applies at every edge. Thus vertex injectivity of  $A$  gives injectivity of every edge-blocked three-site chain.

*Proof.* [✓ Proof](#) Apply the one-edge statement separately to each edge  $e$ . □

**Theorem 15.98** (All edge-blocked chains from singleton regions, cardinal form). [Lean](#) [Lean](#)

[✓ Stmt](#) Assume singleton comparison, union closure for  $\kappa$ , the comparison  $\kappa(M_e) \Rightarrow \text{inj}(T_{A, M_e})$ , and vertex injectivity of  $A$ . If  $2 < |V|$ , then  $V \setminus \{u, v\} \neq \emptyset$  for every edge  $e = (u, v)$ , and these hypotheses give  $\text{inj}(T_{A, V \setminus \{u, v\}})$  and edge-blocked three-site injectivity at every edge.

*Proof.* [✓ Proof](#) For each  $e = (u, v)$ , Theorem 15.41 gives  $V \setminus \{u, v\} \neq \emptyset$ . Apply the preceding all-edge singleton theorem. □

**Definition 15.99** (Finite kernel descent datum). [Lean](#) [✓ Stmt](#) Let  $V$  be a vertex type with decidable equality. A finite kernel descent datum is a predicate  $K$  on finite subsets  $S \subseteq V$  together with implications

$$j \in S \implies K(S) \implies K(S \setminus \{j\}).$$

In the edge-blocking proof,  $K(S)$  is the assertion that the boundary coefficients  $c_\rho$ , with the virtual indices exposed at that stage, give zero after the tensors in  $S$  have been contracted.

**Lemma 15.100** (Finite descent of kernel conditions). [Lean](#) [Lean](#) [✓ Stmt](#) For a finite kernel descent datum and any finite  $R \subseteq V$ ,

$$K(R) \implies K(\emptyset).$$

More generally, if  $K(\emptyset) \implies Q$ , then  $K(R) \implies Q$ .

*Proof.* [✓ Proof](#) Induct on the finite set  $R$ . If  $R = \emptyset$ , there is nothing to prove. Otherwise write  $R = S \cup \{j\}$  with  $j \notin S$ . The deletion implication gives  $K(R) \implies K(S)$ , and the induction hypothesis gives  $K(S) \implies K(\emptyset)$ . □

**Definition 15.101** (Edge-middle kernel descent data). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $e = (u, v)$  and  $M_e = V \setminus \{u, v\}$ . An edge-middle kernel descent datum is indexed by boundary labels  $\rho$  consisting of the two residual endpoint configurations. It assigns to every finitely supported coefficient family  $c = (c_\rho)_\rho$  a finite kernel descent datum  $K_c(S)$  such that

$$\left( \forall \tau, \sum_\rho c_\rho T_{A, M_e}^\rho(\tau) = 0 \right) \implies K_c(M_e),$$

and

$$K_c(\emptyset) \implies \forall \rho, c_\rho = 0.$$

**Theorem 15.102** (Kernel descent gives edge-middle injectivity). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $e = (u, v)$  and  $M_e = V \setminus \{u, v\}$ . If an edge-middle kernel descent datum is given, then  $T_{A, M_e}$  is injective. Consequently, if  $A$  is vertex-injective, the edge-blocked three-site chain at  $e$  is injective.

*Proof.* ✓ Proof Let  $c = (c_\rho)_\rho$  be a finite linear relation among the middle tensor vectors:

$$\forall \tau, \sum_\rho c_\rho T_{A, M_e}^\rho(\tau) = 0.$$

The initial part of the descent datum gives  $K_c(M_e)$ . The finite descent lemma gives  $K_c(M_e) \implies K_c(\emptyset)$ , and the terminal part gives  $\forall \rho, c_\rho = 0$ . Thus every finite relation is trivial, which is  $\text{inj}(T_{A, M_e})$ . Vertex injectivity at  $u$  and  $v$  gives  $\text{inj}(T_{A, u})$  and  $\text{inj}(T_{A, v})$ ; the edge-blocked three-site injectivity at  $e$  follows.  $\square$

**Theorem 15.103** (Edge blocking preserves injectivity). Lean ✓ Stmt *For a tensor map or blocked tensor family  $T$ , write  $\text{inj}(T)$  for the assertion that  $T$  is injective. If  $A$  is vertex-injective and every bond dimension is positive, then for every edge  $e = (u, v)$ ,*

$$\text{inj}(T_{A, u}), \quad \text{inj}(T_{A, V \setminus \{u, v\}}), \quad \text{inj}(T_{A, v}).$$

Here  $T_{A, u}$  and  $T_{A, v}$  are the two endpoint tensor maps, and  $T_{A, V \setminus \{u, v\}}$  is the blocked middle tensor family. This is the precise assertion after the edge-blocking diagram in [MGRPG<sup>+</sup> 18, Section 3]. The positive-bond hypothesis is the source's standing assumption that injective PEPS have nonzero virtual bond spaces. Without it, an empty complement configuration space can make the middle tensor vanish identically; Lemmas 15.79 and 15.80 record this obstruction.

*Proof.* ✓ Proof Fix  $e = (u, v)$  and put  $M_e = V \setminus \{u, v\}$ . From vertex injectivity, construct the edge-middle descent datum required by Theorem 15.102: for each finite relation

$$\forall \tau, \sum_\rho c_\rho T_{A, M_e}^\rho(\tau) = 0,$$

the associated conditions  $K_c(S)$  satisfy  $K_c(M_e)$ ,  $j \in S \implies K_c(S) \implies K_c(S \setminus \{j\})$ , and  $K_c(\emptyset) \implies \forall \rho, c_\rho = 0$ . The descent implication is the local contraction identity: for  $j \in S$ , vertex injectivity at  $j$  gives a left inverse  $L_j$ , and applying  $L_j$  to the relation contracts away that tensor,

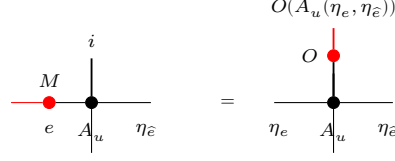
$$j \in S, K_c(S) \xrightarrow{L_j} K_c(S \setminus \{j\}).$$

The contraction at  $j$  uses the factorization

$$\mathcal{C}_e \cong \mathcal{V}_j \times \prod_{\substack{f \neq e \\ f \neq j}} \{0, \dots, \dim A_f - 1\},$$

where  $\mathcal{C}_e$  is the complement virtual-configuration space for the distinguished edge  $e$  and  $\mathcal{V}_j$  is the local virtual configuration space at  $j$ . The terminal step  $K_c(\emptyset) \implies \forall \rho, c_\rho = 0$  isolates each boundary coefficient by surjectivity of the boundary labelling, which is where the positive bond dimensions are used to fill the interior virtual indices of a witness configuration. The kernel descent theorem gives  $\text{inj}(T_{A, M_e})$ . The same theorem, together with vertex injectivity at  $u$  and  $v$ , gives  $\text{inj}(T_{A, u})$ ,  $\text{inj}(T_{A, M_e})$ , and  $\text{inj}(T_{A, v})$  as the edge-blocked three-site injectivity assertion.  $\square$

**Theorem 15.104** (Local virtual insertion realized physically on either endpoint). Lean ✓ Stmt *Let  $A$  be vertex-injective and let  $e = (u, v)$  be an edge. For every matrix  $M$  on the bond space of  $e$ , the one-leg virtual action induced by  $M$  on the local tensor at  $u$ , and likewise at  $v$ , is realized by a physical operator on that endpoint's physical space.*



*Proof.* ✓ Proof Apply the one-edge realization theorem to the two endpoint incidences of the distinguished edge.  $\square$

**Theorem 15.105** (Left-endpoint projected recovery). Lean ✓ Stmt Let  $e = (u, v)$ . A physical operator at the left endpoint pulls back to the one-edge virtual action of  $M^\top$  if and only if its compressed physical action is the canonical realization of that one-edge virtual action.

*Proof.* ✓ Proof This is the projected local recovery equivalence applied to the left incident edge of  $e$ , with the transpose convention used by the left-endpoint coefficient identity.  $\square$

**Theorem 15.106** (Right-endpoint projected recovery). Lean ✓ Stmt Let  $e = (u, v)$ . A physical operator at the right endpoint pulls back to the one-edge virtual action of  $M$  if and only if its compressed physical action is the canonical realization of that one-edge virtual action.

*Proof.* ✓ Proof This is the projected local recovery equivalence applied to the right incident edge of  $e$ .  $\square$

**Theorem 15.107** (Endpoint recovery of a common virtual insertion). Lean ✓ Stmt Let  $e = (u, v)$ . If  $P_u O_1 P_u = \Phi_u T_{u,e}(M^\top) \Psi_u$  and  $P_v O_2 P_v = \Phi_v T_{v,e}(M) \Psi_v$ , then  $\Psi_u O_1 \Phi_u = T_{u,e}(M^\top)$  and  $\Psi_v O_2 \Phi_v = T_{v,e}(M)$ . Here  $T_{w,e}(N)$  denotes the virtual operator on the legs incident to  $w$  which acts by  $N$  on the leg belonging to  $e$  and by the identity on the other incident legs.

*Proof.* ✓ Proof The endpoint equivalences give  $P_u O_1 P_u = \Phi_u T_{u,e}(M^\top) \Psi_u \iff \Psi_u O_1 \Phi_u = T_{u,e}(M^\top)$  and  $P_v O_2 P_v = \Phi_v T_{v,e}(M) \Psi_v \iff \Psi_v O_2 \Phi_v = T_{v,e}(M)$ . The two hypotheses are the left sides of these equivalences.  $\square$

**Theorem 15.108** (Endpoint physical-to-virtual recovery under image preservation). Lean ✓ Stmt Let  $e = (u, v)$ . Suppose that  $P_u O_1 P_u = \Phi_u T_{u,e}(M^\top) \Psi_u$  and  $P_v O_2 P_v = \Phi_v T_{v,e}(M) \Psi_v$ . Suppose also that  $O_1$  maps the image of  $\Phi_u$  into itself and  $O_2$  maps the image of  $\Phi_v$  into itself. Then

$$O_1 \Phi_u = \Phi_u T_{u,e}(M^\top), \quad O_2 \Phi_v = \Phi_v T_{v,e}(M).$$

*Proof.* ✓ Proof The preceding endpoint theorem identifies the two virtual pullbacks with the one-edge actions of  $M^\top$  and  $M$ . The pullback specification gives the projected physical actions, and the image-preservation hypotheses remove the endpoint projectors.  $\square$

**Theorem 15.109** (Right-endpoint physical realization of the inserted coefficient). Lean ✓ Stmt Let  $e = (u, v)$  and let  $\sigma$  be a physical configuration. If a physical operator on  $v$  realizes the one-edge virtual matrix action on the distinguished incident edge, then the inserted-edge coefficient at  $\sigma$  is obtained by leaving the left endpoint and open middle contraction explicit and applying that physical operator to the right endpoint tensor.

*Proof.* ✓ Proof Expand the physical realization on a basis local configuration at the right endpoint. The resulting sum over the right bond index is then reindexed together with the ordinary edge-boundary data.  $\square$

**Theorem 15.110** (Left-endpoint physical realization of the inserted coefficient). [Lean](#) [✓ Stmt](#)  
Let  $e = (u, v)$  and let  $\sigma$  be a physical configuration. If a physical operator on  $u$  realizes the one-edge virtual action of  $M^\top$  on the distinguished incident edge, then the inserted-edge coefficient for  $M$  at  $\sigma$  is obtained by applying that physical operator to the left endpoint tensor and leaving the open middle contraction and the right endpoint explicit.

*Proof.* [✓ Proof](#) Expand the physical realization on a basis local configuration at the left endpoint. The transpose converts the ordinary right bond index and the new left bond index into the coefficient  $M_{ij}$ ; the resulting finite sum is then reindexed with the ordinary edge-boundary data.  $\square$

**Theorem 15.111** (Endpoint physical realizations of the inserted coefficient). [Lean](#) [✓ Stmt](#)  
If the PEPS tensor is vertex-injective, then for every physical configuration  $\sigma$ , each inserted edge matrix is realized by physical operators at the endpoint tensors: the left endpoint realizes  $M^\top$ , the right endpoint realizes  $M$ , and both realizations reproduce the same inserted-edge coefficient at  $\sigma$  in the edge-centered three-site decomposition.

*Proof.* [✓ Proof](#) Apply the local physical-realization theorem at the two endpoints and then use the two coefficient identities above.  $\square$

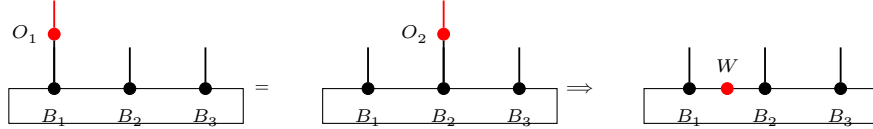
**Theorem 15.112** (Equal neighboring physical actions recover a virtual insertion). [Lean](#) [✓ Stmt](#)  
Let  $e = (u, v)$ . Assume that the edge-blocked three-site chain at  $e$  is injective and that every virtual bond has positive dimension. Let  $O_1, O_2$  be physical operators at the endpoint tensors, and write  $\Phi_u, \Phi_v$  for the two local tensor maps. Suppose that, for every physical configuration  $\sigma$ ,

$$\sum_{\beta} O_1(A_u(\beta_u))_{\sigma_u} C_{\text{mid}}(\sigma; \beta_L, \beta_R) A_v(\beta_v)_{\sigma_v} = \sum_{\beta} A_u(\beta_u)_{\sigma_u} C_{\text{mid}}(\sigma; \beta_L, \beta_R) O_2(A_v(\beta_v))_{\sigma_v}.$$

Then

$$\exists M \in M_{D(e)}(\mathbb{C}), \quad O_1 \Phi_u = \Phi_u T_{u,e}(M^\top), \quad O_2 \Phi_v = \Phi_v T_{v,e}(M).$$

The same implication is represented diagrammatically by the  $O_1, O_2 \mapsto W$  step:



This is the converse part of the insertion-algebra lemma in [MGRPG<sup>+</sup> 18, Section 3], where the recovered matrix is denoted  $W$ .

*Proof.* [✓ Proof](#) Contracting the middle block out of the hypothesis leaves a bond-contracted identity between the two endpoint tensors, holding for every residual boundary configuration. Inverting the right endpoint of this identity writes the action of  $O_1$  on a left tensor vector as a bond-indexed combination of left tensor vectors with the same residual; the combining coefficients are read at a fixed reference right residual and define  $M$ . Inverting the left endpoint writes the action of  $O_2$  on a right tensor vector as a bond-indexed combination of right tensor vectors. The left inverse identifies the two coefficient families, which is the source step  $V = W$ , so the right-endpoint combination is also governed by  $M$ . The reference residuals exist because every virtual bond has positive dimension.  $\square$

**Theorem 15.113** (Injectivity of an inserted edge coefficient). [Lean](#) [✓ Stmt](#) *Let the edge-blocked three-site chain of  $B$  at  $e$  be injective, and assume every virtual bond of  $B$  has positive dimension. If two matrices  $M, M' \in M_{D_B(e)}(\mathbb{C})$  give the same inserted coefficient at every physical configuration,*

$$\forall \sigma, \quad C_B(e, \sigma, M) = C_B(e, \sigma, M'),$$

*then  $M = M'$ .*

*Proof.* [✓ Proof](#) Realize  $M$  on the right endpoint and  $M'^T$  on the left endpoint. The endpoint realizations and the equality  $C_B(e, \sigma, M) = C_B(e, \sigma, M')$  give, for every physical configuration  $\sigma$ ,

$$\sum_{\beta} O_1(B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R) (B_v(\beta_v))_{\sigma_v} = \sum_{\beta} (B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R) O_2(B_v(\beta_v))_{\sigma_v}.$$

Theorem 15.112 gives a single virtual matrix realizing both endpoint operators. Endpoint uniqueness then identifies this matrix with  $M$  and with  $M'$ .  $\square$

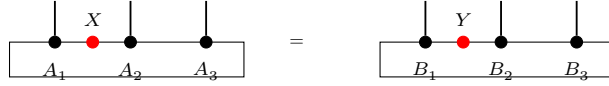
**Definition 15.114** (Edge-blocked insertion algebra correspondence). [Lean](#) [✓ Stmt](#) For two PEPS tensors  $A$  and  $B$  and an edge  $e$ , this property is

$$\exists \Phi_e : M_{D_A(e)}(\mathbb{C}) \simeq_{\text{alg}} M_{D_B(e)}(\mathbb{C}), \quad \forall \sigma, \forall X \in M_{D_A(e)}(\mathbb{C}), \quad C_A(e, \sigma, X) = C_B(e, \sigma, \Phi_e(X)).$$

**Theorem 15.115** (Edge-blocked insertion algebra isomorphism). [Lean](#) [✓ Stmt](#) *Suppose the two edge-blocked three-site chains associated to  $A$  and  $B$  at the edge  $e$  are injective, suppose every virtual bond of  $A$  and  $B$  has positive dimension, and suppose the two PEPS tensors define the same state. Then*

$$\exists \Phi_e : M_{D_A(e)}(\mathbb{C}) \simeq_{\text{alg}} M_{D_B(e)}(\mathbb{C}), \quad \forall \sigma, \forall X \in M_{D_A(e)}(\mathbb{C}), \quad C_A(e, \sigma, X) = C_B(e, \sigma, \Phi_e(X)).$$

*This is the edge-blocked form of the algebra isomorphism  $X \mapsto Y$  in [MGRPG<sup>+</sup> 18, Section 3]:*



*Proof.* [✓ Proof](#) For a matrix  $X \in M_{D_A(e)}(\mathbb{C})$ , realize the insertion of  $X$  at the two endpoints of the first edge-blocked chain by physical operators  $O_1, O_2$ . Write  $C_{\text{mid}}^A$  and  $C_{\text{mid}}^B$  for the open middle weights of the two edge-blocked chains. For every physical configuration  $\sigma$ , endpoint realization of  $X$  and equality of the PEPS states give

$$C_A(e, \sigma, X) = \sum_{\beta} O_1(A_u(\beta_u))_{\sigma_u} C_{\text{mid}}^A(\sigma; \beta_L, \beta_R) (A_v(\beta_v))_{\sigma_v} = \sum_{\beta} O_1(B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R) (B_v(\beta_v))_{\sigma_v}$$

and

$$C_A(e, \sigma, X) = \sum_{\beta} (A_u(\beta_u))_{\sigma_u} C_{\text{mid}}^A(\sigma; \beta_L, \beta_R) O_2(A_v(\beta_v))_{\sigma_v} = \sum_{\beta} (B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R) O_2(B_v(\beta_v))_{\sigma_v}.$$

Therefore, for every physical configuration  $\sigma$ ,

$$\sum_{\beta} O_1(B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R) (B_v(\beta_v))_{\sigma_v} = \sum_{\beta} (B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R) O_2(B_v(\beta_v))_{\sigma_v}.$$



The positive-bond physical-to-virtual recovery applied to the second edge-blocked chain gives a matrix  $Y \in M_{D_B(e)}(\mathbb{C})$  such that, for every physical configuration  $\sigma$ ,

$$C_A(e, \sigma, X) = C_B(e, \sigma, Y).$$

Thus  $X \mapsto Y$  gives the required coefficient correspondence. Denote this correspondence by  $T$ . For every physical configuration  $\sigma$ ,

$$\begin{aligned} C_B(e, \sigma, T(X + X')) &= C_A(e, \sigma, X + X') = C_A(e, \sigma, X) + C_A(e, \sigma, X') \\ &= C_B(e, \sigma, T(X) + T(X')), \end{aligned}$$

and similarly  $T(cX) = cT(X)$ . The multiplicative identity is read from the inserted two-endpoint realization:

$$C_B(e, \sigma, T(XX')) = C_A(e, \sigma, XX') = C_B(e, \sigma, T(X)T(X')).$$

The unit satisfies

$$C_B(e, \sigma, T(1)) = C_A(e, \sigma, 1) = C_B(e, \sigma, 1).$$

Theorem 15.113 identifies the inserted matrices in these coefficient identities. Exchanging  $A$  and  $B$  gives the inverse correspondence, hence a  $\mathbb{C}$ -algebra isomorphism.  $\square$

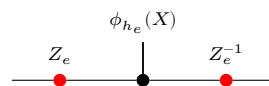
**Theorem 15.116** (Matrix-algebra isomorphisms preserve size). [Lean](#) [✓ Stmt](#) *If the full matrix algebras  $M_D(\mathbb{C})$  and  $M_E(\mathbb{C})$  are isomorphic as  $\mathbb{C}$ -algebras, then  $D = E$ .*

*Proof.* [✓ Proof](#) The isomorphism preserves complex vector-space dimension, and  $\dim_{\mathbb{C}} M_D(\mathbb{C}) = D^2$ ,  $\dim_{\mathbb{C}} M_E(\mathbb{C}) = E^2$ , so  $D = E$ .  $\square$

**Theorem 15.117** (Matrix-algebra isomorphisms are inner after identifying size). [Lean](#) [✓ Stmt](#) *Let  $\varphi : M_D(\mathbb{C}) \rightarrow M_E(\mathbb{C})$  be a  $\mathbb{C}$ -algebra isomorphism. Then  $D = E$ ; let  $h$  denote this equality and  $\iota_h : M_D(\mathbb{C}) \rightarrow M_E(\mathbb{C})$  the canonical index-relabelling isomorphism. There is  $X \in \text{GL}_E(\mathbb{C})$  with  $\varphi(M) = X \iota_h(M) X^{-1}$  for all  $M \in M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) By Theorem 15.116,  $D = E$ , so  $\iota_h$  is defined. Then  $\varphi \circ \iota_h^{-1}$  is a  $\mathbb{C}$ -algebra automorphism of  $M_E(\mathbb{C})$ , and Skolem–Noether (Theorem 3.11) gives  $X \in \text{GL}_E(\mathbb{C})$  with  $(\varphi \circ \iota_h^{-1})(N) = X N X^{-1}$ ; evaluating at  $N = \iota_h(M)$  gives  $\varphi(M) = X \iota_h(M) X^{-1}$ .  $\square$

**Theorem 15.118** (Edge gauge from the insertion algebra isomorphism). [Lean](#) [✓ Stmt](#) *Suppose the edge insertion correspondence is a  $\mathbb{C}$ -algebra isomorphism between the two full matrix algebras on the chosen bond. Then the two bond dimensions on that edge are equal. If  $h_e : D_A(e) = D_B(e)$  is this equality, and if  $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_B(e)}(\mathbb{C})$  is the induced algebra isomorphism, then the insertion correspondence is  $Y = Z_e \phi_{h_e}(X) Z_e^{-1}$  for an invertible edge matrix  $Z_e$ .*



*This is the finite-dimensional matrix-algebra step in [MGRPG<sup>+</sup>18, Section 3].*

*Proof.* [✓ Proof](#) Apply Theorem 15.117 to the full matrix algebras determined by the two edge bond spaces.  $\square$

**Theorem 15.119** (Per-edge gauge family from insertion algebra isomorphisms). [Lean](#) [✓ Stmt](#) Let  $A$  and  $B$  be vertex-injective PEPS tensors with the same PEPS state. Suppose the corresponding virtual bond dimensions are equal and every virtual bond of  $A$  has positive dimension. Then there is a family  $X_e \in \text{GL}(D_A(e))$  such that, for every edge  $e$ , there is a  $\mathbb{C}$ -algebra isomorphism

$$\Phi_e : M_{D_A(e)}(\mathbb{C}) \longrightarrow M_{D_B(e)}(\mathbb{C})$$

satisfying, for every physical configuration  $\sigma$  and matrix  $M$ ,

$$C_A(e, \sigma, M) = C_B(e, \sigma, \Phi_e(M)),$$

and, writing  $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_B(e)}(\mathbb{C})$  for the algebra isomorphism induced by the dimension equality on  $e$ ,

$$\Phi_e(M) = \phi_{h_e}(X_e M X_e^{-1}).$$

*Proof.* [✓ Proof](#) Vertex injectivity and positivity give edge-blocked three-site injectivity for  $A$  and  $B$  at each edge. Theorem 15.115 gives the algebra isomorphism  $\Phi_e$  preserving inserted coefficients. Theorem 15.118 realizes  $\Phi_e$  by conjugation, and the bond-dimension equality transports the conjugating matrix to the  $A$ -side bond.  $\square$

**Definition 15.120** (Factorized local gauge datum). [Lean](#) [✓ Stmt](#) Fix  $A, B$ , a bond-dimension equality, and a vertex  $v$ . Let  $\mathcal{T}_v^A$  be the local tensor map of  $A$ , let  $L_v^A$  be the chosen left inverse of  $\mathcal{T}_v^A$ , and let  $\Pi_v^A = \mathcal{T}_v^A L_v^A$ . The factorized local gauge datum consists of the following two identities for local virtual configurations  $\eta, \eta'$ :

$$\Pi_v^A B_v(\eta, -) = B_v(\eta, -), \quad L_v^A B_v(\eta, -)(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e),$$

with  $X_e \in \text{GL}(D_A(e))$  on each incident edge. This is the local output required from the blocked-middle reduction in [\[MGRPG<sup>+</sup>18, Section 3\]](#).

**Definition 15.121** (Blocked-middle local gauge formula). [Lean](#) [✓ Stmt](#) The blocked-middle local gauge formula at  $v$  is the assertion that there are matrices  $X_e \in \text{GL}(D_A(e))$ , one on each incident edge, such that for all local virtual configurations  $\eta$  and physical indices  $\sigma$ ,

$$B_v(\eta, \sigma) = \sum_{\eta'} \left( \prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

**Theorem 15.122** (Blocked-middle formula implies factorized local gauge). [Lean](#) [✓ Stmt](#) If the blocked-middle local gauge formula holds at  $v$  with matrices  $X_e$ , then the factorized local gauge datum holds at  $v$  with the same matrices.

*Proof.* [✓ Proof](#) For fixed  $\eta$ , set  $c_\eta(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e)$ . The displayed blocked-middle formula is precisely  $B_v(\eta, -) = \mathcal{T}_v^A c_\eta$ . Hence

$$\Pi_v^A B_v(\eta, -) = \Pi_v^A \mathcal{T}_v^A c_\eta = \mathcal{T}_v^A c_\eta = B_v(\eta, -),$$

and

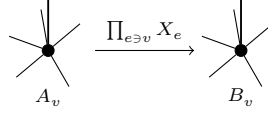
$$L_v^A B_v(\eta, -)(\eta') = L_v^A \mathcal{T}_v^A c_\eta(\eta') = c_\eta(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e).$$

$\square$

**Theorem 15.123** (Blocked-middle formula gives a local gauge formula). [Lean](#) [✓ Stmt](#) Suppose that the edge-centred three-site reduction supplies invertible matrices  $X_e$  on the incident edges of  $v$  such that, for all local virtual configurations  $\eta$  and physical indices  $\sigma$ ,

$$B_v(\eta, \sigma) = \sum_{\eta'} \left( \prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

Then the same family  $X_e$  is a local gauge at  $v$ :

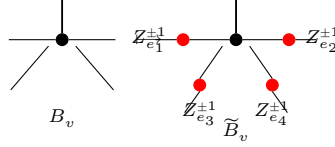


*Proof.* [✓ Proof](#) By the preceding theorem,  $L_v^A B_v(\eta, -)(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e)$ . Since  $\Pi_v^A B_v(\eta, -) = B_v(\eta, -)$  and  $\Pi_v^A = \mathcal{T}_v^A L_v^A$ ,

$$B_v(\eta, \sigma) = \sum_{\eta'} L_v^A B_v(\eta, -)(\eta') A_v(\eta', \sigma) = \sum_{\eta'} \left( \prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

□

**Theorem 15.124** (Absorbing edge gauges into the second PEPS). [Lean](#) [✓ Stmt](#) Given edge gauges  $X_e$ , set  $\tilde{B} = B^X$ . Then  $D_{\tilde{B}} = D_B$  and, for every vertex  $v$ ,  $\tilde{B}_v(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) B_v(\eta', \sigma)$ , where  $M_{ie} = X_e$  or  $M_{ie} = (X_e^{-1})^t$  according to the orientation of the endpoint  $ie$ .



After this absorption, arbitrary insertions of the same matrix on the same edge are compared in the first PEPS and the modified second PEPS.

*Proof.* [✓ Proof](#) With  $\tilde{B} = B^X$ , Theorem 15.11 gives  $D_{\tilde{B}} = D_B$ , and Theorem 15.12 gives  $\tilde{B}_v(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) B_v(\eta', \sigma)$ . □

**Definition 15.125** (Post-absorption inserted-edge comparison). [Lean](#) [✓ Stmt](#) For tensor families  $A, \tilde{B}$ , the post-absorption comparison holds when  $h : D_A = D_{\tilde{B}}$  and the following equality holds. For each edge  $e$ , write  $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_{\tilde{B}}(e)}(\mathbb{C})$  for the matrix-algebra transport induced by  $h_e : D_A(e) = D_{\tilde{B}}(e)$ . Then, for every matrix  $X \in M_{D_A(e)}(\mathbb{C})$  and every physical configuration  $\sigma$ ,

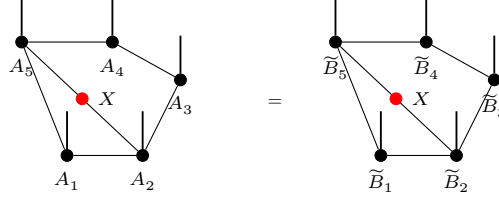
$$C_A(e, X; \sigma) = C_{\tilde{B}}(e, \phi_{h_e}(X); \sigma).$$

**Theorem 15.126** (Post-absorption edge insertion equality). [Lean](#) [✓ Stmt](#) Let  $A$  and  $B$  be vertex-injective PEPS with the same state, assume every bond dimension is positive, and assume the bond-dimension equality  $h : D_A = D_B$ . After the edge gauges have been absorbed into the second tensor family, there are gauges  $Z_e$  and  $\tilde{B} = B^Z$  such that  $D_{\tilde{B}} = D_A$  and, for every edge  $e$ , write  $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_{\tilde{B}}(e)}(\mathbb{C})$  for the matrix-algebra transport induced by the corresponding

equality  $h_e : D_A(e) = D_{\tilde{B}}(e)$ . Then, for every matrix  $X \in M_{D_A(e)}(\mathbb{C})$  and every physical configuration  $\sigma$ ,

$$C_A(e, X; \sigma) = C_{\tilde{B}}(e, \phi_{h_e}(X); \sigma),$$

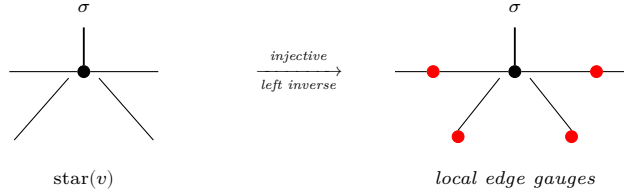
Diagrammatically, this is the equality



from [MGRPG<sup>+</sup>18, Section 3, lines 1037–1065]. The positive-bond hypothesis is the source’s standing assumption that injective PEPS have nonzero virtual bond spaces; it is needed because the edge gauges come from blocking the PEPS around each edge into a three-site injective chain.

*Proof.* ✓ Proof Block the PEPS around each edge into a three-site injective chain and compare the matched matrix insertions: this assigns to every edge an invertible bond matrix  $X_e$  realizing the insertion-algebra transport as conjugation. Absorb the transpose of  $X_e$  into the second tensor family on that edge. The open-edge gauge action then conjugates each inserted matrix by the transpose of the absorbed gauge, while the insertion-algebra transport conjugates by  $X_e$ ; the two conjugations agree because the absorbed gauge is the transpose of  $X_e$ , giving the post-absorption insertion equality.  $\square$

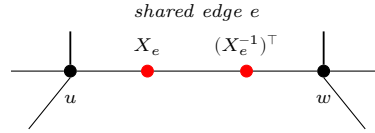
**Theorem 15.127** (Local gauge extraction). Lean Lean ✓ Stmt Under the factorized local gauge hypothesis, the local gauge formula follows. What remains open is to upgrade equality of the edge-blocked coefficients to the blocked-middle formula using the corresponding three-site injective-chain argument. The physical-realization and physical-to-virtual steps in that argument are recorded separately as Theorems 15.104 and 15.112. After Theorem 15.126 supplies the factorized local gauge hypothesis, injectivity on the star neighborhood provides a left inverse that isolates the local edge gauges:



*Proof.* ✓ Proof The factorized local gauge datum already supplies the incident-edge invertible matrices and the factorized local coefficient identity.  $\square$

**Theorem 15.128** (Gauge consistency). Lean ✓ Stmt Suppose  $A$  and  $B$  are vertex-injective PEPS tensors with the same PEPS state, that the corresponding virtual bond dimensions are equal, that every virtual bond of  $A$  has positive dimension, and that the graph is connected. The connectivity assumption is necessary: on a disconnected graph the per-vertex scalars produced by the reduction below cannot be absorbed into edge gauges, since on the empty graph on two vertices the products of the vertex scalars can agree while no edge gauge relates the two tensors. Choose an edge  $e = (u, v)$  and block all other vertices into one middle tensor. The three resulting tensors form an injective chain, so the one-dimensional injective comparison assigns an invertible matrix to the edge  $e$ . Repeating this for every edge and absorbing these matrices into the second tensor

family gives the post-absorption insertion equality of [MGRPG<sup>+</sup>18, Section 3]: for every edge and every matrix, inserting that matrix on the edge in the first PEPS gives the same state as inserting it on the same edge in the modified second PEPS. Applying the local extraction at each vertex then gives one edge gauge family whose endpoint actions transform the first tensor family into the second:



*Proof.* ✓ Proof Absorbing the edge gauges into the second tensor family gives the post-absorption insertion equality. On each vertex with an incident edge, the one-vertex-versus-complement comparison supplies a nonzero scalar relating the first tensor to the absorbed second tensor. Because the state is nonzero, these per-vertex scalars multiply to one, so on the connected graph a spanning-tree construction realizes their reciprocals as the oriented incidence product of one edge-scalar family. Folding those scalars into the inverse of the absorbed edge gauges produces the global gauge. On a one-vertex graph there are no edges and the gauge is trivial.  $\square$

*Remark 15.129.* The PEPS diagrams in this section record the gauge steps used in [MGRPG<sup>+</sup>18, Section 3]. The edge-orientation and vertex-action diagrams fix the endpoint convention for the edge gauges obtained by blocking the PEPS to a three-site injective MPS. The cancellation diagram is the converse contracted identity: inserting opposite endpoint actions on an internal edge leaves the PEPS coefficient unchanged. The edge-gauge absorption diagram records the formation of  $\tilde{B}$  before the post-absorption insertion equality. The local-extraction diagram summarizes the comparison of one-site-enlarged injective regions with injective complements, after the edge gauges have been absorbed into  $\tilde{B}$ . The global-consistency diagram summarizes repeating the edge blocking over all edges, so the two endpoint actions on a shared edge are one edge gauge. The two-injective-tensor diagram records the two-block comparison from [MGRPG<sup>+</sup>18, Section 3], and the final one-vertex-complement diagram records its application to one vertex and its complement. These pictures use the blueprint tensor-dot convention, but each one is attached to the same contraction, insertion, or blocking step as the corresponding argument in the paper.

**Definition 15.130** (Left-identity three-leg residual form). Lean ✓ Stmt For three virtual leg spaces  $\alpha, \beta, \gamma$  and an endomorphism  $Z$  of  $\beta \otimes \gamma$ , define the induced endomorphism of  $\alpha \otimes \beta \otimes \gamma$  by acting as the identity on the  $\alpha$  leg and as  $Z$  on the remaining two legs. This is the  $I_\alpha \otimes Z$  form appearing after the first tensor is inverted in [MGRPG<sup>+</sup>18, Section 3].

**Definition 15.131** (Right-identity three-leg residual form). Lean ✓ Stmt For an endomorphism  $U$  of  $\alpha \otimes \beta$ , define the induced endomorphism of  $\alpha \otimes \beta \otimes \gamma$  by acting as  $U$  on the first two legs and as the identity on the  $\gamma$  leg. This is the  $U \otimes I_\gamma$  form in [MGRPG<sup>+</sup>18, Section 3].

**Definition 15.132** (Middle-identity three-leg residual form). Lean ✓ Stmt For an endomorphism  $W$  of  $\alpha \otimes \gamma$ , define the induced endomorphism of  $\alpha \otimes \beta \otimes \gamma$  by acting as  $W$  on the outer two legs and as the identity on the middle  $\beta$  leg. This is the third residual form in [MGRPG<sup>+</sup>18, Section 3].

**Theorem 15.133** (Scalar reduction of the residual two-tensor gauges). Lean ✓ Stmt In the proof of the generalized two-injective-tensor comparison, applying the inverse of the second injective tensor leaves three possible virtual operators on the exposed legs of the first tensor:

Inverting the first injective tensor then shows that the corresponding residual operator on the three exposed virtual legs has the three decompositions

$$\sum_i I \otimes Z_i^{(1)} \otimes Z_i^{(2)} = \sum_i U_i^{(1)} \otimes U_i^{(2)} \otimes I = \sum_i W_i^{(1)} \otimes I \otimes W_i^{(2)}$$

Equivalently, for nonzero virtual leg spaces  $\alpha, \beta, \gamma$ , if one endomorphism of  $\alpha \otimes \beta \otimes \gamma$  lies simultaneously in the three subspaces  $I_\alpha \otimes \text{End}(\beta \otimes \gamma)$ ,  $\text{End}(\alpha \otimes \beta) \otimes I_\gamma$ , and the analogous subspace with  $I_\beta$  in the middle leg, then it is a scalar multiple of the identity. Thus the residual virtual operators are scalar multiples of the identity. This is the scalar step in the proof of [MGRPG<sup>+</sup> 18, Section 3].

*Proof.* ✓ Proof Write the common operator as  $T$ . The equality  $T = I_\alpha \otimes Z = U \otimes I_\gamma$  gives, for all indices,

$$\delta_{aa'} Z_{bc, b'c'} = U_{ab, a'b'} \delta_{cc'}.$$

If  $c \neq c'$ , the right hand side is zero; choosing  $a = a'$  gives  $Z_{bc, b'c'} = 0$ . Hence  $Z$  is diagonal in the  $\gamma$ -index. Similarly, the equality  $T = I_\alpha \otimes Z = W_{\alpha\gamma} \otimes I_\beta$  gives

$$\delta_{aa'} Z_{bc, b'c'} = W_{ac, a'c'} \delta_{bb'},$$

so  $Z_{bc, b'c'} = 0$  whenever  $b \neq b'$ . Thus  $Z_{bc, b'c'} = 0$  unless  $b = b'$  and  $c = c'$ .

It remains to compare the surviving diagonal entries. From  $\delta_{aa'} Z_{bc, bc} = U_{ab, a'b}$ , the value  $Z_{bc, bc}$  is independent of  $c$ . From the middle-identity equality it is also independent of  $b$ . Therefore  $Z = \lambda I_{\beta \otimes \gamma}$ . Substitution into

$$I_\alpha \otimes Z = U \otimes I_\gamma = W_{\alpha\gamma} \otimes I_\beta$$

gives  $U = \lambda I_{\alpha \otimes \beta}$  and  $W_{\alpha\gamma} = \lambda I_{\alpha \otimes \gamma}$ . □

**Theorem 15.134** (Many-leg scalar extraction). Lean ✓ Stmt Let a finite family of legs be given, each a nonzero finite index set, with invertible matrices  $M_i$  and  $N_i$  on each leg. If

$$\prod_i (M_i)_{p_i q_i} = \prod_i (N_i)_{p_i q_i}$$

for every configuration pair  $(p, q)$ , then there are nonzero scalars  $c_i$  with  $M_i = c_i N_i$  for each leg and  $\prod_i c_i = 1$ .

*Proof.* ✓ Proof Each  $N_i$  is invertible, so each of its rows has a nonzero entry; pick a reference configuration  $(p^0, q^0)$  on which every  $(N_i)_{p_i^0 q_i^0}$  is nonzero. The product identity at  $(p^0, q^0)$  then makes every  $(M_i)_{p_i^0 q_i^0}$  nonzero as well. Fix a leg  $i$  and a pair  $(a, b)$ , and apply the identity to the configuration that agrees with the reference except at leg  $i$ , where it takes  $(a, b)$ . Splitting off leg  $i$  gives

$$(M_i)_{ab} K_i = (N_i)_{ab} L_i,$$

where  $K_i, L_i$  are the products of the other legs at the reference, both nonzero. The reference identity at leg  $i$  relates  $K_i$  and  $L_i$ , and eliminating them yields  $(M_i)_{ab} = c_i (N_i)_{ab}$  with  $c_i = (M_i)_{p_i^0 q_i^0} / (N_i)_{p_i^0 q_i^0}$ , independent of  $(a, b)$ . Taking the product of the scalars over all legs and using the reference identity gives  $\prod_i c_i = 1$ . □

**Theorem 15.135** (Pointwise product cancellation for two injective blocks). [Lean](#) [✓ Stmt](#) Let  $A_1, A_2, B_1, B_2$  be tensors with matching external, shared-boundary, and physical index sets. Assume that the two external boundary spaces and the shared-boundary configuration space are nonempty, and that  $A_1$  and  $A_2$  are injective. Suppose that, for every pair of external indices, every pair of shared-boundary configurations, and every pair of physical indices,

$$A_1(\eta_1, \mu, \sigma_1)A_2(\eta_2, \nu, \sigma_2) = B_1(\eta_1, \mu, \sigma_1)B_2(\eta_2, \nu, \sigma_2).$$

Then there exists  $\lambda \in \mathbb{C}^\times$  such that

$$A_1 = \lambda B_1, \quad A_2 = \lambda^{-1} B_2.$$

*Proof.* [✓ Proof](#) Choose external and shared-boundary indices, and use injectivity of  $A_1$  and  $A_2$  to choose physical indices for which the two chosen coefficients are nonzero. The product equality then shows that the corresponding coefficients of  $B_1$  and  $B_2$  are also nonzero. Define

$$\lambda = \frac{B_2(\eta_2^0, \nu^0, \sigma_2^0)}{A_2(\eta_2^0, \nu^0, \sigma_2^0)}.$$

For every  $(\eta_1, \mu, \sigma_1)$ , the product identity with  $(\eta_2^0, \nu^0, \sigma_2^0)$  fixed gives

$$A_1(\eta_1, \mu, \sigma_1)A_2(\eta_2^0, \nu^0, \sigma_2^0) = B_1(\eta_1, \mu, \sigma_1)B_2(\eta_2^0, \nu^0, \sigma_2^0).$$

Dividing by  $A_2(\eta_2^0, \nu^0, \sigma_2^0) \neq 0$  gives  $A_1 = \lambda B_1$ . Fixing  $(\eta_1^0, \mu^0, \sigma_1^0)$  and substituting  $A_1 = \lambda B_1$  into the product identity gives

$$\lambda B_1(\eta_1^0, \mu^0, \sigma_1^0)A_2(\eta_2, \nu, \sigma_2) = B_1(\eta_1^0, \mu^0, \sigma_1^0)B_2(\eta_2, \nu, \sigma_2).$$

Dividing by  $B_1(\eta_1^0, \mu^0, \sigma_1^0) \neq 0$  gives  $A_2 = \lambda^{-1} B_2$ . □

**Definition 15.136** (Matrix unit for coefficient extraction). [Lean](#) [✓ Stmt](#) For indices  $i$  and  $j$ , let  $E_{ij}$  denote the matrix whose only nonzero entry is the  $(i, j)$  entry, where it is equal to 1.

**Theorem 15.137** (Coefficient extraction for one shared bond). [Lean](#) [✓ Stmt](#) Suppose that the two blocks share a single virtual bond. Then inserting the matrix unit  $E_{\mu, \nu}$  in that bond extracts the corresponding product of tensor coefficients:

$$C_{A_1, A_2}(b, E_{\mu, \nu}; \eta_1, \eta_2, \sigma_1, \sigma_2) = A_1(\eta_1, \mu, \sigma_1)A_2(\eta_2, \nu, \sigma_2).$$

*Proof.* [✓ Proof](#) Since there is only one shared bond, the condition that the two shared configurations agree away from the distinguished bond is vacuous, and

$$C_{A_1, A_2}(b, E_{\mu, \nu}; \eta_1, \eta_2, \sigma_1, \sigma_2) = \sum_{\mu', \nu'} (E_{\mu, \nu})_{\mu'(b), \nu'(b)} A_1(\eta_1, \mu', \sigma_1) A_2(\eta_2, \nu', \sigma_2).$$

The entries of the matrix unit satisfy

$$(E_{\mu, \nu})_{i, j} = \delta_{i, \mu(b)} \delta_{j, \nu(b)}.$$

Thus the only nonzero summand is the one with  $(\mu', \nu') = (\mu, \nu)$ . □

**Theorem 15.138** (One-bond two-injective comparison). [Lean](#) [✓ Stmt](#) Suppose that the two injective blocks have exactly one shared virtual bond and nonempty external boundary spaces. If all matrix insertions on that bond give equal two-block coefficients for the pairs  $(A_1, A_2)$  and  $(B_1, B_2)$ , then there exists  $\lambda \in \mathbb{C}^\times$  such that

$$A_1 = \lambda B_1, \quad A_2 = \lambda^{-1} B_2.$$

*Proof.* ✓ Proof Insert the matrix unit  $E_{\mu,\nu}$  and use the coefficient-extraction theorem to obtain the pointwise product equality

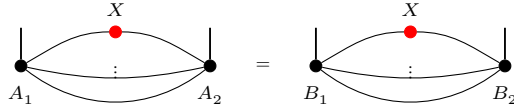
$$A_1(\eta_1, \mu, \sigma_1)A_2(\eta_2, \nu, \sigma_2) = B_1(\eta_1, \mu, \sigma_1)B_2(\eta_2, \nu, \sigma_2)$$

for all indices. The pointwise product cancellation theorem gives the reciprocal scalar proportionality.  $\square$

**Theorem 15.139** (Generalized two-injective-tensor comparison). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)  
[Lean](#) [Lean](#) [✓ Stmt](#) Let  $A_1, A_2, B_1, B_2$  be injective tensors with the same shared virtual bonds and nonempty external boundary spaces, and assume that there is at least one shared virtual bond. For the set  $\mathcal{B}$  of shared bonds, a bond  $b \in \mathcal{B}$ , and  $X \in M_{D_b}(\mathbb{C})$ , set

$$C_{A_1, A_2}(b, X; \eta_1, \eta_2, \sigma_1, \sigma_2) = \sum_{\substack{\mu, \nu \\ \mu|_{\mathcal{B} \setminus \{b\}} = \nu|_{\mathcal{B} \setminus \{b\}}}} X_{\mu_b, \nu_b} A_1(\eta_1, \mu, \sigma_1) A_2(\eta_2, \nu, \sigma_2).$$

Assume  $C_{A_1, A_2}(b, X; \eta_1, \eta_2, \sigma_1, \sigma_2) = C_{B_1, B_2}(b, X; \eta_1, \eta_2, \sigma_1, \sigma_2)$  for every  $b, X, \eta_1, \eta_2, \sigma_1, \sigma_2$ :



*Then*

$$\exists \lambda \in \mathbb{C}^\times, \quad A_1 = \lambda B_1, \quad A_2 = \lambda^{-1} B_2.$$

This is the blueprint form of the two-injective tensor comparison in [MGRP<sup>+</sup>18, Section 3].

*Proof.* ✓ Proof Putting  $X = I_{D_b}$  on every shared bond contracts all bonds diagonally and, by the insertion equality, equates the two fully contracted states. Writing each contracted state as a bipartite operator with the linearly independent Schmidt families supplied by injectivity, the operator-Schmidt uniqueness of [MGRPG<sup>+</sup>18, Section 3] produces an invertible gauge  $g$  on the shared-bond configurations with

$$A_1(\eta_1, \mu, \sigma_1) = \sum_{\nu} g_{\mu, \nu} B_1(\eta_1, \nu, \sigma_1), \quad B_2(\eta_2, \nu, \sigma_2) = \sum_{\mu} g_{\mu, \nu} A_2(\eta_2, \mu, \sigma_2).$$

Substitute both gauge equations into the insertion equality for the matrix unit  $X = E_{p,q}$  on a single bond  $b$ , which frees that bond and contracts the others. Separating the linearly independent families  $B_1$  and  $A_2$  gives, for every bond  $b$ , every pair of endpoints  $p, q$ , and all configurations  $\nu, \rho$ ,

$$[\rho_b = q] g_{\rho^{b \rightarrow p}, \nu} = [\nu_b = p] g_{\rho, \nu^{b \rightarrow q}}, \quad (15.1)$$

where  $\rho^{b \rightarrow p}$  replaces the value of  $\rho$  at  $b$  by  $p$ . This is the bond-wise form of the residual identity of [MGRPG<sup>+</sup>18, Section 3], where it is packaged as the equality  $I_\alpha \otimes Z = U \otimes I_\gamma = W_{\alpha\gamma} \otimes I_\beta$  of the three residual gauges  $Z, U, W$  (Theorem 15.133). Reading (15.1) at the two endpoint choices shows that  $g$  vanishes whenever two configurations differ at a bond and that its diagonal value is invariant under changing any single bond. Hence  $g = \lambda I$  for a scalar  $\lambda$ , and invertibility forces  $\lambda \neq 0$ .

Reinserting  $g = \lambda I$  into the two gauge equations gives  $A_1 = \lambda B_1$  and  $B_2 = \lambda A_2$ , that is  $A_2 = \lambda^{-1} B_2$ .  $\square$



**Definition 15.140** (Region two-block tensor). [Lean](#) [✓ Stmt](#) For a finite region  $R \subseteq V$ , the blocked tensor of  $R$  may be read as a two-block tensor with a one-point external boundary and shared bonds the edges crossing the boundary of  $R$ :

$$B_{A,R}(*, \eta, \sigma) = T_{A,R}^\eta(\sigma).$$

This is the region block used in the one-block-against-complement step of [MGRPG<sup>+</sup>18, Section 3].

**Definition 15.141** (Complement-boundary identification). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The edges crossing the boundary of  $R$  are the same edges crossing the boundary of  $V \setminus R$ . Thus a boundary configuration  $\eta$  for  $R$  determines the boundary configuration  $\eta^c$  for the complement by reading the same edge labels through this identification.

**Definition 15.142** (Complement-region two-block tensor). [Lean](#) [✓ Stmt](#) For a finite region  $R \subseteq V$ , the complement block  $V \setminus R$  is a two-block tensor over the same crossing bonds:

$$B_{A,R^c}(*, \eta, \tau) = T_{A,V \setminus R}^{\eta^c}(\tau).$$

This is the complementary block in the region/complement comparison of [MGRPG<sup>+</sup>18, Section 3].

**Theorem 15.143** (Injectivity of the complement-region two-block tensor). [Lean](#) [✓ Stmt](#) *If the blocked tensor family of  $V \setminus R$  is injective, then the complement-region two-block tensor is injective:*

$$\text{inj}(T_{A,V \setminus R}) \implies \text{inj}(B_{A,R^c}).$$

*Proof.* [✓ Proof](#) The one-point external boundary does not change the coefficient family, and the complement-boundary identification only reindexes the crossing bonds. More explicitly, if  $\eta_1, \eta_2$  are boundary configurations on  $\partial R$ , then

$$\eta_1^c = \eta_2^c \implies \eta_1 = \eta_2.$$

Thus the coefficients of  $T_{A,V \setminus R}$  and  $B_{A,R^c}$  differ only by an injective relabelling of boundary configurations.  $\square$

**Definition 15.144** (Region insertion coefficient). [Lean](#) [✓ Stmt](#) Let  $f$  be an edge crossing the boundary of  $R$ , and let  $M \in M_{D_f}(\mathbb{C})$ . The region insertion coefficient is

$$C_A(R, f, M; \sigma, \tau) = \sum_{\substack{\mu, \nu \\ \mu|_{\partial R \setminus \{f\}} = \nu|_{\partial R \setminus \{f\}}}} M_{\mu_f, \nu_f} T_{A,R}^\mu(\sigma) T_{A,V \setminus R}^{\nu^c}(\tau).$$

It is the coefficient obtained by inserting  $M$  on the boundary bond  $f$  between the region  $R$  and its complement.

**Theorem 15.145** (Region insertion as a two-block insertion). [Lean](#) [✓ Stmt](#) *The abstract two-block insertion coefficient of the pair  $(B_{A,R}, B_{A,R^c})$  is the region insertion coefficient:*

$$C_{B_{A,R}, B_{A,R^c}}(f, M; *, *, \sigma, \tau) = C_A(R, f, M; \sigma, \tau).$$

*Proof.* [✓ Proof](#) Expanding the abstract two-block coefficient gives exactly the two boundary sums in the displayed formula for  $C_A(R, f, M; \sigma, \tau)$ .  $\square$

**Theorem 15.146** (Region insertion under bond-dimension transport). [Lean](#) [✓ Stmt](#) *If a tensor  $B$  is transported along an equality of bond dimensions, then the region insertion coefficient is transported by the induced matrix-algebra identification on the inserted bond:*

$$C_{B^h}(R, f, N; \sigma, \tau) = C_B(R, f, \phi_f(N); \sigma, \tau).$$

*Proof.* [✓ Proof](#) Reindex both boundary-configuration sums by the bond-dimension equivalences. The constraint away from  $f$  and the two blocked-region weights are preserved, while the inserted matrix entry becomes the transported entry  $\phi_f(N)$ .  $\square$

**Theorem 15.147** (Same two-block insertions from region insertions). [Lean](#) [✓ Stmt](#) *Suppose  $A$  and  $B$  have identified bond dimensions. Write  $B^h$  for the transported tensor, and write  $\phi_f$  for the induced transport on the boundary bond  $f$ . If*

$$C_A(R, f, N; \sigma, \tau) = C_B(R, f, \phi_f(N); \sigma, \tau)$$

*for every boundary edge  $f$ , matrix  $N$ , and physical configurations  $\sigma, \tau$ , then the two pairs*

$$(B_{A,R}, B_{A,R^c}), \quad (B_{B^h,R}, B_{B^h,R^c})$$

*have the same one-bond insertion coefficients after the same bond-dimension transport.*

*Proof.* [✓ Proof](#) Rewrite the two abstract insertion coefficients as region insertion coefficients and use the bond-dimension transport formula:

$$C_{B_{A,R}, B_{A,R^c}}(f, N; *, *, \sigma, \tau) = C_A(R, f, N; \sigma, \tau) = C_B(R, f, \phi_f(N); \sigma, \tau) = C_{B_{B^h,R}, B_{B^h,R^c}}(f, N; *, *, \sigma, \tau).$$

$\square$

**Definition 15.148** (Single-vertex two-block tensor). [Lean](#) [✓ Stmt](#) Fix a vertex  $v$ . The single-vertex tensor  $A_v$  is regarded as a two-block tensor whose shared bonds are the edges incident to  $v$ , whose external boundary is a one-point space, and whose physical index is the physical index at  $v$ :

$$T_v(*, \eta, \sigma) = A_v(\eta, \sigma).$$

**Theorem 15.149** (Injectivity of the single-vertex two-block tensor). [Lean](#) [✓ Stmt](#) *If  $A$  is vertex-injective, then the single-vertex two-block tensor at every vertex is injective.*

*Proof.* [✓ Proof](#) Vertex injectivity is linear independence of the coefficient family  $\eta \mapsto A_v(\eta, -)$ . The auxiliary one-point boundary only reindexes this same family, so the two-block injectivity condition is the same linear independence statement.  $\square$

**Definition 15.150** (Complement two-block tensor). [Lean](#) [✓ Stmt](#) For a vertex  $v$ , the complementary block  $V \setminus \{v\}$  defines a two-block tensor on the same incident shared bonds. Its coefficient is the contraction of all tensors away from  $v$ , with the incident bond indices left open at the complement endpoints:

$$T_{v^c}(*, \eta, \tau) = A_{v^c}(\eta, \tau).$$

**Theorem 15.151** (Injectivity of the vertex-complement contraction). [Lean](#) [✓ Stmt](#) *If  $A$  is vertex-injective and every virtual bond space has positive dimension, then the coefficient family*

$$\eta \mapsto (\tau \mapsto A_{v^c}(\eta, \tau))$$

*of the vertex-complement contraction is linearly independent.*

*Proof.* ✓ Proof Let  $c = (c_\eta)_\eta$  be a finite coefficient family on the open star of  $v$ . For a set  $S \subseteq V \setminus \{v\}$ , let  $K_S(c)$  be the assertion that for every virtual configuration  $\zeta_0$  and physical configuration  $\tau$ ,

$$\sum_{\zeta} \mathbf{1}[\zeta_f = \zeta_{0,f} \text{ for every edge } f \text{ whose endpoints are outside } S] c_{\zeta|_{\text{Star}(v)}} \prod_{w \in S} A_w(\zeta|_{\text{Star}(w)}, \tau_w) = 0.$$

The initial vanishing of the complement contraction is exactly  $K_{V \setminus \{v\}}(c)$ . If  $j \in S$  and  $j \neq v$ , the one-sided inverse at the vertex  $j$  separates the local tensor  $A_j$  and gives  $K_S(c) \Rightarrow K_{S \setminus \{j\}}(c)$ . Iterating this implication over the complement vertices gives  $K_\emptyset(c)$ . At  $S = \emptyset$  the exposed virtual indices are all fixed by  $\zeta_0$ ; positive bond dimensions allow  $\zeta_0$  to realize any prescribed star configuration at  $v$ . Hence every coefficient  $c_\eta$  is zero.  $\square$

**Theorem 15.152** (Injectivity of the complement two-block tensor). Lean ✓ Stmt *If  $A$  is vertex-injective and every virtual bond space has positive dimension, then the complement two-block tensor at every vertex is injective.*

*Proof.* ✓ Proof The previous theorem gives linear independence of the coefficient family  $\eta \mapsto A_{v^c}(\eta, -)$ . The two-block tensor only adds a one-point external boundary, so this is the same injectivity condition after the harmless reindexing  $(*, \eta) \leftrightarrow \eta$ .  $\square$

**Theorem 15.153** (One-vertex versus complement comparison). Lean ✓ Stmt *Let the shared bond type and the two external boundary spaces be nonempty. After the edge gauges have been absorbed into the second tensor family  $\tilde{B}$ , block one vertex  $v$  against its complement. Assume that  $A_v, A_{v^c}, \tilde{B}_v, \tilde{B}_{v^c}$  are injective as two-block tensors. The post-absorption insertion equality gives, for every shared bond  $b$ , every matrix  $X \in M_{D_b}(\mathbb{C})$ , and all external and physical indices,*

$$C_{A_v, A_{v^c}}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}) = C_{\tilde{B}_v, \tilde{B}_{v^c}}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}).$$

Then

$$\exists \lambda_v \in \mathbb{C}^\times, \quad A_v = \lambda_v \tilde{B}_v.$$

This is the final local comparison following the post-absorption insertion equality in [MGRPG<sup>+</sup>18, Section 3].

*Proof.* ✓ Proof With  $A_1 = A_v$ ,  $A_2 = A_{v^c}$ ,  $B_1 = \tilde{B}_v$ , and  $B_2 = \tilde{B}_{v^c}$ , the post-absorption equality is precisely

$$\forall b, X, \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}, \quad C_{A_1, A_2}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}) = C_{B_1, B_2}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}).$$

Theorem 15.139 gives

$$\exists \lambda_v \in \mathbb{C}^\times, \quad A_v = \lambda_v \tilde{B}_v, \quad A_{v^c} = \lambda_v^{-1} \tilde{B}_{v^c}.$$

The desired local proportionality is the first equality.  $\square$

**Remark 15.154** (Normal-region form). The normal-region application in [MGRPG<sup>+</sup>18, Section 3] uses the same scalar comparison after comparing two injective regions  $R \subset S = R \cup \{v\}$ . If the region comparisons give  $A_R = \lambda_R \tilde{B}_R$  and  $A_S = \lambda_S \tilde{B}_S$ , then the source-paper diagram reads

$$A_R A_v = A_S = \lambda_S \tilde{B}_S = \lambda_S \tilde{B}_R \tilde{B}_v.$$

Applying the left inverse supplied by injectivity of the block  $A_R = \lambda_R \tilde{B}_R$  gives  $A_v = (\lambda_S / \lambda_R) \tilde{B}_v$ .



and hence the union of the three inside regions, is disjoint from the outside region. In indexed form, the pieces forming  $A$  are disjoint from the pieces forming  $A^c$ , and the pieces forming  $B$  are disjoint from the pieces forming  $B^c$ , both as explicit binary unions and as finite unions over the corresponding selected indexed subfamilies. The indexed four-region family is pairwise disjoint.

*Proof.* ✓ Proof Use the standard disjointness identities for set difference and intersection, together with the reconstruction of  $A \cup B$ . □

**Definition 15.160** (Contraction maps for the union injectivity proof). Lean ✓ Stmt Let  $K$ ,  $E_{012}$ ,  $E_2$ , and  $E_{12}$  be complex vector spaces. A contraction-map family for the proof of [MGRPG<sup>+</sup>18, Section 3] consists of three linear maps  $\mathcal{C}_{\{0,1,2\}} : K \rightarrow E_{012}$ ,  $\mathcal{C}_{\{2\}} : K \rightarrow E_2$ , and  $\mathcal{C}_{\{1,2\}} : K \rightarrow E_{12}$ .

**Definition 15.161** (Contraction chain for the union injectivity proof). Lean ✓ Stmt A contraction chain consists of the three contraction maps of Definition 15.160, together with the three implications used in Lemma 15.162.

**Lemma 15.162** (Kernel criterion for the union contraction chain). Lean Lean ✓ Stmt Let  $P_0 = A \setminus B$ ,  $P_1 = A \cap B$ ,  $P_2 = B \setminus A$ , and  $P_3 = (A \cup B)^c$ . Let  $K$  be the boundary vector space attached to  $P_3$ , and let  $\mathcal{C}_{\{0,1,2\}} : K \rightarrow E_{012}$ ,  $\mathcal{C}_{\{2\}} : K \rightarrow E_2$ , and  $\mathcal{C}_{\{1,2\}} : K \rightarrow E_{12}$  be the three contraction maps in the proof of [MGRPG<sup>+</sup>18, Section 3]. If

$$\mathcal{C}_{\{0,1,2\}}(X) = 0 \implies \mathcal{C}_{\{2\}}(X) = 0, \quad \mathcal{C}_{\{2\}}(X) = 0 \implies \mathcal{C}_{\{1,2\}}(X) = 0,$$

and  $\mathcal{C}_{\{1,2\}}(X) = 0 \implies X = 0$ , then  $\ker \mathcal{C}_{\{0,1,2\}} = \{0\}$ .

*Proof.* ✓ Proof For  $X \in K$  with  $\mathcal{C}_{\{0,1,2\}}(X) = 0$ , apply the three implications in sequence:

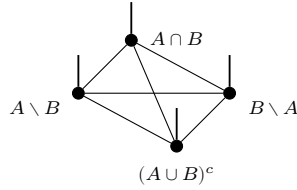
$$\mathcal{C}_{\{0,1,2\}}(X) = 0 \implies \mathcal{C}_{\{2\}}(X) = 0 \implies \mathcal{C}_{\{1,2\}}(X) = 0 \implies X = 0.$$

□

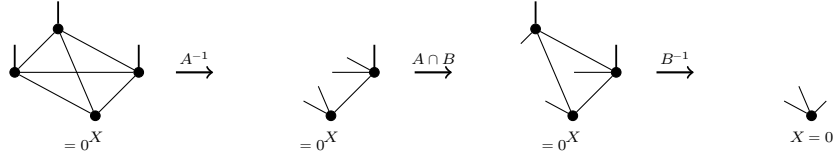
**Lemma 15.163** (Union of injective regions). Not ready Let  $V$  be a finite vertex set. Write  $\text{inj}(R)$  for the assertion that the PEPS tensor blocked to the region  $R \subseteq V$  is injective. For two possibly overlapping regions  $A, B \subseteq V$ ,

$$\text{inj}(A) \wedge \text{inj}(B) \implies \text{inj}(A \cup B).$$

The proof in [MGRPG<sup>+</sup>18, Section 3] blocks the network into four parts  $A \setminus B$ ,  $A \cap B$ ,  $B \setminus A$ , and  $(A \cup B)^c$ , and applies the two available left inverses in sequence:



Taking a boundary tensor  $X$  on  $(A \cup B)^c$  with  $\mathcal{C}_{\{0,1,2\}}(X) = 0$ , injectivity of  $A$  gives  $\mathcal{C}_{\{2\}}(X) = 0$ ; reinserting  $A \cap B$  and applying injectivity of  $B$  then gives  $X = 0$ :



*Proof.* Put  $P_0 = A \setminus B$ ,  $P_1 = A \cap B$ ,  $P_2 = B \setminus A$ , and  $P_3 = (A \cup B)^c$ . Then  $P_0 \cup P_1 = A$ ,  $P_1 \cup P_2 = B$ , and  $P_0 \cup P_1 \cup P_2 = A \cup B$ . Let  $\mathcal{C}_I$  denote contraction of the blocked tensor over the region  $\bigcup_{i \in I} P_i$ . To prove injectivity of  $A \cup B$ , take a boundary tensor  $X$  on  $P_3$  such that  $\mathcal{C}_{\{0,1,2\}}(X) = 0$ . Since  $P_0 \cup P_1 = A$  is injective, its left inverse gives

$$\mathcal{C}_{\{0,1,2\}}(X) = 0 \implies \mathcal{C}_{\{2\}}(X) = 0.$$

The remaining two steps are

$$\begin{aligned} \mathcal{C}_{\{2\}}(X) = 0 &\implies \mathcal{C}_{\{1,2\}}(X) = 0, \\ \mathcal{C}_{\{1,2\}}(X) = 0 &\implies X = 0. \end{aligned}$$

The first implication is obtained by contracting with the tensor on  $P_1$ , and the second is the left inverse for the injective region  $P_1 \cup P_2 = B$ . Thus  $\ker \mathcal{C}_{\{0,1,2\}} = \{0\}$ , and therefore  $P_0 \cup P_1 \cup P_2 = A \cup B$  is injective.  $\square$

**Definition 15.164** (Injectivity predicate for blocked regions). [Lean](#) [Lean](#) [✓ Stmt](#) A region-injectivity hypothesis assigns to each finite vertex region the assertion that the tensor obtained by blocking that region is injective. This predicate is used for regions such as  $R$ ,  $S$ ,  $T$ , and their complements in the normal PEPS proof. The union-closure hypothesis records the conclusion supplied by the source union-of-injective-regions lemma: the union of two injective regions is injective.

**Definition 15.165** (Concrete region-injectivity predicate of a tensor). [Lean](#) [✓ Stmt](#) The region-injectivity predicate of a fixed PEPS tensor  $A$  sends a finite region  $R$  to the assertion that the blocked-region tensor of  $R$  is injective.

**Theorem 15.166** (Kernel descent for blocked regions). [Lean](#) [✓ Stmt](#) Let  $A$  be vertex-injective, and assume that every virtual bond dimension is positive. Then every finite region  $R$  has an injective blocked tensor:

$$\text{inj}(R).$$

The positive-bond restriction is recorded in `docs/paper-gaps/peps_injective_ft_section3_route.tex`.

*Proof.* [✓ Proof](#) Let  $(c_\rho)_\rho$  be a finitely supported coefficient family on the boundary configurations of  $R$ , and assume

$$\sum_\rho c_\rho T_{A,R}^\rho(\tau) = 0 \quad \text{for every physical configuration } \tau.$$

The kernel descent removes the vertices of  $R$  one at a time and preserves the vanishing contraction. After all vertices have been removed, the terminal contraction gives

$$c_\rho = 0 \quad \text{for every boundary configuration } \rho.$$

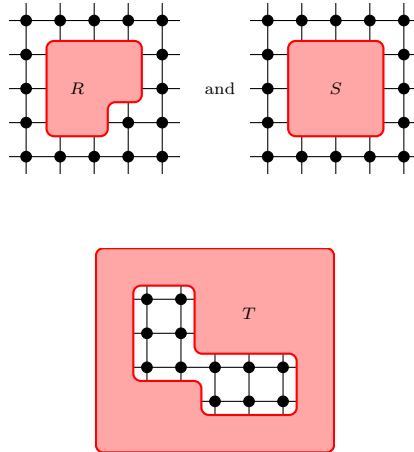
Hence the family  $\{T_{A,R}^\rho\}_\rho$  is linearly independent, which is  $\text{inj}(R)$ .  $\square$

**Theorem 15.167** (Union closure for a vertex-injective tensor). [Lean](#) [✓ Stmt](#) Let  $A$  be vertex-injective with every bond dimension positive. Then the concrete region-injectivity predicate of  $A$  satisfies union closure: for any two finite regions  $R$  and  $S$ ,

$$\text{inj}(R) \wedge \text{inj}(S) \implies \text{inj}(R \cup S).$$

The positive-bond restriction is recorded in `docs/paper-gaps/peps_injective_ft_section3_route.tex`.





**Lemma 15.172** (Basic identities for square-lattice coordinate regions). [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [StmT](#)

Membership in a coordinate rectangle is coordinatewise membership, the contiguous coordinate interval has the expected coordinate inequalities, membership in a contiguous coordinate rectangle is the conjunction of the four bounding inequalities, and membership in  $R$  is the disjunction of membership in the  $2 \times 3$  lower-left piece or in the shifted  $3 \times 2$  upper piece. Membership in  $S$  is the disjunction of membership in the left  $2 \times 3$  piece or in the horizontally shifted  $2 \times 3$  piece. Membership in the local-window  $T$  region is membership in the local  $5 \times 6$  window together with nonmembership in the union of the two displayed edge blocks. Bounded contiguous coordinate rectangles of sizes  $2 \times 3$  and  $3 \times 2$  satisfy, respectively, the predicates for contiguous  $2 \times 3$  and contiguous  $3 \times 2$  square-lattice rectangles. The cardinality of a coordinate rectangle is the product of the two coordinate cardinalities, and the full coordinate rectangle is the whole finite vertex set.

*Proof.* ✓ Proof The coordinate-rectangle statements follow from membership and cardinality identities for Cartesian products of finite coordinate sets. The contiguous-rectangle statement then unfolds the two coordinate intervals. The membership statements for  $R$ ,  $S$ , and the two edge blocks removed from  $T$  additionally unfold membership in a union of two such rectangles, while membership in  $T$  also unfolds the set difference from the local  $5 \times 6$  window. The shape predicates follow by using the displayed lower-left corners as witnesses.  $\square$

**Lemma 15.173** (The normal regions  $R$  and  $S$  are unions of contiguous rectangles). [Lean](#) [Lean](#)

 **Stmnt** *If the displayed  $3 \times 3$  window containing  $R$  and  $S$  lies in the finite square lattice, then  $R$  is the union of a contiguous  $2 \times 3$  rectangle and a contiguous  $3 \times 2$  rectangle, while  $S$  is the union of two contiguous  $2 \times 3$  rectangles.*

*Proof.* ✔ **Proof** This is the defining description of  $R$  and  $S$ , together with the coordinate bounds which make the displayed rectangles contiguous  $2 \times 3$  and  $3 \times 2$  rectangles.  $\square$

**Lemma 15.174** (The edge blocks removed from  $T$  are contiguous rectangles). [Lean](#) [Lean](#)  
[Lean](#) [Stmnt](#) *If the two edge blocks cut out of the displayed  $T$ -region lie in the finite square lattice, then they are respectively a contiguous  $2 \times 3$  rectangle and a contiguous  $3 \times 2$  rectangle.*

*Proof.* ✓ **Proof** This is the defining description of the two removed edge blocks, together with the coordinate bounds which make them valid contiguous rectangles.  $\square$



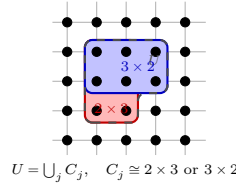
**Lemma 15.175** (The local  $T$ -region and its removed edge blocks fill the window). [Lean](#) [Lean](#)  
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The two edge blocks removed from the displayed local  $T$ -region lie inside the local  $5 \times 6$  coordinate window. They are disjoint from each other and from this local  $T$  region, and their union with it is exactly that local window.*

*Proof.* ✔ Proof This follows by unfolding the definition of  $T$  as the set-theoretic complement of the two displayed edge blocks inside the local window.  $\square$

**Lemma 15.176** (The edge-complementary block fills the finite lattice). [Lean](#) [Lean](#) [Lean](#) [Lean](#)  
 *Stmnt* The finite-lattice complementary block around the edge is disjoint from the two displayed edge blocks, and its union with them is the full finite square lattice. The local-window  $T$  region is contained in this finite-lattice complementary block.

*Proof.* ✓ Proof This is the set-theoretic complement of the two displayed edge blocks in the full finite square lattice. The containment of the local-window  $T$  region follows because its points also avoid the two removed edge blocks.  $\square$

**Definition 15.177** (Rectangular covers of square-lattice regions). [Lean](#) [Lean](#) [Lean](#) [Lean](#)  
[Lean](#) [✓ Stmt](#) A rectangular cover of a square-lattice region is a nonempty finite family of regions whose union is exactly the given region, and each member is a contiguous  $2 \times 3$  or  $3 \times 2$  rectangle. The two cases used in the proof are the cover of the local displayed  $T$ -region and the cover of the finite-lattice complementary block  $A_3$ .



**Lemma 15.178** (The present local  $T$ -window and  $A_3$  cover obstructions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The present origin-based local-window model for the displayed  $T$ -region admits no nonempty finite cover by contained contiguous  $2 \times 3$  and  $3 \times 2$  rectangles whenever the ambient lattice contains the  $5 \times 6$  window. The origin-based rotated local-window model used for vertical edges has the analogous obstruction whenever the ambient lattice contains the  $6 \times 5$  window. The normalized  $5 \times 6$  and  $6 \times 5$  cases are recorded as corollaries. The present origin-based horizontal edge-complement model has the analogous obstruction whenever the ambient lattice contains the  $5 \times 7$  frame; the normalized  $5 \times 7$  case is again recorded as a corollary. The present origin-based vertical edge-complement model has the corresponding obstruction whenever the ambient lattice contains the  $7 \times 5$  frame; the normalized  $7 \times 5$  case is recorded as a corollary.*

*Proof.* ✓ Proof The lower-left point of the local window belongs to the present  $T$  model. Any contiguous  $2 \times 3$  rectangle containing it also contains a point of the removed vertical edge block, while any contiguous  $3 \times 2$  rectangle containing it also contains a point of the removed horizontal edge block. The rotated statement is the same coordinate argument with the two shapes interchanged. The horizontal edge-complement statement uses the same lower-left point, and so does the normalized vertical edge-complement statement. These cases are all instances of the same point-forcing obstruction to an exact rectangular cover.  $\square$

**Lemma 15.179** (Injectivity from a rectangular cover). [Lean](#) [✓ Stmt](#) *If a square-lattice region has a nonempty finite cover by contiguous  $2 \times 3$  and  $3 \times 2$  rectangles, then rectangular injectivity and finite union closure imply that the region is injective.*

*Proof.* [✓ Proof](#) Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family.  $\square$

**Lemma 15.180** (The normal regions  $R$  and  $S$  differ in one site). [Lean](#) [Lean](#) [✓ Stmt](#) *For the displayed regions  $R$  and  $S$  with the same lower-left corner,  $R$  is contained in  $S$ . The difference  $S \setminus R$  is exactly the lower-right site of the  $3 \times 3$  square: in coordinates, the site one obtains by shifting two steps horizontally and zero steps vertically from the lower-left corner.*

*Proof.* [✓ Proof](#) Expand the two displayed unions. The only point of the second  $2 \times 3$  piece of  $S$  which is not already in the union defining  $R$  is its lower-right point.  $\square$

**Definition 15.181** (Normal rectangular injectivity hypotheses). [Lean](#) [Lean](#) [✓ Stmt](#) In the square-lattice normal theorem, every  $2 \times 3$  and  $3 \times 2$  rectangular region is assumed injective. The corresponding abstract hypotheses specify which finite regions have these two shapes and assert injectivity for all of them. In the coordinate square-lattice specialization, these two families are the contiguous coordinate  $2 \times 3$  and  $3 \times 2$  rectangles.

**Lemma 15.182** (Coordinate rectangular injectivity gives abstract rectangular injectivity). [Lean](#) [✓ Stmt](#) *The coordinate square-lattice rectangular injectivity hypotheses determine abstract rectangular injectivity hypotheses, with the two abstract rectangular families taken to be the contiguous coordinate  $2 \times 3$  and  $3 \times 2$  rectangles.*

*Proof.* [✓ Proof](#) This is the direct specialization of the abstract rectangular families to the two contiguous coordinate-rectangle predicates.  $\square$

**Lemma 15.183** (Coordinate rectangular injectivity gives injective rectangles). [Lean](#) [Lean](#) [✓ Stmt](#) *Under the coordinate square-lattice rectangular injectivity hypotheses, every bounded contiguous  $2 \times 3$  rectangle and every bounded contiguous  $3 \times 2$  rectangle is injective.*

*Proof.* [✓ Proof](#) This is the defining rectangular injectivity hypothesis applied to the coordinate witnesses for the two rectangle shapes.  $\square$

**Lemma 15.184** (Rectangular injectivity gives injective  $T$ -edge blocks). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For the displayed local  $T$ -region inside its  $5 \times 6$  coordinate window, if the smaller edge-block bounds  $x_0 + 5 \leq n$  and  $y_0 + 5 \leq m$  hold in the finite  $n \times m$  square lattice, then the coordinate square-lattice rectangular injectivity hypotheses imply that the vertical  $2 \times 3$  block and horizontal  $3 \times 2$  block removed from the window are injective. If, in addition, region injectivity is closed under unions, then the union of these two removed edge blocks is injective.*

*Proof.* [✓ Proof](#) Apply the rectangular injectivity hypotheses to the already identified contiguous rectangle shapes of the two edge blocks. The final assertion is one application of union closure to these two injective blocks.  $\square$

**Lemma 15.185** (Injectivity from a rectangular cover of the local  $T$ -region). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *If the displayed local  $T$ -region has a nonempty finite cover by contiguous  $2 \times 3$  and  $3 \times 2$  rectangles, then rectangular injectivity and finite union closure imply that  $T$  is injective. The rotated vertical local  $T$ -region has the same conditional injectivity criterion.*

*Proof.* [✓ Proof](#) Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family.  $\square$

**Lemma 15.186** (Injectivity from a rectangular cover of the edge complement). [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) *If the finite-lattice complementary block  $A_3$  has a nonempty finite cover by contiguous  $2 \times 3$  and  $3 \times 2$  rectangles, then rectangular injectivity and finite union closure imply that  $A_3$  is injective. The same criterion applies to every vertical edge-complementary block: a nonempty finite cover of that block by contiguous  $2 \times 3$  and  $3 \times 2$  rectangles implies injectivity of the vertical edge-complementary block. In the normalized horizontal-edge  $5 \times 7$  frame, it is enough to give such a cover of the local  $T$ -region, since adjoining the two top-collar  $3 \times 2$  rectangles gives a cover of  $A_3$ .*

*Proof.* [✓ Proof](#) Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family. For the normalized  $5 \times 7$  edge complement, first extend the local  $T$ -cover by the two top-collar rectangles.  $\square$

**Lemma 15.187** (A cover of  $T$  extends to a cover of the edge complement). [Lean](#) [Lean](#) [✓ Stmt](#) *In the normalized horizontal-edge  $5 \times 7$  frame with lower-left corner  $(0, 0)$ , any rectangular cover of the local  $T$ -region extends to a rectangular cover of the edge-complementary block  $A_3$  by adjoining the two top-collar contiguous  $3 \times 2$  rectangles. In the normalized vertical-edge  $7 \times 5$  frame with lower-left corner  $(0, 0)$ , any rectangular cover of the rotated local  $T$ -region extends to a rectangular cover of the vertical edge-complementary block by adjoining the two right-collar contiguous  $2 \times 3$  rectangles.*

*Proof.* [✓ Proof](#) Index the new cover by the old cover indices together with two extra indices for the collar rectangles. For the horizontal block, the decomposition is

$$A_3 = T \cup (\{0, 1, 2\} \times \{5, 6\}) \cup (\{2, 3, 4\} \times \{5, 6\}).$$

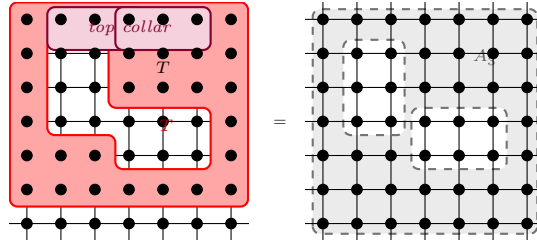
For the vertical block, the decomposition is

$$A_3^v = T^v \cup (\{5, 6\} \times \{0, 1, 2\}) \cup (\{5, 6\} \times \{2, 3, 4\}).$$

$\square$

**Lemma 15.188** (The normalized edge complements have collars). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *In the normalized horizontal-edge  $5 \times 7$  frame, the finite-lattice complementary block  $A_3$  is the local-window  $T$ -region together with two top-collar contiguous  $3 \times 2$  rectangles. Consequently, if the local  $T$ -region is injective, then rectangular injectivity and union closure imply that  $A_3$  is injective. There is also a transposed  $7 \times 5$  horizontal-frame collar identity with two contiguous  $2 \times 3$  right-collar rectangles. For the rotated vertical edge-blocking regions, the corresponding complement is the rotated local  $T$ -region together with the same two right-collar rectangles; hence rotated  $T$ -injectivity, and hence a rectangular cover of rotated  $T$ , implies injectivity of the vertical complementary block.*



*Proof.* [✓ Proof](#) The two set identities follow by expanding the coordinate inequalities. The injectivity statements apply union closure first to the local  $T$ -region and the first collar rectangle, and then to the second collar rectangle.  $\square$





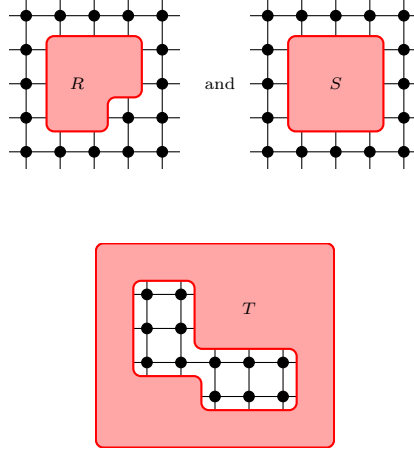
$$u_e \in R_e, \quad v_e \in B_e, \quad R_e \cap B_e = R_e \cap C_e = B_e \cap C_e = \emptyset, \quad R_e \cup B_e \cup C_e = V,$$

*Proof.* ✓ Proof By construction, the one-edge datum at each edge  $e$  is  $d_e$ . The displayed endpoint, disjointness, coverage, and injectivity statements are exactly the corresponding assumptions of that one-edge datum.  $\square$

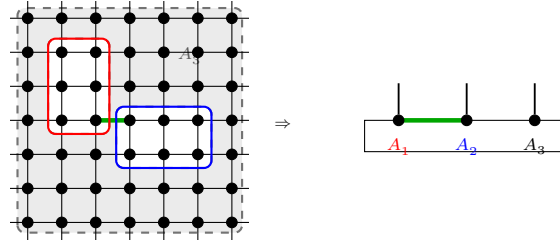
*Proof.* ✓ Proof If  $w = v$ , use the distinguished-site membership and nonmembership assumptions. If  $w \neq v$ , use the equality of the two regions away from the distinguished site.  $\square$

197

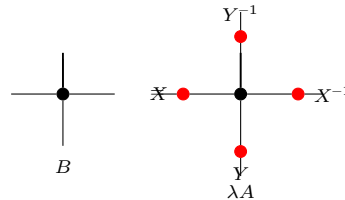
with  $|V| = nm$ , and three finite regions  $R$ ,  $S$ , and  $T$  whose injectivity is assumed directly. It does not by itself assert that these regions are the displayed square-lattice unions, nor that their translated variants give the edge-centred blocking around every edge. Conversely, for the coordinate square lattice with  $n, m \geq 7$ , the rectangular hypotheses, union closure, and a rectangular cover of the displayed  $T$ -region yield this abstract  $R, S, T$  structure, with  $R$  and  $S$  obtained from their rectangular decompositions and  $T$  obtained from the supplied cover. The source regions are:



**Theorem 15.200** (Normal edge blocking to a three-site injective chain). Not ready Under the square-lattice normality hypotheses of [MGRPG<sup>+</sup>18, Theorem 3], the regions  $R$ ,  $S$ , and  $T$ , together with their translated variants, give an edge-centred blocking into red, blue, and complementary injective regions. These are the regions on which the three-site injective-chain insertion argument is applied:



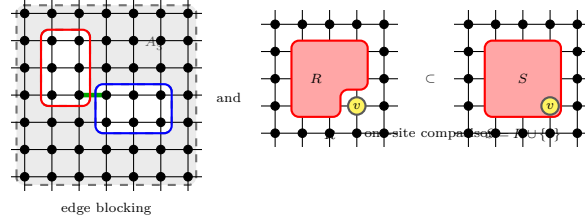
**Theorem 15.201** (Translationally invariant normal gauge absorption). Not ready In the translationally invariant normal square-lattice case, the edge gauges obtained after blocking are represented by one horizontal matrix  $X$  and one vertical matrix  $Y$ , unique up to scalar. Absorbing these matrices into  $B$  gives a tensor  $\tilde{B}$ , equivalently the final local gauge formula of [MGRPG<sup>+</sup>18, Theorem 3]:





**Theorem 15.202** (Fundamental Theorem for normal square-lattice PEPS). Not ready Let  $A$  and  $B$  be translationally invariant normal square-lattice PEPS tensors such that every  $2 \times 3$  and  $3 \times 2$  region is injective. If they generate the same state on an  $n \times m$  region with  $n, m \geq 7$ , then they are related by horizontal and vertical gauges  $X, Y$ , up to a scalar  $\lambda$  satisfying  $\lambda^{nm} = 1$ . This is [MGRPG<sup>+</sup> 18, Theorem 3].

**Definition 15.203** (Normal PEPS blocking hypotheses). Lean Lean Lean Lean Lean ✓ Stmt  
The general normal theorem assumes both the edge-centred blockings into three injective regions and the one-site comparison regions with injective complements. These two assumptions are collected as the normal PEPS blocking hypotheses; the same hypotheses give the corresponding edge and one-site consequences without restating the separate edge and one-site assumptions.



**Theorem 15.204** (Square-lattice normal blocking hypotheses from margin covers). Lean Lean Lean Lean Lean Lean ✓ Stmt Given rectangular injectivity hypotheses, finite union closure, oriented margin-and-cover hypotheses at every edge, and one-site separation, the finite square lattice satisfies the general normal blocking hypotheses. The edge-blocking hypotheses are exactly the margin-cover construction, the one-edge datum at an edge is the datum supplied there, and the one-site separation is exactly the supplied one-site separation. Hence these hypotheses give the three injective regions, the endpoint-disjoint-cover facts at each edge, and the one-site membership formula as consequences of the same normal blocking hypotheses.

*Proof.* ✓ Proof Combine the edge-blocking hypotheses constructed from the margin-cover hypotheses with the given one-site separation hypotheses. □

**Theorem 15.205** (Fundamental Theorem for normal PEPS). Not ready Suppose two normal PEPS generate the same state, can be blocked into three-site injective chains around every edge, and for every site admit two injective regions with injective complements that differ exactly at that site. Then their defining tensors are related by a local gauge, and the gauges are unique up to the scalar freedom allowed by the graph. This is the general normal-PEPS theorem stated after [MGRPG<sup>+</sup> 18, Theorem 3].

**Theorem 15.206** (Fundamental Theorem for normal PEPS, vertex-injective case). Lean ✓ Stmt  
Suppose PEPS tensors  $A$  and  $B$  are vertex injective with positive bond dimensions on a connected graph, generate the same state, and  $A$  satisfies the normal blocking hypotheses. Then their defining tensors are related by a local gauge. This is the vertex-injective specialization of the general normal-PEPS theorem: when each single tensor is already injective the gauge follows from the injective Fundamental Theorem, so the region blocking hypotheses are not used. The general region-injective theorem, in which only blocked regions are injective, remains open; the missing step is the region-level insertion-algebra isomorphism that produces the per-edge gauge after blocking.

*Proof.* ✓ Proof Vertex injectivity, positive bond dimensions, same state, and connectivity are exactly the hypotheses of the injective Fundamental Theorem, which supplies the local gauge. □

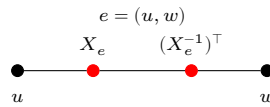


**Definition 15.207** (Oriented endpoint scalar). [Lean](#) [✓ Stmt](#) An edge-scalar family  $c = (c_e)_{e \in E}$  determines the endpoint scalar  $c_{v,e}$  by  $c_{v,e} = c_e$  at the lower endpoint of  $e$  and  $c_{v,e} = c_e^{-1}$  at the upper endpoint.

**Definition 15.208** (Vertex-balanced edge scalars). [Lean](#) [✓ Stmt](#) An edge-scalar family is vertex-balanced when  $\prod_{e \ni v} c_{v,e} = 1$  for every vertex  $v$ .

**Definition 15.209** (Gauge equivalence modulo balanced edge scalars). [Lean](#) [✓ Stmt](#) Two gauge families are equivalent modulo balanced edge scalars if there is a nonzero scalar  $c_e$  on each edge such that the corresponding endpoint actions satisfy  $X_{v,e} = c_{v,e} Y_{v,e}$ , and  $\prod_{e \ni v} c_{v,e} = 1$  for every vertex  $v$ .

**Theorem 15.210** (Multiplication of oriented edge scalars). [Lean](#) [✓ Stmt](#) If two edge-scalar families are multiplied, then their endpoint scalars satisfy  $(cd)_{v,e} = c_{v,e} d_{v,e}$  on every oriented half-edge.



*Proof.* [✓ Proof](#) At the lower endpoint this is ordinary multiplication of the edge scalars. At the upper endpoint,  $(c_e d_e)^{-1} = c_e^{-1} d_e^{-1}$  because the edge scalars lie in  $\mathbb{C}^\times$ .  $\square$

**Theorem 15.211** (Inversion of oriented edge scalars). [Lean](#) [✓ Stmt](#) Inverting an edge-scalar family gives  $(c^{-1})_{v,e} = c_{v,e}^{-1}$  on every oriented endpoint.

*Proof.* [✓ Proof](#) At the lower endpoint,  $(c^{-1})_{v,e} = c_e^{-1} = c_{v,e}^{-1}$ . At the upper endpoint,  $(c^{-1})_{v,e} = (c_e^{-1})^{-1} = c_{v,e}$ .  $\square$

**Theorem 15.212** (Nonzero oriented edge scalars). [Lean](#) [✓ Stmt](#) Every oriented endpoint scalar satisfies  $c_{v,e} \neq 0$ .

*Proof.* [✓ Proof](#) Edge scalars are units, and the inverse of a unit is again a unit.  $\square$

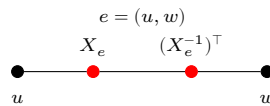
**Theorem 15.213** (Product of balanced edge scalars). [Lean](#) [✓ Stmt](#) The pointwise product of two vertex-balanced edge-scalar families is again vertex-balanced.

*Proof.* [✓ Proof](#) For every vertex  $v$ ,  $\prod_{e \ni v} (cd)_{v,e} = (\prod_{e \ni v} c_{v,e})(\prod_{e \ni v} d_{v,e}) = 1 \cdot 1 = 1$ .  $\square$

**Theorem 15.214** (Inverse of balanced edge scalars). [Lean](#) [✓ Stmt](#) The pointwise inverse of a vertex-balanced edge-scalar family is again vertex-balanced.

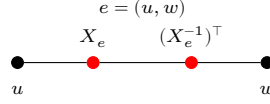
*Proof.* [✓ Proof](#) For every vertex  $v$ ,  $\prod_{e \ni v} (c^{-1})_{v,e} = (\prod_{e \ni v} c_{v,e})^{-1} = 1^{-1} = 1$ .  $\square$

**Theorem 15.215** (Reflexivity of balanced edge-scalar equivalence). [Lean](#) [✓ Stmt](#) Every edge-gauge family is equivalent to itself modulo balanced edge scalars, witnessed by  $c_e = 1$  for every edge  $e$ .



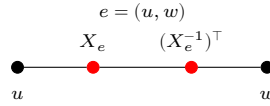
*Proof.* [✓ Proof](#) If  $c_e = 1$ , then  $c_{v,e} = 1$ ,  $X_{v,e} = c_{v,e} X_{v,e}$ , and  $\prod_{e \ni v} c_{v,e} = 1$ .  $\square$

**Theorem 15.216** (Symmetry of balanced edge-scalar equivalence). [Lean](#) [✓ Stmt](#) *If one gauge family differs from another by balanced edge scalars, then the second differs from the first by the inverse balanced edge-scalar family.*



*Proof.* [✓ Proof](#) If  $X_{v,e} = c_{v,e} Y_{v,e}$  with  $c_{v,e} \neq 0$ , then  $Y_{v,e} = c_{v,e}^{-1} X_{v,e}$ . The inverse scalar family is balanced by Theorem 15.214.  $\square$

**Theorem 15.217** (Transitivity of balanced edge-scalar equivalence). [Lean](#) [✓ Stmt](#) *If one gauge family differs from a second by balanced edge scalars, and the second differs from a third by balanced edge scalars, then the first differs from the third by the pointwise product of the two scalar families.*



*Proof.* [✓ Proof](#) From  $X_{v,e} = c_{v,e} Y_{v,e}$  and  $Y_{v,e} = d_{v,e} Z_{v,e}$  one obtains  $X_{v,e} = (c_{v,e} d_{v,e}) Z_{v,e}$ . The product scalar family is balanced by Theorem 15.213.  $\square$

**Theorem 15.218** (The balanced edge-scalar quotient is an equivalence relation). [Lean](#) [✓ Stmt](#) *Gauge equivalence modulo balanced edge scalars is reflexive, symmetric, and transitive.*

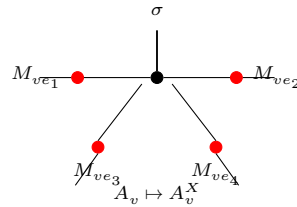
*Proof.* [✓ Proof](#) Reflexivity is witnessed by 1, symmetry by  $c^{-1}$ , and transitivity by  $cd$ .  $\square$

**Definition 15.219** (The balanced edge-scalar quotient relation). [Lean](#) [✓ Stmt](#) The preceding equivalence relation is the quotient relation on edge-gauge families modulo vertex-balanced edge scalars.

**Theorem 15.220** (Balanced edge scalars preserve the local gauge action). [Lean](#) [✓ Stmt](#) *If two gauge families differ by a balanced edge-scalar reweighting, then they produce the same gauged tensor at every vertex.*

*Proof.* [✓ Proof](#) In each summand of the gauged tensor at  $v$ , the edge scalars contribute the factor  $\prod_{e \ni v} c_{v,e} = 1$ . Hence the summand, and therefore the whole local tensor, is unchanged.  $\square$

**Theorem 15.221** (Balanced edge scalars preserve the gauged tensor). [Lean](#) [✓ Stmt](#) *If two gauge families differ by a balanced edge-scalar reweighting, then applying them to the same PEPS tensor produces the same gauged tensor.*



*Proof.* [✓ Proof](#) For every vertex  $v$ , virtual index  $\eta$ , and physical index  $\sigma$ ,  $(A^X)_v(\eta, \sigma) = (A^Y)_v(\eta, \sigma)$  for the two gauged tensors by Theorem 15.220. Hence the two gauged tensors are equal.  $\square$

**Theorem 15.222** (Balanced edge scalars preserve the gauged state). [Lean](#) [✓ Stmt](#) *If two gauge families differ by a balanced edge-scalar reweighting, then the PEPS tensors obtained by applying those gauge families represent the same state.*

*Proof.* [✓ Proof](#) The two gauged tensors are equal, hence all of their state coefficients are equal.  $\square$

**Theorem 15.223** (Gauge uniqueness modulo balanced edge scalars). [Lean](#) [✓ Stmt](#) *Let every bond space be nonzero. If two gauge families relate the same vertex-injective PEPS tensor to the same target tensor, then their edge gauges differ by nonzero scalars  $c_e$  whose oriented product at every vertex is 1. Thus the correct graph quotient is by balanced edge scalars.*

*Proof.* [✓ Proof](#) Combining the two gauge relations gives, at every vertex, equality of the gauged local tensors. Linear independence of the local tensor family promotes this to equality of the products of incident edge-gauge entries over every pair of virtual configurations. Each oriented endpoint gauge is invertible and, since every bond space is nonzero, indexed by a nonempty type; the many-leg scalar extraction then yields, at each vertex, nonzero proportionality constants between the two families whose product is 1. Reading the lower-endpoint constant of each edge defines a single edge scalar  $c_e$ ; the inverse relation transports it to the upper endpoint, and invertibility makes the proportionality constant unique, so the vertex products of the per-vertex constants identify with the oriented products of  $c_e$ . These are all 1, so  $c$  is vertex-balanced.  $\square$

*Remark 15.224.* The local left inverse  $\Psi_v$ , the realization map  $T \mapsto O_T$ , the split at one incident edge, and the partition  $V = \{u\} \sqcup (V \setminus \{u, v\}) \sqcup \{v\}$  isolate the local indices and maps needed for the edge-centered reduction of [\[MGRPG<sup>+</sup>18, Section 3\]](#). The middle-region blocked coefficient is constructed by the preceding definitions, and the local-gauge step is the comparison with the corresponding three-site injective chain. The corrected uniqueness statement is a quotient by balanced edge scalars.

# Chapter 16

## Parent Hamiltonians

This chapter introduces the local ground-space construction and the parent interaction for translation-invariant periodic MPS. We follow the standard setup from [CPGSV21, PGVWC07], with the  $N$ -site Hilbert space identified with functions on the computational basis,

$$\{0, \dots, d-1\}^N \rightarrow \mathbb{C},$$

rather than as an explicit tensor product.

**Definition 16.1** ( $N$ -site Hilbert space). [Lean](#) [✓ Stmt](#) For local dimension  $d$  and chain length  $N$ , the  $N$ -site Hilbert space is

$$\mathcal{H}_N^{(d)} := \{0, \dots, d-1\}^N \rightarrow \mathbb{C}.$$

This is the computational-basis realization used throughout the chapter for periodic-chain states.

### 16.1 Local ground space $G_L(A)$

Fix an MPS tensor  $A = \{A^i\}_{i=0}^{d-1}$  with  $A^i \in M_D(\mathbb{C})$  and a block length  $L \geq 1$ . For a word  $w = (i_1, \dots, i_L)$ , write  $A^w := A^{i_1} \dots A^{i_L}$ . The boundary-condition parametrization is

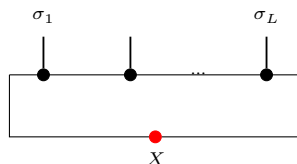
$$\Gamma_L : M_D(\mathbb{C}) \rightarrow (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}), \quad \Gamma_L(X)(\sigma) := \text{tr}(A^\sigma X),$$

where  $A^\sigma$  abbreviates  $A^{\sigma(0)} \dots A^{\sigma(L-1)}$ .

**Definition 16.2** (Ground-space map). [Lean](#) [✓ Stmt](#) The map  $\Gamma_L$  above is a  $\mathbb{C}$ -linear map

$$\Gamma_L : M_D(\mathbb{C}) \rightarrow (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

In tensor-network notation,



the upper chain denotes the word tensor  $A^\sigma$  with open physical indices  $\sigma_1, \dots, \sigma_L$ , and the lower red node denotes the boundary matrix  $X$  inserted on the closing virtual bond before taking the trace. The chain length  $L$  is fixed by the surrounding formula, not by an extra label inside the diagram.

**Lemma 16.3** (Evaluation formula). [Lean](#) [✓ Stmt](#) For every boundary matrix  $X \in M_D(\mathbb{C})$  and every length- $L$  word  $\sigma$ ,

$$\Gamma_L(X)(\sigma) = \text{tr}(A^\sigma X).$$

**Definition 16.4** (Local ground space). [Lean](#) [✓ Stmt](#) The *local ground space* is the image

$$G_L(A) := \text{im}(\Gamma_L) \subseteq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

**Lemma 16.5** (Dimension bound). [Lean](#) [✓ Stmt](#) For every  $L$ ,

$$\dim G_L(A) \leq D^2.$$

*Proof.* [✓ Proof](#) Since  $G_L(A)$  is the range of a linear map from  $M_D(\mathbb{C})$ ,

$$\dim G_L(A) \leq \dim M_D(\mathbb{C}) = D^2.$$

□

**Lemma 16.6** (Ambient-space dimension). [Lean](#) [✓ Stmt](#) The above Hilbert-space realization satisfies

$$\dim(\{0, \dots, d-1\}^L \rightarrow \mathbb{C}) = d^L.$$

**Lemma 16.7** (Nontriviality criterion). [Lean](#) [✓ Stmt](#) If  $d^L > D^2$ , then  $G_L(A)$  is a proper subspace of the local Hilbert space:

$$G_L(A) \neq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

*Proof.* [✓ Proof](#) If  $G_L(A)$  were the whole space, its dimension would be  $d^L$  by Lemma 16.6; Lemma 16.5 gives the contradiction  $d^L \leq D^2$ . □

## 16.2 Parent interaction and chain Hamiltonian

The canonical parent interaction at range  $L$  is

$$h_L(A) = 1 - \Pi_{G_L(A)} = \Pi_{G_L(A)^\perp}, \quad \ker h_L(A) = G_L(A).$$

The sources also allow any positive local interaction with this kernel; the orthogonal projector is the canonical representative used here.

**Definition 16.8** (Hilbert-space copy of the local ground space). [Lean](#) [✓ Stmt](#) Let

$$\iota_L : (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}) \simeq \ell^2(\{0, \dots, d-1\}^L)$$

be the canonical computational-basis identification. The Hilbert-space copy of the local ground space is

$$G_L^{\text{ES}}(A) := \iota_L(G_L(A)) \subseteq \ell^2(\{0, \dots, d-1\}^L).$$

**Lemma 16.9** (Membership in the Hilbert-space copy). [Lean](#) [✓ Stmt](#) For  $v \in \ell^2(\{0, \dots, d-1\}^L)$ ,

$$v \in G_L^{\text{ES}}(A) \iff \iota_L^{-1}v \in G_L(A).$$

*Proof.* ✓ Proof This is the defining equation  $G_L^{\text{ES}}(A) = \iota_L(G_L(A))$ . □

**Definition 16.10** (Canonical parent interaction). Lean ✓ Stmt The parent interaction at range  $L$  is the canonical orthogonal projector

$$h_L(A) := \Pi_{G_L(A)^\perp} = 1 - \Pi_{G_L(A)}, \quad \ker h_L(A) = G_L(A),$$

on the local  $L$ -site space.

**Lemma 16.11** (Parent interaction annihilates ground-space elements). Lean ✓ Stmt For any  $v \in G_L(A)$ , we have  $h_L(A)v = 0$ .

*Proof.* ✓ Proof Since  $h_L(A)$  is the orthogonal projector onto  $G_L(A)^\perp$ , it annihilates every element of  $G_L(A)$ . □

**Definition 16.12** (Restriction to a cyclic interval). Lean ✓ Stmt For  $\sigma \in \{0, \dots, d-1\}^N$  and a starting site  $i$ , write

$$\sigma|_{[i, i+L-1]} \in \{0, \dots, d-1\}^L$$

for the word defined by

$$(\sigma|_{[i, i+L-1]})(r) = \sigma(i + r \bmod N), \quad 0 \leq r < L.$$

**Definition 16.13** (Prescribing a cyclic interval). Lean ✓ Stmt If  $L \leq N$  and  $\tau \in \{0, \dots, d-1\}^L$ , write

$$\sigma^{[i, i+L-1] \leftarrow \tau}$$

for the configuration defined by

$$\sigma^{[i, i+L-1] \leftarrow \tau}(k) = \begin{cases} \tau(r), & k \equiv i + r \pmod{N} \text{ for some } 0 \leq r < L, \\ \sigma(k), & \text{otherwise.} \end{cases}$$

The index  $r$  in the first case is unique because  $L \leq N$ .

**Lemma 16.14** (Extracting a replaced window). Lean ✓ Stmt If  $L \leq N$ , then restricting to the same cyclic interval after prescribing its values recovers the inserted word:

$$(\sigma^{[i, i+L-1] \leftarrow \tau})|_{[i, i+L-1]} = \tau.$$

**Lemma 16.15** (Replacing an extracted window). Lean ✓ Stmt If  $L \leq N$ , then prescribing on a cyclic interval the values already carried by  $\sigma$  leaves  $\sigma$  unchanged:

$$\sigma^{[i, i+L-1] \leftarrow \sigma|_{[i, i+L-1]}} = \sigma.$$

*Proof.* ✓ Proof Case-split on whether the offset  $(k - i + N) \bmod N$  lies in  $[0, L)$ : inside the cyclic interval the prescribed value is the original value of  $\sigma$ ; outside the interval both sides already agree with  $\sigma$ . □

For a periodic chain of length  $N$ , the parent Hamiltonian is the sum of translated copies of the local interaction.

**Definition 16.16** (Translated local term). Lean ✓ Stmt For each site index  $i \in \{0, \dots, N-1\}$ , the translated local term is determined by

$$(h_i \psi)(\sigma) = \begin{cases} [h_L(A)(\tau \mapsto \psi(\sigma^{[i, i+L-1] \leftarrow \tau}))](\sigma|_{[i, i+L-1]}), & L \leq N, \\ 0, & L > N. \end{cases}$$

**Definition 16.17** (Parent Hamiltonian). [Lean](#) [✓ Stmt](#) The chain Hamiltonian is

$$H_N(A, L) := \sum_{i=0}^{N-1} h_i.$$

**Definition 16.18** (Frustration-free condition). [Lean](#) [✓ Stmt](#) A vector  $\psi$  is a frustration-free ground state of the parent Hamiltonian when each local interaction annihilates it:

$$\forall i \in \{0, \dots, N-1\}, \quad h_i \psi = 0.$$

**Lemma 16.19** (MPV window membership). [Lean](#) [✓ Stmt](#) For any window of  $L$  consecutive sites starting at position  $i$  in an  $N$ -site periodic chain ( $L \leq N$ ), the restriction of the MPV state to that window lies in  $G_L(A)$ .

*Proof.* [✓ Proof](#) The boundary matrix is the product of  $A$ -matrices on the complement sites. Trace cyclicity rotates the full  $N$ -site product so that window indices come first, matching the ground-space map definition.  $\square$

**Lemma 16.20** (Local term annihilates the MPV). [Lean](#) [✓ Stmt](#) For each site  $i$ ,  $h_i V^{(N)}(A) = 0$ .

*Proof.* [✓ Proof](#) Combines MPV window membership with the fact that the parent interaction annihilates ground-space elements.  $\square$

**Lemma 16.21** (Parent Hamiltonian annihilates the MPV). [Lean](#) [✓ Stmt](#) For every  $N$ , the MPV state satisfies

$$H_N(A, L) V^{(N)}(A) = 0.$$

*Proof.* [✓ Proof](#) Sum of zeros, since each local term annihilates the MPV.  $\square$

**Lemma 16.22** (Frustration-freeness of the MPV). [Lean](#) [✓ Stmt](#) The MPS vector is a frustration-free ground state: for every site  $i$ ,

$$h_i V^{(N)}(A) = 0.$$

*Proof.* [✓ Proof](#) Immediate from Lemma 16.20.  $\square$

## 16.3 Intersection property and unique ground state

This section develops the intersection property of local ground spaces for injective MPS and the resulting uniqueness of the ground state on the periodic chain.

### 16.3.1 Restriction maps

Given a state  $\psi$  on  $L + 1$  sites, we define two restriction maps.

*Remark 16.23* (Word and restriction identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Appending a last letter to a word corresponds to right multiplication by the corresponding tensor letter, while prepending a first letter corresponds to left multiplication. The restriction maps are the associated linear maps on the Hilbert spaces in the computational-basis realization, together with their pointwise formulas.

**Definition 16.24** (Left restriction). [Lean](#) [✓ Stmt](#) Fix the last physical index  $j$ :

$$\sigma \mapsto \psi(\sigma_1, \dots, \sigma_L, j).$$

**Definition 16.25** (Right restriction). [Lean](#) [✓ Stmt](#) Fix the first physical index  $i$ :

$$\sigma \mapsto \psi(i, \sigma_1, \dots, \sigma_L).$$

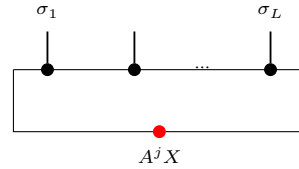
**Definition 16.26** (Left ground membership). [Lean](#) [✓ Stmt](#) A state  $\psi$  on  $L + 1$  sites satisfies the left ground condition if for every  $j$ , the state obtained by fixing the last index to  $j$  lies in  $G_L(A)$ .

**Definition 16.27** (Right ground membership). [Lean](#) [✓ Stmt](#) A state  $\psi$  on  $L + 1$  sites satisfies the right ground condition if for every  $i$ , the state obtained by fixing the first index to  $i$  lies in  $G_L(A)$ .

### 16.3.2 Forward direction: ground space restricts

**Lemma 16.28** (Left restriction). [Lean](#) [✓ Stmt](#) If  $\psi \in G_{L+1}(A)$ , i.e.  $\psi(\sigma) = \text{tr}(A^\sigma X)$  for some  $X$ , then for each  $j$ , the state obtained by fixing the last index to  $j$  lies in  $G_L(A)$  with witness  $A^j X$ .

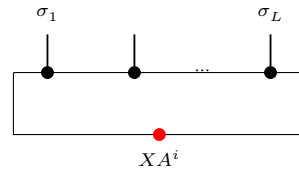
*Proof.* [✓ Proof](#) The restriction simply absorbs the last tensor into the boundary matrix:



Direct computation:  $\psi(\sigma_1, \dots, \sigma_L, j) = \text{tr}(A^{\sigma_1} \dots A^{\sigma_L} \cdot A^j \cdot X) = \Gamma_L(A^j X)(\sigma)$ . □

**Lemma 16.29** (Right restriction). [Lean](#) [✓ Stmt](#) If  $\psi \in G_{L+1}(A)$  with  $\psi(\sigma) = \text{tr}(A^\sigma X)$ , then for each  $i$ , the state obtained by fixing the first index to  $i$  lies in  $G_L(A)$  with witness  $X A^i$ .

*Proof.* [✓ Proof](#) On the right restriction one first rotates the trace and then reads the resulting boundary matrix on the shortened window:



By trace cyclicity:  $\psi(i, \sigma_1, \dots, \sigma_L) = \text{tr}(A^i A^\sigma X) = \text{tr}(A^\sigma \cdot X A^i) = \Gamma_L(X A^i)(\sigma)$ . □

### 16.3.3 Injectivity of the ground-space map

**Theorem 16.30** (Ground-space map injectivity). [Lean](#) [✓ Stmt](#) If  $A$  is injective and  $L \geq 1$ , then  $\Gamma_L$  is injective.

*Proof.* [✓ Proof](#) If  $\Gamma_L(X) = 0$  then  $\text{tr}(A^\sigma X) = 0$  for every length- $L$  word  $\sigma$ . Since  $A$  is injective, the products  $\{A^\sigma\}_\sigma$  span  $M_D(\mathbb{C})$ , so the trace pairing gives  $X = 0$ . □

**Theorem 16.31** (Ground-space map injectivity from word span). [Lean](#) [✓ Stmt](#) If the length- $L$  products  $\{A^\sigma : |\sigma| = L\}$  span  $M_D(\mathbb{C})$ , then  $\Gamma_L$  is injective.

*Proof.* [✓ Proof](#) If  $\Gamma_L(X) = 0$ , then  $\text{tr}(A^\sigma X) = 0$  for every word  $\sigma$  of length  $L$ . The spanning hypothesis turns this into  $\text{tr}(MX) = 0$  for every  $M \in M_D(\mathbb{C})$ , and the trace pairing gives  $X = 0$ . □



**Lemma 16.32** (Ground-space map injectivity after blocking). Lean ✓ Stmt *If the length- $L_0$  products span the full matrix algebra,*

$$\text{span}\{A^\omega : |\omega| = L_0\} = M_D(\mathbb{C}),$$

*then  $\Gamma_{L_0}$  is injective.*

*Proof.* ✓ Proof The hypothesis is precisely the fixed-length spanning condition needed by Theorem 16.31. □

**Theorem 16.33** (Ground-space dimension). Lean ✓ Stmt *If  $A$  is injective and  $L \geq 1$ , then  $\dim G_L(A) = D^2$ .*

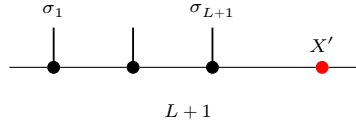
*Proof.* ✓ Proof The upper bound  $\dim G_L(A) \leq D^2$  is Lemma 16.5. Injectivity of  $\Gamma_L$  gives  $\dim G_L(A) = \dim M_D(\mathbb{C}) = D^2$ . □

### 16.3.4 The intersection property

**Theorem 16.34** (One inclusion of the intersection property). Lean ✓ Stmt *If  $A$  is injective and  $L \geq 2$ , then the nontrivial inclusion in the intersection property holds:*

$$G_L^{\text{left}} \cap G_L^{\text{right}} \subseteq G_{L+1}(A).$$

*Proof.* ✓ Proof The “inverting and growing back” argument compares the left and right restrictions on their common  $(L-1)$ -site overlap and then reconstructs the full  $(L+1)$ -site word from a single boundary matrix:



1. From the right ground condition, for each physical index  $i$  there exists a unique  $Y_i \in M_D(\mathbb{C})$  (by injectivity of  $\Gamma_L$ ) such that  $\psi(i, \sigma) = \text{tr}(A^\sigma Y_i)$ .
2. Similarly, from the left ground condition, for each  $j$  there exists a unique  $Z_j$  with  $\psi(\sigma, j) = \text{tr}(A^\sigma Z_j)$ .
3. Matching on the  $(L-1)$ -site overlap and using nondegeneracy of the trace pairing:  $A^j Y_i = Z_j A^i$  for all  $i, j$ .
4. Since  $A$  is injective, the matrices  $\{A^j\}$  span  $M_D(\mathbb{C})$  (Definition 2.28), so  $\mathbb{1} = \sum_j c_j A^j$  for some coefficients  $c_j \in \mathbb{C}$ . Then  $Y_i = \mathbb{1} \cdot Y_i = \sum_j c_j A^j Y_i = \sum_j c_j Z_j A^i = X' A^i$ , where  $X' := \sum_j c_j Z_j$ .
5. By trace cyclicity,  $\psi(\sigma) = \text{tr}(A^\sigma X')$ , so  $\psi \in G_{L+1}(A)$ .

□

**Theorem 16.35** (Intersection property). Lean ✓ Stmt *For injective  $A$  and  $L \geq 2$ :*

$$G_L^{\text{left}} \cap G_L^{\text{right}} = G_{L+1}(A),$$

*equivalently  $\psi \in G_{L+1}(A) \iff \psi \in G_L^{\text{left}} \cap G_L^{\text{right}}$ .*

*Proof.* ✓ Proof Immediate from the two forward-direction lemmas and Theorem 16.34. □

### 16.3.5 Unique ground state

#### Contiguous and cyclic interval restrictions

*Remark 16.36* (Contiguous interval restrictions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) For a non-wrapping interval  $[s, s + M - 1] \subseteq \{0, \dots, N - 1\}$  and a full configuration  $\rho$  supplying the complementary values, the restriction map is

$$(\text{Res}_{s,M}^\rho \psi)(\omega) = \psi(\rho^{[s, s+M-1] \leftarrow \omega}).$$

The listed identities are the equations for  $s = 0$ ,  $M = N$ , for deleting the first or last site, and for first fixing  $K$  sites and then restricting to  $[s + K, s + K + L - 1]$ .

*Remark 16.37* (Cyclic interval restrictions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) On the periodic chain, for a full configuration  $\rho$  supplying the complementary values, put

$$\delta_i(k) := (k + N - i) \bmod N, \quad \rho_{i,L,\omega}^{\text{cyc}}(k) := \begin{cases} \omega(\delta_i(k)), & \delta_i(k) < L, \\ \rho(k), & \delta_i(k) \geq L. \end{cases}$$

The restriction to the cyclic interval beginning at  $i$  is

$$(\text{Res}_{i,L}^\rho \psi)(\omega) = \psi(\rho_{i,L,\omega}^{\text{cyc}}),$$

with all site labels read modulo  $N$ . If  $L \leq N$ , this is the same configuration as  $\rho^{[i, i+L-1] \leftarrow \omega}$ . If two boundary conditions agree whenever  $\delta_i(k) \geq L$ , their restrictions are equal. In particular, filling the interval inside the outside configuration leaves the restriction unchanged:

$$\text{Res}_{i,L}^{\rho_{i,L,\omega}^{\text{cyc}}} \psi = \text{Res}_{i,L}^\rho \psi.$$

For every  $r < L$ , the inserted word is recovered:

$$\rho_{i,L,\omega}^{\text{cyc}}(i + r) = \omega(r).$$

Here site labels are taken modulo  $N$ . The remaining listed identities say that deleting the final site of an  $(L + 1)$ -interval gives the  $L$ -interval beginning at  $i$ ; if  $L + 1 \leq N$ , deleting the first site gives the  $L$ -interval beginning at  $i + 1$ ; and the cyclic formula agrees with the contiguous formula when the interval does not wrap around the ring.

**Lemma 16.38** (One-site restrictions of cyclic-window boundary conditions). [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If a cyclic  $(L + 1)$ -window is represented by a boundary matrix  $Y$ , then deleting one endpoint gives

$$\Gamma_{L+1}(Y)|_{\text{last}=b} = \Gamma_L(A^b Y), \quad \Gamma_{L+1}(Y)|_{\text{first}=a} = \Gamma_L(Y A^a).$$

Hence two adjacent cyclic windows whose restrictions agree on the common length- $L$  overlap have boundary matrices satisfying

$$Y_1 A^a = A^b Y_2.$$

Consequently, for a finite chain of adjacent overlaps indexed by  $0 \leq r < n$ ,

$$Y_r A^{a_r} = A^{b_r} Y_{r+1},$$

one has

$$Y_0 A^{a_0} \dots A^{a_{n-1}} = A^{b_0} \dots A^{b_{n-1}} Y_n.$$

In particular, if the two endpoint words are named by the equations

$$a_r = \rho(i_0 + r), \quad b_r = \rho(i_0 + r + L + 1),$$

with site labels read modulo  $N$ , then the same transport identity is written as

$$Y_0 A^a = A^b Y_n.$$

*Proof.* ✓ Proof The first two identities are the cyclic first- and last-site deletion identities, followed by the corresponding ground-space restriction formulas. For adjacent windows, equality of the completed outside configurations makes the two restricted states on the common overlap equal. Injectivity of the length- $L$  ground-space map then identifies their boundary matrices. Iterating the displayed adjacent identity gives the word-product identity.  $\square$

**Lemma 16.39** (Iterated contiguous-window intersection). Lean ✓ Stmt If an  $N$ -site state satisfies the  $L$ -site ground condition on every non-wrapping contiguous window (with  $L \geq 2$ ), then it lies in  $G_N(A)$ .

*Proof.* ✓ Proof The induction repeatedly merges two neighbouring admissible windows  $[i, i+L-1]$  and  $[i+1, i+L]$  into the larger window  $[i, i+L]$  by applying Theorem 16.34 to their common overlap  $[i+1, i+L-1]$ . Strong induction on window size, combining adjacent windows via Theorem 16.34 at each step.  $\square$

**Definition 16.40** (Suffix restriction). Lean ✓ Stmt Fix a prefix  $u : \{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}$ . The suffix restriction of an  $(K+L)$ -site state  $\psi$  is the  $L$ -site state given by

$$(\text{tailRestrict}_u \psi)(\sigma) = \psi(u, \sigma).$$

**Definition 16.41** (Suffix ground membership). Lean ✓ Stmt A state  $\psi$  on  $K+L$  sites lies in the suffix ground condition if every fixed prefix  $u$  gives a suffix state in  $G_L(A)$ .

**Lemma 16.42** (Ground-space suffix restriction). Lean Lean Lean Lean ✓ Stmt If  $\psi \in G_{K+L}(A)$ , then every fixed-prefix suffix restriction of  $\psi$  lies in  $G_L(A)$ . For a vector in  $G_{L+1}(A)$ , fixing either endpoint gives the expected left or right multiplication of its boundary matrix.

*Proof.* ✓ Proof Write  $\psi(\rho) = \text{tr}(A^\rho X)$  for a boundary matrix  $X$ . Fixing the final site gives the boundary matrix  $A^j X$ , while fixing the first site gives  $X A^i$  by cyclicity of the trace. Fixing a prefix  $u$  gives

$$\psi(u, \sigma) = \text{tr}(A^u A^\sigma X) = \text{tr}(A^\sigma X A^u),$$

so the suffix restriction again has the form of a ground-space vector.  $\square$

**Lemma 16.43** (Left-tail compatibility). Lean ✓ Stmt Assume  $\psi$  on  $K+L_0+1$  sites satisfies the left ground condition on the first  $K+L_0$  sites and the suffix ground condition on every fixed prefix of length  $K$ . Then there exist matrices  $(Z_j)_j$  and  $(Y_u)_u$  such that

$$\psi(\sigma, j) = \text{tr}(A^\sigma Z_j), \quad \psi(u, \tau) = \text{tr}(A^\tau Y_u),$$

and for all  $j$  and  $u$ ,

$$Z_j A^u = A_j Y_u.$$

*Proof.* ✓ Proof Choose  $Z_j$  from the long left-window condition and  $Y_u$  from the suffix ground condition. Restricting the suffix state at its last site gives the boundary matrix  $A_j Y_u$ , while first fixing the last site and then taking the suffix gives  $Z_j A^u$ . The two restrictions are the same because appending the fixed prefix and then the last letter gives the same configuration as appending the last letter to the suffix and then placing the prefix in front. Injectivity of  $\Gamma_{L_0}$ , obtained from  $L_0$ -block injectivity, yields  $Z_j A^u = A_j Y_u$ .  $\square$

**Theorem 16.44** (Inverting step for word compatibility). Lean ✓ Stmt Assume  $A$  is  $L_0$ -block injective with  $L_0 > 0$ . If matrices  $(Z_j)_j$  satisfy

$$Z_j A^\sigma = A_j Y_\sigma$$

for every word  $\sigma$  of length  $K$ , then there exists a matrix  $X$  such that  $Z_j = A_j X$  for all  $j$ .

*Proof.* ✓ Proof For  $K = 1$ , the hypothesis gives  $Z_j A_i = A_j Y_i$  for every letter  $i$ . Every blocked word of length  $L_0$  is nonempty, so this identity extends to the coefficients of the blocked tensor  $A^{[L_0]}$ . Since  $A^{[L_0]}$  is injective, the identity matrix is a linear combination of those blocked coefficients, and substituting this decomposition yields  $Z_j = A_j X$  for a common matrix  $X$ . For larger  $K$ , strip the first letter: for each  $i$ , the matrices  $Z_j A_i$  satisfy the same compatibility hypothesis with words of length  $K - 1$ , so the induction hypothesis reduces the problem to the case  $K = 1$ .  $\square$

**Theorem 16.45** (Growing-back step at the injectivity length). Lean ✓ Stmt Assume  $A$  is  $L_0$ -block injective with  $L_0 > 0$ . If an element  $\psi$  on  $K + L_0 + 1$  sites satisfies the left ground condition on the first  $K + L_0$  sites and the suffix ground condition on every prefix of length  $K$ , then  $\psi \in G_{K+L_0+1}(A)$ .

*Proof.* ✓ Proof Lemma 16.43 produces matrices  $(Z_j)_j$  and  $(Y_u)_u$  with  $Z_j A^u = A_j Y_u$  for every prefix word  $u$  of length  $K$ . Theorem 16.44 gives a common right factor  $X$  such that  $Z_j = A_j X$  for all  $j$ . Therefore each last-site restriction of  $\psi$  agrees with the corresponding last-site restriction of the ground-space vector determined by  $X$ . Since every configuration is obtained by adjoining its final letter to its initial  $(K + L_0)$ -tuple, the two states coincide, so  $\psi$  lies in  $G_{K+L_0+1}(A)$ .  $\square$

**Theorem 16.46** (Open-chain iteration of the intersection property). Lean ✓ Stmt Let  $A$  be  $L_0$ -block injective with  $D \geq 1$  and  $L_0 > 0$ . If an  $N$ -site state satisfies the ground-space constraint on every non-wrapping contiguous interval of length  $L_0 + 1$ , for every fixed choice of the complementary sites, with  $N \geq L_0 + 1$ , then it lies in  $G_N(A)$ .

*Proof.* ✓ Proof Induct on the extra length beyond  $L_0 + 1$ . The induction hypothesis gives the long left-window ground condition after deleting the last site, while Definition 16.40 and the contiguous-interval restriction identities above identify every fixed-prefix suffix with one of the assumed length- $L_0 + 1$  intervals. Theorem 16.45 then grows back the rightmost site.  $\square$

**Lemma 16.47** (Open-chain iteration for a sequence of subspaces). Lean ✓ Stmt Let  $S_m \subseteq (\mathbb{C}^d)^{\otimes m}$  be subspaces, and let  $0 < L \leq N$ . Suppose that every non-wrapping length- $L$  interval of  $\psi$ , for every fixed choice of the complementary sites, lies in  $S_L$ . Suppose also that, for every  $m \geq L$ ,

$$\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}.$$

Then  $\psi \in S_N$ .

*Proof.* ✓ Proof Induct on the length  $m$  of a non-wrapping interval. The case  $m = L$  is the assumed local condition. If all length- $m$  intervals lie in  $S_m$ , then the two one-site restrictions of a length- $m + 1$  interval are adjacent length- $m$  intervals. Hence the length- $m + 1$  interval lies in  $\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d$ , and the displayed identity places it in  $S_{m+1}$ . Taking  $m = N$  and the interval starting at the first site gives  $\psi \in S_N$ .  $\square$

**Lemma 16.48** (Cyclic-window range antitonicity). Lean Lean ✓ Stmt On a nonempty periodic chain, longer cyclic-window constraints imply all shorter cyclic-window constraints: if  $L' \leq L \leq N$ , then  $\mathcal{G}_{N,L}(A) \subseteq \mathcal{G}_{N,L'}(A)$ .

*Proof.* ✓ Proof Fix a cyclic window of length  $L+1$  and restrict its last site. The resulting length- $L$  vector is again in the ground space, so one-site peeling gives the inclusion for consecutive ranges. Iteration gives the general antitonicity.  $\square$

**Theorem 16.49** (Open-chain containment from cyclic local constraints). Lean ✓ Stmt *Let  $A$  be  $L_0$ -block injective with  $L_0 > 0$  and  $D \geq 1$ . If an  $N$ -site periodic-chain state satisfies all cyclic ground-space constraints at some range  $L_0 + 1 \leq L \leq N$ , then it lies in the open-chain ground space  $G_N(A)$ .*

*Proof.* ✓ Proof Lemma 16.48 reduces the cyclic hypotheses to range  $L_0 + 1$ . On every non-wrapping interval, the cyclic and contiguous restriction maps agree, so Theorem 16.46 applies.  $\square$

**Lemma 16.50** (Boundary-crossing compatibility at the last site). Lean ✓ Stmt *Assume  $A$  is  $L_0$ -block injective with  $L_0 > 0$  and  $M \geq L_0$ . If the reduced cyclic interval used in closing the boundary at the last site has boundary matrices  $Y_\tau$ , then for every physical letter  $j$  one has*

$$C_\tau^+ A^j X = Y_\tau A^j, \quad (16.1)$$

where  $C_\tau^+$  is the complementary word seen by that cyclic interval.

*Proof.* ✓ Proof The trace identity for this cyclic interval is tested against all words of length  $L_0$ . Injectivity of  $\Gamma_{L_0}$ , which follows from  $L_0$ -block injectivity, strips the test word and gives the displayed matrix identity (16.1).  $\square$

**Lemma 16.51** (Boundary-crossing compatibility at the opposite boundary position). Lean ✓ Stmt *Assume  $A$  is  $L_0$ -block injective with  $L_0 > 0$  and  $M \geq L_0$ . If the opposite reduced cyclic interval used in closing the boundary has boundary matrices  $Y_\tau$ , then*

$$X A^j C_\tau^- = A^j Y_\tau$$

for every physical letter  $j$ , where  $C_\tau^-$  is the complementary word seen from the opposite cyclic position.

*Proof.* ✓ Proof Factor the cyclic word at the opposite cyclic position, rotate the trace, and use injectivity of  $\Gamma_{L_0}$  to remove the length- $L_0$  test word.  $\square$

**Theorem 16.52** (The two boundary-crossing compatibilities). Lean ✓ Stmt *Let  $A$  be  $L_0$ -block injective with  $L_0 > 0$  and  $D \geq 1$ . Suppose  $\psi \in \mathcal{G}_{N,L}(A)$ ,  $N \geq 2$ ,  $L_0 < L \leq N$ , and  $\psi = \Gamma_N(X)$  for a boundary matrix  $X$ . Then there are two families of boundary matrices  $Y_\tau^+$  and  $Y_\tau^-$ , indexed by fixed complement values  $\tau$ , such that for every physical letter  $j$ ,*

$$C_\tau^+ A^j X = Y_\tau^+ A^j, \quad X A^j C_\tau^- = A^j Y_\tau^-, \quad (16.2)$$

where  $C_\tau^+$  and  $C_\tau^-$  are the complementary words exposed by the two opposite cyclic intervals used in closing the boundary, both of reduced length  $L_0 + 1$ .

*Proof.* ✓ Proof First use Lemma 16.48 to reduce the cyclic constraints from range  $L$  to range  $L_0 + 1$ . Then apply the two cyclic-interval compatibility lemmas at the two boundary positions to obtain the displayed one-sided identities (16.2).  $\square$

**Lemma 16.53** (Boundary-crossing equations for  $\Gamma_{M+1}(X)$ ). Lean ✓ Stmt *Let  $A$  be  $L_0$ -block injective with  $L_0 > 0$ ,  $D \geq 1$ , and  $L_0 \leq M$ . Suppose  $\psi = \Gamma_{M+1}(X)$ , and suppose matrices  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions*

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Then, for every physical letter  $j$  and boundary condition  $\tau$ ,

$$A^{\tau_{L_0}} \dots A^{\tau_{M-1}} A^j X = Y_M(\tau) A^j, \quad X A^j A^{\tau_1} \dots A^{\tau_{M-L_0}} = A^j Y_{M+1-L_0}(\tau). \quad (16.3)$$

*Proof.* [✓ Proof](#) Substitute  $\psi = \Gamma_{M+1}(X)$  in the two cyclic restrictions at the last site and at the opposite boundary-crossing support. Lemmas 16.50 and 16.51 then give the two equations in (16.3).  $\square$

**Lemma 16.54** (One-sided products for a common complement word). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $D \geq 1$ , and  $L_0 \leq M$ . Suppose  $\psi = \Gamma_{M+1}(X)$ , and suppose matrices  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

For every boundary letter  $\eta$ , every physical letter  $j$ , and every complementary word  $\mu$ ,

$$Y_M(\tau_\eta^+(\mu)) A^j = A^\mu A^j X, \quad X A^j A^\mu = A^j Y_{M+1-L_0}(\tau_\eta^-(\mu)).$$

*Proof.* [✓ Proof](#) Substitute  $\tau_\eta^+(\mu)$  and  $\tau_\eta^-(\mu)$  into Lemma 16.53. The exposed-complement identities of Lemma 16.55 identify both complementary products with  $A^\mu$ .  $\square$

**Lemma 16.55** (Common complement word for the two boundary supports). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On a chain of length  $N = M + 1$ , with  $L_0 \leq M$ , fix a physical letter  $\eta$  used outside the complementary sites and a word  $\mu$  of length  $M + 1 - (L_0 + 1)$ . Write  $\tau_\eta^+(\mu)$  for the boundary condition at the support crossing the last site and  $\tau_\eta^-(\mu)$  for the boundary condition at the opposite boundary-crossing support. Their exposed complements are both exactly  $\mu$ .

*Proof.* [✓ Proof](#) Put the same letter  $\eta$  on the remaining sites. Fill the physical sites  $L_0, \dots, M - 1$  with  $\mu$  for the support crossing the last site, and fill the sites  $1, \dots, M - L_0$  with  $\mu$  for the opposite boundary-crossing support. The two complement-extraction identities are then immediate from the index arithmetic. The same outside-site calculation gives, for any  $\rho$  with  $\rho_{k+L_0} = \mu_k$ ,

$$\text{Res}_{M, L_0+1}^\rho \psi = \text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)} \psi,$$

and, for any  $\rho$  with  $\rho_{k+1} = \mu_k$ ,

$$\text{Res}_{M+1-L_0, L_0+1}^\rho \psi = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)} \psi.$$

$\square$

**Lemma 16.56** (Boundary equations from a common complement word). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective, with  $L_0 > 0$ , and let  $L_0 \leq M$ . Suppose two length- $(L_0 + 1)$  restrictions of a state  $\psi$  at the same cyclic support are represented by

$$\text{Res}_{i, L_0+1}^\rho \psi = \Gamma_{L_0+1}(Y_\rho), \quad \text{Res}_{i, L_0+1}^\tau \psi = \Gamma_{L_0+1}(Y_\tau).$$

If these two restrictions are equal, then for every physical letter  $j$ ,

$$Y_\rho A^j = Y_\tau A^j.$$

In particular, the last-site boundary-crossing condition satisfies

$$\rho_{k+L_0} = \mu_k \implies Y_\rho A^j = Y_{\tau_\eta^+(\mu)} A^j,$$

and the opposite boundary-crossing condition satisfies

$$\rho_{k+1} = \mu_k \implies Y_\rho A^j = Y_{\tau_\eta(\mu)} A^j.$$

Consequently, for every word  $\sigma$  of length  $L_0$ , the two displayed implications remain true after right multiplication by  $A^\sigma$ .

*Proof.* ✓ **Proof** Apply the first-letter restriction to both equal length- $(L_0+1)$  restrictions. The two resulting length- $L_0$  vectors are represented by  $\Gamma_{L_0}(Y_\rho A^j)$  and  $\Gamma_{L_0}(Y_{\tau_\eta(\mu)} A^j)$ , and block injectivity makes  $\Gamma_{L_0}$  injective. The two displayed special cases follow from Lemma 16.55. The length- $L_0$  word forms are obtained by multiplying these equations on the right by  $A^\sigma$ .  $\square$

**Lemma 16.57** (Reduction to compatible boundary conditions). Lean ✓ **Stmt** Let  $A$  be  $L_0$ -block-injective, with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M+1$ , and let the matrices  $Y_i(\tau)$  represent its length- $(L_0+1)$  cyclic restrictions:

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a boundary letter  $\eta$  and a complementary word  $\mu$ . Let  $\rho^+$  and  $\rho^-$  be two boundary conditions satisfying

$$\rho_{k+L_0}^+ = \mu_k, \quad \rho_{k+1}^- = \mu_k.$$

If the adjacent-window comparison gives, for every physical letter  $j$  and every word  $\sigma$  of length  $L_0$ ,

$$Y_M(\rho^+) A^j A^\sigma = Y_{M+1-L_0}(\rho^-) A^j A^\sigma,$$

then the same product equation holds for  $\tau_\eta^+(\mu)$  and  $\tau_\eta^-(\mu)$ :

$$Y_M(\tau_\eta^+(\mu)) A^j A^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j A^\sigma.$$

*Proof.* ✓ **Proof** By Lemma 16.56,

$$Y_M(\rho^+) A^j = Y_M(\tau_\eta^+(\mu)) A^j, \quad Y_{M+1-L_0}(\rho^-) A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j.$$

Multiplying on the right by  $A^\sigma$  and applying the hypothesis gives the displayed product equation.  $\square$

**Theorem 16.58** (Pointwise reduction to compatible boundary conditions). Lean ✓ **Stmt** Let  $A$  be  $L_0$ -block-injective, with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M+1$ , and let the matrices  $Y_i(\tau)$  represent its length- $(L_0+1)$  cyclic restrictions:

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a boundary letter  $\eta$  and a complementary word  $\mu$ . Suppose that, for each physical letter  $j$  and word  $\sigma$  of length  $L_0$ , boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  satisfy, for every complementary position  $k$ ,

$$\rho_{j,\sigma}^+(k+L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k+1) = \mu_k.$$

If, for every  $j$  and  $\sigma$ ,

$$Y_M(\rho_{j,\sigma}^+) A^j A^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-) A^j A^\sigma,$$

then, for every  $j$  and  $\sigma$ ,

$$Y_M(\tau_\eta^+(\mu)) A^j A^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j A^\sigma.$$

*Proof.* ✓ **Proof** Fix  $j$  and  $\sigma$ . Lemma 16.56 gives

$$Y_M(\rho_{j,\sigma}^+)A^j = Y_M(\tau_\eta^+(\mu))A^j, \quad Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Multiply the two identities on the right by  $A^\sigma$  and use the assumed product equation for the chosen pair  $j, \sigma$ . □

**Lemma 16.59** (Endpoint words at the closing boundary). Lean ✓ **Stmt** *Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M + 1$ . Suppose that matrices  $Y_r(\rho)$ , for  $0 \leq r \leq L_0 - 1$ , represent the length- $(L_0 + 1)$  cyclic restrictions beginning at the sites  $M + 1 - L_0 + r$ :*

$$\text{Res}_{M+1-L_0+r, L_0+1}^\rho \psi = \Gamma_{L_0+1}(Y_r(\rho)).$$

Then

$$Y_0(\rho)A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}} = A^{\rho_1} \dots A^{\rho_{L_0-1}}Y_{L_0-1}(\rho).$$

For  $L_0 = 1$ , both products are empty.

*Proof.* ✓ **Proof** Use Lemma 16.38 for the  $L_0 - 1$  adjacent restrictions starting at  $i_0 = M + 1 - L_0$ . For  $0 \leq r < L_0 - 1$ , the two endpoint words are determined by

$$i_0 + r = M + 1 - L_0 + r, \quad i_0 + r + L_0 + 1 \equiv r + 1 \pmod{M + 1},$$

Substituting these two words in the iterated transport identity gives the displayed product equation. □

**Lemma 16.60** (Boundary-condition product at the closing boundary). Lean Lean ✓ **Stmt** *Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M + 1$ , and suppose matrices  $Y_i(\rho)$  represent the length- $(L_0 + 1)$  cyclic restrictions*

$$\text{Res}_{i, L_0+1}^\rho(\psi) = \Gamma_{L_0+1}(Y_i(\rho)).$$

Then, for every boundary condition  $\rho$ ,

$$Y_{M+1-L_0}(\rho)A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}} = A^{\rho_1} \dots A^{\rho_{L_0-1}}Y_M(\rho).$$

For  $L_0 = 1$ , both products are empty. The same equation remains true after multiplying both sides on the right by any matrix  $R$ .

*Proof.* ✓ **Proof** Apply Lemma 16.38 to the  $L_0 - 1$  restrictions beginning at  $M + 1 - L_0 + r$ , with

$$Y_r(\rho) = Y_{M+1-L_0+r}(\rho).$$

With  $i_0 = M + 1 - L_0$ , the site labels satisfy

$$i_0 + r = M + 1 - L_0 + r, \quad i_0 + r + L_0 + 1 \equiv r + 1 \pmod{M + 1}$$

hence, for  $r = 0, \dots, L_0 - 2$ ,

$$Y_{M+1-L_0+r}(\rho)A^{\rho_{M+1-L_0+r}} = A^{\rho_{r+1}}Y_{M+2-L_0+r}(\rho).$$

Multiplying these  $L_0 - 1$  equations from left to right gives the displayed product equation. The right-multiplied form follows by multiplying this equality by  $R$ . □



**Lemma 16.61** (Transport followed by the last-boundary equation). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ , and let  $L_0 \leq M$ . Let  $\psi = \Gamma_{M+1}(X)$ , and suppose that matrices  $Y_i(\rho)$  represent all length- $(L_0 + 1)$  cyclic restrictions of  $\psi$ . Then, for every boundary condition  $\rho$  and every physical letter  $j$ ,

$$Y_{M+1-L_0}(\rho) A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}} A^j = A^{\rho_1} \dots A^{\rho_{L_0-1}} A^{\rho_{L_0}} \dots A^{\rho_{M-1}} A^j X.$$

For  $\rho = \tau_\eta^-(\mu)$  this is the transport identity supplied by the adjacent-window argument. It is not the remaining padded identity, where the factor after  $Y_{M+1-L_0}(\tau_\eta^-(\mu))$  is  $A^j A^\sigma$ .

*Proof.* [✓ Proof](#) Lemma 16.60 gives

$$Y_{M+1-L_0}(\rho) A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}} A^j = A^{\rho_1} \dots A^{\rho_{L_0-1}} Y_M(\rho) A^j.$$

The last-boundary one-sided equation for  $\psi = \Gamma_{M+1}(X)$  gives

$$Y_M(\rho) A^j = A^{\rho_{L_0}} \dots A^{\rho_{M-1}} A^j X.$$

Substitution gives the displayed formula. □

**Lemma 16.62** (Long boundary-condition product at the closing boundary). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M + 1$ , and suppose matrices  $Y_i(\rho)$  represent the length- $(L_0 + 1)$  cyclic restrictions

$$\text{Res}_{i, L_0+1}^\rho(\psi) = \Gamma_{L_0+1}(Y_i(\rho)).$$

Then, for every boundary condition  $\rho$ ,

$$Y_M(\rho) A^{\rho_M} A^{\rho_0} \dots A^{\rho_{M-L_0}} = A^{\rho_{L_0}} \dots A^{\rho_M} A^{\rho_0} Y_{M+1-L_0}(\rho).$$

Each side has  $M + 2 - L_0$  one-site factors, with site labels read modulo  $M + 1$ .

*Proof.* [✓ Proof](#) Apply Lemma 16.38 to the  $M + 2 - L_0$  restrictions beginning at  $M + r$  modulo  $M + 1$ . With  $i_0 = M$ , the site labels satisfy

$$i_0 + r \equiv M + r, \quad i_0 + r + L_0 + 1 \equiv L_0 + r \pmod{M + 1}.$$

The final window is

$$i_0 + M + 2 - L_0 \equiv M + 1 - L_0 \pmod{M + 1}.$$

Substituting these site labels in the iterated product identity gives the displayed equation. □

**Lemma 16.63** (One matrix  $Y_\mu$  from compared boundary matrices). [Lean](#) [✓ Stmt](#) Fix a physical letter  $\eta$  used outside the complementary sites. Suppose the two one-sided boundary matrix families  $Y^+$  and  $Y^-$  have been extracted from the cyclic windows. If, for every word on the complementary sites  $\mu$ , these boundary matrices agree on the boundary conditions built from  $\eta$ ,

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-,$$

then there is a single family  $Y_\mu$  satisfying

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu$$

for all letters  $a, b$ .

*Proof.* ✓ Proof Define  $Y_\mu$  to be the first boundary matrix evaluated on the boundary condition  $\tau_\eta^+(\mu)$ . Lemma 16.55 rewrites the first cyclic-window complement as  $\mu$ , while the assumed comparison at  $\tau_\eta^-(\mu)$  rewrites the opposite boundary matrix to the same  $Y_\mu$ .  $\square$

**Lemma 16.64** (Commutation from two one-sided equations). Lean ✓ Stmt Suppose a single family of boundary matrices  $Y_\mu$ , indexed by words  $\mu$  on the complementary sites, satisfies the two one-sided identities

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu \quad (16.4)$$

for every pair of physical letters  $a, b$ . Then  $X$  commutes with every word product  $A^a A^\mu A^b$  obtained by adjoining one letter on each side of the word  $A^\mu$ .

*Proof.* ✓ Proof Multiply the second identity in (16.4) on the right by  $A^b$  and the first identity on the left by  $A^a$ :

$$X A^a A^\mu A^b = A^a Y_\mu A^b = A^a A^\mu A^b X.$$

$\square$

**Theorem 16.65** (Boundary matrix commutation from periodic closure). Lean ✓ Stmt Let  $A$  be injective with  $D \geq 1$ , and let  $\psi = \Gamma_N(X)$  for some boundary matrix  $X \in M_D(\mathbb{C})$ . If every cyclic restriction of length  $L$  of  $\psi$  lies in  $G_L(A)$ , with  $N \geq 2$  and  $1 < L \leq N$ , then, for every  $j = 0, \dots, d-1$ ,

$$X A^j = A^j X$$

*Proof.* ✓ Proof The periodic boundary condition produces matrix identities of the form  $XMW = MY$  for arbitrary test matrices  $M$  and complementary words  $W$ . Spanning first over the length- $(L-1)$  tails and then over the complementary words, injectivity forces  $(XM - MX)W = 0$  for every  $W$ , hence  $XM = MX$  for every matrix  $M$ . In particular,  $X$  commutes with each letter  $A^j$ .  $\square$

**Definition 16.66** (MPV submodule). Lean ✓ Stmt The subspace spanned by the MPS vector  $V^{(N)}(A)(\sigma) = \text{tr}(A^{\sigma_0} \dots A^{\sigma_{N-1}})$ .

**Lemma 16.67** (The MPV is the ground-space map at the identity). Lean ✓ Stmt For every chain length  $N$ ,

$$V^{(N)}(A) = \Gamma_N(\mathbb{1}).$$

**Lemma 16.68** (MPV lies in the ground space). Lean ✓ Stmt  $V^{(N)}(A) \in G_N(A)$  for every  $N$ , with boundary matrix  $X = I$ .

*Proof.* ✓ Proof  $V^{(N)}(A)(\sigma) = \text{tr}(A^\sigma) = \text{tr}(A^\sigma \cdot I) = \Gamma_N(I)(\sigma)$ .  $\square$

**Definition 16.69** (Unique ground state). Lean ✓ Stmt A submodule  $S$  has a unique ground state if  $\dim S = 1$ .

**Theorem 16.70** (One-dimensionality criterion). Lean ✓ Stmt For finite-dimensional  $S$ ,  $\dim S = 1$  iff there exists a nonzero  $\psi_0 \in S$  such that every  $\psi \in S$  is of the form  $c\psi_0$ .

*Proof.* ✓ Proof Apply the standard finite-dimensional criterion that a nonzero space has dimension one exactly when all of its vectors are proportional to a single nonzero vector.  $\square$

**Definition 16.71** (Periodic chain ground space). [Lean](#) [✓ Stmt](#) For  $N > 0$  and  $L \leq N$ , define the *periodic chain ground space*

$$\mathcal{G}_{N,L}(A) := \{\psi \in (\{0, \dots, d-1\}^N \rightarrow \mathbb{C}) : \forall i \in \{0, \dots, N-1\}, \psi|_{[i, i+L-1]} \in G_L(A)\},$$

where the condition  $\psi|_{[i, i+L-1]} \in G_L(A)$  means that for every choice of physical indices outside the window, the restriction of  $\psi$  to that window lies in  $G_L(A)$ . When  $N = 0$  or  $L > N$ , set  $\mathcal{G}_{N,L}(A) := \top$  by convention.

**Lemma 16.72** (Cyclic-window witnesses inside the chain ground space). [Lean](#) [✓ Stmt](#) If  $0 < N$ ,  $L \leq N$ , and  $\psi \in \mathcal{G}_{N,L}(A)$ , then for every cyclic window and every outside configuration there is a boundary matrix  $Y$  such that the corresponding restricted  $L$ -site state is  $\Gamma_L(Y)$ .

*Proof.* [✓ Proof](#) This is the defining cyclic-window condition for membership in  $\mathcal{G}_{N,L}(A)$ , together with the definition of the local ground space as the range of  $\Gamma_L$ .  $\square$

**Lemma 16.73** (The MPV satisfies every cyclic window constraint). [Lean](#) [✓ Stmt](#) If  $0 < N$  and  $L \leq N$ , then

$$V^{(N)}(A) \in \mathcal{G}_{N,L}(A).$$

**Theorem 16.74** (MPV nonvanishing for block-injective tensors). [Lean](#) [✓ Stmt](#) If  $A$  is  $L_0$ -block-injective with  $L_0 > 0$ ,  $D \geq 1$ , and  $N \geq L_0 + 1$ , then  $V^{(N)}(A) \neq 0$  in the  $N$ -site space.

*Proof.* [✓ Proof](#) Suppose for contradiction that  $V^{(N)}(A) = 0$ , i.e.  $\text{tr}(A^\sigma) = 0$  for every word  $\sigma$  of length  $N$ . For any words  $w$  and  $u$  with  $|w| = N - L_0$  and  $|u| = L_0$ , we have  $\text{tr}(A^w A^u) = 0$ . Since  $A$  is  $L_0$ -block-injective,  $\{A^u : |u| = L_0\}$  spans  $M_D(\mathbb{C})$ . By nondegeneracy of the trace pairing,  $A^w = 0$  for every  $|w| = N - L_0$ . Repeating the argument with words of length  $N - 2L_0$  (since  $|w'| + L_0 = N - L_0$ , the product  $A^{w'} A^{u_1}$  has length  $N - L_0$ , so  $A^{w'} A^{u_1} = 0$  by the previous step) and iterating, we eventually reach  $\text{tr}(A^u) = 0$  for  $|u| = L_0$ , which gives  $I = 0$ , a contradiction.  $\square$

### 16.3.6 Block-word stripping

**Theorem 16.75** (Common right factor from block-word compatibility). [Lean](#) [✓ Stmt](#) Assume  $A$  is  $L_0$ -block-injective with  $L_0 > 0$ . Let  $\{Z_u\}_{|u|=L_0}$  be a family indexed by words of length  $L_0$ . If for some  $K \geq 0$  and every suffix word  $w$  of length  $K$  there exists  $Y_w \in M_D(\mathbb{C})$  such that, for every word  $u$  of length  $L_0$ ,

$$Z_u A^w = A^u Y_w$$

then there exists  $X \in M_D(\mathbb{C})$  with  $Z_u = A^u X$  for every word  $u$  of length  $L_0$ .

*Proof.* [✓ Proof](#) Induct on  $K$ . For  $K = 0$  the empty suffix gives  $Z_u = A^u Y_\emptyset$ . For the successor step, fix the first physical index and apply the induction hypothesis to the shortened suffix. This reduces the statement to letter compatibility on the right. Extending that letter compatibility to all length- $L_0$  words and decomposing  $\mathbb{1}$  in the spanning family  $\{A^u : |u| = L_0\}$  yields a common factor  $X$ .  $\square$

**Theorem 16.76** (Block-word commutation from long-word commutation). [Lean](#) [✓ Stmt](#) If  $A$  is  $L_0$ -block-injective with  $L_0 > 0$ ,  $m \geq L_0$ , and  $X \in M_D(\mathbb{C})$  commutes with every length- $m$  product  $A^\omega$ , then  $X$  already commutes with every length- $L_0$  product  $A^u$ .

*Proof.* [✓ Proof](#) Write each length- $m$  word as a concatenation  $uw$  with  $|u| = L_0$  and  $|w| = m - L_0$ . The hypothesis gives

$$XA^u A^w = A^u A^w X = A^u (A^w X).$$

Thus the family  $Z_u := XA^u$  satisfies the hypothesis of Theorem 16.75. One obtains  $XA^u = A^u R$  for all  $|u| = L_0$ . Decomposing  $1$  in the span of the length- $L_0$  words shows  $R = X$ , hence  $XA^u = A^u X$ .  $\square$

**Theorem 16.77** (Centrality from long-word commutation). [Lean](#) [✓ Stmt](#) *If  $A$  is  $L_0$ -block-injective with  $L_0 > 0$ ,  $m \geq L_0$ , and  $X \in M_D(\mathbb{C})$  commutes with every length- $m$  word product, then  $X$  commutes with every matrix in  $M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) By Theorem 16.76,  $X$  commutes with every length- $L_0$  word product. These products span  $M_D(\mathbb{C})$  by block injectivity, so linearity gives  $XM = MX$  for every  $M \in M_D(\mathbb{C})$ .  $\square$

**Lemma 16.78** (Amplifying fixed-length word commutation). [Lean](#) [✓ Stmt](#) *If a boundary matrix  $X$  commutes with every word product of length  $m$ , then it commutes with every word product whose length is a multiple  $qm$ .*

*Proof.* [✓ Proof](#) Proceed by induction on  $q$ . For  $q = 0$ , the claim is trivial. For the inductive step, let a word of length  $(q+1)m$  be given and write it as a length- $m$  prefix followed by a length- $qm$  suffix. The prefix commutes with  $X$  by hypothesis, and the suffix commutes by the induction hypothesis.  $\square$

**Lemma 16.79** (Padded complement annihilation). [Lean](#) [Lean](#) [✓ Stmt](#) *If  $A$  is  $L_0$ -block-injective,  $q \geq 1$ , and  $k \leq qL_0$ , then the two one-sided annihilation statements are:*

$$ZA^w = 0 \quad (|w| = k) \quad \implies \quad Z = 0.$$

*The left-handed form is*

$$A^w Z = 0 \quad (|w| = k) \quad \implies \quad Z = 0.$$

*Proof.* [✓ Proof](#) The hypothesis  $\text{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$  implies the same spanning statement at every positive multiple  $qL_0$ . Each length- $qL_0$  word factors as a length- $k$  prefix followed by padding, so  $Z$  annihilates all generators of the full span and hence annihilates the identity. For the left-handed form, factor each length- $qL_0$  word as a prefix followed by a length- $k$  suffix and use the same spanning argument.  $\square$

**Lemma 16.80** (One-sided boundary witnesses are unique). [Lean](#) [Lean](#) [✓ Stmt](#) *Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ . If two boundary matrices have the same product with every one-site tensor on one fixed side, then the two boundary matrices are equal.*

*Proof.* [✓ Proof](#) If  $Y_1 A^j = Y_2 A^j$  for every physical index  $j$ , then for every nonempty word  $w$ , induction on the word gives

$$Y_1 A^w = Y_2 A^w$$

Since  $A$  is  $L_0$ -block-injective,

$$\text{span}\{A^w : |w| = L_0\} = M_D(\mathbb{C}).$$

Thus the left multiplication maps  $M \mapsto Y_1 M$  and  $M \mapsto Y_2 M$  are equal on  $M_D(\mathbb{C})$ , and evaluating on the identity gives  $Y_1 = Y_2$ . The other one-sided statement is the same argument, with right multiplication in place of left multiplication.  $\square$

**Lemma 16.81** (Generator commutation from long-word commutation). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 16.77, one has  $XA^i = A^iX$  for every physical index  $i$ .

*Proof.* [✓ Proof](#) Apply Theorem 16.77 with  $M = A^i$ . □

**Theorem 16.82** (MPS-line containment from long-word commutation). [Lean](#) [✓ Stmt](#) Let  $A$  be injective after blocking  $L_0 > 0$  sites, and let  $m \geq L_0$ . If a boundary matrix  $X$  commutes with every length- $m$  word product, then

$$\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}.$$

*Proof.* [✓ Proof](#) Theorem 16.77 makes  $X$  commute with the whole matrix algebra. The center of  $M_D(\mathbb{C})$  consists of scalar matrices, so  $X = c\mathbb{1}$ , and therefore  $\Gamma_N(X) = cV^{(N)}(A)$ . □

**Theorem 16.83** (MPS-line containment from positive-length commutation). [Lean](#) [✓ Stmt](#) Let  $A$  be injective after blocking  $L_0 > 0$  sites. If  $X$  commutes with every word product of some positive length  $m$ , then

$$\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}.$$

*Proof.* [✓ Proof](#) Lemma 16.78 amplifies the length- $m$  commutation hypothesis to length  $L_0 m \geq L_0$ . Then Theorem 16.82 applies. □

**Theorem 16.84** (MPS-line containment from two one-sided equations). [Lean](#) [✓ Stmt](#) Let  $A$  be injective after blocking  $L_0 > 0$  sites. Suppose there are matrices  $Y_\mu$  such that

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu$$

for every pair of physical letters  $a, b$ . Then

$$\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}.$$

*Proof.* [✓ Proof](#) The identities in (16.4) give commutation with all words of length  $|\mu| + 2$ , a positive length. The positive-length containment theorem then applies. □

**Theorem 16.85** (Boundary equality implies containment in the MPS line). [Lean](#) [✓ Stmt](#) Let  $A$  be injective after blocking  $L_0 > 0$  sites, and fix a physical letter  $\eta$  used outside the complementary sites. Suppose that, for every boundary condition  $\tau$  and every physical letter  $j$ ,

$$C_\tau^+ A^j X = Y_\tau^+ A^j, \quad X A^j C_\tau^- = A^j Y_\tau^-,$$

where  $C_\tau^+$  and  $C_\tau^-$  are the complementary words exposed by the two boundary-crossing cyclic windows. Suppose also that the two boundary matrices satisfy

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-$$

for every word  $\mu$  on the complementary sites, then

$$\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}.$$

*Proof.* [✓ Proof](#) Lemma 16.63 gives a family  $Y_\mu$  satisfying

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu.$$

Theorem 16.84 applies. □

**Theorem 16.86** (Closure-property step for periodic chains). [Lean](#) [✓ Stmt](#) Let  $A$  be injective after blocking  $L_0 > 0$  sites and let  $D \geq 1$ . Assume  $N \geq 2$  and  $L_0 < L \leq N$ . Suppose that, for every physical letter  $\eta$  used in the two boundary conditions and every boundary matrix obtained from an  $N$ -site chain ground state, the identities

$$A^\mu A^j X = Y_{\tau_\eta^+(\mu)}^+ A^j, \quad X A^j A^\mu = A^j Y_{\tau_\eta^-(\mu)}^-$$

hold for every physical letter  $j$  and every complementary word  $\mu$ , and are supplemented, for every  $\mu$ , by

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-.$$

Then

$$\mathcal{G}_{N,L}(A) \subseteq \text{span}\{V^{(N)}(A)\}.$$

*Proof.* [✓ Proof](#) The cyclic-to-open-chain reduction writes any chain ground state as  $\Gamma_N(X)$ . The two boundary-crossing cyclic windows give

$$A^\mu A^j X = Y_{\tau_\eta^+(\mu)}^+ A^j, \quad X A^j A^\mu = A^j Y_{\tau_\eta^-(\mu)}^-.$$

Together with

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-,$$

Theorem 16.85 finishes. □

**Lemma 16.87** (Products with one-site tensors determine the witness). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ . Fix two matrix families  $Y^+$  and  $Y^-$  and the two reindexed boundary conditions  $\tau_\eta^+(\mu)$  and  $\tau_\eta^-(\mu)$ . If, for every physical letter  $j$ ,

$$Y_{\tau_\eta^+(\mu)}^+ A^j = Y_{\tau_\eta^-(\mu)}^- A^j,$$

then

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-.$$

*Proof.* [✓ Proof](#) Apply Lemma 16.80 to the two matrices  $Y_{\tau_\eta^+(\mu)}^+$  and  $Y_{\tau_\eta^-(\mu)}^-$ . □

**Lemma 16.88** (First-letter restrictions determine a vector). [Lean](#) [✓ Stmt](#) Let  $R_j$  be restriction to first letter  $j$ :

$$(R_j \phi)(\sigma_1, \dots, \sigma_L) = \phi(j, \sigma_1, \dots, \sigma_L).$$

For  $\phi, \psi \in (\mathbb{C}^d)^{\otimes(L+1)}$ ,

$$(\forall j, R_j \phi = R_j \psi) \implies \phi = \psi.$$

*Proof.* [✓ Proof](#) At  $(\sigma_0, \sigma_1, \dots, \sigma_L)$  the required equality is

$$\phi(\sigma_0, \sigma_1, \dots, \sigma_L) = (R_{\sigma_0} \phi)(\sigma_1, \dots, \sigma_L) = (R_{\sigma_0} \psi)(\sigma_1, \dots, \sigma_L) = \psi(\sigma_0, \sigma_1, \dots, \sigma_L).$$

□

**Lemma 16.89** (Auxiliary restriction equality for the closure property). [Lean](#) [✓ Stmt](#) Let  $L_0 > 0$  and  $L_0 \leq M$ , and set

$$B_{\eta,\mu}^+(\psi) = \text{Res}_{M,L_0+1}^{\tau_\eta^+(\mu)}(\psi), \quad B_{\eta,\mu}^-(\psi) = \text{Res}_{M+1-L_0,L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

With  $R_j$  as in Lemma 16.88,

$$(\forall j, R_j B_{\eta,\mu}^+(\psi) = R_j B_{\eta,\mu}^-(\psi)) \implies B_{\eta,\mu}^+(\psi) = B_{\eta,\mu}^-(\psi).$$

*Proof.* ✓ Proof Apply Lemma 16.88 to the two length- $(L_0 + 1)$  restrictions, using the displayed family of first-letter equalities.  $\square$

**Lemma 16.90** (Closed-boundary restriction from right products). Lean ✓ Stmt Let  $A$  be an MPS tensor, let  $L_0 > 0$ , and let  $L_0 \leq M$ . Suppose the two length- $(L_0 + 1)$  restrictions at the closed boundary are represented by boundary matrices  $Y_{\eta, \mu}^+$  and  $Y_{\eta, \mu}^-$ :

$$B_{\eta, \mu}^+(\psi) = \Gamma_{L_0+1}(Y_{\eta, \mu}^+), \quad B_{\eta, \mu}^-(\psi) = \Gamma_{L_0+1}(Y_{\eta, \mu}^-).$$

If, for every physical letter  $j$ ,

$$Y_{\eta, \mu}^+ A^j = Y_{\eta, \mu}^- A^j,$$

then

$$B_{\eta, \mu}^+(\psi) = B_{\eta, \mu}^-(\psi).$$

*Proof.* ✓ Proof Fix  $j$  and take the first-letter restriction of the two displayed length- $(L_0 + 1)$  restrictions. These are

$$\Gamma_{L_0}(Y_{\eta, \mu}^+ A^j), \quad \Gamma_{L_0}(Y_{\eta, \mu}^- A^j).$$

The one-site product equality gives, for every  $j$ ,

$$\Gamma_{L_0}(Y_{\eta, \mu}^+ A^j) = \Gamma_{L_0}(Y_{\eta, \mu}^- A^j).$$

Lemma 16.88 identifies the two original restrictions.  $\square$

**Lemma 16.91** (Restriction form of the closure property). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ , let  $L_0 \leq M$ , and let

$$\psi \in \mathcal{G}_{M+1, L_0+1}(A), \quad \psi = \Gamma_{M+1}(X).$$

For every boundary letter  $\eta$  and complementary word  $\mu$ , the local coordinate restriction equality representing the boundary-closing step of [CPGSV21, lines 2078–2079] is

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

*Proof.* Not ready The window-witness lemma gives matrices  $Y_M(\tau_\eta^+(\mu))$  and  $Y_{M+1-L_0}(\tau_\eta^-(\mu))$  such that

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \Gamma_{L_0+1}(Y_M(\tau_\eta^+(\mu))), \quad \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi) = \Gamma_{L_0+1}(Y_{M+1-L_0}(\tau_\eta^-(\mu))).$$

Lemma 16.107 gives, for every  $j$ ,

$$Y_M(\tau_\eta^+(\mu)) A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j.$$

Lemma 16.90 identifies the two length- $(L_0 + 1)$  restrictions.  $\square$

**Lemma 16.92** (Auxiliary product from closing-boundary restrictions). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M + 1$ , and let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Suppose that, for every outside letter  $\eta$ ,

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Then there are boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  such that, for every complementary position  $k$ ,

$$\rho_{j,\sigma}^+(k + L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k + 1) = \mu_k,$$

and, for every physical letter  $j$  and every word  $\sigma$  of length  $L_0$ ,

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

*Proof.* ✓ Proof For the product indexed by  $j$  and  $\sigma$ , specialize the preceding equality at the outside letter  $\eta = j$ . Take  $\rho_{j,\sigma}^+ = \tau_j^+(\mu)$  and  $\rho_{j,\sigma}^- = \tau_j^-(\mu)$ . The complement equations are the two identities of Lemma 16.55. Fix  $j$ . Applying the first-letter restriction to the displayed equality gives

$$\Gamma_{L_0}(Y_M(\tau_j^+(\mu))A^j) = \Gamma_{L_0}(Y_{M+1-L_0}(\tau_j^-(\mu))A^j),$$

where the two sides are identified by Lemma 16.38. By Lemma 16.32,

$$Y_M(\tau_j^+(\mu))A^j = Y_{M+1-L_0}(\tau_j^-(\mu))A^j.$$

Right multiplication by  $A^\sigma$  gives the asserted product equation. □

**Lemma 16.93** (Auxiliary product from right products). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M + 1$ , and let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i,L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a complementary word  $\mu$  of length  $M + 1 - (L_0 + 1)$ . Suppose that, for every boundary letter  $\eta$  and every physical letter  $j$ ,

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Then there are boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  such that, for every complementary position  $k$ ,

$$\rho_{j,\sigma}^+(k + L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k + 1) = \mu_k,$$

and, for every physical letter  $j$  and every word  $\sigma$  of length  $L_0$ ,

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

*Proof.* ✓ Proof For each boundary letter  $\eta$ , Lemma 16.90 gives

$$\text{Res}_{M,L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0,L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Applying these equalities, Lemma 16.92 gives the displayed boundary conditions and product equation. □

**Lemma 16.94** (Opposite-boundary comparison after multiplication by words). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Fix a complementary word  $\mu$ , a matrix  $X$ , and a family of matrices  $Y_i(\tau)$ . Suppose that, for every boundary letter  $\eta$ , every physical letter  $j$ , and every word  $\sigma$  of length  $L_0$ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma = A^\mu A^j X A^\sigma.$$

Then, for every  $\eta$  and  $j$ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j = A^\mu A^j X.$$



*Proof.* ✓ **Proof** Fix  $\eta$  and  $j$ , and set

$$Z_{\eta,j} = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j - A^\mu A^j X.$$

The hypothesis gives, for every  $\sigma$  with  $|\sigma| = L_0$ ,

$$Z_{\eta,j}A^\sigma = 0.$$

Since  $A$  is  $L_0$ -block-injective, the products  $A^\sigma$  with  $|\sigma| = L_0$  span the full matrix algebra. Applying Lemma 16.79 gives  $Z_{\eta,j} = 0$ . □

**Lemma 16.95** (Opposite-boundary comparison after left multiplication by words). Lean ✓ **Stmt** Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Fix a complementary word  $\mu$ , a matrix  $X$ , and a family of matrices  $Y_i(\tau)$ . Suppose that, for every boundary letter  $\eta$ , every physical letter  $j$ , and every pair of words  $\sigma, \alpha$  of length  $L_0$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

Then, for every  $\eta, j$ , and  $\sigma$ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma = A^\mu A^j X A^\sigma.$$

*Proof.* ✓ **Proof** Fix  $\eta, j$ , and  $\sigma$ , and set

$$Z_{\eta,j,\sigma} = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma - A^\mu A^j X A^\sigma.$$

The hypothesis says that  $A^\alpha Z_{\eta,j,\sigma} = 0$  for every word  $\alpha$  of length  $L_0$ . The left-handed form of Lemma 16.79 gives  $Z_{\eta,j,\sigma} = 0$ . □

**Lemma 16.96** (Auxiliary product from the opposite-boundary comparison). Lean ✓ **Stmt** Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M+1$ , and let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i,L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a complementary word  $\mu$  and a matrix  $X$ . Suppose that, for every boundary letter  $\eta$  and every physical letter  $j$ ,

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X.$$

Suppose also that, for every  $\eta, j$ , and every word  $\sigma$  of length  $L_0$ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma = A^\mu A^j X A^\sigma.$$

Then there are boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  such that, for every complementary position  $k$ ,

$$\rho_{j,\sigma}^+(k + L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k + 1) = \mu_k,$$

and, for every physical letter  $j$  and every word  $\sigma$  of length  $L_0$ ,

$$Y_M(\rho_{j,\sigma}^+)A^j A^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^j A^\sigma.$$

*Proof.* ✓ Proof Lemma 16.94 gives, for every  $\eta$  and  $j$ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j = A^\mu A^j X.$$

Combining this equation with

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X$$

gives

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Lemma 16.93 then gives the displayed boundary conditions and auxiliary product. □

**Lemma 16.97** (Auxiliary product from the left-multiplied opposite-boundary comparison).

Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M + 1$ , and let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a complementary word  $\mu$  and a matrix  $X$ . Suppose that, for every boundary letter  $\eta$  and every physical letter  $j$ ,

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X.$$

Suppose also that, for every  $\eta, j$ , and every pair of words  $\sigma, \alpha$  of length  $L_0$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

Then there are boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  satisfying the complementary-word equations

$$\rho_{j,\sigma}^+(k + L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k + 1) = \mu_k$$

and the product equation

$$Y_M(\rho_{j,\sigma}^+)A^j A^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^j A^\sigma.$$

*Proof.* ✓ Proof Lemma 16.95 gives

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma = A^\mu A^j X A^\sigma.$$

Lemma 16.96 then gives the displayed boundary conditions and product equation. □

**Lemma 16.98** (Auxiliary product from the left-multiplied closure-property comparison). Lean

✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $L_0 \leq M$ , and  $D \geq 1$ . Let

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation, and let  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions of  $\psi$ . Fix a complementary word  $\mu$ . Suppose that, for every boundary letter  $\eta$ , physical letter  $j$ , and length- $L_0$  words  $\sigma, \alpha$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

Then there are boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  satisfying the complementary-word equations

$$\rho_{j,\sigma}^+(k + L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k + 1) = \mu_k$$

and the product equation

$$Y_M(\rho_{j,\sigma}^+)A^j A^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^j A^\sigma.$$

*Proof.* ✓ Proof Lemma 16.54 gives the last-boundary equation

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X.$$

The assumed left-multiplied coordinate equation and Lemma 16.97 then give the displayed boundary conditions and product equation.  $\square$

**Lemma 16.99** (Equality of the boundary-closing cyclic restrictions). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $L_0 \leq M$ , and  $D \geq 1$ . Let

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation. Assume that for all  $i$  and  $\tau$ ,

$$\text{Res}_{i, L_0+1}^\tau(\psi) \in G_{L_0+1}(A).$$

Fix a complementary word  $\mu$ . For every boundary letter  $\eta$ , choose the two local boundary conditions so that, for each complementary index  $k$ ,

$$\tau_\eta^+(\mu)_{k+L_0} = \mu_k, \quad \tau_\eta^-(\mu)_{k+1} = \mu_k,$$

and put the letter  $\eta$  on the remaining sites. These two choices have the same complementary word. The notation  $\tau_\eta^\pm(\mu)$  is a local coordinate device for the boundary-closing comparison; it is not notation used in the source. In these coordinates the boundary-closing step is the equality

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

*Proof.* Not ready  $\square$

**Lemma 16.100** (First-letter products from equal boundary restrictions). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $L_0 \leq M$ . Let  $\psi$  be a vector on  $M+1$  sites, and suppose the two length- $(L_0+1)$  boundary restrictions are represented by

$$\Gamma_{L_0+1}(Y_M(\tau_\eta^+(\mu))), \quad \Gamma_{L_0+1}(Y_{M+1-L_0}(\tau_\eta^-(\mu))).$$

If these two restrictions are equal, then for every physical letter  $j$ ,

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Thus the first-letter restriction of the preceding equality is precisely the displayed product equation.

*Proof.* ✓ Proof Restricting both sides to the first physical letter  $j$  gives

$$\Gamma_{L_0}(Y_M(\tau_\eta^+(\mu))A^j) = \Gamma_{L_0}(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j).$$

Since  $A$  is  $L_0$ -block-injective,  $\Gamma_{L_0}$  is injective, and the displayed product equality follows.  $\square$

**Lemma 16.101** (Left-multiplied comparison from equal boundary restrictions). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $L_0 \leq M$ . Let  $\psi$  be a vector on  $M+1$  sites, and let  $Y_i(\tau)$  represent the length- $(L_0+1)$  cyclic restrictions of  $\psi$ . Assume that for every boundary letter  $\eta$ ,

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Then for every boundary letter  $\eta$ , physical letter  $j$ , and length- $L_0$  words  $\alpha, \sigma$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu))A^j A^\sigma).$$

*Proof.* ✓ Proof The preceding lemma gives

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Multiplying on the left by  $A^\alpha$  and on the right by  $A^\sigma$  gives the desired comparison.  $\square$

**Lemma 16.102** (Left-multiplied comparison of boundary matrices). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $L_0 \leq M$ , and  $D \geq 1$ . Let

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation, and let  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions of  $\psi$ . Fix a complementary word  $\mu$ . Then for every boundary letter  $\eta$ , physical letter  $j$ , and length- $L_0$  words  $\alpha, \sigma$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu))A^j A^\sigma).$$

*Proof.* Not ready The representations by the matrices  $Y_i(\tau)$  supply the local ground-space hypothesis for the preceding lemma. The preceding boundary-restriction equality gives

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Applying the left-multiplied comparison for equal boundary restrictions gives

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu))A^j A^\sigma).$$

$\square$

**Lemma 16.103** (Left-multiplied boundary-closing comparison). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $L_0 \leq M$ , and  $D \geq 1$ . Let

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation, and let  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions of  $\psi$ . Fix a complementary word  $\mu$ . Then for every boundary letter  $\eta$ , physical letter  $j$ , and length- $L_0$  words  $\alpha, \sigma$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

*Proof.* The comparison of the two cyclic restrictions gives

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu))A^j A^\sigma).$$

The last-boundary one-sided equation gives

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X.$$

Substitution gives the displayed identity.  $\square$

**Lemma 16.104** (Letterwise restriction comparison at the closing boundary). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ , let  $L_0 \leq M$ , and let

$$\psi \in \mathcal{G}_{M+1, L_0+1}(A), \quad \psi = \Gamma_{M+1}(X).$$

Let  $R_j$  denote restriction to first physical letter  $j$ . For every outside letter  $\eta$ , complementary word  $\mu$ , and physical letter  $j$ , one has

$$R_j \text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = R_j \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

This is the letterwise coordinate form of the local closure-property comparison corresponding to [CPGSV21, lines 2078–2079].

*Proof.* Not ready Let  $Y_i(\tau)$  be the window witnesses of Lemma 16.72. By Lemma 16.107,

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Fixing the first physical letter in the two length- $(L_0 + 1)$  windows gives

$$\begin{aligned} R_j \text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) &= \Gamma_{L_0}(Y_M(\tau_\eta^+(\mu))A^j), \\ R_j \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi) &= \Gamma_{L_0}(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j). \end{aligned}$$

The two restrictions are therefore equal. □

**Lemma 16.105** (Auxiliary boundary-condition product equation). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ . Let  $\psi$  be a state on the chain of length  $M + 1$  with open-chain representation  $\psi = \Gamma_{M+1}(X)$ . Let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

If  $L_0 \leq M$ , then for every complementary word  $\mu$  there are boundary conditions  $\rho_{j, \sigma}^+$  and  $\rho_{j, \sigma}^-$ , depending on the physical letter  $j$  and the word  $\sigma$  of length  $L_0$ , such that, for every complementary position  $k$ ,

$$\rho_{j, \sigma}^+(k + L_0) = \mu_k, \quad \rho_{j, \sigma}^-(k + 1) = \mu_k$$

and, for every  $j$  and  $\sigma$ ,

$$Y_M(\rho_{j, \sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j, \sigma}^-)A^jA^\sigma$$

*Proof.* Not ready Lemma 16.54 gives

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X.$$

By Lemma 16.96, it remains to prove, for every boundary letter  $\eta$ , every physical letter  $j$ , and every word  $\sigma$  of length  $L_0$ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma = A^\mu A^j X A^\sigma.$$

□

**Lemma 16.106** (Product equality after closing the boundary). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ . Let  $\psi \in \mathcal{G}_{M+1, L_0+1}(A)$  have an open-chain representation  $\psi = \Gamma_{M+1}(X)$ . Let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

If  $L_0 \leq M$ , then for every boundary letter  $\eta$ , complementary word  $\mu$ , physical letter  $j$ , and word  $\sigma$  of length  $L_0$ ,

$$Y_M(\tau_\eta^+(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma.$$

*Proof.* Not ready Lemma 16.105 gives boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  satisfying

$$\rho_{j,\sigma}^+(k + L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k + 1) = \mu_k$$

and

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

Theorem 16.58 replaces these auxiliary conditions by the displayed boundary conditions:

$$Y_M(\tau_\eta^+(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma,$$

which is the product equality needed after closing the boundary. □

**Lemma 16.107** (Right-product equality after closing the boundary). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ . Let  $\psi \in \mathcal{G}_{M+1, L_0+1}(A)$  have an open-chain representation  $\psi = \Gamma_{M+1}(X)$ . Let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

If  $L_0 \leq M$ , then for every letter  $\eta$  used in the two boundary conditions, every word  $\mu$  on the complementary sites, and every physical letter  $j$ ,

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

This is the coordinate one-site product equality used to compare the two boundary matrices after closing the boundary.

*Proof.* Not ready Fix  $j$  and set

$$Z_j = Y_M(\tau_\eta^+(\mu))A^j - Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Lemma 16.106 gives, after subtracting the two sides, for every word  $\sigma$  of length  $L_0$ ,

$$Z_j A^\sigma = 0.$$

Since  $A$  is  $L_0$ -block-injective,

$$\text{span}\{A^\sigma : |\sigma| = L_0\} = M_D(\mathbb{C}).$$

Applying Lemma 16.79 gives  $Z_j = 0$ , hence

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

□

**Theorem 16.108** (Boundary conditions agree in the closure property). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ , let  $N \geq 2$  and  $L_0 < L \leq N$ , and let  $\psi \in \mathcal{G}_{N, L}(A)$  have an open-chain representation  $\psi = \Gamma_N(X)$ . Suppose  $Y^+$  and  $Y^-$  are two families of matrices obtained from the two boundary-crossing local constraints used in the closure property, and that they satisfy, for every physical letter  $j$  and every boundary condition  $\tau$ ,

$$A^{\tau_{L_0} \cdots \tau_{N-2}} A^j X = Y_\tau^+ A^j, \quad X A^j A^{\tau_1 \cdots \tau_{N-L_0-1}} = A^j Y_\tau^-.$$

For every physical letter  $\eta \in \{0, \dots, d-1\}$  and every word  $\mu$  on the complementary sites, define two boundary conditions by

$$\tau_\eta^+(\mu)_i = \begin{cases} \mu_{i-L_0}, & L_0 \leq i < N-1, \\ \eta, & \text{otherwise,} \end{cases} \quad \tau_\eta^-(\mu)_i = \begin{cases} \mu_{i-1}, & 1 \leq i < N-L_0, \\ \eta, & \text{otherwise.} \end{cases}$$

Then the two families agree on these two boundary conditions:

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-.$$

This is the boundary-matrix comparison required by the closure property.

*Proof.* Not ready The reduced cyclic-window compatibilities at the boundary give, for every physical letter  $j$ ,

$$A^\mu A^j X = Y_{\tau_\eta^+(\mu)}^+ A^j, \quad X A^j A^\mu = A^j Y_{\tau_\eta^-(\mu)}^-.$$

By Lemma 16.87, it is enough to prove, for every  $j$ ,

$$Y_{\tau_\eta^+(\mu)}^+ A^j = Y_{\tau_\eta^-(\mu)}^- A^j.$$

Write  $N = M + 1$ . The two boundary-crossing supports start at  $M$  and  $M + 1 - L_0$ . Let  $Y_i(\tau)$  denote the cyclic-window witness supplied by Lemma 16.72 at the window beginning at  $i$ . The first step identifies the two abstract boundary-condition families with the two boundary-crossing witnesses:

$$Y_{\tau_\eta^+(\mu)}^+ = Y_M(\tau_\eta^+(\mu)), \quad Y_{\tau_\eta^-(\mu)}^- = Y_{M+1-L_0}(\tau_\eta^-(\mu)).$$

Together with the preceding identifications, Lemma 16.107 gives, for every physical letter  $j$ ,

$$Y_{\tau_\eta^+(\mu)}^+ A^j = Y_{\tau_\eta^-(\mu)}^- A^j.$$

Lemma 16.87 then gives the desired equality of boundary matrices. □

**Theorem 16.109** (Chain ground space is spanned by the MPS vector in the injective case).

Lean ✓ Stmt If  $A$  is injective,  $D \geq 1$ ,  $N \geq 2$ , and  $1 < L \leq N$ , then

$$\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}.$$

*Proof.* ✓ Proof Let  $\psi \in \mathcal{G}_{N,L}(A)$ . The non-wrapping window conditions and Lemma 16.39 imply  $\psi \in G_N(A)$ , so  $\psi = \Gamma_N(X)$  for some boundary matrix  $X$ . The periodic boundary condition now applies Theorem 16.65 and forces  $X$  to commute with every letter  $A^j$ . Since  $A$  is injective, the letters generate the full matrix algebra, hence  $X$  is scalar. Therefore  $\psi$  is proportional to  $V^{(N)}(A)$ , while the reverse inclusion is Lemma 16.73. □

**Theorem 16.110** (Closure-property step toward uniqueness of the ground state). Lean ✓ Stmt If  $A$  is normal,  $D \geq 1$ , and injective after blocking  $L_0 > 0$  sites, with  $N \geq 2$  and  $L_0 < L \leq N$ , then

$$\mathcal{G}_{N,L}(A) \subseteq \text{span}\{V^{(N)}(A)\}.$$

*Proof.* Not ready For a chain ground state, the open-chain containment gives  $\psi = \Gamma_N(X)$ . The two boundary-crossing local constraints give

$$A^\mu A^j X = Y_{\tau_\eta^+(\mu)}^+ A^j, \quad X A^j A^\mu = A^j Y_{\tau_\eta^-(\mu)}^-.$$

The remaining closed-boundary comparison is the equality

$$Y_{\tau_{\eta}^+(\mu)}^+ = Y_{\tau_{\eta}^-(\mu)}^-.$$

These equations imply

$$\psi \in \text{span}\{V^{(N)}(A)\}.$$

□

**Theorem 16.111** (Chain ground space is spanned by the MPS vector in the normal case).

[Lean](#) [✓ Stmt](#) *If  $A$  is normal,  $D \geq 1$ , and injective after blocking  $L_0 > 0$  sites, with  $N \geq 2$  and  $L_0 < L \leq N$ , then*

$$\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}.$$

**Theorem 16.112** (Unique ground state on the periodic chain). [Lean](#) [✓ Stmt](#) *For an injective tensor  $A$  with  $D \geq 1$ , if  $N \geq 2$  and  $1 < L \leq N$ , then the periodic chain ground space is one-dimensional.*

*Proof.* [✓ Proof](#) By Theorem 16.109, the ground space is the line  $\text{span}\{V^{(N)}(A)\}$ . It remains to show that the MPV is nonzero. If  $V^{(N)}(A) = 0$ , then Lemma 16.67 gives

$$\Gamma_N(\mathbb{1}) = \Gamma_N(0).$$

Since  $A$  is injective and  $N > 0$ , Theorem 16.30 forces  $\mathbb{1} = 0$ , a contradiction. Therefore the spanning line is one-dimensional. □

**Theorem 16.113** (Unique ground state for block-injective tensors). [Lean](#) [✓ Stmt](#) *If  $A$  is  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ , the parent Hamiltonian with interaction range  $2L_0$  has a unique ground state on every periodic chain of  $N \geq 2L_0$  sites.*

**Theorem 16.114** (Unique ground state for normal tensors at range  $L_0 + 1$ ). [Lean](#) [✓ Stmt](#) *If  $A$  is normal (hence  $L_0$ -block-injective for some  $L_0 > 0$ ),  $D \geq 1$ , then the parent Hamiltonian with interaction range  $L_0 + 1$  has a unique ground state on every periodic chain of  $N \geq L_0 + 1$  sites.*

## 16.4 Commuting parent Hamiltonians

A parent Hamiltonian has *commuting* local terms when all translated interaction projectors commute with each other.

**Definition 16.115** (Commuting parent Hamiltonian). [Lean](#) [✓ Stmt](#) The parent Hamiltonian with block length  $L$  on  $N$  sites has *commuting* local terms if  $h_i h_j = h_j h_i$  for all  $i, j \in \{0, \dots, N-1\}$ . See Ref. [CPGSV16, Definition 3.9].

**Definition 16.116** (Nearest-neighbor commuting parent Hamiltonian). [Lean](#) [✓ Stmt](#) A *nearest-neighbor commuting parent Hamiltonian* (NNCPH) is a commuting parent Hamiltonian with block length  $L = 2$ . See Ref. [CPGSV16, Definition 3.9].

*Remark 16.117* (Elementary consequences of the commuting definition). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The nearest-neighbor condition is simply the length-2 specialization of the general commuting condition; the latter is symmetric in the two sites, and it implies that the full parent Hamiltonian commutes with each local interaction term.





*Proof.* ✓ Proof Take  $P_{AX} = P_{XB} = P_K$ . Their commutativity is immediate, and the product identity is exactly the assumed idempotence of  $P_K$ . □

**Lemma 16.124** (Cancellation for commuting idempotents). Lean ✓ Stmt *If  $P$  and  $Q$  are commuting idempotents, then*

$$Q \circ (1 - P \circ Q) \circ P = 0.$$

*Proof.* ✓ Proof Expand the middle factor, use commutativity to rewrite  $QPQP$  as  $QPPQ$ , and then apply idempotence of  $P$  and  $Q$ . □

**Theorem 16.125** (Idempotence of the product projector). Lean ✓ Stmt *If  $P_K = P_{AX} \circ P_{XB}$  is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_K = P_K.$$

*Proof.* ✓ Proof Since  $P_{AX}$  and  $P_{XB}$  are idempotent and commute, their product  $P_{AX} \circ P_{XB}$  is idempotent. Substituting  $P_K = P_{AX} \circ P_{XB}$  gives the claim. □

**Theorem 16.126** (Absorption by the left projector). Lean ✓ Stmt *If  $P_K = P_{AX} \circ P_{XB}$  is a commuting parent Hamiltonian decomposition, then*

$$P_{AX} \circ P_K = P_K.$$

*Proof.* ✓ Proof Using  $P_K = P_{AX} \circ P_{XB}$  and the idempotence of  $P_{AX}$ ,

$$P_{AX} \circ P_K = P_{AX} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

**Theorem 16.127** (Absorption by the right projector). Lean ✓ Stmt *If  $P_K = P_{AX} \circ P_{XB}$  is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_{XB} = P_K.$$

*Proof.* ✓ Proof Using  $P_K = P_{AX} \circ P_{XB}$  and the idempotence of  $P_{XB}$ ,

$$P_K \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

**Theorem 16.128** (Cross-absorption by the right projector). Lean ✓ Stmt *If  $P_K = P_{AX} \circ P_{XB}$  is a commuting parent Hamiltonian decomposition, then*

$$P_{XB} \circ P_K = P_K.$$

*Proof.* ✓ Proof By commutativity,  $P_{XB} \circ P_K = P_{XB} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB}$ , and idempotence of  $P_{XB}$  yields  $P_{XB} \circ P_K = P_K$ . □

**Theorem 16.129** (Cross-absorption by the left projector). Lean ✓ Stmt *If  $P_K = P_{AX} \circ P_{XB}$  is a commuting parent Hamiltonian decomposition, then*

$$P_K \circ P_{AX} = P_K.$$

*Proof.* ✓ Proof By commutativity,  $P_K \circ P_{AX} = P_{AX} \circ P_{XB} \circ P_{AX} = P_{AX} \circ P_{AX} \circ P_{XB}$ , and idempotence of  $P_{AX}$  yields  $P_K \circ P_{AX} = P_K$ . □

**Theorem 16.130** (Reverse product identity). [Lean](#) [✓ Stmt](#) If  $P_K = P_{AX} \circ P_{XB}$  is a commuting parent Hamiltonian decomposition, then

$$P_{XB} \circ P_{AX} = P_K.$$

*Proof.* [✓ Proof](#) This is immediate from commutativity of  $P_{AX}$  and  $P_{XB}$  together with the identity  $P_{AX} \circ P_{XB} = P_K$ .  $\square$

**Theorem 16.131** (Commuting complementary projectors). [Lean](#) [✓ Stmt](#) If  $P_K = P_{AX} \circ P_{XB}$  is a commuting parent Hamiltonian decomposition and

$$Q_{AX} = 1 - P_{AX}, \quad Q_{XB} = 1 - P_{XB},$$

then

$$Q_{AX} \circ Q_{XB} = Q_{XB} \circ Q_{AX}.$$

*Proof.* [✓ Proof](#) Expanding both products and using  $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$  shows that the complementary projectors commute.  $\square$

**Theorem 16.132** (Commuting parent Hamiltonian implies decorrelation). [Lean](#) [✓ Stmt](#) If a subspace  $K$  has a commuting parent Hamiltonian decomposition  $P_K = P_{AX} \circ P_{XB}$  with  $[P_{AX}, P_{XB}] = 0$ , and  $A$ -observables commute with  $P_{XB}$  while  $B$ -observables commute with  $P_{AX}$ , then regions  $A$  and  $B$  are decorrelated w.r.t.  $K$ . This is the backward direction of [CPGSV16, Appendix D, Section D.2, Proposition D.3].

*Proof.* [✓ Proof](#) The key identity is  $P_{XB} \circ (1 - P_{AX} \circ P_{XB}) \circ P_{AX} = 0$ . Using the commutation hypotheses to slide  $O_A$  past  $P_{XB}$  and  $O_B$  past  $P_{AX}$ , the five-fold composition factors through this zero term.  $\square$

**Theorem 16.133** (Commuting parent Hamiltonian implies decorrelation). [Lean](#) [✓ Stmt](#) Corollary of Theorem 16.132: if  $P_K$  has a commuting parent Hamiltonian decomposition and observables respect locality, then regions  $A$  and  $B$  are decorrelated w.r.t.  $K$ .

**Lemma 16.134** (Frustration-free identity). [Lean](#) [✓ Stmt](#) For any endomorphisms  $P$  and  $Q$ ,

$$(1 - P) + (1 - Q) - (1 - P) \circ (1 - Q) = 1 - P \circ Q.$$

*Proof.* [✓ Proof](#) Expanding the left-hand side and cancelling gives the result.  $\square$

**Theorem 16.135** (Commuting parent Hamiltonian identity). [Lean](#) [✓ Stmt](#) If  $P_K = P_{AX} \circ P_{XB}$  is a commuting parent Hamiltonian decomposition, then

$$Q_{AX} + Q_{XB} - Q_{AX} \circ Q_{XB} = 1 - P_K,$$

where  $Q_{AX} = 1 - P_{AX}$  and  $Q_{XB} = 1 - P_{XB}$  are the complementary projectors. This is the frustration-free form of the parent Hamiltonian from [CPGSV16, Appendix D, Section D.2].

*Proof.* [✓ Proof](#) Applying Lemma 16.134 to  $P_{AX}$  and  $P_{XB}$ , and substituting  $P_{AX} \circ P_{XB} = P_K$ .  $\square$

**Theorem 16.136** (Ground-space membership). [Lean](#) [✓ Stmt](#) If  $P_K = P_{AX} \circ P_{XB}$  is a commuting parent Hamiltonian decomposition, then

$$P_K v = v \iff P_{AX} v = v \text{ and } P_{XB} v = v.$$

This is the algebraic form of equation (D.2) from [CPGSV16]:  $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$ .

*Proof.* [✓ Proof](#) ( $\Rightarrow$ ) If  $P_K v = v$ , then  $P_{AX}(P_K v) = (P_{AX} \circ P_K)v = P_K v = v$  by left absorption, and similarly for  $P_{XB}$ .

( $\Leftarrow$ ) If  $P_{AX} v = v$  and  $P_{XB} v = v$ , then  $P_K v = (P_{AX} \circ P_{XB})v = P_{AX}(P_{XB} v) = P_{AX} v = v$ .  $\square$

## 16.6 Spectral gap

A frustration-free Hamiltonian is *gapped* if the first excited eigenvalue is bounded away from zero uniformly in the chain length  $N$ . For MPS parent Hamiltonians, the spectral gap follows from a martingale criterion due to Kastoryano and Lucia [KL18].

**Definition 16.137** (Transported parent ground space). Lean ✓ Stmt Denote by  $\mathcal{G}_{N,L}^{\text{ES}}(A)$  the parent ground space after identifying the coefficient space with its Euclidean  $\ell^2$ -model.

**Definition 16.138** (Transported parent Hamiltonian). Lean ✓ Stmt Denote by  $H_{N,L}^{\text{ES}}(A)$  the conjugate of  $H_N(A, L)$  by the canonical identification between the coefficient space and its Euclidean avatar.

**Theorem 16.139** (Euclidean transport of the parent ground space). Lean ✓ Stmt *Under the canonical  $\ell^2$  transport of the  $N$ -site state space, the transported parent ground space is exactly the kernel of the transported parent Hamiltonian.*

*Proof.* ✓ Proof This is an immediate unraveling of the definitions: both constructions are obtained from the same parent Hamiltonian by conjugating with the canonical identification between the site space and its Euclidean-space avatar, so the transported ground space is precisely the transported kernel. □

**Lemma 16.140** (Membership in the transported parent ground space). Lean ✓ Stmt *A vector belongs to the transported parent ground space exactly when the transported parent Hamiltonian annihilates it.*

*Remark 16.141* (Euclidean local projector ingredients). Lean Lean Lean Lean ✓ Stmt Let  $P_L^{\text{ES}}(A)$  denote the orthogonal projector onto  $(G_L^{\text{ES}}(A))^\perp$  on the Euclidean  $L$ -site space. For a cyclic window starting at  $i$  and an outside configuration  $\tau$ , the transported restriction map  $R_{i,\tau}$  evaluates an  $N$ -site Euclidean vector on that window. The transported local term  $h_{i,\text{ES}}$  is the corresponding  $N$ -site local projector, and  $R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}$  is the basic positive summand in its cyclic-averaging formula.

**Lemma 16.142** (Positivity of the Euclidean local summands). Lean Lean Lean ✓ Stmt *The Euclidean local projector  $P_L^{\text{ES}}(A)$  is positive. Consequently, every summand  $R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}$  is positive, and so is the finite sum over  $\tau$ .*

*Proof.* ✓ Proof Orthogonal projectors are positive. Conjugation by  $R_{i,\tau}$  preserves positivity, and finite sums of positive operators remain positive. □

**Lemma 16.143** (Cyclic averaging for the transported ES local terms). Lean Lean ✓ Stmt *The transported parent Hamiltonian is the sum of the transported local terms  $H_{N,\text{ES}} = \sum_i h_{i,\text{ES}}$ , and each transported local term admits the exact cyclic-averaging formula*

$$h_{i,\text{ES}} = d^{-L} \sum_{\tau} R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}. \quad (16.5)$$

*Proof.* ✓ Proof Unfold the transported local term and compare it configuration-by-configuration with the cyclic restrictions. The  $d^{-L}$  factor compensates for the  $d^L$  choices of the window coordinates in the ambient configuration parameter  $\tau$ . □

**Lemma 16.144** (Positivity of the transported ES Hamiltonian). Lean Lean ✓ Stmt *Each transported local term  $h_{i,\text{ES}}$  is positive, hence the full transported parent Hamiltonian  $H_{N,\text{ES}}$  is positive.*

*Proof.* ✓ Proof Apply the averaging formula (16.5):  $h_{i,\text{ES}}$  is a non-negative scalar multiple of a finite sum of positive conjugates of  $P_L^{\text{ES}}(A)$ , hence is itself positive. Summing over  $i$  yields positivity of  $H_{N,\text{ES}}$ .  $\square$

**Lemma 16.145** (Symmetric-projection structure of transported ES local terms). Lean Lean  
✓ Stmt *The Euclidean local projector  $P_L^{\text{ES}}(A)$  is a symmetric projection, and every transported local term  $h_{i,\text{ES}}$  on the  $N$ -site chain is a symmetric projection.*

*Proof.* ✓ Proof The local operator  $P_L^{\text{ES}}(A)$  is an orthogonal projection onto  $(G_L^{\text{ES}}(A))^\perp$ . For  $L \leq N$ , restricting  $h_{i,\text{ES}}v$  to a cyclic window gives  $P_L^{\text{ES}}(A)$  applied to the corresponding restriction of  $v$ ; applying  $h_{i,\text{ES}}$  a second time therefore applies  $(P_L^{\text{ES}}(A))^2 = P_L^{\text{ES}}(A)$  on that window. The positivity result above supplies symmetry, and the case  $L > N$  is the zero projection by definition.  $\square$

**Lemma 16.146** (Kernel of the Euclidean parent interaction). Lean ✓ Stmt *The Euclidean local interaction  $P_L^{\text{ES}}(A)$  annihilates exactly the Euclidean local ground space:*

$$P_L^{\text{ES}}(A)v = 0 \iff v \in G_L^{\text{ES}}(A).$$

*Proof.* ✓ Proof Since  $P_L^{\text{ES}}(A)$  is the orthogonal projector onto  $(G_L^{\text{ES}}(A))^\perp$ , its kernel is the original subspace  $G_L^{\text{ES}}(A)$ .  $\square$

**Lemma 16.147** (Local-term kernel via cyclic restrictions). Lean Lean ✓ Stmt *For  $L \leq N$ , a transported local term  $h_{i,\text{ES}}$  annihilates a vector  $v$  iff every boundary-filled restriction of  $v$  to the cyclic  $L$ -window starting at  $i$  lies in  $G_L^{\text{ES}}(A)$ .*

*Proof.* ✓ Proof Restricting  $h_{i,\text{ES}}v$  to the same cyclic window gives  $P_L^{\text{ES}}(A)$  applied to the corresponding restriction of  $v$ . Lemma 16.146 identifies the vanishing of this local projector with membership in  $G_L^{\text{ES}}(A)$ .  $\square$

**Theorem 16.148** (Adjacent local kernels realize the intersection property). Lean Lean Lean  
✓ Stmt *If  $A$  is injective and  $L \geq 2$ , then on an  $(L+1)$ -site chain the kernels of the two adjacent transported  $L$ -site local terms are exactly the transported  $(L+1)$ -site MPS ground space:*

$$h_{0,\text{ES}}v = 0 \text{ and } h_{1,\text{ES}}v = 0 \iff v \in G_{L+1}^{\text{ES}}(A).$$

*Proof.* ✓ Proof Lemma 16.147 translates the two local kernel assumptions into the left and right  $L$ -site ground conditions. The open-chain intersection property, Theorem 16.34, then grows back the shared  $(L-1)$ -site overlap to the  $(L+1)$ -site ground space. The converse uses the forward restriction lemmas for ground-space vectors and the same kernel–restriction characterization.  $\square$

**Lemma 16.149** (Non-overlap positivity for transported ES local terms). Lean Lean Lean Lean  
Lean ✓ Stmt *Let  $L \leq N$  and let  $h_{i,\text{ES}}, h_{j,\text{ES}}$  be transported local terms whose cyclic  $L$ -windows are site-disjoint: no site of the periodic chain has cyclic offset  $< L$  from both  $i$  and  $j$ . Then the terms commute pointwise: for every vector  $v$ ,*

$$h_{i,\text{ES}}(h_{j,\text{ES}}v) = h_{j,\text{ES}}(h_{i,\text{ES}}v),$$

*and their ordered cross term is non-negative:*

$$0 \leq \text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle.$$

*Proof.* ✓ Proof In coordinates,  $h_{i,\text{ES}}$  applies  $P_L^{\text{ES}}(A)$  only to the  $i$ -window variables and leaves the complementary variables fixed. Site disjointness gives two replacement identities: replacing the  $i$ -window does not change the extracted  $j$ -window, and the  $i$ - and  $j$ -window replacements commute. Therefore the two embedded local operators commute. The preceding projection-geometry identity for commuting symmetric projections gives the non-negative ordered cross term.  $\square$

**Lemma 16.150** (Quadratic form to norm bound). Lean ✓ Stmt Let  $H$  be a positive semidefinite self-adjoint operator on a finite-dimensional complex Hilbert space, and suppose that, for all  $v$ ,

$$\gamma \langle v, Hv \rangle \leq \langle Hv, Hv \rangle$$

i.e.  $H^2 \geq \gamma H$  as a quadratic form, for some  $\gamma > 0$ . Then for every  $v$  in the orthogonal complement of  $\ker H$ ,

$$\gamma \|v\| \leq \|Hv\|.$$

In particular, every positive eigenvalue of  $H$  is at least  $\gamma$ . The positive-semidefiniteness hypothesis is essential: without it,  $H = -\text{Id}$  with  $\gamma = 2$  satisfies the quadratic-form inequality vacuously but violates the norm bound.

*Proof.* ✓ Proof By the finite-dimensional spectral theorem,  $H$  diagonalises in an orthonormal eigenbasis with real eigenvalues  $\lambda_k$ . For a unit eigenvector  $\psi_k$  with eigenvalue  $\lambda_k$  the hypothesis gives  $\gamma \lambda_k \leq \lambda_k^2$ , whence  $\lambda_k \leq 0$  or  $\lambda_k \geq \gamma$ . Since  $H$  is positive semidefinite (its quadratic form is  $\geq 0$ ), the negative case is excluded and every nonzero eigenvalue satisfies  $\lambda_k \geq \gamma$ . Writing  $v = \sum_k c_k \psi_k$  and projecting onto  $(\ker H)^\perp$  keeps only terms with  $\lambda_k \geq \gamma$ , so  $\|Hv\|^2 = \sum_k |c_k|^2 \lambda_k^2 \geq \gamma^2 \sum_k |c_k|^2 = \gamma^2 \|v\|^2$ .  $\square$

**Lemma 16.151** (Parent-Hamiltonian quadratic-form reduction). Lean Lean ✓ Stmt Let  $H_{N,L}^{\text{ES}}(A)$  be the transported parent Hamiltonian. If a constant  $\gamma > 0$  satisfies the uniform quadratic-form estimate

$$\gamma \operatorname{Re} \langle H_{N,L}^{\text{ES}}(A)v, v \rangle \leq \operatorname{Re} \langle H_{N,L}^{\text{ES}}(A)v, H_{N,L}^{\text{ES}}(A)v \rangle \quad (16.6)$$

for every  $N \geq 2L$  and every vector  $v$ , then

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|$$

for every  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ . In particular, with  $L > 1$  and  $\gamma = 1/(4L)$ , the explicit gap-bound conclusion follows once this same quadratic-form estimate (16.6) is supplied.

*Proof.* ✓ Proof The transported Hamiltonian is positive by Lemma 16.144, and its transported ground space is its kernel by Theorem 16.139. Therefore the abstract spectral-theorem conversion of Lemma 16.150 applies directly. The explicit constant is positive because  $L > 1$  implies  $4L > 0$ .  $\square$

**Remark 16.152** (Projection-geometry identities for row-sum arguments). Lean Lean Lean Lean ✓ Stmt The projection-geometry reduction uses the following elementary Hilbert-space identities: the diagonal identity  $\operatorname{Re} \langle Pv, Pv \rangle = \operatorname{Re} \langle Pv, v \rangle$  for symmetric projections; nonnegativity of this quadratic form; nonnegativity of  $\operatorname{Re} \langle Pv, Qv \rangle$  for commuting symmetric projections  $P, Q$ ; the Cauchy–Schwarz conversion from  $\|PQv\| \leq c\|Pv\|$  to  $\operatorname{Re} \langle Pv, Qv \rangle \geq -c \operatorname{Re} \langle Pv, v \rangle$ ; finite-sum expansions of  $\operatorname{Re} \langle (\sum_i P_i)v, v \rangle$  and  $\operatorname{Re} \langle (\sum_i P_i)v, (\sum_i P_i)v \rangle$ ; the diagonal/off-diagonal split of the ordered double sum; and the aggregate off-diagonal estimate obtained from row sums  $\sum_{j \neq i} c_{ij} \leq 1$ .

**Lemma 16.153** (Ordered projection row-sum quadratic-form reduction). Lean ✓ Stmt Let  $P_i$  be a finite family of symmetric projections and let  $H = \sum_i P_i$ . If the ordered off-diagonal forms obey

$$\operatorname{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) c_{ij} \operatorname{Re}\langle P_i v, v \rangle \quad (i \neq j),$$

with row sums  $\sum_{j \neq i} c_{ij} \leq 1$  and  $\gamma \leq 1$ , then  $H^2 \geq \gamma H$  as a quadratic form.

*Proof.* ✓ Proof Expand  $\operatorname{Re}\langle H v, H v \rangle$  as an ordered double sum. Symmetric idempotence identifies each diagonal term with  $\operatorname{Re}\langle P_i v, v \rangle$ , while the row-summable ordered cross-term bounds control the off-diagonal contribution by  $-(1 - \gamma) \sum_i \operatorname{Re}\langle P_i v, v \rangle$ .  $\square$

**Lemma 16.154** (Finite-overlap projection row-sum reduction). Lean ✓ Stmt Let  $P_i$  be a finite family of symmetric projections and let  $\gamma \leq 1$ . Suppose an interaction predicate  $i \sim j$  has at most  $m$  off-diagonal neighbours in each row for some positive integer  $m$ . If noninteracting ordered cross terms are non-negative and, on each interacting pair,

$$\operatorname{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) \frac{1}{m} \operatorname{Re}\langle P_i v, v \rangle,$$

then  $H = \sum_i P_i$  satisfies  $H^2 \geq \gamma H$  as a quadratic form.

*Proof.* ✓ Proof Choose  $c_{ij} = 1/m$  on interacting pairs and  $c_{ij} = 0$  otherwise. The row cardinality bound gives  $\sum_{j \neq i} c_{ij} \leq 1$ ; noninteracting nonnegativity supplies the zero-coefficient cross-term estimates. Apply Lemma 16.153.  $\square$

**Lemma 16.155** (Finite-overlap projection norm-compression reduction). Lean Lean ✓ Stmt Let  $P_i$  be a finite family of symmetric projections and let  $\gamma \leq 1$ . Suppose an interaction predicate has at most  $m > 0$  off-diagonal neighbours in each row, noninteracting ordered cross terms are non-negative, and interacting pairs satisfy the norm-compression estimate

$$\|P_i P_j v\| \leq (1 - \gamma) \frac{1}{m} \|P_i v\|. \quad (16.7)$$

Then  $H = \sum_i P_i$  satisfies  $H^2 \geq \gamma H$  as a quadratic form. The same conclusion holds from a separate compression coefficient  $\eta$  whenever  $\eta \leq (1 - \gamma)/m$ .

*Proof.* ✓ Proof Cauchy–Schwarz and symmetric idempotence turn the norm-compression estimate into  $\operatorname{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) m^{-1} \operatorname{Re}\langle P_i v, v \rangle$ . If a separate coefficient  $\eta$  is used, the assumption  $\eta \leq (1 - \gamma)/m$  first weakens that estimate to (16.7). The finite-overlap row-sum reduction then applies.  $\square$

**Lemma 16.156** (Parent-Hamiltonian ordered row-sum quadratic-form reduction). Lean ✓ Stmt For a fixed chain length  $N$ , suppose the transported local terms  $h_{i,\text{ES}}$  satisfy the ordered row-sum estimate

$$\operatorname{Re}\langle h_{i,\text{ES}} v, h_{j,\text{ES}} v \rangle \geq -(1 - \gamma) c_{ij} \operatorname{Re}\langle h_{i,\text{ES}} v, v \rangle \quad (i \neq j), \quad (16.8)$$

with  $\sum_{j \neq i} c_{ij} \leq 1$  and  $\gamma \leq 1$ . Then the transported parent Hamiltonian satisfies

$$\gamma \operatorname{Re}\langle H_{N,L}^{\text{ES}} v, v \rangle \leq \operatorname{Re}\langle H_{N,L}^{\text{ES}} v, H_{N,L}^{\text{ES}} v \rangle$$

for every vector  $v$ .

*Proof.* ✓ Proof Lemma 16.145 supplies the symmetric-projection hypothesis for the family  $P_i = h_{i,\text{ES}}$ . Apply Lemma 16.153 to this family and use  $H_{N,L}^{\text{ES}} = \sum_i h_{i,\text{ES}}$ .  $\square$



**Lemma 16.157** (Parent-Hamiltonian gap from ordered local row bounds). [Lean](#) [✓ Stmt](#) Assume uniformly for every  $N \geq 2L$  that the transported local terms satisfy the ordered row-sum estimate (16.8) with  $\gamma = 1/(4L)$ . If  $L > 1$ , then  $1/(4L) > 0$  and, for every  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ ,

$$\frac{1}{4L} \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

*Proof.* [✓ Proof](#) The fixed-chain row-sum reduction supplies the quadratic-form estimate with  $\gamma = 1/(4L)$  for every admissible  $N$ . The positivity and kernel identification of the transported Hamiltonian then allow Lemma 16.151 to convert this quadratic-form estimate into the norm lower bound on the orthogonal complement of the ground space.  $\square$

**Lemma 16.158** (Parent-Hamiltonian finite-overlap Friedrichs quadratic reduction). [Lean](#) [✓ Stmt](#) For a fixed chain length  $N$ , let  $m$  be a positive integer and let  $\gamma \leq 1$ . Suppose an overlap predicate on local windows has at most  $m$  overlapping off-diagonal windows in each row. If non-overlapping local terms have non-negative ordered cross terms, overlapping local terms satisfy the finite-overlap Friedrichs estimate

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1-\gamma) \frac{1}{m} \text{Re}\langle h_{i,\text{ES}}v, v \rangle,$$

and the transported local terms are symmetric projections, then the transported Hamiltonian satisfies  $H_{N,L}^{\text{ES}^2} \geq \gamma H_{N,L}^{\text{ES}}$  as a quadratic form.

*Proof.* [✓ Proof](#) Apply the finite-overlap projection reduction with  $P_i = h_{i,\text{ES}}$ .  $\square$

**Lemma 16.159** (Finite-overlap norm-compression reduction). [Lean](#) [Lean](#) [✓ Stmt](#) For a fixed chain length  $N$ , let  $m > 0$  and  $\gamma \leq 1$ . Suppose an overlap predicate on local windows has at most  $m$  overlapping off-diagonal windows in each row, non-overlapping local terms have non-negative ordered cross terms, and overlapping local terms satisfy

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq (1-\gamma) \frac{1}{m} \|h_{i,\text{ES}}v\|.$$

Then  $H_{N,L}^{\text{ES}^2} \geq \gamma H_{N,L}^{\text{ES}}$  as a quadratic form. The same conclusion holds from a separate coefficient  $\eta$  satisfying  $\eta \leq (1-\gamma)/m$ .

*Proof.* [✓ Proof](#) Apply the finite-overlap norm-compression projection reduction to the family  $P_i = h_{i,\text{ES}}$ . If a separate coefficient  $\eta$  is used, first weaken the compression estimate using  $\eta \leq (1-\gamma)/m$ .  $\square$

**Definition 16.160** (Cyclic-window support). [Lean](#) [✓ Stmt](#) For a length- $L$  cyclic window on  $\{0, \dots, N-1\}$  starting at  $i$ , its support is

$$S_i = \{i, i+1, \dots, i+L-1\} \pmod{N},$$

with repeated visits counted once.

**Lemma 16.161** (Cyclic-window support offsets). [Lean](#) [✓ Stmt](#) If  $L \leq N$ , then a site  $k$  belongs to the cyclic support  $S_i$  exactly when its cyclic offset from  $i$  is below  $L$ :

$$k \in S_i \iff (k-i) \bmod N < L.$$

*Proof.* [✓ Proof](#) Write  $k = i + r \pmod{N}$  for a representative offset  $r$ . Membership in  $S_i$  gives an offset  $r < L$ , and conversely an offset below  $L$  gives the corresponding point of the support.  $\square$



**Definition 16.162** (Cyclic-window overlap). [Lean](#) [✓ Stmt](#) Two length- $L$  cyclic windows overlap, written  $i \sim_L j$ , when their supports meet:  $S_i \cap S_j \neq \emptyset$ .

**Lemma 16.163** (Concrete non-overlap positivity for cyclic supports). [Lean](#) [Lean](#) [✓ Stmt](#) If  $L \leq N$  and the cyclic supports  $S_i$  and  $S_j$  do not meet, then the transported local ES terms have non-negative ordered cross term:

$$0 \leq \text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle.$$

*Proof.* [✓ Proof](#) The support-offset characterization converts failure of  $S_i \cap S_j \neq \emptyset$  into site-disjointness of the two cyclic windows. The disjoint-window positivity lemma then applies.  $\square$

**Lemma 16.164** (Cyclic-window overlap row cardinality). [Lean](#) [✓ Stmt](#) If  $2L \leq N$  and  $L > 1$ , then for every  $i \in \{0, \dots, N-1\}$ ,

$$\#\{j \neq i : i \sim_L j\} \leq 2(L-1).$$

*Proof.* [✓ Proof](#) If  $S_i$  and  $S_j$  meet, choose offsets  $a, b < L$  with  $i + a \equiv j + b \pmod{N}$ . Since  $2L \leq N$ , the offset from  $i$  to  $j$  is either the clockwise distance  $a - b \in \{1, \dots, L-1\}$  or the counterclockwise distance  $b - a \in \{1, \dots, L-1\}$ ; the zero distance is excluded by  $j \neq i$ . Thus every overlapping off-diagonal start lies among the  $L-1$  clockwise starts or the  $L-1$  counterclockwise starts.  $\square$

**Lemma 16.165** (Parent-Hamiltonian gap from finite-overlap Friedrichs data). [Lean](#) [✓ Stmt](#) Fix  $L > 1$  and set  $m = 2(L-1)$ . Suppose uniformly for every  $N \geq 2L$  that the hypotheses of the fixed-chain finite-overlap Friedrichs reduction hold with  $\gamma = 1/(4L)$  and this value of  $m$ . Then  $1/(4L) > 0$ , and the explicit norm lower bound with constant  $1/(4L)$  follows for vectors in  $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ .

*Proof.* [✓ Proof](#) The fixed-chain finite-overlap reduction gives the required quadratic-form estimate for every admissible  $N$ . Lemma 16.151 converts that estimate to the norm lower bound.  $\square$

**Lemma 16.166** (Parent-Hamiltonian gap from cyclic-window Friedrichs data). [Lean](#) [✓ Stmt](#) Fix  $L > 1$  and let  $i \sim_L j$  mean that the cyclic supports of the length- $L$  windows at  $i$  and  $j$  meet. Suppose uniformly for every  $N \geq 2L$  that overlapping off-diagonal terms satisfy

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1 - \frac{1}{4L}) \frac{1}{2(L-1)} \text{Re}\langle h_{i,\text{ES}}v, v \rangle. \quad (16.9)$$

Then  $1/(4L) > 0$ , and the explicit norm lower bound with constant  $1/(4L)$  follows for vectors in  $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ .

*Proof.* [✓ Proof](#) Lemma 16.164 gives the row-cardinality bound  $m = 2(L-1)$  for the concrete cyclic-window overlap relation, and Lemma 16.145 supplies the symmetric projection structure of each transported local term. If two cyclic supports do not meet, the corresponding windows are site-disjoint, so Lemma 16.163 gives the required non-negative ordered cross term. Apply Lemma 16.165 with these facts and the overlapping-window estimate (16.9).  $\square$

**Lemma 16.167** (Parent-Hamiltonian gap from overlapping cyclic Friedrichs data). [Lean](#) [✓ Stmt](#) Fix  $L > 1$  and let  $i \sim_L j$  mean that the cyclic supports of the length- $L$  windows at  $i$  and  $j$  meet. Suppose uniformly for every  $N \geq 2L$  that overlapping off-diagonal terms satisfy

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1 - \frac{1}{4L}) \frac{1}{2(L-1)} \text{Re}\langle h_{i,\text{ES}}v, v \rangle. \quad (16.10)$$

Then the explicit norm lower bound with constant  $1/(4L)$  follows for vectors in  $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ .

*Proof.* ✓ Proof Apply Lemma 16.166. Its non-overlap positivity and finite-row hypotheses are already internal to that cyclic-window reduction; the overlapping estimate (16.10) supplies its remaining ordered cross-term condition.  $\square$

**Lemma 16.168** (Parent-Hamiltonian gap from overlapping cyclic norm-compression data). Lean ✓ Stmt Fix  $L > 1$ . Suppose uniformly for every  $N \geq 2L$  and every overlapping off-diagonal pair that

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq (1 - \frac{1}{4L}) \frac{1}{2(L-1)} \|h_{i,\text{ES}}v\|.$$

Then the explicit norm lower bound with constant  $1/(4L)$  follows for vectors in  $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ .

*Proof.* ✓ Proof The norm-compression condition is converted to the ordered Friedrichs lower bound by the projection-geometry Cauchy-Schwarz lemma. The preceding overlapping-cyclic Friedrichs reduction then gives the norm lower bound.  $\square$

**Lemma 16.169** (Parent-Hamiltonian gap from a cyclic overlap compression constant). Lean ✓ Stmt Fix  $L > 1$ , and let  $m = 2(L-1)$ . Suppose uniformly for every  $N \geq 2L$  and every overlapping off-diagonal pair that

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \eta \|h_{i,\text{ES}}v\|.$$

If  $\gamma > 0$ ,  $\gamma \leq 1$ , and  $\eta \leq (1 - \gamma)/m$ , then  $0 < \gamma$  and, for every admissible  $N$  and every  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ ,

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

The assertion  $0 < \gamma$  repeats the hypothesis, in the same shape as the strict statement below. In particular, if  $\eta \geq 0$  and  $\eta m < 1$ , then  $0 < 1 - \eta m$  and, for every admissible  $N$  and every  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ ,

$$(1 - \eta m) \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

*Proof.* ✓ Proof The cyclic-window cardinality lemma gives at most  $m = 2(L-1)$  overlapping off-diagonal neighbours in each row, while the non-overlap positivity lemma handles disjoint supports. The finite-overlap norm-compression reduction gives the quadratic-form estimate with the chosen  $\gamma$ , and the positive quadratic-form criterion converts it to the norm lower bound. The strict form is the same statement with  $\gamma = 1 - \eta m$ .  $\square$

**Theorem 16.170** (Martingale criterion for spectral gap). Not ready Consider a frustration-free Hamiltonian

$$H = \sum_{i=1}^n h_i$$

with projector interaction terms ( $h_i^2 = h_i$ ). Suppose that for every pair of overlapping terms  $h_i, h_j$  there exist constants  $c_{ij} \geq 0$  and  $\gamma > 0$  satisfying

$$h_i h_j + h_j h_i \geq -c_{ij}(1 - \gamma)(h_i + h_j), \tag{16.11}$$

and suppose further that  $c_{ij} = c_{ji}$  for every overlapping pair  $(i, j)$  and that  $\sum_{j: j \neq i} c_{ij} \leq 1$  for every  $i$ . Then the smallest positive eigenvalue of  $H$  satisfies  $\lambda_{\min}^+(H) \geq \gamma$ .

*Proof.* **Not ready** The relevant geometry is a family of translated length- $L$  projector windows on the ring; two terms  $h_i$  and  $h_j$  interact only when their supports overlap, equivalently when the cyclic distance  $d_N(i, j)$  is smaller than  $L$ . From  $h_i^2 = h_i$ ,

$$H^2 = \sum_i h_i + \sum_{\substack{i < j \\ \text{overlapping}}} (h_i h_j + h_j h_i) + \sum_{\substack{i < j \\ \text{non-overlapping}}} (h_i h_j + h_j h_i).$$

Non-overlapping projectors commute ( $h_i h_j = h_j h_i$ ), so their product  $h_i h_j$  is itself a projector and hence PSD; thus the last sum is  $\geq 0$ . For the overlapping sum, the martingale condition (16.11) gives  $h_i h_j + h_j h_i \geq -c_{ij}(1 - \gamma)(h_i + h_j)$ . Summing over unordered pairs  $\{i, j\}$ :

$$\sum_{\{i, j\}, \text{overlap}} (h_i h_j + h_j h_i) \geq -(1 - \gamma) \sum_{\{i, j\}, \text{overlap}} c_{ij} (h_i + h_j).$$

Collecting terms for each  $h_i$  and using the symmetry assumption  $c_{ij} = c_{ji}$ :

$$\sum_{\{i, j\}, \text{overlap}} c_{ij} (h_i + h_j) = \sum_i \left( \sum_{\substack{j \neq i \\ \text{overlap}}} c_{ij} \right) h_i \leq H,$$

where the last inequality uses the row-sum hypothesis  $\sum_{j \neq i} c_{ij} \leq 1$ . Combining,  $H^2 \geq H - (1 - \gamma)H = \gamma H$ . For any eigenvector with  $H\psi = \lambda\psi$  and  $\lambda > 0$ , this gives  $\lambda^2 \geq \gamma\lambda$ , hence  $\lambda \geq \gamma$  [KL18].  $\square$

**Lemma 16.171** (Martingale condition for MPS). **Not ready** Let  $A$  be injective and fix an interaction range  $L \geq 2$  as in Definition 16.10. Let  $h_L(A) := \Pi_{G_L(A)^\perp}$  be the  $L$ -site parent interaction, and for each chain length and site  $i$  let  $h_i$  denote its translate as in Definition 16.16. Then for every pair of overlapping terms  $h_i$  and  $h_j$  with  $0 < d_N(i, j) < L$ , where  $d_N(i, j) := \min(|i - j|, N - |i - j|)$  is the cyclic distance on the ring (i.e. whose  $L$ -site supports share at least one site under periodic boundary conditions), the martingale condition of Theorem 16.170 holds with constants  $c_{ij}$  and  $\gamma > 0$  depending only on  $A$  (not on  $N$ ). Cirac–Perez-García–Schuch–Verstraete [CPGSV21, Section IV.C] obtain this uniformity by the martingale method, invoking finite-chain-state estimates to control the constants. For the explicit cyclic-window constants displayed here, the remaining quantitative step is the overlapping-window Friedrichs-angle bound in Theorem 16.172; the preceding lemmas reduce the martingale inequality and the spectral gap to that estimate. (When  $L = 1$  the interaction terms have disjoint supports, so there are no overlapping pairs and the spectral gap follows directly without the martingale argument.)

*Proof.* **Not ready** Two terms  $h_i, h_j$  with  $0 < d_N(i, j) < L$  have supports overlapping in  $L - d_N(i, j)$  sites (where  $d_N$  is the cyclic distance on the  $N$ -site ring). Since  $A$  is injective, the intersection property (Theorem 16.34) gives the qualitative identity  $\ker(h_i) \cap \ker(h_j) = G_{[i, j+L-1]}(A)$ . Because the local Hilbert space is finite-dimensional, the subspaces  $\ker(h_i)$  and  $\ker(h_j)$  live in a fixed finite-dimensional space. The intersection identity shows that  $\ker(h_i) \cap \ker(h_j)$  is exactly  $G_{[i, j+L-1]}(A)$ ; in particular there is no non-trivial vector in one kernel that is orthogonal to the other yet lies in both. Hence the Friedrichs angle  $\theta_{d_N(i, j)}$  between  $\ker(h_i)$  and  $\ker(h_j)$  is strictly positive (for the fixed injective tensor  $A$ ). Because the angle depends only on the overlap pattern  $d_N(i, j) \in \{1, \dots, L - 1\}$  and not on the absolute position, the number of distinct angles is at most  $L - 1$ . Let  $\alpha := \max_{1 \leq s < L} \cos \theta_s < 1$  be the worst-case cosine of the Friedrichs angles.

*Row-sum bound.* Since  $L \geq 2$ , each site  $i$  overlaps with at most  $2(L - 1)$  neighbours (those  $j$  with  $0 < d_N(i, j) < L$ ). Set  $c_{ij} := \frac{1}{2(L-1)}$  for every overlapping pair and  $c_{ij} := 0$  otherwise.

These constants depend only on the cyclic distance  $d_N(i, j)$  and therefore satisfy  $c_{ij} = c_{ji}$ . Then  $\sum_{j \neq i} c_{ij} \leq 2(L-1) \cdot \frac{1}{2(L-1)} = 1$ , satisfying the row-sum hypothesis of Theorem 16.170.

*Explicit cyclic-window constant.* With the row coefficient above and

$$\gamma_L = \frac{1}{4L},$$

the martingale condition is the fixed cyclic-window estimate

$$h_i h_j + h_j h_i \geq -\left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} (h_i + h_j)$$

for every overlapping off-diagonal pair. A sufficient form, used below, is the one-sided norm-compression estimate

$$\|h_i h_j v\| \leq \left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} \|h_i v\|.$$

The qualitative Friedrichs-angle conclusion  $\alpha < 1$  does not by itself imply this numerical bound; the missing step is the uniform overlapping-window estimate with the displayed coefficient. Thus the remaining comparison is whether the principal-angle estimates cited in [FNW92, Nac96, KL18] supply this cyclic-window inequality, with constants independent of  $N$ , under the convention used here.  $\square$

**Theorem 16.172** (Explicit Friedrichs-angle gap bound). Lean ✓ Stmt *For an injective tensor  $A$  and a range  $L > 1$ , the theorem asserts the analytic estimate with the explicit constant*

$$\gamma_L := \frac{1}{4L} > 0.$$

*Concretely, it asserts that if  $N \geq 2L$  and  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ , then*

$$\gamma_L \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

*Proof.* Not ready The local averaging, positivity, and symmetric-projection statements are now established in Lemmas 16.143, 16.144, and 16.145. Lemma 16.151 reduces the present theorem to the remaining MPS-specific task: deriving the uniform quadratic-form estimate. Lemma 16.157 establishes the finite-sum algebra from ordered cross-term row bounds, and Lemmas 16.158 and 16.165 state the common finite-overlap specialization with  $m = 2(L-1)$ . Lemma 16.149 supplies the non-overlap positivity statement for site-disjoint windows, and Lemma 16.163 translates this to the concrete support-disjoint cyclic windows. Lemma 16.164 gives the finite-overlap row-cardinality bound. Lemma 16.166 combines these ingredients with the finite-overlap reduction, so the remaining analytic task is the Friedrichs-angle estimate for overlapping cyclic windows. Lemma 16.167 states the same overlap-only formulation, and Lemma 16.168 states a sufficient norm-compression formulation.  $\square$

**Theorem 16.173** (Spectral gap for MPS parent Hamiltonians). Lean ✓ Stmt *For an injective tensor  $A$  and a range  $L > 1$ , there exists  $\gamma > 0$  such that for every  $N \geq 2L$  and every vector  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$  one has*

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

*By Lemma 16.150, this implies the usual uniform lower bound on every nonzero eigenvalue of the parent Hamiltonian.*

*Proof.* Not ready This follows immediately from Theorem 16.172 by taking as  $\gamma$  the explicit constant produced there, which yields the required existential spectral-gap constant.  $\square$

## 16.7 Ground space of non-injective MPS

For an MPS with more than one block in canonical form, after blocking to block-injective canonical form, the physical–virtual correspondence is restricted to block-diagonal boundary conditions. In the notation of [CPGSV21, lines 2113–2117], set

$$\tilde{A}^\omega := (A_1^\omega, \dots, A_g^\omega) \in \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Then block injectivity is the statement

$$\forall X = (X_j)_{j=1}^g \in \bigoplus_{j=1}^g M_{D_j}(\mathbb{C}), \quad \exists (c_\omega(X))_\omega, \quad X = \sum_\omega c_\omega(X) \tilde{A}^\omega.$$

Equivalently, with the same coefficient family,

$$\forall j, \quad X_j = \sum_\omega c_\omega(X) A_j^\omega.$$

The parent-Hamiltonian theorem in [CPGSV21, lines 2121–2128] is

$$N \geq L_0 + 1 \implies \ker H_N = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

The statements below record the corresponding cyclic-chain ground-space formulation; the passage to  $\ker H_N$  is kept separate.

**Definition 16.174** (Chain ground space for the block-sum tensor). Not ready For a basis of normal tensors  $A = (A_j)_{j=1}^g$  and nonzero weights  $\mu_j$ , let

$$B := \bigoplus_{j=1}^g \mu_j A_j.$$

The periodic chain ground space considered in this section is

$$\mathcal{G}_{N,L}(B).$$

**Lemma 16.175** (Block-sum chain ground space). Not ready By definition, for  $B = \bigoplus_{j=1}^g \mu_j A_j$  one has

$$\mathcal{G}_{N,L}(B) = \mathcal{G}_{N,L}\left(\bigoplus_{j=1}^g \mu_j A_j\right).$$

**Definition 16.176** (Vector space generated by a basis of normal tensors). Not ready For a basis of normal tensors  $A_1, \dots, A_g$ , the vector space generated by their length- $N$  matrix product vectors is

$$\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

**Definition 16.177** (Finite blockwise physical–virtual correspondence). Lean ✓ Stmt For a fixed blocked length  $m$ , the finite-length blockwise physical–virtual correspondence is

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = m\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Equivalently,

$$\forall X = (X_j)_j \in \bigoplus_{j=1}^g M_{D_j}(\mathbb{C}), \quad \exists (c_\omega(X))_{|\omega|=m}, \quad (X_j)_{j=1}^g = \sum_{|\omega|=m} c_\omega(X) (A_1^\omega, \dots, A_g^\omega).$$

This is the fixed-length form of the block-injective correspondence of [CPGSV21, lines 2113–2117].

**Lemma 16.178** (Each normal-tensor vector lies in the chain ground space). Not ready *If  $1 < L$  and  $N \geq L + 1$ , then for each normal tensor  $A_j$  in the basis the vector  $V^{(N)}(A_j)$  lies in  $\mathcal{G}_{N,L}(\bigoplus_k \mu_k A_k)$ .*

*Proof.* Not ready Write  $\Gamma_N^B$  for the length- $N$  boundary-to-state map of the tensor  $B$ , and let  $\iota_j(Z)$  be the block-diagonal matrix with  $Z$  in the  $j$ -th block and 0 elsewhere. The direct-sum evaluation gives

$$\Gamma_N^{\bigoplus_k \mu_k A_k}(\iota_j(\mu_j^{-N} X)) = \Gamma_N^{A_j}(X).$$

Taking  $X = I$  gives the vector  $V^{(N)}(A_j)$ , and Lemma 16.73 applied to the individual block gives its membership in  $\mathcal{G}_{N,L}(\bigoplus_k \mu_k A_k)$ .  $\square$

**Lemma 16.179** (Inclusion of a block ground space). Not ready *For each block index  $j$  with  $\mu_j \neq 0$ , the chain ground space of the block  $A_j$  embeds into the chain ground space of the direct-sum tensor:*

$$\mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right).$$

*Proof.* Not ready The cyclic-window definition of the chain ground space reduces the inclusion to the local block-embedding statement that each ground vector of  $A_j$  on the window of length  $L$  lifts to the direct sum by the identity

$$\Gamma_L^{\bigoplus_k \mu_k A_k}(\iota_j(\mu_j^{-L} X)) = \Gamma_L^{A_j}(X).$$

Here  $\iota_j$  inserts a matrix in the  $j$ -th diagonal block and has zero off that block. Thus  $\iota_j(\mu_j^{-L} X)$  has exactly the local vector of the  $j$ -th block on that window.  $\square$

**Lemma 16.180** (Forward inclusion from the block ground spaces). Not ready *Assume  $\mu_j \neq 0$  for every block index  $j$ . The linear sum of the blockwise periodic-chain ground spaces is contained in the direct-sum tensor's periodic-chain ground space:*

$$\sum_j \mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right),$$

where  $j$  ranges over all block indices.

*Proof.* Not ready Take the linear sum of the inclusions

$$\mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right)$$

over all block indices  $j$ .  $\square$

**Lemma 16.181** (Forward inclusion for the direct-sum tensor). Not ready Let  $A^i = \bigoplus_j \mu_j A_j^i$  be in block-injective canonical form, where  $A_1, \dots, A_g$  is a basis of normal tensors. The linear sum of the blockwise periodic-chain ground spaces is contained in the periodic chain ground space of the block-sum tensor:

$$\sum_j \mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right).$$

*Proof.* Not ready The block-injective canonical-form hypothesis supplies  $\mu_j \neq 0$  for every  $j$ , and the block-sum chain ground space is  $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$  by definition.  $\square$

**Lemma 16.182** (Block-diagonal commutant reduction from span extension). Not ready Let  $S$  be a family of virtual matrices. If each sector projection satisfies

$$P_j \in \text{span } S$$

and a boundary matrix  $X$  satisfies

$$XM = MX \quad (M \in S),$$

then, for every sector  $j$ ,

$$XP_j = P_j X,$$

and hence

$$P_j X P_k = 0 \quad (j \neq k).$$

Thus  $X$  is block diagonal with respect to the sector decomposition.

*Proof.* Not ready The equations  $XM = MX$  are linear in  $M$ , so they hold for every  $M \in \text{span } S$ . Substituting  $M = P_j$  gives  $XP_j = P_j X$ . Multiplying this identity on the left by  $P_j$  and on the right by  $P_k$  with  $j \neq k$  gives  $P_j X P_k = 0$ .  $\square$

**Lemma 16.183** (Block projections from product-algebra span). Not ready Let  $T_a = (T_a(i))_i$  be a family of tuples spanning the full product algebra  $\prod_i M_{n_i}(\mathbb{C})$ , and let  $c_i \neq 0$  be nonzero scalars. Then, for every sector  $k$ , the sector projection  $P_k$  lies in the span of the scaled dependent block-diagonal matrices

$$\bigoplus_i c_i T_a(i).$$

*Proof.* Not ready Apply the linear map  $(M_i)_i \mapsto \bigoplus_i c_i M_i$  to the full-span product-algebra element with  $(c_k)^{-1}I$  in the  $k$ -th component and zero in every other component. Its image is exactly  $P_k$ .  $\square$

**Lemma 16.184** (Sector projections from blockwise word span). Not ready If the simultaneous length- $m$  block-word tuples

$$\omega \mapsto (A_j^\omega)_j$$

span the full product algebra  $\prod_j M_{D_j}(\mathbb{C})$  and every  $\mu_j$  is nonzero, then the pulled-back length- $m$  word products of  $\bigoplus_j \mu_j A_j$  span every virtual sector projection  $P_j$ . Consequently, any boundary matrix commuting with all those direct-sum words has block-diagonal pulled-back form, and its off-block entries vanish.

*Proof.* Not ready First embed the product algebra linearly as dependent block-diagonal matrices, scaling the  $j$ -th block by  $\mu_j^m$ . Since  $\mu_j^m \neq 0$ , the product-algebra span contains the tuple with  $(\mu_j^m)^{-1}I$  in one component and zero elsewhere; its scaled diagonal image is exactly  $P_j$ . The word-evaluation formula for the direct-sum tensor identifies these scaled diagonal matrices with the pulled-back direct-sum word products, and the commutant conclusion follows from Lemma 16.182.  $\square$

**Definition 16.185** (Pairwise block-separating equations). Lean Lean ✓ Stmt Fix a finite target set  $T$  of block indices. A length- $S$  block-separating equation for  $k$  against  $T$  is a choice of coefficients  $c_\omega$  such that

$$\sum_{|\omega|=S} c_\omega A_\ell^\omega = \begin{cases} I, & \ell = k, \\ 0, & \ell \in T. \end{cases}$$

The pairwise version requires this equation for each ordered pair of distinct blocks, with  $T = \{j\}$ .

**Definition 16.186** (Pair trace-separation criterion). Lean Lean Lean Lean Lean Lean ✓ Stmt For two blocks  $A$  and  $B$ , the length- $S$  pair word tuple is  $w \mapsto (A^w, B^w)$ . The pair product-span condition says that these tuples span  $M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C})$ . Equivalently, in the trace-dual formulation, the only pair of test matrices  $(\Delta_A, \Delta_B)$  whose trace pairing with every length- $S$  tuple vanishes is  $(0, 0)$ . The cumulative variant replaces one fixed length by all word lengths at most  $S$ , and the all-length variant allows every finite word.

**Lemma 16.187** (Three-block trace reduction from a nonzero middle matrix). Lean Lean ✓ Stmt Let  $A$  be  $L$ -block injective, and let  $\Delta_A \neq 0$ . Suppose that, for all words  $u, v, t$  of length  $L$ ,

$$\text{tr}(\Delta_A A_v A_t A_u) + \text{tr}(\Delta_B B_v B_t B_u) = 0.$$

Then for every  $Z \in M_{D_A}(\mathbb{C})$  there is a matrix  $W \in M_{D_B}(\mathbb{C})$  such that, for all words  $t$  of length  $L$ ,

$$\text{tr}(Z A_t) + \text{tr}(W B_t) = 0.$$

This is the trace-dual algebraic step in the two-block case of the direct-sum lemma of [PGVWC07].

*Proof.* ✓ Proof Since  $\Delta_A \neq 0$ , Lemma 8.11 writes  $Z$  as a linear combination of matrices  $A_u \Delta_A A_v$ . For one such term, cyclicity of the trace rewrites the three-block relation as

$$\text{tr}(A_u \Delta_A A_v A_t) + \text{tr}(B_u \Delta_B B_v B_t) = 0.$$

Taking the same linear combination of the matrices  $B_u \Delta_B B_v$  gives the required  $W$ , and linearity gives the general case.  $\square$

**Lemma 16.188** (Finite-chain image inclusion from a three-block relation). Lean Lean Lean Lean ✓ Stmt Under the hypotheses of Lemma 16.187, the length- $L$  image spaces satisfy

$$\mathcal{G}_L^A \subseteq \mathcal{G}_L^B.$$

If the corresponding nonzero middle matrix exists on both sides, then  $\mathcal{G}_L^A = \mathcal{G}_L^B$ .

*Proof.* ✓ Proof Write a vector in  $\mathcal{G}_L^A$  as  $t \mapsto \text{tr}(A_t Z)$ . The preceding lemma supplies  $W$  with  $\text{tr}(Z A_t) + \text{tr}(W B_t) = 0$  for every word  $t$ . Cyclicity of the trace gives

$$\text{tr}(A_t Z) = \text{tr}(Z A_t) = -\text{tr}(W B_t) = \text{tr}(B_t (-W)),$$

so the vector lies in  $\mathcal{G}_L^B$ . The equality statement follows by applying the same argument with the two blocks exchanged.  $\square$



**Lemma 16.189** (Dimension step for two-block direct-sum image spaces). [Lean](#) [Lean](#) [Lean](#)

**Lemma 16.188** *If  $A$  and  $B$  are length- $L$  block-injective, the map  $Z \mapsto (t \mapsto \text{tr}(A_t Z))$  has image dimension  $D_A^2$ , and similarly for  $B$ . Therefore an inclusion  $\mathcal{G}_L^A \subseteq \mathcal{G}_L^B$ , together with  $D_B \leq D_A$ , forces  $D_A = D_B$  and  $\mathcal{G}_L^A = \mathcal{G}_L^B$ . In particular, the three-block trace relation from Lemma 16.188 gives this equality whenever the source-paper size ordering is available. If the inclusion comes from the strictly larger block,  $D_B < D_A$  then the same dimension step already contradicts the existence of the three-block trace relation. More generally, if an external source input rules out the simultaneous conclusion  $D_A = D_B$  and  $\mathcal{G}_L^A = \mathcal{G}_L^B$ , then the three-block image spaces  $\mathcal{G}_{3L}^A$  and  $\mathcal{G}_{3L}^B$  have zero intersection. With injectivity at the three-block length, the strict-size branch itself already gives homogeneous pair trace separation at length  $3L$ . One such source input is non-proportionality of the periodic MPS vectors at a sufficiently long chain length, equivalently distinctness of the corresponding spaces  $\text{CV}^{(N)}(A)$ : equality of the local image spaces gives equality of the periodic-chain constraint spaces, and injective uniqueness identifies those spaces with the two one-dimensional MPS spans. For same-dimensional separated BNT blocks, the non-equivalence condition supplies such non-proportional MPS vectors at arbitrarily large chain lengths; the direct-sum injectivity hypotheses remain explicit. Zero intersection of the two three-block image spaces also gives homogeneous pair trace separation at that length once the boundary-to-chain maps at the three-block length are injective. Combining the equal-dimension and unequal-dimension branches over all ordered block pairs gives one common three-block trace-separation length, and hence the positive-length blockwise word span required by the block-separation argument. Since each block in block-injective canonical form is one-site injective, one may take  $L = 2$ ; injectivity at lengths 2 and 6 then follows by passing from a fixed injective length to positive multiples. This length-6 trace separation is equivalently a full homogeneous pair span; concatenating one-letter words propagates fullness to all later homogeneous lengths, so the direct-sum witness also gives a uniform finite pair-span period window. Combining this window with the conditional period-window homogenization theorem gives the homogeneous blockwise word span without separately assuming homogeneous identity pairs. The same conclusion applies to the corresponding normal canonical-form hypotheses with a basis of normal tensors once one-site injectivity is supplied explicitly, since those hypotheses carry the irreducibility and normalization assumptions used in the direct-sum comparison.*

*Proof.* ✓ Proof Block injectivity makes the boundary-to-chain map injective, so  $\dim \mathcal{G}_L^A = D_A^2$  and  $\dim \mathcal{G}_L^B = D_B^2$ . Inclusion gives  $D_A^2 \leq D_B^2$ , while  $D_B \leq D_A$  gives the reverse inequality. Hence  $D_A = D_B$ , and equality of dimensions inside the inclusion gives equality of the image spaces. In the strict-size branch this contradicts  $D_B < D_A$ . For the conditional directness statement, a vector in the intersection of the two three-block image spaces has two boundary representations. If the first boundary matrix is zero, the vector is zero. Otherwise the difference of the two coefficient formulas is exactly the homogeneous three-block trace relation, so the dimension step gives the forbidden collapse. In the equal-size branch, equal local image spaces impose the same cyclic local constraints on a periodic chain. The injective uniqueness theorem then identifies both cyclic ground spaces with the spans of their MPV states, contradicting non-proportionality of those states.  $\square$

**Lemma 16.190** (All-length pair trace separation has finite cumulative length). [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)  
[Lean](#) [✓ Stmt](#) *If no nonzero trace functional vanishes on all finite pair words, then there is a finite  $S$  such that vanishing on every word of length at most  $S$  already forces the test pair to be zero. For a finite family of ordered block pairs, these bounds can be replaced by one common bound.*





*Proof.* [✓ Proof](#) The Euclidean division of  $n - s$  by  $p$  gives  $n = s + r + qp$  with  $0 \leq r < p$ . The assumed residue-window span gives the full product algebra at length  $s + r$ . Applying homogeneous padding with the length- $p$  span once gives the full product algebra at length  $s + r + p$ ; iterating gives the full product algebra at length  $s + r + qp$ .  $\square$

**Lemma 16.197** (Blockwise word span from block-separating equations). [Lean](#) [Lean](#) [✓ Stmt](#) *Let  $J$  be a finite set of  $r$  blocks, and let  $(A_j)_{j \in J}$  be a finite family of tensors with common block injectivity at length  $L$ . Suppose there are length- $S$  block-separating equations: for each block  $k$ ,*

$$\sum_{|\omega|=S} c_{\omega}^{(k)} A_j^{\omega} = \begin{cases} I, & j = k, \\ 0, & j \neq k. \end{cases}$$

*Then the simultaneous length- $(L + S)$  block-word evaluations span  $\prod_j M_{D_j}(\mathbb{C})$ . If the block-separating equations are first obtained from pairwise block-separating equations at length  $S$ , the same conclusion holds at length  $L + (r - 1)S$ .*

*Proof.* [✓ Proof](#) Block injectivity writes each target block matrix as a length- $L$  linear combination of word evaluations. For a target tuple  $(M_j)_j$ , write  $M_k$  in this form in block  $k$ , multiply by the block-separating equation for  $k$ , and sum over  $k$ . Thus, if  $M_k = \sum_{|u|=L} a_u^{(k)} A_k^u$  and  $\sum_{|v|=S} b_v^{(k)} A_j^v = \delta_{j,k} I$ , then for every block  $j$ ,

$$M_j = \sum_{k \in J} \sum_{|u|=L} \sum_{|v|=S} a_u^{(k)} b_v^{(k)} A_j^u A_j^v.$$

The right side is a linear combination of length- $(L + S)$  word tuples, so these tuples span the full product algebra. If one starts with pairwise separation instead, first use Lemma 16.194 to multiply the pairwise equations into the full equations.  $\square$

**Lemma 16.198** (BNT direct-sum input gives a finite blockwise product span). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let  $A^1, \dots, A^r$  be separated BNT blocks satisfying the irreducibility, left-canonicity, and normalized self-overlap hypotheses used in the direct-sum comparison. Assume the direct-sum injectivity hypotheses*

$$\text{span}\{A_u^j : |u| = L\} = M_{D_j}(\mathbb{C}), \quad \text{span}\{A_v^j : |v| = L + (L + L)\} = M_{D_j}(\mathbb{C}),$$

*for every block  $j$ , with  $1 < L$ . Then for every ordered pair  $k \neq j$  there are coefficients  $c_{\omega}^{k,j}$  such that*

$$\sum_{|\omega|=L+(L+L)} c_{\omega}^{k,j} A_{\omega}^k = I, \quad \sum_{|\omega|=L+(L+L)} c_{\omega}^{k,j} A_{\omega}^j = 0.$$

*Consequently the simultaneous block-word tuples at length*

$$L + (r - 1)(L + (L + L))$$

*span the full product algebra*

$$\prod_{j=1}^r M_{D_j}(\mathbb{C}).$$

*Proof.* [✓ Proof](#) Lemma 16.189 gives homogeneous trace separation at length  $L + (L + L)$  for every ordered pair of distinct BNT blocks. Lemma 16.193 converts these trace-separation equations into the displayed pairwise block-separating equations. Multiplying the equations for the  $r - 1$  blocks different from  $k$  gives a block-separating equation for  $k$ , and Lemma 16.197 then supplies the displayed product span.  $\square$

**Lemma 16.199** (Blockwise boundary compatibility from traces). [Lean](#) [✓ Stmt](#) Let  $A^1, \dots, A^g$  be block tensors with the following common word-span property: for a fixed  $m$ , the simultaneous tuples

$$w \mapsto (A_w^1, \dots, A_w^g), \quad |w| = m,$$

span the full product algebra  $\prod_j M_{D_j}(\mathbb{C})$ . Suppose that matrices  $C_a^j, D_b^j \in M_{D_j}(\mathbb{C})$  satisfy, for all physical indices  $a, b$  and all words  $w$  of length  $m$ ,

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

Then

$$A_b^j C_a^j = D_b^j A_a^j$$

for every block  $j$  and all  $a, b$ .

*Proof.* [✓ Proof](#) For fixed  $a, b$ , set

$$\Delta_j = A_b^j C_a^j - D_b^j A_a^j.$$

The assumed trace equality says that

$$\sum_j \text{tr}(\Delta_j A_w^j) = 0$$

for every word  $w$  of length  $m$ . Since the tuples  $(A_w^1, \dots, A_w^g)$  span the product algebra, the product trace pairing is nondegenerate and hence every  $\Delta_j$  is zero.  $\square$

**Lemma 16.200** (Normalized boundary matrix identities). [Lean](#) [✓ Stmt](#) Let  $(A_a)_{a \in I}$ ,  $(C_a)_{a \in I}$ , and  $(D_b)_{b \in I}$  be matrices in  $M_D(\mathbb{C})$ , indexed by a finite physical alphabet. If

$$\sum_a A_a A_a^\dagger = \mathbb{1}, \quad A_b C_a = D_b A_a$$

for all  $a, b \in I$ , and if

$$E = \sum_a C_a A_a^\dagger,$$

then

$$D_b = A_b E, \quad A_b C_a = A_b E A_a$$

for all  $a, b \in I$ .

*Proof.* [✓ Proof](#) Multiply  $D_b$  by the normalization:

$$D_b = D_b \sum_a A_a A_a^\dagger = \sum_a D_b A_a A_a^\dagger = \sum_a A_b C_a A_a^\dagger = A_b E.$$

Substituting this in  $A_b C_a = D_b A_a$  gives  $A_b C_a = A_b E A_a$ .  $\square$

**Lemma 16.201** (Left-boundary summands are ground-space vectors). [Lean](#) [✓ Stmt](#) Let  $A^1, \dots, A^g$  be block tensors, and let  $C_a^j, E^j \in M_{D_j}(\mathbb{C})$  satisfy

$$A_b^j C_a^j = A_b^j E^j A_a^j$$

for every block  $j$  and all physical indices  $a, b$ . For a word  $\sigma = (\sigma_1, \dots, \sigma_{n+2})$ , define

$$\alpha_j(\sigma) = \text{tr}(A_{\sigma_{n+2}}^j C_{\sigma_1}^j A_{\sigma_2}^j \cdots A_{\sigma_{n+1}}^j).$$

Then

$$\sum_j \alpha_j \in \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ Proof Fix  $j$ . By the assumed boundary identity and cyclicity of the trace,

$$\mathrm{tr}\left(A_{\sigma_{n+2}}^j C_{\sigma_1}^j A_{\sigma_2}^j \cdots A_{\sigma_{n+1}}^j\right) = \mathrm{tr}\left(A_{\sigma_1}^j A_{\sigma_2}^j \cdots A_{\sigma_{n+2}}^j E^j\right).$$

Thus  $\alpha_j$  is the image of  $E^j$  under the parametrization of  $G_{n+2}(A^j)$ . Hence each  $\alpha_j$  lies in  $G_{n+2}(A^j)$ , and the finite sum lies in the supremum of these subspaces.  $\square$

**Lemma 16.202** (Left-boundary trace decompositions lie in the block ground spaces). Lean  
✓ Stmt Let  $A^1, \dots, A^g$  be block tensors with common word-span separation at length  $n$ , and suppose that

$$\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$$

for every block  $j$ . Suppose a vector  $\psi \in (\mathbb{C}^d)^{\otimes(n+2)}$  has the left-boundary trace decomposition

$$\psi = \sum_j \alpha_j, \quad \alpha_j(i_1, \dots, i_{n+2}) = \mathrm{tr}\left(A_{i_{n+2}}^j C_{i_1}^j A_{i_2}^j \cdots A_{i_{n+1}}^j\right),$$

and suppose that the two trace decompositions agree for every middle word:

$$\sum_j \mathrm{tr}\left(A_b^j C_a^j A_w^j\right) = \sum_j \mathrm{tr}\left(D_b^j A_a^j A_w^j\right).$$

Then

$$\psi \in \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ Proof The trace-decomposition equality and the common word-span hypothesis give

$$A_b^j C_a^j = D_b^j A_a^j$$

for every  $j, a, b$ . By the normalization,

$$D_b^j = D_b^j \sum_a A_a^j A_a^{j\dagger} = \sum_a A_b^j C_a^j A_a^{j\dagger} = A_b^j E^j, \quad E^j := \sum_a C_a^j A_a^{j\dagger}.$$

Hence  $A_b^j C_a^j = A_b^j E^j A_a^j$ . Lemma 16.201 then gives

$$\psi = \sum_j \alpha_j \in \bigvee_j G_{n+2}(A^j).$$

$\square$

**Lemma 16.203** (One-step block intersection from block restrictions). Lean ✓ Stmt Let  $A^1, \dots, A^g$  be block tensors with common word-span separation at length  $n$ , and suppose

$$\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$$

for every block  $j$ . If  $\psi \in (\mathbb{C}^d)^{\otimes(n+2)}$  satisfies

$$\psi(a, -) \in \bigvee_j G_{n+1}(A^j), \quad \psi(-, b) \in \bigvee_j G_{n+1}(A^j)$$

for every physical index  $a, b$ , then

$$\psi \in \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ Proof Choose finite decompositions of the fixed-boundary vectors:

$$\psi(a, i_2, \dots, i_{n+2}) = \sum_j \text{tr}(A_{i_2}^j \cdots A_{i_{n+2}}^j C_a^j),$$

and

$$\psi(i_1, \dots, i_{n+1}, b) = \sum_j \text{tr}(A_{i_1}^j \cdots A_{i_{n+1}}^j D_b^j).$$

Evaluating these two decompositions on  $(a, w, b)$ , where  $w$  has length  $n$ , gives

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

The first decomposition also gives

$$\psi = \sum_j \alpha_j, \quad \alpha_j(i_1, \dots, i_{n+2}) = \text{tr}(A_{i_{n+2}}^j C_{i_1}^j A_{i_2}^j \cdots A_{i_{n+1}}^j).$$

Lemma 16.202 therefore gives

$$\psi \in \bigvee_j G_{n+2}(A^j).$$

□

**Lemma 16.204** (One-step block intersection characterization). Lean ✓ Stmt Under the hypotheses of Lemma 16.203, an  $(n+2)$ -site vector satisfies

$$\psi \in \bigvee_j G_{n+2}(A^j)$$

if and only if

$$\psi(a, -) \in \bigvee_j G_{n+1}(A^j), \quad \psi(-, b) \in \bigvee_j G_{n+1}(A^j)$$

for every physical index  $a, b$ .

*Proof.* ✓ Proof The reverse implication is Lemma 16.203. For the forward implication, write

$$\psi = \sum_j \gamma_j, \quad \gamma_j \in G_{n+2}(A^j).$$

Each  $\gamma_j$  has fixed-boundary restrictions in  $G_{n+1}(A^j)$ :

$$\gamma_j(a, -) \in G_{n+1}(A^j), \quad \gamma_j(-, b) \in G_{n+1}(A^j).$$

Taking the finite sum over  $j$  gives the two restrictions of  $\psi$  in  $\bigvee_j G_{n+1}(A^j)$ . □

**Lemma 16.205** (Subspace form of the one-step block intersection). Lean ✓ Stmt Under the hypotheses of Lemma 16.203, set

$$S_n := \bigvee_j G_{n+1}(A^j).$$

Then

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ Proof Lemma 16.204 gives the displayed equivalence for each vector  $\psi$ . Interpreting the two sides of that equivalence as subspaces gives the stated equality.  $\square$

**Lemma 16.206** (Period-window form of the block intersection identity). Lean ✓ Stmt Let  $p > 0$ . Suppose the length- $p$  simultaneous block-word tuples span  $\prod_j M_{D_j}(\mathbb{C})$ , and suppose that for every  $0 \leq r < p$ , the length- $s + r$  simultaneous block-word tuples span  $\prod_j M_{D_j}(\mathbb{C})$ . Assume also the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I$$

for every block  $j$ . Then there is a length  $N$  such that, for every  $n \geq N$ , if

$$S_n := \bigvee_j G_{n+1}(A^j),$$

then

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ Proof Lemma 16.196 gives a length  $N$  such that the simultaneous block-word tuples span  $\prod_j M_{D_j}(\mathbb{C})$  at every length  $n \geq N$ . For each such  $n$ , Lemma 16.205 applies and gives the displayed subspace equality.  $\square$

**Lemma 16.207** (BNT direct-sum input gives one block-intersection step). Lean ✓ Stmt Under the hypotheses of Lemma 16.198, assume in addition the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I$$

for every block  $j$ . Let

$$n = L + (r - 1)(L + (L + L)), \quad S_n = \bigvee_j G_{n+1}(A^j).$$

Then

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ Proof Lemma 16.198 gives

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C}).$$

Substituting this span into Lemma 16.205, together with the normalization, gives the stated subspace equality.  $\square$

**Lemma 16.208** (BNT direct-sum input with a block-injectivity window). Lean ✓ Stmt Under the hypotheses of Lemma 16.198, set

$$S = L + (L + L), \quad q = (r - 1)S.$$

Suppose that  $p > 0$  and let  $s_0$  be a starting length. Assume that, for every block  $j$ ,

$$\text{span}\{A_u^j : |u| = p\} = M_{D_j}(\mathbb{C}),$$



and that, for every  $0 \leq s < p + q$  and every block  $j$ ,

$$\text{span}\{A_v^j : |v| = s_0 + s\} = M_{D_j}(\mathbb{C}).$$

Assume also the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I$$

for every block  $j$ . Then there is  $N$  such that, for every  $n \geq N$ , if

$$S_n := \bigvee_j G_{n+1}(A^j),$$

then

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ Proof The direct-sum input gives, for each ordered pair of distinct blocks, length- $S$  equations which are the identity on the first block and zero on the second. Multiplying these equations over the  $r - 1$  other blocks gives a block-separating equation of total length  $q$ . Combining this equation with block injectivity at length  $p$  gives the full product span at length  $p + q$ . Combining it with the assumed block-injectivity window gives the full product span at lengths  $s_0 + q + s$ , for  $0 \leq s < p + q$ . The period-window form of the block-intersection identity then gives the displayed equality for all sufficiently large  $n$ .  $\square$

**Lemma 16.209** (Normalized block-separating equations give all large product spans). Lean

Lean ✓ Stmt Let  $A^1, \dots, A^r$  be blocks satisfying the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I$$

and assume that every block is injective at length  $L$ :

$$\text{span}\{A_u^j : |u| = L\} = M_{D_j}(\mathbb{C}).$$

If there are block-separating equations of length  $S$ , then, for every  $n \geq L + (r - 1)S$ ,

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C}).$$

In particular, for separated BNT blocks, the same conclusion holds with

$$S = L + (L + L), \quad n \geq L + (r - 1)(L + (L + L)).$$

*Proof.* ✓ Proof By Theorem 8.82, the normalization propagates the length- $L$  span equality to every larger homogeneous length:

$$m \geq L \implies \text{span}\{A_u^j : |u| = m\} = M_{D_j}(\mathbb{C}).$$

For  $n \geq L + (r - 1)S$ , take  $m = n - (r - 1)S$ . Combining the length- $m$  block span with the  $r - 1$  block-separating equations gives the displayed product span at length  $n$ . In the BNT case, Lemma ?? supplies the block-separating equations with  $S = L + (L + L)$ .  $\square$

**Lemma 16.210** (Product span gives directness of block image spaces). [Lean](#) [Lean](#) [✓ Stmt](#)  
 Suppose

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C}).$$

Then the sum of the image spaces  $G_n(A^j)$  is direct: if

$$\phi_j \in G_n(A^j), \quad \sum_j \phi_j = 0,$$

then  $\phi_j = 0$  for every  $j$ . In particular, for the normalized BNT input above, this directness holds for every

$$n \geq L + (r-1)(L + (L+L)).$$

*Proof.* [✓ Proof](#) Write

$$\phi_j(\sigma) = \text{tr}(A_\sigma^j X_j).$$

Evaluating  $\sum_j \phi_j = 0$  on each word  $w$  of length  $n$  gives

$$\sum_j \text{tr}(A_w^j X_j) = 0.$$

Since  $\text{tr}(A_w^j X_j) = \text{tr}(X_j A_w^j)$ , the product-span equation gives, for every  $(M_1, \dots, M_r) \in \prod_j M_{D_j}(\mathbb{C})$ ,

$$\sum_j \text{tr}(X_j M_j) = 0.$$

Taking only the  $j$ -th component nonzero and using nondegeneracy of the trace pairing yields  $X_j = 0$ , hence  $\phi_j = 0$ . The BNT statement follows by substituting Lemma 16.209.  $\square$

**Lemma 16.211** (Normalized BNT input gives the large-length block intersection). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let  $A^1, \dots, A^r$  be separated BNT blocks satisfying the irreducibility, left-canonicity, and normalized self-overlap hypotheses used in the direct-sum comparison. Suppose  $1 < L$ , every block is injective at length  $L$ ,

$$\text{span}\{A_u^j : |u| = L\} = M_{D_j}(\mathbb{C}),$$

and every block satisfies the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Set

$$N = L + (r-1)(L + (L+L)).$$

For every  $n \geq N$ , set

$$S_n := \bigvee_j G_{n+1}(A^j),$$

and the two consecutive sums are internal direct sums:

$$\bigvee_j G_{n+1}(A^j) = \bigoplus_j G_{n+1}(A^j), \quad \bigvee_j G_{n+2}(A^j) = \bigoplus_j G_{n+2}(A^j).$$

With this notation,

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ **Proof** Lemma 16.209 gives

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C})$$

for every  $n \geq N$ . Substituting this product-span equation into Lemma 16.205, together with the normalization equation, gives the displayed intersection identity. Lemma 16.210 at lengths  $n+1$  and  $n+2$  gives the two directness statements.  $\square$

**Lemma 16.212** (Normalized BNT input gives an eventual block intersection). Lean ✓ **Stmt** Let  $A^1, \dots, A^r$  be separated BNT blocks satisfying the irreducibility, left-canonicity, and normalized self-overlap hypotheses used in the direct-sum comparison. Suppose  $1 < L$ , every block is injective at length  $L$ ,

$$\text{span}\{A_u^j : |u| = L\} = M_{D_j}(\mathbb{C}),$$

and every block satisfies the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Then there is a length  $N$  such that, for every  $n \geq N$ , with

$$S_n := \bigvee_j G_{n+1}(A^j),$$

one has

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ **Proof** Apply Lemma 16.211 and take

$$N = L + (r-1)(L + (L+L)).$$

$\square$

**Theorem 16.213** (Block-diagonal intersection identity). Not ready Let  $A_1, \dots, A_g$  be the blocks in a block-injective canonical form, and assume that their MPV families are pairwise different:

$$j \neq k \implies \exists n \geq 1, V^{(n)}(A_j) \neq V^{(n)}(A_k).$$

Let  $L_0$  be a common injectivity length. In the multi-block range of [PGVWC07, Theorem 12], namely

$$g \geq 2, \quad L_0 \geq 1, \quad L \geq 3(g-1)(L_0+1)+1,$$

the block-diagonal intersection property says that, for every  $m \geq L$ ,

$$\mathbb{C}^d \otimes \left( \bigoplus_{j=1}^g G_m(A_j) \right) \cap \left( \bigoplus_{j=1}^g G_m(A_j) \right) \otimes \mathbb{C}^d = \bigoplus_{j=1}^g G_{m+1}(A_j).$$

For nonzero weights  $\mu_j$  and every  $N \geq L + L_0$ , the corresponding periodic chain ground spaces satisfy

$$\mathcal{G}_{N,L} \left( \bigoplus_{j=1}^g \mu_j A_j \right) = \sum_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

*Proof.* Not ready The source proof takes a vector  $\psi$  in the left-hand side and writes its two boundary descriptions as

$$\psi = \sum_{j=1}^g \alpha_j = \sum_{j=1}^g \beta_j, \quad \alpha_j \in \mathbb{C}^d \otimes G_m(A_j), \quad \beta_j \in G_m(A_j) \otimes \mathbb{C}^d.$$

In the source notation, the two components have boundary matrices  $C_{i_1}^j$  and  $D_{i_{m+1}}^j$ : the first family leaves the initial physical index open, and the second family leaves the final physical index open.

$$\begin{aligned} \alpha_j &= \sum_{i_1, \dots, i_{m+1}} \text{tr}(A_{i_{m+1}}^j C_{i_1}^j A_{i_2}^j \cdots A_{i_m}^j) |i_1 \cdots i_{m+1}\rangle, \\ \beta_j &= \sum_{i_1, \dots, i_{m+1}} \text{tr}(D_{i_{m+1}}^j A_{i_1}^j A_{i_2}^j \cdots A_{i_m}^j) |i_1 \cdots i_{m+1}\rangle. \end{aligned}$$

The direct-sum separation lemma used in [PGVWC07, Theorem 12], together with injectivity of each block, gives the blockwise matrix identity

$$A_{i_{m+1}}^j C_{i_1}^j = D_{i_{m+1}}^j A_{i_1}^j$$

for every  $j$ ,  $i_1$ , and  $i_{m+1}$ . For fixed  $j$ , set

$$E^j := \sum_a C_a^j A_a^{j\dagger}.$$

In the normalization used in the source proof,  $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$ , so Lemma 16.200 gives

$$D_b^j = \sum_a D_b^j A_a^j A_a^{j\dagger} = \sum_a A_b^j C_a^j A_a^{j\dagger} = A_b^j E^j.$$

Hence

$$A_b^j C_a^j = A_b^j E^j A_a^j$$

for every  $a, b$ , and cyclicity of the trace gives

$$\alpha_j = \sum_{i_1, \dots, i_{m+1}} \text{tr}(A_{i_1}^j \cdots A_{i_m}^j A_{i_{m+1}}^j E^j) |i_1 \cdots i_{m+1}\rangle = \Gamma_{m+1}^{A_j}(E^j) \in G_{m+1}(A_j).$$

Since  $\psi = \sum_j \alpha_j$ , this gives

$$\psi \in \bigoplus_j G_{m+1}(A_j).$$

For one block the preceding implication is

$$\mathbb{C}^d \otimes G_m(A_j) \cap G_m(A_j) \otimes \mathbb{C}^d \subseteq G_{m+1}(A_j).$$

The reverse inclusion is blockwise:

$$G_{m+1}(A_j) \subseteq \mathbb{C}^d \otimes G_m(A_j) \cap G_m(A_j) \otimes \mathbb{C}^d.$$

Thus

$$\mathbb{C}^d \otimes G_m(A_j) \cap G_m(A_j) \otimes \mathbb{C}^d = G_{m+1}(A_j).$$

With

$$S_m := \bigoplus_{j=1}^g G_m(A_j),$$

the preceding calculation is the open-segment recursion

$$\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}$$

for every  $m \geq L$ . For  $B = \bigoplus_j \mu_j A_j$  with every  $\mu_j \neq 0$ ,

$$G_L(B) = S_L.$$

The passage from the open-segment recursion to the periodic chain is the separate closure step in [PGVWC07, Theorem 12], used under  $N \geq L + L_0$ :

$$N \geq L + L_0 \implies \mathcal{G}_{N,L}(B) = \sum_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

□

**Lemma 16.214** (Conditional reduction to block-diagonal boundary conditions). Not ready *Let  $A^i = \bigoplus_j \mu_j A_j^i$ , and assume  $1 < L$ ,  $N \geq L + 1$ , and  $0 < m$ . Suppose that every virtual sector projection  $P_j$  lies in the pulled-back span of the length- $m$  words  $A^\omega$ . If a boundary matrix  $X$  commutes with all those words, then the commutant reduction gives*

$$X = \bigoplus_j X_j.$$

Therefore the vector obtained by applying the ground-space map to  $X$  lies in  $\sum_j \mathcal{G}_{N,L}(A_j)$ . Consequently, if every vector in  $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$  admits such a commuting boundary matrix representation, as in the direct-sum analogue of Theorem 16.65, then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \sum_j \mathcal{G}_{N,L}(A_j),$$

and together with the already-proved forward inclusion this gives equality. Composing the reverse inclusion with blockwise injective uniqueness gives the conditional containment in the vector space generated by the basis of normal tensors. This is a conditional form of the block-diagonal boundary-condition step in [CPGSV21, lines 2126–2128].

*Proof.* Not ready The commutant reduction makes the pulled-back boundary matrix block diagonal. Restricting the same word-commutation identities to a diagonal block and using the injectivity of that block shows, by the scalar-commutant lemma, that each diagonal block is scalar. Write  $\Gamma_N^B$  for the length- $N$  boundary-to-state map of the tensor  $B$ . For a block-diagonal boundary matrix,

$$\Gamma_N^{\bigoplus_j \mu_j A_j} \left( \bigoplus_j X_j \right) = \sum_j \mu_j^N \Gamma_N^{A_j}(X_j).$$

If  $X_j = \lambda_j I$ , this is a finite sum of scalar multiples of the vectors  $V^{(N)}(A_j)$ . The reverse inclusion and equality statements are reformulations of this calculation, combined with the forward inclusion and the blockwise uniqueness theorem. □

**Lemma 16.215** (Block-diagonal boundary reduction from blockwise word span). Not ready Let  $A^i = \bigoplus_j \mu_j A_j^i$  be in block-injective canonical form, and assume  $1 < L$ ,  $N \geq L + 1$ , and  $0 < m$ . If

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = m\} = \prod_j M_{D_j}(\mathbb{C})$$

and every vector in  $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$  comes with a boundary matrix representation commuting with all length- $m$  direct-sum words, then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \sum_j \mathcal{G}_{N,L}(A_j).$$

Combined with the forward inclusion, this gives the corresponding conditional equality. In the parent-ground-space formulation, the same hypotheses imply containment in  $\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$ . The blockwise word span and boundary representation are the displayed hypotheses.

*Proof.* Not ready The blockwise word-span hypothesis gives

$$P_j \in \text{span}\left\{\left(\bigoplus_r \mu_r A_r\right)^\omega : |\omega| = m\right\}.$$

If  $X$  commutes with all displayed word products, then  $XP_j = P_jX$  for every  $j$ , hence  $X = \bigoplus_j X_j$ . The direct-sum boundary formula

$$\Gamma_N^{\bigoplus_j \mu_j A_j}\left(\bigoplus_j X_j\right) = \sum_j \mu_j^N \Gamma_N^{A_j}(X_j)$$

gives the reverse inclusion, and the blockwise uniqueness theorem gives the stated span containment.  $\square$

**Lemma 16.216** (Block-diagonal boundary reduction from block-separating equations). Not ready Let  $A^i = \bigoplus_j \mu_j A_j^i$  be in block-injective canonical form, with  $A_1, \dots, A_g$  a basis of normal tensors. Assume  $1 < L$  and  $N \geq L + 1$ , and suppose that for each block  $k$  there are coefficients  $c_\omega^{(k)}$  such that

$$\sum_{|\omega|=S} c_\omega^{(k)} A_j^\omega = \begin{cases} I, & j = k, \\ 0, & j \neq k. \end{cases}$$

If every vector in  $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$  has a boundary matrix representation commuting with all length- $(1 + S)$  direct-sum words, then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \sum_j \mathcal{G}_{N,L}(A_j).$$

Together with the forward inclusion, this gives the corresponding conditional equality. In the parent-ground-space formulation, the same hypotheses imply containment in  $\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$ . This is the block-injective ground-space argument of [CPGSV21, lines 2120–2128], where the re-growing step is restricted to block-diagonal boundary conditions.

*Proof.* Not ready In block-injective canonical form, each block is one-site injective, hence one-block injective by Lemma 2.30. The displayed separation equations imply

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = 1 + S\} = \prod_j M_{D_j}(\mathbb{C}).$$

This is Lemma 16.197 with  $L = 1$ . Therefore Lemma 16.215 applies with  $m = 1 + S$ . Its conclusion is the displayed containment, and the equality follows after adding the forward inclusion.  $\square$

**Theorem 16.217** (Containment in the space generated by a basis of normal tensors). Not ready *If  $A^i = \bigoplus_j \mu_j A_j^i$  is in block-injective canonical form, where  $A_1, \dots, A_g$  is a basis of normal tensors, assume  $g \geq 2$  and  $\mu_j \neq 0$  for every  $j$ . Let  $L_0 \geq 1$  be a common injectivity length for the blocks, and suppose*

$$L_0 < L, \quad L \geq 3(g-1)(L_0+1)+1, \quad N \geq L+L_0,$$

*then*

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

*Proof.* Not ready The length hypotheses are the conservative range of [PGVWC07, Theorem 12]. The shorter range stated in [CPGSV21, lines 2126–2128] requires the strengthened block-diagonal re-growing step. Since  $g \geq 2$  and  $L_0 \geq 1$ ,

$$L \geq 3(g-1)(L_0+1)+1 \geq 3L_0+4 > L_0, \quad N \geq L+L_0 \geq L+1,$$

Because  $A_1, \dots, A_g$  is a basis of normal tensors, the block MPV families are non-redundant:

$$j \neq k \implies \exists n \geq 1, V^{(n)}(A_j) \neq V^{(n)}(A_k).$$

Thus the ambient hypotheses of the block-diagonal intersection theorem are available. Theorem 16.213 gives the periodic decomposition

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \sum_j \mathcal{G}_{N,L}(A_j).$$

Theorem 16.111 gives

$$\mathcal{G}_{N,L}(A_j) = \text{span}\{V^{(N)}(A_j)\}$$

for every  $j$ . Substituting these one-block identities into the preceding sum gives the displayed containment.  $\square$

**Theorem 16.218** (Chain ground space generated by a basis of normal tensors). Not ready *If  $A^i = \bigoplus_j \mu_j A_j^i$  is in block-injective canonical form, where  $A_1, \dots, A_g$  is a basis of normal tensors, assume  $g \geq 2$  and  $\mu_j \neq 0$  for every  $j$ . Let  $L_0 \geq 1$  be a common injectivity length for the blocks, and suppose*

$$L_0 < L, \quad L \geq 3(g-1)(L_0+1)+1, \quad N \geq L+L_0,$$

*then*

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

*Proof.* Not ready We prove the two inclusions separately. The inclusion

$$\mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right) \subseteq \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$$

is exactly Theorem 16.217. For the reverse inclusion, the nonzero-weight hypothesis and Lemma 16.178 show that each vector  $V^{(N)}(A_j)$  lies in  $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ . Since the chain ground space is a linear subspace, it contains every linear combination of such vectors, and hence contains  $\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$ . Thus

$$\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\} \subseteq \mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right),$$

and the two spaces are equal. □

*Remark.* A separate kernel-identification statement for these direct-sum block families would convert the chain-ground-space statement into the corresponding statement for  $\ker(H_N)$ .



## Chapter 17

# Exponential Decay of Correlations

This chapter derives the transfer-matrix expression for thermodynamic-limit correlators of a normal MPS and the resulting exponential decay estimate. The key mechanism is spectral: after removing the leading eigenvalue contribution, the connected part is governed by subleading eigenvalues of the transfer map.

When the MPS tensor generates a parent Hamiltonian (Chapter 16), the correlator quantifies spatial correlations in the ground state. The exponential decay rate is set by the spectral properties of the transfer map, i.e. by the modulus of its subleading eigenvalues.

The spectral analysis of the transfer map and the mixed transfer operator, including the eigenvalue bound  $\rho(F_{AB}) \leq 1$ , the strict transfer-operator gap  $\rho(F_{AB}) < 1$  for non-equivalent blocks, and the resulting overlap decay, is developed in Chapter 7. This chapter focuses on the *correlator* side: one-point and two-point expectations, their spectral decomposition, and quantitative bounds on decay rates.

### 17.1 Connected correlators from the transfer map

**Definition 17.1** (One-point expectation). [Lean](#) [✓ Stmt](#) Fix an MPS tensor and a right fixed point of its transfer map. Write the tensor as  $A$  and the fixed point as  $\rho^R$ . Under the hypotheses of Theorem 6.11, that fixed point exists and is unique up to scaling; we normalize it so that  $\text{tr}(\rho^R) = 1$ . For a local observable  $X \in M_D(\mathbb{C})$  define

$$\langle X_0 \rangle := \text{tr}(X \rho^R).$$

**Definition 17.2** (Two-point expectation at distance  $n$ ). [Lean](#) [✓ Stmt](#) For observables  $X, Y \in M_D(\mathbb{C})$  and  $n \geq 0$ ,

$$\langle X_0 Y_n \rangle := \text{tr}(Y \mathcal{E}_A^n(X \rho^R)).$$

Equivalently: insert  $X$  at site 0, propagate by  $\mathcal{E}_A^n$ , then insert  $Y$  and take the trace.

**Definition 17.3** (Connected correlator). [Lean](#) [✓ Stmt](#) The connected two-point function is

$$C(X, Y; n) := \langle X_0 Y_n \rangle - \langle X_0 \rangle \langle Y_0 \rangle.$$

## 17.2 Spectral expansion and exponential decay

**Theorem 17.4** (Sum of exponentials (spectral expansion interface)). [Lean](#) [✓ Stmt](#) If coefficients  $c_j \in \mathbb{C}$  and eigenvalues  $\lambda_j \in \mathbb{C}$  ( $j = 1, \dots, D^2 - 1$ ) satisfy, for all  $n \geq 0$ ,

$$C(X, Y; n) = \sum_{j=1}^{D^2-1} c_j \lambda_j^n,$$

then the connected correlator equals the stated sum of exponentials. The source [\[CPGSV21, Section II.B.3\]](#) states that the connected correlator is a sum of  $D^2 - 1$  pure exponentials  $C(X, Y; n) = \sum_{j \geq 2}^{D^2} c_{XY}(j) \lambda_j^n$ , which always exists for a normal MPS via the transfer-map spectral decomposition.

*Proof.* [✓ Proof](#) Given the expansion hypothesis, the equality is immediate. The source [\[CPGSV21, Section II.B.3\]](#) derives this expansion from the spectral decomposition of  $\mathcal{E}_A^n$  restricted to the complement of the fixed-point eigenspace: the leading eigenvalue  $\lambda_1 = 1$  contributes  $\langle X_0 \rangle \langle Y_0 \rangle$ , which is subtracted in the connected correlator, leaving the sum over subleading eigenvalues.  $\square$

**Theorem 17.5** (Exponential decay bound (interface)). [Lean](#) [✓ Stmt](#) If a constant  $C_{XY} \in \mathbb{R}$  and  $\lambda_2 \in \mathbb{C}$  satisfy, for all  $n \geq 0$ ,

$$|C(X, Y; n)| \leq C_{XY} |\lambda_2|^n,$$

then the connected correlator satisfies that exponential decay bound. The source [\[CPGSV21, Section II.B.3\]](#) obtains this by combining the sum-of-exponentials expansion with the triangle inequality, and Sec. 4 provides the transfer-map gap condition  $|\lambda_j| < 1$  for the subleading eigenvalues.

*Proof.* [✓ Proof](#) Given the bound hypothesis, the estimate is immediate. The source [\[CPGSV21, Section II.B.3\]](#) obtains this bound by applying the triangle inequality to the sum-of-exponentials expansion and using the transfer-map gap condition  $|\lambda_j| \leq |\lambda_2| < 1$  for all subleading eigenvalues.  $\square$

*Remark 17.6* (Transfer-map spectral hypothesis). The hypothesis “ $|\lambda_2| < 1$ ” in Theorem 17.5 is a transfer-map gap condition: all non-leading eigenvalues of  $\mathcal{E}_A$  lie strictly inside the unit disk. It is distinct from the Hamiltonian spectral gap of Chapter 16. The complementary transfer-map gap  $\rho(\mathcal{E}_A - P) < 1$  is established for primitive channels (Theorem 4.51, Chapter 4). The mixed-transfer gap needed for block separation is supplied for irreducible trace-preserving tensors (Theorem 7.30), and quantitative single-tensor transfer-map bounds for injective tensors are recorded in Theorem 17.11. Thus the analytical input needed to make Theorems 17.4 and 17.5 unconditional is already present; what remains is extracting the spectral expansion coefficients  $c_j, \lambda_j$  from the eigendecomposition of  $\mathcal{E}_A$  (Issue #1447).

**Definition 17.7** (Correlation length). [Lean](#) [✓ Stmt](#) The correlation length associated with a subleading eigenvalue  $\lambda_2$  of the transfer map ( $|\lambda_2| < 1$ ) is

$$\xi := -\frac{1}{\log |\lambda_2|}.$$

**Lemma 17.8** (Correlation length is positive). [Lean](#) [✓ Stmt](#) If  $|\lambda_2| < 1$  and  $\lambda_2 \neq 0$ , then  $\xi > 0$ .

*Proof.* [✓ Proof](#) Since  $0 < |\lambda_2| < 1$ , we have  $\log |\lambda_2| < 0$ , so  $\xi = -1/\log |\lambda_2| > 0$ .  $\square$

### 17.3 Quantitative transfer-map gap bounds

The transfer-map gap theorems in Chapter 4 establish  $\rho(\mathcal{E}_A - P) < 1$  for primitive channels qualitatively. This section gives the *quantitative* refinements with explicit constants and rates for the single-tensor transfer map  $\mathcal{E}_A$ .

**Theorem 17.9** (Exponential convergence of injective primitive channels). [Lean] [✓ Stmt] Fix an injective, trace-preserving MPS tensor  $A$  with positive definite fixed point  $\rho$ . Let

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$$

be the corresponding fixed-point projection. Then there exist  $C > 0$  and  $0 < \delta \leq 1$  such that for all  $n \geq 0$  and  $X \in M_D(\mathbb{C})$ ,

$$\|\mathcal{E}_A^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|.$$

The transfer-map gap  $\delta$  exists by primitivity (Theorem 4.51, the complementary transfer-map gap).

*Proof.* [✓ Proof] Injectivity implies primitivity of the transfer map. The complementary spectral radius  $\rho(\mathcal{E}_A - P) < 1$  then follows from the primitive transfer-map gap theorem for the complementary map. The Gelfand formula yields a geometric bound  $\|(\mathcal{E}_A - P)^n\| \leq C(1 - \delta)^n$  for suitable constants, giving the claimed estimate.  $\square$

**Theorem 17.10** (Correlation length bound). [Lean] [✓ Stmt] For an injective trace-preserving MPS tensor, there exist  $C > 0$  and  $\xi > 0$  such that for all  $n \geq 0$  and all traceless  $X \in M_D(\mathbb{C})$  (i.e.  $\text{tr}(X) = 0$ ),

$$\|\mathcal{E}_A^n(X)\| \leq C e^{-n/\xi} \|X\|.$$

Traceless matrices lie in  $\ker P$ , where  $P$  is the fixed-point projection. Since  $\mathcal{E}_A - P$  has spectral radius strictly less than 1, the Gelfand formula gives exponential decay at rate  $\xi = -1/\log \rho(\mathcal{E}_A - P)$ .

*Proof.* [✓ Proof] For traceless  $X$  we have  $P(X) = 0$ , so  $\mathcal{E}_A^n(X) = (\mathcal{E}_A - P)^n(X)$ . The exponential convergence theorem gives a geometric bound  $C(1 - \delta)^n \|X\|$ . Rewriting  $(1 - \delta)^n = e^{-n/\xi}$  with  $\xi = -1/\log(1 - \delta)$  yields the claimed estimate.  $\square$

**Theorem 17.11** (Transfer-map gap from injectivity). [Lean] [✓ Stmt] For an injective trace-preserving MPS tensor, there exists  $\delta > 0$  such that every eigenvalue  $\mu \neq 1$  of the transfer map satisfies  $|\mu| \leq 1 - \delta$ .

*Proof.* [✓ Proof] Injectivity implies eventual full Kraus rank. This yields primitivity, hence a complementary transfer-map gap. Since the transfer map has only finitely many eigenvalues, one may choose a uniform  $\delta > 0$  separating 1 from the remaining spectrum.  $\square$

### 17.4 Relation to parent Hamiltonians

*Remark 17.12.* When the MPS tensor  $A$  generates a parent Hamiltonian  $H_N(A, L)$  (Definition 16.17) whose ground state is the matrix product vector (Lemma 16.21), the exponential decay bound (Theorem 17.5) shows that ground-state correlations decay with a finite correlation length  $\xi$ .

## 17.5 Renormalization fixed points and zero correlation length

**Definition 17.13** (Renormalization fixed point). [Lean](#) [✓ Stmt](#) An MPS tensor  $A$  is a *renormalization fixed point* (RFP) when its transfer map is idempotent,  $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$ . See [CPGSV16, Definition 3.2].

**Theorem 17.14** (Kraus-isometry criterion for RFP). [Lean](#) [✓ Stmt](#) Suppose there exists an isometry  $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d^2}$ , written as coefficients  $V_{(i_1, i_2), j}$ , such that

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1, i_2), j} A^j \quad (17.1)$$

for all  $i_1, i_2 \in \{0, \dots, d-1\}$ . Then  $A$  is a renormalization fixed point. This is the backward direction of [CPGSV16, Theorem 3.1].

*Proof.* [✓ Proof](#) Expanding  $\mathcal{E}_A^2$  gives a double Kraus sum indexed by pairs  $(i_1, i_2)$ . Substituting the decomposition (17.1) and using the isometry relation  $V^\dagger V = 1$  collapses the double sum to the single Kraus family  $\{A^j\}$ , hence  $\mathcal{E}_A^2 = \mathcal{E}_A$ .  $\square$

**Definition 17.15** (Local orthogonality). [Lean](#) [✓ Stmt](#) A single BNT block is *locally orthogonal* when its self-transfer map is idempotent. For a single tensor this coincides with the RFP condition (Definition 17.13); in the full multi-block BNT setting, local orthogonality additionally requires that mixed transfer operators vanish. See [CPGSV16, Definition 3.5].

**Theorem 17.16** (RFP normal tensor is injective). [Lean](#) [✓ Stmt](#) If  $A$  is a nonzero normal RFP tensor, then  $A$  is injective.

*Proof.* [✓ Proof](#) The RFP condition  $\mathcal{E}_A^2 = \mathcal{E}_A$  implies that the cumulative span  $\text{span}\{A^w : |w| \leq n\}$  stabilises after one step. Since  $A$  is normal (hence has an injective blocked form), the one-step span already equals  $M_D(\mathbb{C})$ , so  $A$  is injective.  $\square$

**Theorem 17.17** (Left-canonical normal RFP tensor is injective). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . Let  $A$  be a normal renormalization fixed point such that

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then  $A$  is injective. This is the left-canonical injectivity step in [CPGSV16, Appendix B].

*Proof.* [✓ Proof](#) The normalization  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  forces at least one matrix  $A^i$  to be nonzero, since otherwise the left-hand side would vanish. Hence  $A$  is a nonzero normal renormalization fixed point, and Theorem 17.16 applies.  $\square$

**Theorem 17.18** (Unitary diagonal fixed point for a left-canonical normal RFP). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . Let  $A$  be a normal renormalization fixed point such that

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then there exist a unitary  $U$  and a diagonal positive definite matrix  $\Lambda$  such that, for  $B^i := U^\dagger A^i U$ ,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \quad \text{and} \quad \mathcal{E}_B(\Lambda) = \Lambda.$$

This is the diagonal fixed-point reduction in [CPGSV16, Appendix B].

*Proof.* ✓ Proof By Theorem 17.17, the tensor  $A$  is injective. Therefore its transfer map is irreducible, and hence  $A$  is irreducible as a tensor. Applying Theorem 9.20 gives a unitary conjugate  $B$  for which the trace-preserving normalization remains valid and a diagonal positive definite matrix  $\Lambda$  with  $\mathcal{E}_B(\Lambda) = \Lambda$ .  $\square$

**Theorem 17.19** (Rank-one classification for RFP transfer maps). Lean ✓ Stmt *Let  $A$  be an injective left-canonical RFP tensor with positive-definite fixed point  $\rho$  of the transfer map  $\mathcal{E}_A$ . Then  $\mathcal{E}_A = P_\rho$ , where  $P_\rho(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$  is the rank-one fixed-point projection.*

*Proof.* ✓ Proof The exponential convergence bound gives  $\|\mathcal{E}_A^n(X) - P_\rho(X)\| \leq C(1 - \delta)^n \|X\|$ . Idempotence of  $\mathcal{E}_A$  implies  $\mathcal{E}_A^{n+1} = \mathcal{E}_A$  for all  $n \geq 0$ , so  $\|\mathcal{E}_A(X) - P_\rho(X)\| \leq C(1 - \delta)^{n+1} \|X\|$  for all  $n$ . Taking  $n \rightarrow \infty$  gives  $\mathcal{E}_A = P_\rho$ .  $\square$

**Theorem 17.20** (Full structural form for RFP tensors). Lean ✓ Stmt *A normal tensor  $A$  in canonical form  $II$  that is a renormalization fixed point admits a decomposition  $A^i = X \Lambda U^i X^{-1}$ , where  $\Lambda$  is diagonal positive and the family  $U = (U^i)$  is an isometry in the physical indices.*

*Proof.* ✓ Proof Combine the diagonal fixed-point reduction with the rank-one classification of injective left-canonical RFP tensors. One then writes the resulting rank-one projector in a canonical Kraus form, applies Kraus freedom to identify the physical-index family with an isometry  $U$ , and transports the decomposition back through the conjugation.  $\square$

**Theorem 17.21** (Canonical-form blocks are injective). Lean ✓ Stmt *For a family of tensors in canonical form, every block is injective. This is a direct projection of the injectivity field of the canonical-form predicate; see [CPGSV16, Section 2.3].*

*Proof.* ✓ Proof Immediate from the `block_injective` field of `IsCanonicalForm`.  $\square$

**Theorem 17.22** (RG flow convergence for canonical-form tensors). Lean ✓ Stmt *For a tensor in canonical form with blocks  $(A_k)$  and weights  $(\mu_k)$ , the iterated blocking  $\mathcal{E}_{A_k}^{2^n}$  converges pointwise to an idempotent linear map  $\mathcal{E}_\infty$  as  $n \rightarrow \infty$ . That is, there exists  $\mathcal{E}_\infty$  with  $\mathcal{E}_\infty^2 = \mathcal{E}_\infty$  such that  $\mathcal{E}_{A_k}^{2^n}(\rho) \rightarrow \mathcal{E}_\infty(\rho)$  for all density matrices  $\rho$ . This is [CPGSV16, Appendix B].*

*Proof.* ✓ Proof Each block is injective and left-canonical, so its transfer map is a primitive channel with a unique positive-definite fixed point  $\rho_0$ . The exponential convergence bound  $\|\mathcal{E}^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|$  (Theorem 17.9) then gives pointwise convergence  $\mathcal{E}^n(X) \rightarrow P(X)$ , and composing with the subsequence  $2^n \rightarrow \infty$  yields the result. The witness  $\mathcal{E}_\infty = P$  is the rank-one fixed-point projection, which is idempotent.  $\square$

*Remark 17.23.* When the transfer map is idempotent ( $\mathcal{E}^2 = \mathcal{E}$ ), connected correlations become separation-independent for all separations  $n \geq 1$ . Informally, this corresponds to zero correlation length ( $\xi = 0$ ). The full spectral equivalence (idempotence iff vanishing subleading spectrum) is deferred here.

**Definition 17.24** (Correlations independent of distance). Lean ✓ Stmt *For a positive definite right fixed point  $\rho^R > 0$ , an MPS tensor has *correlations independent of distance* (CID) when for all local observables  $X$  and  $Y$  and all separations  $n, m \geq 1$ ,*

$$C(X, Y; n) = C(X, Y; m).$$

**Definition 17.25** (Zero correlation length). Lean ✓ Stmt *An MPS tensor has *zero correlation length* when it is both locally orthogonal and correlations are independent of distance. See [CPGSV16, Definition 3.6].*

**Theorem 17.26** (CID implies RFP). Lean ✓ Stmt *If an MPS tensor admits a positive definite right fixed point of its transfer map and has correlations independent of distance, then its transfer map is idempotent. This is the reverse direction of [CPGSV16, Theorem 3.8].*

*Proof.* ✓ Proof Trace nondegeneracy and invertibility of the fixed point reduce idempotence to equality of the connected correlator at separations  $n = 2$  and  $n = 1$ . □

**Theorem 17.27** (Zero correlation length iff idempotent transfer). Lean ✓ Stmt *An MPS tensor has zero correlation length (i.e. is both locally orthogonal and CID) if and only if its transfer map is idempotent. This is [CPGSV16, Theorem 3.8].*

*Proof.* ✓ Proof The forward direction extracts local orthogonality (which equals idempotence by definition). The reverse uses  $\mathcal{E}^n = \mathcal{E}$  for  $n \geq 1$  to show the connected correlator is separation-independent. □

**Theorem 17.28** (Canonical-form blocks satisfy RFP iff ZCL). Lean ✓ Stmt *For a tensor in canonical form, each block is a renormalization fixed point if and only if it has zero correlation length. This is the RFP–ZCL part of [CPGSV16, Theorem 3.10].*

*Proof.* ✓ Proof Apply Theorem 17.27 to the chosen canonical-form block. □

# Chapter 18

## Concrete Examples

This chapter collects concrete MPS tensors from [CPGSV21, Section III]. Each example is specified by its physical dimension  $d$ , bond dimension  $D$ , and explicit matrix family  $\{A^i\}$ . Key structural properties (injectivity, RFP status, symmetry) are stated for each tensor.

### 18.1 The GHZ state

The Greenberger–Horne–Zeilinger (GHZ) state is the prototypical non-injective, renormalization-fixed-point MPS.

**Definition 18.1** (GHZ tensor). [Lean](#) [✓ Stmt](#) Set  $d = D = 2$ . The GHZ tensor is the family  $\{A^0, A^1\} \subset M_2(\mathbb{C})$  defined by

$$A^0 = |0\rangle\langle 0| = \text{diag}(1, 0), \quad A^1 = |1\rangle\langle 1| = \text{diag}(0, 1).$$

For a system of  $N$  sites the resulting MPV is the GHZ state

$$|\text{GHZ}_N\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}.$$

**Definition 18.2** (Two-site computational vector). [Lean](#) [✓ Stmt](#) For  $a, b \in \{0, 1\}$ , the vector

$$|ab\rangle \in (\{0, 1\}^2 \rightarrow \mathbb{C})$$

is the indicator of the configuration

$$(\sigma_1, \sigma_2) = (a, b).$$

**Definition 18.3** (Constant product vector). [Lean](#) [✓ Stmt](#) For  $a \in \{0, 1\}$  and  $N \geq 0$ ,

$$|a\rangle^{\otimes N}(\sigma_1, \dots, \sigma_N) = \begin{cases} 1, & (\sigma_1, \dots, \sigma_N) = (a, \dots, a), \\ 0, & \text{otherwise.} \end{cases}$$

**Theorem 18.4** (GHZ transfer map is idempotent). [Lean](#) [✓ Stmt](#) The transfer map of the GHZ tensor satisfies  $\mathcal{E}_A^2 = \mathcal{E}_A$ .

*Proof.* [✓ Proof](#) The transfer map acts by  $\mathcal{E}_A(X)_{ij} = \delta_{ij}X_{ij}$  (extraction of the diagonal). This operation is idempotent.  $\square$

**Theorem 18.5** (GHZ tensor is an RFP). [Lean](#) [✓ Stmt](#) *The GHZ tensor is a renormalization fixed point.*

*Proof.* [✓ Proof](#) Immediate from Theorem 18.4.  $\square$

**Theorem 18.6** (GHZ tensor is not injective). [Lean](#) [✓ Stmt](#) *The GHZ tensor is not injective:  $\text{span}_{\mathbb{C}}\{A^0, A^1\}$  is the two-dimensional space of diagonal matrices in  $M_2(\mathbb{C})$ , which does not equal  $M_2(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) Every element of  $\text{span}_{\mathbb{C}}\{A^0, A^1\}$  has zero off-diagonal entries, so the matrix with all entries equal to 1 is not in the span.  $\square$

**Theorem 18.7** ( $\mathbb{Z}_2$  on-site symmetry of the GHZ tensor). [Lean](#) [✓ Stmt](#) *Let  $\mathbb{Z}_2 = \{1, \sigma_x\}$  act on  $\mathbb{C}^2$  by the Pauli-X matrix  $\sigma_x$ . The GHZ tensor is on-site symmetric under this action: for each  $g \in \mathbb{Z}_2$ , the twisted tensor  $\tilde{A}_g$  defines the same MPV family as  $A$ , i.e.  $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$ .*

*Proof.* [✓ Proof](#) The identity element is trivial. For the generator  $g = \sigma_x$ , the twisted tensor satisfies  $\tilde{A}_g^i = A^{1-i}$ , i.e. the two matrices are swapped. Conjugation by  $\sigma_x$  implements the same swap, giving gauge equivalence and hence the same MPV family.  $\square$

**Definition 18.8** (Virtual  $\mathbb{Z}_2$  representation of the GHZ tensor). [Lean](#) [✓ Stmt](#) *The virtual  $\mathbb{Z}_2$  representation on the bond space is the homomorphism  $\mathbb{Z}_2 \rightarrow M_2(\mathbb{C})$  sending the identity to  $I$  and the generator to  $\sigma_z = \text{diag}(1, -1)$ .*

**Theorem 18.9** (Virtual  $\mathbb{Z}_2$  symmetry of the GHZ tensor). [Lean](#) [✓ Stmt](#) *The matrix  $\sigma_z$  commutes with every GHZ matrix  $A^i$ . A non-scalar matrix commuting with all  $A^i$  is the hallmark of a non-injective ( $\mathbb{Z}_2$ -injective) tensor: the on-site symmetry of Theorem 18.7 swaps the two one-dimensional injective blocks  $A^0 \leftrightarrow A^1$ , while  $\sigma_z$  acts on each block by a phase.*

*Proof.* [✓ Proof](#) Both  $A^0$  and  $A^1$  are diagonal, so each commutes with the diagonal matrix  $\sigma_z$ .  $\square$

**Theorem 18.10** (GHZ order parameter is a transfer-map fixed point). [Lean](#) [✓ Stmt](#) *The order-parameter operator  $\sigma_z$  is a fixed point of the GHZ transfer map,  $\mathcal{E}_A(\sigma_z) = \sigma_z$ .*

*Proof.* [✓ Proof](#) Since  $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$  and both  $A^i$  are diagonal, the claim is the entrywise computation  $\mathcal{E}_A(\sigma_z) = \sigma_z$ .  $\square$

*Remark 18.11.* A primitive (injective) tensor has a one-dimensional transfer-map fixed-point space, spanned by a positive matrix that equals the identity only in a normalized gauge. That the GHZ transfer map fixes the additional non-scalar operator  $\sigma_z$  is the transfer-matrix signature of the extra non-decaying mode behind its long-range ferromagnetic correlations.

## 18.2 The AKLT state

The Affleck–Kennedy–Lieb–Tasaki (AKLT) state is the canonical example of a normal (block-injective) MPS with non-trivial symmetry-protected topological (SPT) order.

**Definition 18.12** (AKLT tensor). [Lean](#) [✓ Stmt](#) Set  $d = 3$  (spin-1) and  $D = 2$ . The AKLT tensor is the family  $\{A^0, A^1, A^2\} \subset M_2(\mathbb{C})$  defined via the Pauli matrices by

$$A^0 = \frac{1}{\sqrt{3}}\sigma_z, \quad A^1 = \frac{\sqrt{2}}{\sqrt{3}}|0\rangle\langle 1|, \quad A^2 = -\frac{\sqrt{2}}{\sqrt{3}}|1\rangle\langle 0|,$$



where  $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$ , so equivalently  $A^1 = \frac{1}{\sqrt{3}}\sigma_+$  and  $A^2 = -\frac{1}{\sqrt{3}}\sigma_-$  for  $\sigma_+ = \sqrt{2}|0\rangle\langle 1|$  and  $\sigma_- = \sqrt{2}|1\rangle\langle 0|$ . Equivalently, the matrices are the Clebsch–Gordan coefficients coupling spin-1 and spin- $\frac{1}{2}$  to spin- $\frac{1}{2}$ .

**Theorem 18.13** (AKLT tensor is not injective). [Lean](#) [✓ Stmt](#) *The AKLT tensor is not injective at a single site:  $\text{span}_{\mathbb{C}}\{A^0, A^1, A^2\}$  is the three-dimensional space of traceless matrices  $\mathfrak{sl}(2, \mathbb{C})$ , which is a proper subspace of  $M_2(\mathbb{C})$ .*

**Theorem 18.14** (AKLT tensor is normal). [Lean](#) [✓ Stmt](#) *The transfer map of the AKLT tensor has a unique eigenvalue of maximal modulus (a simple spectral-radius eigenvalue); in particular the AKLT tensor is normal. Concretely, the length-2 blocked tensor is injective.*

**Theorem 18.15** ( $\mathbb{Z}_2 \times \mathbb{Z}_2$  on-site symmetry of the AKLT tensor). [Lean](#) [✓ Stmt](#) *The AKLT tensor is on-site symmetric under the  $\mathbb{Z}_2 \times \mathbb{Z}_2$  subgroup of  $\text{SO}(3)$  generated by two commuting  $\pi$ -rotations about orthogonal axes, acting on the spin-1 physical index.*

**Theorem 18.16** (Anticommuting virtual gauges of the AKLT symmetry). [Lean](#) [✓ Stmt](#) *The two virtual matrices that implement the symmetry of Theorem 18.15, namely  $i\sigma_y$  and  $\sigma_z$ , anti-commute.*

**Definition 18.17** (AKLT factor system). [Lean](#) [✓ Stmt](#) *The AKLT factor system on  $\mathbb{Z}_2 \times \mathbb{Z}_2$  is  $\omega(g, h) = (-1)^{(g_1+g_2)h_1}$  (the value  $-1$  when  $g_1 + g_2 = 1$  and  $h_1 = 1$ , and  $1$  otherwise). It is the factor system of the explicit projective representation with virtual gauges  $i\sigma_y$  and  $\sigma_z$ ; the diagonal term  $g_1 h_1$  relative to the cluster case reflects  $(i\sigma_y)^2 = -I$ .*

**Lemma 18.18** (AKLT commutator phase). [Lean](#) [✓ Stmt](#) *The commutator phase of the AKLT factor system on the two generators is  $-1$ .*

**Theorem 18.19** (The AKLT factor system is a 2-cocycle). [Lean](#) [✓ Stmt](#) *The AKLT factor system is a genuine 2-cocycle, being the factor system of an explicit projective representation, so it represents an element of  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1))$ .*

**Theorem 18.20** (The AKLT  $\mathbb{Z}_2 \times \mathbb{Z}_2$  phase is non-trivial). [Lean](#) [✓ Stmt](#) *The AKLT factor system (Theorem 18.19), built from the anticommuting gauges  $i\sigma_y$  and  $\sigma_z$  (Theorem 18.16), is not a coboundary: its commutator phase on the two generators is  $-1$ . Its class in  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$  is therefore non-trivial, so the AKLT chain realizes a non-trivial SPT phase rather than a trivial product state.*

**Remark 18.21.** This  $\mathbb{Z}_2 \times \mathbb{Z}_2$  is the smallest subgroup of  $\text{SO}(3)$  that still carries a non-trivial 2-cocycle. That minimality, and the full continuous classification over  $\text{SO}(3)$ , are not formalized (Remark 18.25).

**Definition 18.22** (Cartesian AKLT tensor). [Lean](#) [✓ Stmt](#) *The Cartesian presentation of the AKLT tensor has  $d = 3$ ,  $D = 2$ , and  $A^x = \sigma_x$ ,  $A^y = \sigma_y$ ,  $A^z = \sigma_z$ , the three Pauli matrices. This is the rotationally natural form used in [CPGSV21] (around line 1159); it is the AKLT state in the Cartesian basis of the spin-1 site, the counterpart of the  $|m\rangle$ -basis form of Definition 18.12.*

**Theorem 18.23** (The spin- $\frac{1}{2}$  cover is surjective onto  $\text{SO}(3)$ ). [Lean](#) [✓ Stmt](#) *Every rotation  $R \in \text{SO}(3)$  is realized by conjugating the Pauli vector by some  $U \in \text{SU}(2)$ : there is  $U \in \text{SU}(2)$  with  $(\frac{1}{2} \text{tr}(\sigma_i U \sigma_j U^{-1}))_{ij} = R$ . Equivalently, the adjoint double cover  $\text{SU}(2) \rightarrow \text{SO}(3)$  is surjective. The proof factors  $R = R_z(\alpha) R_x(\beta) R_z(\gamma)$  by its  $ZXZ$  Euler decomposition and maps each coordinate rotation to an explicit  $\text{SU}(2)$  generator:  $\text{diag}(e^{-i\theta/2}, e^{i\theta/2}) \mapsto R_z(\theta)$  and  $\begin{pmatrix} \cos \frac{\beta}{2} & -i \sin \frac{\beta}{2} \\ -i \sin \frac{\beta}{2} & \cos \frac{\beta}{2} \end{pmatrix} \mapsto R_x(\beta)$ .*

**Theorem 18.24** (SO(3) on-site symmetry of the AKLT tensor). [Lean](#) [✓ Stmt](#) *The Cartesian AKLT tensor is on-site symmetric under the spin-1 (defining) representation of SO(3) on the physical index.*

*Remark 18.25.* The bond index carries the projective spin- $\frac{1}{2}$  representation: the gauge for a rotation is either of its two SU(2) preimages, so the assignment is a representation only up to sign and factors through the double cover  $\text{SU}(2) \rightarrow \text{SO}(3)$ . Its class is the non-trivial element of  $H^2(\text{SO}(3), \text{U}(1)) \cong \mathbb{Z}_2$ , witnessing the SPT phase.

### 18.3 The cluster state

The cluster state is a universal resource for measurement-based quantum computation and provides another example of a non-trivial SPT phase.

**Definition 18.26** (Cluster-state tensor). [Lean](#) [✓ Stmt](#) Set  $d = D = 2$ . The cluster-state tensor is the family  $\{A^0, A^1\} \subset M_2(\mathbb{C})$  defined by

$$A^0 = |+\rangle\langle 0| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad A^1 = |-\rangle\langle 1| = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix},$$

where  $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ . The review [\[CPGSV21\]](#) writes the reflected convention  $A^0 = |0\rangle\langle +|$ ,  $A^1 = |1\rangle\langle -|$ ; the transpose taken here is the same state read in the opposite direction and carries the same SPT order.

**Theorem 18.27** (Cluster-state tensor is not injective). [Lean](#) [✓ Stmt](#) *The cluster-state tensor is not injective:  $\text{span}_{\mathbb{C}}\{A^0, A^1\} = \text{span}_{\mathbb{C}}\{|+\rangle\langle 0|, |-\rangle\langle 1|\}$  has dimension 2, not  $\dim M_2(\mathbb{C}) = 4$ .*

*Proof.* [✓ Proof](#) Both  $A^0$  and  $A^1$  are rank-one matrices with linearly independent column spaces, so they are linearly independent. Their span is two-dimensional, which is strictly less than  $\dim M_2(\mathbb{C}) = 4$ .  $\square$

**Theorem 18.28** (Cluster-state tensor is normal). [Lean](#) [✓ Stmt](#) *Although the cluster-state tensor is not injective at a single site, its length-2 blocking is injective: the four length-2 products span  $M_2(\mathbb{C})$ . In particular the cluster-state tensor is normal.*

*Remark 18.29.* Although the cluster-state tensor is not injective at length 1, the length-2 blocked tensor is injective. Using  $\langle 0|\pm\rangle = \frac{1}{\sqrt{2}}$ ,  $\langle 1|+\rangle = \frac{1}{\sqrt{2}}$ , and  $\langle 1|-\rangle = -\frac{1}{\sqrt{2}}$ , the four products  $A^{ij} = A^i A^j$  are

$$A^{00} = \frac{1}{\sqrt{2}}|+\rangle\langle 0|, \quad A^{01} = \frac{1}{\sqrt{2}}|+\rangle\langle 1|, \quad A^{10} = \frac{1}{\sqrt{2}}|-\rangle\langle 0|, \quad A^{11} = -\frac{1}{\sqrt{2}}|-\rangle\langle 1|. \quad (18.1)$$

Since  $\{|+\rangle, |-\rangle\}$  and  $\{|0\rangle, |1\rangle\}$  are bases, these four rank-one matrices are linearly independent and span  $M_2(\mathbb{C})$ .

**Theorem 18.30** ( $\mathbb{Z}_2 \times \mathbb{Z}_2$  on-site symmetry of the blocked cluster-state tensor). [Lean](#) [✓ Stmt](#) *The length-2 blocked cluster-state tensor is on-site symmetric under  $\mathbb{Z}_2 \times \mathbb{Z}_2$ . The two generators act on the 4-dimensional blocked physical space  $(\mathbb{C}^2)^{\otimes 2}$  by  $\sigma_x \otimes I$  and  $I \otimes \sigma_x$ , which commute and each square to the identity, defining a linear representation of  $\mathbb{Z}_2 \times \mathbb{Z}_2$ .*

**Theorem 18.31** (Anticommuting virtual gauges of the cluster-state symmetry). [Lean](#) [✓ Stmt](#) *The virtual matrices implementing the two generators of Theorem 18.30, namely  $V_1 = \sigma_z$  and  $V_2 = \sigma_x$ , anticommute ( $V_1 V_2 = -V_2 V_1$ ).*

**Definition 18.32** (Cluster-state factor system). [Lean](#) [✓ Stmt](#) The cluster-state factor system on  $\mathbb{Z}_2 \times \mathbb{Z}_2$  is  $\omega(g, h) = (-1)^{g_2 h_1}$  (the value  $-1$  when  $g_2 = 1$  and  $h_1 = 1$ , and  $1$  otherwise). It is the factor system of the explicit projective representation with virtual gauges  $\sigma_z$  and  $\sigma_x$ , both squaring to the identity, so there is no diagonal term.

**Lemma 18.33** (Cluster-state commutator phase). [Lean](#) [✓ Stmt](#) The commutator phase of the cluster-state factor system on the two generators is  $-1$ .

**Theorem 18.34** (The cluster-state factor system is a 2-cocycle). [Lean](#) [✓ Stmt](#) The cluster-state factor system is a genuine 2-cocycle, being the factor system of an explicit projective representation, so it represents an element of  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1))$ .

**Theorem 18.35** (The cluster-state  $\mathbb{Z}_2 \times \mathbb{Z}_2$  phase is non-trivial). [Lean](#) [✓ Stmt](#) The cluster-state factor system (Theorem 18.34), built from the anticommuting gauges  $\sigma_z$  and  $\sigma_x$  (Theorem 18.31), is not a coboundary: its commutator phase on the two generators is  $-1$ . Its class in  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$  is therefore non-trivial, so the cluster state realizes a non-trivial SPT phase.

**Theorem 18.36** (The cluster state has string order under its  $\mathbb{Z}_2 \times \mathbb{Z}_2$  symmetry). [Lean](#) [✓ Stmt](#) For every element  $g$  of  $\mathbb{Z}_2 \times \mathbb{Z}_2$ , the length-2 blocked cluster-state tensor has string order in the sense of Definition 12.38 for the twist by  $g$ , with the maximally mixed boundary state  $\Lambda = \frac{1}{2}\mathbb{1}$  as its stationary boundary state.

*Proof.* [✓ Proof](#) The blocked tensor is injective (Remark 18.29) and on-site symmetric under  $\mathbb{Z}_2 \times \mathbb{Z}_2$  (Theorem 18.30), and each group element acts by a real symmetric involutive permutation matrix, hence unitarily. The maximally mixed state  $\Lambda = \frac{1}{2}\mathbb{1}$  is positive definite and has trace 1. The four blocked matrices in (18.1) satisfy  $\sum_i A^i (A^i)^\dagger = \mathbb{1}$  and  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , so the transfer map is unital and  $\Lambda$  is a fixed point of the adjoint transfer map. These are exactly the hypotheses of Theorem 12.61, which then yields string order for every  $g$ . The cluster state is thus the first explicit witness of that criterion.  $\square$

## 18.4 Parent Hamiltonian statements for the examples

The parent-Hamiltonian construction of [CPGSV21, lines 1988–2000] is recorded here in the form

$$\mathcal{G}_L(A) = \left\{ \sum_{i_1, \dots, i_L} \text{tr}(A^{i_1} \dots A^{i_L} X) |i_1 \dots i_L\rangle : X \in M_D(\mathbb{C}) \right\}.$$

Thus

$$\ker h_L = \mathcal{G}_L(A), \quad H_N^{(L)} = \sum_i (h_L)_i.$$

We write

$$\text{GS}(H) = \ker(H - E_0(H)I)$$

for the ground eigenspace.

**Theorem 18.37** (GHZ two-site parent space). [Lean](#) [✓ Stmt](#) For the GHZ tensor  $A$  of Definition 18.1,

$$\mathcal{G}_2(A) = \text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}.$$

*Proof.* ✓ Proof By definition,

$$\Gamma_2(X)(\sigma_1, \sigma_2) = \text{tr}(A^{\sigma_1} A^{\sigma_2} X).$$

For the GHZ matrices  $A^0 = \text{diag}(1, 0)$  and  $A^1 = \text{diag}(0, 1)$ , whenever  $\sigma_1 \neq \sigma_2$ ,

$$A^{\sigma_1} A^{\sigma_2} = 0.$$

Hence

$$\Gamma_2(X) = X_{00}|00\rangle + X_{11}|11\rangle,$$

and the image of  $\Gamma_2$  is  $\text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}$ . □

**Theorem 18.38** (Constant GHZ product vectors satisfy the cyclic constraints). Lean ✓ Stmt

For  $N \geq 2$  and  $a \in \{0, 1\}$ ,

$$|a\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\text{GHZ}}).$$

*Proof.* ✓ Proof For a cyclic nearest-neighbour window and outside assignment  $\tau$ , set

$$R_{i,\tau} := |a\rangle^{\otimes N}|_{[i,i+1],\tau}.$$

For  $b, c \in \{0, 1\}$  with  $b \neq c$ , the completed configuration is not constant, hence

$$R_{i,\tau}(b, c) = 0.$$

Therefore

$$R_{i,\tau} \in \{v : v(0, 1) = v(1, 0) = 0\} = \text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\} = \mathcal{G}_2(A_{\text{GHZ}}),$$

so  $|a\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\text{GHZ}})$ . □

**Theorem 18.39** (GHZ periodic nearest-neighbour ground space). Lean ✓ Stmt For  $N \geq 2$ , the common nearest-neighbour cyclic ground space of the GHZ tensor satisfies

$$\mathcal{G}_{N,2}(A_{\text{GHZ}}) = \text{span}_{\mathbb{C}}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\}.$$

This is the cyclic-window form of the two-fold degeneracy in [CPGSV21, line 2205].

*Proof.* ✓ Proof Let  $\psi \in \mathcal{G}_{N,2}(A_{\text{GHZ}})$ . For every cyclic nearest-neighbour window and outside assignment  $\tau$ ,

$$\psi|_{[i,i+1],\tau} \in \mathcal{G}_2(A_{\text{GHZ}}) = \text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}.$$

Taking the outside assignment from the word  $\sigma$  gives

$$\psi|_{[i,i+1],\sigma}(\sigma_i, \sigma_{i+1}) = \psi(\sigma).$$

Hence a mixed nearest-neighbour word has zero coefficient:

$$\sigma_i \neq \sigma_{i+1} \implies \psi(\sigma) = 0.$$

A nonconstant binary cyclic word has a mixed edge,

$$\sigma \notin \{(0, \dots, 0), (1, \dots, 1)\} \implies \exists i, \sigma_i \neq \sigma_{i+1}.$$

Therefore

$$\psi = \psi(0, \dots, 0)|0\rangle^{\otimes N} + \psi(1, \dots, 1)|1\rangle^{\otimes N}.$$

Conversely, Theorem 18.38 gives

$$|0\rangle^{\otimes N}, |1\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\text{GHZ}}),$$

and hence their span lies in  $\mathcal{G}_{N,2}(A_{\text{GHZ}})$ . □

*Remark 18.40* (GHZ parent interaction). For the GHZ tensor of Definition 18.1,

$$\mathcal{G}_2(A) = \text{span}\{|00\rangle, |11\rangle\}, \quad h_{\text{GHZ}} = |01\rangle\langle 01| + |10\rangle\langle 10| = I - |00\rangle\langle 00| - |11\rangle\langle 11|.$$

The periodic nearest-neighbour parent Hamiltonian therefore satisfies

$$H_{\text{GHZ},N} = \sum_i (h_{\text{GHZ}})_i, \quad \ker H_{\text{GHZ},N} = \text{span}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\},$$

using the cyclic-window equality

$$\mathcal{G}_{N,2}(A_{\text{GHZ}}) = \text{span}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\}$$

from Theorem 18.39. This is the ferromagnetic Ising parent Hamiltonian [CPGSV21, lines 1194–1196 and line 2205]; see also [FGSW<sup>+</sup>15, Section 3] for the displayed two-site space and interaction.

*Remark 18.41* (Cluster-state stabilizer convention). The one-dimensional cluster tensor in Definition 18.26 is the reflected convention of the tensor displayed in [CPGSV21, lines 2364–2369]. The sourced stabilizer is

$$K_i^{ZZZ} = \sigma_i^z \sigma_{i+1}^x \sigma_{i+2}^z.$$

The corresponding positive parent interaction is

$$h_i^{ZZZ} = \frac{1}{2}(I - K_i^{ZZZ}), \quad H_{\text{cl},N}^{ZZZ} = \sum_i h_i^{ZZZ}, \quad \dim \ker H_{\text{cl},N}^{ZZZ} = 1$$

[PGVWC07, local TeX lines 374–387]. Thus an  $XZX$  formula should be used for this tensor only after establishing an equivalence such as

$$UH_{\text{cl},N}^{ZZZ}U^\dagger = H_{\text{cl},N}^{XZX} \quad \text{or} \quad TH_{\text{cl},N}^{ZZZ}T^{-1} = H_{\text{cl},N}^{XZX}.$$

*Remark 18.42* (AKLT parent Hamiltonian and edge modes). For the AKLT tensor of Definition 18.12, with  $\mathcal{G}_{[a,b]}$  denoting the local ground space on sites  $a, \dots, b$ ,

$$L_0 = 2, \quad \mathcal{G}_{[1,2]} \cap \mathcal{G}_{[2,3]} = \mathcal{G}_{[1,3]}$$

[CPGSV21, line 2095]. The displayed local spin-chain Hamiltonian is

$$H_{\text{AKLT},N} = \sum_i \left( \vec{S}_i \cdot \vec{S}_{i+1} + \frac{1}{3}(\vec{S}_i \cdot \vec{S}_{i+1})^2 \right),$$

up to an additive constant [PGVWC07, local TeX lines 340–349]; [SPGC11, lines 2126–2130]. The spin-2 projector form is related to the polynomial Hamiltonian by the spin-addition identity

$$P_{i,i+1}^{(2)} = \frac{1}{2} \left( \vec{S}_i \cdot \vec{S}_{i+1} + \frac{1}{3}(\vec{S}_i \cdot \vec{S}_{i+1})^2 + \frac{2}{3}I \right)$$

for two spin-1 particles. For open boundary conditions,

$$\dim \text{GS}(H_{\text{AKLT},N}^{\text{open}}) = 4, \quad \dim(\text{GS}(H_{\text{AKLT},N}^{\text{open}}) \cap \mathcal{H}_{S=0}) = 1$$

[CPGSV21, lines 1169–1174].

## Chapter 19

# Channel Representations and Normal Forms

This chapter collects the representation material for finite-dimensional quantum channels from [Wol12, Chapter 2]: the Choi–Jamiołkowski isomorphism, Kraus representation theorems, Stinespring and Naimark dilations, ordered completely positive maps, Radon–Nikodym and open-system representations, trace-pairing expansions, SVD and Lorentz normal forms, and the channel determinant. These results are important for the later channel theory, but they are not prerequisites for the matrix-product-state Fundamental Theorem.

### 19.1 Representations

The following definitions support the Choi–Jamiołkowski isomorphism (Theorem 19.6) and the representation theorems of [Wol12, Chapter 2].

**Definition 19.1** (Partial traces). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $X \in M_{d,d'}(\mathbb{C})$  be a bipartite matrix indexed by

$$(\{0, \dots, d-1\} \times \{0, \dots, d'-1\})^2.$$

The *left partial trace*  $\text{tr}_A(X)$  and *right partial trace*  $\text{tr}_B(X)$  are the  $d' \times d'$  and  $d \times d$  matrices defined by

$$(\text{tr}_A(X))_{ij} = \sum_k X_{(k,i),(k,j)}, \quad (\text{tr}_B(X))_{ij} = \sum_k X_{(i,k),(j,k)}.$$

**Definition 19.2** (Maximally entangled state). [Lean](#) [✓ Stmt](#) The *maximally entangled state* on  $\mathbb{C}^d \otimes \mathbb{C}^d$  is the rank-one projector

$$|\Omega\rangle\langle\Omega|, \quad |\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle.$$

Concretely,  $(|\Omega\rangle\langle\Omega|)_{(i_1, i_2), (j_1, j_2)} = \frac{1}{d} \delta_{i_1 j_1} \delta_{i_2 j_2}$  and  $\text{tr}(|\Omega\rangle\langle\Omega|) = 1$ .

**Definition 19.3** (Tensor product of a map with identity). [Lean](#) [✓ Stmt](#) Given a linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ , the *tensor extension*  $T \otimes \text{id}$  acts on bipartite matrices  $X \in M_{D \times D}(\mathbb{C})$  by applying  $T$  to each “slice”:

$$(T \otimes \text{id})(X)_{(i_1, i_2), (j_1, j_2)} = (T(X^{(i_2, j_2)}))_{i_1 j_1},$$

where  $X_{ab}^{(i_2, j_2)} = X_{(a, i_2), (b, j_2)}$  is the bipartite slice.

## 19.2 Choi–Jamiołkowski isomorphism

**Definition 19.4** (Choi matrix). [Lean](#) [✓ Stmt](#) The *Choi matrix* of a linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is

$$\tau = (T \otimes \text{id})(|\Omega\rangle\langle\Omega|) \in M_{D \times D}(\mathbb{C}).$$

Concretely,  $\tau_{(i_1, i_2), (j_1, j_2)} = \frac{1}{D} (T(E_{i_2 j_2}))_{i_1 j_1}$  where  $E_{i_2 j_2}$  is the matrix unit. This is the Choi–Jamiołkowski convention of [Wol12, Proposition 2.1].

**Theorem 19.5** (Choi matrix of the identity map). [Lean](#) [✓ Stmt](#) The identity channel has Choi matrix  $|\Omega\rangle\langle\Omega|$ .

*Proof.* [✓ Proof](#) Apply the identity map in the definition of the Choi matrix.  $\square$

**Theorem 19.6** (Complete positivity iff the Choi matrix is positive semidefinite). [Lean](#) [✓ Stmt](#) A linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is completely positive (in the Kraus sense) if and only if its Choi matrix  $\tau \geq 0$ .

*Proof.* [✓ Proof](#) ( $\Rightarrow$ ) If  $T(X) = \sum_i K_i X K_i^\dagger$ , then  $\tau = \sum_i |v_i\rangle\langle v_i|$  where  $(v_i)_{(a,b)} = c K_i(a, b)$  and  $c = 1/\sqrt{D}$ . This is a sum of rank-one PSD matrices.

( $\Leftarrow$ ) If  $\tau \geq 0$ , write  $\tau = \sum_m |v_m\rangle\langle v_m|$  by spectral decomposition, define  $K_m(a, b) = v_m(a, b)/c$ , and recover  $T(X) = \sum_m K_m X K_m^\dagger$  by linearity on matrix units.  $\square$

**Theorem 19.7** (Trace preservation implies the partial-trace condition). [Lean](#) [✓ Stmt](#) If  $T$  is trace-preserving, then  $\text{tr}_A(\tau) = \frac{1}{D} \mathbb{1}_D$ .

*Proof.* [✓ Proof](#)  $(\text{tr}_A(\tau))_{ij} = \text{tr}(T(\Omega^{(ij)})) = \text{tr}(\Omega^{(ij)})$  by trace preservation, where  $\Omega^{(ij)} = \frac{1}{D} E_{ij}$  is the  $(i, j)$ -slice of  $|\Omega\rangle\langle\Omega|$ . Taking the trace gives  $\frac{1}{D} \delta_{ij}$ .  $\square$

**Theorem 19.8** (Hermiticity preservation and the Choi matrix). [Lean](#) [✓ Stmt](#) The Choi matrix  $\tau$  is Hermitian if and only if  $T(B^\dagger) = T(B)^\dagger$  for all  $B \in M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) The Choi matrix entries satisfy  $\tau_{(a,i),(b,j)} = \frac{1}{D} (T(E_{ij}))_{ab}$ . Hermiticity of  $\tau$  says  $\overline{\tau_{(b,j),(a,i)}} = \tau_{(a,i),(b,j)}$ , which unravels to  $\overline{T(E_{ji})_{ba}} = T(E_{ij})_{ab}$ , i.e.,  $T(E_{ij}^\dagger) = T(E_{ij})^\dagger$ . By linearity this extends to all  $B$ .  $\square$

**Theorem 19.9** (Trace of the Choi matrix of a trace-preserving map). [Lean](#) [✓ Stmt](#) If  $T$  is trace-preserving, then  $\text{tr}(\tau) = 1$ .

*Proof.* [✓ Proof](#)  $\text{tr}(\tau) = \text{tr}(\text{tr}_A(\tau)) = \text{tr}(\frac{1}{D} \mathbb{1}_D) = 1$ .  $\square$

## 19.3 Representation corollaries for channel decompositions

The next three corollaries follow from the Choi/Kraus/Stinespring representations already developed above. They correspond to [Wol12, Propositions 2.2–2.4].

**Theorem 19.10** (Polarization of sesquilinear sandwiches). [Lean](#) [✓ Stmt](#) For any  $A, B, X \in M_D(\mathbb{C})$ ,

$$4AXB^\dagger = (A+B)X(A+B)^\dagger - (A-B)X(A-B)^\dagger + i(A+iB)X(A+iB)^\dagger - i(A-iB)X(A-iB)^\dagger.$$

Each of the four summands on the right is a single-Kraus CP map, so every sesquilinear sandwich decomposes into a signed complex linear combination of CP maps.

*Proof.* ✓ Proof Working entrywise, reduces to the scalar polarization identity  $4\alpha\bar{\delta} = \sum_{k=0}^3 i^k(\alpha + i^k\beta)(\gamma + i^k\delta)$  in  $\mathbb{C}$ , which holds after substituting  $i^2 = -1$ . □

**Theorem 19.11** (CP decomposition of sandwich sums). Lean ✓ Stmt For any finite Kraus-like families  $\{A_i\}, \{B_i\}$ , the map  $T(X) = \sum_i A_i X B_i^\dagger$  satisfies  $4T = T_1 - T_2 + iT_3 - iT_4$  with each  $T_k$  a CP map given explicitly by a single-side Kraus sum of the combined families.

*Proof.* ✓ Proof Apply the polarization identity summand-by-summand. □

**Theorem 19.12** (No information without disturbance). Lean ✓ Stmt Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be linear. If  $T(|v\rangle\langle v|) = |v\rangle\langle v|$  for every  $v \in \mathbb{C}^D$ , then  $T = \text{id}$ . In particular, a quantum channel preserving every pure state equals the identity.

*Proof.* ✓ Proof Rank-one outer products  $|u\rangle\langle v|$  polarize into rank-one self-outer-products:

$$4|u\rangle\langle v| = \sum_{k=0}^3 i^k |u + i^k v\rangle\langle u + i^k v|,$$

so  $T$  preserves every  $|u\rangle\langle v|$ . Since the matrix units  $E_{ij} = |e_i\rangle\langle e_j|$  span  $M_D(\mathbb{C})$ ,  $T$  equals the identity on a spanning set, hence everywhere. □

**Definition 19.13** (Pure-state ensemble density). Lean ✓ Stmt Given a finite family  $\{\psi_i\}_{i \in \iota}$  of (unnormalized) vectors in  $\mathbb{C}^D$ , its *pure-ensemble density* is  $\rho = \sum_i |\psi_i\rangle\langle\psi_i|$ . Weights  $p_i \geq 0$  with  $\sum p_i = 1$  can be absorbed by replacing  $\psi_i \mapsto \sqrt{p_i}\psi_i$ .

**Theorem 19.14** (Isometric mixing preserves the density). Lean ✓ Stmt If two pure-state ensembles  $\{\psi_i\}_{i \in \iota_1}$  and  $\{\phi_j\}_{j \in \iota_2}$  are related by an isometric mixing matrix  $V \in \mathbb{C}^{\iota_1 \times \iota_2}$  with  $V^\dagger V = \mathbb{1}$  and  $\psi_i = \sum_j V_{ij}\phi_j$ , then they induce the same pure-ensemble density operator. This is the sufficient direction of the Hughston–Jozsa–Wootters theorem; the converse is Theorem 19.15.

*Proof.* ✓ Proof Expanding and applying the orthogonality relation  $\sum_i V_{ij}\overline{V_{ij'}} = \delta_{jj'}$  from  $V^\dagger V = \mathbb{1}$ :

$$\sum_i |\psi_i\rangle\langle\psi_i| = \sum_{j,j'} \left( \sum_i V_{ij}\overline{V_{ij'}} \right) |\phi_j\rangle\langle\phi_{j'}| = \sum_j |\phi_j\rangle\langle\phi_j|.$$

□

**Theorem 19.15** (Hughston–Jozsa–Wootters converse). Lean ✓ Stmt If two pure-state ensembles  $\{\psi_i\}_{i \in \iota_1}$  and  $\{\phi_j\}_{j \in \iota_2}$  induce the same pure-ensemble density operator and  $|\iota_2| \leq |\iota_1|$ , then there exists a tall isometric mixing matrix  $V \in \mathbb{C}^{\iota_1 \times \iota_2}$  with  $V^\dagger V = \mathbb{1}$  and  $\psi_i = \sum_j V_{ij}\phi_j$ . The cardinality hypothesis is what makes  $V$  a tall isometry; the symmetric case  $|\iota_1| \leq |\iota_2|$  follows by swapping the roles of the ensembles.

*Proof.* ✓ Proof Embed each vector  $\psi_i, \phi_j$  as the 0-th column of a  $D \times D$  matrix with zeros elsewhere; call these  $K_i, L_j \in M_D(\mathbb{C})$ . A direct entry-wise computation shows that for any  $X \in M_D(\mathbb{C})$  both  $\sum_i K_i X K_i^\dagger$  and  $\sum_j L_j X L_j^\dagger$  collapse to  $X_{00} \cdot \rho$ , so the density equality  $\rho_\psi = \rho_\phi$  forces the two Kraus families to define the same CP map. Theorem 19.26 then supplies an isometry  $V$  with  $V^\dagger V = \mathbb{1}$  and  $K_i = \sum_j V_{ij} L_j$ ; reading the equation off at column 0 recovers the vector relation  $\psi_i = \sum_j V_{ij}\phi_j$ . □

**Theorem 19.16** (Hughston–Jozsa–Wootters equivalence). Lean ✓ Stmt Under the cardinality hypothesis  $|\iota_2| \leq |\iota_1|$ , two pure-state ensembles induce the same density operator iff they are related by a tall isometric mixing matrix.

*Proof.* ✓ Proof Combine the two directions above. □



## 19.4 Kraus representation theorem

**Theorem 19.17** (Trace preservation implies Kraus normalization). [Lean](#) [✓ Stmt](#) If  $T(X) = \sum_i K_i X K_i^\dagger$  is trace-preserving, then  $\sum_i K_i^\dagger K_i = \mathbb{1}$ .

*Proof.* [✓ Proof](#) For any  $N$ ,  $\text{tr}((\sum_i K_i^\dagger K_i) N) = \sum_i \text{tr}(K_i^\dagger K_i N) = \sum_i \text{tr}(K_i N K_i^\dagger) = \text{tr}(T(N)) = \text{tr}(N)$ . Non-degeneracy of the trace pairing forces  $\sum_i K_i^\dagger K_i = \mathbb{1}$ .  $\square$

**Theorem 19.18** (Kraus normalization implies trace preservation). [Lean](#) [✓ Stmt](#) If  $\sum_i K_i^\dagger K_i = \mathbb{1}$ , then  $T(X) = \sum_i K_i X K_i^\dagger$  is trace-preserving.

*Proof.* [✓ Proof](#)  $\text{tr}(T(X)) = \sum_i \text{tr}(K_i X K_i^\dagger) = \sum_i \text{tr}(K_i^\dagger K_i X) = \text{tr}((\sum_i K_i^\dagger K_i) X) = \text{tr}(X)$ .  $\square$

**Theorem 19.19** (Unitality implies Kraus unital normalization). [Lean](#) [✓ Stmt](#) If  $T(X) = \sum_i K_i X K_i^\dagger$  satisfies  $T(\mathbb{1}) = \mathbb{1}$ , then

$$\sum_i K_i K_i^\dagger = \mathbb{1}.$$

*Proof.* [✓ Proof](#) Evaluate the Kraus formula at the identity matrix.  $\square$

**Theorem 19.20** (Unitary freedom in Kraus operators). [Lean](#) [✓ Stmt](#) Let  $U$  be a unitary  $r \times r$  matrix ( $U^\dagger U = \mathbb{1}$ ) and suppose  $K_j = \sum_\ell U_{j\ell} \tilde{K}_\ell$ . Then  $\{K_j\}$  and  $\{\tilde{K}_\ell\}$  define the same Kraus map: for every  $X \in M_D(\mathbb{C})$ ,

$$\sum_j K_j X K_j^\dagger = \sum_\ell \tilde{K}_\ell X \tilde{K}_\ell^\dagger.$$

*Proof.* [✓ Proof](#) Substitute the combination formula, expand the triple sum over  $j, \ell, \ell'$ , swap summation order, and use the entry-wise form of  $U^\dagger U = \mathbb{1}$  to collapse to  $\ell = \ell'$  diagonal terms.  $\square$

**Theorem 19.21** (Isometric mixing preserves the Kraus map). [Lean](#) [✓ Stmt](#) Let  $W$  be an isometry between two Kraus index spaces, so  $W^\dagger W = \mathbb{1}$ . If

$$K_j = \sum_\ell W_{j\ell} \tilde{K}_\ell,$$

then the Kraus families  $\{K_j\}$  and  $\{\tilde{K}_\ell\}$  define the same completely positive map.

*Proof.* [✓ Proof](#) Expand the Kraus sums, interchange the finite summations, and use  $W^\dagger W = \mathbb{1}$  to collapse the coefficient matrix.  $\square$

**Theorem 19.22** (The transition matrix of orthonormal Kraus families is unitary). [Lean](#) [✓ Stmt](#) Let  $\{K_j\}_{j=0}^{r-1}$  and  $\{K'_j\}_{j=0}^{r-1}$  be two Hilbert–Schmidt orthonormal Kraus families, and suppose

$$K_j = \sum_{\ell=0}^{r-1} U_{j\ell} K'_\ell.$$

Then the transition matrix  $U$  is unitary:  $U^\dagger U = \mathbb{1}_r$ .

*Proof.* [✓ Proof](#) Expand  $\text{tr}(K_j^\dagger K_i)$  using the change of basis. Hilbert–Schmidt orthonormality of both Kraus families identifies the Gram matrix with the identity, yielding the matrix identity  $U U^\dagger = \mathbb{1}_r$ , hence also  $U^\dagger U = \mathbb{1}_r$ .  $\square$

**Theorem 19.23** (Dual map equality from primal map equality). [Lean](#) [✓ Stmt](#) If two Kraus families satisfy  $\sum_{\alpha} B_{\alpha} X B_{\alpha}^{\dagger} = \sum_j A_j X A_j^{\dagger}$  for all  $X$ , then  $\sum_{\alpha} B_{\alpha}^{\dagger} Y B_{\alpha} = \sum_j A_j^{\dagger} Y A_j$  for all  $Y$ .

*Proof.* [✓ Proof](#) Use the trace pairing: for all  $X$ ,  $\text{tr}(X^{\dagger}(\sum B_{\alpha}^{\dagger} Y B_{\alpha})) = \text{tr}((\sum B_{\alpha} X B_{\alpha}^{\dagger})Y) = \text{tr}((\sum A_j X A_j^{\dagger})Y) = \text{tr}(X^{\dagger}(\sum A_j^{\dagger} Y A_j))$ . Nondegeneracy of the trace pairing gives the result.  $\square$

**Theorem 19.24** (Equal Stinespring Gramians). [Lean](#) [✓ Stmt](#) If two Kraus families define the same CPM, then  $\sum_{\alpha} B_{\alpha}^{\dagger} B_{\alpha} = \sum_j A_j^{\dagger} A_j$ .

*Proof.* [✓ Proof](#) Specialise Theorem 19.23 at  $Y = \mathbb{1}$ .  $\square$

**Theorem 19.25** (Rectangular Kraus freedom). [Lean](#) [✓ Stmt](#) If  $\sum_{\alpha} B_{\alpha} X B_{\alpha}^{\dagger} = \sum_j A_j X A_j^{\dagger}$  for all  $X$  and  $r_2 \leq r_1$  (where  $\{B_{\alpha}\}_{\alpha=0}^{r_1-1}$  is the larger family and  $\{A_j\}_{j=0}^{r_2-1}$  the smaller), then there exists a rectangular isometry  $V$  ( $r_1 \times r_2$ ,  $V^{\dagger}V = \mathbb{1}_{r_2}$ ) such that  $B_{\alpha} = \sum_j V_{\alpha j} A_j$ .

*Proof.* [✓ Proof](#) Pad the smaller family to size  $r_1$ , establish Gram matrix equality  $M_B^{\dagger} M_B = M_{A'}^{\dagger} M_{A'}$  from dual map equality, construct a partial isometry via inner product preservation on the range, and extend to a full isometry  $V$  with  $V^{\dagger}V = \mathbb{1}_{r_2}$ .  $\square$

**Theorem 19.26** (Rectangular Kraus freedom with general finite index types). [Lean](#) [✓ Stmt](#) Variant of Theorem 19.25 with general finite index sets  $\iota_1, \iota_2$  in place of standard index sets of cardinalities  $r_1$  and  $r_2$ .

*Proof.* [✓ Proof](#) Reindex by choosing enumerations of those finite index sets and apply Theorem 19.25.  $\square$

**Theorem 19.27** (Isometric freedom of Kraus representations). [Lean](#) [✓ Stmt](#) Let  $\{B_{\alpha}\}_{\alpha \in \iota_1}$  and  $\{A_j\}_{j \in \iota_2}$  be finite Kraus families with  $|\iota_2| \leq |\iota_1|$ . Then the following are equivalent:

1.  $\sum_{\alpha} B_{\alpha} X B_{\alpha}^{\dagger} = \sum_j A_j X A_j^{\dagger}$  for all  $X \in M_D(\mathbb{C})$ .
2. There exists a matrix  $V = (V_{\alpha j})$  with  $V^{\dagger}V = \mathbb{1}$  such that  $B_{\alpha} = \sum_j V_{\alpha j} A_j$  for every  $\alpha$ .

*Proof.* [✓ Proof](#) The implication (1)  $\Rightarrow$  (2) is Theorem 19.26. Conversely, expand  $\sum_{\alpha} B_{\alpha} X B_{\alpha}^{\dagger}$  using  $B_{\alpha} = \sum_j V_{\alpha j} A_j$ , interchange the finite sums, and use  $V^{\dagger}V = \mathbb{1}$  to collapse the coefficient matrix to the diagonal.  $\square$

**Theorem 19.28** (Unitary freedom of Kraus representations). [Lean](#) [✓ Stmt](#) Let  $\{B_{\alpha}\}_{\alpha \in \iota}$  and  $\{A_j\}_{j \in \iota}$  be two Kraus families with the same finite index set. Then the following are equivalent:

1.  $\sum_{\alpha} B_{\alpha} X B_{\alpha}^{\dagger} = \sum_j A_j X A_j^{\dagger}$  for all  $X \in M_D(\mathbb{C})$ .
2. There exists a unitary matrix  $U = (U_{\alpha j})$  such that  $B_{\alpha} = \sum_j U_{\alpha j} A_j$  for every  $\alpha$ .

*Proof.* [✓ Proof](#) Apply Theorem 19.27 with  $|\iota_1| = |\iota_2|$ . In the square case an isometry matrix is unitary, and the converse is immediate.  $\square$

**Remark 19.29** (Choi rank and minimal Kraus cardinality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) If a completely positive map admits a Kraus representation with exactly  $r$  operators, then its Choi matrix is a sum of  $r$  rank-one outer products. Consequently,

$$\text{rank}(\tau_E) \leq r.$$

Conversely, diagonalizing the positive semidefinite Choi matrix and keeping only the nonzero eigenvalue summands produces a Kraus family with exactly  $\text{rank}(\tau_E)$  operators. Hence the Choi rank is precisely the minimal Kraus cardinality.

*Proof.* ✓ Proof For the forward implication, expand the Choi matrix using the Kraus formula and write  $\tau_E = \sum_j v_j v_j^\dagger$  with one vector  $v_j$  for each Kraus operator. The column space of this sum lies in the span of the  $r$  vectors  $\{v_j\}$ , so the rank is at most  $r$ . For the converse, use the spectral decomposition of the positive semidefinite Choi matrix, discard the zero-eigenvalue terms, and rescale the remaining eigenvectors into Kraus operators.  $\square$

## 19.5 Stinespring dilation

**Definition 19.30** (Stinespring isometry). Lean ✓ Stmt Given Kraus operators  $\{K_j\}_{j=0}^{r-1}$  with  $K_j \in M_D(\mathbb{C})$ , the *Stinespring isometry*  $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$  is the  $(D \cdot r) \times D$  matrix

$$V_{(i,j),k} = (K_j)_{ik}.$$

This implements  $V = \sum_j K_j \otimes |j\rangle$ .

**Lemma 19.31** (Entry formula for the Stinespring isometry). Lean ✓ Stmt For indices  $i, k \in \{0, \dots, D-1\}$  and  $j \in \{0, \dots, r-1\}$ ,

$$V_{(i,j),k} = (K_j)_{ik}.$$

**Theorem 19.32** (Stinespring Gram identity). Lean ✓ Stmt The Stinespring isometry satisfies

$$V^\dagger V = \sum_{j=0}^{r-1} K_j^\dagger K_j.$$

**Theorem 19.33** (Stinespring dual representation). Lean ✓ Stmt For Kraus operators  $\{K_j\}$  and  $A \in M_D(\mathbb{C})$ ,

$$V^\dagger (A \otimes \mathbb{1}_r) V = \sum_j K_j^\dagger A K_j = T^*(A).$$

This is the Stinespring representation of the dual (Heisenberg picture) map  $T^*$ .

*Proof.* ✓ Proof Compute  $(V^\dagger (A \otimes \mathbb{1}_r) V)_{ab}$  by summing over the product index  $(i, j)$ ; the  $\delta_{jl}$  from  $\mathbb{1}_r$  collapses the sum over  $l$ , giving  $\sum_j K_j^\dagger A K_j$ .  $\square$

**Theorem 19.34** (The Stinespring isometry condition iff trace preservation). Lean ✓ Stmt  $V^\dagger V = \mathbb{1}_D$  if and only if  $\sum_j K_j^\dagger K_j = \mathbb{1}_D$ . In particular,  $V$  is an isometry precisely when the Kraus map is trace-preserving.

*Proof.* ✓ Proof  $(V^\dagger V)_{ab} = \sum_{(i,j)} \overline{V_{(i,j),a}} V_{(i,j),b} = \sum_j \sum_i \overline{(K_j)_{ia}} (K_j)_{ib} = (\sum_j K_j^\dagger K_j)_{ab}$ .  $\square$

**Theorem 19.35** (Stinespring Schrödinger representation). Lean ✓ Stmt The Kraus map  $T(\rho) = \sum_j K_j \rho K_j^\dagger$  equals the partial trace over the dilation space:

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \text{tr}_r(V \rho V^\dagger).$$

This is the Schrödinger-picture Stinespring representation.

*Proof.* [✓ Proof](#) Expand  $(V\rho V^\dagger)_{(i,k),(j,k)}$  and sum over  $k$  to recover  $\sum_k K_k \rho K_k^\dagger$ .  $\square$

**Theorem 19.36** (Existence of a Stinespring dilation). [Lean](#) [✓ Stmt](#) *Every completely positive map admits an ancilla dimension  $r$ , Kraus operators  $\{K_j\}_{j=0}^{r-1}$ , and the concrete representation  $\pi(A) = A \otimes \mathbb{1}_r$  such that*

$$E(A) = V^\dagger \pi(A) V.$$

*Proof.* [✓ Proof](#) Choose a Kraus representation of  $E$  and apply the explicit Heisenberg-picture Stinespring formula.  $\square$

**Theorem 19.37** (CPTP maps admit isometric Stinespring dilations). [Lean](#) [✓ Stmt](#) *Every quantum channel admits a Stinespring dilation whose matrix  $V$  is an isometry, equivalently  $V^\dagger V = \mathbb{1}$ .*

*Proof.* [✓ Proof](#) Choose Kraus operators for the channel and use trace preservation to obtain the Stinespring isometry condition  $V^\dagger V = \mathbb{1}$ .  $\square$

## 19.6 Ordered CP maps, Radon–Nikodym, and open-system representation

The next three results form the structural part of [Wol12, Theorems 2.3–2.5]: the ordered CP-map theorem, the Radon–Nikodym theorem for completely positive maps, and the open-system representation. They refine the Stinespring dilation by tracking how two CP maps related by domination or decomposition sit inside a common dilation space.

**Definition 19.38** (CP partial order). [Lean](#) [✓ Stmt](#) For linear maps  $S, T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ , write  $T \leq S$  (in the CP order) when  $S - T$  is completely positive.

**Definition 19.39** (Block-top contraction). [Lean](#) [✓ Stmt](#) For natural numbers  $r, s$ , let  $C = C_{r,s} \in \mathbb{C}^{r \times (r+s)}$  be the rectangular  $r \times (r+s)$  matrix whose rows are the first  $r$  rows of the identity on  $\mathbb{C}^{r+s}$ :  $C_{ij} = 1$  if  $j = i < r$  and 0 otherwise.

**Lemma 19.40** ( $C$  is a co-isometry). [Lean](#) [✓ Stmt](#) *The block-top matrix satisfies  $CC^\dagger = \mathbb{1}_r$ .*

*Proof.* [✓ Proof](#) Direct entrywise computation:  $(CC^\dagger)_{ii'} = \sum_j C_{ij} \overline{C_{i'j}}$  collapses to  $\delta_{ii'}$ .  $\square$

**Lemma 19.41** ( $C$  is a contraction). [Lean](#) [✓ Stmt](#) *The block-top matrix satisfies  $C^\dagger C \leq \mathbb{1}_{r+s}$ .*

*Proof.* [✓ Proof](#) The matrix  $C^\dagger C$  is diagonal with a 1 on the first  $r$  coordinates and a 0 on the last  $s$ , so  $\mathbb{1} - C^\dagger C$  is positive semidefinite.  $\square$

**Lemma 19.42** (Intertwining of Stinespring isometries). [Lean](#) [✓ Stmt](#) *Let  $K : \{0, \dots, r-1\} \rightarrow M_D(\mathbb{C})$  and  $L : \{0, \dots, s-1\} \rightarrow M_D(\mathbb{C})$  be two Kraus families. Write  $K \mathrel{++} L$  for their concatenation as a family indexed by  $\{0, \dots, r+s-1\}$ . Then*

$$V_K = (\mathbb{1}_D \otimes C_{r,s}) V_{K \mathrel{++} L}.$$

*Proof.* [✓ Proof](#) Entrywise expansion of both sides: the Kronecker product with  $C_{r,s}$  picks out the first  $r$  coordinates of the dilation space, matching the action of  $V_{K \mathrel{++} L}$  on those coordinates, which is precisely  $V_K$ .  $\square$

**Theorem 19.43** (Ordered CP maps via Stinespring contractions). [Lean](#) [✓ Stmt](#) Let  $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be CP maps with  $T_1 \leq T_2$ . Then there exist an ancilla dimension  $m$ , Heisenberg-form Stinespring matrices  $V_1, V_2$  realizing  $T_1, T_2$  via  $T_i(A) = V_i^\dagger(A \otimes \mathbb{1})V_i$ , and a contraction  $\tilde{C}$  on the dilation space with  $\tilde{C}^\dagger \tilde{C} \leq \mathbb{1}$  such that

$$V_1 = (\mathbb{1}_D \otimes \tilde{C}) V_2.$$

This is [\[Wol12, Theorem 2.3\]](#).

*Proof.* [✓ Proof](#) Choose a Heisenberg-form Kraus family  $K_1$  of  $T_1$  with Stinespring matrix  $V_1 = V_{K_1}$ , and Kraus operators  $L$  of the CP map  $T_2 - T_1$  in Schrödinger orientation. Let  $L^\dagger$  denote the conjugate-transposed family. Set  $m = r_1 + s$  and let  $K_2 = K_1 ++ L^\dagger$ . Then  $T_2(A) = T_1(A) + (T_2 - T_1)(A) = \sum_i (K_1^i)^\dagger A K_1^i + \sum_j L_j A L_j^\dagger$ , and the latter equals  $\sum_i (K_1^i)^\dagger A K_1^i + \sum_j (L_j^\dagger)^\dagger A L_j^\dagger$  which is the Heisenberg form for  $K_2$ , yielding  $T_2(A) = V_{K_2}^\dagger(A \otimes \mathbb{1})V_{K_2}$ . Taking  $\tilde{C} = C_{r_1, s}$  the intertwining  $V_{K_1} = (\mathbb{1}_D \otimes \tilde{C})V_{K_2}$  is Lemma 19.42, and  $\tilde{C}^\dagger \tilde{C} \leq \mathbb{1}$  is Lemma 19.41.  $\square$

**Definition 19.44** (Block-diagonal projectors on the dilation space). [Lean](#) [Lean](#) [✓ Stmt](#) For  $r, s$ , let  $P_{\text{top}} = C_{r, s}^\dagger C_{r, s}$  and  $P_{\text{bot}} = \mathbb{1} - P_{\text{top}}$ . Both are PSD (for  $P_{\text{bot}}$ , by Lemma 19.41) and  $P_{\text{top}} + P_{\text{bot}} = \mathbb{1}$ .

**Theorem 19.45** (Radon–Nikodym theorem for CP maps). [Lean](#) [✓ Stmt](#) Let  $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be completely positive. There exist an ancilla dimension  $m$ , a Kraus family  $K$  with Stinespring matrix  $V = V_K$ , and positive semidefinite operators  $P_1, P_2 : \mathbb{C}^m \rightarrow \mathbb{C}^m$  with  $P_1 + P_2 = \mathbb{1}_m$  such that, for  $i = 1, 2$  and every  $A \in M_D(\mathbb{C})$ ,

$$T_i(A) = V^\dagger(A \otimes P_i)V.$$

This is [\[Wol12, Theorem 2.4\]](#).

*Proof.* [✓ Proof](#) Take Heisenberg-form Kraus families  $K_1$  for  $T_1$  and Schrödinger-form  $L$  for  $T_2$ , and form  $K = K_1 ++ L^\dagger$  on  $\mathbb{C}^{r_1+s}$ . Choose  $P_1 = P_{\text{top}}$ ,  $P_2 = P_{\text{bot}}$  as in Definition 19.44. The Kronecker identity  $A \otimes (C^\dagger C) = (\mathbb{1} \otimes C)^\dagger (A \otimes \mathbb{1})(\mathbb{1} \otimes C)$  rewrites  $V^\dagger(A \otimes P_{\text{top}})V$  as  $((\mathbb{1} \otimes C)V)^\dagger (A \otimes \mathbb{1})((\mathbb{1} \otimes C)V)$ , which equals  $V_{K_1}^\dagger(A \otimes \mathbb{1})V_{K_1} = T_1(A)$  by the intertwining Lemma 19.42. For the complementary block,  $A \otimes P_{\text{bot}} = A \otimes \mathbb{1} - A \otimes P_{\text{top}}$  and sandwiching by  $V$  yields  $(T_1 + T_2)(A) - T_1(A) = T_2(A)$ .  $\square$

**Theorem 19.46** (Open-system representation of quantum channels). [Lean](#) [✓ Stmt](#) Every quantum channel  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  admits a Stinespring isometry  $V$  on an ancilla space  $\mathbb{C}^r$  such that

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \text{tr}_r(V \rho V^\dagger).$$

*Proof.* [✓ Proof](#) Apply the existence of an isometric Stinespring dilation (Theorem 19.37) and the Schrödinger-picture identity (Theorem 19.35).  $\square$

**Theorem 19.47** (Open-system representation, partial-trace form). [Lean](#) [✓ Stmt](#) Every quantum channel  $T$  admits an isometric Stinespring dilation  $V$  such that  $T(\rho) = \text{tr}_E(V \rho V^\dagger)$ .

*Proof.* [✓ Proof](#) Rewrite the componentwise identity of Theorem 19.46 as a partial trace over the ancilla factor.  $\square$

**Remark 19.48** (Unitary form). The fully unitary open-system form  $T(\rho) = \text{tr}_E(U(\rho \otimes |\phi\rangle\langle\phi|)U^\dagger)$  of [\[Wol12, Theorem 2.5\]](#) follows from the isometric form by extending the Stinespring isometry  $V$  to a unitary  $U$  on  $\mathbb{C}^D \otimes \mathbb{C}^r$ ; this uses orthonormal basis extension and is left to a later theorem.

## 19.7 POVMs and Naimark dilation

**Definition 19.49** (Positive operator-valued measure). [Lean](#) [✓ Stmt](#) A *positive operator-valued measure* with  $n$  outcomes on  $\mathbb{C}^D$  is a family  $\{E_i\}_{i=0}^{n-1}$  of positive semidefinite operators on  $\mathbb{C}^D$  satisfying the resolution of identity

$$\sum_{i=0}^{n-1} E_i = \mathbb{1}_D.$$

**Definition 19.50** (Naimark Kraus square roots). [Lean](#) [Lean](#) [✓ Stmt](#) For a POVM  $\{E_i\}$ , each effect  $E_i \geq 0$  admits a square-root factorisation  $E_i = M_i^\dagger M_i$  with  $M_i \in M_D(\mathbb{C})$ . The operators  $M_i$  are the *Naimark Kraus square roots*.

**Theorem 19.51** (Naimark square roots satisfy the Kraus normalization). [Lean](#) [✓ Stmt](#) For a POVM  $\{E_i\}$  with square roots  $E_i = M_i^\dagger M_i$ , one has

$$\sum_i M_i^\dagger M_i = \mathbb{1}.$$

*Proof.* [✓ Proof](#) Sum the identities  $E_i = M_i^\dagger M_i$  and use the defining resolution of the identity for the POVM.  $\square$

**Definition 19.52** (Naimark isometry). [Lean](#) [✓ Stmt](#) The *Naimark isometry*  $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$  of a POVM is the Stinespring-type construction

$$V = \sum_{i=0}^{n-1} M_i \otimes |i\rangle, \quad V_{(a,i),k} = (M_i)_{a,k}.$$

**Definition 19.53** (Naimark projective measurement). [Lean](#) [Lean](#) [✓ Stmt](#) The *Naimark projectors* on  $\mathbb{C}^D \otimes \mathbb{C}^n$  are

$$P_i = \mathbb{1}_D \otimes |i\rangle\langle i|, \quad (P_i)_{(a,b),(c,d)} = \delta_{a,c} \delta_{b,i} \delta_{d,i}.$$

**Theorem 19.54** (Naimark isometry condition). [Lean](#) [✓ Stmt](#) The Naimark isometry satisfies  $V^\dagger V = \mathbb{1}_D$ .

*Proof.* [✓ Proof](#) The Stinespring Gram identity gives  $V^\dagger V = \sum_i M_i^\dagger M_i = \sum_i E_i = \mathbb{1}_D$  by the resolution of identity.  $\square$

**Theorem 19.55** (Naimark projectors form a projective measurement). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The Naimark projectors satisfy  $P_i^2 = P_i$ ,  $P_i^\dagger = P_i$ ,  $P_i P_j = 0$  for  $i \neq j$ , and  $\sum_i P_i = \mathbb{1}_{\mathbb{C}^D \otimes \mathbb{C}^n}$ .

*Proof.* [✓ Proof](#) Each identity follows from direct entrywise computation using the delta-function form of  $(P_i)_{(a,b),(c,d)}$ .  $\square$

**Theorem 19.56** (Naimark dilation). [Lean](#) [✓ Stmt](#) Every POVM  $\{E_i\}_{i=0}^{n-1}$  arises as a projective measurement on a dilation: for the isometry  $V$  and projectors  $P_i$  above,

$$E_i = V^\dagger P_i V.$$

*Proof.* [✓ Proof](#) Computing entrywise,  $(V^\dagger P_i V)_{k,\ell} = \sum_a \overline{(M_i)_{a,k}} (M_i)_{a,\ell} = (M_i^\dagger M_i)_{k,\ell} = (E_i)_{k,\ell}$ .  $\square$

**Theorem 19.57** (Existential Naimark dilation). [Lean](#) [✓ Stmt](#) For every POVM  $\{E_i\}$  on  $\mathbb{C}^D$  there exist a dilation dimension  $r$ , an isometry  $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$ , and a projective measurement  $\{P_i\}$  on the dilation satisfying  $E_i = V^\dagger P_i V$ .

*Proof.* [✓ Proof](#) Take  $r = n$  and the explicit witnesses  $V = \sum_i M_i \otimes |i\rangle$  and  $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$ .  $\square$

**Definition 19.58** (Naimark dilation as a structure). [Lean](#) [✓ Stmt](#) A *Naimark dilation* of a POVM  $\{E_i\}$  consists of an isometry  $V$  into a larger Hilbert space together with a projective measurement  $\{P_i\}$  there such that  $E_i = V^\dagger P_i V$  for all  $i$ .

**Theorem 19.59** (Canonical dilation satisfies the Naimark dilation axioms). [Lean](#) [✓ Stmt](#) The explicit isometry  $V = \sum_i M_i \otimes |i\rangle$  and projectors  $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$  satisfy the defining axioms of a Naimark dilation.

*Proof.* [✓ Proof](#) Combine the previously proved identities  $V^\dagger V = \mathbb{1}_D$ ,  $P_i^2 = P_i$ ,  $P_i^\dagger = P_i$ ,  $P_i P_j = 0$  for  $i \neq j$ ,  $\sum_i P_i = \mathbb{1}$ , and  $V^\dagger P_i V = E_i$ .  $\square$

**Theorem 19.60** (Concrete uniqueness for canonical projectors). [Lean](#) [✓ Stmt](#) Let  $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$  satisfy  $V^\dagger P_i V = E_i$  for the canonical projectors  $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$ . Then there exists an isometry  $W$  on the dilated space such that  $V = W V_0$ , where  $V_0$  is the canonical Naimark isometry of  $\{E_i\}$ .

*Proof.* [✓ Proof](#) For each outcome  $i$ , the  $i$ -th block of  $V$  has Gram matrix  $E_i$ . Comparing with the canonical square root  $M_i$  gives  $V_i^\dagger V_i = M_i^\dagger M_i$ , hence  $V_i = U_i M_i$  for a unitary  $U_i$  on  $\mathbb{C}^D$ . Assemble the  $U_i$  block-diagonally to obtain an isometry  $W = \bigoplus_i U_i$  on  $\mathbb{C}^D \otimes \mathbb{C}^n$ , and then  $V = W \sum_i M_i \otimes |i\rangle = W V_0$ .  $\square$

**Definition 19.61** (POVM from a PSD resolution of identity on a dilation). [Lean](#) [✓ Stmt](#) Given an isometry  $V : \mathbb{C}^D \rightarrow \mathbb{C}^{d'}$  with  $V^\dagger V = \mathbb{1}_D$  and a family  $\{P_i\}_{i=0}^{n-1}$  of positive semidefinite operators on  $\mathbb{C}^{d'}$  summing to the identity, the pulled-back operators

$$E_i := V^\dagger P_i V$$

form a POVM. In particular, any projective measurement on the dilation (a special case of a PSD resolution of identity) pulls back to a POVM.

**Definition 19.62** (Quantum instrument). [Lean](#) [✓ Stmt](#) A *quantum instrument* with  $n$  outcomes is a family  $\{\Phi_i\}_{i=0}^{n-1}$  of completely positive maps on  $M_D(\mathbb{C})$  whose sum  $\sum_i \Phi_i$  is trace-preserving.

**Definition 19.63** (Instrument operations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Associated to an instrument are the total channel  $\sum_i \Phi_i$ , the unnormalized update  $\rho \mapsto \Phi_i(\rho)$  for each outcome  $i$ , the outcome probability  $p_i(\rho) = \text{tr}(\Phi_i(\rho))$ , and the normalized posterior state  $\Phi_i(\rho)/p_i(\rho)$  whenever  $p_i(\rho) \neq 0$ .

**Theorem 19.64** (Instrument total map is a channel). [Lean](#) [✓ Stmt](#) The total map  $\sum_i \Phi_i$  of an instrument is a quantum channel.

*Proof.* [✓ Proof](#) Complete positivity of the sum follows from closure of CP maps under finite addition; trace preservation is built into the definition.  $\square$

**Theorem 19.65** (Conservation of probability for instruments). [Lean](#) [✓ Stmt](#) For every state  $\rho \in M_D(\mathbb{C})$ , the outcome probabilities  $p_i(\rho) := \text{tr}(\Phi_i(\rho))$  sum to  $\text{tr}(\rho)$ .

*Proof.* ✓ **Proof** Linearity of trace and trace preservation of  $\sum_i \Phi_i$  give  $\sum_i \text{tr}(\Phi_i(\rho)) = \text{tr}((\sum_i \Phi_i)(\rho)) = \text{tr}(\rho)$ .  $\square$

**Theorem 19.66** (Non-negativity of instrument probabilities). Lean ✓ **Stmt** If  $\rho \in M_D(\mathbb{C})$  is positive semidefinite, then each outcome probability  $p_i(\rho) = \text{tr}(\Phi_i(\rho))$  is a non-negative real number.

*Proof.* ✓ **Proof** Each  $\Phi_i$  is completely positive, hence positive, so  $\Phi_i(\rho)$  is positive semidefinite and its trace is a non-negative real.  $\square$

## 19.8 Trace-pairing expansion in transfer-matrix form

**Definition 19.67** (Trace-self-dual basis). Lean ✓ **Stmt** A basis  $\{\sigma_i\}_i$  of  $M_D(\mathbb{C})$  is *trace-self-dual* when its coordinate functionals are given by trace pairing:

$$X = \sum_i \text{tr}(\sigma_i X) \sigma_i.$$

**Definition 19.68** (Trace-pairing coefficients). Lean ✓ **Stmt** For a linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  and a trace-self-dual basis  $\{\sigma_i\}$ , define coefficients

$$t_{ij} := \text{tr}(\sigma_i T(\sigma_j)).$$

**Theorem 19.69** (Trace-pairing expansion of a linear map). Lean ✓ **Stmt** If  $\{\sigma_i\}$  is trace-self-dual, then every linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  admits the expansion

$$T(\rho) = \sum_i \sum_j t_{ij} \text{tr}(\sigma_j \rho) \sigma_i, \quad t_{ij} = \text{tr}(\sigma_i T(\sigma_j)).$$

*Proof.* ✓ **Proof** Expand  $\rho$  and each  $T(\sigma_j)$  in the chosen basis and substitute. The trace-self-dual identity replaces coordinates by traces, giving the stated double-sum formula.  $\square$

## 19.9 SVD normal form (existence)

[Wol12, Section 2.3] discusses two families of normal forms for the transfer-matrix representation of a quantum channel: the *singular value decomposition* (SVD) and, for qubit channels, the *Lorentz normal form* obtained by  $\text{SL}(2, \mathbb{C})$  filtering operations. The SVD normal form is proved below. The Lorentz normal form theorem (Theorem 19.81) requires the compactness and minimisation argument for  $\text{SL}(2, \mathbb{C})$  filterings in [Wol12, Propositions 2.9 and 2.11]; see Section 19.10.

**Theorem 19.70** (SVD for positive semidefinite matrices). Lean ✓ **Stmt** Every positive semidefinite matrix  $M \in M_D(\mathbb{C})$  admits a decomposition

$$M = U \text{diag}(\sigma) U^\dagger,$$

where  $U \in \mathcal{U}(D)$  is unitary and  $\sigma : \{0, \dots, D-1\} \rightarrow \mathbb{R}_{\geq 0}$  has non-negative entries.

*Proof.* ✓ **Proof** The spectral theorem for Hermitian matrices provides an orthonormal eigenbasis with real eigenvalues. Positive semidefiniteness forces those eigenvalues to be non-negative, and the decomposition is written in SVD form with  $U$  the eigenvector unitary and  $\sigma$  the eigenvalue sequence.  $\square$



**Theorem 19.71** (SVD existence for invertible matrices). Lean ✓ Stmt Every invertible complex square matrix  $M \in M_D(\mathbb{C})$  admits a singular value decomposition

$$M = U \operatorname{diag}(\sigma) V^\dagger,$$

with  $U, V \in \mathcal{U}(D)$  unitary and  $\sigma_i > 0$  for all  $i$ .

*Proof.* ✓ Proof Apply the spectral theorem to the positive definite matrix  $M^\dagger M$  to obtain  $M^\dagger M = V \operatorname{diag}(\lambda) V^\dagger$  with  $\lambda_i > 0$ . Set  $\sigma_i := \sqrt{\lambda_i}$  and  $\Sigma := \operatorname{diag}(\sigma)$ ; since  $\Sigma$  is a real positive diagonal, it is self-adjoint and invertible. Define  $U := MV\Sigma^{-1}$ . Then  $U^\dagger U = \Sigma^{-1} V^\dagger (M^\dagger M) V \Sigma^{-1} = \Sigma^{-1} \operatorname{diag}(\lambda) \Sigma^{-1} = \mathbb{1}$ , so  $U$  is unitary, and  $U \Sigma V^\dagger = MV(\Sigma^{-1} \Sigma) V^\dagger = MVV^\dagger = M$ .  $\square$

**Theorem 19.72** (SVD representation of a transfer matrix). Lean ✓ Stmt Every invertible transfer matrix  $\widehat{T}$  of a linear super-operator  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  admits a singular value decomposition  $\widehat{T} = U \operatorname{diag}(\sigma) V^\dagger$  with  $U, V$  unitary on  $\mathbb{C}^D \otimes \mathbb{C}^D$  and  $\sigma_i > 0$ .

*Proof.* ✓ Proof Direct specialisation of Theorem 19.71 to the index type  $\{0, \dots, D-1\} \times \{0, \dots, D-1\}$  used by the transfer-matrix representation.  $\square$

*Remark 19.73* (Sorted / unique singular values). Theorem 19.71 produces the singular values as an unordered family, without the usual convention that  $\sigma_i$  are sorted in non-increasing order or the statement that they are uniquely determined by  $M$ . Downstream applications in [Wol12, Section 2.3], such as trace-norm identities, polar decomposition, and the Lorentz normal form, will typically require the sorted / unique variant; that refinement is future work.

## 19.10 Lorentz normal form

This section states the existence theorems from [Wol12, Section 2.3].

**Definition 19.74** (SL-filtering operation). An *SL-filtering* for  $D \times D$  matrices is a completely positive map of the form  $\Phi(X) = SXS^\dagger$  where  $S \in M_D(\mathbb{C})$  satisfies  $\det S = 1$ . Such maps are invertible and have Kraus rank 1.

**Definition 19.75** (Doubly-stochastic map). A linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is *doubly-stochastic* if  $T(\mathbb{1}) \propto \mathbb{1}$  and the reduced density matrix  $\operatorname{tr}_1[\tau]$  of its Choi matrix  $\tau = (T \otimes \operatorname{id})(|\Omega\rangle\langle\Omega|)$  is proportional to the identity. This is the normal form in [Wol12, Proposition 2.8].

**Lemma 19.76** (Infimum attainment for SL-filterings). For a positive-definite Choi matrix  $\tau$ , the infimum of  $\operatorname{tr}[(S_2 \otimes_k S_1) \tau (S_2 \otimes_k S_1)^\dagger]$  over  $S_1, S_2 \in M_D(\mathbb{C})$  with  $\det S_1 = \det S_2 = 1$  is attained. **Not yet proved**—requires compactness of bounded subsets of  $\operatorname{SL}(D, \mathbb{C})$  and the extreme-value theorem.

**Theorem 19.77** (Generic normal form for CP maps). Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a completely positive map whose Choi matrix is positive-definite (equivalently,  $T$  has full Kraus rank). Then there exist SL-filterings  $\Phi_1, \Phi_2$  such that  $\Phi_2 \circ T \circ \Phi_1$  is doubly-stochastic. **Not yet proved**—proof blocked on Lemma 19.76.

**Definition 19.78** (Diagonal Lorentz normal form). For a completely positive, trace-preserving qubit channel  $T' : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ , write  $\widehat{T'}$  for its matrix in the normalized Pauli basis  $\{\sigma_0/\sqrt{2}, \sigma_1/\sqrt{2}, \sigma_2/\sqrt{2}, \sigma_3/\sqrt{2}\}$ . The channel is in *diagonal Lorentz normal form* when it is unital and every off-diagonal entry of  $\widehat{T'}$  is zero.

**Definition 19.79** (Non-diagonal Lorentz normal form). For a completely positive, trace-preserving qubit channel  $T' : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ , write  $\widehat{T}'$  in the normalized Pauli basis. The channel is in *non-diagonal Lorentz normal form* when, for some  $x \in [0, 1]$ ,

$$\widehat{T}' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & x/\sqrt{3} & 0 & 0 \\ 0 & 0 & x/\sqrt{3} & 0 \\ 2/3 & 0 & 0 & 1/3 \end{pmatrix}.$$

Trace preservation supplies the first row of the displayed matrix.

**Definition 19.80** (Singular Lorentz normal form). For a completely positive, trace-preserving qubit channel  $T' : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ , write  $\widehat{T}'$  in the normalized Pauli basis. The channel is in *singular Lorentz normal form* when

$$\widehat{T}' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

equivalently, every input state is mapped to  $(1 + \sigma_3)/2$ . Trace preservation supplies the first row of the displayed matrix.

**Theorem 19.81** (Lorentz normal form for qubit channels). *For every qubit channel  $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ , there exist  $\text{SL}(2, \mathbb{C})$ -filterings  $\Phi_1, \Phi_2$  such that the filtered channel  $T' = \Phi_2 \circ T \circ \Phi_1$  is in one of the three Lorentz normal forms: diagonal, non-diagonal, or singular. **Not yet proved**—proof blocked on Lemma 19.76 and the Lorentz group classification of  $\text{SL}(2, \mathbb{C})$  orbits.*

## 19.11 Determinant of a quantum channel

**Definition 19.82** (Channel determinant). [Lean](#) [✓ Stmt](#) For a linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ , the *channel determinant*  $\det T$  is the determinant of the matrix of  $T$  with respect to the standard matrix-unit basis of  $M_D(\mathbb{C})$ . This is the quantity studied in [Wol12, Section 6.1].

*Remark 19.83* (Channel determinant as an operator determinant). [Lean](#) [✓ Stmt](#) The channel determinant agrees with the ordinary determinant of the linear endomorphism  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ .

**Definition 19.84** (Unitary channel). [Lean](#) [Lean](#) [✓ Stmt](#) For a unitary matrix  $U \in \mathcal{U}(D)$ , the *unitary channel* is  $T(\rho) = U\rho U^\dagger$ . It is automatically a quantum channel.

**Theorem 19.85** (Unitary channels have determinant one). [Lean](#) [Lean](#) [✓ Stmt](#) *If  $T(\rho) = U\rho U^\dagger$  is a unitary channel, then  $\det T = 1$  and hence  $|\det T| = 1$ .*

*Proof.* [✓ Proof](#) Vectorization identifies  $T$  with a Kronecker product  $\overline{U} \otimes U$ , whose determinant is  $\overline{\det U}^D (\det U)^D = 1$ . □

**Theorem 19.86** (Determinant bound for positive trace-preserving maps). [Lean](#) [Lean](#) [✓ Stmt](#) *For any positive trace-preserving map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ ,  $|\det T| \leq 1$ . This is [Wol12, Theorem 6.1(1)].*

*Proof.* ✓ Proof Every eigenvalue  $\mu$  of  $T$  satisfies  $|\mu| \leq 1$  (positivity and trace preservation force spectral radius  $\leq 1$ ). Since  $\det T$  is the product of the eigenvalues (counted with algebraic multiplicity), the bound  $|\det T| = \prod_i |\mu_i| \leq 1$  follows by induction.  $\square$

**Theorem 19.87** (Determinant one iff the channel is unitary). Lean Lean ✓ Stmt For a CPTP map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ ,

$$|\det T| = 1 \iff \exists U \in \mathcal{U}(D), \quad T(\rho) = U\rho U^\dagger.$$

This is [Wol12, Theorem 6.1(2)].

*Proof.* ✓ Proof Reverse direction is Theorem 19.85. Forward direction:  $|\det T| = 1$  together with  $|\mu| \leq 1$  for every eigenvalue  $\mu$  (from trace preservation and positivity) forces every eigenvalue to satisfy  $|\mu| = 1$ . Hence  $T$  is bijective. A bijective CPTP map is a  $*$ -automorphism: the Heisenberg dual  $T^*(Y) = \sum_i K_i^\dagger Y K_i$  is unital (since  $T$  is TP), so the Kadison–Schwarz inequality  $T^*(Y^\dagger Y) \geq T^*(Y)^\dagger T^*(Y)$  becomes an equality for every  $Y$ . Hence  $T^*$ , and therefore  $T$ , preserves products. By the Skolem–Noether theorem, every  $*$ -automorphism of  $M_D(\mathbb{C})$  has the form  $T(\rho) = U\rho U^\dagger$  for some unitary  $U$ . Kraus freedom then gives  $K_i = c_i U$  with  $\sum_i |c_i|^2 = 1$ .  $\square$

## Chapter 20

# Quantum Dynamical Semigroups

This chapter develops the theory of one-parameter semigroups of quantum channels and characterizes their generators via the GKSL (Gorini–Kossakowski–Sudarshan–Lindblad) theorem, following [Wol12, Chapter 7].

Section 20.1 establishes definitions and the basic exponential form (Proposition 20.14). Section 20.2 gives the Duhamel formula and perturbation bound. Section 20.3 develops the generator theory and proves the GKSL theorem.

### 20.1 Dynamical semigroups and the exponential form

**Definition 20.1** (Dynamical semigroup). [Lean](#) [✓ Stmt](#) A family of linear maps  $T : \mathbb{R} \rightarrow (M_D(\mathbb{C}) \rightarrow_{\ell} M_D(\mathbb{C}))$  is a *dynamical semigroup* if

1. (semigroup law)  $T_{t+s} = T_t \circ T_s$  for all  $t, s \geq 0$ ; and
2. (initial condition)  $T_0 = \text{id}$ .

This is [Wol12, Equation (7.1)].

**Definition 20.2** (Continuous dynamical semigroup). [Lean](#) [✓ Stmt](#) A dynamical semigroup  $T$  is *norm-continuous* if the map  $t \mapsto T_t$  is continuous in the operator norm topology. In finite dimension this coincides with strong continuity.

**Definition 20.3** (Exponential semigroup). [Lean](#) [Lean](#) [✓ Stmt](#) Given a generator  $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$ , the *exponential semigroup* is

$$T_t := e^{tL} = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}.$$

**Theorem 20.4** (Semigroup law for the exponential). [Lean](#) [Lean](#) [✓ Stmt](#) For all  $t, s \in \mathbb{R}$ ,  $e^{(t+s)L} = e^{tL} \cdot e^{sL}$ .

*Proof.* [✓ Proof](#) Since  $(t \cdot L)$  commutes with  $(s \cdot L)$ , this follows from the exponential addition formula for commuting elements in a Banach algebra.  $\square$

**Theorem 20.5** (Exponential semigroup at time 0). [Lean](#) [✓ Stmt](#) The continuous-linear-map exponential satisfies  $e^{0L} = \mathbb{1}$ .

*Proof.* [✓ Proof](#) The exponential of the zero endomorphism is the identity.  $\square$

**Theorem 20.6** (Continuity of the exponential semigroup). [Lean](#) [Lean](#) [✓ Stmt](#) *The map  $t \mapsto e^{tL}$  is continuous in the operator norm topology.*

*Proof.* [✓ Proof](#) The exponential is analytic (convergence radius =  $\infty$ ), hence continuous.  $\square$

**Theorem 20.7** (Derivative of the exponential semigroup). [Lean](#) [✓ Stmt](#) *For all  $t \in \mathbb{R}$ ,*

$$\frac{d}{dt}e^{tL} = e^{tL} \cdot L.$$

*This is [Wol12, Equation (7.2)].*

*Proof.* [✓ Proof](#) Chain rule applied to the composition  $\mathbb{R} \xrightarrow{t \mapsto t \cdot L} \text{End}(M_D(\mathbb{C})) \xrightarrow{\exp} \text{End}(M_D(\mathbb{C}))$ .  $\square$

**Theorem 20.8** (Derivative at the origin). [Lean](#) [✓ Stmt](#) *One has*

$$\left. \frac{d}{dt}e^{tL} \right|_{t=0} = L.$$

*Proof.* [✓ Proof](#) This is Theorem 20.7 at  $t = 0$  together with Theorem 20.5.  $\square$

**Theorem 20.9** (Generator uniqueness). [Lean](#) [✓ Stmt](#) *If  $e^{tL} = e^{tL'}$  for all  $t \geq 0$ , then  $L = L'$ .*

*Proof.* [✓ Proof](#) Both semigroups agree on  $[0, \infty)$ , so their derivatives within  $[0, \infty)$  at  $t = 0$  agree. Since  $[0, \infty)$  has unique differentials at 0,  $L = L'$ .  $\square$

**Theorem 20.10** (CLM and linear-map forms coincide). [Lean](#) [✓ Stmt](#) *The linear-map exponential semigroup and the continuous-linear-map exponential semigroup agree under the canonical identification of endomorphisms with continuous linear endomorphisms.*

*Proof.* [✓ Proof](#) This is a direct unpacking of the definition of the linear-map exponential semigroup.  $\square$

**Theorem 20.11** (Pointwise derivative formula). [Lean](#) [✓ Stmt](#) *For every  $X \in M_D(\mathbb{C})$  and every  $t \in \mathbb{R}$ ,*

$$\frac{d}{dt}(e^{tL}(X)) = e^{tL}(L(X)).$$

*Proof.* [✓ Proof](#) Evaluate the derivative of the continuous-linear-map semigroup at  $X$  and transport the statement through Theorem 20.10.  $\square$

**Theorem 20.12** (Linear-map exponential semigroup at time 0). [Lean](#) [✓ Stmt](#) *The linear-map exponential semigroup satisfies  $e^{0L} = \mathbb{1}$ .*

*Proof.* [✓ Proof](#) Transport Theorem 20.5 through Theorem 20.10.  $\square$

**Theorem 20.13** (Exponential semigroup as a dynamical semigroup). [Lean](#) [✓ Stmt](#) *The family  $t \mapsto e^{tL}$  satisfies the semigroup law and the initial condition, hence is a dynamical semigroup.*

*Proof.* [✓ Proof](#) Transport the continuous-linear-map semigroup law from Theorem 20.4 through Theorem 20.10; the initial condition is Theorem 20.12.  $\square$

**Theorem 20.14** (Every continuous semigroup is exponential). [Lean](#) [✓ Stmt](#) Every norm-continuous dynamical semigroup  $T$  on  $M_D(\mathbb{C})$  is of the form

$$T_t = e^{tL}$$

for some generator  $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$  and every  $t \geq 0$ . This is [Wol12, Proposition 7.1].

*Proof.* [✓ Proof](#) Since  $T_0 = \mathbb{1}$  and  $T$  is continuous, the integral  $M_\varepsilon = \int_0^\varepsilon T_s ds$  is invertible for small  $\varepsilon > 0$ . Using the semigroup property,  $T_t = M_\varepsilon^{-1}(M_{t+\varepsilon} - M_t)$  is differentiable for  $t \geq 0$ . ODE uniqueness then gives  $T_t = e^{tL}$  for all  $t \geq 0$ .  $\square$

## 20.2 Perturbation theory

**Theorem 20.15** (Duhamel formula). [Lean](#) [✓ Stmt](#) Let  $\Delta = L' - L$ . For  $t \geq 0$ ,

$$e^{tL'} - e^{tL} = \int_0^t e^{(t-s)L} \Delta e^{sL'} ds.$$

This is [Wol12, Lemma 7.1].

*Proof.* [✓ Proof](#) Define  $f(s) = e^{(t-s)L} e^{sL'}$ . Then  $f'(s) = e^{(t-s)L} \Delta e^{sL'}$  (Theorem 20.7), and  $e^{tL'} - e^{tL} = f(t) - f(0) = \int_0^t f'(s) ds$ .  $\square$

**Theorem 20.16** (Perturbation bound). [Lean](#) [✓ Stmt](#) For  $t \geq 0$ ,

$$\|e^{tL'} - e^{tL}\| \leq t \|\Delta\| \cdot \sup_{s \in [0, t]} \|e^{sL}\| \cdot \sup_{s \in [0, t]} \|e^{sL'}\|,$$

where  $\Delta = L' - L$ . This is [Wol12, Corollary 7.1].

*Proof.* [✓ Proof](#) Apply the triangle inequality and submultiplicativity of the operator norm to the Duhamel integral (Theorem 20.15).  $\square$

**Corollary 20.17** (Perturbation bound under unit-norm control). [Lean](#) [✓ Stmt](#) For  $t \geq 0$ , if  $\|e^{sL}\| \leq 1$  and  $\|e^{sL'}\| \leq 1$  for every  $s \in [0, t]$ , then

$$\|e^{tL'} - e^{tL}\| \leq t \|L' - L\|.$$

*Proof.* [✓ Proof](#) Apply Theorem 20.16 and bound both suprema by 1.  $\square$

**Lemma 20.18** (Dyson-series summability from factorial majorant). [Lean](#) [✓ Stmt](#) Assume a factorial majorant of Dyson–Phillips iterates:

$$\|D_n(t)\| \leq M \frac{(t \|L' - L\| M)^n}{n!}.$$

Then the Dyson series is summable in operator norm.

*Proof.* [✓ Proof](#) This is the Weierstrass M-test step, using the scalar convergence of  $\sum_n a^n/n!$  and lifting via norm-bounded summability.  $\square$

**Theorem 20.19** (Factorial norm bound for Dyson iterates). [Lean](#) [✓ Stmt](#) For  $s \in [0, t]$  and  $M = \sup_{u \in [0, t]} \|e^{uL}\|$ ,

$$\|\tilde{T}_s^{(n)}\| \leq M \frac{(s \|\Delta\| M)^n}{n!}.$$

*Proof.* ✓ Proof Induction on  $n$ . The base case uses the semigroup supremum bound. The inductive step bounds the integrand pointwise by  $M\|\Delta\| \cdot M(u\|\Delta\|M)^n/n!$  and integrates  $\int_0^s u^n du = s^{n+1}/(n+1)$ , recovering the factorial.  $\square$

**Corollary 20.20** (Dyson series summability). Lean ✓ Stmt *The Dyson–Phillips series  $\sum_n \tilde{T}_t^{(n)}$  converges in operator norm for every  $t \geq 0$ .*

*Proof.* ✓ Proof Compose the factorial norm bound (Theorem 20.19) with the Weierstrass M-test step (Lemma 20.18).  $\square$

**Corollary 20.21** (Factorial norm bound at the final time). Lean ✓ Stmt *For  $t \geq 0$ , setting  $s = t$  in Theorem 20.19 gives*

$$\|\tilde{T}_t^{(n)}\| \leq M \frac{(t\|\Delta\|M)^n}{n!},$$

where  $M = \sup_{u \in [0, t]} \|e^{uL}\|$ .

*Proof.* ✓ Proof This is the special case  $s = t$  of Theorem 20.19.  $\square$

**Theorem 20.22** (Continuity of Dyson iterates). Lean ✓ Stmt *Each Dyson iterate  $t \mapsto \tilde{T}_t^{(n)}$  is continuous in the time parameter.*

*Proof.* ✓ Proof By induction on  $n$ . The base case is continuity of the matrix exponential. For the inductive step, the parametric integral  $t \mapsto \int_0^t e^{(t-s)L} \Delta \tilde{T}_s^{(n)} ds$  is continuous by the continuous-parameter primitive theorem; the pasting lemma extends this to all of  $\mathbb{R}$  since  $\tilde{T}_t^{(n+1)} = 0$  for  $t \leq 0$ .  $\square$

**Theorem 20.23** (Factorial bound on the Dyson remainder). Lean ✓ Stmt *For  $s \in [0, t]$ ,  $M = \sup_{u \in [0, t]} \|e^{uL}\|$ , and  $M' = \sup_{u \in [0, t]} \|e^{uL'}\|$ :*

$$\|T'_s - \sum_{n < N} \tilde{T}_s^{(n)}\| \leq M' \frac{(s\|\Delta\|M)^N}{N!}.$$

*Proof.* ✓ Proof Induction on  $N$ . The base case is the semigroup supremum bound. The inductive step uses the telescoping remainder identity  $R_{N+1}(s) = \int_0^s e^{(s-u)L} \Delta R_N(u) du$  (derived from the Duhamel formula), bounds the integrand pointwise, and integrates to recover the factorial.  $\square$

**Theorem 20.24** (Dyson series identity [Wol12, Equation (7.13)]). Lean ✓ Stmt *For  $t \geq 0$ , the Dyson–Phillips series equals the perturbed semigroup:*

$$\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}.$$

*Proof.* ✓ Proof The partial-sum remainder  $\|e^{tL'} - \sum_{n < N} \tilde{T}_t^{(n)}\| \leq M' (t\|\Delta\|M)^N/N!$  tends to zero as  $N \rightarrow \infty$ , since  $c^N/N! \rightarrow 0$  for any constant  $c$ . Together with the summability of the series (Corollary 20.20), this identifies the sum via uniqueness of limits.  $\square$

**Corollary 20.25** (Tsum form of the Dyson series). Lean ✓ Stmt  $\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}$  (*tsum form*).

*Proof.* ✓ Proof Immediate from Theorem 20.24.  $\square$

## 20.3 GKSL/Lindblad generators

This section characterizes generators of semigroups of completely positive and trace-preserving maps. The central result is the GKSL theorem (Theorem 20.54), which shows that such generators have the standard Lindblad form.

### 20.3.1 Generator decomposition and conditional complete positivity

**Definition 20.26** (Generator decomposition). Lean ✓ Stmt A *generator decomposition* consists of a completely positive map  $\phi : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$  and a matrix  $\kappa \in M_d(\mathbb{C})$ , defining the linear map

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger.$$

This is [Wol12, Equation (7.14)].

**Definition 20.27** (Conditional complete positivity). Lean ✓ Stmt A linear map  $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$  is *conditionally completely positive* (CCP) if it admits a generator decomposition, i.e. there exist a CP map  $\phi$  and a matrix  $\kappa$  such that  $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ . This is condition 1 of [Wol12, Proposition 7.2].

**Definition 20.28** (Trace-annihilating map). Lean ✓ Stmt A linear map  $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$  is *trace-annihilating* if  $\text{tr}(L(\rho)) = 0$  for all  $\rho \in M_d(\mathbb{C})$ . This is the infinitesimal version of trace preservation.

**Definition 20.29** (Trace constraint). Lean ✓ Stmt A generator decomposition  $(\phi, \kappa)$  satisfies the *trace constraint* if  $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$ , where  $\phi^*$  is the Hilbert–Schmidt adjoint. This is the condition in [Wol12, Equation (7.20)].

**Theorem 20.30** (Trace constraint implies trace-annihilating). Lean ✓ Stmt If  $(\phi, \kappa)$  satisfies  $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$ , then  $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$  is trace-annihilating.

*Proof.* ✓ Proof Let  $\phi(\rho) = \sum_i K_i \rho K_i^\dagger$ . By the cyclic property of trace,  $\text{tr}(L(\rho)) = \text{tr}((\sum_i K_i^\dagger K_i)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = \text{tr}((\kappa + \kappa^\dagger)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = 0$ . □

### 20.3.2 The Lindblad form

**Definition 20.31** (Lindblad form). Lean ✓ Stmt A *Lindblad form* consists of a Hermitian matrix  $H = H^\dagger$  (the Hamiltonian) and a family of matrices  $\{L_j\}_{j=0}^{r-1}$  (the Lindblad operators), defining the linear map

$$L(\rho) = i[\rho, H] + \sum_j \left( L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\}_+ \right),$$

where  $[A, B] = AB - BA$  and  $\{A, B\}_+ = AB + BA$ . This is [Wol12, Equation (7.21)].

**Theorem 20.32** (Lindblad form is trace-annihilating). Lean ✓ Stmt Every Lindblad form defines a trace-annihilating linear map.

*Proof.* ✓ Proof The Hamiltonian part satisfies  $\text{tr}(i[\rho, H]) = 0$  by the cyclic property. Each dissipator term satisfies  $\text{tr}(L_j \rho L_j^\dagger) = \text{tr}(L_j^\dagger L_j \rho) = \text{tr}(\rho L_j^\dagger L_j)$ , so the three terms sum to zero. □

**Theorem 20.33** (Lindblad form equals generator decomposition). Lean ✓ Stmt A Lindblad form with Hamiltonian  $H$  and operators  $\{L_j\}$  defines the same linear map as the generator decomposition  $(\phi, \kappa)$  with  $\phi(\rho) = \sum_j L_j \rho L_j^\dagger$  and  $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$ . This is [Wol12, Equation (7.24)].



*Proof.* ✓ Proof Compute  $\kappa^\dagger = -iH + \frac{1}{2} \sum_j L_j^\dagger L_j$  using  $H^\dagger = H$  and  $(A^\dagger A)^\dagger = A^\dagger A$ . Setting  $S = \sum_j L_j^\dagger L_j$ , expand  $\phi(\rho) - \kappa\rho - \rho\kappa^\dagger = \sum_j L_j \rho L_j^\dagger - iH\rho - \frac{1}{2}S\rho + i\rho H - \frac{1}{2}\rho S$ , which is  $i[\rho, H] + \sum_j (L_j \rho L_j^\dagger - \frac{1}{2}L_j^\dagger L_j \rho - \frac{1}{2}\rho L_j^\dagger L_j)$ .  $\square$

**Theorem 20.34** (Lindblad form is CCP). Lean ✓ Stmt *Every Lindblad form defines a conditionally completely positive map.*

*Proof.* ✓ Proof By Theorem 20.33, the Lindblad form equals a generator decomposition with  $\phi$  CP.  $\square$

**Definition 20.35** (Commutator form of Lindblad equation). Lean ✓ Stmt *The commutator form of the Lindblad equation writes the generator as*

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_j ([L_j, \rho L_j^\dagger] + [L_j \rho, L_j^\dagger]),$$

where  $[A, B] = AB - BA$ . This is [Wol12, Equation (7.22)].

**Theorem 20.36** (Commutator form equals standard Lindblad form). Lean ✓ Stmt *The commutator form (Definition 20.35) defines the same linear map as the standard Lindblad form (Definition 20.31).*

*Proof.* ✓ Proof Expanding the commutator brackets  $[L_j, \rho L_j^\dagger]$  and  $[L_j \rho, L_j^\dagger]$  yields the standard dissipator terms.  $\square$

### 20.3.3 Characterization of conditional complete positivity

**Theorem 20.37** (Conditional complete positivity implies projected Choi positivity). Lean ✓ Stmt *If  $L$  is CCP and  $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$ , then*

$$P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0.$$

*This is [Wol12, Proposition 7.2].*

*Proof.* ✓ Proof Write  $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ . The left- and right-multiplication terms are annihilated by  $P$ , so the projected Choi matrix reduces to the projected Choi matrix of  $\phi$ , which is positive.  $\square$

**Theorem 20.38** (Projected Choi positivity implies conditional complete positivity). Lean ✓ Stmt *If  $L$  is Hermiticity-preserving and  $P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0$  where  $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$ , then  $L$  is CCP. This is [Wol12, Proposition 7.2].*

*Proof.* ✓ Proof Decompose the Hermitian Choi matrix as  $\tau_L = Q - |\psi\rangle\langle\Omega| - |\Omega\rangle\langle\psi|$  with  $Q \geq 0$  supported on  $|\Omega\rangle^\perp$ . The Choi–Jamiołkowski isomorphism applied to  $Q$  gives the CP map  $\phi$ , and  $(\kappa \otimes \mathbb{1})|\Omega\rangle = |\psi\rangle$  defines  $\kappa$ .  $\square$

### 20.3.4 Completely positive semigroups and conditionally completely positive generators

**Theorem 20.39** (CP semigroups have CCP generators). Lean ✓ Stmt *If  $e^{tL}$  is CP for all  $t \geq 0$ , then  $L$  is CCP. This is [Wol12, Proposition 7.3].*

*Proof.* ✓ Proof From  $(e^{tL} \otimes \text{id})(|\Omega\rangle\langle\Omega|) \geq 0$ , expand at infinitesimal  $t$  and project onto  $P$  to get  $P(L \otimes \text{id})(|\Omega\rangle\langle\Omega|)P \geq 0$ . Then Theorem 20.38 gives CCP. □

**Theorem 20.40** (CCP generators yield CP semigroups). Lean ✓ Stmt *If  $L$  is CCP, then  $e^{tL}$  is CP for all  $t \geq 0$ . This is [Wol12, Proposition 7.3].*

*Proof.* ✓ Proof Approximate  $e^{tL}$  by the completely positive Euler steps

$$\rho \mapsto (\mathbb{1} - h\kappa)\rho(\mathbb{1} - h\kappa)^\dagger + h\phi(\rho), \quad h = \frac{t}{n+1}.$$

Each step is CP, hence so are its powers. These powers converge in operator norm to  $e^{tL}$ , and the cone of CP maps is closed. □

**Theorem 20.41** (CP semigroups iff CCP generators). Lean ✓ Stmt  *$e^{tL}$  is CP for all  $t \geq 0$  if and only if  $L$  is CCP.*

*Proof.* ✓ Proof Combine Theorems 20.39 and 20.40. □

### 20.3.5 Freedom in generator representation

**Theorem 20.42** (Traceless Kraus operators exist). Lean ✓ Stmt *Assume  $d > 0$ . Given any Kraus operators  $\{K_j\}$ , there exist  $c_j \in \mathbb{C}$  such that  $K'_j = K_j + c_j\mathbb{1}$  is traceless. This is part of [Wol12, Proposition 7.4].*

*Proof.* ✓ Proof Set  $c_j = -\text{tr}(K_j)/d$ . □

**Theorem 20.43** (Shift invariance of generators). Lean ✓ Stmt *If  $K'_j = K_j + c_j\mathbb{1}$  and  $\kappa'$  is adjusted per [Wol12, Equation (7.19)], then  $(\phi', \kappa')$  and  $(\phi, \kappa)$  define the same generator.*

*Proof.* ✓ Proof Direct algebraic expansion. □

### 20.3.6 Uniqueness of the traceless Lindblad form

**Definition 20.44** (Traceless Kraus operators). Lean ✓ Stmt A Lindblad form has *traceless Kraus operators* if  $\text{tr}(L_j) = 0$  for every  $j$ .

**Theorem 20.45** (Uniqueness of the completely positive part). Lean ✓ Stmt *If two Lindblad forms  $F, F'$  induce the same generator and both have traceless Kraus operators, then their CP parts agree:  $\phi = \phi'$ . This is part of [Wol12, Proposition 7.4(2)].*

*Proof.* ✓ Proof Compare projected Choi matrices and use Choi injectivity. □

**Theorem 20.46** (Uniqueness of the drift term modulo phase). Lean ✓ Stmt *If two Lindblad forms  $F, F'$  induce the same generator and both have traceless Kraus operators, then their drift matrices differ by an imaginary scalar:  $\kappa' = \kappa + i\lambda\mathbb{1}$  for some  $\lambda \in \mathbb{R}$ . This is part of [Wol12, Proposition 7.4(2)].*

*Proof.* ✓ Proof After identifying  $\phi = \phi'$  (Theorem 20.45), the generator equality forces  $\Delta\kappa \cdot \rho + \rho \cdot (\Delta\kappa)^\dagger = 0$  for all  $\rho$ . Setting  $\rho = \mathbb{1}$  gives  $\Delta\kappa + (\Delta\kappa)^\dagger = 0$  (skew-Hermiticity), which converts the equation to  $[\Delta\kappa, \rho] = 0$  for all  $\rho$ . The scalar commutant lemma gives  $\Delta\kappa = c \cdot \mathbb{1}$ , and skew-Hermiticity forces  $c = i\lambda$  for some  $\lambda \in \mathbb{R}$ . □

**Theorem 20.47** (Combined uniqueness of the traceless Lindblad form). Lean ✓ Stmt *If two Lindblad forms  $F, F'$  induce the same generator and both have traceless Kraus operators, then  $\phi = \phi'$  and  $\kappa' = \kappa + i\lambda\mathbb{1}$  for some  $\lambda \in \mathbb{R}$ .*

*Proof.* ✓ Proof Combine Theorems 20.45 and 20.46. □

### 20.3.7 Trace-annihilation and trace preservation

**Theorem 20.48** (Trace-annihilating generators yield trace-preserving semigroups). [Lean](#) [✓ Stmt](#)  
*If  $L$  is trace-annihilating, then  $e^{tL}$  is trace-preserving for every  $t \in \mathbb{R}$ .*

*Proof.* [✓ Proof](#) For fixed  $\rho$ , the function  $f(t) = \text{tr}(e^{tL}(\rho))$  has derivative  $f'(t) = \text{tr}(L(e^{tL}(\rho))) = 0$ , so  $f$  is constant and  $\text{tr}(e^{tL}(\rho)) = f(t) = f(0) = \text{tr}(\rho)$ .  $\square$

**Theorem 20.49** (Trace-preserving semigroups have trace-annihilating generators). [Lean](#) [✓ Stmt](#)  
*If  $e^{tL}$  is trace-preserving for every  $t \geq 0$ , then  $L$  is trace-annihilating.*

*Proof.* [✓ Proof](#) Differentiate the identity  $\text{tr}(e^{tL}(\rho)) = \text{tr}(\rho)$  at  $t = 0$  from the right.  $\square$

### 20.3.8 The GKSL/Lindblad theorem

**Definition 20.50** (GKSL generator). [Lean](#) [✓ Stmt](#) A linear map  $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$  is a *GKSL generator* if  $e^{tL}$  is a quantum channel (CPTP) for all  $t \geq 0$ .

**Theorem 20.51** (A trace-constrained generator decomposition gives a GKSL generator). [Lean](#) [✓ Stmt](#)  
*If  $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$  with  $\phi$  CP and  $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$ , then  $L$  is a GKSL generator. This is [Wol12, Theorem 7.1, Equation (7.20)].*

*Proof.* [✓ Proof](#) CP follows from Theorem 20.40. The trace constraint implies trace-annihilation by Theorem 20.30, so Theorem 20.48 gives trace preservation.  $\square$

**Theorem 20.52** (GKSL generators admit a trace-constrained generator decomposition). [Lean](#) [✓ Stmt](#)  
*If  $L$  is a GKSL generator, then there exist a CP map  $\phi$  and a matrix  $\kappa$  such that*

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$$

and  $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$ .

*Proof.* [✓ Proof](#) Apply Theorem 20.39 to obtain a CCP decomposition. Then apply Theorem 20.49 to the trace-preserving part of the semigroup and rewrite the resulting trace-annihilation condition as  $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$ .  $\square$

**Theorem 20.53** (GKSL iff conditional complete positivity and trace annihilation). [Lean](#) [✓ Stmt](#)  
 *$L$  is a GKSL generator if and only if  $L$  is CCP and trace-annihilating.*

*Proof.* [✓ Proof](#) Forward: apply Theorem 20.39 to the CP part and Theorem 20.49 to the TP part. Reverse: combine Theorem 20.40 with Theorem 20.48.  $\square$

**Theorem 20.54** (GKSL iff Lindblad form). [Lean](#) [✓ Stmt](#)  *$L$  is a GKSL generator if and only if*

$$L(\rho) = i[\rho, H] + \sum_j \left( L_j \rho L_j^\dagger - \frac{1}{2} \{ L_j^\dagger L_j, \rho \}_+ \right)$$

with  $H = H^\dagger$ . This is [Wol12, Theorem 7.1].

*Proof.* [✓ Proof](#) Forward: apply Theorem 20.52 to write  $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$  with trace constraint. Then choose traceless Kraus operators by Theorem 20.42 and rewrite  $\kappa$  as  $iH + \frac{1}{2}\phi^*(\mathbb{1})$ ; Theorem 20.33 gives the Lindblad formula. Reverse: Theorems 20.34 and 20.32 show that every Lindblad form satisfies the right-hand side of Theorem 20.53.  $\square$

### 20.3.9 Kossakowski matrix form

**Definition 20.55** (Kossakowski form). [Lean](#) [✓ Stmt](#) A *Kossakowski form* consists of  $H = H^\dagger$ , a finite family of matrices  $\{F_k\}_{k=0}^{n-1}$ , and a positive semidefinite matrix  $C \in M_n(\mathbb{C})$ , defining

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_{k,l} C_{lk} ([F_k, \rho F_l^\dagger] + [F_k \rho, F_l^\dagger]).$$

In Wolf's formula one takes  $\{F_k\}$  to be a basis of traceless matrices; here we keep only the algebraic data needed for the conversion to Lindblad form.

**Theorem 20.56** (Kossakowski form iff Lindblad form). [Lean](#) [✓ Stmt](#) A linear map admits a Kossakowski form iff it admits a Lindblad form.

*Proof.* [✓ Proof](#) Forward: diagonalize  $C = B^\dagger B$  and set  $L_j = \sum_k B_{jk} F_k$ . Reverse: keep the same Hamiltonian and family  $F_k = L_k$ , and take  $C = 1$ .  $\square$

## 20.4 Primitivity and irreducibility of QDS

### 20.4.1 Auxiliary spectral and semigroup lemmas

**Definition 20.57** (Quantum dynamical semigroup). [Lean](#) [✓ Stmt](#) A quantum dynamical semigroup is a norm-continuous dynamical semigroup whose time slices are quantum channels for every  $t \geq 0$ .

**Theorem 20.58** (Semigroup power identity). [Lean](#) [✓ Stmt](#) For  $t \geq 0$  and  $n \in \mathbb{N}$ , one has  $T_{nt} = (T_t)^n$ .

**Theorem 20.59** (Eigenvector propagation under powers). [Lean](#) [✓ Stmt](#) If  $E(v) = \mu v$ , then  $E^n(v) = \mu^n v$  for all  $n \in \mathbb{N}$ .

**Lemma 20.60** (Density matrices are nonzero). [Lean](#) [✓ Stmt](#) Every density matrix is nonzero.

**Theorem 20.61** (Compression invariance is stable under powers). [Lean](#) [✓ Stmt](#) If a corner compression is invariant under  $E$ , it remains invariant under every power  $E^n$ .

**Theorem 20.62** (Generator eigenvectors along the exponential semigroup). [Lean](#) [✓ Stmt](#) If  $L(X) = \mu X$ , then  $e^{tL}(X) = e^{t\mu} X$  for all  $t \in \mathbb{R}$ .

**Theorem 20.63** (Spectral mapping for semigroup generators). [Lean](#) [✓ Stmt](#) If  $\mu$  is in the spectrum of  $L$ , then  $e^{t\mu}$  is in the spectrum of  $e^{tL}$ .

**Theorem 20.64** (Root-of-unity rigidity for phases). [Lean](#) [✓ Stmt](#) If  $e^{it\theta}$  is a root of unity for every  $t > 0$ , then  $\theta = 0$ .

**Theorem 20.65** (Peripheral generator eigenvalues are purely imaginary). [Lean](#) [✓ Stmt](#) Peripheral spectral points of the generator have vanishing real part.

**Theorem 20.66** (Peripheral cardinality bound). [Lean](#) [✓ Stmt](#) The number of peripheral eigenvalues is bounded by  $\dim(M_D(\mathbb{C}))$ .

**Theorem 20.67** (Bounded order under peripheral power-closure). [Lean](#) [✓ Stmt](#) If all powers of a peripheral eigenvalue remain peripheral, then some positive power not exceeding  $\dim(M_D(\mathbb{C}))$  equals 1.

**Theorem 20.68** (Peripheral powers stay peripheral under irreducibility). [Lean](#) [✓ Stmt](#) *For an irreducible channel with faithful fixed point, every power of a peripheral eigenvalue is again peripheral.*

**Theorem 20.69** (Continuous-linear-map power evaluation). [Lean](#) [✓ Stmt](#) *Evaluating powers of a linear map agrees with powers in its continuous linear realization.*

**Theorem 20.70** (Spectral-radius criterion from eigenvalue bounds). [Lean](#) [✓ Stmt](#) *If all eigenvalues satisfy  $|\lambda| < 1$ , then the spectral radius is  $< 1$ .*

**Theorem 20.71** (Primitive powers annihilate trace-zero matrices). [Lean](#) [✓ Stmt](#) *If  $E$  is an irreducible primitive channel with faithful fixed density  $\sigma$ , then  $E^n(X) \rightarrow 0$  for every matrix  $X$  with  $\text{tr}(X) = 0$ .*

*Proof.* [✓ Proof](#) Decompose  $E^n$  into the fixed-point projection and its complementary part. On the complement, every eigenvalue has modulus  $< 1$ , so the complementary powers tend to 0; the trace-zero hypothesis removes the fixed-point component.  $\square$

**Theorem 20.72** (Fixed points of an irreducible channel are trace multiples). [Lean](#) [✓ Stmt](#) *If  $E$  is an irreducible channel with faithful fixed density  $\sigma$ , then every nonzero fixed point  $V$  of  $E$  satisfies  $V = \text{tr}(V)\sigma$ .*

*Proof.* [✓ Proof](#) Subtract the trace multiple  $\text{tr}(V)\sigma$  and apply the vanishing theorem for trace-zero fixed points of an irreducible channel.  $\square$

**Corollary 20.73** (Nonzero fixed points have nonzero trace). [Lean](#) [✓ Stmt](#) *Every nonzero fixed point of an irreducible channel has nonzero trace.*

*Proof.* [✓ Proof](#) Otherwise the preceding theorem would force the fixed point to vanish.  $\square$

**Theorem 20.74** (Peripheral eigenvectors become periodic fixed points). [Lean](#) [✓ Stmt](#) *For every positive time  $t$ , a quantum dynamical semigroup irreducible at every positive time has the following property: every peripheral eigenvalue  $\mu$  of  $T_t$  admits a nonzero eigenvector  $V$  and a positive integer  $p$  such that  $T_t(V) = \mu V$  and  $T_{pt}(V) = V$ .*

*Proof.* [✓ Proof](#) Peripheral powers remain peripheral for an irreducible channel with faithful fixed point. The bounded-order lemma then gives  $\mu^p = 1$ , and the semigroup power identity turns the eigenvector equation for  $T_t$  into a fixed-point equation for  $T_{pt}$ .  $\square$

**Theorem 20.75** (Peripheral eigenvectors with nonzero trace). [Lean](#) [✓ Stmt](#) *For every positive time  $t$ , under the same hypotheses every peripheral eigenvalue of  $T_t$  has a nonzero eigenvector with nonzero trace.*

*Proof.* [✓ Proof](#) Apply Theorem 20.74 and then use the previous corollary at the irreducible time  $pt$ .  $\square$

**Theorem 20.76** (Trace-preserving eigenvectors force eigenvalue 1). [Lean](#) [✓ Stmt](#) *If  $E$  is trace-preserving,  $E(V) = \mu V$ , and  $\text{tr}(V) \neq 0$ , then  $\mu = 1$ .*

*Proof.* [✓ Proof](#) Take traces in  $E(V) = \mu V$  and use trace preservation.  $\square$

**Theorem 20.77** (Peripheral collapse under irreducibility at all times). [Lean](#) [✓ Stmt](#) *If a quantum dynamical semigroup is irreducible at every positive time, then every peripheral eigenvalue of every positive-time slice equals 1.*

*Proof.* ✓ Proof Combine Theorem 20.75 with Theorem 20.76. □

**Theorem 20.78** (Primitivity from irreducibility at all times). Lean ✓ Stmt *If a quantum dynamical semigroup is irreducible at every positive time, then every positive-time slice is primitive.*

*Proof.* ✓ Proof The unique faithful fixed density gives the one-dimensional peripheral spectrum criterion, and Theorem 20.77 collapses the peripheral set to  $\{1\}$ . □

**Remark 20.79** (Semigroup irreducible/primitive equivalence). ✓ Stmt The semigroup analog of the irreducible/primitive equivalence starts from one irreducible time slice, constructs a primitive smaller time step, and lifts the conclusion to the full semigroup (Theorems 20.95 and 20.96).

**Remark 20.80** (Reduction to a smaller time step for semigroups). ✓ Stmt Starting from one irreducible time  $t_0$ , we construct a positive time step  $u = t_0/N$  with  $T_{t_0} = (T_u)^N$ , prove peripheral eigenvalue collapse at time  $u$ , and deduce primitivity of  $T_u$ .

**Theorem 20.81** (The stationary density is fixed for all times). Lean ✓ Stmt *If  $t_0 > 0$ ,  $T_{t_0}$  is irreducible, and  $\sigma$  is its unique fixed density matrix, then  $T_u(\sigma) = \sigma$  for every  $u \geq 0$ .*

*Proof.* ✓ Proof By semigroup commutativity,  $T_{t_0}(T_u(\sigma)) = T_u(T_{t_0}(\sigma)) = T_u(\sigma)$ , and uniqueness of the fixed density at time  $t_0$  identifies  $T_u(\sigma)$  with  $\sigma$ . □

**Theorem 20.82** (Fixed points at the irreducible time are trace multiples). Lean ✓ Stmt *If  $T_{t_0}$  is irreducible with faithful fixed density  $\sigma$ , then every fixed point  $V$  of  $T_{t_0}$  satisfies  $V = \text{tr}(V)\sigma$ .*

*Proof.* ✓ Proof This is the fixed-point trace-scalar theorem for the irreducible channel  $T_{t_0}$ . □

**Definition 20.83** (Residual slice index). Lean ✓ Stmt For  $u \geq 0$  and  $s > 0$ , the residual slice index is

$$m_n = \left\lfloor \frac{nu}{s} \right\rfloor.$$

**Definition 20.84** (Residual slice time). Lean ✓ Stmt For  $u \geq 0$  and  $s > 0$ , the residual slice time is the remainder

$$r_n = s \text{ fract} \left( \frac{nu}{s} \right).$$

**Theorem 20.85** (Residual slice decomposition). Lean Lean ✓ Stmt *For  $u \geq 0$  and  $s > 0$ , one has  $r_n \in [0, s]$  and*

$$nu = m_n s + r_n.$$

*Proof.* ✓ Proof This is the floor-plus-fraction decomposition of  $nu/s$ , multiplied by  $s$ . □

**Theorem 20.86** (A residual slice can vanish). Lean ✓ Stmt *Suppose  $u \geq 0$  and  $s > 0$ , and that  $T_s(\delta) = \delta$  and  $T_{nu}(\delta) \rightarrow 0$  as  $n \rightarrow \infty$ . Then there exists  $a \in [0, s]$  such that  $T_a(\delta) = 0$ .*

*Proof.* ✓ Proof The residual slice times lie in the compact interval  $[0, s]$ , so a subsequence converges to some  $a \in [0, s]$ . The residual decomposition rewrites  $T_{nu}(\delta)$  in terms of  $T_{r_n}(\delta)$ , and continuity of the semigroup identifies the limit with  $T_a(\delta)$ . □

**Theorem 20.87** (Reduced-time eigenvector has nonzero trace). Lean ✓ Stmt *Under the reduced-time hypotheses  $T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!}$ , if  $\mu$  is a peripheral eigenvalue of  $T_u$ , then  $T_u$  admits a  $\mu$ -eigenvector with nonzero trace.*

**Theorem 20.88** (Peripheral collapse at the reduced time step). Lean ✓ Stmt *Under the same hypotheses, every peripheral eigenvalue of  $T_u$  equals 1.*

**Theorem 20.89** (Primitivity at the reduced time step). [Lean](#) [✓ Stmt](#) *Under the same hypotheses,  $T_u$  is primitive.*

**Theorem 20.90** (Irreducibility descends to a reduced time step). [Lean](#) [✓ Stmt](#) *If  $T_{t_0} = (T_u)^N$  and  $T_{t_0}$  is irreducible, then  $T_u$  is irreducible.*

*Proof.* [✓ Proof](#) Any invariant compression for  $T_u$  remains invariant under powers, hence also for  $T_{t_0}$ ; irreducibility of  $T_{t_0}$  rules this out.  $\square$

**Theorem 20.91** (Existence of an irreducible reduced time step). [Lean](#) [✓ Stmt](#) *If  $t_0 > 0$ ,  $T_{t_0}$  is irreducible, and  $\sigma$  is fixed for all times, then there exists a positive time step  $u$  such that*

$$T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!},$$

*the slice  $T_u$  is a channel,  $T_u$  is irreducible, and  $T_u(\sigma) = \sigma$ .*

*Proof.* [✓ Proof](#) Take  $u = t_0/N$  with  $N = (\dim M_D(\mathbb{C}))!$ . The semigroup power identity gives the factorization of  $T_{t_0}$ , channelity comes from the semigroup hypotheses, and Theorem 20.90 gives irreducibility of  $T_u$ .  $\square$

**Theorem 20.92** (Fixed points of the primitive reduced time step are trace-scalars). [Lean](#) [✓ Stmt](#) *If  $T_u$  is primitive and  $\sigma$  is the common fixed density, then every fixed point of any positive-time slice  $T_s$  has the form  $\tau = \text{tr}(\tau)\sigma$ .*

**Theorem 20.93** (Existence of a primitive reduced time step). [Lean](#) [✓ Stmt](#) *If  $T_{t_0}$  is irreducible for some  $t_0 > 0$ , then there exists a positive reduced time step  $u$  such that  $T_u$  is primitive.*

**Theorem 20.94** (Irreducibility propagates to all positive times). [Lean](#) [✓ Stmt](#) *If  $T_{t_0}$  is irreducible for some  $t_0 > 0$ , then every positive-time slice  $T_s$  is irreducible.*

*Proof.* [✓ Proof](#) Let  $\sigma$  be the unique faithful fixed density of  $T_{t_0}$ . By Theorem 20.81, it is fixed for all times, and Theorem 20.82 makes the fixed-point space of  $T_{t_0}$  one-dimensional. Choose a primitive reduced time step by Theorem 20.93; then the trace-scalar description of fixed points from Theorem 20.92 gives the uniqueness criterion for irreducibility at every positive time.  $\square$

**Theorem 20.95** (An irreducible semigroup is primitive). [Lean](#) [✓ Stmt](#) *Let  $T_t = e^{tL}$  be a quantum dynamical semigroup. If  $T_{t_0}$  is irreducible for some  $t_0 > 0$ , then  $T_t$  is primitive (and irreducible) for every  $t > 0$ . This is [Wol12, Proposition 7.5].*

*Proof.* [✓ Proof](#) First apply Theorem 20.94 to propagate irreducibility from the single time  $t_0$  to every positive time. Then apply Theorem 20.78 to conclude primitivity of every positive-time slice.  $\square$

**Theorem 20.96** (Irreducibility iff primitivity for a quantum dynamical semigroup). [Lean](#) [✓ Stmt](#) *For a quantum dynamical semigroup  $T_t = e^{tL}$ :*

$$(\exists t_0 > 0, T_{t_0} \text{ irreducible}) \iff (\forall t > 0, T_t \text{ primitive and irreducible}).$$

*This is [Wol12, Proposition 7.5].*

*Proof.* [✓ Proof](#) Forward: apply Theorem 20.95. Reverse: take  $t_0 = 1$  (or any positive time); the right-hand side already includes irreducibility of  $T_t$  for every  $t > 0$ .  $\square$



## 20.5 Kernel of the adjoint Liouvillian

**Definition 20.97** (Faithful stationary state). [Lean](#) [✓ Stmt](#) A generator  $L$  has a faithful stationary state if there exists a positive definite density matrix  $\rho_0$  with  $L(\rho_0) = 0$ .

**Definition 20.98** (Adjoint dissipator). [Lean](#) [✓ Stmt](#) For a Lindblad operator  $L_j$ , the adjoint dissipator on observables is

$$\mathcal{D}_{L_j}^*(A) = L_j^\dagger A L_j - \frac{1}{2} L_j^\dagger L_j A - \frac{1}{2} A L_j^\dagger L_j.$$

**Definition 20.99** (Adjoint generator). [Lean](#) [✓ Stmt](#) The adjoint generator of a Lindblad form is

$$L^*(A) = i[H, A] + \sum_j \mathcal{D}_{L_j}^*(A).$$

**Theorem 20.100** (Trace-pairing characterization of the adjoint generator). [Lean](#) [✓ Stmt](#) For every density matrix  $\rho$  and every observable  $A$ ,

$$\text{tr}(\rho L^*(A)) = \text{tr}(L(\rho) A).$$

*Proof.* [✓ Proof](#) Expand the Schrödinger and Heisenberg formulas and apply cyclicity of the trace term by term.  $\square$

**Definition 20.101** (Commutant of the Lindblad data). [Lean](#) [✓ Stmt](#) The commutant of a Lindblad form is the set of matrices commuting with  $H$ , with every Lindblad operator  $L_j$ , and with every adjoint  $L_j^\dagger$ .

**Definition 20.102** (Adjoint kernel). [Lean](#) [✓ Stmt](#) The adjoint kernel is the set of observables  $A$  satisfying  $L^*(A) = 0$ .

**Theorem 20.103** (Commutant lies in the adjoint kernel). [Lean](#) [Lean](#) [✓ Stmt](#) If an observable commutes with  $H$ , with every  $L_j$ , and with every  $L_j^\dagger$ , then it lies in  $\ker(L^*)$ .

*Proof.* [✓ Proof](#) The Hamiltonian commutator term vanishes, and each adjoint dissipator  $\mathcal{D}_{L_j}^*(A)$  is zero under the commutation hypotheses.  $\square$

**Theorem 20.104** (Adjoint kernel lies in the commutant). [Lean](#) [✓ Stmt](#) Let  $L$  be a GKSL generator in Lindblad form with Hamiltonian  $H$  and Lindblad operators  $\{L_j\}$ . Define the adjoint generator  $L^*$  by  $\text{tr}(\rho L^*(A)) = \text{tr}(L(\rho) A)$ . If  $L$  has a faithful stationary state  $\rho_0 > 0$  with  $L(\rho_0) = 0$ , then for any  $A$ :

$$L^*(A) = 0 \implies [A, H] = [A, L_j] = [A, L_j^\dagger] = 0 \quad \forall j.$$

That is,  $\ker(L^*) \subseteq \{H, L_j, L_j^\dagger\}'$ . This is [\[Wol12, Theorem 7.2\]](#).

*Proof.* [✓ Proof](#) From  $L^*(A) = 0$  and  $L^*(A^\dagger) = 0$ , compute  $L^*(A^\dagger A) = \sum_j [A, L_j]^\dagger [A, L_j]$ . Faithfulness of  $\rho_0$  and  $\text{tr}(\rho_0 L^*(A^\dagger A)) = 0$  force  $[A, L_j] = 0$  for all  $j$ . Similarly  $[A, L_j^\dagger] = 0$ , and then  $L^*(A) = 0$  directly gives  $[A, H] = 0$ .  $\square$

**Theorem 20.105** (Adjoint kernel equals the commutant). [Lean](#) [✓ Stmt](#) Under the same hypotheses as [Theorem 20.104](#),

$$\ker(L^*) = \{H, L_j, L_j^\dagger\}',$$

i.e. the adjoint kernel equals the commutant of  $\{H, L_j, L_j^\dagger\}$ . This is [\[Wol12, Theorem 7.2\]](#).

*Proof.* [✓ Proof](#) The inclusion  $\supseteq$  is the easy direction (commutant elements are in the kernel by a direct calculation). The reverse  $\subseteq$  is [Theorem 20.104](#).  $\square$



## 20.6 Reducibility of quantum dynamical semigroups

The four equivalent reducibility conditions of [Wol12, Proposition 7.6] are as follows.

**Definition 20.106** (Nontrivial projection). [Lean](#) [✓ Stmt](#) A projection is nontrivial if it is orthogonal and neither 0 nor  $\mathbb{1}$ .

**Definition 20.107** (Rank-deficient fixed density). [Lean](#) [✓ Stmt](#) Condition (1): there exists a density matrix  $\rho_0$  with nontrivial kernel such that  $e^{tL}(\rho_0) = \rho_0$  for all  $t \geq 0$ .

**Definition 20.108** (Rank-deficient kernel element). [Lean](#) [✓ Stmt](#) Condition (2): there exists a density matrix  $\rho_0$  with nontrivial kernel satisfying  $L(\rho_0) = 0$ .

**Definition 20.109** (Invariant compression). [Lean](#) [✓ Stmt](#) Condition (3): there exists a nontrivial orthogonal projector  $P$  such that  $e^{tL}(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$  for all  $t \geq 0$ .

**Definition 20.110** (Block-upper-triangular Lindblad form). [Lean](#) [✓ Stmt](#) Condition (4): there exists a nontrivial projector  $P$  and a Lindblad form  $(H, \{L_j\})$  for  $L$  such that  $(1 - P)L_jP = 0$  and  $(1 - P)\kappa P = 0$  for all  $j$ , where  $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$ .

**Definition 20.111** (Generator-preserved compression). [Lean](#) [✓ Stmt](#) For a projection  $P$ , the generator preserves the compression  $PM_d(\mathbb{C})P$  if

$$P L(PXP) P = L(PXP)$$

for every  $X \in M_d(\mathbb{C})$ .

**Definition 20.112** (Reducible quantum dynamical semigroup). [Lean](#) [✓ Stmt](#) A quantum dynamical semigroup is reducible if it has a nontrivial invariant compression.

**Theorem 20.113** (Semigroup invariance implies generator invariance). [Lean](#) [✓ Stmt](#) If the semigroup preserves  $PM_d(\mathbb{C})P$  for all non-negative times, then the generator preserves the same compression.

*Proof.* [✓ Proof](#) Differentiate the identity  $Pe^{tL}(PXP)P = e^{tL}(PXP)$  at  $t = 0$ . □

**Theorem 20.114** (Generator invariance implies semigroup invariance). [Lean](#) [✓ Stmt](#) If the generator preserves  $PM_d(\mathbb{C})P$ , then so does the exponential semigroup  $e^{tL}$  for every  $t \geq 0$ .

*Proof.* [✓ Proof](#) Apply the compression map termwise to the exponential series. Every iterate of  $L$  preserves the compression, so the whole series does as well. □

**Theorem 20.115** (Reducibility as a generator-level criterion). [Lean](#) [✓ Stmt](#) A quantum dynamical semigroup is reducible if and only if its generator preserves some nontrivial compression.

*Proof.* [✓ Proof](#) Combine Theorems 20.113 and 20.114. □

**Theorem 20.116** (Rank-deficient fixed densities and kernel elements). [Lean](#) [✓ Stmt](#) A rank-deficient density matrix is a fixed point of  $e^{tL}$  for all  $t \geq 0$  if and only if it lies in  $\ker L$ . This is [Wol12, Proposition 7.6, (1) $\Leftrightarrow$ (2)].

*Proof.* [✓ Proof](#) The equivalence  $L(\rho) = 0 \iff e^{tL}(\rho) = \rho$  for all  $t \geq 0$  follows from the generator-semigroup correspondence: differentiate at  $t = 0$  for the forward direction, and exponentiate for the reverse. □

**Theorem 20.117** (A rank-deficient fixed density yields an invariant compression). [Lean](#) [✓ Stmt](#) *If  $L$  is a GKSL generator and there exists a rank-deficient fixed density matrix, then the semigroup preserves a nontrivial compression. This is [Wol12, Proposition 7.6, (1) $\Rightarrow$ (3)].*

*Proof.* [✓ Proof](#) Let  $\rho_0$  be a rank-deficient fixed density. Let  $P$  be the support projection of  $\rho_0$ . Since  $\rho_0$  is rank-deficient,  $P \neq \mathbb{1}$ ; since  $\rho_0 \neq 0$ ,  $P \neq 0$ . Channel positivity and trace preservation force  $e^{tL}$  to preserve  $PM_D(\mathbb{C})P$ .  $\square$

**Theorem 20.118** (Invariant compression yields block-upper-triangular Lindblad data). [Lean](#) [✓ Stmt](#) *If a GKSL generator preserves a nontrivial compression, then its Lindblad data can be chosen block-upper-triangular with respect to that compression.*

*Proof.* [✓ Proof](#) First pass from semigroup invariance to generator invariance by Theorem 20.113. The generator-level vanishing condition forces  $(1 - P)L_j P = 0$  and  $(1 - P)\kappa P = 0$  through the sum-of-squares identity.  $\square$

**Theorem 20.119** (Block-upper-triangular Lindblad data yield invariant compression). [Lean](#) [✓ Stmt](#) *If the Lindblad data are block-upper-triangular with respect to a nontrivial projection, then the associated semigroup preserves that compression.*

*Proof.* [✓ Proof](#) The block-upper-triangular relations imply generator-level compression invariance; then apply Theorem 20.114.  $\square$

**Theorem 20.120** (Invariant compression and triangular Lindblad data). [Lean](#) [Lean](#) [✓ Stmt](#) *For a GKSL generator  $L$ , the semigroup preserves a nontrivial compression if and only if the Lindblad data can be chosen block-upper-triangular. This is [Wol12, Proposition 7.6, (3) $\Leftrightarrow$ (4)].*

*Proof.* [✓ Proof](#) Combine Theorems 20.119 and 20.118.  $\square$

**Theorem 20.121** (Triangular Lindblad data give deficient kernels). [Lean](#) [✓ Stmt](#) *If the Lindblad data of a GKSL generator are block-upper-triangular with respect to some nontrivial projector  $P$ , then there exists a density matrix  $\rho_0$  with nontrivial kernel satisfying  $L(\rho_0) = 0$ . This is [Wol12, Proposition 7.6, (4) $\Rightarrow$ (2)].*

*Proof.* [✓ Proof](#) The block-upper-triangular structure implies  $L$  preserves the compression  $PM_d(\mathbb{C})P$ . Therefore the semigroup  $e^{tL}$  also preserves  $PM_d(\mathbb{C})P$  for all  $t \geq 0$ . By the Brouwer fixed-point theorem,  $e^{(1/m)L}$  has a fixed density matrix  $\rho_m$  supported in  $PM_d(\mathbb{C})P$  for each  $m$ . Passing to a subsequential limit gives a density matrix  $\rho_0$  with  $P\rho_0 P = \rho_0$  and  $L(\rho_0) = 0$ . Since  $P \neq \mathbb{1}$ , the matrix  $\rho_0$  has nontrivial kernel.  $\square$

**Theorem 20.122** (Equivalent formulations of reducibility). [Lean](#) [✓ Stmt](#) *For a GKSL generator  $L : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ , the four conditions*

1. rank-deficient fixed density,
2. rank-deficient kernel element,
3. invariant compression, and
4. block-upper-triangular Lindblad form

*are equivalent. This is [Wol12, Proposition 7.6].*

*Proof.* [✓ Proof](#) The cycle  $(1) \leftrightarrow (2)$ ,  $(1) \rightarrow (3)$ ,  $(3) \leftrightarrow (4)$ , and  $(4) \rightarrow (2)$  closes the equivalence chain.  $\square$

### 20.6.1 Sufficient conditions for non-reducibility

**Definition 20.123** (Lindblad span). [Lean](#) [✓ Stmt](#) The  $\mathbb{C}$ -linear span of the Lindblad operators  $\{L_j\}$ .

**Definition 20.124** (Hermitian-closed Lindblad span). [Lean](#) [✓ Stmt](#) The span of the Lindblad operators is closed under Hermitian conjugation, i.e. it forms a  $*$ -subspace of  $M_d(\mathbb{C})$ .

**Definition 20.125** (Trivial Lindblad commutant). [Lean](#) [✓ Stmt](#) The commutant of the Lindblad family  $\{L_j\}'$  equals  $\mathbb{C} \cdot \mathbb{1}$ , i.e. the Lindblad operators act irreducibly on  $M_d(\mathbb{C})$ .

**Definition 20.126** (Kossakowski rank). [Lean](#) [✓ Stmt](#) The minimal number of Lindblad operators across all GKSL representations of  $L$ , which equals the rank of the Kossakowski matrix in the orthonormal-basis representation.

**Theorem 20.127** ( $\kappa$  block vanishing from generator compression). [Lean](#) [✓ Stmt](#) Let  $(H, \{L_j\})$  be a Lindblad form with  $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$ , and let  $P$  be an orthogonal projection such that the generator preserves the compression  $PM_d(\mathbb{C})P$ . If  $(1 - P)L_jP = 0$  for every  $j$ , then  $(1 - P)\kappa P = 0$ .

*Proof.* [✓ Proof](#) From  $L = \varphi - \kappa(\cdot) - (\cdot)\kappa^\dagger$  and  $(1 - P)L(P) = 0$ : the  $(1 - P)\varphi(P)$  term vanishes because each  $(1 - P)L_jP = 0$ , and the  $(1 - P)(P\kappa^\dagger)$  term vanishes because  $(1 - P)P = 0$ . What remains is  $(1 - P)\kappa P = 0$ .  $\square$

**Theorem 20.128** (Full algebra generation forbids block-upper-triangular form). [Lean](#) [✓ Stmt](#) Let  $(H, \{L_j\})$  be a Lindblad form. If the algebra generated by  $\{L_j\}$  and  $\kappa$  is the entire matrix algebra  $M_d(\mathbb{C})$ , then the Lindblad data do not admit a block-upper-triangular decomposition.

*Proof.* [✓ Proof](#) Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection  $P$ . By Theorem 20.127,  $(1 - P)\kappa P = 0$ . Induction on elements of the subalgebra shows  $(1 - P)AP = 0$  for every generator. Since the generated algebra is all of  $M_d(\mathbb{C})$ , every matrix satisfies  $(1 - P)AP = 0$ , forcing  $P = 0$  or  $P = \mathbb{1}$  — a contradiction.  $\square$

**Theorem 20.129** (Hermitian span with trivial commutant forbids triangular form). [Lean](#) [✓ Stmt](#) If the Lindblad span is closed under Hermitian conjugation and its commutant is  $\mathbb{C} \cdot \mathbb{1}$ , then the Lindblad data do not admit a block-upper-triangular decomposition.

*Proof.* [✓ Proof](#) Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection  $P$ . The block vanishing  $(1 - P)L_jP = 0$  and Hermitian closure give  $PL_j(1 - P) = 0$  as well, so  $[P, L_j] = 0$  for all  $j$ . But then  $P$  lies in the commutant of the Lindblad span, contradicting triviality.  $\square$

**Theorem 20.130** (Large Kossakowski rank forbids block-upper-triangular form). [Lean](#) [✓ Stmt](#) If the Kossakowski rank satisfies

$$\text{rank}(C) + d \geq d^2 + 1,$$

then no block-upper-triangular Lindblad form exists.

*Proof.* [✓ Proof](#) A block-upper-triangular traceless Lindblad family lies in a proper subspace whose dimension is at most  $d^2 - d$ . The formal rank minimization argument then contradicts the large-rank bound.  $\square$

**Theorem 20.131** (No block-upper-triangular Lindblad form implies non-reducibility). [Lean](#) [✓ Stmt](#) *If a GKSL generator admits no block-upper-triangular Lindblad form, then the associated quantum dynamical semigroup is not reducible.*

*Proof.* [✓ Proof](#) A reducible semigroup would have an invariant compression, and Theorem 20.118 would then produce a block-upper-triangular Lindblad form.  $\square$

**Theorem 20.132** (Full algebra generation implies non-reducibility). [Lean](#) [✓ Stmt](#) *If the algebra generated by  $\{L_j\}$  and  $\kappa$  is the entire matrix algebra  $M_d(\mathbb{C})$ , then the QDS is not reducible. This is [Wol12, Corollary 7.2(1)].*

*Proof.* [✓ Proof](#) By Theorem 20.128, no block-upper-triangular decomposition exists. Then apply Theorem 20.131.  $\square$

**Theorem 20.133** (Hermitian span and trivial commutant imply non-reducibility). [Lean](#) [✓ Stmt](#) *If the Lindblad span is Hermitian-closed and its commutant is trivial, then the QDS is not reducible. This is [Wol12, Corollary 7.2(2)].*

*Proof.* [✓ Proof](#) By Theorem 20.129, no block-upper-triangular decomposition exists. Then apply Theorem 20.131.  $\square$

**Theorem 20.134** (Large Kossakowski rank implies non-reducibility). [Lean](#) [✓ Stmt](#) *If  $\text{rank}(C) > d^2 - d$ , then the QDS is not reducible. This is [Wol12, Corollary 7.2(3)].*

*Proof.* [✓ Proof](#) The rank hypothesis rules out block-upper-triangular Lindblad data by Theorem 20.130. Then apply Theorem 20.131.  $\square$

## Chapter 21

# Operator Convexity and Jensen Inequalities

This chapter collects the operator convexity/concavity results, trace inequalities, the operator Jensen inequality for positive maps, and the Lieb concavity theorem. These are standard results in matrix analysis ([Bha97, Chapter V]; [Wol12, Theorems 5.12 and 5.13]; Hansen–Pedersen’s Jensen inequality [HP82]; and Lieb’s concavity theorem [Lie73]). They are provided here as cited inputs, since the corresponding matrix-analysis proofs for real powers  $x \mapsto x^p$  on the Loewner order and for the matrix logarithm are not developed elsewhere in this document.

### 21.1 Diagonal Jensen inequality

**Lemma 21.1** (Diagonal Jensen inequality). Lean ✓ Stmt *Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be convex on  $[0, \infty)$ , let  $A$  be a positive semidefinite matrix on a finite-dimensional space indexed by  $n$ , and let  $v \in \mathbb{C}^n$  satisfy  $\langle v, v \rangle = 1$ . Then*

$$f(\Re \langle v, Av \rangle) \leq \Re \langle v, f(A)v \rangle,$$

where  $f(A)$  is defined by the Hermitian continuous functional calculus.

*Proof.* ✓ Proof Write  $A = U \operatorname{diag}(\mu) U^*$  by the spectral theorem and set  $w = U^*v$ . Since  $U$  is unitary,  $p_i := |w_i|^2$  satisfies  $\sum_i p_i = \|v\|^2 = 1$ , so  $(p_i)$  is a probability distribution on  $n$ . The eigenvalues  $\mu_i$  lie in  $[0, \infty)$  because  $A$  is positive semidefinite. A direct computation yields  $\Re \langle v, Av \rangle = \sum_i p_i \mu_i$  and  $\Re \langle v, f(A)v \rangle = \sum_i p_i f(\mu_i)$ , so the scalar Jensen inequality applied to the weights  $(p_i)$  and points  $(\mu_i)$  gives the conclusion. See [Bha97, Chapter V].  $\square$

### 21.2 Trace concavity and convexity of matrix powers

**Theorem 21.2** (Trace concavity of real powers). Lean Lean ✓ Stmt *For  $p \in [0, 1]$ , PSD matrices  $A_1, A_2$ , and  $t \in [0, 1]$ :*

$$t \Re \operatorname{tr}(A_1^p) + (1 - t) \Re \operatorname{tr}(A_2^p) \leq \Re \operatorname{tr}((tA_1 + (1 - t)A_2)^p).$$

*Follows from operator concavity of  $x \mapsto x^p$  composed with trace monotonicity on the Loewner order.*

*Proof.* ✓ Proof Let  $A = tA_1 + (1-t)A_2$  and let  $\{\psi_j\}$  with eigenvalues  $\mu_j \geq 0$  be an orthonormal eigenbasis of  $A$ . A direct computation gives  $\Re \operatorname{tr}(A^p) = \sum_j \mu_j^p$  and  $\mu_j = t a_j + (1-t) b_j$  where  $a_j = \Re \langle \psi_j, A_1 \psi_j \rangle$ ,  $b_j = \Re \langle \psi_j, A_2 \psi_j \rangle$ . Scalar concavity of  $x \mapsto x^p$  on  $[0, \infty)$  yields  $\mu_j^p \geq t a_j^p + (1-t) b_j^p$ , and the diagonal Jensen inequality (Lemma 21.1, concave form) gives  $a_j^p \leq \Re \langle \psi_j, A_1^p \psi_j \rangle$  (and similarly for  $b_j$ ). Summing over  $j$  and using  $\sum_j \Re \langle \psi_j, M \psi_j \rangle = \Re \operatorname{tr} M$  yields the conclusion. See [Bha97, Chapter V].  $\square$

**Theorem 21.3** (Trace convexity of real powers). Lean Lean ✓ Stmt For  $p \in [1, 2]$ , PSD matrices  $A_1, A_2$ , and  $t \in [0, 1]$ :

$$\Re \operatorname{tr}((tA_1 + (1-t)A_2)^p) \leq t \Re \operatorname{tr}(A_1^p) + (1-t) \Re \operatorname{tr}(A_2^p).$$

*Follows from operator convexity of  $x \mapsto x^p$  composed with trace monotonicity on the Loewner order.*

*Proof.* ✓ Proof Identical eigenbasis reduction as in Theorem 21.2, with scalar convexity (rather than concavity) of  $x \mapsto x^p$  on  $[0, \infty)$  for  $p \geq 1$  and the convex form of the diagonal Jensen inequality.  $\square$

## 21.3 Operator Jensen inequality for positive maps

**Definition 21.4** (Finite-POVM dilation construction). Lean Lean ✓ Stmt For a finite family of matrices  $\{C_i\}_{i \in \iota}$  and a defect block  $S$ , one forms an isometric dilation whose rows are the adjoints of the matrices  $C_i$  together with the adjoint of  $S$ . One also forms the scalar block-diagonal matrix whose  $\iota$ -blocks carry prescribed weights and whose defect block carries a single scalar.

**Lemma 21.5** (Compression auxiliary for Jensen). Lean ✓ Stmt Let  $Y$  be positive definite and let  $W$  be an isometry. Then

$$(W^* Y W)^{-1} \leq W^* Y^{-1} W. \quad (21.1)$$

*Proof.* ✓ Proof Apply the Schur-complement criterion to the block matrix  $\begin{pmatrix} Y^{-1} & W \\ W^* & W^* Y W \end{pmatrix}$ . The lower-right Schur complement is zero, hence positive semidefinite, and compressing the resulting upper-left Schur complement by  $W$  gives the inequality (21.1).  $\square$

**Lemma 21.6** (Finite-POVM compression identities). Lean Lean Lean Lean Lean Lean ✓ Stmt If the defect block satisfies

$$SS^* = \mathbb{1} - \sum_{i \in \iota} C_i C_i^*, \quad (21.2)$$

then the dilation is an isometry, compressing the weighted scalar block-diagonal matrix gives

$$\sum_{i \in \iota} w_i C_i C_i^* + t SS^*,$$

the defect relation rewrites as

$$\sum_{i \in \iota} C_i C_i^* + SS^* = \mathbb{1},$$

and strict positivity of all weights together with  $t > 0$  implies positive definiteness of the scalar block-diagonal matrix. If all weights are nonzero and the defect scalar is nonzero, the inverse diagonal has reciprocal weights, and compressing this inverse gives

$$\sum_{i \in \iota} w_i^{-1} C_i C_i^* + t^{-1} SS^*.$$

*Proof.* ✓ Proof Expand the block matrix multiplication entrywise. The defect identity (21.2) gives the isometry relation after summing the  $\iota$ -blocks and the defect block, and the weighted compression identity follows from the same expansion with the scalar diagonal inserted. Positive definiteness uses strict positivity of every diagonal entry, while the inverse formula uses the nonvanishing of every diagonal entry.  $\square$

**Lemma 21.7** (Finite-POVM resolvent inequality). Lean ✓ Stmt Let  $w_i \geq 0$  and  $t > 0$ . If the defect block satisfies

$$SS^* = \mathbb{1} - \sum_{i \in \iota} C_i C_i^*,$$

then

$$\left( \sum_{i \in \iota} w_i C_i C_i^* + t \mathbb{1} \right)^{-1} \leq \sum_{i \in \iota} (w_i + t)^{-1} C_i C_i^* + t^{-1} SS^*. \quad (21.3)$$

*Proof.* ✓ Proof Apply the compression inequality to the positive definite scalar block-diagonal matrix with entries  $w_i + t$  and defect entry  $t$ . The compression identities rewrite the compressed matrix as  $\sum_i w_i C_i C_i^* + t \mathbb{1}$  and its inverse compression as the upper bound (21.3).  $\square$

**Theorem 21.8** (Jensen for concave real powers). Lean Not ready For a positive subunital map  $T$ , a positive semidefinite matrix  $A$ , and  $p \in [0, 1]$ :

$$T(A^p) \leq (TA)^p.$$

This is [Wol12, Theorem 5.13] for concave  $f(x) = x^p$ .

**Theorem 21.9** (Map form of Jensen for concave real powers). Lean ✓ Stmt For a positive subunital map  $T$ , a positive semidefinite matrix  $A$ , and  $p \in [0, 1]$ , the inequality  $T(A^p) \leq (TA)^p$  holds.

*Proof.* ✓ Proof Direct application of the axiom.  $\square$

**Theorem 21.10** (Jensen for convex real powers). Lean Not ready For a positive subunital map  $T$ , a positive semidefinite matrix  $A$ , and  $p \in [1, 2]$ :

$$(TA)^p \leq T(A^p).$$

This is [Wol12, Theorem 5.11] for the subunital operator-convex case  $f(x) = x^p$ .

**Theorem 21.11** (Map form of Jensen for convex real powers). Lean ✓ Stmt For a positive subunital map  $T$ , a positive semidefinite matrix  $A$ , and  $p \in [1, 2]$ , the inequality  $(TA)^p \leq T(A^p)$  holds.

*Proof.* ✓ Proof Direct application of the axiom.  $\square$

**Theorem 21.12** (Jensen for concave log). Lean Not ready For a positive unital map  $T$  and positive-definite  $A$ :

$$T(\log A) \leq \log(TA).$$

This is [Wol12, Theorem 5.13] for  $f = \log$ . Requires unitality ( $T(\mathbb{1}) = \mathbb{1}$ ), not merely subunitality.

**Theorem 21.13** (Map form of Jensen for the concave logarithm). Lean ✓ Stmt For a positive unital map  $T$  and a positive-definite matrix  $A$ , the inequality  $T(\log A) \leq \log(TA)$  holds.

*Proof.* ✓ Proof Direct application of the axiom.  $\square$

## 21.4 Lieb concavity theorem

**Theorem 21.14** (Lieb concavity). Lean Not ready For  $s \in [0, 1]$ , any matrix  $K$ , and positive-definite matrices  $A_1, A_2, B_1, B_2$ , the map  $(A, B) \mapsto \text{tr}(K^\dagger A^s K B^{1-s})$  is jointly concave:

$$t \Re \text{tr}(K^\dagger A_1^s K B_1^{1-s}) + (1-t) \Re \text{tr}(K^\dagger A_2^s K B_2^{1-s}) \leq \Re \text{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K (tB_1 + (1-t)B_2)^{1-s}). \quad (21.4)$$

See [Lie73, And79].

**Theorem 21.15** (Map form of Lieb concavity). Lean ✓ Stmt For  $s \in [0, 1]$ , any matrix  $K$ , and positive-definite matrices  $A_1, A_2, B_1, B_2$ , the function  $(A, B) \mapsto \Re \text{tr}(K^\dagger A^s K B^{1-s})$  is jointly concave, satisfying the inequality (21.4) of Theorem 21.14.

*Proof.* ✓ Proof Direct application of the axiom. □

**Corollary 21.16** (Lieb concavity,  $K = \mathbb{1}$  case). Lean ✓ Stmt For  $s \in [0, 1]$ ,  $t \in [0, 1]$ , and positive-definite matrices  $A_1, A_2, B_1, B_2$ , the map  $(A, B) \mapsto \Re \text{tr}(A^s B^{1-s})$  is jointly concave:

$$t \Re \text{tr}(A_1^s B_1^{1-s}) + (1-t) \Re \text{tr}(A_2^s B_2^{1-s}) \leq \Re \text{tr}((tA_1 + (1-t)A_2)^s (tB_1 + (1-t)B_2)^{1-s}).$$

Obtained from Theorem 21.15 by taking  $K = \mathbb{1}$ .

*Proof.* ✓ Proof Specialize Theorem 21.15 at  $K = \mathbb{1}$  and simplify  $\mathbb{1}^\dagger = \mathbb{1}$  together with  $\mathbb{1} \cdot X = X \cdot \mathbb{1} = X$ . □

**Corollary 21.17** (Concavity of Lieb's map in the first argument). Lean ✓ Stmt For  $s \in [0, 1]$ ,  $t \in [0, 1]$ , matrix  $K$ , and positive-definite matrices  $A_1, A_2, B$ , the map  $A \mapsto \Re \text{tr}(K^\dagger A^s K B^{1-s})$  is concave:

$$t \Re \text{tr}(K^\dagger A_1^s K B^{1-s}) + (1-t) \Re \text{tr}(K^\dagger A_2^s K B^{1-s}) \leq \Re \text{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K B^{1-s}).$$

*Proof.* ✓ Proof Specialize Theorem 21.15 at  $B_1 = B_2 = B$ , so that the convex combination collapses:  $tB + (1-t)B = B$ . □

**Corollary 21.18** (Concavity of Lieb's map in the second argument). Lean ✓ Stmt For  $s \in [0, 1]$ ,  $t \in [0, 1]$ , matrix  $K$ , and positive-definite matrices  $A, B_1, B_2$ , the map  $B \mapsto \Re \text{tr}(K^\dagger A^s K B^{1-s})$  is concave:

$$t \Re \text{tr}(K^\dagger A^s K B_1^{1-s}) + (1-t) \Re \text{tr}(K^\dagger A^s K B_2^{1-s}) \leq \Re \text{tr}(K^\dagger A^s K (tB_1 + (1-t)B_2)^{1-s}).$$

*Proof.* ✓ Proof Specialize Theorem 21.15 at  $A_1 = A_2 = A$ , so that the convex combination collapses:  $tA + (1-t)A = A$ . □



# Chapter 22

## Quantum Entropy

This chapter develops the von Neumann entropy and its basic properties for finite-dimensional quantum systems.

### 22.1 Von Neumann entropy

**Definition 22.1** (Von Neumann entropy). [Lean](#) [✓ Stmt](#) For a Hermitian matrix  $\rho \in M_D(\mathbb{C})$  with eigenvalues  $\lambda_0, \dots, \lambda_{D-1}$ , the *von Neumann entropy* is

$$S(\rho) = - \sum_{i=0}^{D-1} \lambda_i \log \lambda_i,$$

where  $0 \log 0 := 0$ .

**Theorem 22.2** (Entropy is non-negative). [Lean](#) [✓ Stmt](#) For any density matrix  $\rho$ ,  $S(\rho) \geq 0$ .

*Proof.* [✓ Proof](#) Each eigenvalue  $\lambda_i$  of a density matrix satisfies  $0 \leq \lambda_i \leq 1$ , and  $-x \log x \geq 0$  on  $[0, 1]$ .  $\square$

**Lemma 22.3** (Eigenvalue sum). [Lean](#) [✓ Stmt](#) The eigenvalues of a density matrix sum to 1.

*Proof.* [✓ Proof](#) Follows from  $\text{tr}(\rho) = \sum_i \lambda_i = 1$ .  $\square$

**Theorem 22.4** (Eigenvalue bound). [Lean](#) [✓ Stmt](#) Each eigenvalue of a density matrix lies in  $[0, 1]$ .

*Proof.* [✓ Proof](#) Non-negativity comes from positive semidefiniteness. The upper bound follows because the eigenvalues are non-negative and sum to 1.  $\square$

**Theorem 22.5** (Entropy upper bound). [Lean](#) [✓ Stmt](#) For a density matrix  $\rho \in M_D(\mathbb{C})$  with  $D \geq 1$ ,

$$S(\rho) \leq \log D.$$

*Proof.* [✓ Proof](#) By Jensen's inequality applied to the concave function  $-x \log x$ , the entropy is maximized when all eigenvalues equal  $1/D$ .  $\square$

**Lemma 22.6** (Entropy from characteristic polynomial roots). [Lean](#) [✓ Stmt](#) *The von Neumann entropy of a Hermitian matrix is the  $-x \log x$  sum over the real parts of the roots of its characteristic polynomial  $\chi_\rho$ :*

$$S(\rho) = \sum_{\lambda \in \text{roots}(\chi_\rho)} (-\text{Re}(\lambda) \log \text{Re}(\lambda)).$$

*Proof.* [✓ Proof](#) The eigenvalues of a Hermitian matrix are exactly the roots of its characteristic polynomial, counted with multiplicity, so the eigenvalue sum defining  $S(\rho)$  equals the displayed sum over roots.  $\square$

**Theorem 22.7** (Entropy is invariant under reindexing). [Lean](#) [✓ Stmt](#) *Let  $\rho$  be a Hermitian matrix indexed by a finite set  $J$ , and let  $e : I \rightarrow J$  be a bijection from a finite set  $I$ . The reindexed matrix on  $I$  with entries  $\rho_{e(i) e(j)}$  has the same von Neumann entropy as  $\rho$ .*

*Proof.* [✓ Proof](#) By Lemma 22.6 the entropy depends only on the characteristic polynomial. Reindexing conjugates  $\rho$  by a permutation matrix, so

$$\chi_{(\rho_{e(i) e(j)})} = \chi_\rho,$$

and the entropies coincide.  $\square$

**Theorem 22.8** (Entropy of  $AB$  equals entropy of  $BA$ ). [Lean](#) [✓ Stmt](#) *For matrices  $A \in M_{m \times n}(\mathbb{C})$  and  $B \in M_{n \times m}(\mathbb{C})$  such that  $AB$  and  $BA$  are Hermitian,*

$$S(AB) = S(BA).$$

*Proof.* [✓ Proof](#) The rectangular characteristic-polynomial identity gives

$$X^n \chi_{AB} = X^m \chi_{BA}.$$

Hence  $AB$  and  $BA$  have the same nonzero eigenvalues, with multiplicity. The roots at 0 contribute nothing because  $0 \log 0 = 0$ , so Lemma 22.6 gives  $S(AB) = S(BA)$ .  $\square$

## 22.2 Tripartite partial traces

**Definition 22.9** (Partial trace over  $A$ ). [Lean](#) [✓ Stmt](#) For a tripartite matrix  $\rho_{ABC}$  on  $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \otimes \mathbb{C}^{d_C}$ , the partial trace over  $A$  is

$$(\text{tr}_A \rho_{ABC})_{(b_1, c_1)(b_2, c_2)} = \sum_{a=0}^{d_A-1} (\rho_{ABC})_{(a, b_1, c_1)(a, b_2, c_2)}.$$

**Definition 22.10** (Partial trace over  $C$ ). [Lean](#) [✓ Stmt](#) The partial trace over  $C$ :

$$(\text{tr}_C \rho_{ABC})_{(a_1, b_1)(a_2, b_2)} = \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a_1, b_1, c)(a_2, b_2, c)}.$$

**Definition 22.11** (Partial trace over  $AC$ ). [Lean](#) [✓ Stmt](#) The partial trace over  $A$  and  $C$ :

$$(\text{tr}_{AC} \rho_{ABC})_{b_1 b_2} = \sum_{a=0}^{d_A-1} \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a, b_1, c)(a, b_2, c)}.$$

**Lemma 22.12** (Partial trace preserves Hermiticity). [Lean](#) [✓ Stmt](#) If  $\rho_{ABC}$  is Hermitian, then  $\text{tr}_A(\rho_{ABC})$ ,  $\text{tr}_C(\rho_{ABC})$ , and  $\text{tr}_{AC}(\rho_{ABC})$  are all Hermitian. The same holds for bipartite partial traces  $\text{tr}_A(\rho_{AB})$  and  $\text{tr}_B(\rho_{AB})$ .

*Proof.* [✓ Proof](#) Follows from  $\overline{\rho_{ji}} = \rho_{ij}$  applied entry-wise inside the summation defining each partial trace.  $\square$

## 22.3 Strong subadditivity

**Theorem 22.13** (Strong subadditivity). [Lean](#) [Not ready](#) For a tripartite density matrix  $\rho_{ABC}$ ,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

**Definition 22.14** (SSA equality). [Lean](#) [✓ Stmt](#) A tripartite density matrix  $\rho_{ABC}$  satisfies SSA equality if

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

**Definition 22.15** (Quantum Markov decomposition). [Lean](#) [✓ Stmt](#) A Hayashi Markov decomposition of a tripartite state  $\rho_{ABC}$  consists of a finite direct-sum decomposition

$$H_B \cong \bigoplus_j H_{B_j^L} \otimes H_{B_j^R},$$

together with a unitary change of basis on  $B$ , a probability vector  $(p_j)_j$ , and density matrices  $\rho_{AB_j^L}$  and  $\rho_{B_j^R C}$  such that, in the adapted basis, the state becomes

$$\bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R C}.$$

The terminology follows Hayashi’s presentation of quantum Markov structure [\[Hay06\]](#); the block decomposition used by the MPDO argument is the structure theorem of [\[HJPW04\]](#).

**Theorem 22.16** (Hayashi SSA-equality characterization). [Lean](#) [Not ready](#) For a tripartite density matrix  $\rho_{ABC}$ ,

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$$

holds if and only if  $\rho_{ABC}$  admits a quantum Markov decomposition on the middle subsystem  $B$ .

This equality criterion is cited here as an input for the later MPDO arguments. It is the equality criterion needed by Appendix C of [arXiv:1606.00608](#). The equality conditions are reviewed in [\[Hay06, Rus02\]](#); the structural block-decomposition formulation used for MPDOs appears in [\[HJPW04\]](#).

## 22.4 Mutual information

**Definition 22.17** (Mutual information). [Lean](#) [✓ Stmt](#) The quantum mutual information of a bipartite state  $\rho_{AB}$  is

$$I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

**Theorem 22.18** (Mutual information is non-negative). [Lean](#) [✓ Stmt](#) For any bipartite density matrix  $\rho_{AB}$ ,  $I(A:B) \geq 0$ .

*Proof.* [✓ Proof](#) Apply strong subadditivity (Theorem [22.13](#)) with trivial  $B$  (one-dimensional middle system). The SSA inequality  $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$  reduces to subadditivity  $S(\rho_{AC}) \leq S(\rho_A) + S(\rho_C)$ , which gives  $I(A:C) \geq 0$ .  $\square$

## 22.5 Entropy formulations

This section states the entropy formulations used later in the development: von Neumann entropy, strong subadditivity, quantum Markov decomposition, and mutual information. These statements are cited from the standard entropy literature and supply the entropy-theoretic input for the later MPDO arguments.

**Definition 22.19** (Entropy formulation of von Neumann entropy). [Lean](#) [✓ Stmt](#) This formulation has the same value  $S(\rho) = -\sum_i \lambda_i \log \lambda_i$  as in Definition 22.1.

**Theorem 22.20** (Entropy formulation of strong subadditivity). [Lean](#) [✓ Stmt](#) For any tripartite density matrix  $\rho_{ABC}$  on  $A \otimes B \otimes C$ ,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

*This formulation introduces no new axiom: it is the same strong-subadditivity statement as Theorem 22.13. The underlying axiom's proof in [LR73] is deferred to the umbrella entropy issue.*

**Definition 22.21** (Entropy formulation of quantum Markov decomposition). [Lean](#) [✓ Stmt](#) This is the entropy formulation of the Hayashi Markov decomposition from Definition 22.15.

**Theorem 22.22** (SSA equality iff quantum Markov decomposition). [Lean](#) [✓ Stmt](#) For any tripartite density matrix  $\rho_{ABC}$ , equality in strong subadditivity holds if and only if  $\rho_{ABC}$  admits a quantum Markov decomposition on the middle subsystem  $B$ .

*This formulation introduces no new axiom: it is the same equality criterion as Theorem 22.16.*

**Definition 22.23** (Entropy formulation of mutual information). [Lean](#) [✓ Stmt](#) This formulation has the same value  $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$  as in Definition 22.17.

**Theorem 22.24** (Subadditivity from trivial-middle SSA). [Lean](#) [✓ Stmt](#) For a tripartite density matrix  $\rho_{ABC}$  with  $\dim B = 1$ ,

$$S(\rho_{ABC}) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

*Proof.* [✓ Proof](#) Apply Theorem 22.20 with one-dimensional middle subsystem. The reduced middle state has trace 1 by Theorem 22.30, hence its entropy vanishes by Theorem 22.31.  $\square$

## 22.6 Mutual information: monotonicity and area-law bound

The two inequalities below are the downstream MPDO-facing consequences of the entropy inequalities in this chapter. The monotonicity inequality is the strong-subadditivity content underlying the MPDO mutual-information monotonicity  $I_L \leq I_{L+1}$  (arXiv:1606.00608, Proposition C.1); the elementary area-law bound is the single-site entropy bound underlying the MPDO area-law bound  $I_L \leq 4 \log D$  (arXiv:1606.00608, cited from the Wolf area-law bound). Both inequalities follow directly from strong subadditivity (taken as an axiom) and the single-system  $S(\rho) \leq \log D$  bound; neither introduces a new axiom.

**Theorem 22.25** (Entropy nonnegativity for finite index sets). [Lean](#) [✓ Stmt](#) For any PSD Hermitian matrix  $\rho$  with  $\text{tr}(\rho) = 1$  on an arbitrary finite index set,  $S(\rho) \geq 0$ . This is the analog of Theorem 22.2 for arbitrary finite index sets: it does not require the index set to be  $\{0, \dots, D-1\}$ , so it applies to bipartite density matrices on  $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ .

*Proof.* ✓ Proof The eigenvalues of a PSD matrix are non-negative, and eigenvalues of a trace-1 Hermitian matrix with non-negative eigenvalues are bounded above by 1 (each single eigenvalue is at most the total sum). Apply  $x \log x \leq 0$  on  $[0, 1]$  pointwise and sum.  $\square$

**Theorem 22.26** (Mutual information is monotone under enlargement). Lean ✓ Stmt For any tripartite density matrix  $\rho_{ABC}$  on  $A \otimes B \otimes C$ ,

$$I(A:B)_{\text{tr}_C \rho_{ABC}} \leq I(A:BC)_{\rho_{ABC}},$$

where the left-hand side is the bipartite mutual information of the reduced state  $\rho_{AB} = \text{tr}_C(\rho_{ABC})$  and the right-hand side is evaluated by expanding  $I(A:BC) = S(\rho_A) + S(\rho_{BC}) - S(\rho_{ABC})$ .

*Proof.* ✓ Proof Expanding both sides in entropy form and cancelling the common  $S(\rho_A)$  term, the inequality reduces to strong subadditivity  $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$ . The bipartite  $B$ -reduced state of  $\rho_{AB}$  matches the tripartite  $B$ -reduced state  $\text{tr}_{AC}(\rho_{ABC})$ , ensuring the  $S(\rho_B)$  term is the one supplied by Theorem 22.20.  $\square$

**Theorem 22.27** (Elementary area-law bound on mutual information). Lean ✓ Stmt For a bipartite density matrix  $\rho_{AB}$  on  $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$  with  $d_A, d_B \geq 1$  whose single-system reduced states are obtained by partial trace,

$$I(A:B) \leq \log d_A + \log d_B.$$

*Proof.* ✓ Proof The reduced states are density matrices because partial trace preserves positivity and trace. Combine  $S(\rho_A) \leq \log d_A$  and  $S(\rho_B) \leq \log d_B$  (Theorem 22.5) with  $S(\rho_{AB}) \geq 0$  (Theorem 22.25).  $\square$

## 22.7 Trivial-factor corollaries

The partial traces and von Neumann entropy are introduced in Chapter 22. This supplement collects elementary trace-preservation identities and the dimension-1 entropy bound that follow directly from those definitions. They are listed as separate results because each proof requires only unfolding definitions and re-indexing finite sums.

**Theorem 22.28** (Partial trace over  $A$  preserves the full trace). Lean ✓ Stmt For any tripartite matrix  $\rho_{ABC}$ ,

$$\text{tr}(\rho_{ABC}) = \text{tr}(\text{tr}_A(\rho_{ABC})).$$

*Proof.* ✓ Proof Unfolding the definitions,

$$\text{tr}(\text{tr}_A(\rho_{ABC})) = \sum_{b,c} \sum_a (\rho_{ABC})_{(a,b,c)(a,b,c)} = \text{tr}(\rho_{ABC}).$$

$\square$

**Theorem 22.29** (Partial trace over  $C$  preserves the full trace). Lean ✓ Stmt For any tripartite matrix  $\rho_{ABC}$ ,

$$\text{tr}(\rho_{ABC}) = \text{tr}(\text{tr}_C(\rho_{ABC})).$$

*Proof.* ✓ Proof Unfolding the definitions,

$$\text{tr}(\text{tr}_C(\rho_{ABC})) = \sum_{a,b} \sum_c (\rho_{ABC})_{(a,b,c)(a,b,c)} = \text{tr}(\rho_{ABC}).$$

$\square$

**Theorem 22.30** (Partial trace over  $AC$  preserves the full trace). Lean ✓ Stmt For any tripartite matrix  $\rho_{ABC}$ ,

$$\mathrm{tr}(\rho_{ABC}) = \mathrm{tr}(\mathrm{tr}_{AC}(\rho_{ABC})).$$

*Proof.* ✓ Proof Unfolding the definitions and interchanging the outer sums,

$$\mathrm{tr}(\mathrm{tr}_{AC}(\rho_{ABC})) = \sum_b \sum_{a,c} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \mathrm{tr}(\rho_{ABC}).$$

□

**Theorem 22.31** (Entropy vanishes in dimension 1). Lean ✓ Stmt If  $\rho \in M_1(\mathbb{C})$  is Hermitian and  $\mathrm{tr}(\rho) = 1$ , then

$$S(\rho) = 0.$$

*Proof.* ✓ Proof The unique eigenvalue equals the trace, hence equals 1, and  $-1 \cdot \log 1 = 0$ :

$$\lambda_0 = \mathrm{tr}(\rho) = 1, \quad S(\rho) = -\lambda_0 \log \lambda_0 = -1 \cdot \log 1 = 0.$$

□

## Chapter 23

# Matrix Product Operators and Density Operators

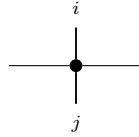
Fix a physical dimension  $d$  and a bond dimension  $D$ . An MPO tensor carries two physical indices, one ket index and one bra index, together with the virtual left and right indices.

### 23.1 MPO tensors

**Definition 23.1** (MPO tensor). [Lean](#) [✓ Stmt](#) An *MPO tensor* is a family of matrices

$$\{M^{ij}\}_{i,j=0}^{d-1}, \quad M^{ij} \in M_D(\mathbb{C}),$$

indexed by a ket index  $i$  and a bra index  $j$ . Diagrammatically,



The black node denotes the tensor  $M$ , the horizontal legs are virtual, the upper leg is the ket index  $i$ , and the lower leg is the bra index  $j$ .

**Definition 23.2** (Doubled-index MPS view). [Lean](#) [✓ Stmt](#) By combining the pair  $(i, j)$  into a single physical index in  $\{0, \dots, d^2 - 1\}$ , one obtains an MPS tensor with physical dimension  $d^2$  and bond dimension  $D$ :

$$M_{(i,j)}^{\text{MPS}} = M^{ij},$$

identifying  $\{0, \dots, d-1\} \times \{0, \dots, d-1\}$  with  $\{0, \dots, d^2-1\}$  via the product encoding.

**Definition 23.3** (Word evaluation). [Lean](#) [✓ Stmt](#) For words  $u = (i_1, \dots, i_N)$  and  $v = (j_1, \dots, j_N)$  in  $\{0, \dots, d-1\}^N$ , the *word evaluation* is

$$M^{u,v} := M^{i_1 j_1} M^{i_2 j_2} \dots M^{i_N j_N} \in M_D(\mathbb{C}).$$

The empty pair of words evaluates to  $\mathbb{1}_D$ , and word pairs of different lengths evaluate to 0.

**Lemma 23.4** (Word-evaluation factorization). [Lean](#) [✓ Stmt](#) For words  $u_1, v_1$  of equal length and arbitrary words  $u_2, v_2$ , word evaluation factors as

$$M^{u_1 u_2, v_1 v_2} = M^{u_1, v_1} M^{u_2, v_2}.$$

*Proof.* [✓ Proof](#) We argue by induction on the common length of  $u_1$  and  $v_1$ . For the empty prefix,

$$M^{u_2, v_2} = \mathbb{1}_D M^{u_2, v_2}.$$

If  $u_1 = iu$  and  $v_1 = jv$ , then

$$M^{iuu_2, jvv_2} = M^{ij} M^{uu_2, vv_2} = M^{ij} M^{u, v} M^{u_2, v_2} = M^{iu, jv} M^{u_2, v_2},$$

using associativity of matrix multiplication.  $\square$

**Lemma 23.5** (Trace cyclicity of the closed MPO word). [Lean](#) [✓ Stmt](#) Let  $a, b \in \{0, \dots, d-1\}$ , and let  $u, v$  be words of equal length. Writing  $au, bv$  for adding the letters at the beginning and  $ua, vb$  for adding them at the end,

$$\text{tr}(M^{au, bv}) = \text{tr}(M^{ua, vb}).$$

*Proof.* [✓ Proof](#) The factorization lemma gives

$$M^{ua, vb} = M^{u, v} M^{ab}, \quad M^{au, bv} = M^{ab} M^{u, v}.$$

Hence

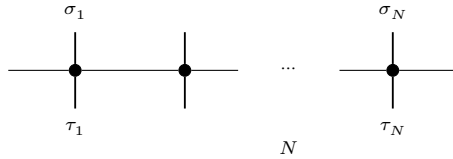
$$\text{tr}(M^{ua, vb}) = \text{tr}(M^{u, v} M^{ab}) = \text{tr}(M^{ab} M^{u, v}) = \text{tr}(M^{au, bv}),$$

by cyclicity of the trace.  $\square$

**Definition 23.6** (MPO operator family). [Lean](#) [✓ Stmt](#) The operator generated by  $M$  on  $N$  sites is the matrix  $\rho^{(N)}(M)$  indexed by configurations  $\sigma, \tau \in \{0, \dots, d-1\}^N$  and given by

$$\rho^{(N)}(M)_{\sigma, \tau} = \text{tr}(M^{\sigma, \tau}).$$

Diagrammatically,



each black node denotes the same local tensor  $M$ , and the matrix entry  $\rho^{(N)}(M)_{\sigma, \tau}$  is obtained by closing the outer virtual legs cyclically and taking the resulting trace.

**Definition 23.7** (Cyclic-shift reindexing). [Lean](#) [✓ Stmt](#) The *cyclic shift* of configurations on  $N$  sites is the bijection that relabels each site by one position,  $\sigma \mapsto \sigma \circ r$ , where  $r$  is the cyclic rotation  $i \mapsto i + 1$  of  $\{0, \dots, N-1\}$ .

**Theorem 23.8** (Translation invariance of the periodic MPDO). [Lean](#) [✓ Stmt](#) The operator  $\rho^{(N)}(M)$  is invariant under the simultaneous cyclic shift of its bra and ket configurations:

$$\rho^{(N)}(M)_{r\sigma, r\tau} = \rho^{(N)}(M)_{\sigma, \tau}.$$



*Proof.* ✓ Proof The entry is a closed bond trace,  $\rho^{(N)}(M)_{\sigma, \tau} = \text{tr}(M^{\sigma_1 \tau_1} \dots M^{\sigma_N \tau_N})$ . Shifting both configurations by  $r$  rotates the matrix product by one factor, and the trace is invariant under that cyclic rotation,

$$\text{tr}(M^{\sigma_2 \tau_2} \dots M^{\sigma_N \tau_N} M^{\sigma_1 \tau_1}) = \text{tr}(M^{\sigma_1 \tau_1} \dots M^{\sigma_N \tau_N}),$$

so  $\rho^{(N)}(M)_{r\sigma, r\tau} = \rho^{(N)}(M)_{\sigma, \tau}$ . □

**Definition 23.9** (Hermitian MPO tensor). Lean ✓ Stmt An MPO tensor is *Hermitian* if, for all  $i, j \in \{0, \dots, d-1\}$ ,

$$M^{ij} = (M^{ji})^\dagger$$

## 23.2 Transfer maps

**Definition 23.10** (MPO transfer map). Lean ✓ Stmt The *transfer map* associated to  $M$  is the linear map  $\mathcal{E}_M : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  defined by

$$\mathcal{E}_M(X) = \sum_{i,j=0}^{d-1} M^{ij} X (M^{ij})^\dagger.$$

**Lemma 23.11** (Identification with the doubled-index MPS transfer map). Lean ✓ Stmt The transfer map of an MPO tensor agrees with the transfer map of its doubled-index MPS view.

*Proof.* ✓ Proof After identifying the single physical index of the doubled tensor with a pair  $(i, j)$ , the single sum defining the MPS transfer map is exactly the double sum defining  $\mathcal{E}_M$ . □

**Theorem 23.12** (Positivity of the MPO transfer map). Lean ✓ Stmt If  $X \geq 0$ , then  $\mathcal{E}_M(X) \geq 0$ .

*Proof.* ✓ Proof By Lemma 23.11,  $\mathcal{E}_M$  is the transfer map of the doubled-index MPS tensor. The corresponding MPS transfer map is positive, so  $\mathcal{E}_M(X) \geq 0$  whenever  $X \geq 0$ . □

## 23.3 Zero correlation length

**Definition 23.13** (Zero correlation length for MPO tensors). Lean ✓ Stmt An MPO tensor  $M$  has *zero correlation length* when its transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

This is the mixed-state analog of the renormalization fixed-point condition for MPS tensors; see [CPGSV16, definition of MPDO zero correlation length, lines 736–741] and [CPGSV21, Section II.E.2, lines 937–939].

**Theorem 23.14** (MPO ZCL iff doubled-index MPS RFP). Lean ✓ Stmt An MPO tensor has zero correlation length if and only if its doubled-index MPS view is a renormalization fixed point.

*Proof.* ✓ Proof By Lemma 23.11, the transfer map of  $M$  agrees with the transfer map of its doubled-index MPS view. Thus  $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$  holds exactly when the transfer map of that doubled-index MPS tensor is idempotent, which is the RFP condition. □

## 23.4 MPDOs and LPDOs

**Definition 23.15** (MPDO). [Lean](#) [✓ Stmt](#) An MPO tensor is an *MPDO* if, for every  $N \in \mathbb{N}$ ,

$$\rho^{(N)}(M) \geq 0$$

This is the global positivity condition of [CPGSV16, Section 4].

**Definition 23.16** (LPDO). [Lean](#) [✓ Stmt](#) An MPO tensor is an *LPDO* if there exist a Kraus dimension  $d_K$ , an inner bond dimension  $D'$ , a bond-space identification  $e : \{0, \dots, D-1\} \simeq \{0, \dots, D'-1\} \times \{0, \dots, D'-1\}$  (equivalently,  $D$  and  $(D')^2$  have the same cardinality), and matrices  $A^{(i,k)} \in M_{D'}(\mathbb{C})$  such that, for all physical indices  $i, j$ ,

$$M^{ij} = \left( \sum_{k=0}^{d_K-1} A^{(i,k)} \otimes \overline{A^{(j,k)}} \right)_{e,e}$$

where  $\otimes$  denotes the Kronecker product,  $\overline{(\cdot)}$  denotes entrywise complex conjugation, and  $(\cdot)_{e,e}$  denotes reindexing along  $e$  back to  $\{0, \dots, D-1\}$ . See [CPGSV16, Section 4.3].

**Theorem 23.17** (LPDOs are MPDOs). [Lean](#) [✓ Stmt](#) Every LPDO tensor is an MPDO.

*Proof.* [✓ Proof](#) By the mixed-product property  $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ , the  $N$ -site product decomposes as a sum of Kronecker products after transporting along the bond-space identification  $e$ :

$$M^{\sigma_1 \tau_1} \dots M^{\sigma_N \tau_N} = \left( \sum_{\kappa} P_{\kappa} \otimes \overline{Q_{\kappa}} \right) \Big|_e,$$

where  $P_{\kappa} = A^{(\sigma_1, \kappa_1)} \dots A^{(\sigma_N, \kappa_N)}$  and  $Q_{\kappa} = A^{(\tau_1, \kappa_1)} \dots A^{(\tau_N, \kappa_N)}$ . Since the trace is invariant under the reindexing by  $e$ , we apply  $\text{tr}(X \otimes Y) = \text{tr}(X) \text{tr}(Y)$  to obtain

$$\rho^{(N)}(M)_{\sigma, \tau} = \sum_{\kappa} \text{tr}(P_{\kappa}) \overline{\text{tr}(Q_{\kappa})} = \sum_{\kappa} \psi_{\kappa}(\sigma) \overline{\psi_{\kappa}(\tau)},$$

where  $\psi_{\kappa}(\sigma) = \text{tr}(P_{\kappa})$ . Hence  $\rho^{(N)}(M) = \sum_{\kappa} |\psi_{\kappa}\rangle \langle \psi_{\kappa}| \geq 0$ . □

**Definition 23.18** (Local purification RFP condition). [Lean](#) [✓ Stmt](#) An MPO tensor  $M$  satisfies the local purification RFP condition if it is an LPDO whose purifying tensor  $A$ , viewed as a matrix product state tensor on the combined spin–ancilla index, is a pure-state renormalization fixed point. Thus

$$M^{ij} = \left( \sum_k A^{(i,k)} \otimes \overline{A^{(j,k)}} \right)_{e,e}.$$

This local condition is motivated by the purification tensor formula [CPGSV16, lines 744–747], but is weaker than the source purification RFP definition; the latter remains open.

**Theorem 23.19** (A local purification RFP generates MPDOs). [Lean](#) [✓ Stmt](#) Every tensor satisfying the local purification RFP condition generates matrix product density operators.

*Proof.* [✓ Proof](#) Dropping the renormalization-fixed-point condition on the purifying tensor leaves the local purification structure, so  $M$  is an LPDO; the conclusion follows from Theorem 23.17. □

## 23.5 Pure-state recovery inside the MPO formalism

**Definition 23.20** (MPO transfer-map idempotence). [Lean](#) [✓ Stmt](#) An MPO tensor  $M$  has *transfer-map idempotence* if

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

**Definition 23.21** (Trace-preserving completely positive map). [Lean](#) [✓ Stmt](#) A linear map  $\mathcal{S}$  between matrix algebras is *trace-preserving completely positive* if it has a Kraus form  $\mathcal{S}(X) = \sum_i A_i X A_i^\dagger$  with  $\sum_i A_i^\dagger A_i = I$ . The Kraus operators may be rectangular, so the input and output dimensions need not agree.

**Definition 23.22** (One- and two-site physical closure maps). [Lean](#) [Lean](#) [✓ Stmt](#) For an MPO tensor  $M$ , the one- and two-site physical operators obtained by closing one or two tensors with a virtual operator  $X$  are  $M_1(X)_{ij} = \text{tr}(M^{ij}X)$  and  $M_2(X)_{(i_1 i_2)(j_1 j_2)} = \text{tr}(M^{i_1 j_1} M^{i_2 j_2} X)$ , both linear in  $X$ .

**Definition 23.23** (Renormalization fixed point via trace-preserving maps). [Lean](#) [✓ Stmt](#) An MPO tensor  $M$  is a *renormalization fixed point* if there exist two trace-preserving completely positive maps  $\mathcal{S}, \mathcal{T}$  on the physical indices such that  $\mathcal{S}[M_2(X)] = M_1(X)$  and  $\mathcal{T}[M_1(X)] = M_2(X)$  for every virtual operator  $X$ , where  $M_1(X)_{ij} = \text{tr}(M^{ij}X)$  and  $M_2(X)_{(i_1 i_2)(j_1 j_2)} = \text{tr}(M^{i_1 j_1} M^{i_2 j_2} X)$  are the one- and two-site physical operators obtained by closing one or two tensors with  $X$ . This condition is strictly stronger than transfer-map idempotence for general mixed states, and coincides with it for pure states. This is [CPGSV16, Definition 4.1, line 657].

**Theorem 23.24** (Transfer-map idempotence iff MPO ZCL). [Lean](#) [✓ Stmt](#) *MPO transfer-map idempotence is exactly zero correlation length.*

*Proof.* [✓ Proof](#) Both conditions assert  $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$ ; they are definitionally identical.  $\square$

**Theorem 23.25** (Transfer-map idempotence iff doubled-index MPS RFP). [Lean](#) [✓ Stmt](#) *MPO transfer-map idempotence is equivalent to the pure-state RFP condition for the doubled-index MPS tensor.*

*Proof.* [✓ Proof](#) Unfold transfer-map idempotence to its definitional equivalent, MPO ZCL, then apply Theorem 23.14.  $\square$

**Definition 23.26** (Diagonal pure-state embedding as an MPO). [Lean](#) [✓ Stmt](#) An MPS tensor  $A = \{A^i\}_{i=0}^{d-1}$  determines an MPO tensor by placing  $A^i$  on the diagonal in the bra and ket indices:

$$M^{ij} = \delta_{ij} A^i.$$

Equivalently,  $M^{ii} = A^i$  and  $M^{ij} = 0$  for  $i \neq j$ .

**Theorem 23.27** (Transfer map of the diagonal pure-state embedding). [Lean](#) [✓ Stmt](#) *The transfer map of the diagonal MPO associated to  $A$  agrees with the original MPS transfer map:*

$$\mathcal{E}_{A^{\text{MPO}}} = \mathcal{E}_A.$$

*Proof.* [✓ Proof](#) In the double sum defining  $\mathcal{E}_{A^{\text{MPO}}}$ , all off-diagonal terms vanish because  $M^{ij} = 0$  for  $i \neq j$ , while the diagonal terms are exactly  $A^i X (A^i)^\dagger$ . Thus only the sum over  $i$  remains, which is the defining formula for  $\mathcal{E}_A$ .  $\square$

**Theorem 23.28** (Pure-state recovery of transfer-map idempotence). [Lean](#) [✓ Stmt](#) For an MPS tensor  $A$ , the diagonal MPO embedding has idempotent transfer map if and only if  $A$  is an RFP:

$$\mathcal{E}_{\text{AMPO}} \circ \mathcal{E}_{\text{AMPO}} = \mathcal{E}_{\text{AMPO}} \iff A \text{ is RFP.}$$

*Proof.* [✓ Proof](#) Both conditions assert idempotence of the corresponding transfer map. Theorem 23.27 identifies those transfer maps, so the two predicates are equivalent.  $\square$

**Theorem 23.29** (Pure-state recovery of zero correlation length). [Lean](#) [✓ Stmt](#) For an MPS tensor  $A$ , the diagonal MPO embedding has idempotent transfer map if and only if  $A$  has zero correlation length.

*Proof.* [✓ Proof](#) Combine Theorem 23.28 with the characterization of zero correlation length by idempotence of the MPS transfer map (Theorem 17.27).  $\square$

## 23.6 Simple local structure from SAL and ZCL

For the simple MPDO case in Appendix C.2 of [CPGSV16], the strong-area-law hypothesis is used locally through the equality case of strong subadditivity for a normalized three-site reduced state.

**Definition 23.30** (Local  $\eta$ -structure for a three-site reduced state). [Lean](#) [✓ Stmt](#) For a three-site density operator  $\rho_{ABC}$ , the local  $\eta$ -structure is the quantum Markov decomposition on the middle subsystem  $B$  produced by equality in strong subadditivity. Concretely, after a unitary change of basis on  $B$ , one has a finite direct-sum decomposition

$$B = \bigoplus_k (b_1^{(k)} \otimes b_2^{(k)})$$

such that

$$\rho_{ABC} = \bigoplus_k \rho_{Ab_1}^{(k)} \otimes \rho_{b_2C}^{(k)}.$$

**Theorem 23.31** (SAL implies local  $\eta$ -structure). [Lean](#) [✓ Stmt](#) If a normalized three-site reduced state satisfies the equality case of strong subadditivity — the local entropy form of the simple-MPDO strong area law — then it admits the local  $\eta$ -structure of Definition 23.30.

*Proof.* [✓ Proof](#) This is exactly the forward implication of the Hayashi equality characterization: equality in strong subadditivity yields a quantum Markov decomposition on the middle subsystem.  $\square$

**Definition 23.32** (Injective inverse-map layer for a simple MPO tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A simple MPO tensor  $K$  is called injective if its doubled-index MPS tensor is injective. For such a tensor, the chosen decomposition map of the doubled-index MPS furnishes a concrete inverse tensor  $K^{-1}$  together with right- and left-physical realization maps that turn virtual bond insertions back into local physical operators.

**Theorem 23.33** (Inverse-map identities for an injective simple MPO tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For an injective simple MPO tensor, contracting  $K^{-1}$  with  $K$  recovers the matrix units on the virtual bond space. Likewise, every right virtual insertion admits a physical realization, this realization is multiplicative, and the analogous left-insertion statement also holds.

**Definition 23.34** (Explicit neighboring  $\eta$ -operator data). [Lean](#) [Lean](#) [✓ Stmt](#) Fix a Hayashi decomposition witness  $h_\eta$ . An explicit neighboring  $\eta$ -family is a dependent family  $(\eta_{k,h})$  indexed by sector pairs, where  $\eta_{k,h}$  acts on the neighboring bond space  $B_k^R \otimes B_h^L$ . Requiring positivity for each pair gives the corresponding positive  $\eta$ -data.

**Definition 23.35** (Trace matrices of explicit  $\eta$ -data). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Every explicit neighboring  $\eta$ -family determines a sector-by-sector trace matrix. We give both its complex-valued form and the real-part version, which is the direct data for the real Perron–Frobenius matrix  $T$  used in the rank-one step.

**Theorem 23.36** (Positivity of bipartite partial traces). [Lean](#) [Lean](#) [✓ Stmt](#) If  $\rho$  is positive semidefinite on a bipartite space  $A \otimes B$ , then both reduced operators  $\text{tr}_A \rho$  and  $\text{tr}_B \rho$  are positive semidefinite.

*Proof.* [✓ Proof](#) Write each partial trace as the finite sum, over the traced index, of a principal submatrix of  $\rho$ . Principal submatrices of a positive semidefinite matrix are positive semidefinite, and finite sums of positive semidefinite matrices remain positive semidefinite.  $\square$

**Definition 23.37** (Hayashi–Markov extraction of explicit  $\eta$ -operators). [Lean](#) [✓ Stmt](#) Every Hayashi decomposition witness  $h_\eta$  canonically produces an explicit neighboring  $\eta$ -family by Kronecker-multiplying the sector-reduced states on the neighboring bond spaces:

$$\eta_{k,h} := \text{tr}_C(\rho_{b_2C}^{(k)}) \otimes \text{tr}_A(\rho_{Ab_1}^{(h)}).$$

Positive semidefiniteness of each  $\eta_{k,h}$  follows from the fact that partial traces preserve positivity and that positive semidefiniteness is closed under Kronecker products. This is the sector-reduced extraction layer; the remaining connection to Appendix C.2 is the identification of this family with the inverse-map construction in the original simple-MPDO tensor coordinates.

**Theorem 23.38** (Trace matrix of the Hayashi–Markov  $\eta$ -operators). [Lean](#) [Lean](#) [✓ Stmt](#) The trace matrix induced by the extracted family is entrywise  $T_{k,h} = 1$  in both its complex-valued form and its real-valued form, matching the normalization  $T_{k,h} = a_k b_h$  used in  $a_k = b_h = 1$ .

*Proof.* [✓ Proof](#) Each  $\rho_{b_2C}^{(k)}$  is a density matrix, hence has unit trace; the partial trace  $\text{tr}_C$  preserves the trace. The analogous statement holds for  $\text{tr}_A \rho_{Ab_1}^{(h)}$ . Multiplicativity of the trace under Kronecker products gives  $T_{k,h} = 1 \cdot 1 = 1$ , and taking real parts gives the real-valued statement.  $\square$

**Theorem 23.39** (Entrywise nonnegativity of the real trace matrix). [Lean](#) [✓ Stmt](#) Positivity of the neighboring operators gives entrywise nonnegativity of the real trace matrix. Matrix-level positive semidefiniteness of  $T$  is a separate structural condition.

*Proof.* [✓ Proof](#) The trace of a positive semidefinite neighboring operator is non-negative in the complex order. Taking real parts gives the stated entrywise nonnegativity.  $\square$

**Definition 23.40** (Auxiliary matrix predicates for the rank-one step). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The auxiliary predicates state three matrix-theoretic ingredients used in the rank-one step of Appendix C.2, Lemma C.5: constant traces of all positive powers, existence of a rank-one factorization  $T = ab^\top$ , and the conditional Perron–Frobenius implication from primitivity together with constant trace powers to such a factorization.

This conditional implication is not a globally provable theorem. As a universal statement over primitive non-negative matrices it is false; a concrete  $3 \times 3$  counterexample appears in the

archive. The corrected criterion proved here adds positive semidefiniteness and trace normalization, which provide the diagonalizability needed to turn the trace-power condition into a rank-one factorization.

**Theorem 23.41** (Trace square of a Hermitian matrix). [Lean](#) [✓ Stmt](#) *Let  $T$  be a finite Hermitian matrix over a real or complex scalar field, with Hermitian eigenvalues  $\lambda_i$ . Then*

$$\text{tr}(T^2) = \sum_i \lambda_i^2.$$

*Proof.* [✓ Proof](#) The spectral theorem writes  $T$  as a unitary conjugate of the diagonal matrix of its Hermitian eigenvalues. Squaring commutes with this conjugation, and cyclicity of trace reduces the identity to the trace of the squared diagonal matrix.  $\square$

**Theorem 23.42** (PSD trace-power rank-one criterion). [Lean](#) [Lean](#) [✓ Stmt](#) *Let  $T$  be a finite real square matrix. If  $T$  is positive semidefinite, normalized by  $\text{tr}(T) = 1$ , and has constant traces on all positive powers, then  $T$  has a rank-one factorization. Consequently, under the same positive-semidefinite and trace-normalization hypotheses, the conditional rank-one criterion used in Lemma C.5 follows.*

*Proof.* [✓ Proof](#) The spectral theorem diagonalizes the positive semidefinite matrix with non-negative real eigenvalues. The trace gives that their sum is 1, while the second trace moment gives that the sum of their squares is also 1. Hence exactly one eigenvalue is equal to 1 and all others vanish, so the diagonal form is rank one. Unitary conjugation preserves the existence of an outer-product factorization.  $\square$

**Theorem 23.43** (SAL and ZCL imply rank-one  $T$  (conditional form)). [Lean](#) [✓ Stmt](#) *Let  $T$  be a finite real square matrix. Assume that  $T$  is primitive, that  $\text{tr}(T) = 1$ , and that every positive power of  $T$  has the same trace as  $T$ . If one further supplies the Perron–Frobenius hypothesis that this particular primitive non-negative matrix with constant trace powers is rank one, then*

$$T = ab^\top$$

for some vectors  $a$  and  $b$ , with normalization

$$a \cdot b = 1.$$

*Proof.* [✓ Proof](#) The Perron–Frobenius hypothesis gives the factorization  $T = ab^\top$ . Taking traces and using  $\text{tr}(ab^\top) = a \cdot b$  converts the normalization  $\text{tr}(T) = 1$  into  $a \cdot b = 1$ .  $\square$

**Theorem 23.44** (PSD-corrected rank-one  $T$  criterion). [Lean](#) [✓ Stmt](#) *Let  $T$  be a finite real square matrix. Assume that  $T$  is primitive, positive semidefinite, normalized by  $\text{tr}(T) = 1$ , and has constant traces on all positive powers. Then*

$$T = ab^\top$$

for some vectors  $a$  and  $b$ , with  $a \cdot b = 1$ .

*Proof.* [✓ Proof](#) The positive-semidefinite matrix criterion diagonalizes  $T$ , uses the first two trace moments to show that exactly one eigenvalue is nonzero, and hence supplies the auxiliary rank-one criterion. The conditional Lemma C.5 statement then gives the factorization and trace normalization.  $\square$

## 23.7 Horizontal and Vertical Canonical Forms

**Definition 23.45** (Horizontal canonical form for an MPO). [Learn] Not ready An MPO tensor is in *horizontal canonical form* if its doubled-index MPS tensor generates the same MPV family as a block-diagonal tensor  $\bigoplus_k \mu_k A_k$  whose blocks are injective, left-canonical, and have nonzero weights  $\mu_k$ .

**Definition 23.46** (Vertical canonical form for an MPO). [Lean] [Not ready] An MPO tensor is in *vertical canonical form* if its diagonal MPS tensor  $\{M^{ii}\}_{i=0}^{d-1}$  generates the same MPV family as a block-diagonal tensor obtained from a basis of normal tensors by repeating each block according to a multiplicity and weighting each copy by a positive scalar. This is the flattened form of the decomposition  $\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha}$  from [CPGSV16, Proposition 4.13].

*Remark 23.47* (Finite witnesses for block-injective canonical form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Let  $(A_k)$  be a family of block tensors, and fix a length  $L$ . Write  $A_k^w$  for the length- $L$  word evaluation of  $A_k$  along a word  $w$ . The finite-length spanning hypothesis says that the tuple-valued evaluations

$$w \mapsto (A_k^w)_k$$

span the full product algebra of block matrices. The corresponding finite-length separation property asserts that there exists a length  $L$  such that if, for every word  $w$  of length  $L$ ,

$$\sum_k \text{tr}(\Delta_k A_k^w) = 0$$

then these pairings vanish on a spanning set for the product algebra, so the induced linear functional on the product algebra is zero; by nondegeneracy of the product trace pairing, every  $\Delta_k$  vanishes. In other words, a single finite blocking length already separates the blocks by their word evaluations. Combined with injectivity, left-canonicity, and nonzero block weights, this finite-length separation yields the horizontal canonical-form data of the block-diagonal tensor.

The entrywise scalar family of block-matrix coefficients encodes the same finite-length data. Linear independence of that scalar family implies the product-algebra spanning statement, hence a block-injective canonical-form witness. The same criterion therefore gives horizontal canonical-form data directly. The duplicate-scalar counterexample shows that this finite-length witness is strictly stronger than blockwise injectivity, left-canonicity, and nonzero weights alone: two identical scalar blocks satisfy those hypotheses, but the block-injective canonical-form separation condition fails and no blocking length makes the entrywise family linearly independent.

The block-injectivity proposition in [CPGSV16, lines 340–345] gives a concrete sufficient criterion for the spanning condition: after a common blocking length each block is block-injective, and after a further finite blocking length one can form linear combinations of word evaluations which equal the identity on one chosen block and vanish on all others. Concatenating an injective prefix with this finite block-separation data produces the product-algebra spanning condition, hence the horizontal canonical-form data.

**Theorem 23.48** (Finite block-separation criterion). [Lean] ✓ Stmt *If a block family admits the finite-length data described in the block-injectivity proposition of [CPGSV16, lines 340–345], then it is in block-injective canonical form.*

*Proof.* ✓ Proof This is the direct implication from the finite-length data to the finite-length trace-separation statement. What remains open in the full block-injectivity proposition of [CPGSV16, lines 340–345] is the derivation of this finite witness from separated canonical-form / BNT data.

☐



**Lemma 23.49** (Blockwise equality from agreement on the MPV family). Lean ✓ Stmt Let  $\bigoplus_k \mu_k A_k$  be a block-diagonal tensor whose blocks are injective, left-canonical, weighted by nonzero scalars, and jointly block-injective in the sense that for some blocking length  $L$  the tuple of trace pairings against length- $L$  block words is faithful across all blocks simultaneously. If two physical operators induce the same first-site action on every MPV generated by this tensor, then their first-site insertions agree on each block  $A_k$ .

*Proof.* ✓ Proof This is Lemma L of [CPGSV16, Appendix A], using the finite criterion from Theorem 23.48. Expanding the hypothesis over a word of length  $L + 1$  beginning with the fixed site and folding each block contribution into a trace pairing against the corresponding first-site insertion tensor yields, upon taking the  $Y$ - $Z$  difference, a tuple of block matrices whose trace pairings against every length- $L$  block word vanish. The block-injective canonical-form hypothesis then forces each block difference to be zero, and dividing by the nonzero weight  $\mu_k^{L+1}$  concludes the blockwise equality of the two first-site insertions.  $\square$

**Theorem 23.50** (Horizontal canonical form implies vertical canonical form). Not ready Every MPDO in horizontal canonical form is also in vertical canonical form.

*Proof.* Starting from the horizontal canonical-form decomposition, one uses the MPDO positivity condition to compare the first-site action of the doubled-index tensor on the diagonal sector. Lemma 23.49 then shows that the diagonal tensor splits along the same normal blocks, with the coefficients rearranged into positive scalar weights. Flattening those positive diagonal coefficients produces the required vertical canonical-form decomposition.  $\square$

## 23.8 Fusion isometries

The next definitions isolate the transfer-map content underlying the fusion-isometry picture of [CPGSV16, Section 4.5]: they record when the blocked transfer map factors through its support algebra as a retract, which is the idempotence criterion, not the physical fusion isometry of two tensors into one. For each blocked size  $n \geq 1$ , the doubled-index blocked tensor of an MPO has a blocked transfer map  $E_n$  acting on bond-space matrices, and the support algebra is modeled by a subspace through which  $E_n$  factors as a retract.

**Definition 23.51** (Transfer-map fusion-isometry datum). Lean ✓ Stmt A *transfer-map fusion-isometry datum* at blocked size  $n$  for an MPO tensor consists of a subspace  $\mathcal{A}_n \subseteq M_D(\mathbb{C})$  together with linear maps

$$T_n : M_D(\mathbb{C}) \rightarrow \mathcal{A}_n, \quad S_n : \mathcal{A}_n \rightarrow M_D(\mathbb{C})$$

such that

$$T_n \circ S_n = \text{id}_{\mathcal{A}_n}, \quad S_n \circ T_n = E_n,$$

where  $E_n$  denotes the blocked transfer map.

**Definition 23.52** (Fusion-isometry RFP predicate). Lean ✓ Stmt An MPO tensor  $M$  satisfies the *fusion-isometry formulation* of the renormalization fixed-point condition when for every blocked size  $n \geq 1$  there exists transfer-map fusion-isometry data for the blocked transfer map  $E_n$ .

**Theorem 23.53** (Fusion-isometry data imply transfer-map idempotence). Lean Lean ✓ Stmt A one-site fusion-isometry datum implies transfer-map idempotence. Hence the fusion-isometry formulation, which supplies such data at every positive blocked size, implies the same condition.



*Proof.* [✓ Proof](#) Apply the definition at blocked size  $n = 1$ . If  $E_1 = S_1 \circ T_1$  and  $T_1 \circ S_1 = \text{id}$ , then

$$E_1^2 = S_1 T_1 S_1 T_1 = S_1 (T_1 S_1) T_1 = S_1 T_1 = E_1.$$

Since  $E_1$  is the original transfer map, this is exactly transfer-map idempotence.  $\square$

**Theorem 23.54** (One-site fusion retract criterion). [Lean](#) [✓ Stmt](#) *A one-site transfer-map fusion-isometry datum exists if and only if the MPO transfer map is idempotent.*

*Proof.* [✓ Proof](#) The forward implication is the retract calculation  $E_1^2 = S_1 T_1 S_1 T_1 = S_1 T_1$ . Conversely, if  $E_1$  is idempotent, then it factors through its range, giving the required one-site datum.  $\square$

**Theorem 23.55** (Transfer-map fusion criterion for idempotence). [Lean](#) [✓ Stmt](#) *The fusion-isometry formulation is equivalent to transfer-map idempotence.*

*Proof.* [✓ Proof](#) The forward implication is Theorem 23.53. Conversely, if the transfer map  $E$  is idempotent, then every positive blocked transfer map equals  $E$  and therefore factors through its range. This range-factorization gives the required fusion-isometry data.  $\square$

**Corollary 23.56** (Pure-state recovery). [Lean](#) [✓ Stmt](#) *For the diagonal MPO associated to a pure MPS tensor, the fusion-isometry formulation reduces to the usual pure-state RFP condition.*

*Proof.* [✓ Proof](#) Combine Theorem 23.55 with the diagonal pure-state recovery theorem for the MPO predicate.  $\square$

**Theorem 23.57** (All-blocked fusion reduces to one-site fusion). [Lean](#) [✓ Stmt](#) *The fusion-isometry formulation for all positive blocked sizes is equivalent to the existence of a one-site transfer-map fusion-isometry datum.*

*Proof.* [✓ Proof](#) Combine the one-site criterion with the equivalence between the all-blocked fusion formulation and transfer-map idempotence.  $\square$

## 23.9 Algebra structure

**Definition 23.58** (Algebra-structure data for an MPDO). [Lean](#) [✓ Stmt](#) A family of support  $*$ -algebras  $\{\mathcal{A}_n\}_{n \geq 0}$ , one at every blocking size, equipped with blocking maps  $m_n : \mathcal{A}_n \otimes \mathcal{A}_n \rightarrow \mathcal{A}_{2n}$  and inclusion maps  $\iota_n : \mathcal{A}_n \rightarrow \mathcal{A}_{n+1}$  such that, after viewing all elements inside the ambient bond-space matrix algebra,  $m_n$  is ordinary matrix multiplication and  $\iota_n$  is the ambient inclusion.

**Definition 23.59** (Algebra-structure RFP predicate). [Lean](#) [✓ Stmt](#) An MPO tensor  $M$  satisfies the *algebra-structure formulation* used here when there exist algebra-structure data whose support algebra at every positive blocking size coincides with the fixed-point algebra of the adjoint blocked transfer map. This is a nontrivial algebraic condition, but it is still weaker than the full coefficient formulation of [CPGSV16, Theorem 4.14(ii)]; that formulation also includes the coefficient family  $c_{\alpha, \beta, \gamma}^{(L)}$ .

**Theorem 23.60** (RFP yields a stationary algebra tower under a faithful fixed point). [Lean](#) [✓ Stmt](#) *Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If the transfer map is idempotent, then there exists a stationary family of support algebras, all equal to the fixed-point algebra of the adjoint transfer map, with multiplication given by matrix multiplication and inclusions given by the identity.*

*Proof.* [✓ Proof](#) Under the trace-preserving normalization and the faithful fixed point, Wolf's fixed-point-algebra theorem gives a  $*$ -subalgebra of adjoint fixed points. Idempotence identifies every positive blocked transfer map with the original one, so the same fixed-point algebra works at every blocking size.  $\square$

**Theorem 23.61** (Fusion yields a stationary algebra tower under a faithful fixed point). [Lean](#)

[✓ Stmt](#) Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If the transfer-map fusion criterion holds, then the MPO satisfies the algebra-structure formulation of the renormalization fixed-point condition.

*Proof.* [✓ Proof](#) Combine Theorem 23.53 with Theorem 23.60.  $\square$

**Theorem 23.62** (Stabilized blocked adjoint fixed points yield algebra data). [Lean](#) [✓ Stmt](#)

Assume the doubled tensor is trace-preserving and the transfer map admits a positive-definite fixed point. If, for every positive blocked size  $n$ , the adjoint fixed-point space of the blocked transfer map agrees with the adjoint fixed-point space of the original transfer map, then the algebra-structure formulation holds.

*Proof.* [✓ Proof](#) Use the adjoint fixed-point algebra of the original transfer map as a constant support-algebra tower. The stabilization hypothesis identifies that same algebra with the adjoint fixed-point space of every positive blocked transfer map, so the stationary family satisfies the definition.  $\square$

This does not give the full converse direction of [CPGSV16, Theorem 4.14]: the compatibility condition above only sees stabilized adjoint fixed-point algebras, not the coefficient/BNT data. In particular, Theorem IV.13(ii) also requires the special diagonal matrices  $\chi_{\alpha,\beta,\gamma}$ . The coefficient data attached to the algebra tower are: a chosen basis of each support algebra, coordinate maps into a finite scalar family, and the corresponding multiplication / inclusion coefficient arrays.

**Definition 23.63** (Blocked coefficients for the algebra tower). [Lean](#) [✓ Stmt](#) For each blocked size  $n$ , once a basis of  $\mathcal{A}_n$  is chosen, every element of  $\mathcal{A}_n$  is identified with a finite coefficient family on that basis.

**Theorem 23.64** (Reconstruction from blocked coefficients). [Lean](#) [✓ Stmt](#) Reconstructing a coefficient family gives the corresponding basis expansion  $\sum_i a_i b_i$  in  $\mathcal{A}_n$ .

*Proof.* [✓ Proof](#) This is exactly the finite-basis reconstruction formula for the chosen basis of  $\mathcal{A}_n$ .  $\square$

**Definition 23.65** (Blocked multiplication coefficients). [Lean](#) [✓ Stmt](#) The blocking multiplication map  $m_n$  determines a coefficient family for the product of any two chosen basis elements of  $\mathcal{A}_n$ , now viewed in the basis of  $\mathcal{A}_{2n}$ .

**Theorem 23.66** (Blocked multiplication reconstructs from its coefficients). [Lean](#) [✓ Stmt](#) In the ambient bond-space matrix algebra, the product of two chosen basis elements of  $\mathcal{A}_n$  is the sum of the basis elements of  $\mathcal{A}_{2n}$  weighted by these extracted coefficients.

*Proof.* [✓ Proof](#) Apply the reconstruction formula of Theorem 23.64 to the element  $m_n(b_i, b_j) \in \mathcal{A}_{2n}$ .  $\square$

**Theorem 23.67** (Compatible adjoint fixed points reconstruct from blocked coefficients). [Lean](#)

[✓ Stmt](#) If the algebra tower is compatible with an MPO tensor  $M$ , then every adjoint blocked fixed point of the blocked transfer map  $E_n$  belongs to  $\mathcal{A}_n$  and is recovered exactly from its extracted coefficient family.

*Proof.* [✓ Proof](#) Compatibility identifies  $\mathcal{A}_n$  with the adjoint fixed-point algebra of  $E_n$ , so the fixed point first becomes an element of  $\mathcal{A}_n$  and then Theorem 23.64 reconstructs it.  $\square$

**Theorem 23.68** (Adjoint fixed points descend through the algebra tower). [Lean](#) [✓ Stmt](#) *If an MPO tensor satisfies the algebra-structure formulation, then every adjoint fixed point of the blocked transfer map  $E_n$  at a positive blocking size  $n$  is already an adjoint fixed point of the unblocked transfer map  $E_1$ .*

*Proof.* [✓ Proof](#) Compatibility at size  $n$  places  $X$  in  $\mathcal{A}_n$ . The inclusion map  $\iota_n$  lifts  $X$  into  $\mathcal{A}_{n+1}$ , and since  $\iota_n$  is just an inclusion it does not change the underlying operator, so we continue to denote the lifted element by  $X$ . Thus compatibility at size  $n + 1$  yields  $E_{n+1}^\dagger X = X$ . Rewriting  $E_{n+1} = E_n \circ E_1$  and applying  $(A \circ B)^\dagger = B^\dagger \circ A^\dagger$  together with the hypothesis  $E_n^\dagger X = X$  extracts  $E_1^\dagger X = X$ .  $\square$

**Lemma 23.69** (Reverse inclusion of adjoint fixed points). [Lean](#) [✓ Stmt](#) *Every adjoint fixed point of the unblocked transfer map  $E_1$  is an adjoint fixed point of every blocked transfer map  $E_n$ .*

*Proof.* [✓ Proof](#) Induction on  $n$  using  $E_{n+1} = E_n \circ E_1$  and  $(A \circ B)^\dagger = B^\dagger \circ A^\dagger$ ; the base case is  $E_0 = \text{id}$ . This step does not require the algebra-structure data.  $\square$

**Lemma 23.70** (Blocked powers of adjoint eigenvectors). [Lean](#) [✓ Stmt](#) *If  $E = E_1$  and  $E^\dagger X = \lambda X$ , then  $E_n^\dagger X = \lambda^n X$  for every  $n$ .*

*Proof.* [✓ Proof](#) The case  $n = 0$  is  $E_0 = \text{id}$ . For the induction step,  $E_{n+1} = E_n \circ E_1$ , hence

$$E_{n+1}^\dagger X = E_1^\dagger E_n^\dagger X = E_1^\dagger (\lambda^n X) = \lambda^n E_1^\dagger X = \lambda^{n+1} X.$$

$\square$

**Theorem 23.71** (Finite-order adjoint eigenvalues for algebra towers). [Lean](#) [✓ Stmt](#) *Assume the algebra-structure formulation holds. If  $X \neq 0$ ,  $E_1^\dagger X = \lambda X$ , and  $\lambda^n = 1$  for some  $n > 0$ , then  $\lambda = 1$ .*

*Proof.* [✓ Proof](#) By Lemma 23.70,  $E_n^\dagger X = \lambda^n X = X$ . The descent theorem gives  $E_1^\dagger X = X$ . Thus  $\lambda X = X$ , so  $(\lambda - 1)X = 0$ . Since  $X \neq 0$ , one obtains  $\lambda = 1$ .  $\square$

**Theorem 23.72** (Adjoint fixed-point equality for algebra towers). [Lean](#) [✓ Stmt](#) *If an MPO tensor satisfies the algebra-structure formulation, then for every positive blocking size  $n$  the adjoint fixed points of  $E_n$  are exactly the adjoint fixed points of  $E_1$ .*

*Proof.* [✓ Proof](#) Combine Theorem 23.68 with Lemma 23.69.  $\square$

**Theorem 23.73** (Algebra data give the stationary fixed-point tower). [Lean](#) [✓ Stmt](#) *Under the trace-preserving normalization and a positive-definite fixed point, any algebra-structure witness forces the stationary adjoint fixed-point tower itself to be compatible with the MPO tensor.*

*Proof.* [✓ Proof](#) Apply the adjoint fixed-point equality obtained from the algebra-structure predicate to the stationary tower constructed from the faithful fixed point.  $\square$

*Remark 23.74* (Why algebra data do not imply fusion data). Write  $E = E_1$  for the one-site transfer map. The present algebra predicate gives, for every  $n > 0$ ,

$$\text{Fix}((E^n)^\dagger) = \text{Fix}(E^\dagger).$$

Hence a vector  $X$  with  $E^\dagger X = \lambda X$  and  $\lambda^n = 1$  lies in  $\text{Fix}(E^\dagger)$ , so  $(\lambda - 1)X = 0$  and  $\lambda = 1$  by Theorem 23.71. It does not imply  $E^2 = E$ . For instance, if  $\Pi_{\text{diag}}$  is the projection onto diagonal matrices and  $0 < \varepsilon < 1$ , set

$$E(X) = \varepsilon X + (1 - \varepsilon) \Pi_{\text{diag}}(X),$$

so that

$$E^n(X) = \Pi_{\text{diag}}(X) + \varepsilon^n(X - \Pi_{\text{diag}}(X)), \quad E^2 - E = (\varepsilon^2 - \varepsilon)(\text{id} - \Pi_{\text{diag}}).$$

The fixed-point algebra is diagonal for every  $n > 0$ , but  $E^2 \neq E$ .

The missing source step is the coefficient argument in [CPGSV16, Appendix C.4]. One needs the same-length product law

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma \text{tr}(\chi_{\alpha,\beta,\gamma}^L)O_L(M_\gamma) \quad (23.1)$$

and the length-one idempotent trace identity

$$m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma})m_\alpha m_\beta. \quad (23.2)$$

The proof of [CPGSV16, Theorem 4.14] uses (23.1) and (23.2), positivity, and the simple-matrix lemma [CPGSV16, lines 1155–1163] to give

$$\{\nu_{\gamma,k}\}_k = \{\mu_{\alpha,k_1}\mu_{\beta,k_2}\chi_{\alpha,\beta,\gamma,k_3}\}_{\alpha,\beta,k_1,k_2,k_3}, \quad \text{tr}(\nu_\gamma) = m_\gamma. \quad (23.3)$$

The reconstruction maps in [CPGSV16, lines 2065–2085] then use

$$\widetilde{R}_2 \left( \bigoplus_\gamma X_\gamma \right) = \bigoplus_\gamma \frac{\nu_\gamma}{m_\gamma} \otimes X_\gamma, \quad \widetilde{R}_1 \left( \bigoplus_\gamma X_\gamma \right) = \bigoplus_\gamma \frac{\mu_\gamma}{m_\gamma} \otimes X_\gamma. \quad (23.4)$$

Thus  $\text{tr}(\nu_\gamma) = m_\gamma$  and the defining equality  $m_\gamma = \text{tr}(\mu_\gamma)$  normalize the factors  $\nu_\gamma/m_\gamma$  and  $\mu_\gamma/m_\gamma$ , respectively, in  $T = \widetilde{R}_2 \widetilde{T} R_1$  and  $S = \widetilde{R}_1 \widetilde{S} R_2$ . The scalar Newton–Girard power-sum identity needed to obtain the trace-power form is already available as Theorem 10.19; what remains is the MPDO construction of the BNT-label witness carrying (23.3) and (23.4).

### 23.9.1 Diagonal $\chi$ -matrices and the trace-power formula

The following statements isolate the form of [CPGSV16, Theorem 4.14(ii)]: the structure coefficients of the length- $L$  operator algebra are trace-powers of positive diagonal matrices  $\chi_{\alpha,\beta,\gamma}$ . They state the coefficient identity and the elementary trace-power identity for diagonal matrices.

**Definition 23.75** (Diagonal  $\chi$  family). [Lean] ✓ Stmt A family of diagonal complex matrices  $\chi_{\alpha,\beta,\gamma}$  indexed by ordered triples drawn from a common index type, each of a specified size. This is the matrix  $\chi_{\alpha,\beta,\gamma}$  of [CPGSV16, Theorem 4.14(ii)].

**Theorem 23.76** (Trace of  $\chi^L$  equals the sum of  $L$ -th powers of entries). [Lean] ✓ Stmt For every triple  $(\alpha, \beta, \gamma)$ , if  $\chi_{\alpha,\beta,\gamma}$  has size  $r_{\alpha,\beta,\gamma}$ , then for every  $L \geq 0$ ,

$$\text{tr}(\chi_{\alpha,\beta,\gamma}^L) = \sum_{k=1}^{r_{\alpha,\beta,\gamma}} \chi_{\alpha,\beta,\gamma,k}^L,$$

where  $\chi_{\alpha,\beta,\gamma,k}$  are the diagonal entries.

*Proof.* ✓ Proof Diagonal matrices raise to powers entrywise and the trace of a diagonal matrix is the sum of its entries.  $\square$

**Definition 23.77** (Trace-power form of a structure-coefficient family). Lean ✓ Stmt A binary compatibility predicate between an abstract structure-coefficient family  $c_{\alpha,\beta,\gamma}^{(L)}$  and a diagonal  $\chi$  family: the pair  $(c, \chi)$  is said to be in *trace-power form* when

$$c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L) = \sum_{k=1}^{r_{\alpha,\beta,\gamma}} \chi_{\alpha,\beta,\gamma,k}^L$$

for every  $L \geq 0$  and every triple  $(\alpha, \beta, \gamma)$ , where  $r_{\alpha,\beta,\gamma}$  is the size of  $\chi_{\alpha,\beta,\gamma}$ . This is the formal analog of the target identity of [CPGSV16, Theorem 4.14(ii)]; the existential version—“there exists  $\chi$  for which  $(c, \chi)$  has trace-power form”—is obtained by quantifying over  $\chi$ .

**Definition 23.78** (BNT-label coefficient family). Lean Lean Lean Lean ✓ Stmt The same-length coefficient system  $c_{\alpha,\beta,\gamma}^{(L)}$  from Theorem IV.13(ii), indexed by fixed BNT labels  $\alpha, \beta, \gamma$ . The coefficient is attached to the length- $L$  product  $O_L(M_\alpha)O_L(M_\beta)$ , not to the blocked-basis multiplication  $\mathcal{A}_n \times \mathcal{A}_n \rightarrow \mathcal{A}_{2n}$ . When a diagonal  $\chi$ -family has already been constructed, it also canonically determines the coefficient family  $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L) = \sum_r \chi_{\alpha,\beta,\gamma,r}^L$ , as in [CPGSV16, Appendix C.4].

**Definition 23.79** (BNT-label operator family). Lean ✓ Stmt A BNT-label operator family records, for each chain length  $L$ , the operators  $O_L(M_\alpha)$  indexed by the fixed BNT labels  $\alpha$ . The ambient algebra at length  $L$  is left abstract here; the later comparison step must relate it to the chosen blocked support algebras.

**Definition 23.80** (Same-length BNT product form). Lean ✓ Stmt A BNT-label operator family and a BNT-label coefficient system have same-length product form when, for every positive length  $L$ ,

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma} c_{\alpha,\beta,\gamma}^{(L)} O_L(M_\gamma).$$

This is the product identity appearing in Theorem IV.13(ii), stated independently of the blocked-basis multiplication  $\mathcal{A}_n \times \mathcal{A}_n \rightarrow \mathcal{A}_{2n}$ .

**Theorem 23.81** (Same-length BNT product expansion). Lean ✓ Stmt *Under same-length BNT product form, the product of two length- $L$  BNT-label operators is the corresponding finite linear combination of length- $L$  BNT-label operators.*

*Proof.* ✓ Proof This is exactly the defining equality of same-length product form.  $\square$

**Definition 23.82** (BNT-label trace-scalar family). Lean ✓ Stmt A BNT-label trace-scalar family records the scalars  $m_\alpha = \text{tr}(\mu_\alpha)$  indexed by the fixed BNT labels  $\alpha$ , as in the idempotent condition of Theorem IV.13(ii).

**Definition 23.83** (BNT idempotent coefficient form). Lean ✓ Stmt A BNT-label trace-scalar family and a BNT-label coefficient system have idempotent coefficient form when, for every BNT label  $\gamma$ ,

$$m_\gamma = \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta.$$

This is the idempotent condition in Theorem IV.13(ii), stated separately from the trace-power formula.

**Theorem 23.84** (BNT idempotent coefficient expansion). [Lean](#) [✓ Stmt](#) Under BNT idempotent coefficient form, each  $m_\gamma$  is the corresponding finite linear combination of products  $m_\alpha m_\beta$  with the length-one coefficients  $c_{\alpha,\beta,\gamma}^{(1)}$ .

*Proof.* [✓ Proof](#) This is exactly the defining equality of BNT idempotent coefficient form.  $\square$

**Definition 23.85** (Blocked-basis BNT-label assignment). [Lean](#) [Lean](#) [✓ Stmt](#) The BNT-label assignment consists, for every positive blocked length  $n$ , of a source label map  $\sigma_n$  from the chosen basis of  $\mathcal{A}_n$  and a target label map  $\tau_n$  from the chosen basis of  $\mathcal{A}_{2n}$  to the fixed BNT labels. It also gives the combined label map  $\sigma_n \sqcup \tau_n$  on the disjoint union of these two blocked bases.

**Definition 23.86** (BNT comparison with blocked coefficients). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A comparison between the BNT-label coefficients and the chosen blocked-basis coefficients consists of a BNT-label assignment together with the equality

$$c_{i,j,k}^{(n)} = c_{\sigma_n(i),\sigma_n(j),\tau_n(k)}^{(n)}.$$

This is a blocked-basis comparison: the left hand side is expanded in the chosen basis of  $\mathcal{A}_{2n}$ , so the statement is not itself the same-length BNT product law. This records the coefficient-comparison statement separately from the construction of the label maps from the MPDO tensor.

**Theorem 23.87** (Blocked coefficients are pulled back from BNT labels). [Lean](#) [✓ Stmt](#) Under a BNT comparison with blocked coefficients, each chosen blocked-basis multiplication coefficient is the corresponding BNT-label coefficient.

*Proof.* [✓ Proof](#) This is exactly the defining equality of the comparison.  $\square$

**Definition 23.88** (Positive-length BNT-label trace-power form). [Lean](#) [✓ Stmt](#) A BNT-label coefficient system and a diagonal  $\chi$ -family are compatible when, for every positive length  $L$ ,

$$c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L).$$

The same matrix  $\chi_{\alpha,\beta,\gamma}$  is used for all positive  $L$ .

**Theorem 23.89** (BNT-label coefficient as a positive-length trace power). [Lean](#) [✓ Stmt](#) Under positive-length BNT-label trace-power form, for every  $L > 0$ ,  $c_{\alpha,\beta,\gamma}^{(L)}$  is the trace of  $\chi_{\alpha,\beta,\gamma}^L$ .

*Proof.* [✓ Proof](#) Combine the defining positive-length trace-power identity with the trace formula for powers of a diagonal matrix.  $\square$

**Theorem 23.90** (Canonical BNT coefficients from a  $\chi$ -family). [Lean](#) [✓ Stmt](#) The coefficient family canonically determined by a diagonal  $\chi$ -family satisfies the positive-length trace-power condition with respect to the same  $\chi$ -family.

*Proof.* [✓ Proof](#) This is immediate from the definition of the canonical coefficient family.  $\square$

**Definition 23.91** (Positive BNT-label  $\chi$  trace-power witness). [Lean](#) [Lean](#) [✓ Stmt](#) A positive BNT-label  $\chi$  witness consists of a length-independent diagonal  $\chi_{\alpha,\beta,\gamma}$ -family, positivity of every diagonal entry, and the positive-length trace-power identity for the BNT-label coefficients. This is the coefficient statement of Theorem IV.13(ii); constructing the witness from an MPDO tensor and comparing it to blocked bases remain separate obligations. If the coefficient family is chosen canonically from the same  $\chi$ -family, positivity of the diagonal entries alone gives such a witness.

**Theorem 23.92** (Positive BNT-label witness gives trace powers). [Lean](#) [✓ Stmt](#) *A positive BNT-label  $\chi$  witness gives, for every  $L > 0$ , the formula  $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ .*

*Proof.* [✓ Proof](#) Apply the trace reformulation of the positive-length trace-power identity from the witness.  $\square$

**Theorem 23.93** (Same-length BNT product with chi-trace coefficients). [Lean](#) [✓ Stmt](#) *If the same-length BNT product identity holds and the BNT-label coefficients come from a positive length-independent  $\chi$ -witness, then for every  $L > 0$ ,*

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma} \text{tr}(\chi_{\alpha,\beta,\gamma}^L)O_L(M_\gamma).$$

*Proof.* [✓ Proof](#) Substitute the positive-length trace-power formula into the same-length BNT product expansion.  $\square$

**Theorem 23.94** (Same-length BNT product for canonical chi coefficients). [Lean](#) [✓ Stmt](#) *If the same-length BNT product identity holds with the coefficient family canonically determined by a diagonal  $\chi$ -family, then for every  $L > 0$ ,*

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma} \text{tr}(\chi_{\alpha,\beta,\gamma}^L)O_L(M_\gamma).$$

*Proof.* [✓ Proof](#) Substitute the canonical trace-power formula for the coefficients into the same-length BNT product expansion.  $\square$

**Theorem 23.95** (BNT idempotent coefficients as chi traces). [Lean](#) [✓ Stmt](#) *If the BNT trace scalars satisfy the idempotent coefficient condition and the BNT-label coefficients come from a positive length-independent  $\chi$ -witness, then*

$$m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma})m_\alpha m_\beta.$$

*Proof.* [✓ Proof](#) Substitute the length-one trace-power formula into the idempotent coefficient identity.  $\square$

**Theorem 23.96** (BNT idempotent coefficients for canonical chi coefficients). [Lean](#) [✓ Stmt](#) *If the BNT trace scalars satisfy the idempotent coefficient condition with the coefficient family canonically determined by a diagonal  $\chi$ -family, then*

$$m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma})m_\alpha m_\beta.$$

*Proof.* [✓ Proof](#) Substitute the length-one canonical trace formula into the idempotent coefficient identity.  $\square$

**Theorem 23.97** (Blocked coefficients inherit BNT trace powers). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *If the chosen blocked-basis coefficients are compared with the fixed BNT-label coefficient system, and that BNT-label system has a positive length-independent  $\chi$ -witness, then each positive-length blocked coefficient is the trace power of the corresponding BNT-label  $\chi$ -matrix:*

$$c_{i,j,k}^{(n)} = \text{tr}(\chi_{\sigma_n(i),\sigma_n(j),\tau_n(k)}^n).$$



*Proof.* ✔ Proof Use the comparison equality to replace the blocked-basis coefficient by the corresponding BNT-label coefficient, and then apply the BNT-label trace-power formula. In the special case where the compared BNT-label coefficient family is already the canonical family determined by  $\chi$ , the same conclusion follows directly from the canonical trace formula.  $\square$

**Definition 23.98** (Blocked  $\chi$  family for multiplication coefficients). [Lean](#) [Stm](#) A dependent diagonal matrix family indexed by the blocked multiplication triples  $(n, i, j, k)$ , where  $i$  and  $j$  label basis elements of  $\mathcal{A}_n$  and  $k$  labels a basis element of  $\mathcal{A}_{2n}$ . This is a blocked-basis analog of the uniform BNT-label family in [CPGSV16, Theorem 4.14(ii)]; the remaining uniformity gap is recorded in docs/paper-gaps/cpgsv17\_blocked\_chi\_uniformity.tex.

**Theorem 23.99** (Trace of a blocked  $\chi$  power). [Lean] [✓ Stmt] For each blocked multiplication triple  $(n, i, j, k)$  and every exponent  $L$ , the trace of the  $L$ -th power of its diagonal matrix is the sum of the  $L$ -th powers of the corresponding diagonal entries.

*Proof.* ✓ Proof Diagonal matrices raise to powers entrywise and the trace of a diagonal matrix is the sum of its entries. □

**Definition 23.100** (Trace-power form for blocked multiplication coefficients). [Lean](#) [✓ Stmt](#) The blocked multiplication coefficients are in trace-power form when, for every positive blocked size  $n$  and every multiplication triple  $(i, j, k)$ , the coefficient  $c_{i,j,k}^{(n)}$  equals

$$\sum_r \chi_{n,i,j,k,r}^n.$$

Equivalently, by Theorem 23.99, it is the trace of the  $n$ -th power of the corresponding diagonal matrix. The exponent  $n$  is the source blocking length of the two factors in  $\mathcal{A}_n$ ; the product is expanded in the basis of  $\mathcal{A}_{2n}$ .

**Theorem 23.101** (Blocked coefficient as a trace power). Lean ✓ Stmt Under trace-power form for the blocked multiplication coefficients, for every positive blocked size  $n$ , the coefficient  $c_{i,j,k}^{(n)}$  is the trace of the  $n$ -th power of the diagonal matrix attached to  $(n, i, j, k)$ .

*Proof.* ✓ Proof Combine the defining trace-power identity with the trace formula for powers of a diagonal matrix. □

**Definition 23.102** (Positive blocked  $\chi$  trace-power witness). [Lean](#) [✓ Stmt](#) A positive blocked  $\chi$  trace-power witness consists of a blocked  $\chi$  family, positivity of all its diagonal entries, and the positive-length trace-power identity for the blocked multiplication coefficients. This is the blocked-basis analog of the positive diagonal matrices in [CPGSV16, Theorem 4.14(ii)]; the uniform BNT-label gap is recorded in docs/paper-gaps/cpgsv17\_blocked\_chi\_uniformity.tex.

**Theorem 23.103** (BNT witnesses give blocked  $\chi$  witnesses). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)  
[Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the chosen blocked-basis coefficients are compared with a fixed BNT-label coefficient system, and that this BNT-label system has a positive length-independent  $\chi$ -witness. Then the chosen blocked bases carry a positive blocked  $\chi$  trace-power witness. The construction is obtained by pulling back the BNT-label  $\chi_{\alpha,\beta,\gamma}$ -matrices along the source and target label maps.

*Proof.*  **Proof** Reindex the positive BNT-label  $\chi$ -family along the blocked-basis label map. Positivity is preserved by reindexing, and the trace-power identity follows by substituting the blocked-basis comparison into the BNT-label trace-power formula.  $\square$





**Theorem 23.110** (BNT theorem data give the chi-trace product law). [Lean](#) [✓ Stmt](#) *BNT-label theorem data give the same-length product expansion with coefficients written as traces of powers of the length-independent  $\chi_{\alpha,\beta,\gamma}$ -matrices.*

*Proof.* [✓ Proof](#) Apply the chi-trace product law to the product identity and the positive BNT-label  $\chi$ -witness belonging to the theorem data.  $\square$

**Theorem 23.111** (BNT theorem data give the chi-trace idempotent law). [Lean](#) [✓ Stmt](#) *BNT-label theorem data give the idempotent scalar identity with the length-one coefficients written as traces of the corresponding  $\chi$ -matrices.*

*Proof.* [✓ Proof](#) Apply the chi-trace idempotent law to the idempotent condition and the positive BNT-label  $\chi$ -witness belonging to the theorem data.  $\square$

**Theorem 23.112** (BNT theorem data give blocked coefficient trace powers). [Lean](#) [✓ Stmt](#) *BNT-label theorem data give the blocked-basis coefficient trace-power formula obtained by pulling the BNT-label  $\chi_{\alpha,\beta,\gamma}$ -matrices back along the blocked-basis comparison maps.*

*Proof.* [✓ Proof](#) Apply the blocked-basis coefficient trace-power theorem to the comparison and the positive BNT-label  $\chi$ -witness belonging to the theorem data.  $\square$

**Theorem 23.113** (BNT theorem data give a positive blocked  $\chi$  witness). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *BNT-label theorem data give a positive blocked-basis  $\chi$  trace-power witness by pulling the length-independent BNT-label  $\chi$ -family back along the blocked-basis comparison maps.*

*Proof.* [✓ Proof](#) Apply the BNT-to-blocked witness construction to the comparison and the positive BNT-label  $\chi$ -witness belonging to the theorem data.  $\square$

**Theorem 23.114** (BNT theorem data give blocked trace-power form). [Lean](#) [✓ Stmt](#) *The pulled-back blocked-basis  $\chi$ -family obtained from BNT-label theorem data satisfies the blocked trace-power predicate.*

*Proof.* [✓ Proof](#) This is the trace-power part of the positive blocked-basis  $\chi$ -witness obtained from the theorem data.  $\square$

**Theorem 23.115** (BNT theorem data give positive blocked chi families). [Lean](#) [✓ Stmt](#) *The pulled-back blocked-basis  $\chi$ -family obtained from BNT-label theorem data has positive diagonal entries.*

*Proof.* [✓ Proof](#) This is the positivity part of the positive blocked-basis  $\chi$ -witness obtained from the theorem data.  $\square$

**Theorem 23.116** (BNT theorem data give positive pulled-back blocked entries). [Lean](#) [✓ Stmt](#) *The pulled-back blocked-basis  $\chi$ -entries obtained from BNT-label theorem data are positive.*

*Proof.* [✓ Proof](#) This is the positivity part of the positive blocked-basis  $\chi$ -witness obtained from the theorem data.  $\square$

**Theorem 23.117** (BNT theorem data give pulled-back blocked trace powers). [Lean](#) [✓ Stmt](#) *The positive-length blocked-basis coefficients are traces of powers of the pulled-back blocked-basis  $\chi$ -matrices obtained from BNT-label theorem data.*

*Proof.* [✓ Proof](#) Apply the trace-power identity of the positive blocked-basis  $\chi$ -witness obtained from the theorem data.  $\square$





**Theorem 23.132** (BNT theorem witnesses give pulled-back blocked trace powers). [Lean](#) [✓ Stmt](#) *The positive-length blocked-basis coefficients are traces of powers of the pulled-back blocked-basis  $\chi$ -matrices obtained from an existential BNT-label theorem witness.*

*Proof.* [✓ Proof](#) Apply the trace-power identity of the positive blocked-basis  $\chi$ -witness obtained from the existential witness.  $\square$

**Theorem 23.133** (Positive blocked  $\chi$  witness gives trace powers). [Lean](#) [✓ Stmt](#) *A positive blocked  $\chi$  trace-power witness gives, for every positive blocked size  $n$ , the formula  $c_{i,j,k}^{(n)} = \text{tr}(\chi_{n,i,j,k}^n)$ .*

*Proof.* [✓ Proof](#) Apply the trace reformulation of the trace-power identity from the witness.  $\square$

## 23.10 Commuting form and GSNNCH

**Definition 23.134** (Commuting-form data for an MPDO). [Lean](#) [✓ Stmt](#) For a fixed chain length  $N \geq 2$ , this consists of a positive semidefinite two-site operator  $B$  such that its translated copies on the periodic chain commute pairwise.

**Definition 23.135** (Commuting-form property). [Lean](#) [✓ Stmt](#) An MPO tensor has the commuting-form property if for every  $N \geq 2$  its  $N$ -site operator is a positive scalar multiple of the product of the translated copies of some commuting two-site operator  $B$ .

**Definition 23.136** (GSNNCH). [Lean](#) [✓ Stmt](#) An MPO tensor is a Gibbs state of a nearest-neighbor commuting Hamiltonian if every finite-chain operator admits the equivalent positive-operator presentation  $\rho^{(N)} = c \prod_i B_{i,i+1}$  with  $c > 0$  and commuting nearest-neighbor factors. The projector-limit convention in [CPGSV16] identifies this with the exponential Gibbs form.

**Theorem 23.137** (GSNNCH iff commuting form). [Lean](#) [✓ Stmt](#) *For an MPO tensor, the GSNNCH condition is equivalent to the existence of commuting-form data on every finite chain.*

*Proof.* [✓ Proof](#) A positive normalization constant converts commuting-form data into a GSNNCH witness, and every GSNNCH witness remembers its underlying commuting-form data.  $\square$

**Definition 23.138** (GSNNCH with zero correlation length). [Lean](#) [✓ Stmt](#) This is the conjunction of the GSNNCH condition with MPO zero correlation length, matching item (iii) of the simple-MPDO equivalence in [CPGSV16].

**Theorem 23.139** (GSNNCH-ZCL iff commuting form with ZCL). [Lean](#) [✓ Stmt](#) *The GSNNCH-ZCL branch is equivalent to the conjunction of the commuting-form property with MPO zero correlation length.*

*Proof.* [✓ Proof](#) Unfold the definition of GSNNCH with zero correlation length and apply Theorem 23.137.  $\square$

**Definition 23.140** (Translation-invariant two-site bond). [Lean](#) [✓ Stmt](#) This records a single positive two-site bond together with its translated copies on every finite periodic chain, requiring those translated copies to commute pairwise.

**Definition 23.141** ( $\eta$ -local commuting-form data). [Lean](#) [✓ Stmt](#) This defines the eta-local form of Appendix C.2: the translation-invariant positive two-site bond data together with the requirement that the corresponding product realizes the MPO at every finite length.

**Definition 23.142** (Finite-chain data from a translation-invariant bond). [Lean](#) [✓ Stmt](#) A translation-invariant positive two-site bond determines finite-chain commuting-form data at every length  $N \geq 2$ .

**Theorem 23.143** (Component identities for finite-chain bond data). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The finite-chain data associated to a translation-invariant bond retains the original local bond and the supplied lower bound  $2 \leq N$ , and the factor at site  $i$  is the translated copy of that local two-site bond.*

*Proof.* [✓ Proof](#) These are the component identities for the finite-chain construction. □

**Theorem 23.144** ( $\eta$ -local structure carries commuting form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The  $\eta$ -local structure itself determines a commuting-form witness for every chain length  $N \geq 2$ . In particular, the translated nearest-neighbor bonds commute and the MPDO at length  $N$  is a positive scalar multiple of their product. This is the commuting-form half of [CPGSV16, Proposition C.8], taking the  $\eta$ -local data as given; deriving that data from the saturation of the area law is the deferred step recorded in Remark 23.165.*

*Proof.* [✓ Proof](#) At length  $N$ , apply the commuting two-site bond and realization identity provided by the  $\eta$ -local structure. □

**Theorem 23.145** ( $\eta$ -local structure gives GSNNCH). [Lean](#) [✓ Stmt](#) *The same  $\eta$ -local structure gives a GSNNCH witness on every finite periodic chain.*

*Proof.* [✓ Proof](#) Evaluate the local structure at each chain length  $N \geq 2$ . □

**Theorem 23.146** ( $\eta$ -local structure with ZCL gives GSNNCH–ZCL). [Lean](#) [Lean](#) [✓ Stmt](#) *Once the  $\eta$ -local structure gives the commuting nearest-neighbor product form, adding zero correlation length gives the GSNNCH–ZCL case of the simple-MPDO equivalence.*

*Proof.* [✓ Proof](#) Use the commuting-form witness carried by the  $\eta$ -local structure and combine it with the ZCL hypothesis. □

**Theorem 23.147** ( $\eta$ -local structure implies commuting form). [Lean](#) [✓ Stmt](#) *Once the explicit neighboring operators  $\eta_{k,h}$  have been assembled into an  $\eta$ -local structure, the global commuting-form property follows.*

*Proof.* [✓ Proof](#) This is the commuting-form consequence already carried by the  $\eta$ -local structure. □

**Theorem 23.148** ( $\eta$ -local structure implies GSNNCH). [Lean](#) [✓ Stmt](#) *Once the explicit neighboring operators  $\eta_{k,h}$  have been assembled into the  $\eta$ -local structure, the GSNNCH condition follows.*

*Proof.* [✓ Proof](#) This is the GSNNCH witness already carried by the  $\eta$ -local structure. □

**Remark 23.149** (Entropy-side construction). The remaining entropy-side step is the construction of a single  $\eta$ -local product form from  $\text{SAL}(K)$ . The local  $\eta_{k,h}$  are already supplied by Definition 23.37. What remains is the tensor-coordinate assembly

$$B = \sum_{k,h} \text{Id}_{H_L^{(k)}} \otimes \eta_{k,h} \otimes \text{Id}_{H_R^{(h)}}, \quad B \geq 0, \quad (23.5)$$



and, for every  $N \geq 2$ ,

$$\sigma^{(N)}(K) = c_N \prod_{n=1}^N B_{n,n+1}, \quad c_N > 0, \quad [B_{i,i+1}, B_{j,j+1}] = 0. \quad (23.6)$$

(23.5) and (23.6) are exactly the  $\eta$ -local product hypothesis; after that, Theorem 23.147 gives the commuting-form conclusion. Zero correlation length enters only in the subsequent RFP and GSNCH–ZCL conclusions.

## 23.11 Blocked-RFP construction for simple MPDOs

**Definition 23.150** (Local simple-MPDO structure). [Lean](#) [✓ Stmt](#) This structure records the local Appendix C.2 ingredients: a normalized three-site reduced state with equality in strong subadditivity, the resulting local  $\eta$ -structure, and the rank-one conclusion for the auxiliary primitive matrix  $T$ , supplied here through the conditional rank-one criterion of Theorem 23.43.

**Theorem 23.151** (Local simple-MPDO structure from SAL–ZCL hypotheses). [Lean](#) [Lean](#) [✓ Stmt](#) The hypotheses of Lemmas C.2, C.4, and C.5 determine the local simple-MPDO structure. There are two forms. The conditional form assumes the rank-one implication for  $T$ . The corrected form assumes instead that  $T \geq 0$  as a matrix, with  $\text{tr}(T) = 1$  and  $\text{tr}(T^m) = 1$  for every  $m > 0$ , and then applies Theorem 23.44.

**Definition 23.152** (Simple-MPDO blocked-RFP structure). [Lean](#) [✓ Stmt](#) For an MPO tensor  $K$ , this records the local Appendix C.2 ingredients, the commuting-form property, and zero correlation length. The local component consists of the hypotheses used in the entropy-side argument for commuting form.

**Theorem 23.153** (Blocked-RFP structure from SAL–ZCL and commuting form). [Lean](#) [Lean](#) [✓ Stmt](#) The SAL–ZCL hypotheses together with a commuting-form witness determine the simple-MPDO blocked-RFP structure. In the PSD-corrected form, the matrix input is  $T \geq 0$ ,  $\text{tr}(T) = 1$ , and  $\text{tr}(T^m) = 1$  for all  $m > 0$ , rather than an abstract Perron–Frobenius rank-one assumption for  $T$ .

**Theorem 23.154** (Blocked-RFP structure from  $\eta$ -local structure). [Lean](#) [✓ Stmt](#) Given the local Appendix C.2 data, ZCL, and an  $\eta$ -local product  $\sigma^{(N)}(K) = c_N \prod_n B_{n,n+1}$  with  $c_N > 0$ ,  $B_{n,n+1} \geq 0$ , and  $[B_{i,i+1}, B_{j,j+1}] = 0$ , one obtains the simple-MPDO blocked-RFP structure for  $K$ .

*Proof.* [✓ Proof](#) The  $\eta$ -local product gives the commuting-form property for  $K$ ; the local Appendix C.2 hypotheses and ZCL fill the other two fields.  $\square$

**Theorem 23.155** (Blocked-RFP structure from SAL–ZCL and  $\eta$ -local structure). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose Lemma C.2 gives the Markov decomposition, Lemma C.4 gives the local  $\eta$ -structure and a primitive matrix  $T$  with  $\text{tr}(T) = 1$ , and ZCL together with Lemma C.5 gives  $\text{tr}(T^m)$  independent of  $m$  and the conditional rank-one criterion for  $T$ . Suppose also that the assembled  $\eta$ -local structure gives  $\sigma^{(N)}(K) = c_N \prod_{n=1}^N B_{n,n+1}$  with  $c_N > 0$ ,  $B_{n,n+1} \geq 0$ , and  $[B_{i,i+1}, B_{j,j+1}] = 0$  for all  $i, j$ . Then these hypotheses determine the simple-MPDO blocked-RFP structure for  $K$ . In the PSD-corrected form, the conditional rank-one criterion is replaced by  $T \geq 0$ ,  $\text{tr}(T) = 1$ , and  $\text{tr}(T^m) = 1$  for all  $m > 0$ .

*Proof.* [✓ Proof](#) First form the local structure from the Lemma C.2 Markov decomposition, the Lemma C.4 local  $\eta$ -structure and primitivity statement, and the Lemma C.5 conditional rank-one criterion for  $T$ . The  $\eta$ -local product formula supplies the commuting-form component, and ZCL supplies the remaining component of the blocked-RFP structure.  $\square$

**Theorem 23.156** (Zero correlation length gives a blocked fusion-isometry witness). [Lean](#) [✓ Stmt](#) *If an MPO tensor  $K$  has zero correlation length, then it has a blocked fusion-isometry witness at size 2 in the sense of Definition 23.51.*

*Proof.* [✓ Proof](#) Zero correlation length is transfer-map idempotence. Applying the converse direction of Theorem 23.55 to that RFP witness yields fusion-isometry data at every positive blocked size; evaluating it at size 2 produces the blocked witness.  $\square$

**Theorem 23.157** (Blocked-RFP data give a blocked fusion-isometry witness). [Lean](#) [✓ Stmt](#) *Given the blocked-RFP data for an MPO tensor  $K$ , one obtains a blocked fusion-isometry witness at size 2.*

*Proof.* [✓ Proof](#) Apply the zero-correlation-length theorem to the ZCL component of the blocked-RFP data.  $\square$

**Lemma 23.158** (Blocked-RFP data give the fusion formulation). [Lean](#) [✓ Stmt](#) *Given the blocked-RFP data for an MPO tensor  $K$ , one obtains the transfer-map fusion formulation of idempotence.*

*Proof.* [✓ Proof](#) The implication used here is

$$\text{ZCL}(K) \implies \text{RFP}(K) \implies \text{RFP}_{\text{fusion}}(K).$$

$\square$

**Theorem 23.159** (Simple-MPDO RFP chain from blocked-RFP data). [Lean](#) [✓ Stmt](#) *Given the blocked-RFP data for an MPO tensor  $K$ , one obtains the GSNNCH-with-ZCL case, the blocked fusion-isometry witness at size 2, and the transfer-map fusion formulation of idempotence.*

*Proof.* [✓ Proof](#) The commuting-form and zero-correlation-length assumptions give the GSNNCH-with-ZCL case. Zero correlation length also gives the blocked fusion-isometry witness and the transfer-map fusion formulation.  $\square$

**Lemma 23.160** (Size-2 fusion witness from  $\eta$ -local structure). [Lean](#) [✓ Stmt](#) *If the  $\eta$ -local structure has been constructed for an MPO tensor  $K$ , then zero correlation length gives the blocked fusion-isometry witness at size 2.*

*Proof.* [✓ Proof](#) The implication used here is

$$\text{ZCL}(K) \implies \text{RFP}(K) \implies \text{FusionWitness}_2(K).$$

$\square$

**Lemma 23.161** (Fusion formulation from  $\eta$ -local structure). [Lean](#) [✓ Stmt](#) *If the  $\eta$ -local structure has been constructed for an MPO tensor  $K$ , then zero correlation length gives the transfer-map fusion formulation of idempotence.*

*Proof.* [✓ Proof](#) This is the implication

$$\text{ZCL}(K) \implies \text{RFP}(K) \implies \text{RFP}_{\text{fusion}}(K).$$

$\square$

**Theorem 23.162** (Simple-MPDO RFP chain from  $\eta$ -local structure). [Lean](#) [Lean](#) [✓ Stmt](#) *If the  $\eta$ -local structure has been constructed for an MPO tensor  $K$ , then zero correlation length gives the GSNNCH-with-ZCL case, the blocked fusion-isometry witness at size 2, and the transfer-map fusion formulation of idempotence.*



*Proof.* ✓ Proof The  $\eta$ -local structure supplies the commuting nearest-neighbor product form. The remaining two implications are

$$\text{ZCL}(K) \implies \text{RFP}(K) \implies \text{RFP}_{\text{fusion}}(K) \quad \text{and} \quad \text{ZCL}(K) \implies \text{RFP}(K) \implies \text{FusionWitness}_2(K).$$

□

**Theorem 23.163** (Simple-MPDO RFP chain from SAL–ZCL and  $\eta$ -local structure). Lean Lean ✓ Stmt Assume the local SAL–ZCL hypotheses of Lemmas C.2, C.4, and C.5 and the assembled  $\eta$ -local product form, with the conditional rank-one criterion for  $T$ ,  $\sigma^{(N)}(K) = c_N \prod_{n=1}^N B_{n,n+1}$  with  $c_N > 0$ ,  $B_{n,n+1} \geq 0$ , and  $[B_{i,i+1}, B_{j,j+1}] = 0$ . Then  $K$  is GSNNCH with ZCL, admits a blocked fusion-isometry witness at size 2, and satisfies the transfer-map fusion formulation of idempotence. The PSD-corrected form uses  $T \geq 0$  together with  $\text{tr}(T^m) = 1$  for all  $m > 0$  in place of the conditional matrix rank-one hypothesis.

*Proof.* ✓ Proof Combine the local SAL–ZCL hypotheses and the  $\eta$ -local product form into the blocked-RFP structure, then apply Theorem 23.159. □

**Theorem 23.164** (Simple-MPDO RFP chain from commuting form and ZCL). Lean ✓ Stmt From the commuting-form property and zero correlation length one simultaneously recovers the GSNNCH-with-ZCL criterion, the blocked fusion-isometry witness at size 2, and the transfer-map fusion formulation of idempotence.

*Proof.* ✓ Proof The commuting-form and zero-correlation-length hypotheses give the GSNNCH–ZCL branch via Theorem 23.139. The blocked witness is Theorem 23.156. The full transfer-map fusion formulation follows by applying the converse direction of Theorem 23.55 to zero correlation length, which is transfer-map idempotence. □

*Remark 23.165* (SAL–ZCL implication for the GSNNCH–ZCL criterion). Not ready The implication from SAL and ZCL alone to the conclusion of Appendix C.2 requires a derivation of the commuting-form property from the local simple-MPDO structure of Lemmas C.2, C.4, and C.5. Once that derivation is available, Theorem 23.164 gives the GSNNCH–ZCL branch and the blocked fusion-isometry witness.

## 23.12 Saturation of the area law

**Definition 23.166** (Block entropy of an MPDO). Lean ✓ Stmt For a fixed system size  $N$ , a block length  $L \leq N$ , and an MPO tensor  $M$  such that  $\rho^{(N)}(M)$  is positive semidefinite, the  $L$ -block entropy  $S_L^{(N)}(M)$  is the von Neumann entropy of the reduced state of the first  $L$  of  $N$  spins of the normalized state  $\sigma^{(N)}(M) = \rho^{(N)}(M) / \text{tr}[\rho^{(N)}(M)]$ .

**Definition 23.167** (Mutual information of an MPDO). Lean ✓ Stmt For the same fixed system size  $N$ , block length  $L \leq N$ , and positive semidefiniteness of  $\rho^{(N)}(M)$ , the mutual information between a block of  $L$  spins and the rest is  $I_L = S_L^{(N)}(M) + S_{N-L}^{(N)}(M) - S_N^{(N)}(M)$ .

**Definition 23.168** (Saturation of the area law). Lean ✓ Stmt An MPO tensor  $M$  that generates an MPDO and whose finite-chain operators  $\rho^{(N)}(M)$  all have nonzero trace (so the normalized states are well-defined) saturates the area law if  $I_L = I_{L+1}$  for all  $1 \leq L < \lfloor N/2 \rfloor$  and all  $N$ ; that is, the mutual information is constant,  $I_1 = I_2 = \dots = I_{\lfloor N/2 \rfloor}$ . This is [CPGSV16, Definition 4.6, line 811].

**Proposition 23.169** (Monotonicity of the mutual information). [Lean](#) [✓ Stmt](#) Let  $M$  generate an MPDO whose finite-chain operator  $\rho^{(N)}(M)$  is positive semidefinite with nonzero trace. For every block length  $L$  with  $2L + 1 \leq N$ ,

$$I_L \leq I_{L+1}.$$

The implication

$$1 \leq L < \lfloor N/2 \rfloor \implies 2L + 1 \leq N$$

shows that this contains the range of [\[CPGSV16, Proposition 4.5, line 801\]](#).

*Proof.* [✓ Proof](#) Apply strong subadditivity to the reduced state of the first  $N - L$  spins, split into three contiguous segments of lengths 1,  $L$ , and  $N - 2L - 1$ . Tracing the appropriate segments yields the four block entropies  $S_{N-L}$ ,  $S_L$ ,  $S_{L+1}$ , and  $S_{N-L-1}$ , where the cyclic invariance of  $\rho^{(N)}(M)$  identifies the entropy of a contiguous block with that of any block of the same length. Strong subadditivity then gives  $S_{N-L} + S_L \leq S_{L+1} + S_{N-L-1}$ , which rearranges to  $I_L \leq I_{L+1}$  after subtracting  $S_N$ .  $\square$

**Definition 23.170** (Block entropy and area-law saturation for pure states). [Lean](#) [Lean](#) [✓ Stmt](#) For an MPS tensor  $A$ , a system size  $N$ , and a block length  $L \leq N$ , the  $L$ -block entropy  $S_L^{(N)}(A)$  is the von Neumann entropy of the reduced state of the first  $L$  spins of

$$\sigma^{(N)}(A) = \frac{|V^{(N)}(A)\rangle\langle V^{(N)}(A)|}{\|V^{(N)}(A)\|^2}.$$

The tensor saturates the area law if  $S_L^{(N)}(A) = S_{L+1}^{(N)}(A)$  for all  $1 \leq L < \lfloor N/2 \rfloor$  and all  $N$ . This is [\[CPGSV16, Definition 3.13, line 600\]](#).

**Definition 23.171** (Purification tensor). [Lean](#) [✓ Stmt](#) For an MPS tensor  $A$  on bond space  $\mathbb{C}^D$ , define the associated MPO tensor  $M$  on bond space  $\mathbb{C}^{D^2}$  by

$$M^{ij} = A^i \otimes \bar{A}^j.$$

This is the single-ancilla purification tensor associated with [\[CPGSV16, eq. \(line 752\)\]](#).

**Lemma 23.172** (Pure state as a purification MPDO). [Lean](#) [✓ Stmt](#) For an MPS tensor  $A$ , the pure-state operator  $|V^{(N)}(A)\rangle\langle V^{(N)}(A)|$  equals the MPDO generated by the tensor  $M^{ij} = A^i \otimes \bar{A}^j$ , acting on a bond space of dimension  $D^2$ :

$$\rho^{(N)}(A \otimes \bar{A}) = |V^{(N)}(A)\rangle\langle V^{(N)}(A)|.$$

This is the single-ancilla case of the purification picture of [\[CPGSV16, eq. \(line 752\)\]](#); it lets the MPDO block-entropy theory apply to pure-state block entropies.

*Proof.* [✓ Proof](#) The word product of doubled-tensor entries is the Kronecker product of the bra word product and the conjugated ket word product (mixed-product property of the Kronecker product), so its trace factors as  $\overline{V^{(N)}(A)}(\tau) V^{(N)}(A)(\sigma)$ , the matrix element of  $|V^{(N)}(A)\rangle\langle V^{(N)}(A)|$ .  $\square$

**Lemma 23.173** (Block entropy of the purification tensor). [Lean](#) [✓ Stmt](#) For an MPS tensor  $A$ , system size  $N$ , and block length  $L \leq N$ , the MPDO block entropy of the tensor  $M^{ij} = A^i \otimes \bar{A}^j$  agrees with the pure-state block entropy:

$$S_L^{(N)}(M) = S_L^{(N)}(A).$$

*Proof.* ✓ Proof Lemma 23.172 gives  $\rho^{(N)}(M) = |V^{(N)}(A)\rangle\langle V^{(N)}(A)|$ , hence

$$\sigma^{(N)}(M) = \frac{\rho^{(N)}(M)}{\text{tr}[\rho^{(N)}(M)]} = \frac{|V^{(N)}(A)\rangle\langle V^{(N)}(A)|}{\|V^{(N)}(A)\|^2} = \sigma^{(N)}(A).$$

Therefore the reduced block states agree:

$$\rho_L^{(N)}(M) = \text{tr}_{N \setminus L}[\sigma^{(N)}(M)] = \text{tr}_{N \setminus L}[\sigma^{(N)}(A)] = \rho_L^{(N)}(A).$$

Thus

$$S_L^{(N)}(M) = S(\rho_L^{(N)}(M)) = S(\rho_L^{(N)}(A)) = S_L^{(N)}(A).$$

□

**Lemma 23.174** (Schmidt symmetry of the block entropy). Lean ✓ Stmt For an MPS tensor  $A$  and a block length  $L \leq N$ , the block entropies of a block and its complement coincide:

$$S_L^{(N)}(A) = S_{N-L}^{(N)}(A).$$

*Proof.* ✓ Proof The reduced state of the first  $L$  spins is  $\rho_L \propto WW^\dagger$ , where  $W$  is the matrix of amplitudes  $\langle uw | V^{(N)}(A) \rangle$  with  $u$  the first  $L$  spins and  $w$  the remaining  $N - L$ . By cyclic invariance of the amplitudes, the reduced state of the complement is  $\propto \overline{W}^\dagger \overline{W}$ . Since  $WW^\dagger$  and  $W^\dagger W$  share the same nonzero spectrum, and complex conjugation preserves the (real) eigenvalues of a Hermitian matrix, the two reduced states have equal von Neumann entropy. □

**Proposition 23.175** (Pure-state area-law monotonicity). Lean ✓ Stmt Let  $A$  be an MPS tensor with  $V^{(N)}(A) \neq 0$ . For every block length  $L$  with  $2L + 1 \leq N$  (covering  $1 \leq L < \lfloor N/2 \rfloor$ ),

$$S_L^{(N)}(A) \leq S_{L+1}^{(N)}(A).$$

This is the pure-state area law of [CPGSV16, Section 3, line 599].

*Proof.* ✓ Proof Apply the block-entropy form of strong subadditivity (the inequality underlying Proposition 23.169) to the purification MPDO of  $A$ , giving  $S_{N-L} + S_L \leq S_{L+1} + S_{N-L-1}$ . By the Schmidt symmetry  $S_{N-L} = S_L$  and  $S_{N-L-1} = S_{L+1}$ , this reduces to  $S_L \leq S_{L+1}$ . □

# Bibliography

- [And79] T. Ando. Concavity of certain maps on positive definite matrices and applications to Hadamard products. *Linear Algebra and Its Applications*, 26:203–241, 1979.
- [Bha97] Rajendra Bhatia. *Matrix Analysis*, volume 169 of *Graduate Texts in Mathematics*. Springer, New York, 1997.
- [CGW11] Xie Chen, Zheng-Cheng Gu, and Xiao-Gang Wen. Classification of gapped symmetric phases in one-dimensional spin systems. *Physical Review B*, 83(3):035107, 2011.
- [CPGSV16] J. I. Cirac, D. Pérez-García, N. Schuch, and F. Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: *Ann. Phys.* **378** (2017) 100–149, 2016.
- [CPGSV17] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product unitaries: Structure, symmetries, and topological invariants. *Journal of Statistical Mechanics: Theory and Experiment*, 2017(8):083105, 2017.
- [CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.
- [DICCSPG17] Gemma De las Cuevas, J. Ignacio Cirac, Norbert Schuch, and David Pérez-García. Irreducible forms of matrix product states: Theory and applications. *Journal of Mathematical Physics*, 58(12):121901, 2017.
- [EHK78] David E. Evans and Raphael Høegh-Krohn. Spectral properties of positive maps on  $C^*$ -algebras. *Journal of the London Mathematical Society*, 17(2):345–355, 1978.
- [FGSW<sup>+</sup>15] Carlos Fernández-González, Norbert Schuch, Michael M. Wolf, J. Ignacio Cirac, and David Pérez-García. Frustration free gapless Hamiltonians for Matrix Product States. *Communications in Mathematical Physics*, 333:299–333, 2015.
- [FNW92] M. Fannes, B. Nachtergaele, and R. F. Werner. Finitely correlated states on quantum spin chains. *Communications in Mathematical Physics*, 144(3):443–490, 1992.
- [Hay06] Masahito Hayashi. *Quantum Information: An Introduction*. Springer, Berlin, Heidelberg, 2006.
- [HJPW04] Patrick Hayden, Richard Jozsa, Dénes Petz, and Andreas Winter. Structure of states which satisfy strong subadditivity of quantum entropy with equality. *Communications in Mathematical Physics*, 246(2):359–374, 2004.

- [HP82] Frank Hansen and Gert K. Pedersen. Jensen’s inequality for operators and lowner’s theorem. *Mathematische Annalen*, 258(3):229–241, 1982.
- [Jac09] Nathan Jacobson. *Basic Algebra II*. Dover Publications, 2009. Unaltered reprint of the 2nd edition (W. H. Freeman, 1989).
- [KL18] Michael J. Kastoryano and Angelo Lucia. Divide and conquer method for proving gaps of frustration free Hamiltonians. *Journal of Statistical Mechanics: Theory and Experiment*, 2018(3):033105, 2018.
- [Lie73] Elliott H. Lieb. Convex trace functions and the Wigner–Yanase–Dyson conjecture. *Advances in Mathematics*, 11(3):267–288, 1973.
- [LR73] Elliott H. Lieb and Mary Beth Ruskai. Proof of the strong subadditivity of quantum-mechanical entropy. *Journal of Mathematical Physics*, 14(12):1938–1941, 1973.
- [MGRPG<sup>+</sup>18] András Molnár, José Garre-Rubio, David Pérez-García, Norbert Schuch, and J. Ignacio Cirac. Normal projected entangled pair states generating the same state. *New Journal of Physics*, 20(11):113017, 2018.
- [Nac96] Bruno Nachtergaele. The spectral gap for some spin chains with discrete symmetry breaking. *Communications in Mathematical Physics*, 175(3):565–606, 1996.
- [PGVWC07] D. Pérez-García, F. Verstraete, M. M. Wolf, and J. I. Cirac. Matrix product state representations. *Quantum Information & Computation*, 7(5):401–430, 2007.
- [PGWS<sup>+</sup>08] D. Pérez-García, M. M. Wolf, M. Sanz, F. Verstraete, and J. I. Cirac. String order and symmetries in quantum spin lattices. *Physical Review Letters*, 100(16):167202, 2008.
- [Rus02] Mary Beth Ruskai. Inequalities for quantum entropy: A review with conditions for equality. *Journal of Mathematical Physics*, 43(9):4358–4375, 2002.
- [SPGC11] Norbert Schuch, David Pérez-García, and Ignacio Cirac. Classifying quantum phases using matrix product states and projected entangled pair states. *Physical Review B*, 84(16):165139, 2011.
- [SPGWC10] M. Sanz, D. Pérez-García, M. M. Wolf, and J. I. Cirac. A quantum version of Wielandt’s inequality. *IEEE Transactions on Information Theory*, 56(9):4668–4673, 2010.
- [Wol12] Michael M. Wolf. Quantum channels & operations: Guided tour. Lecture notes, 2012. Originally distributed via TU Munich foswiki; archival PDF at <https://mediatum.ub.tum.de/doc/1099654/1099654.pdf>.